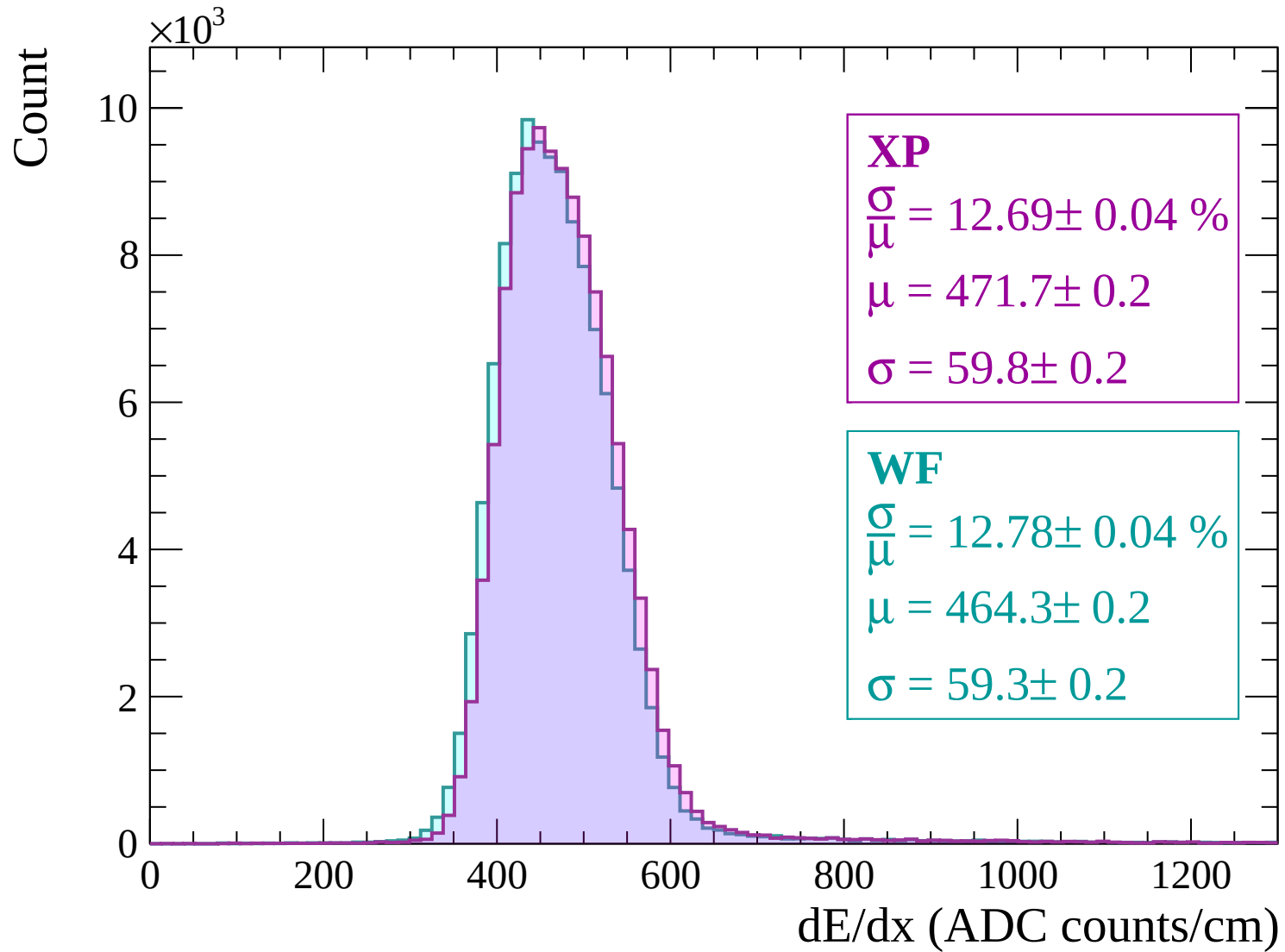
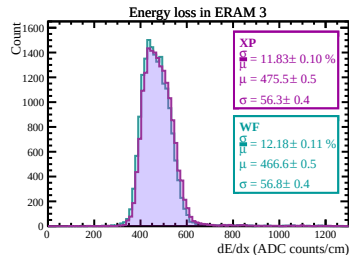
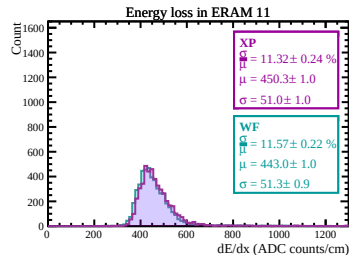
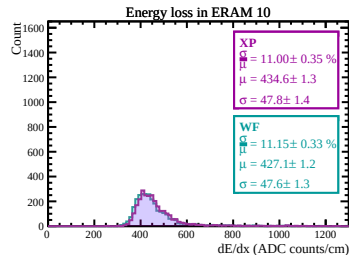
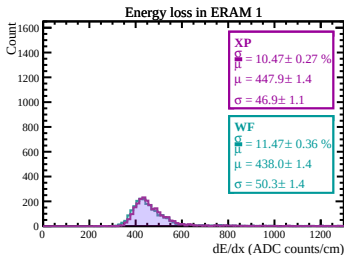
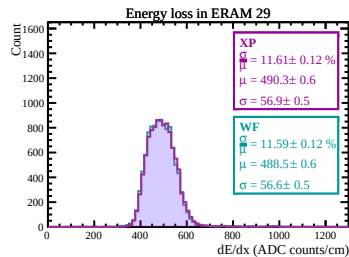
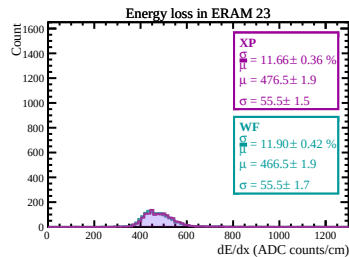
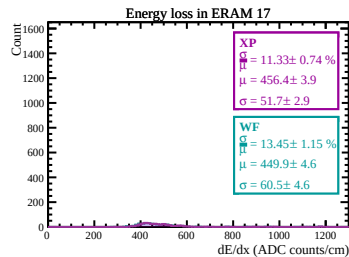
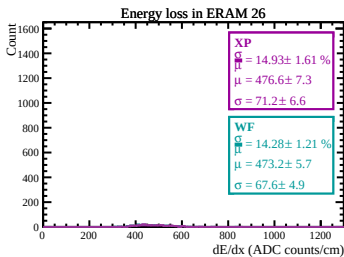
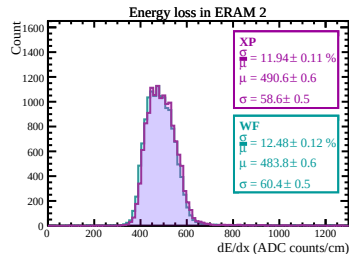
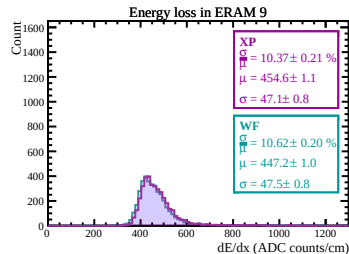
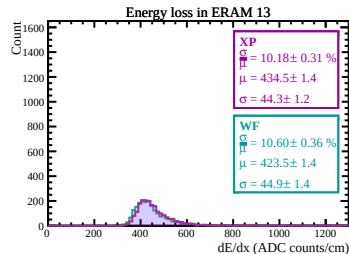
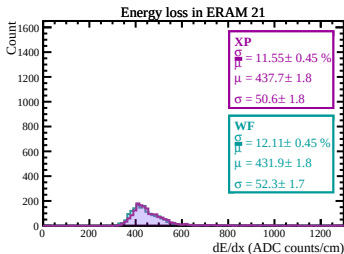
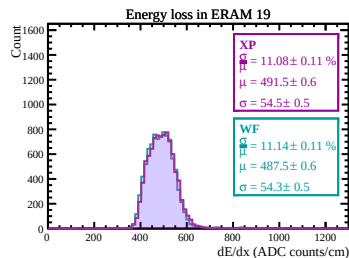
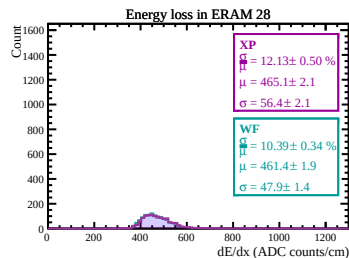
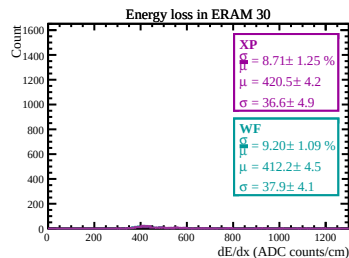
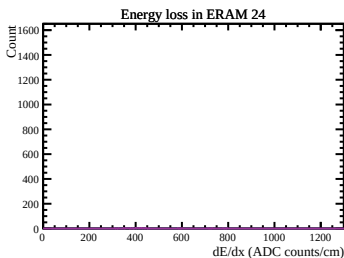
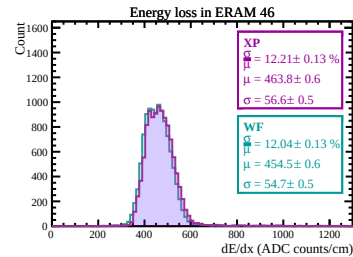
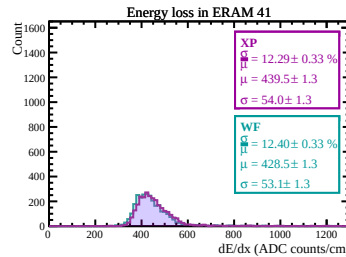
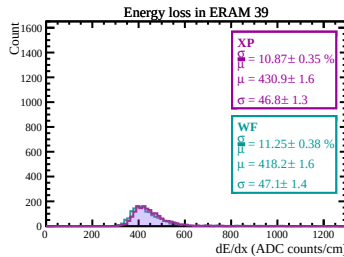
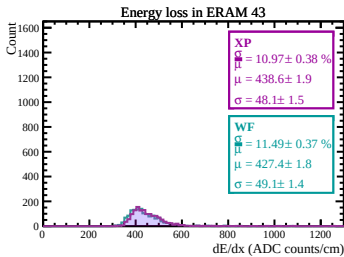
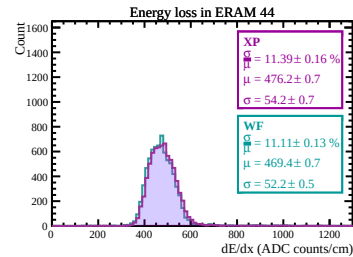
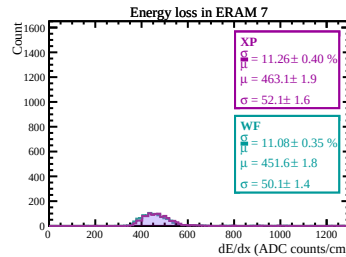
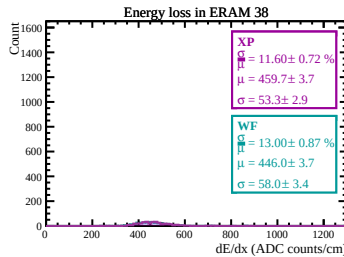
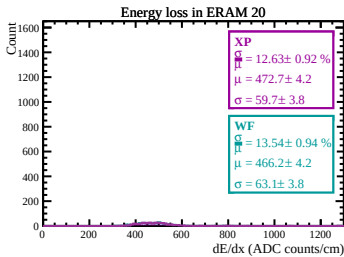
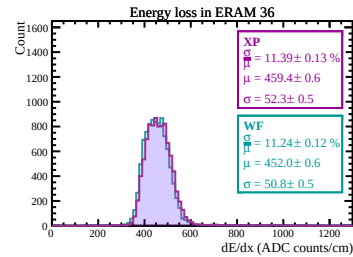
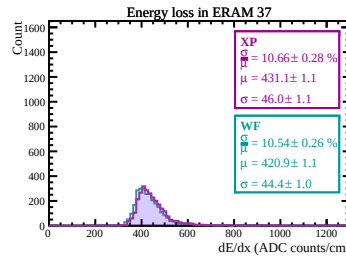
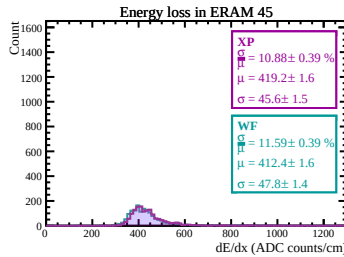
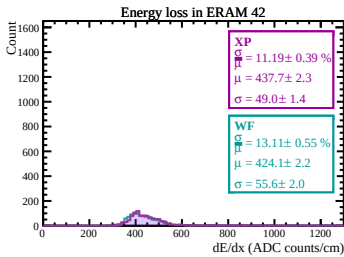
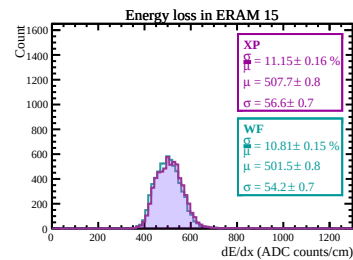
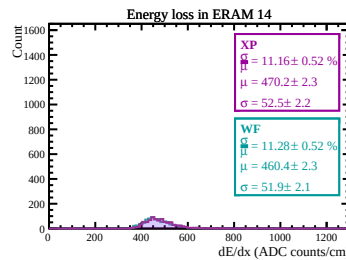
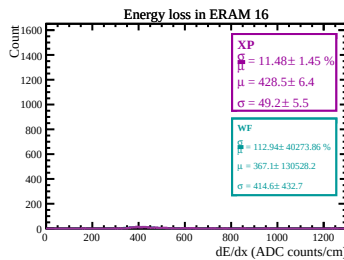
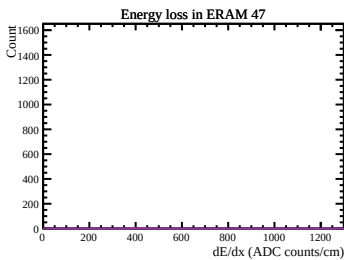
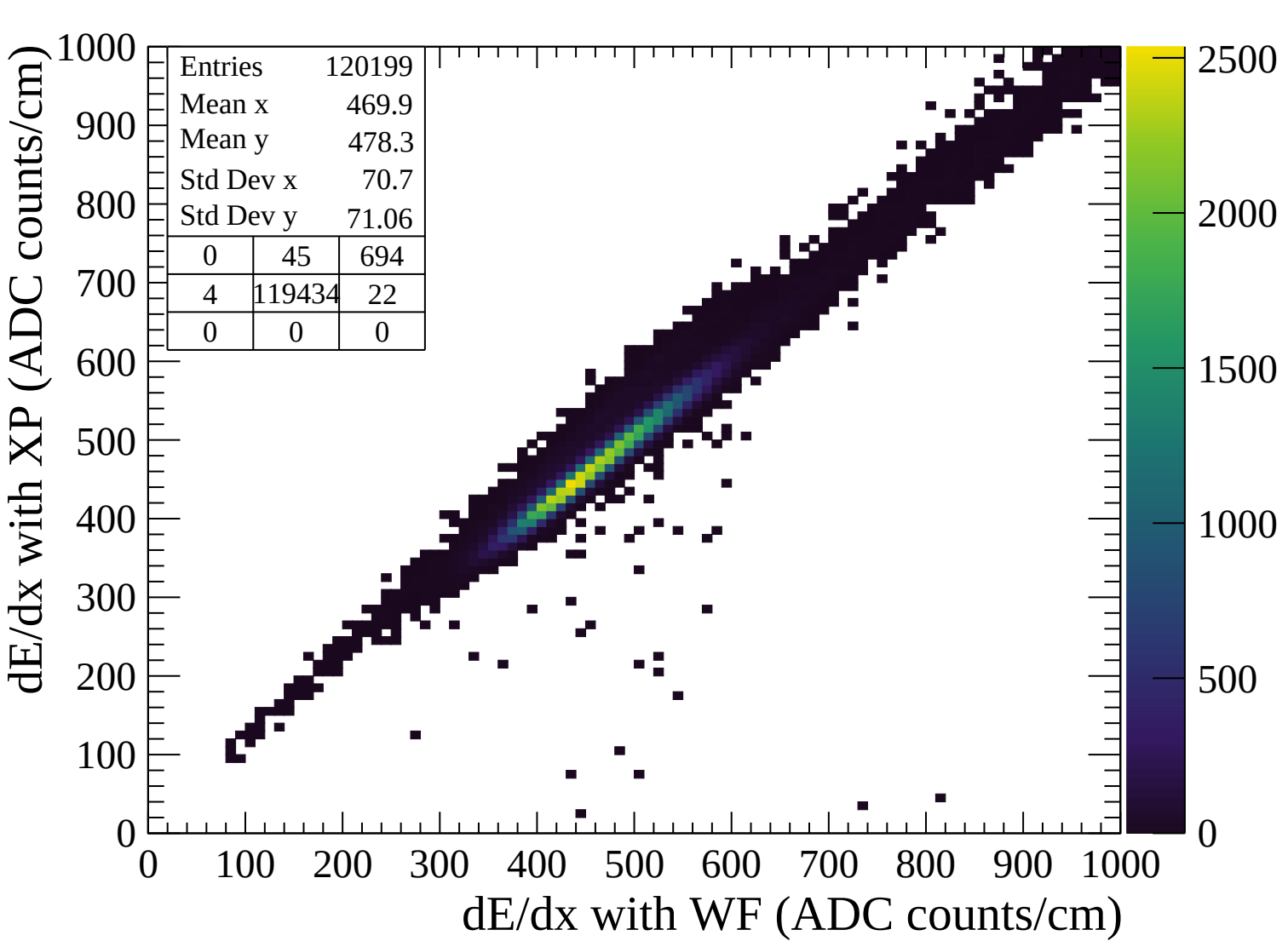
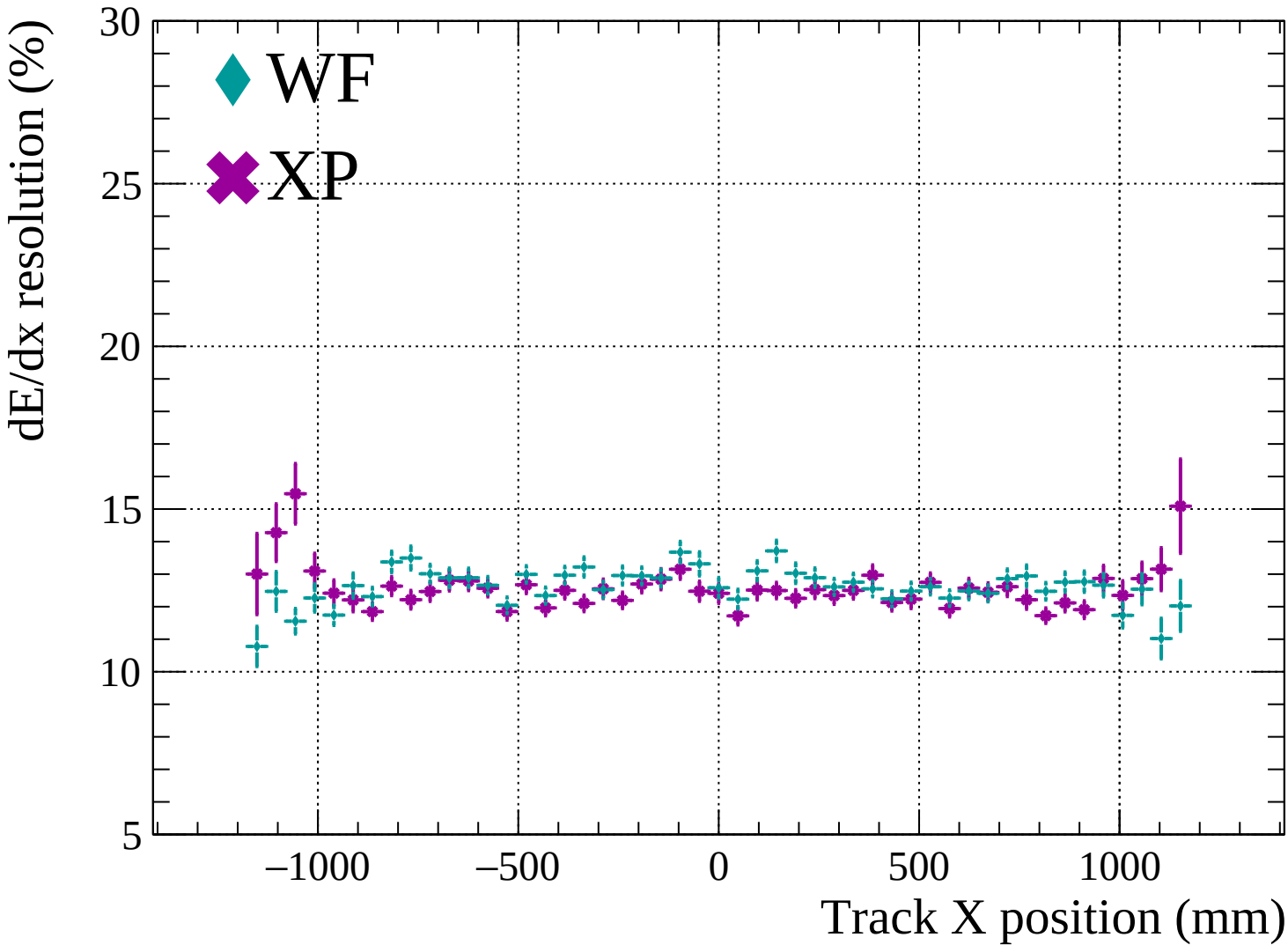


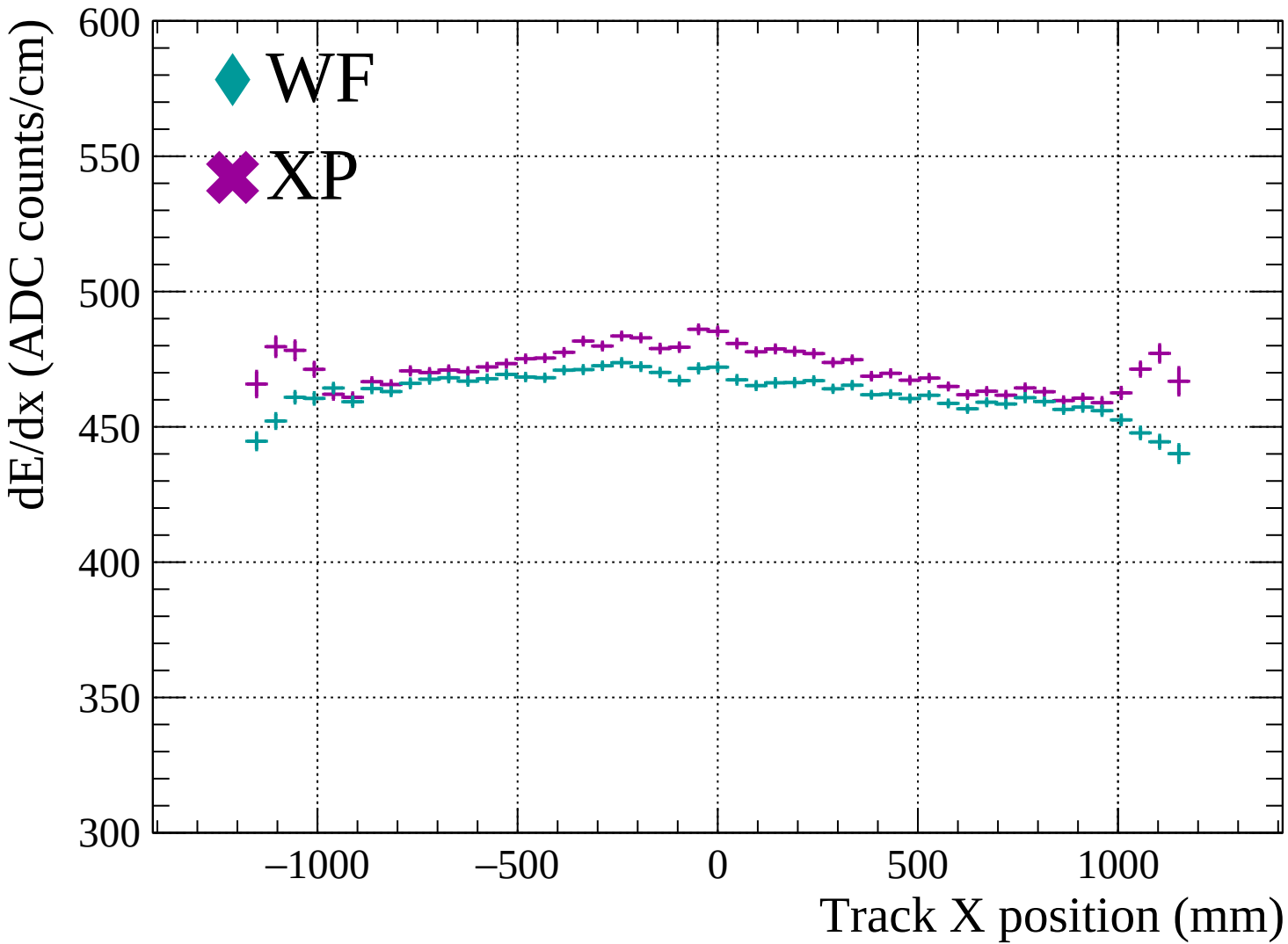


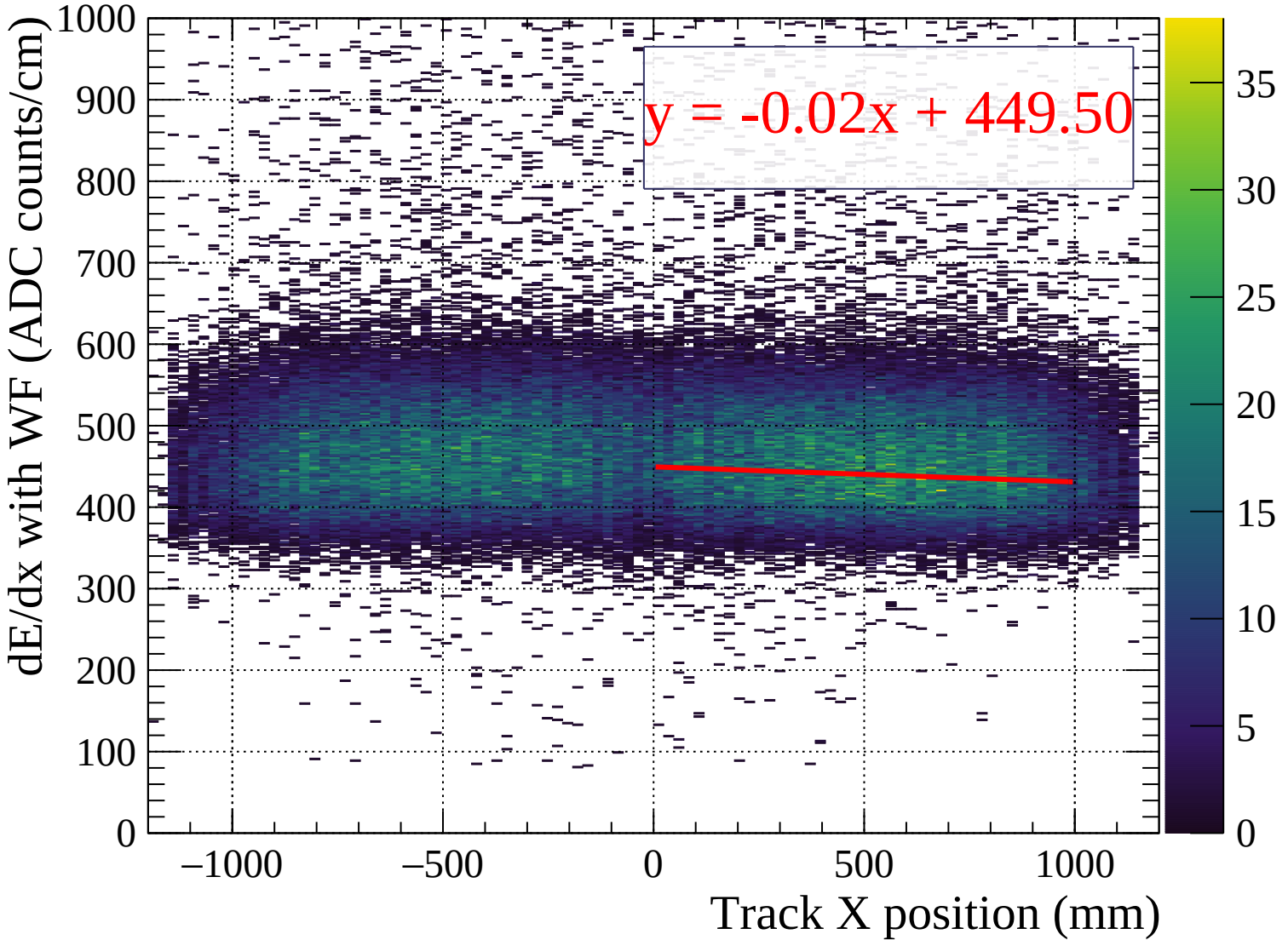


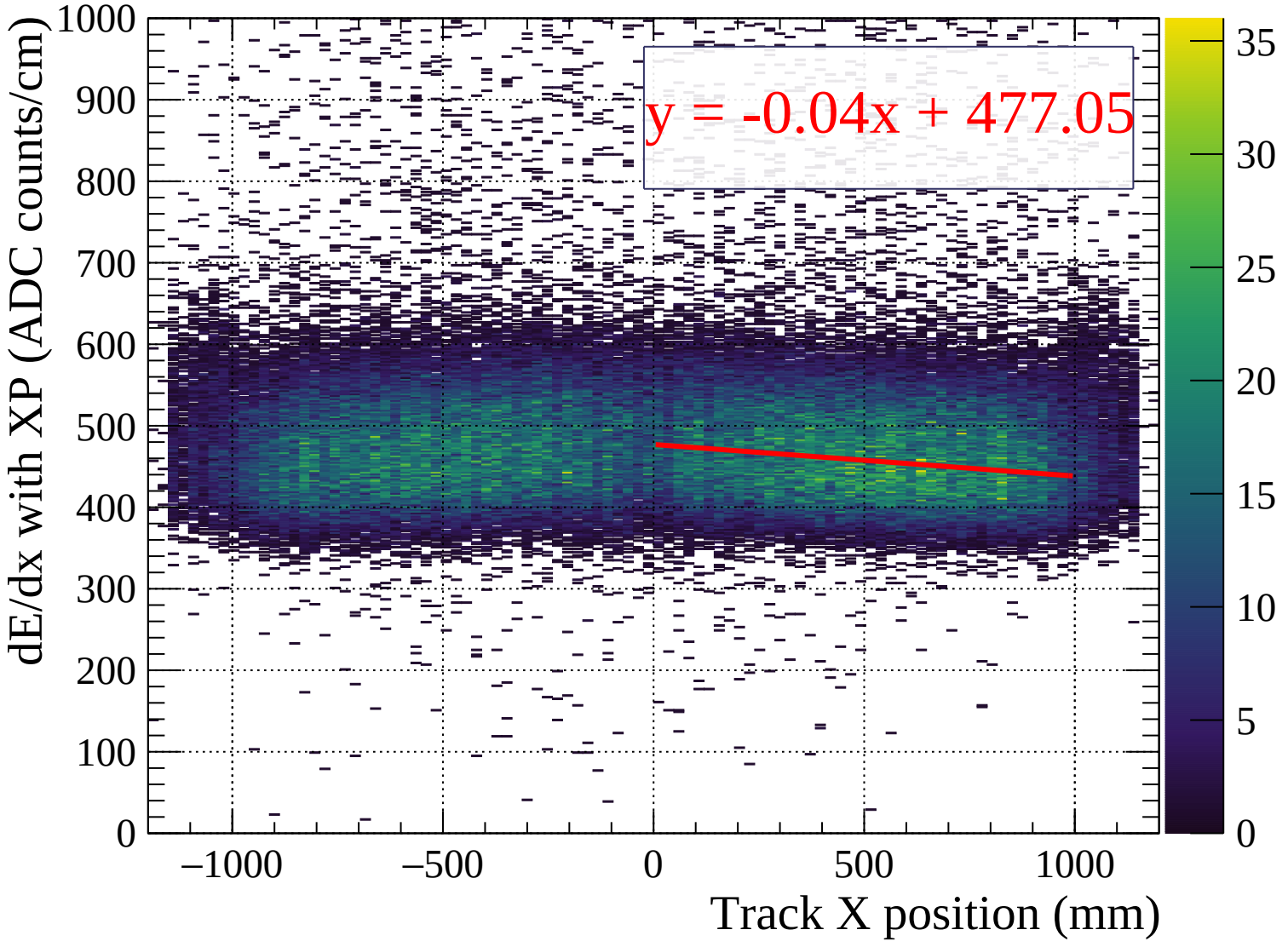


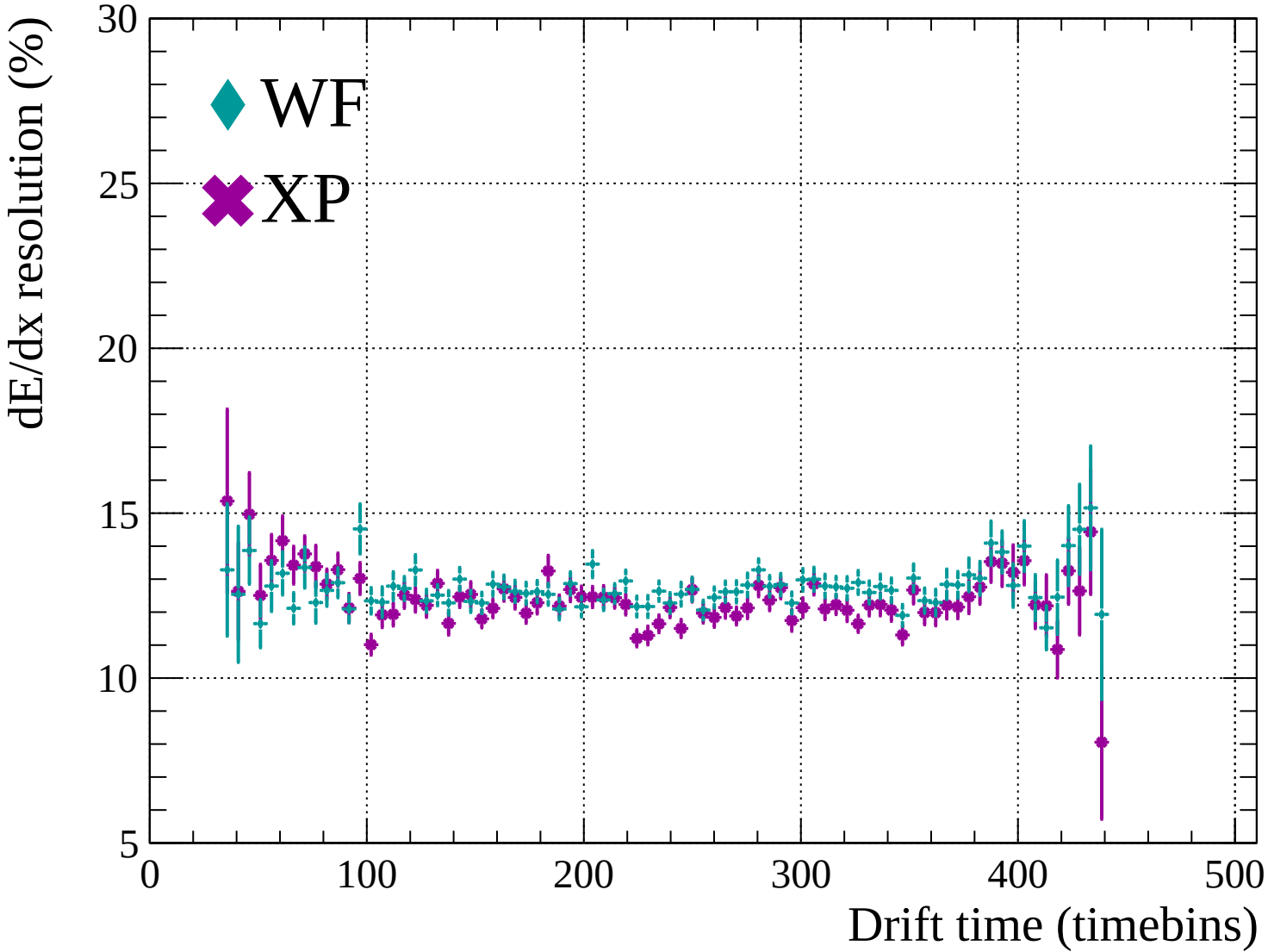


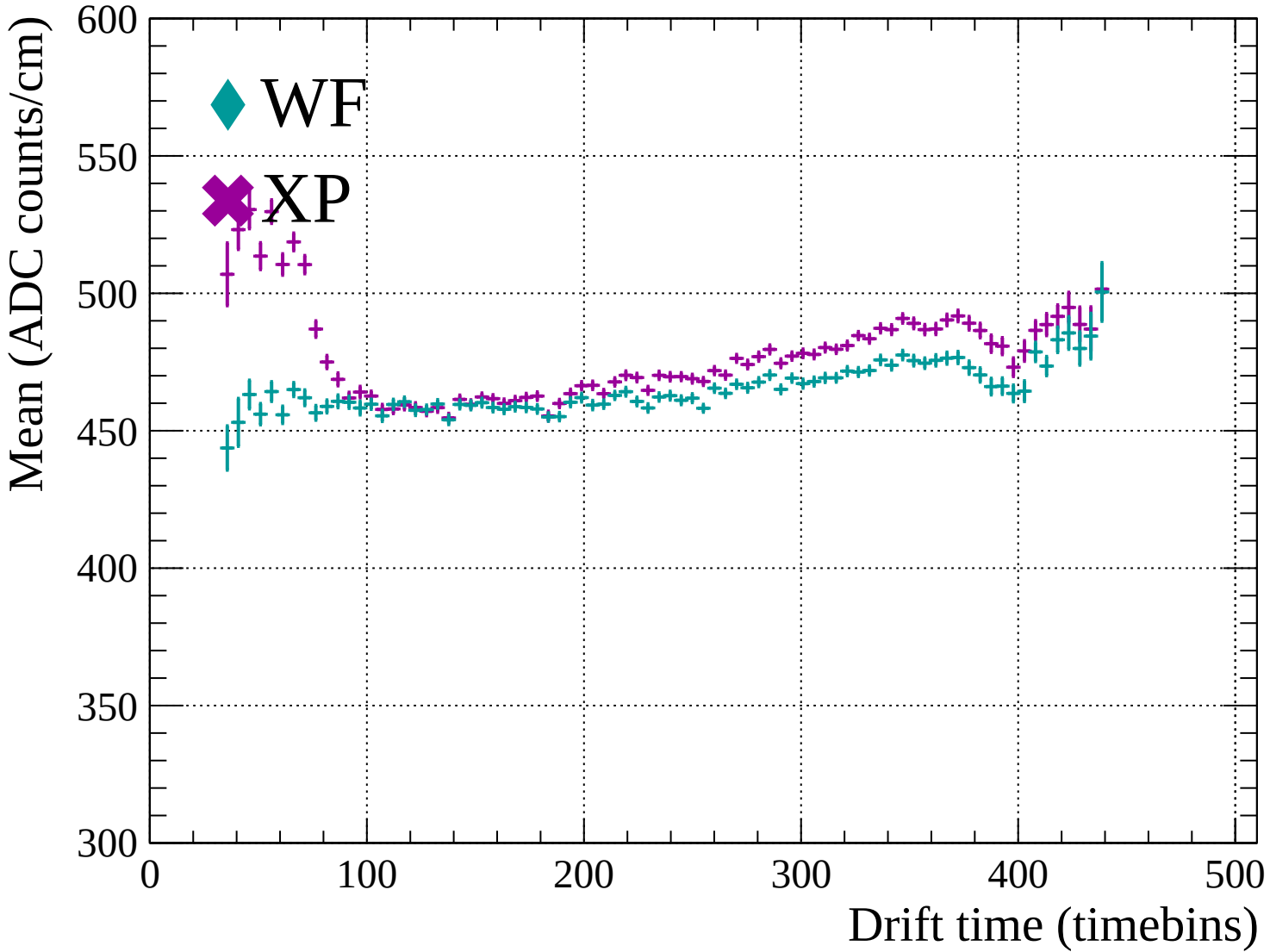


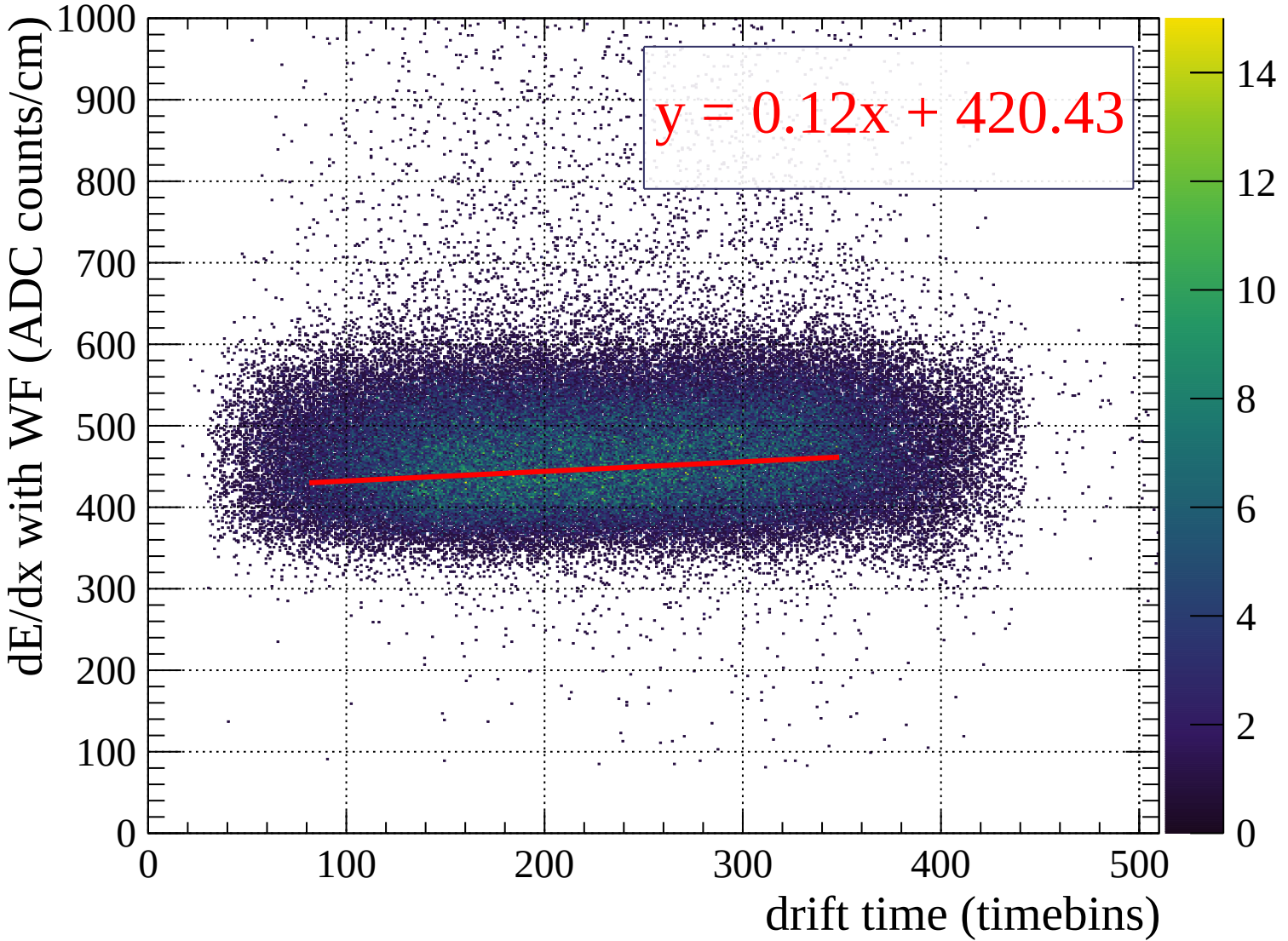


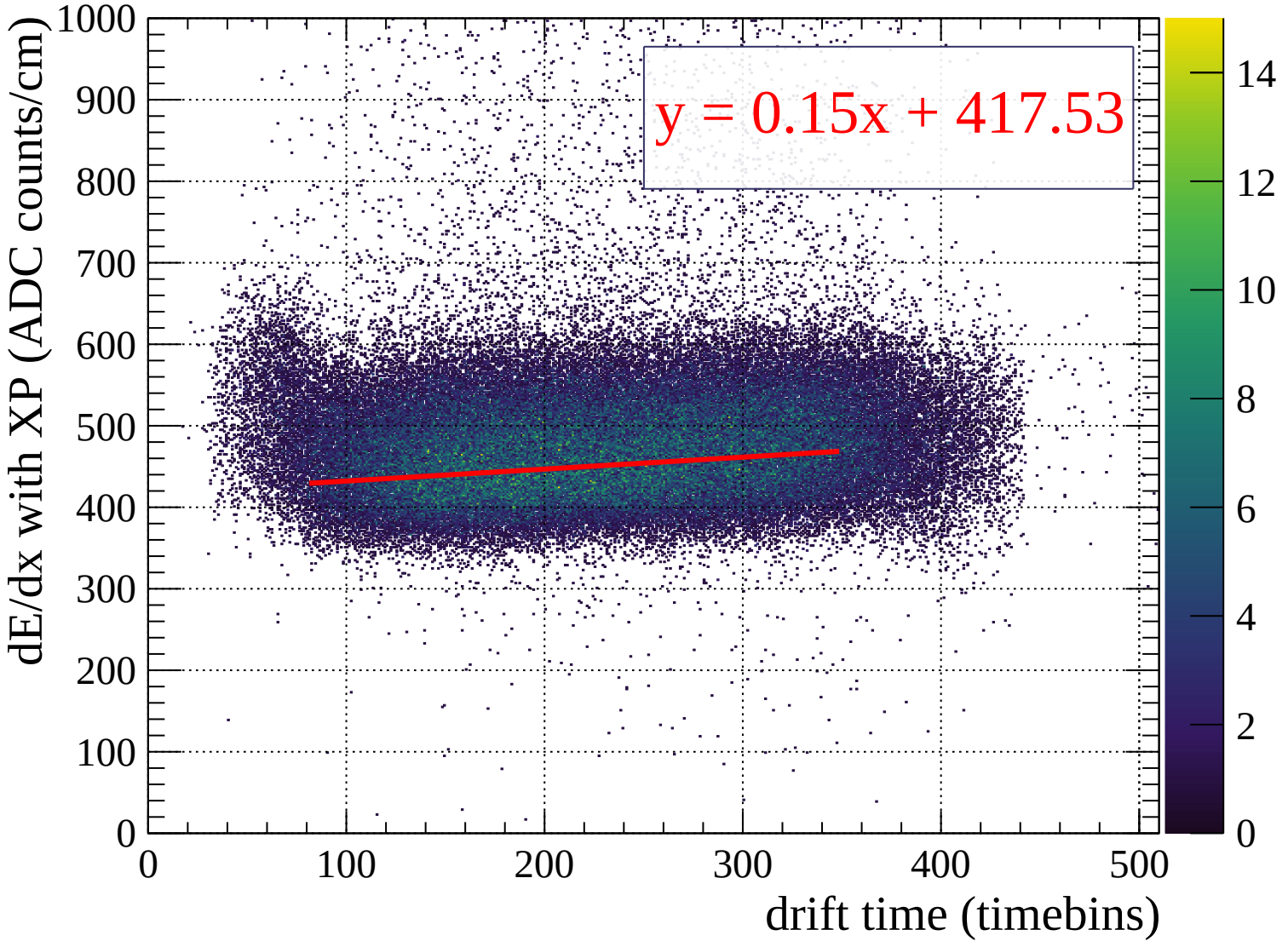


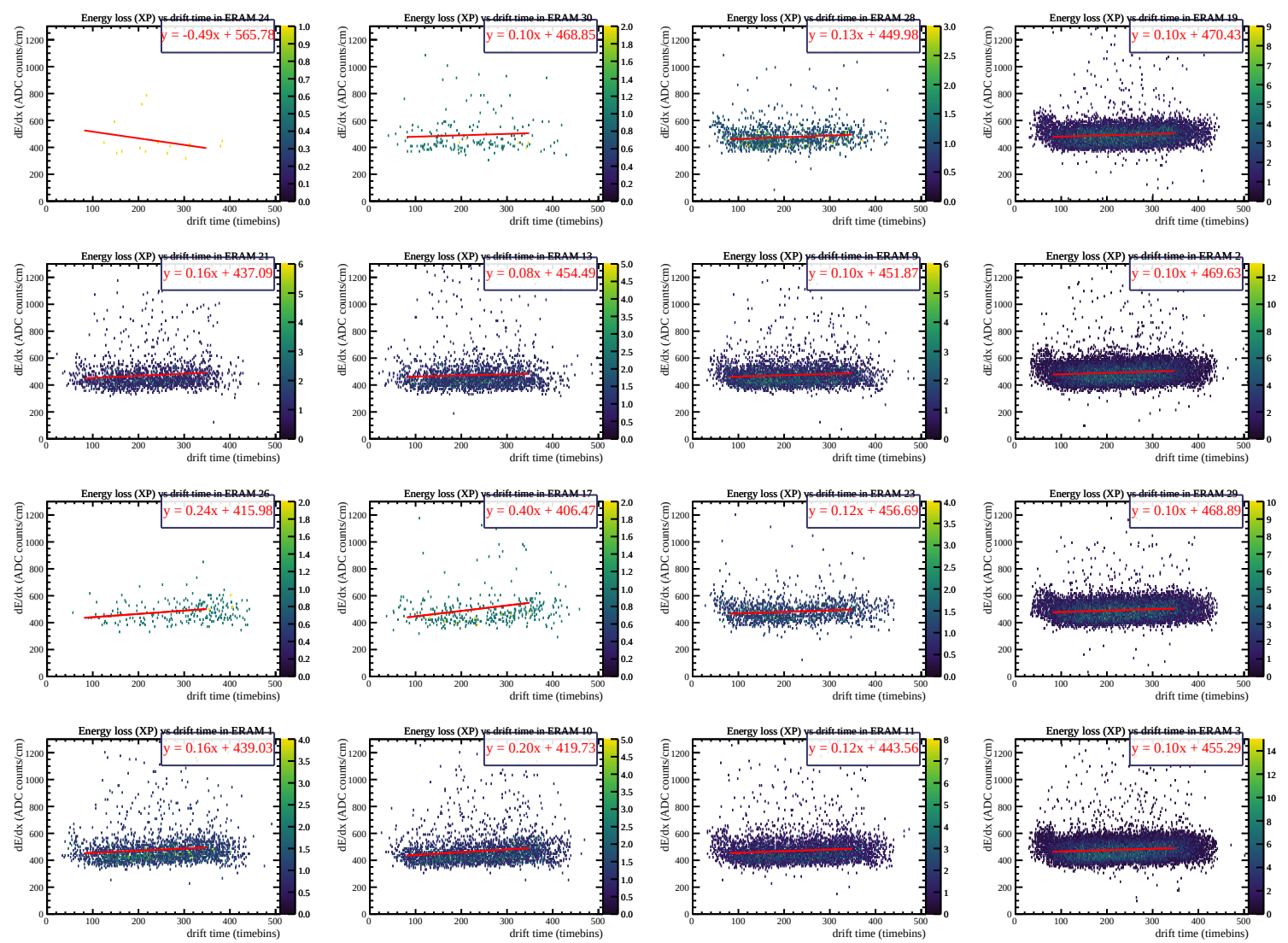


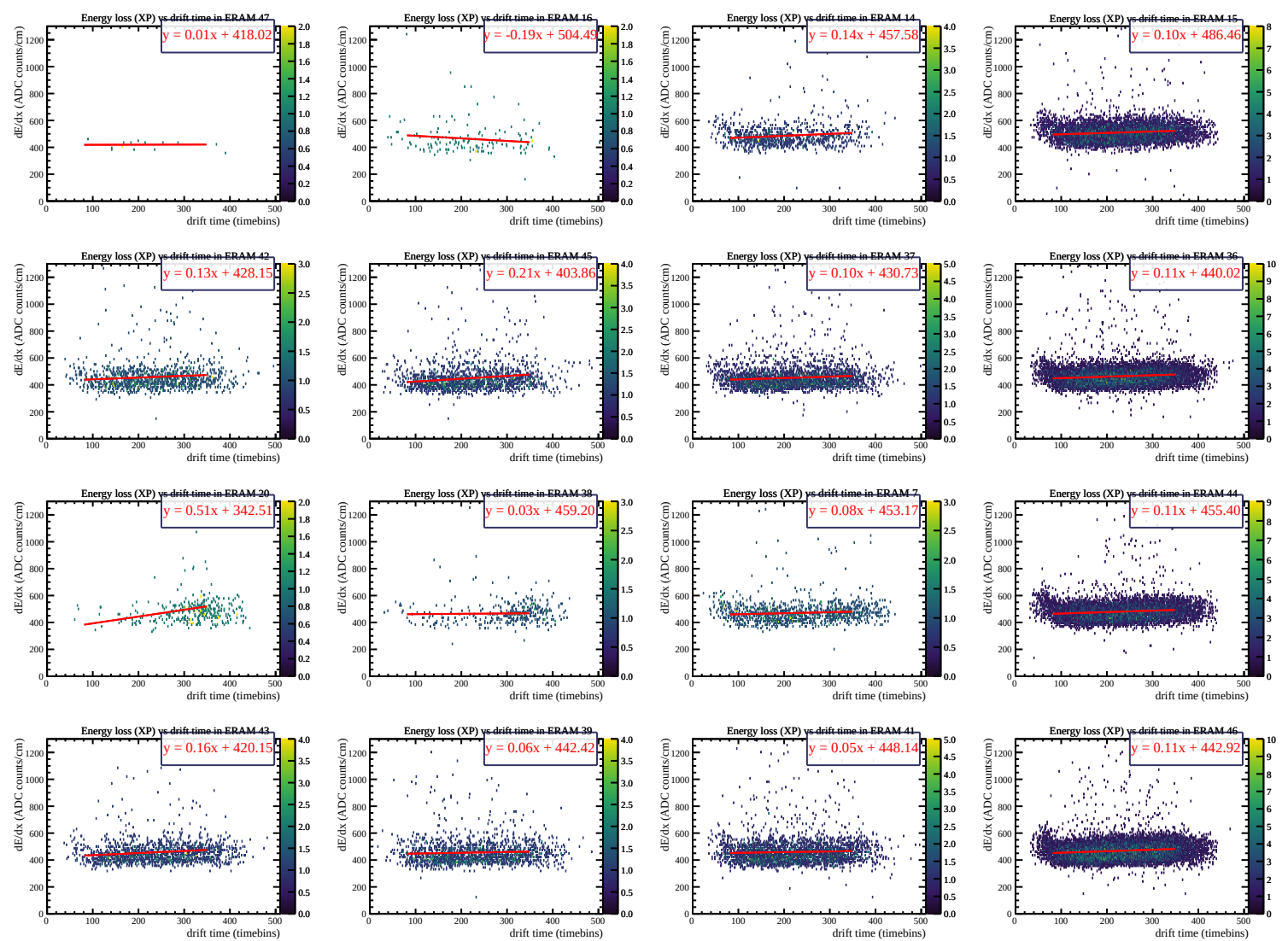


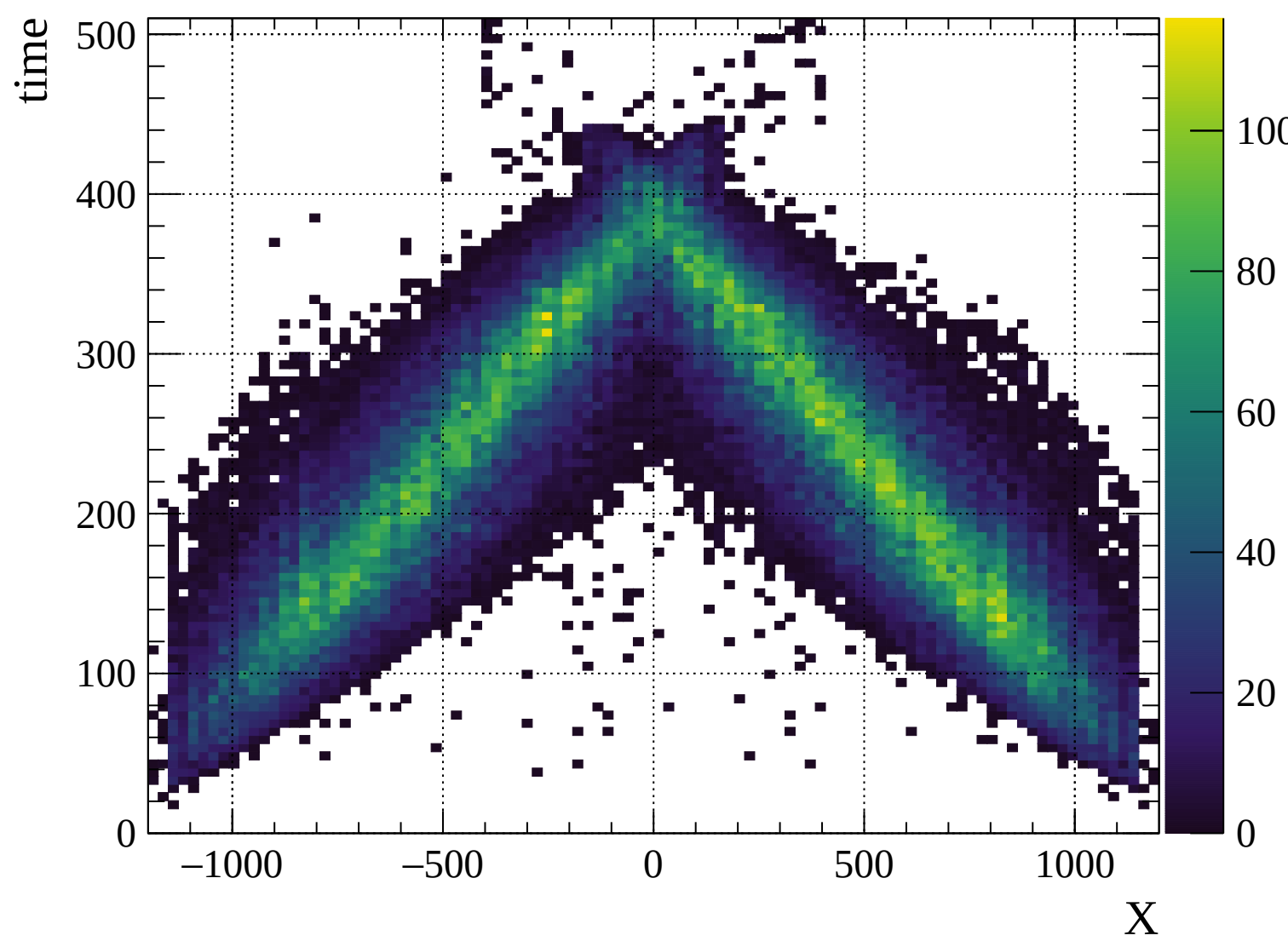


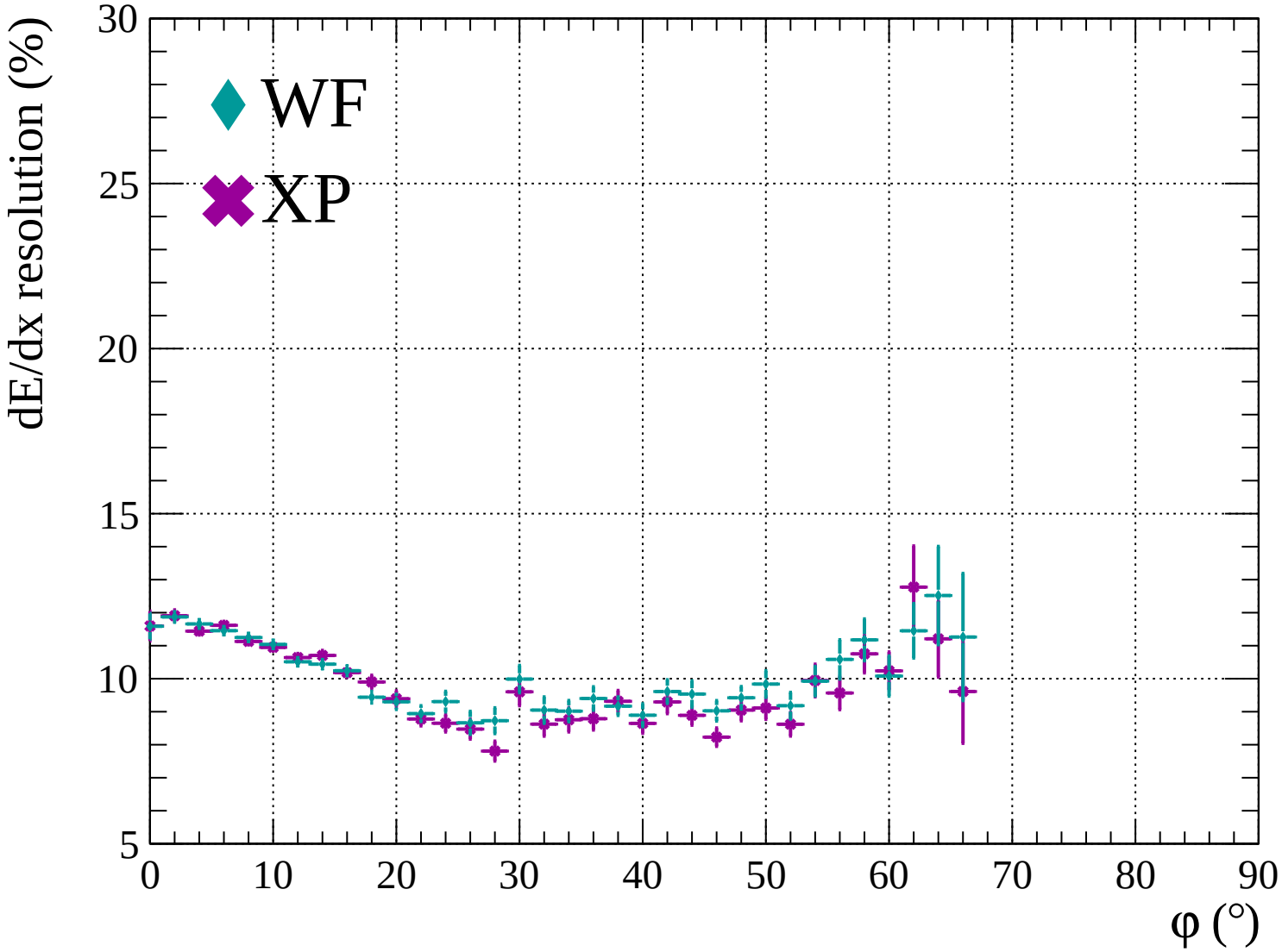


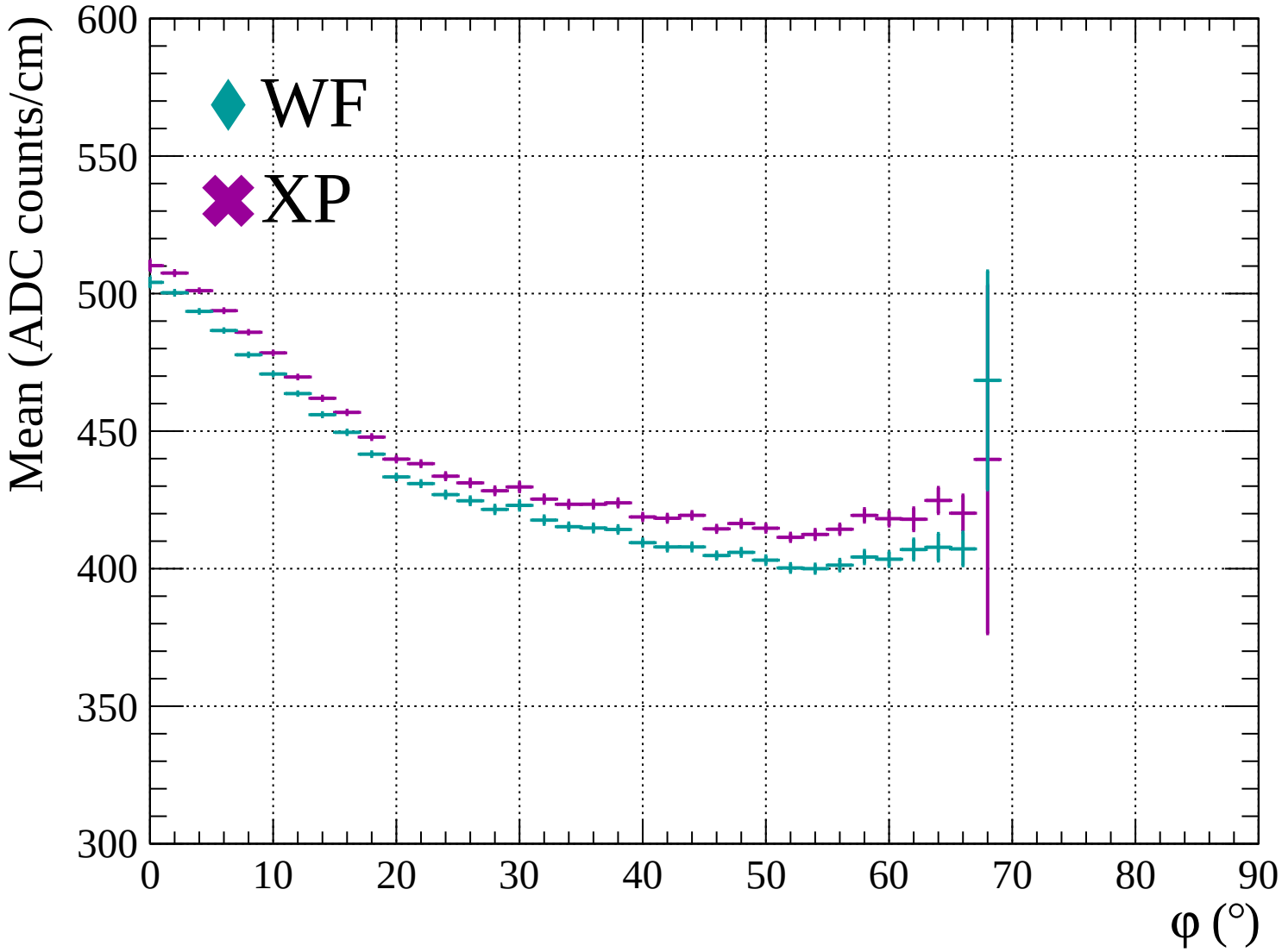


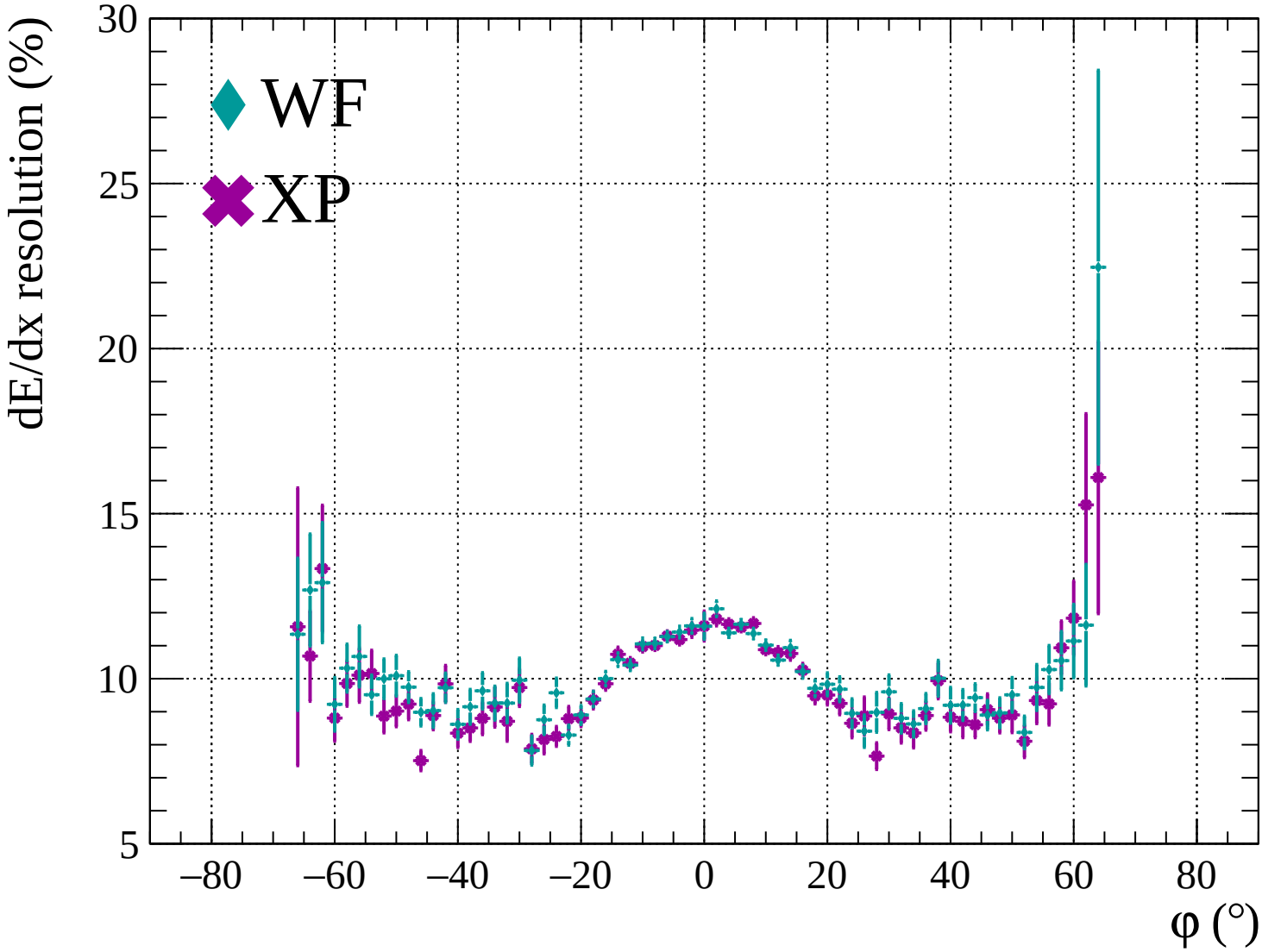


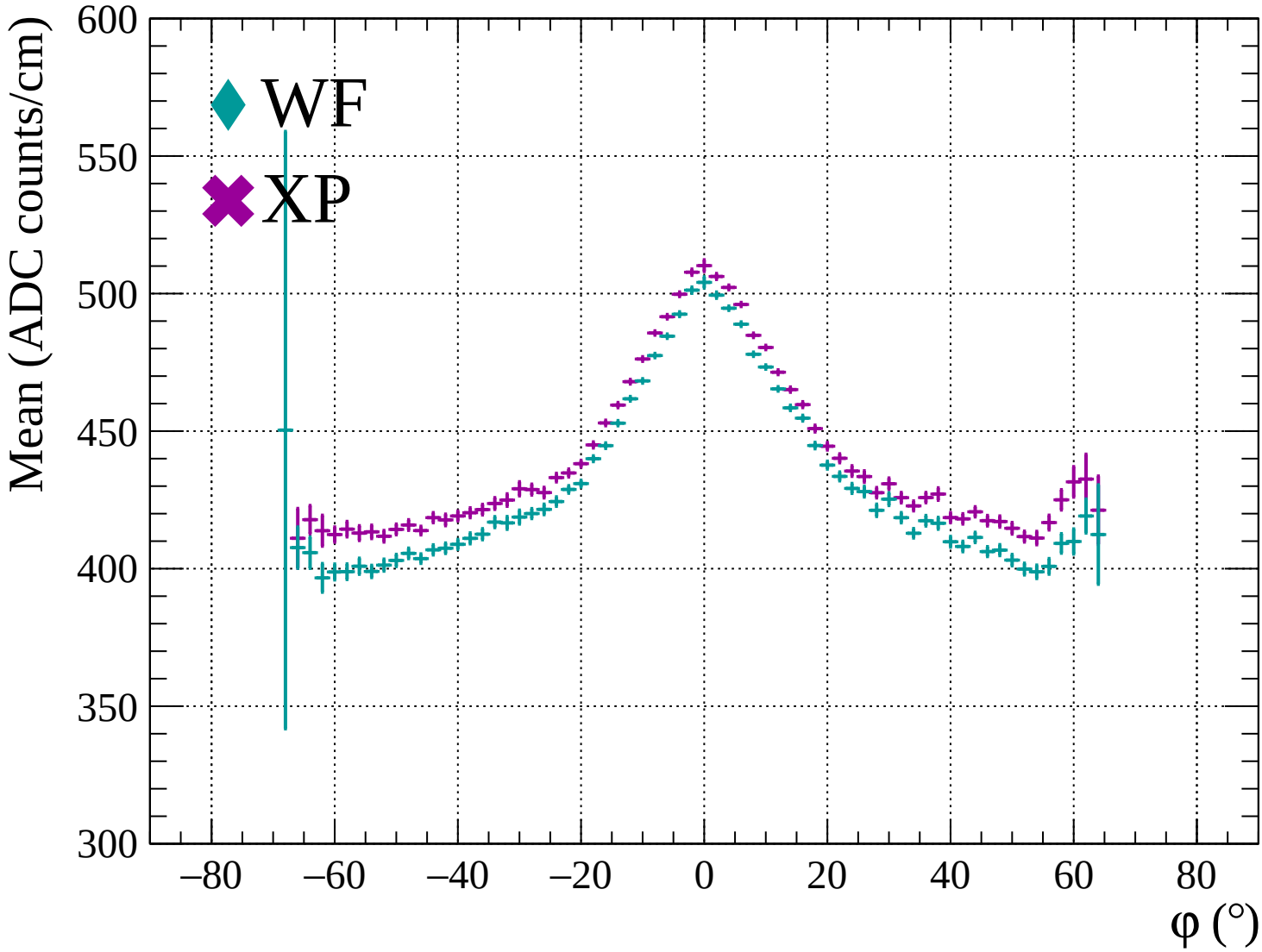


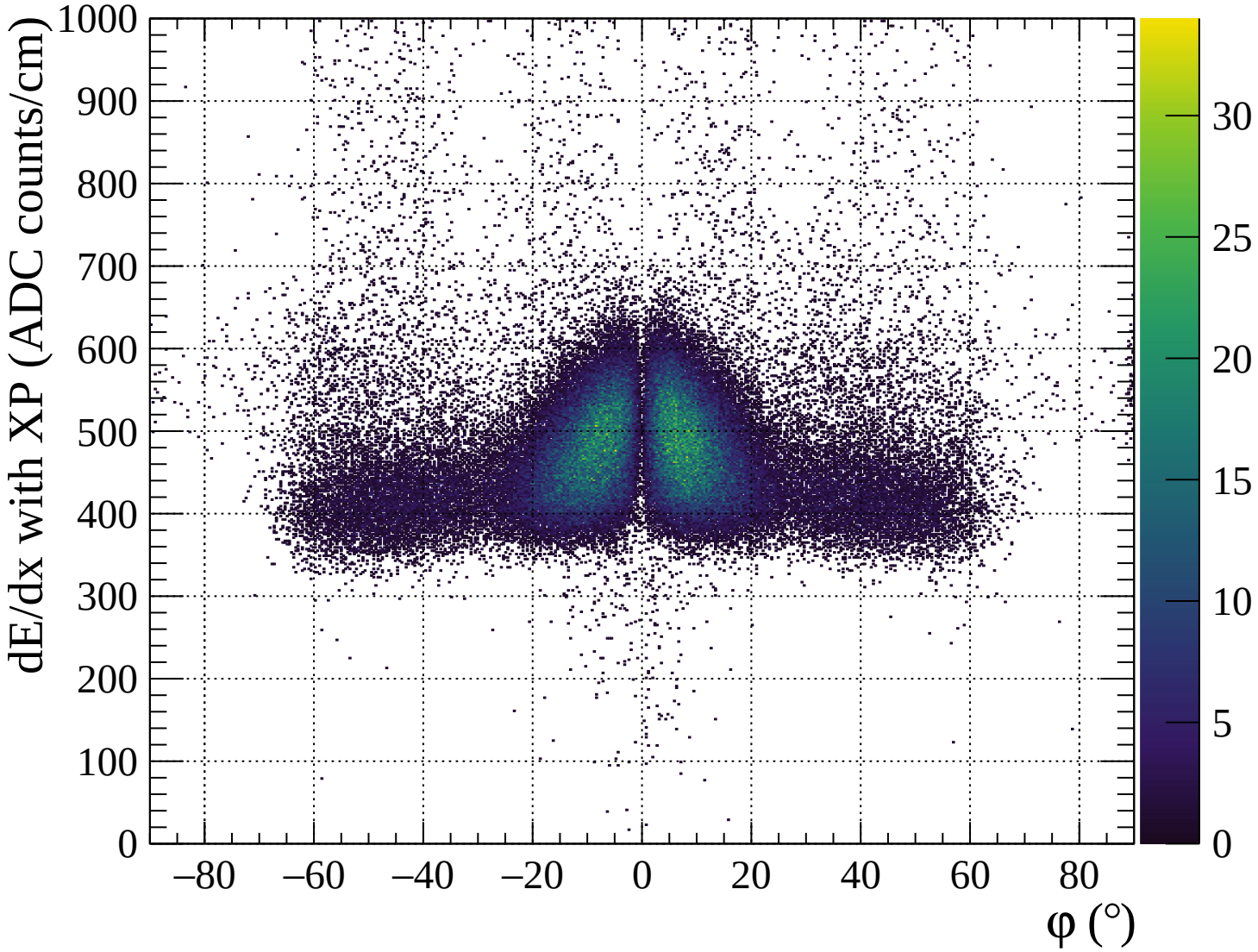


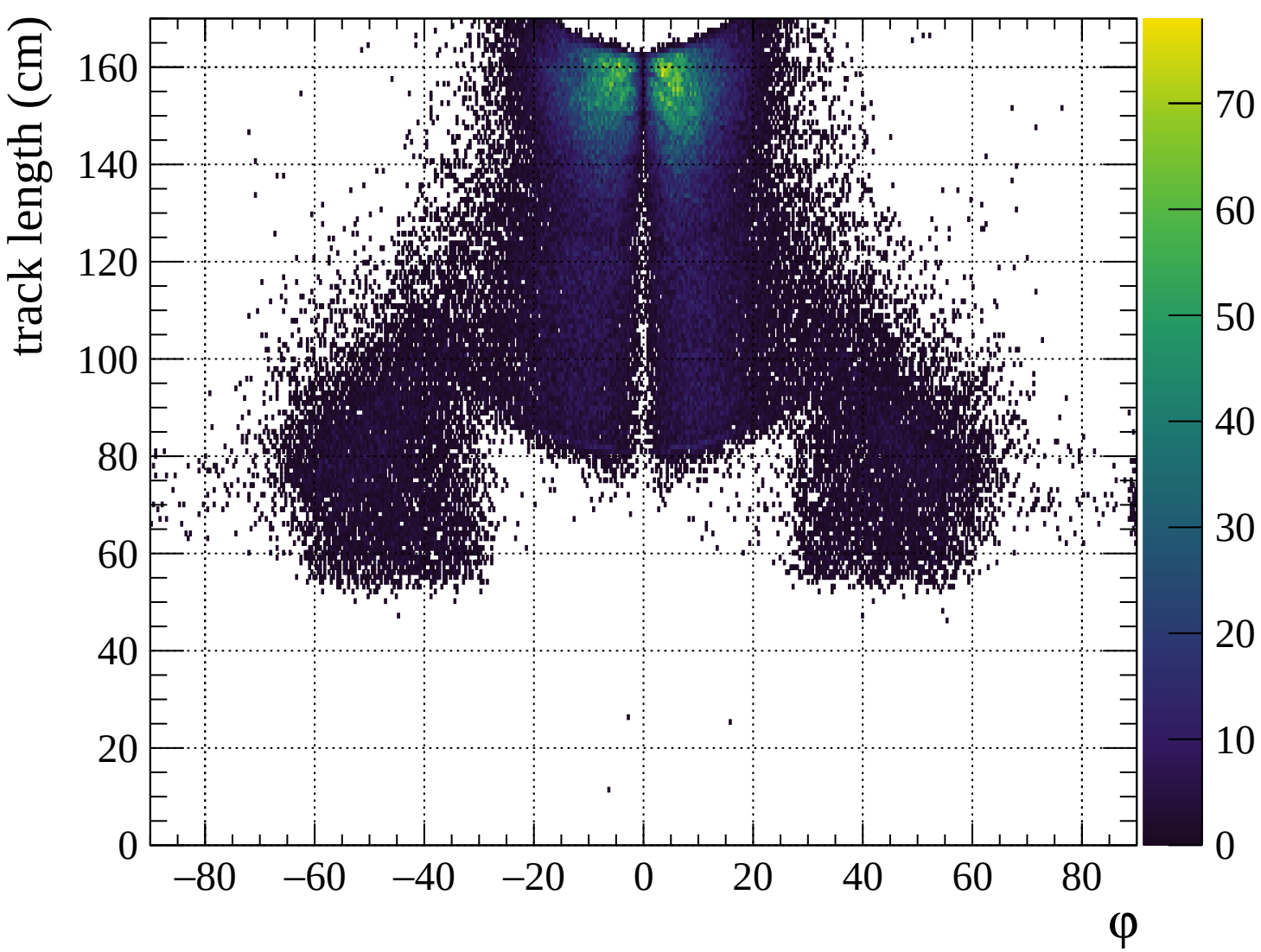


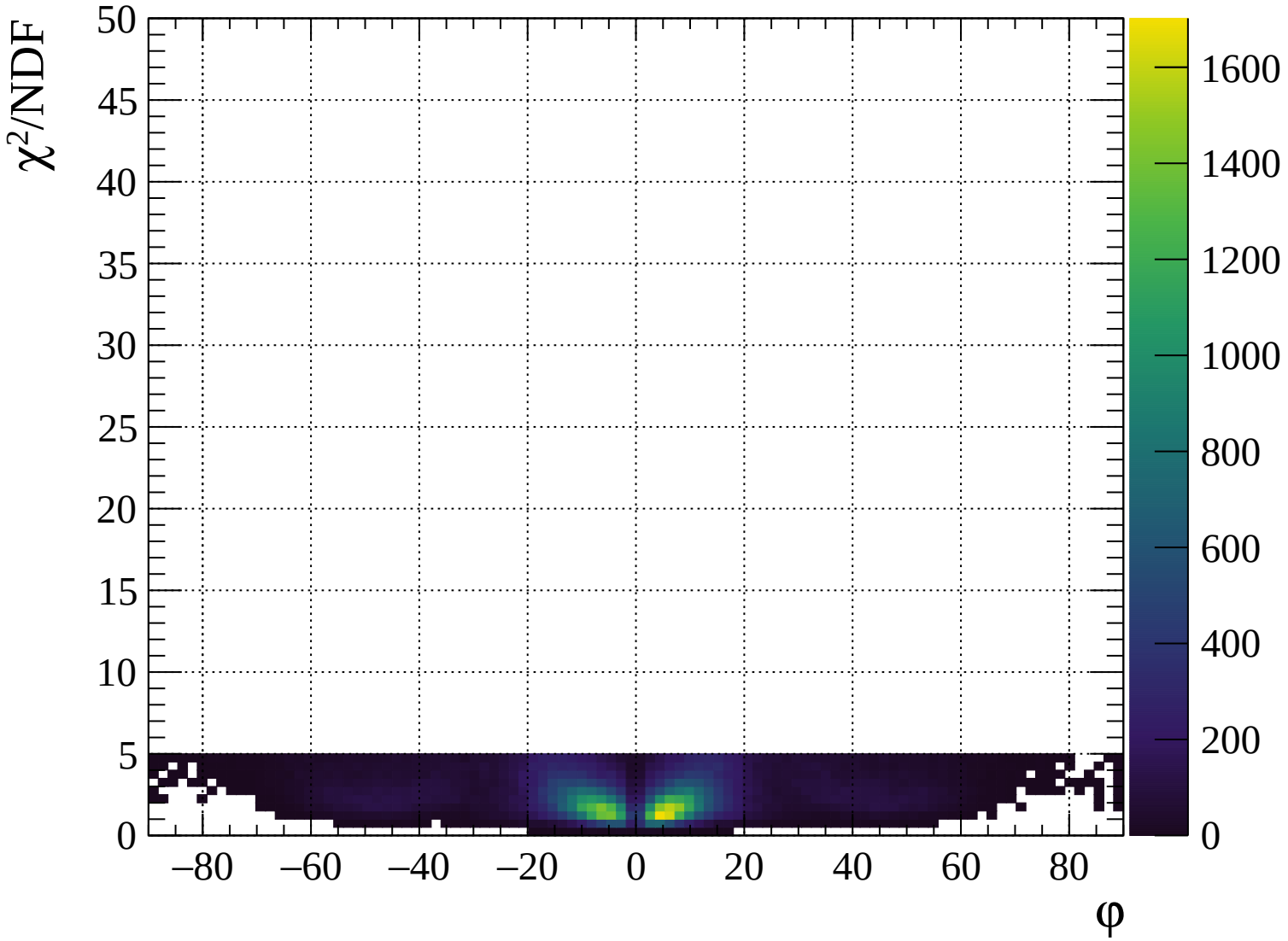


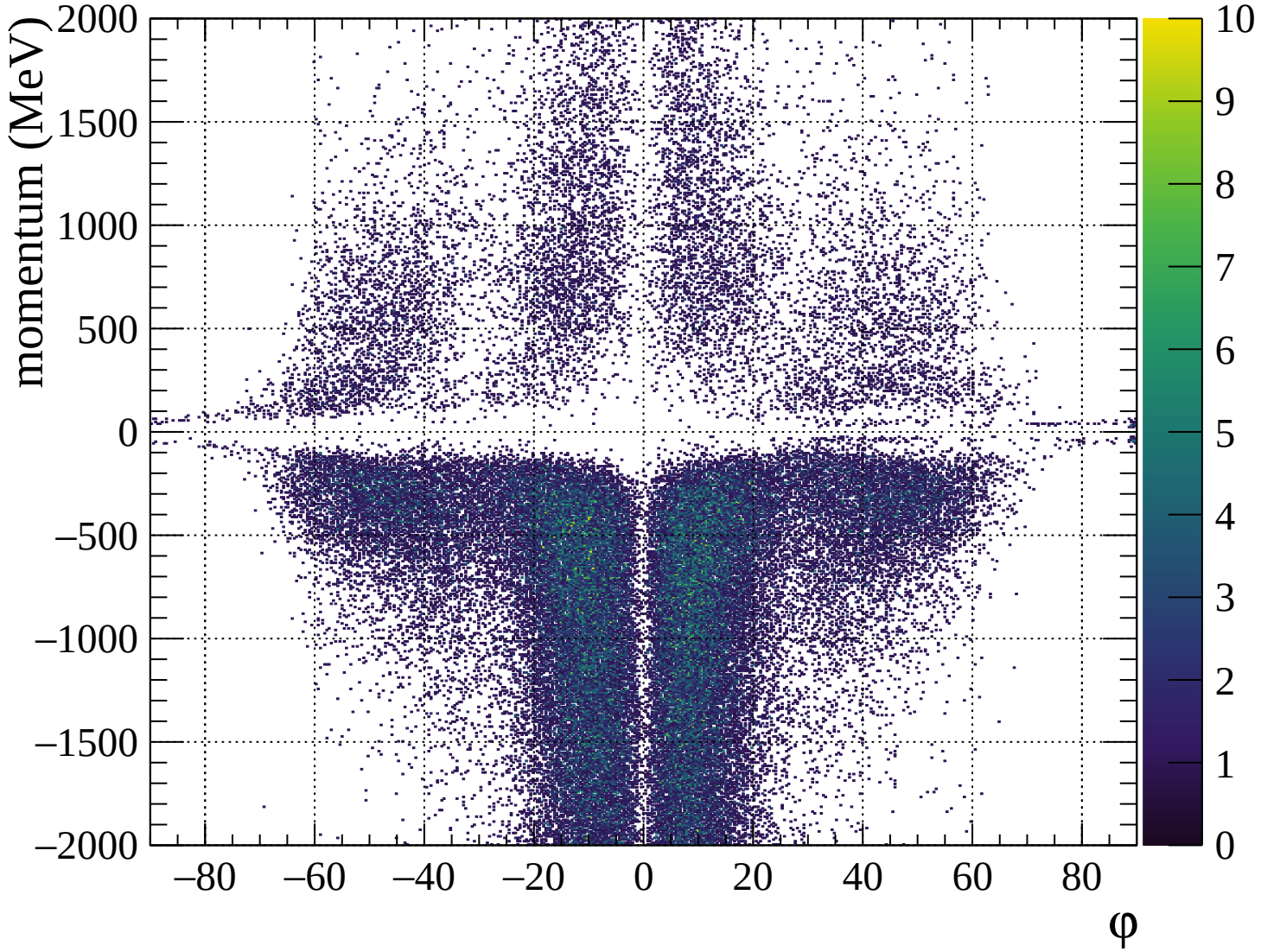


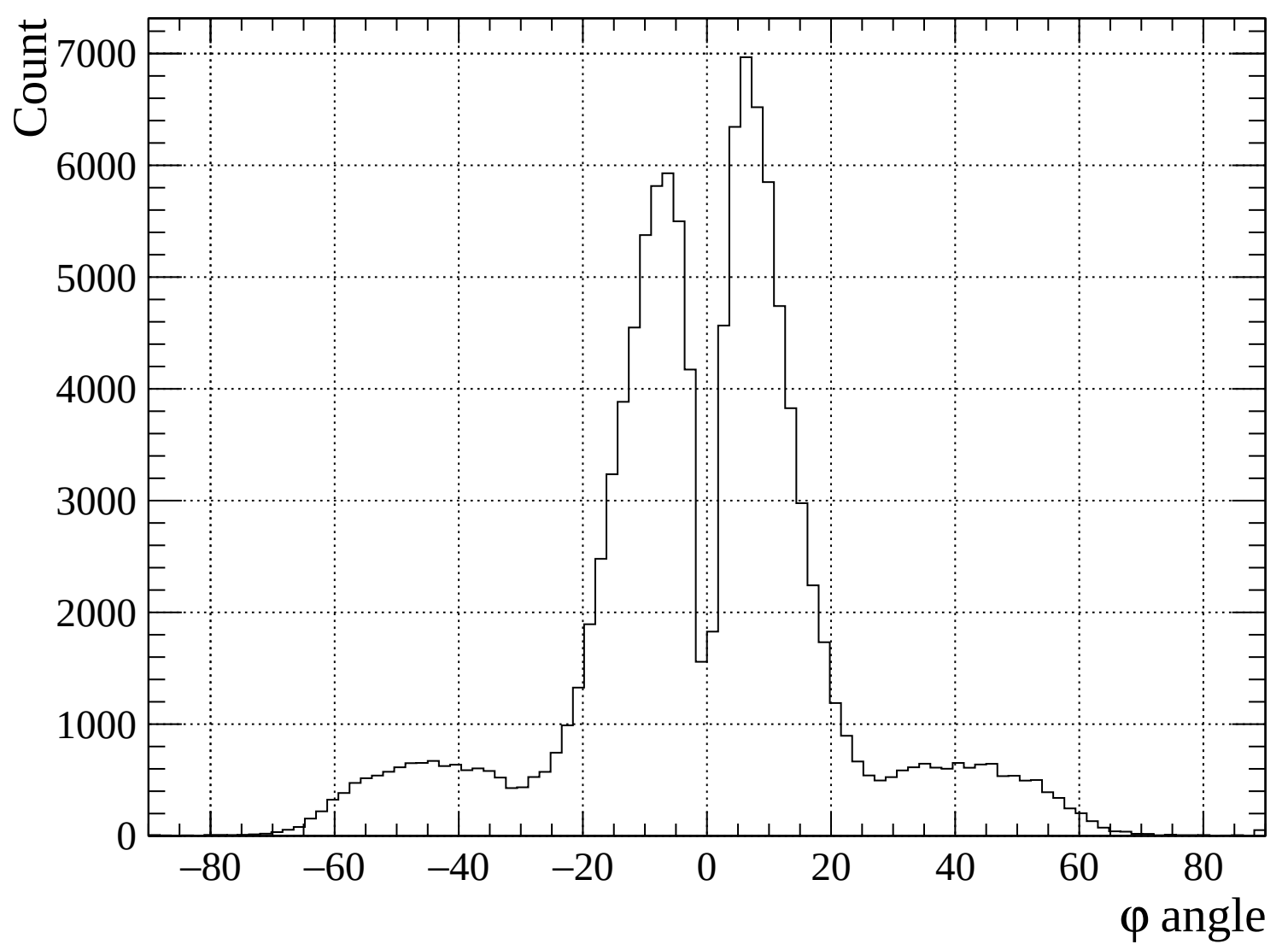


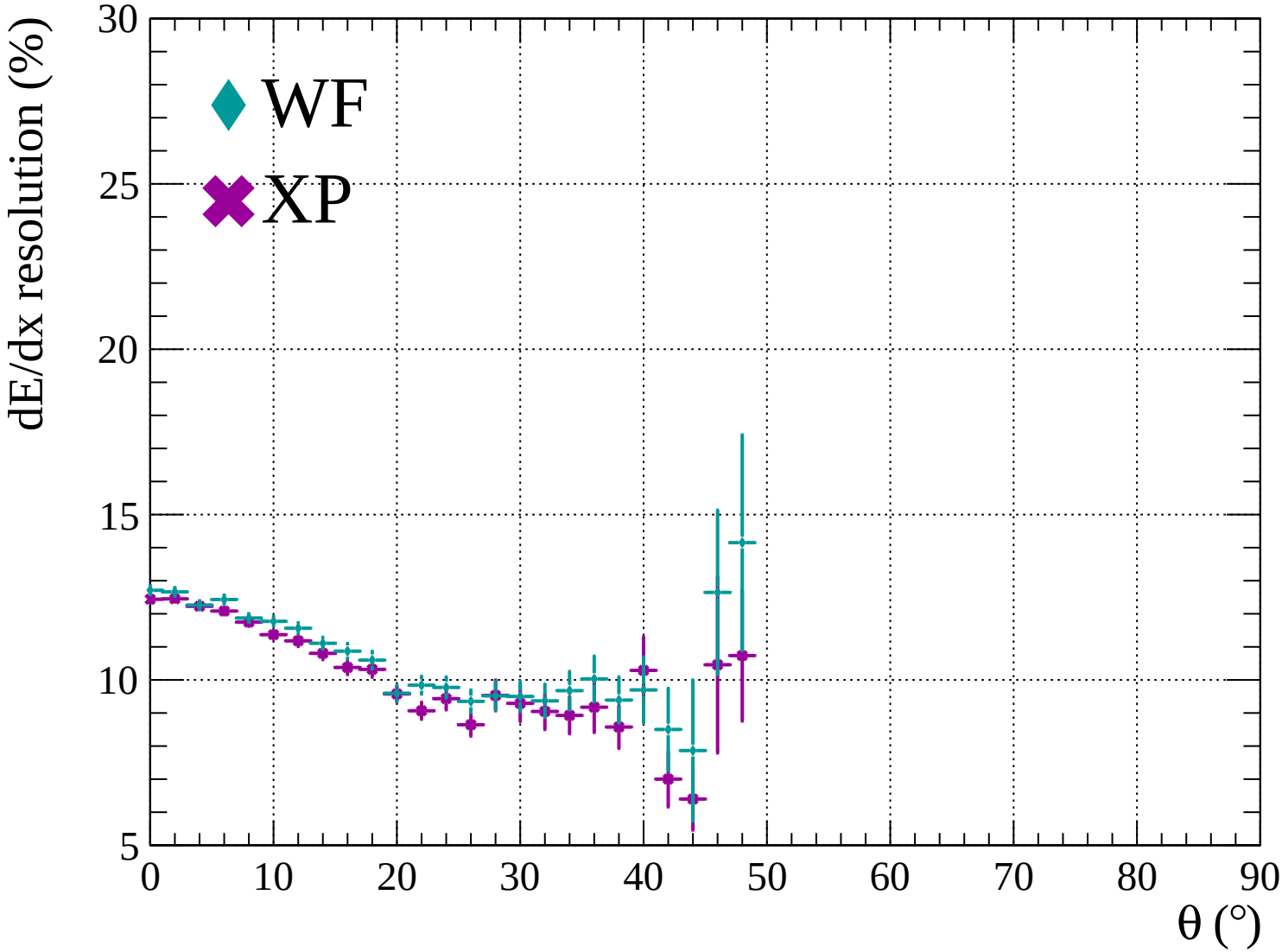


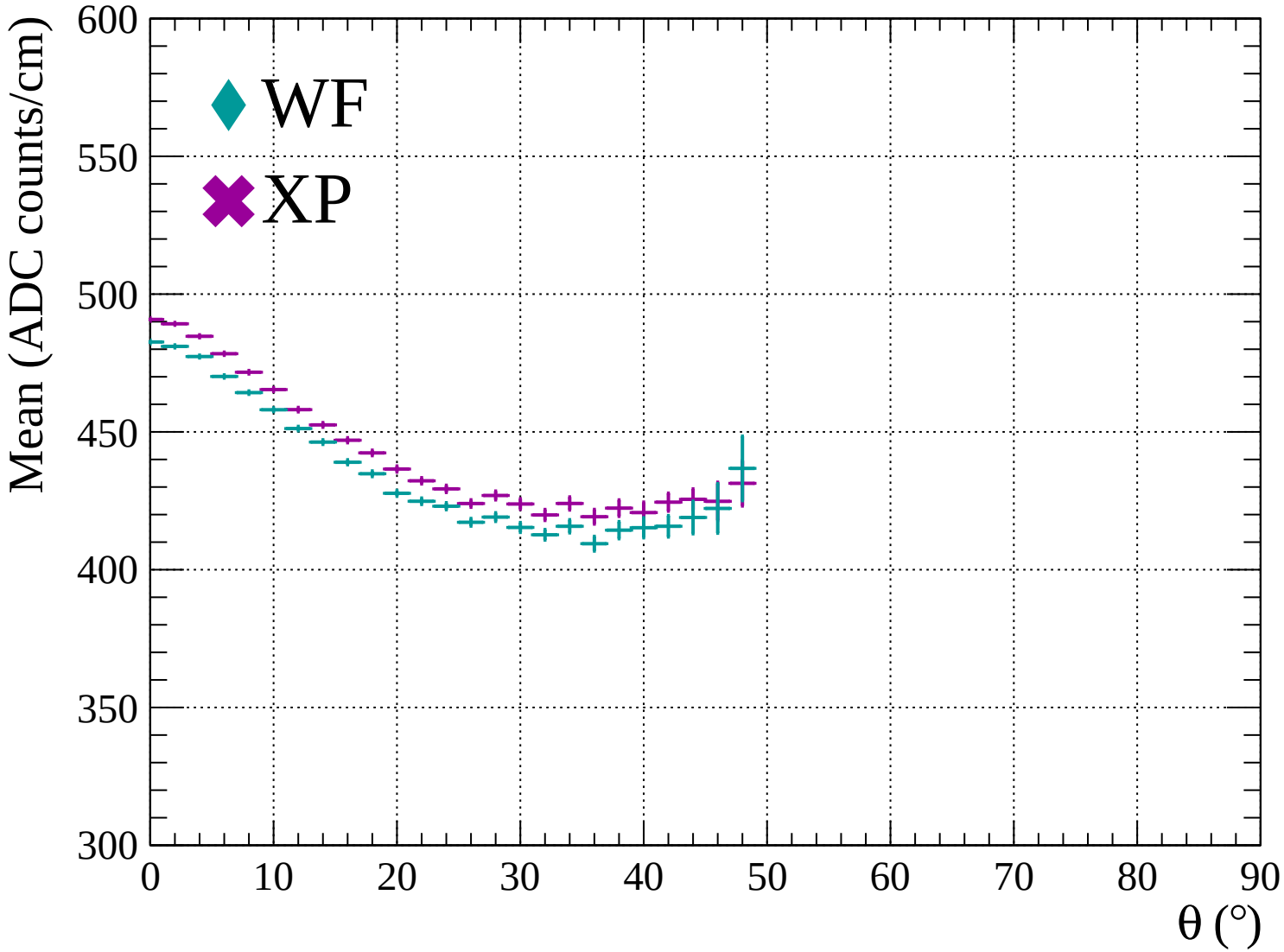


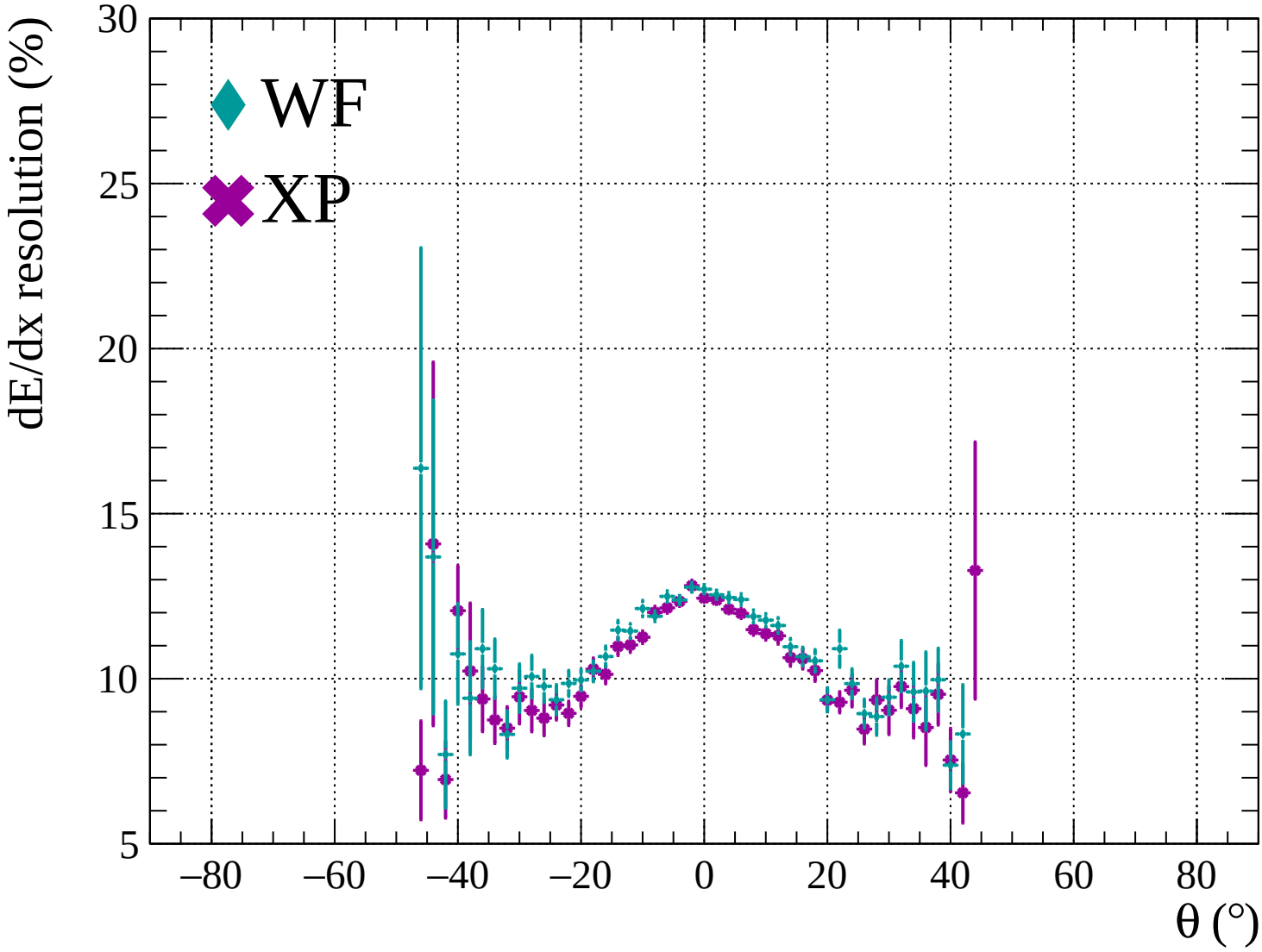


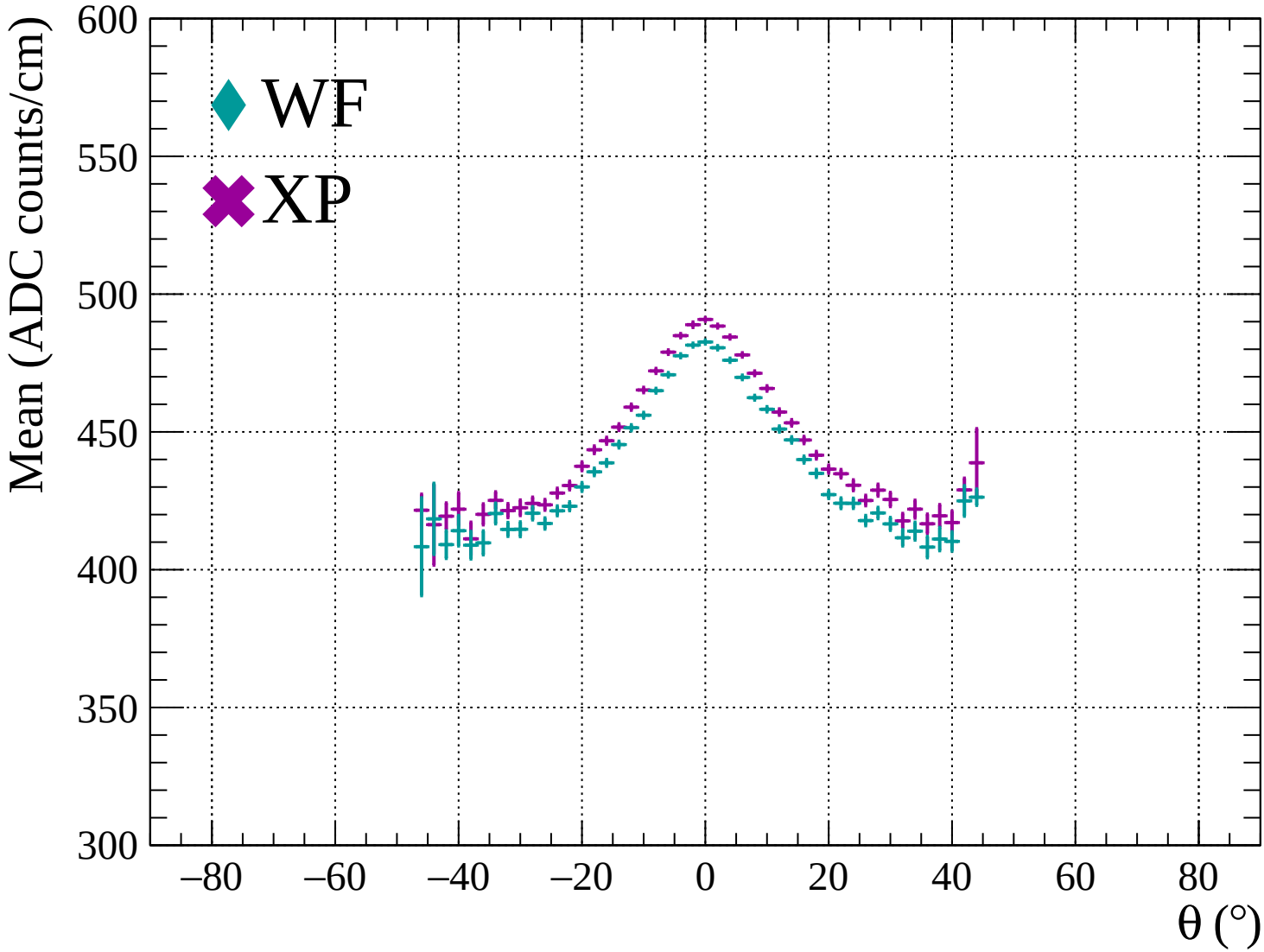


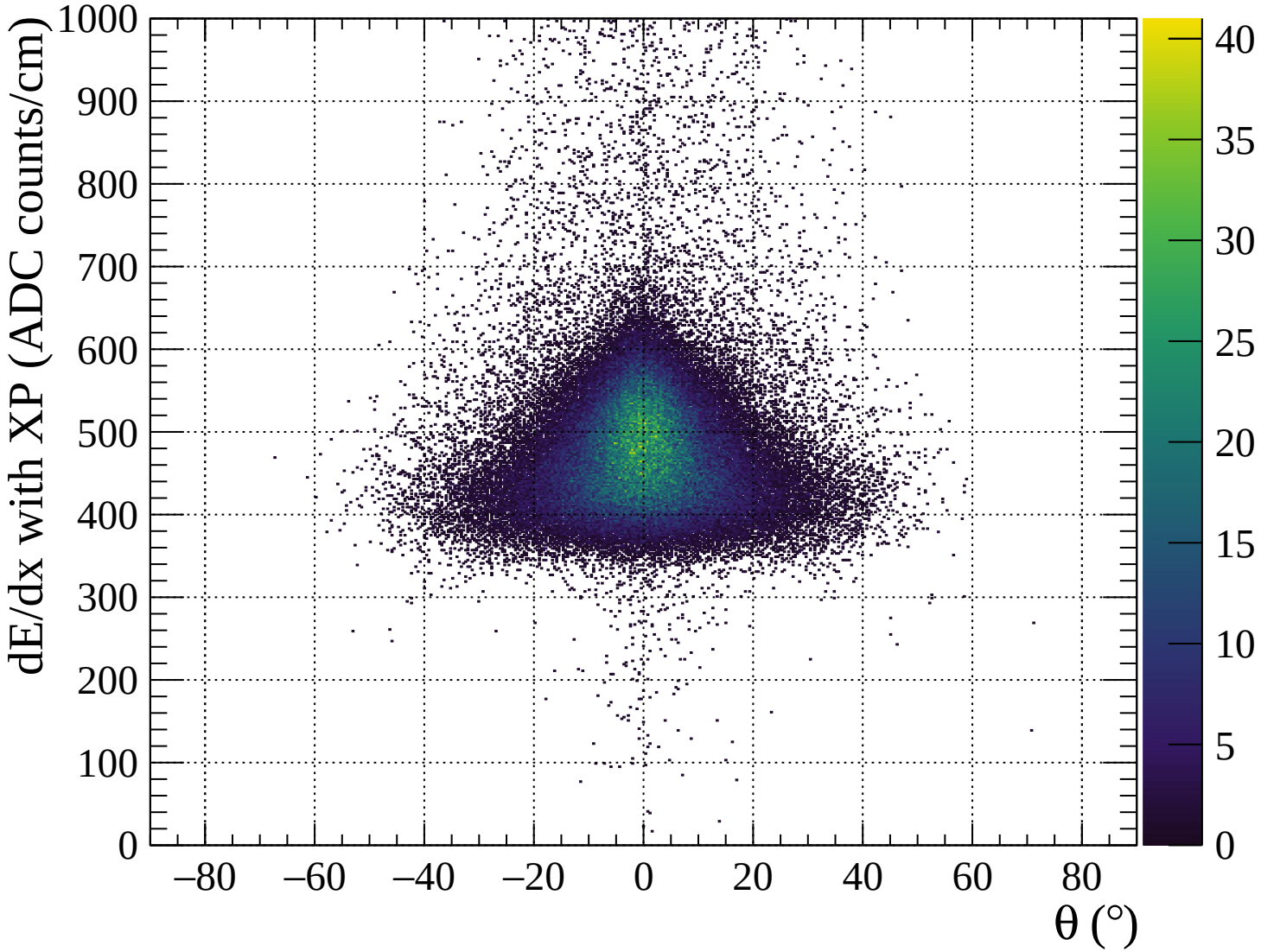


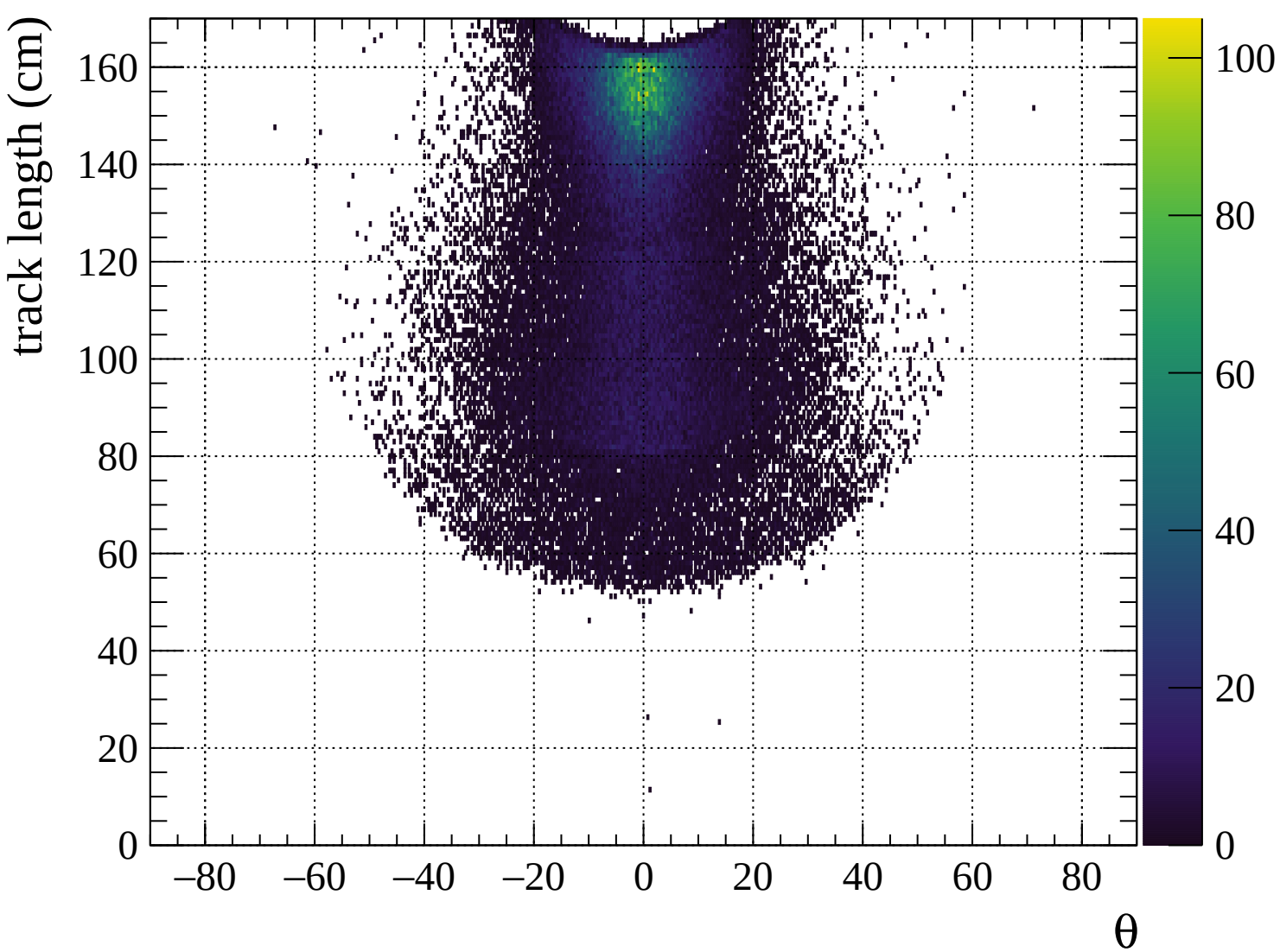


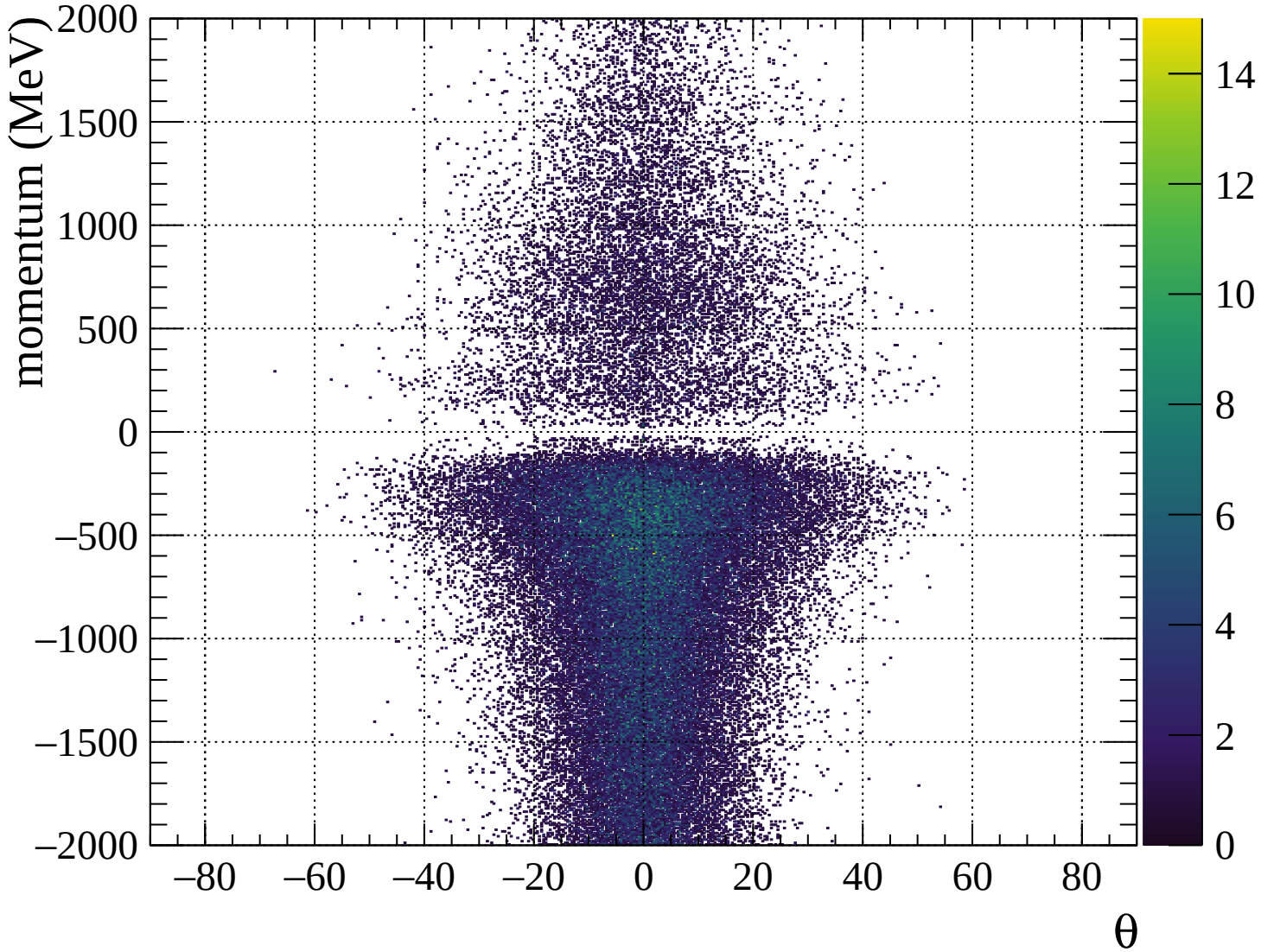


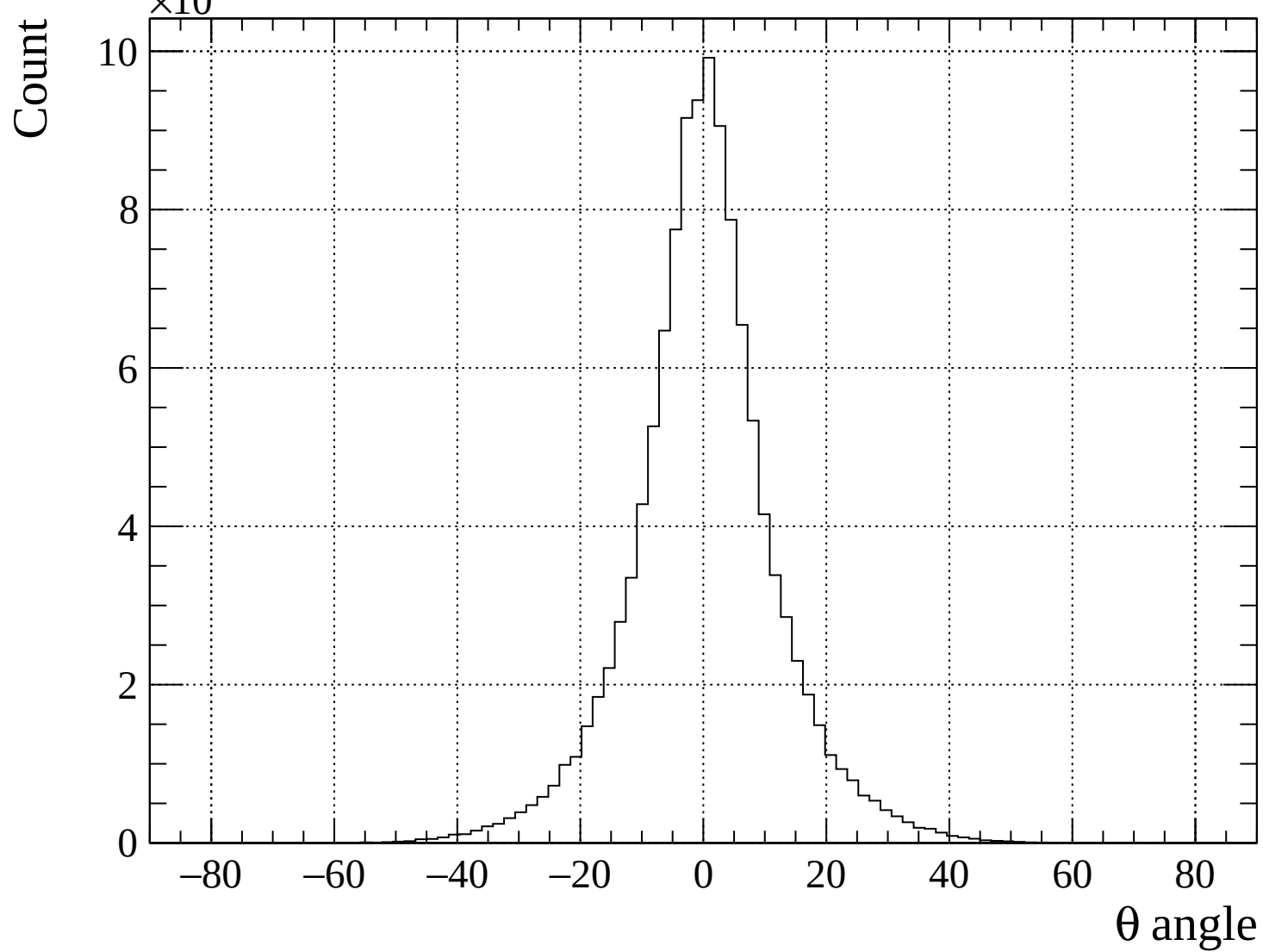


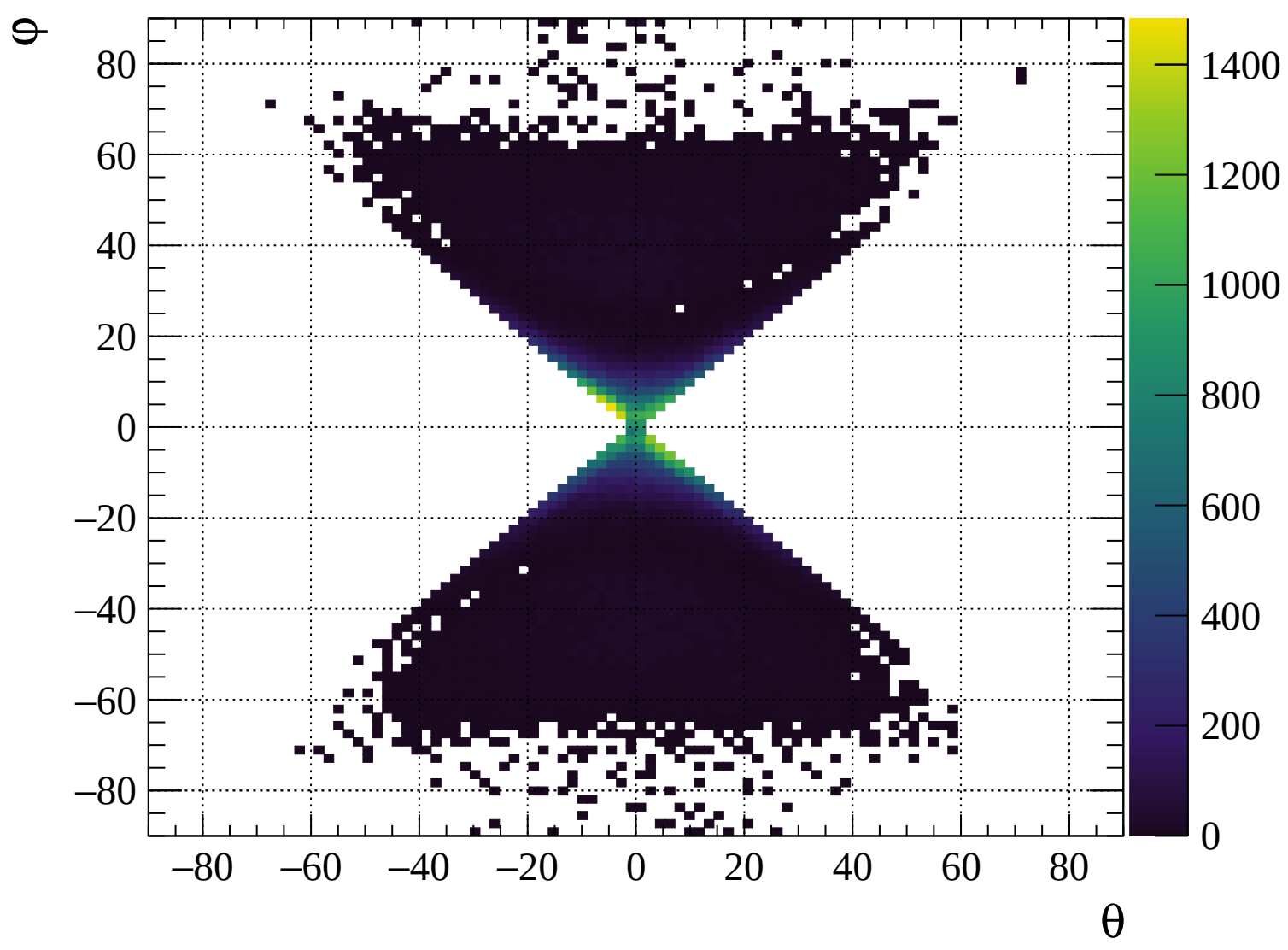


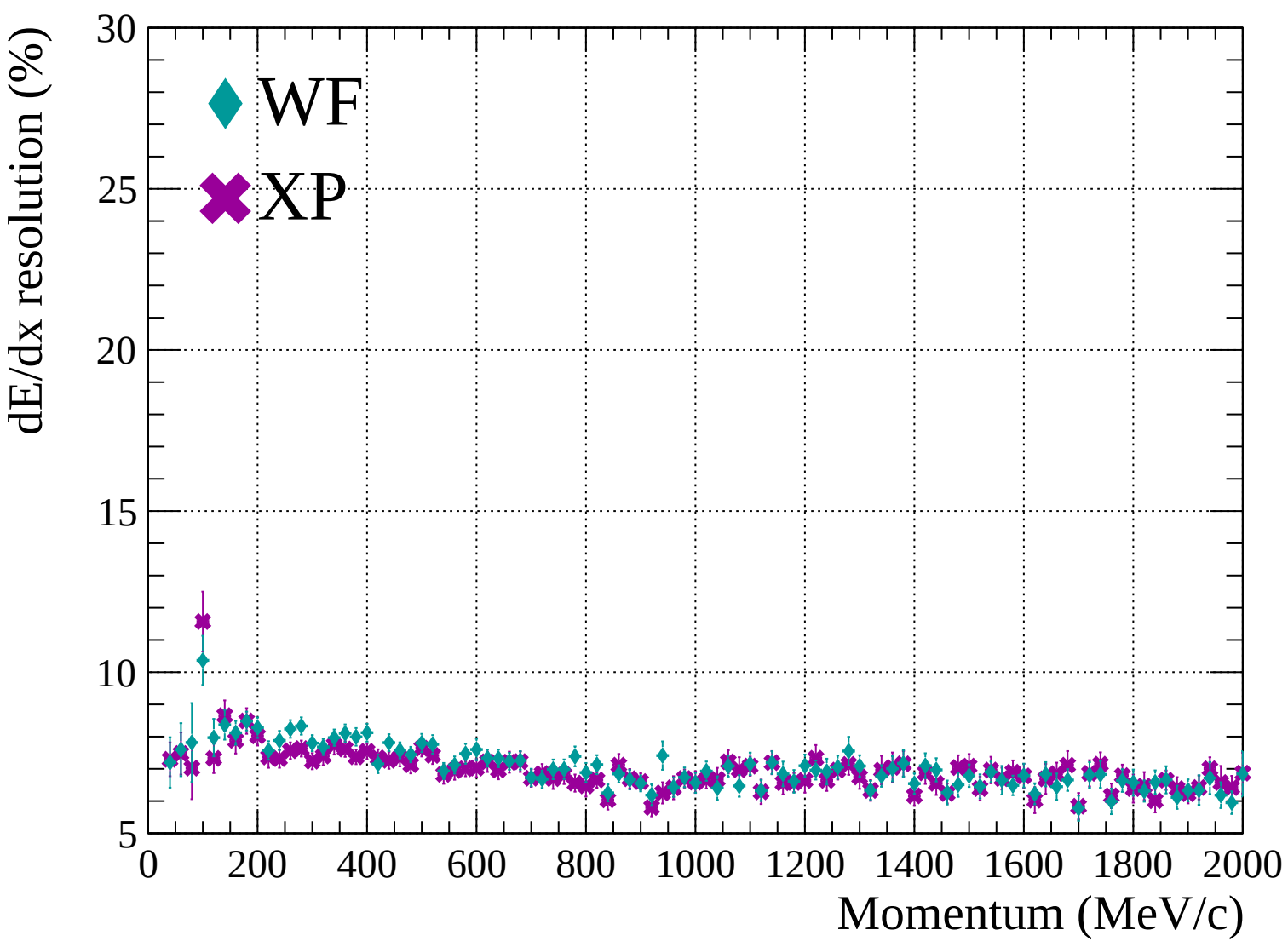


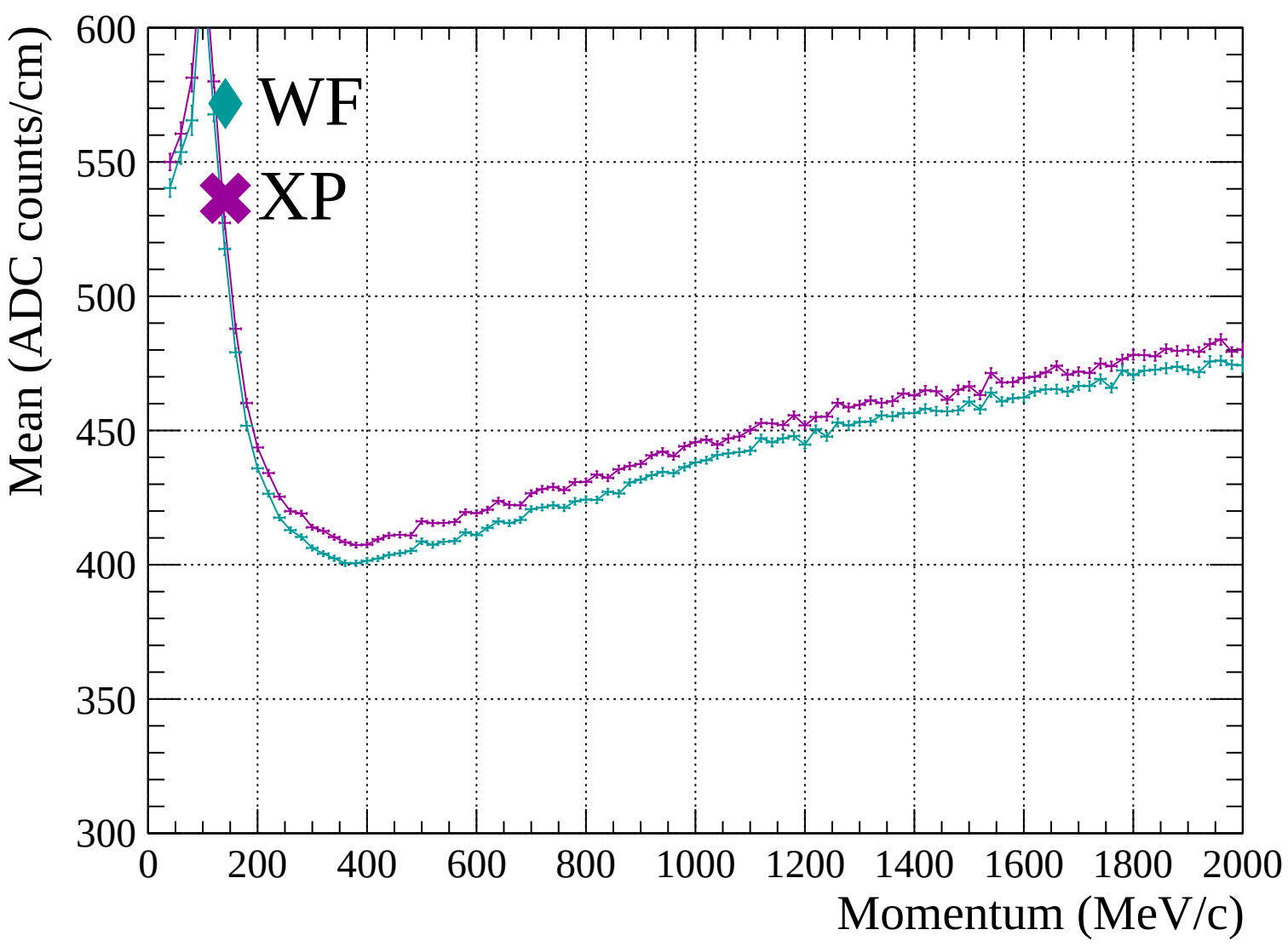


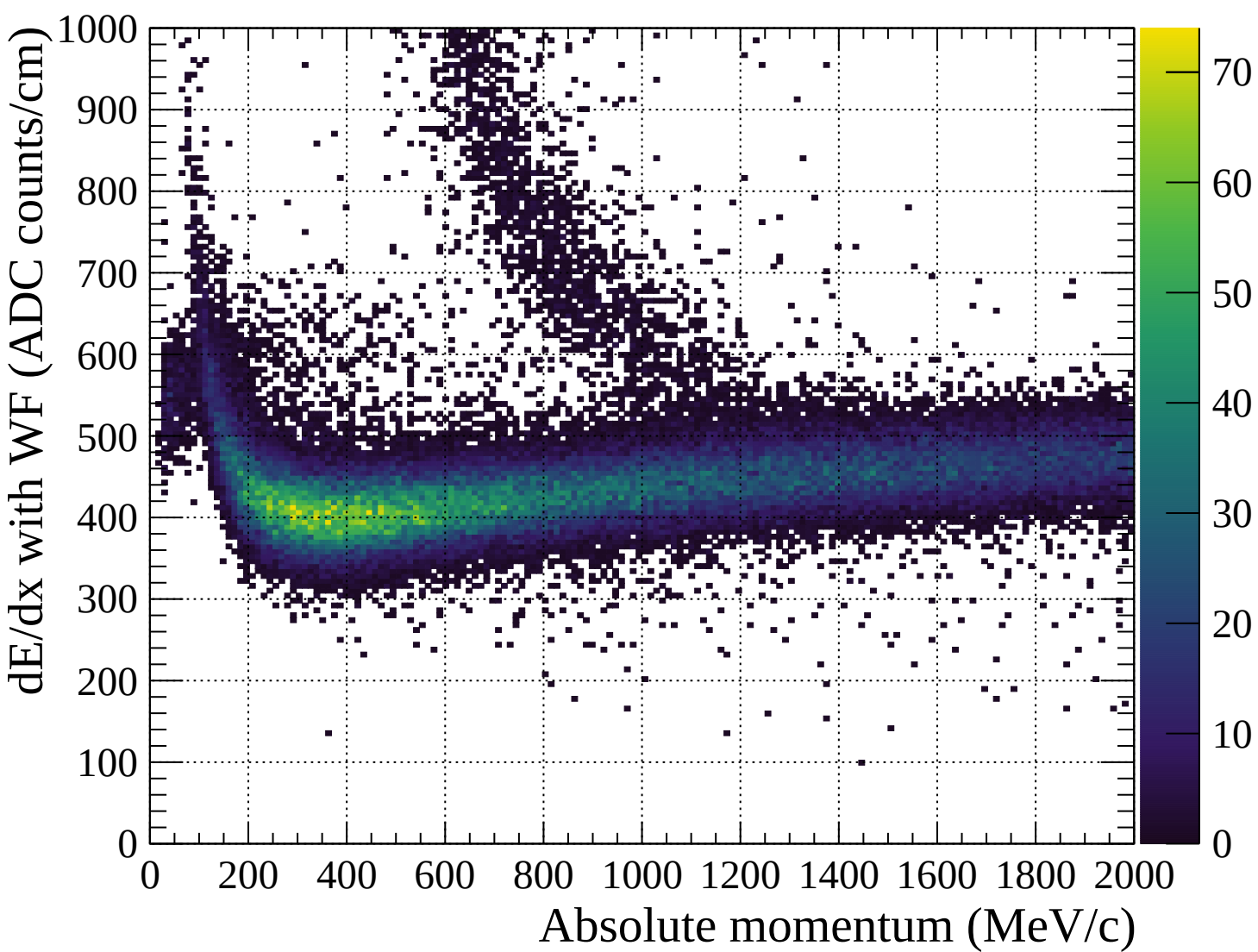


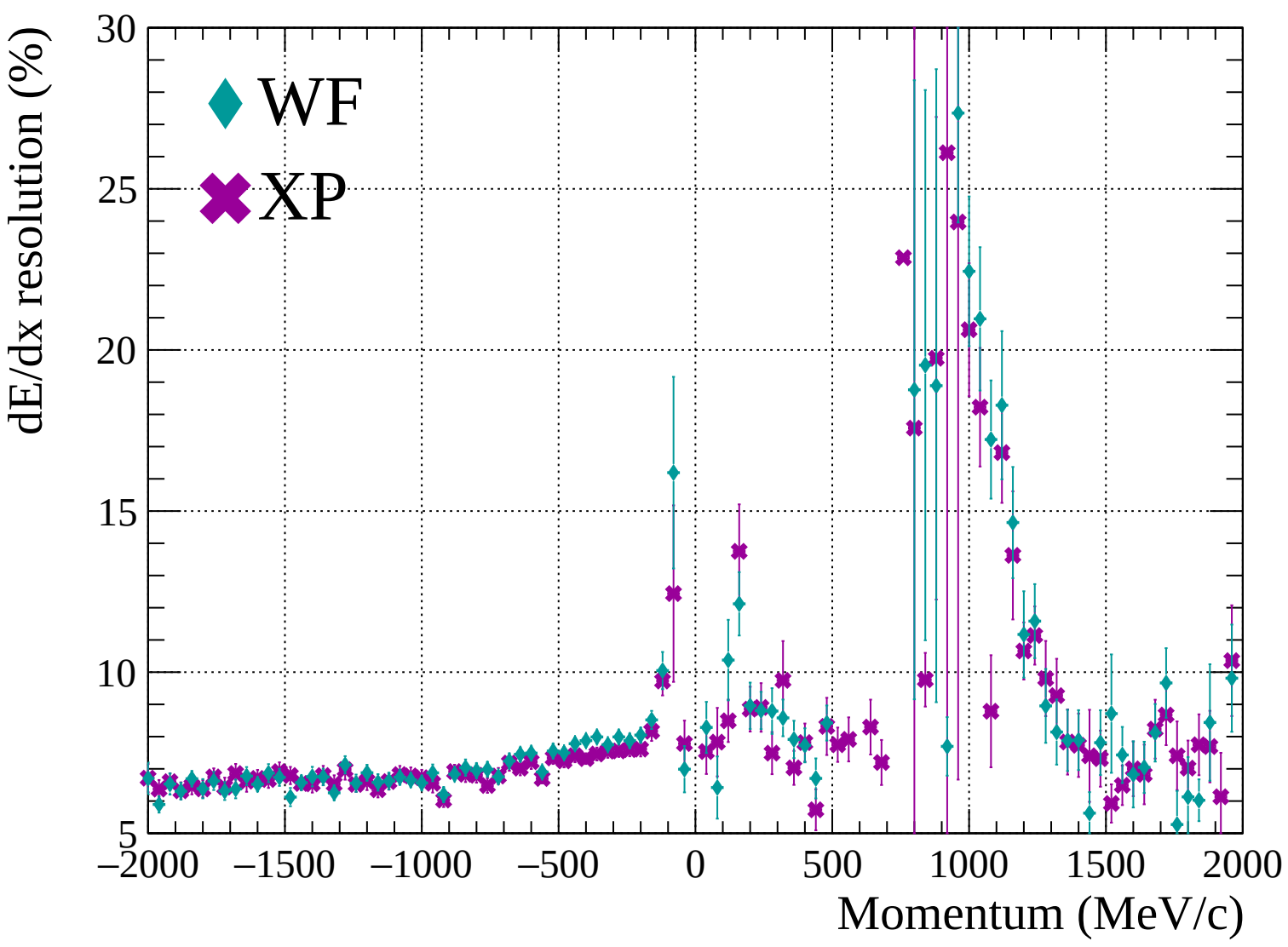


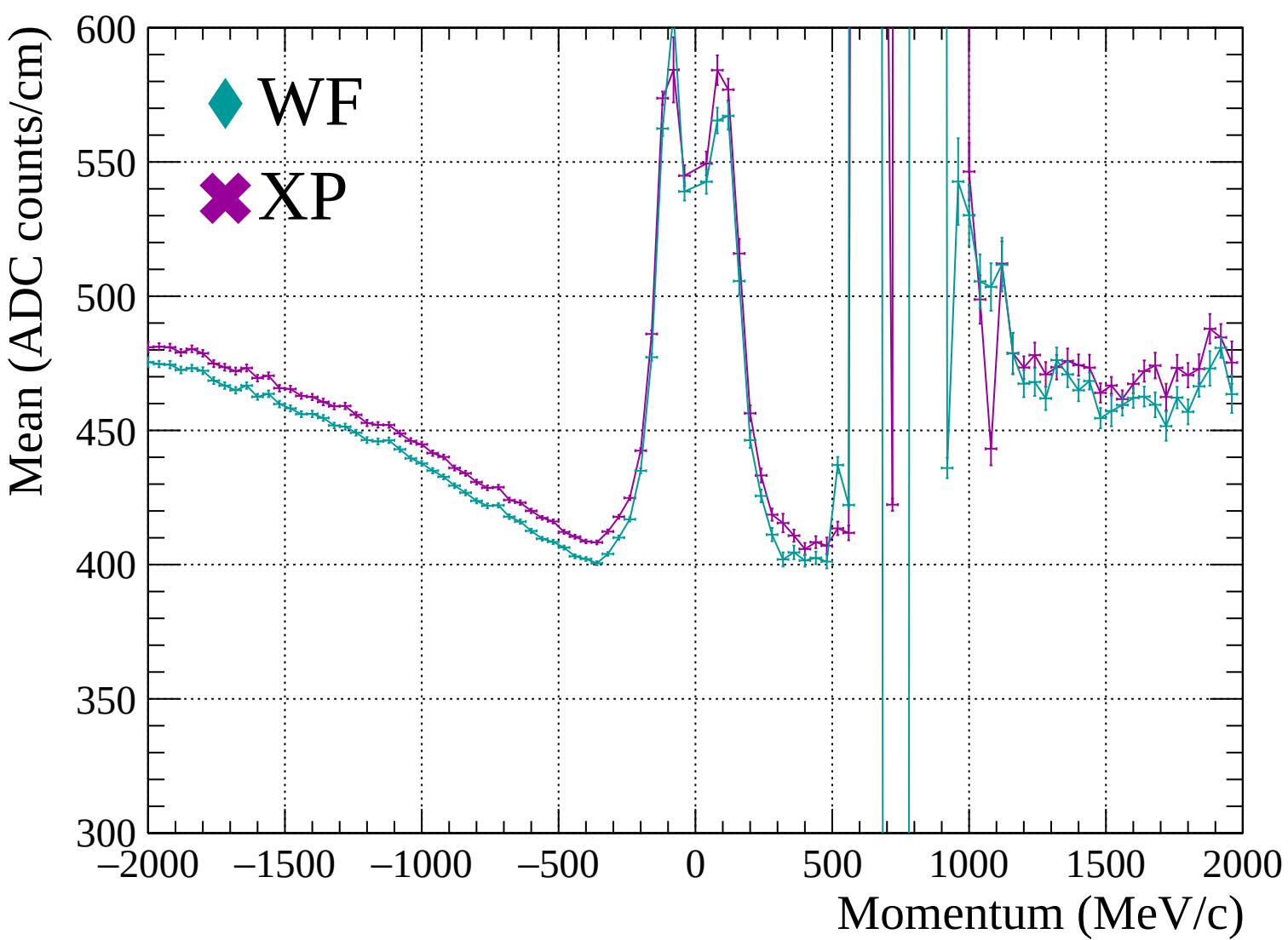


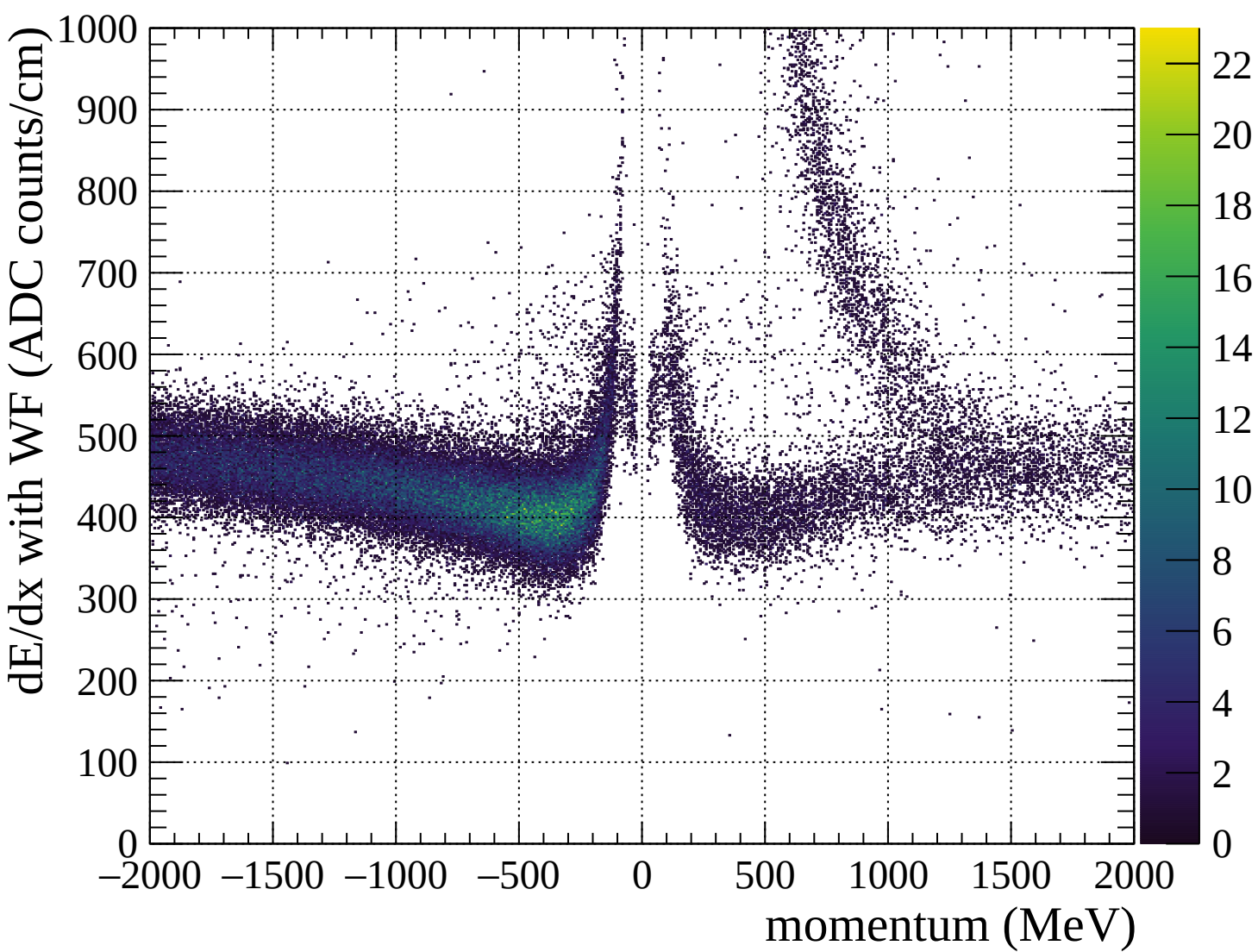


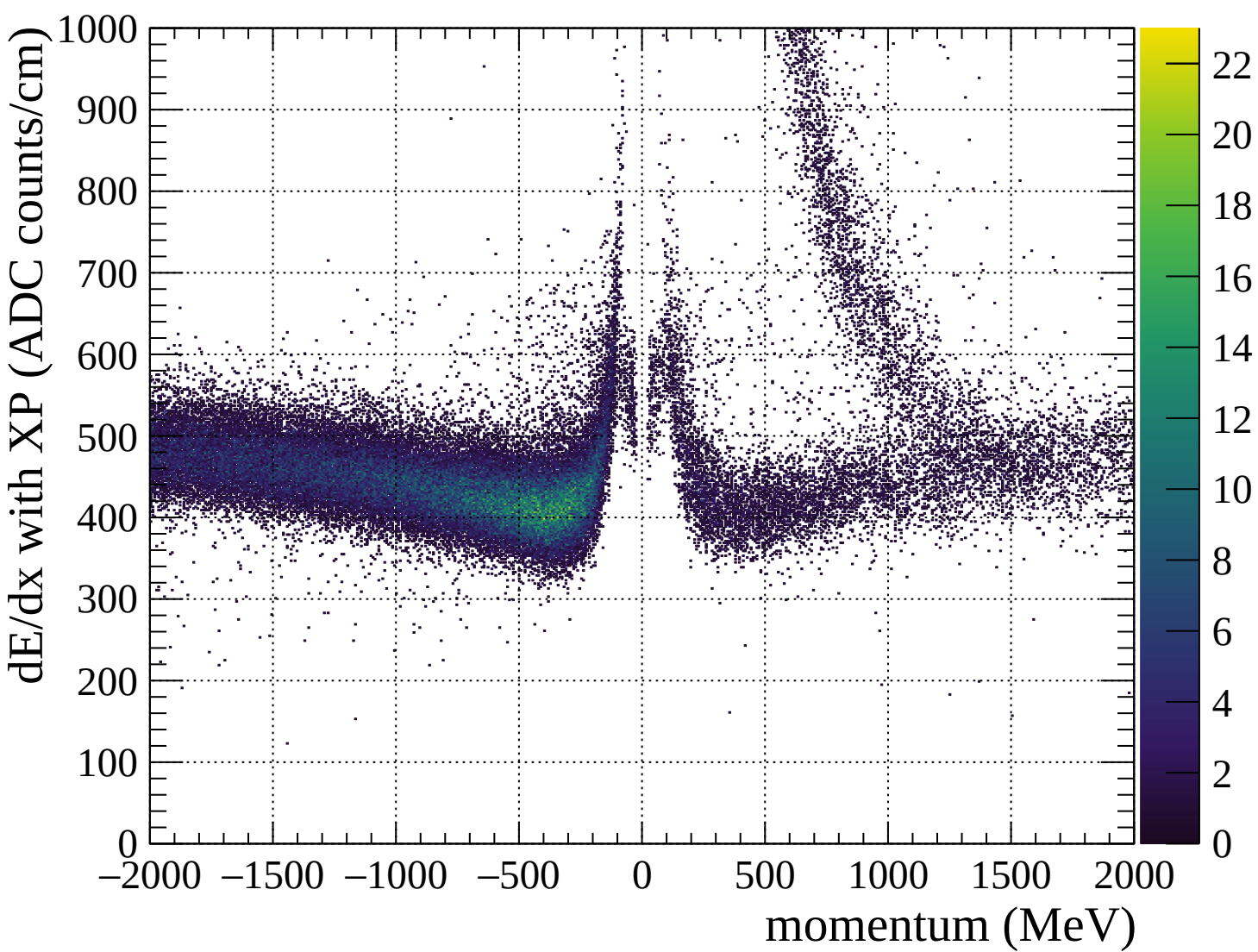


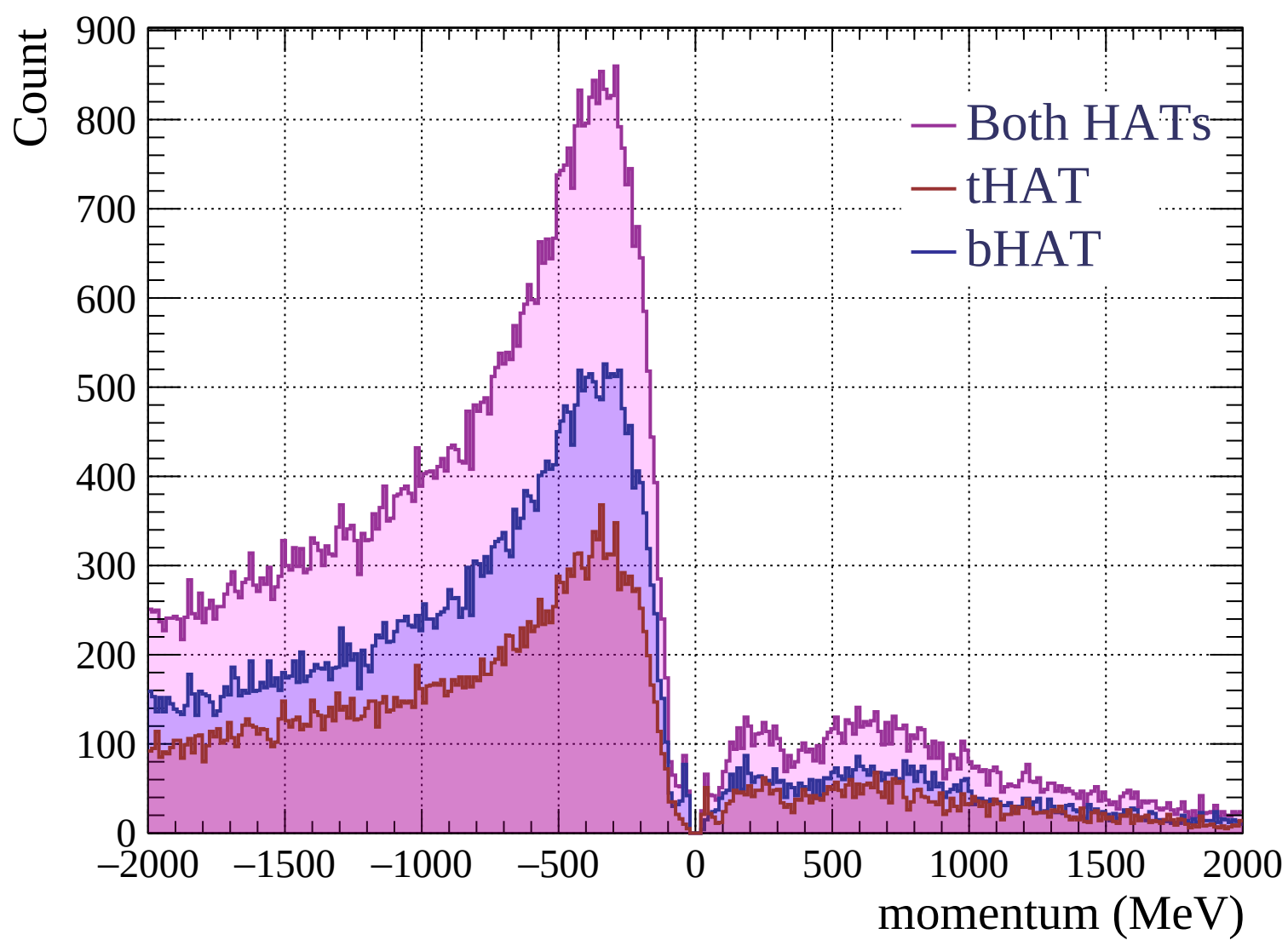


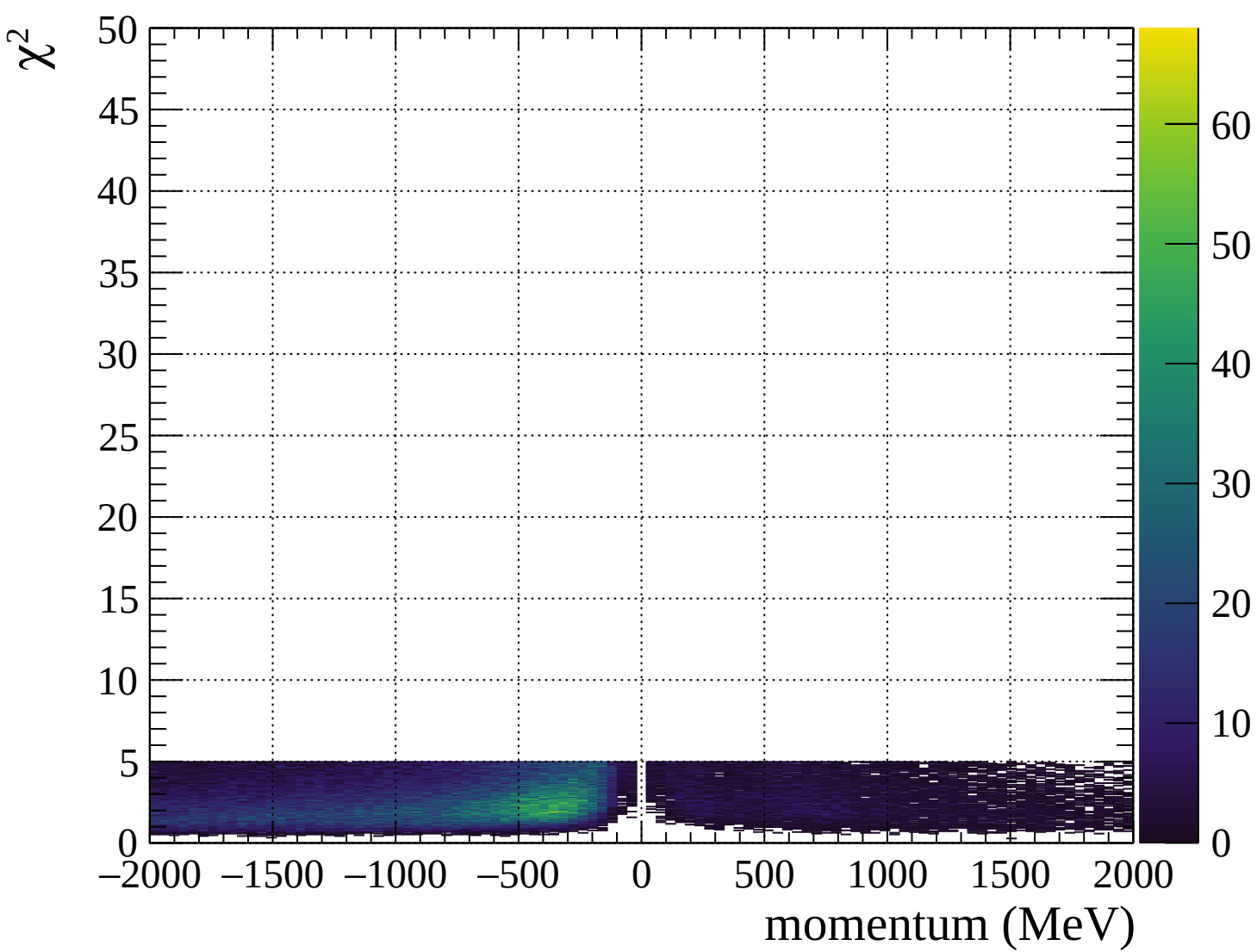




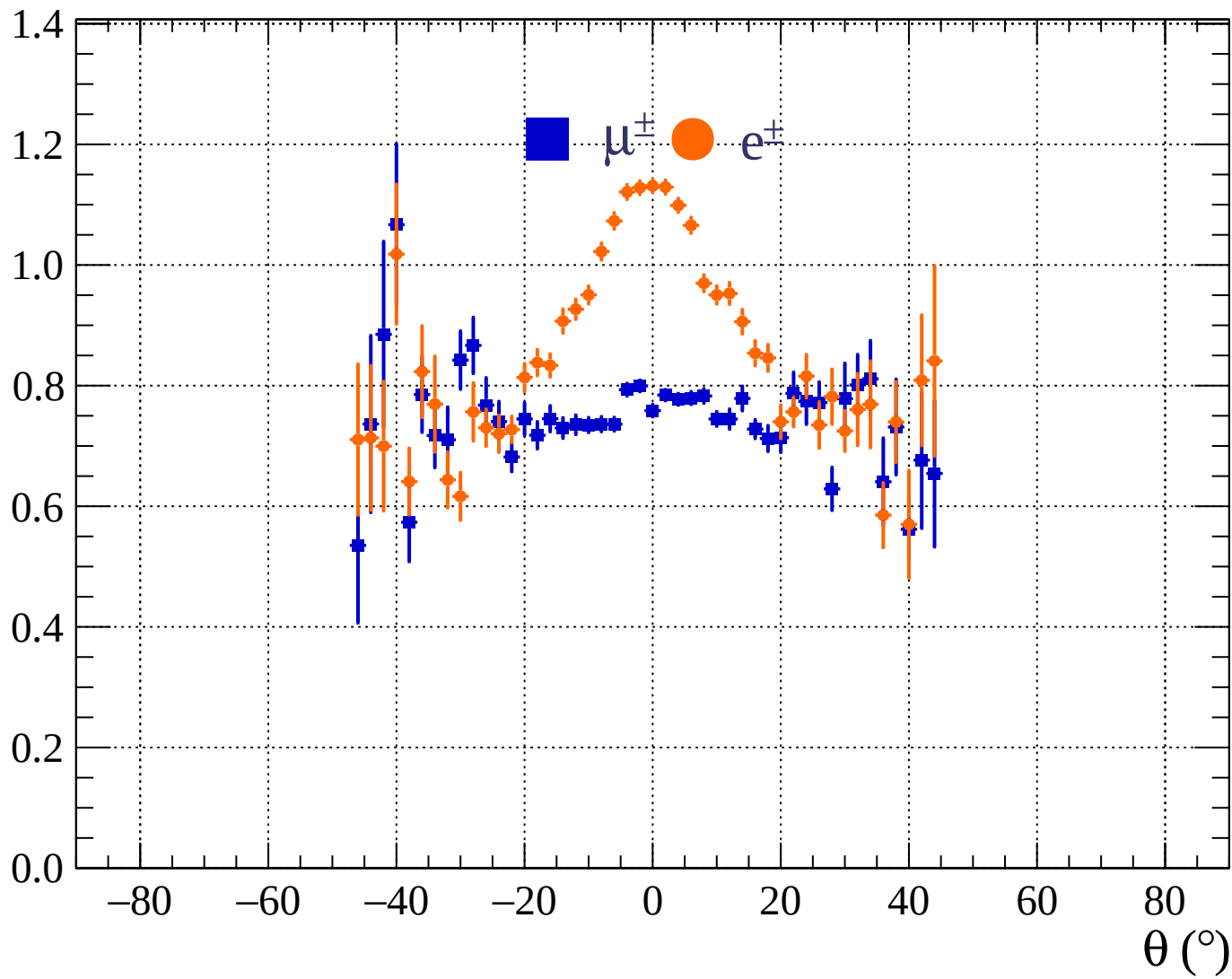


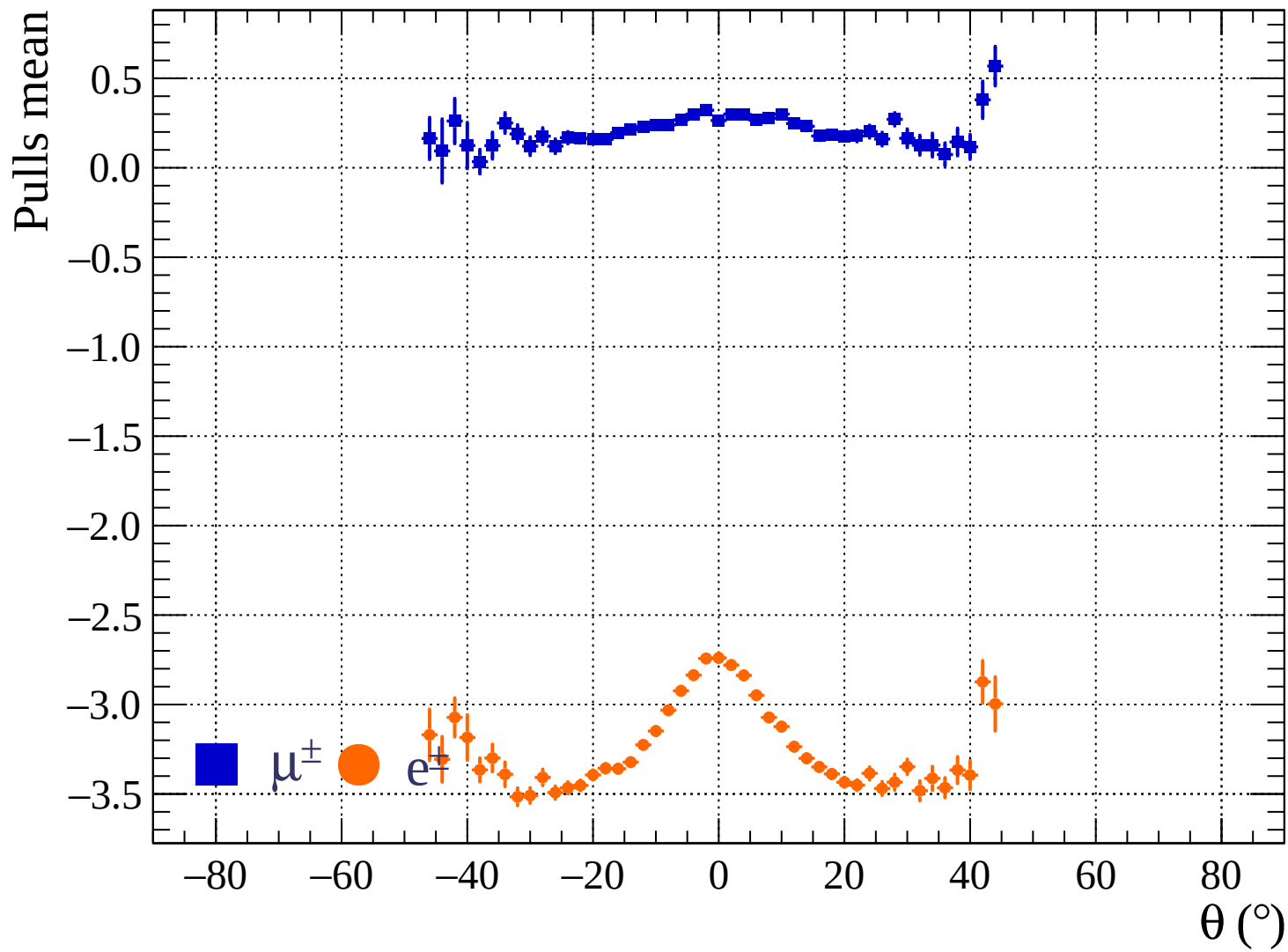




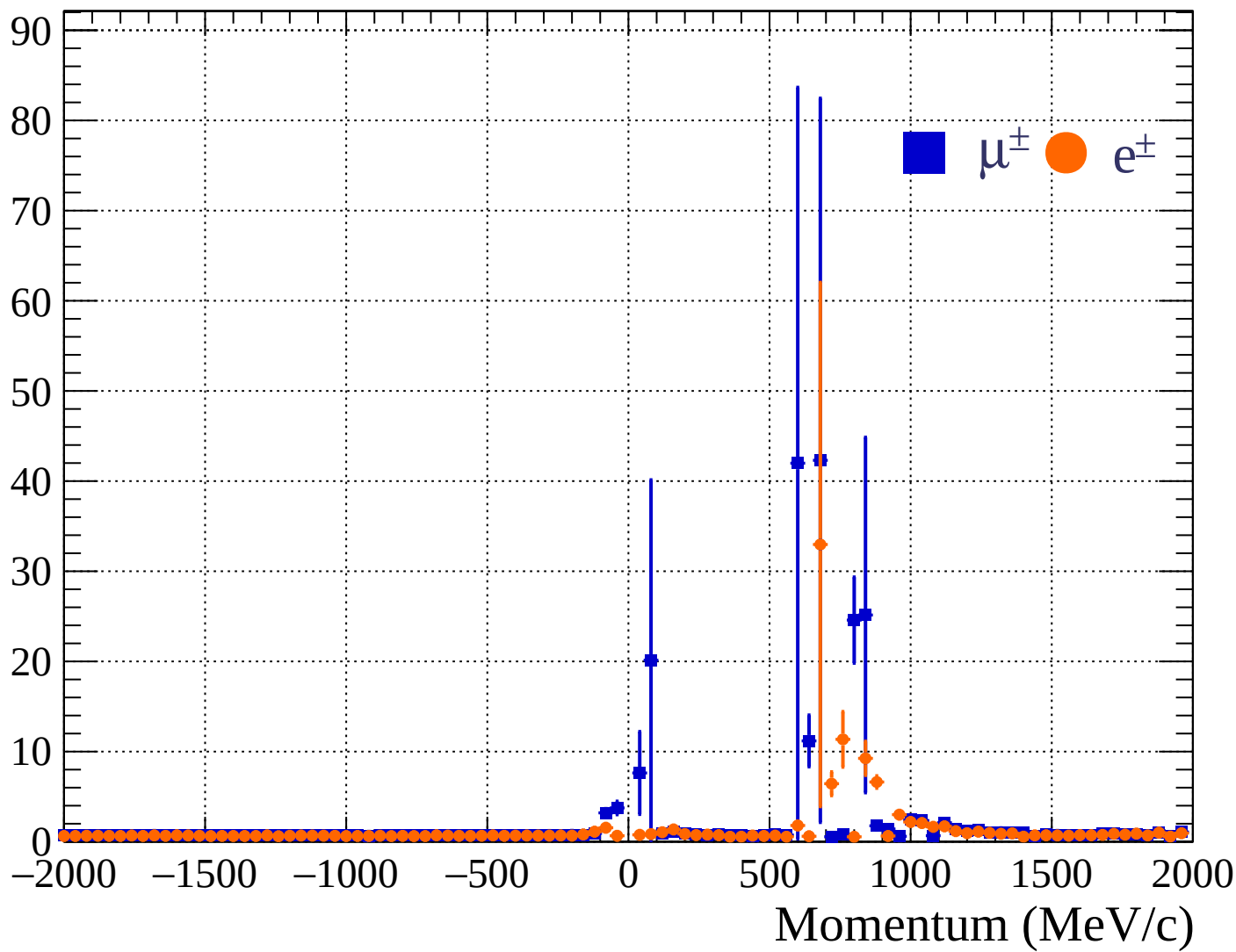


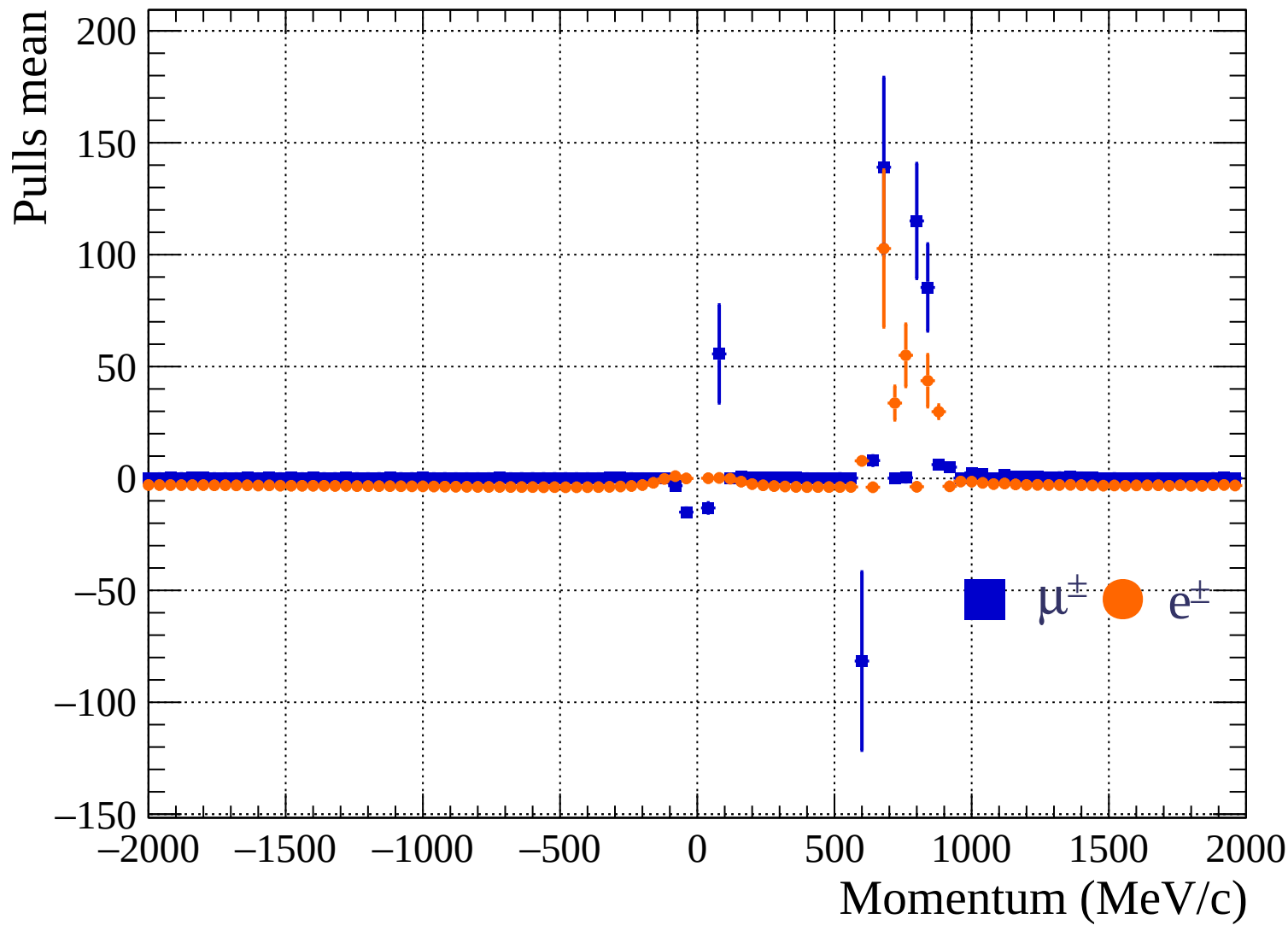
Pulls standard deviation

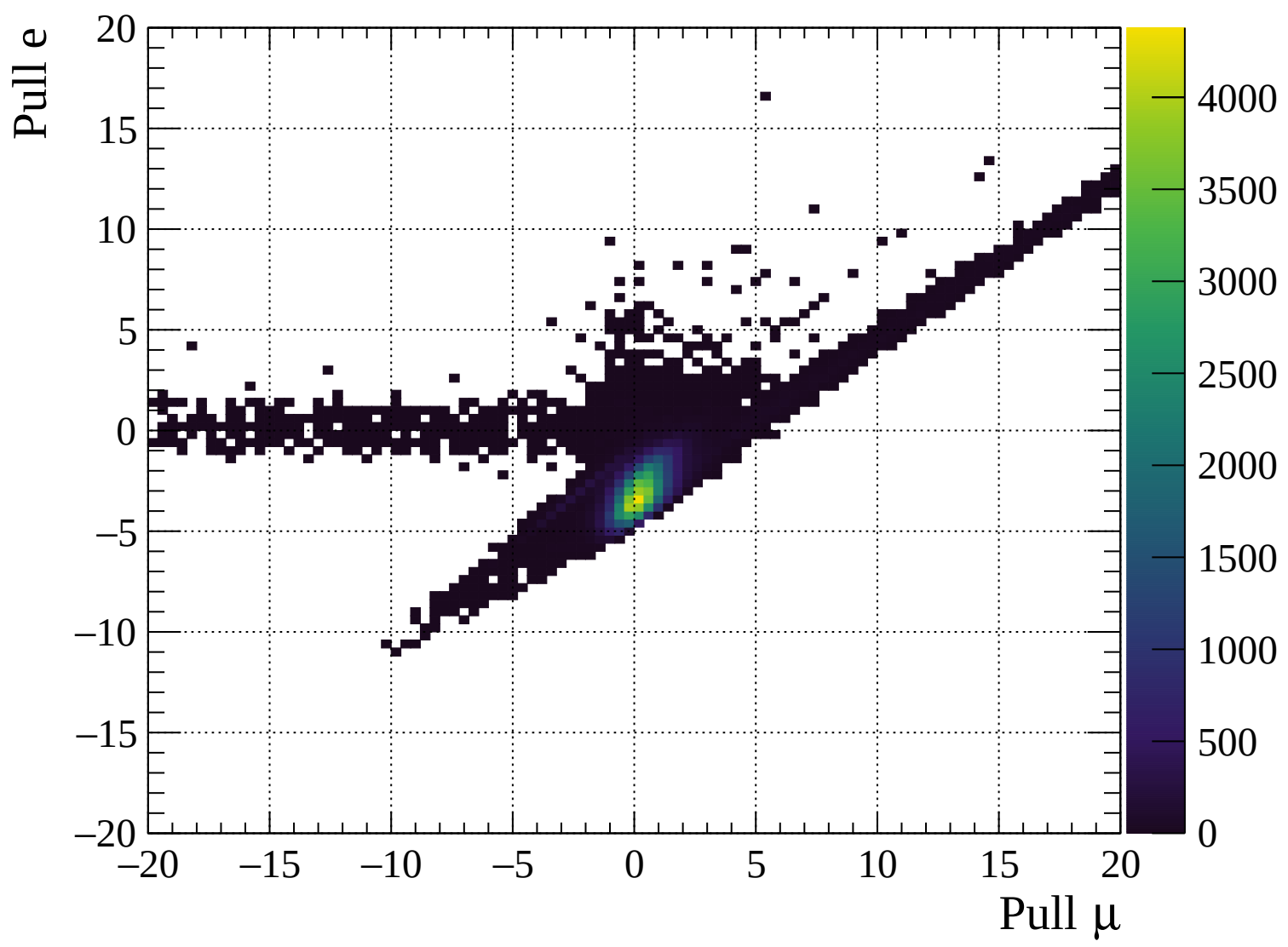


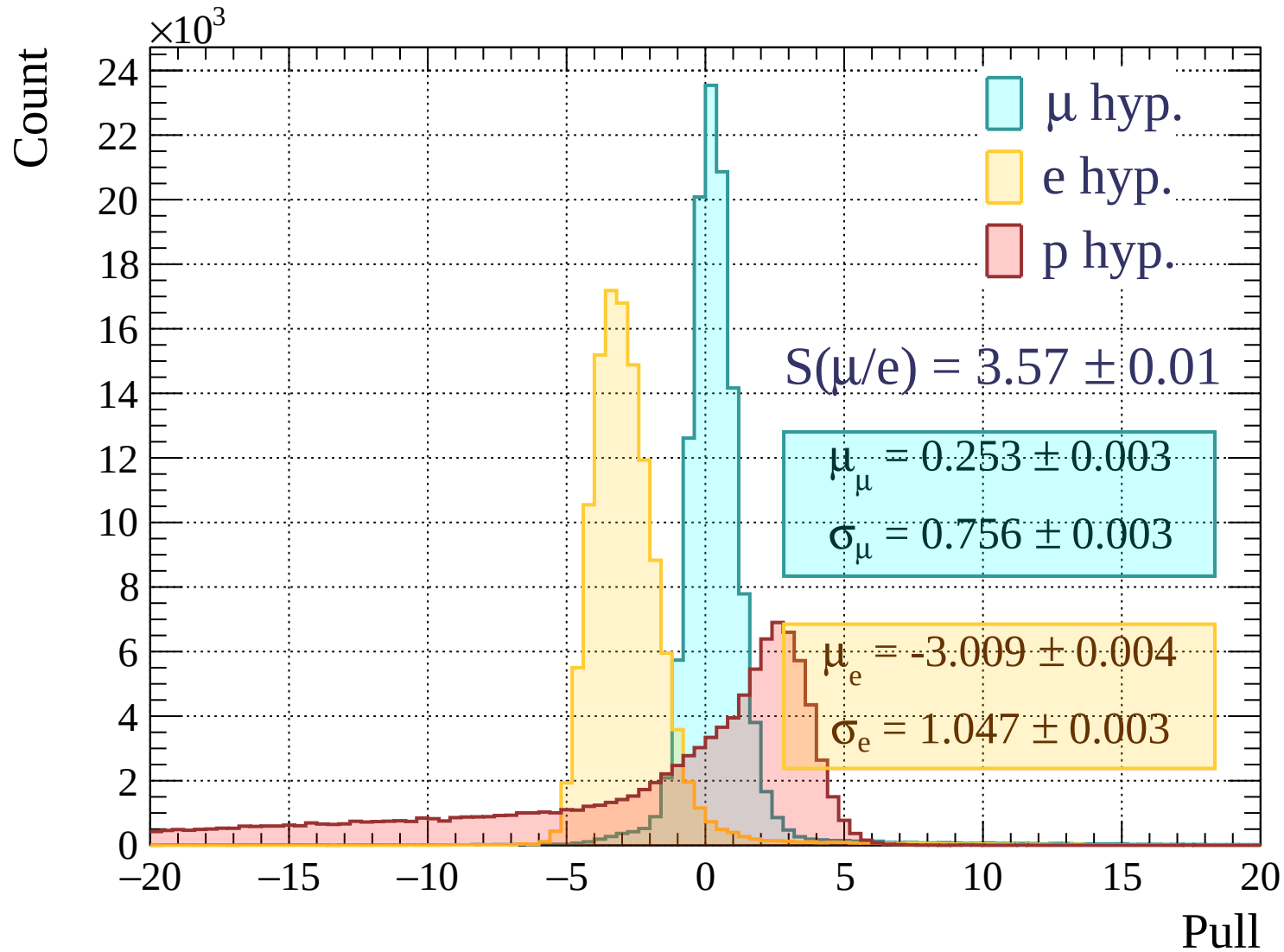


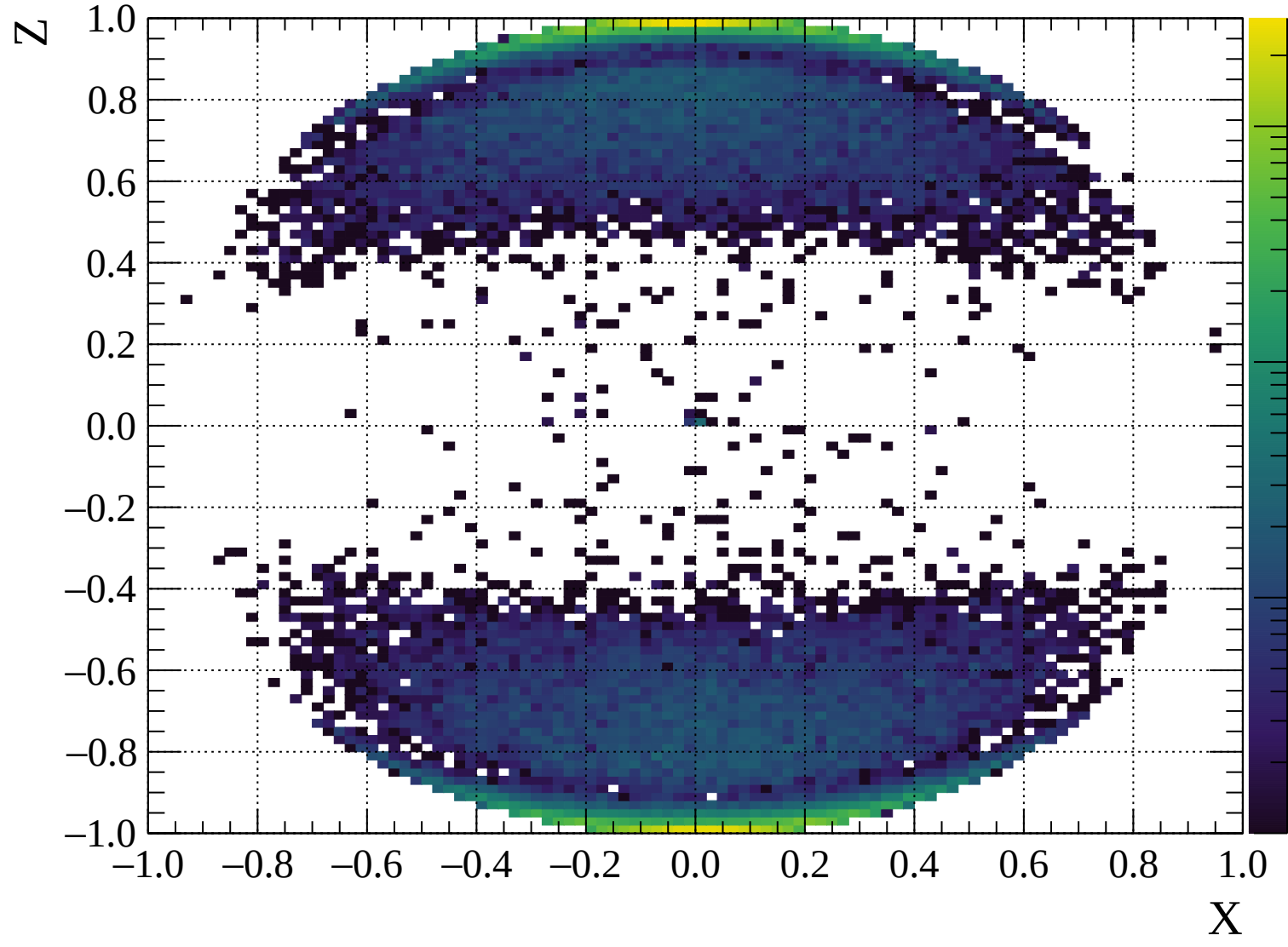
Pulls standard deviation

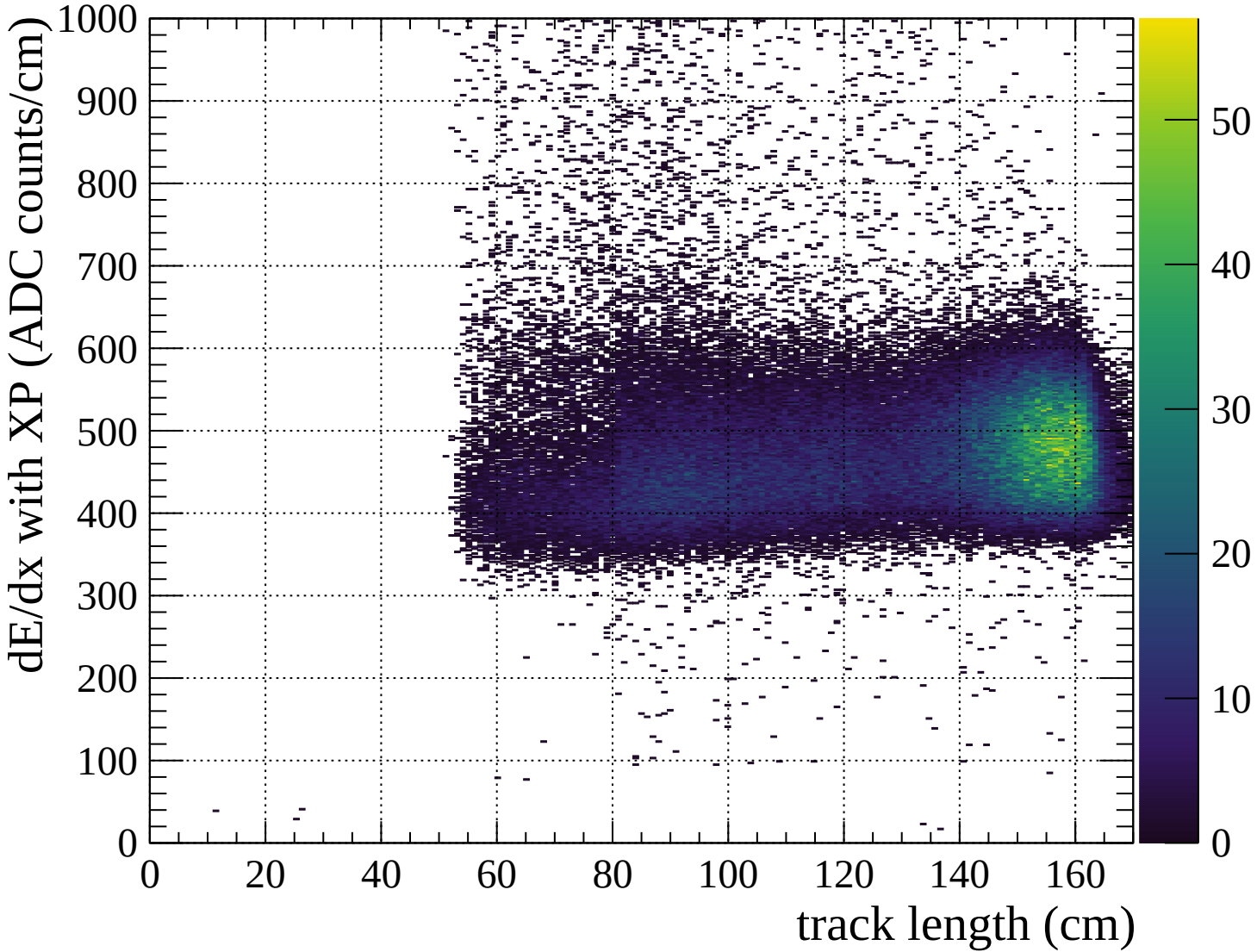


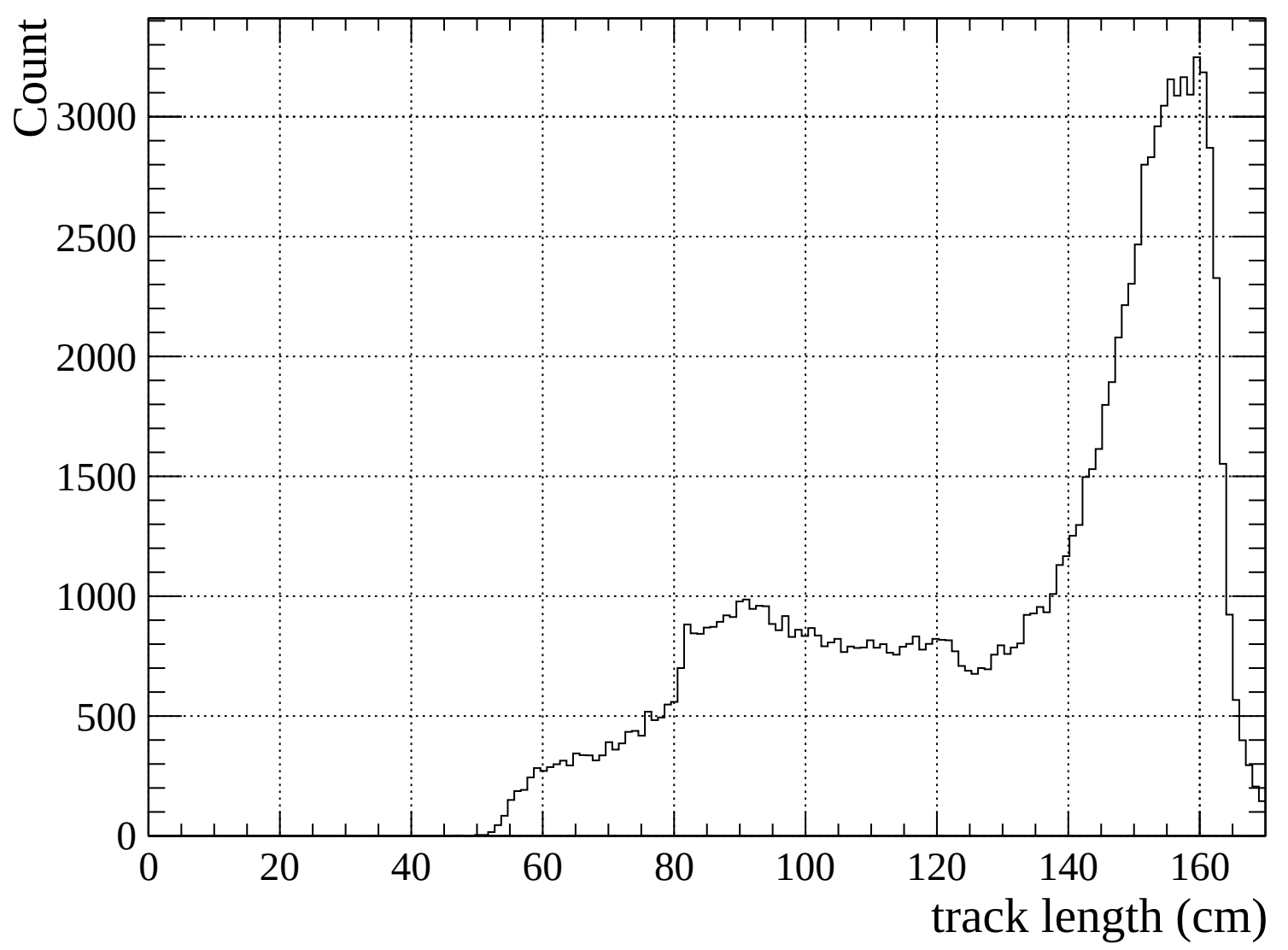


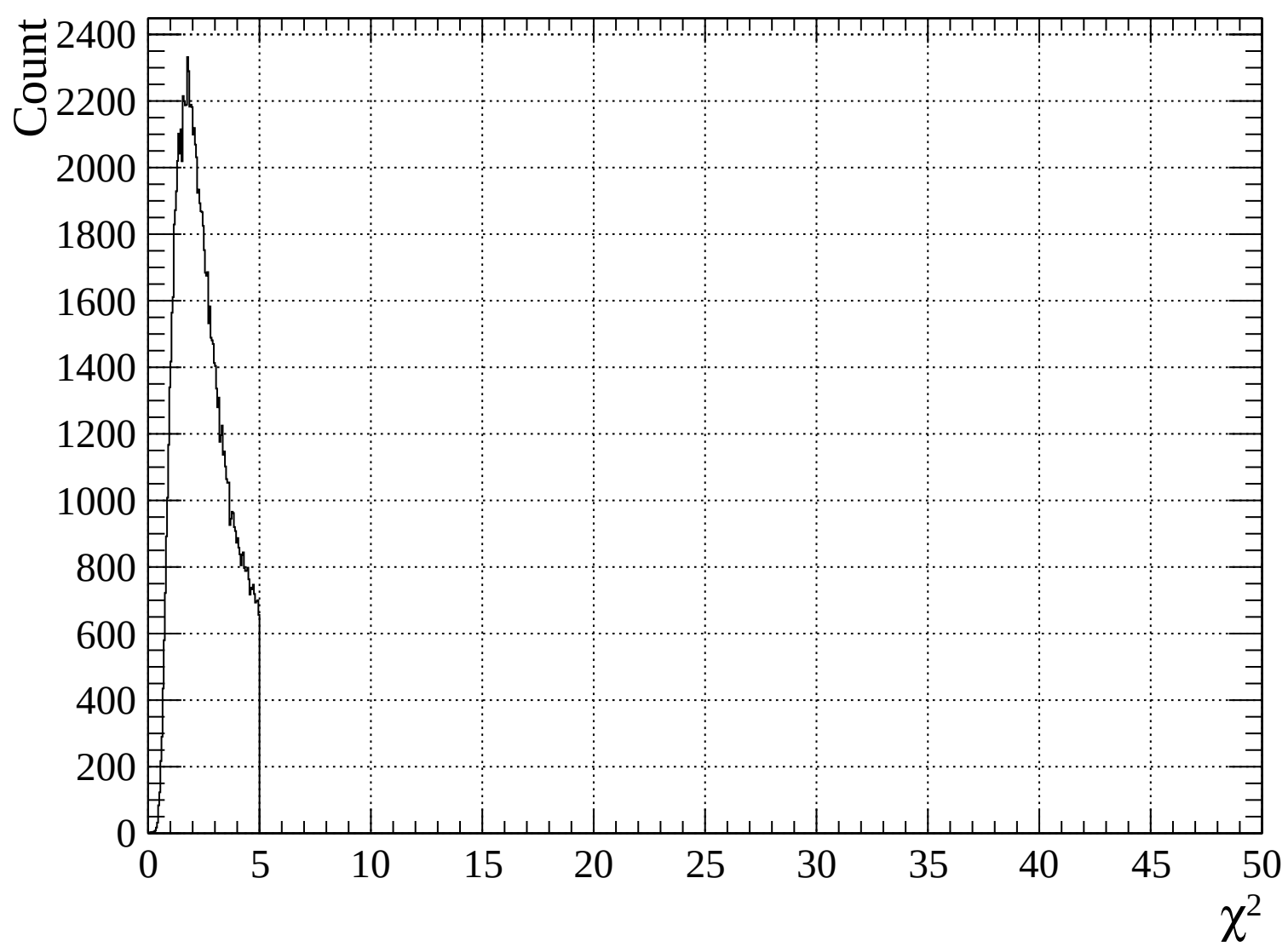


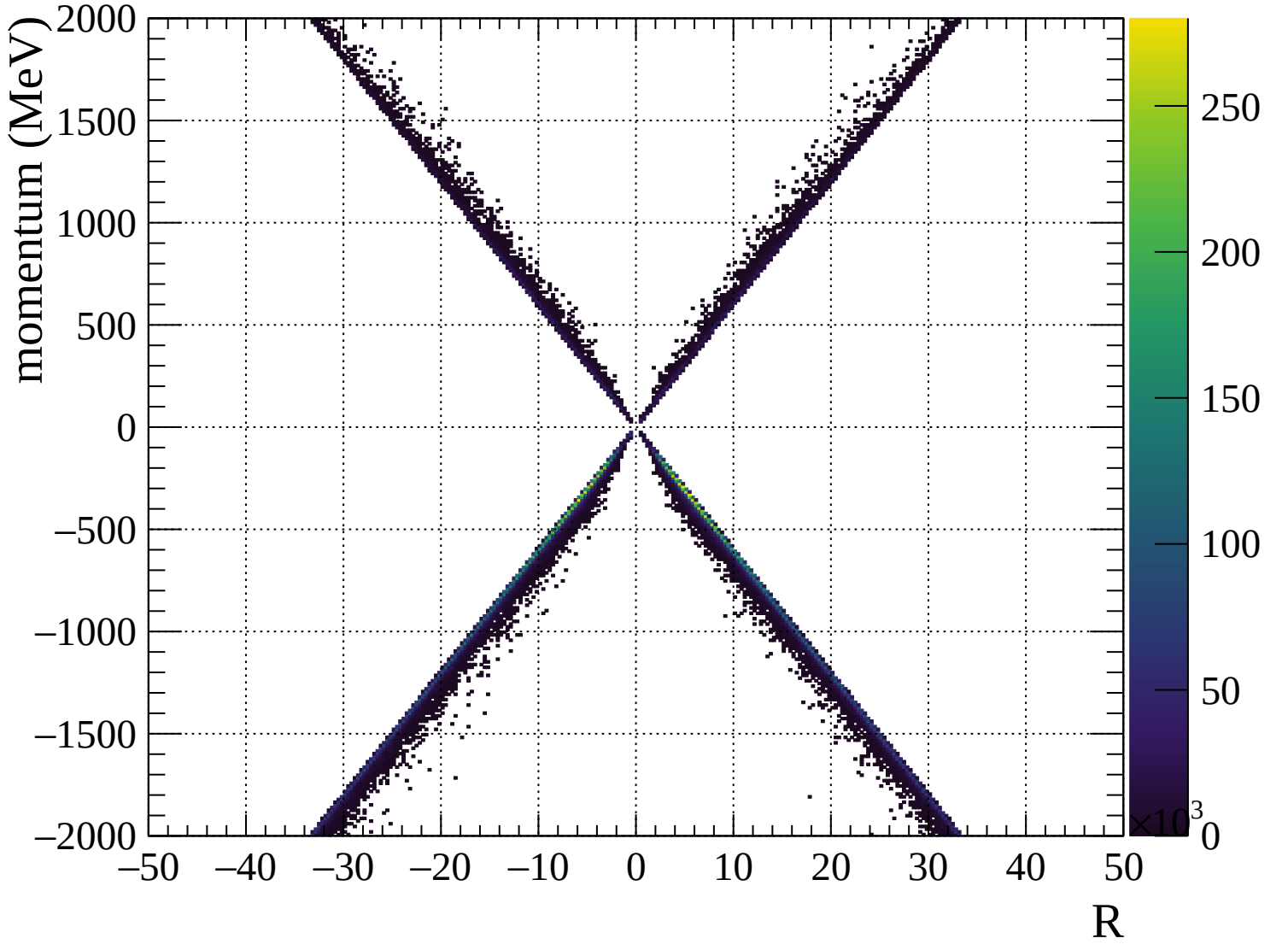


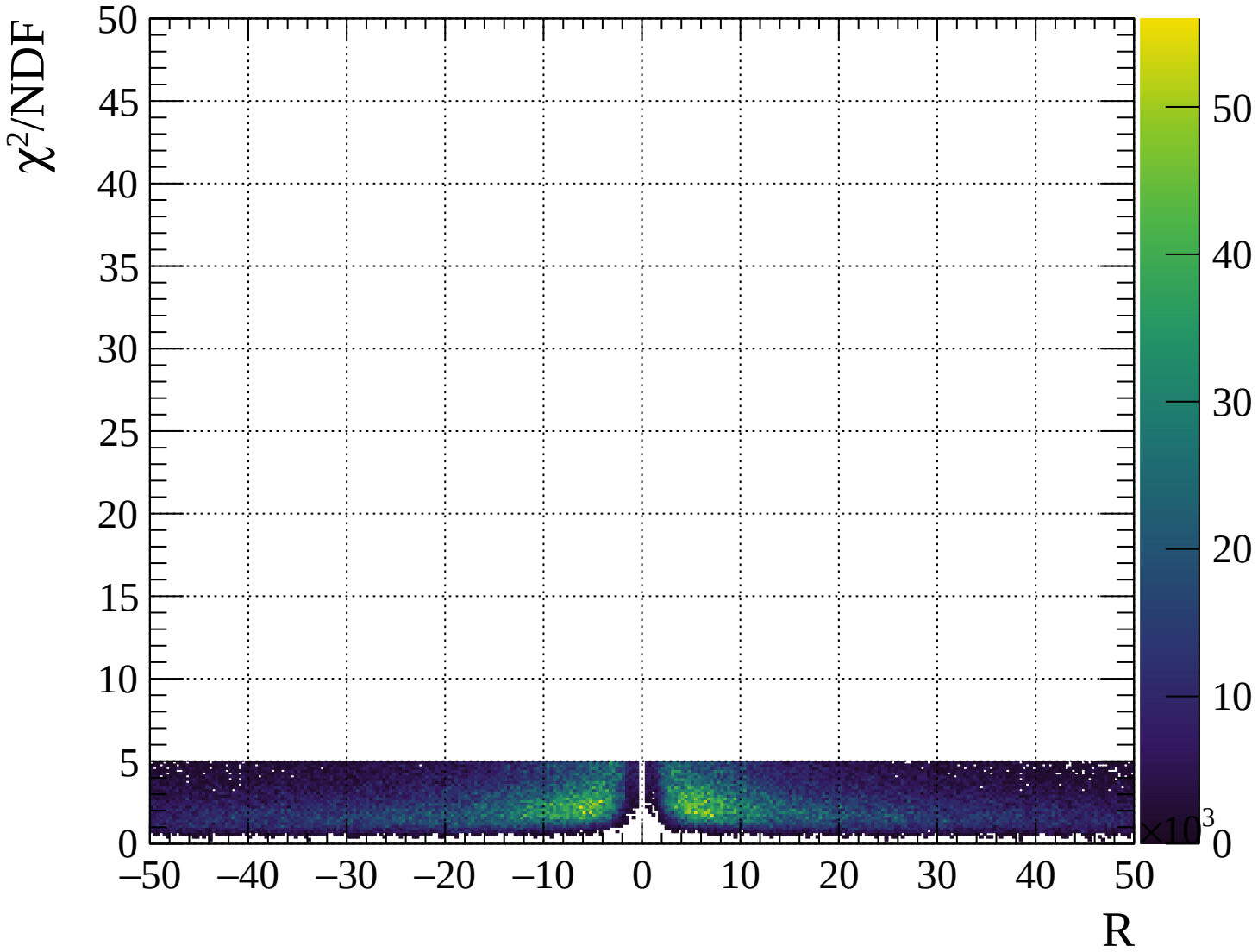






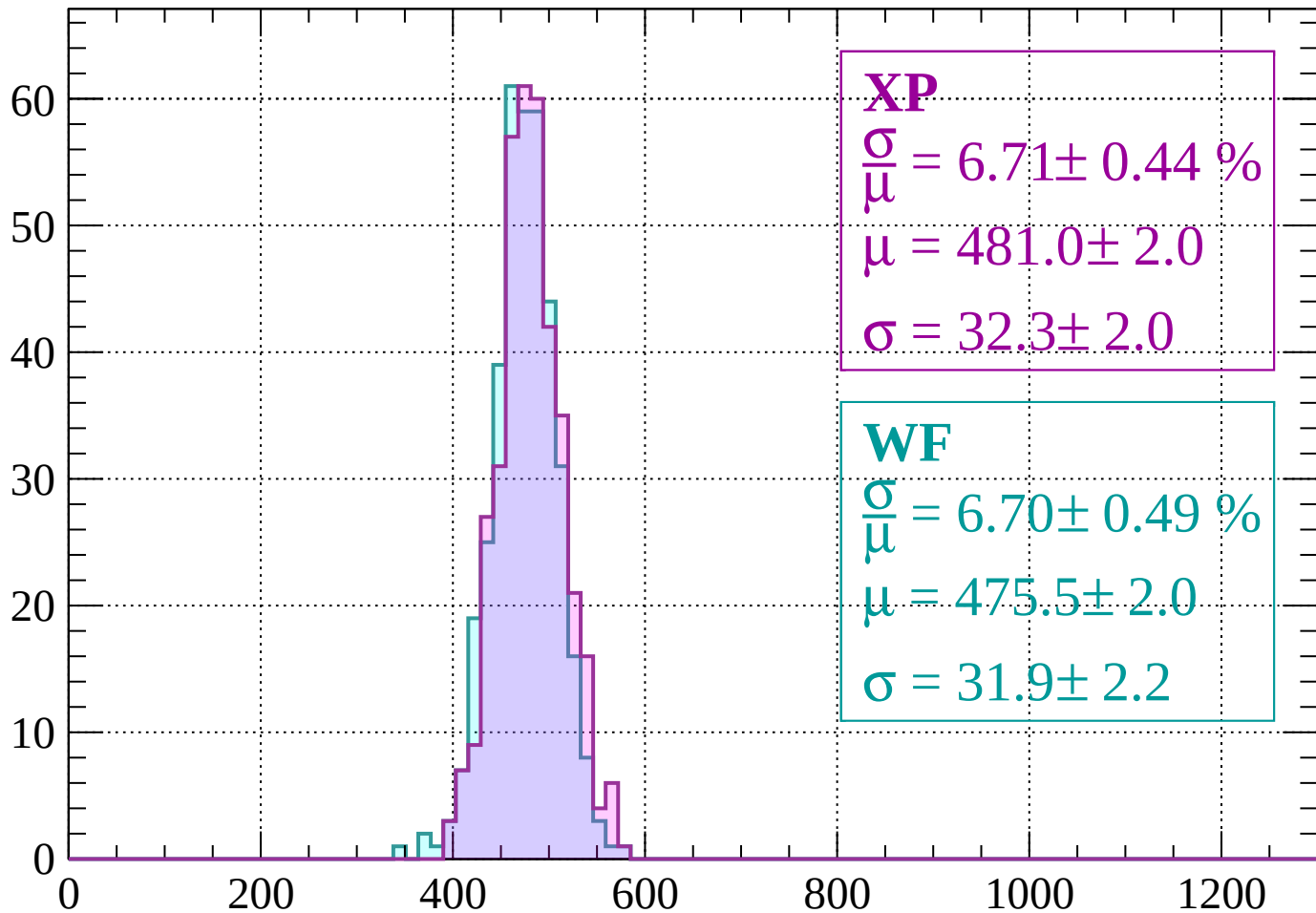






Energy loss |  $-2000 < p < -1960$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.71 \pm 0.44 \%$$

$$\mu = 481.0 \pm 2.0$$

$$\sigma = 32.3 \pm 2.0$$

**WF**

$$\frac{\sigma}{\mu} = 6.70 \pm 0.49 \%$$

$$\mu = 475.5 \pm 2.0$$

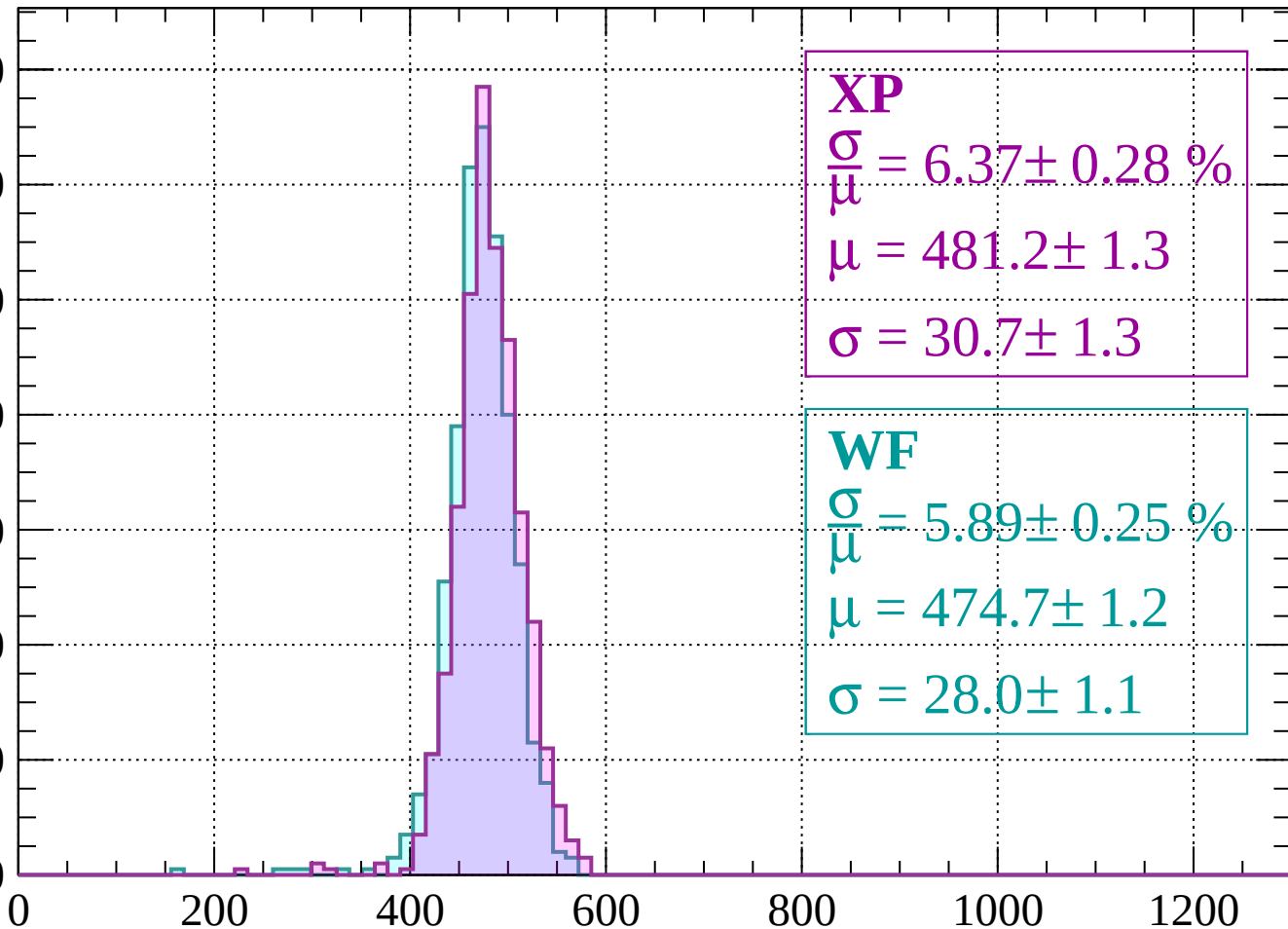
$$\sigma = 31.9 \pm 2.2$$

dE/dx (ADC counts/cm)

# Energy loss | -1960 < p < -1920

Count

140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.37 \pm 0.28 \%$$

$$\mu = 481.2 \pm 1.3$$

$$\sigma = 30.7 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 5.89 \pm 0.25 \%$$

$$\mu = 474.7 \pm 1.2$$

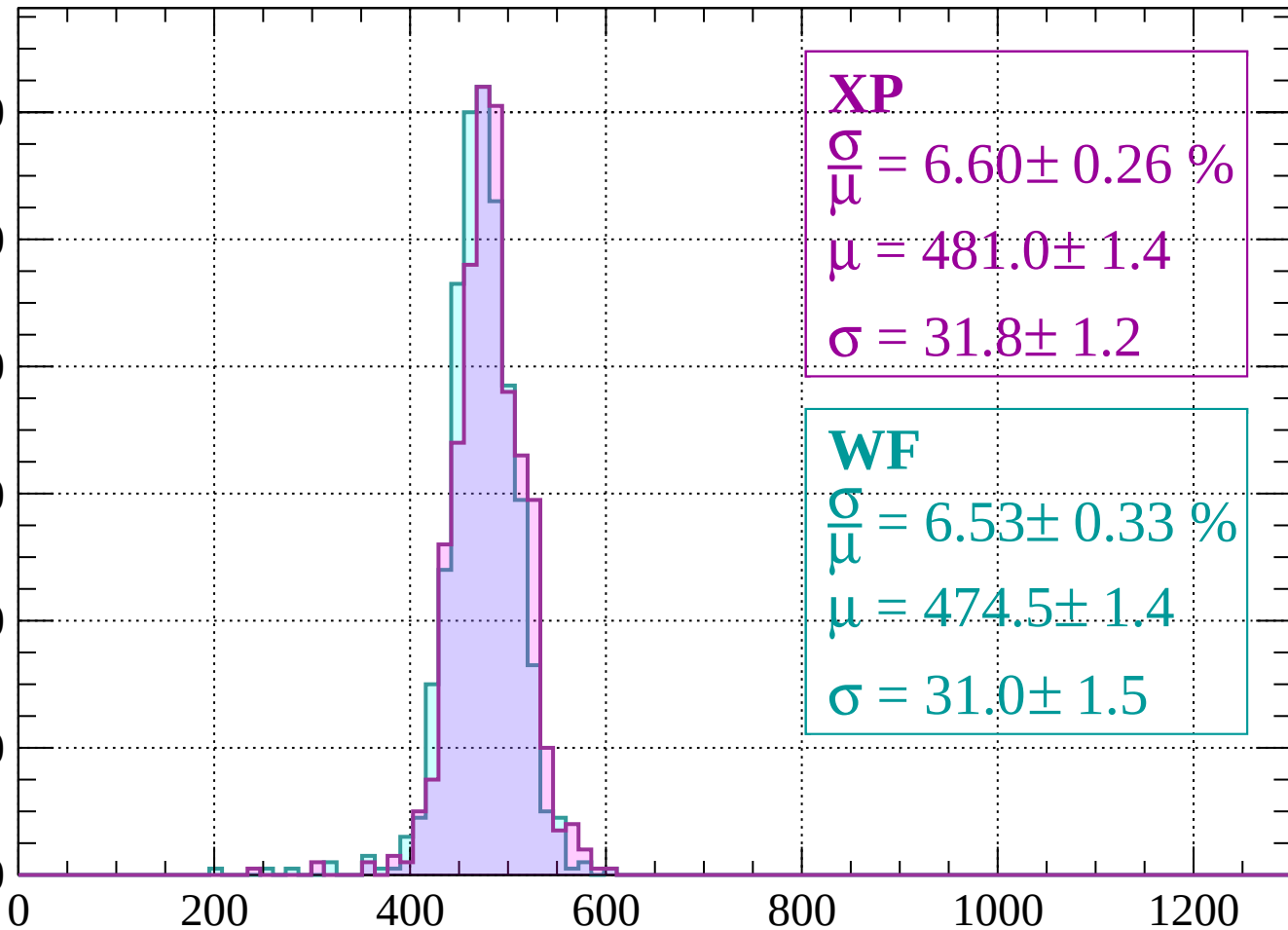
$$\sigma = 28.0 \pm 1.1$$

dE/dx (ADC counts/cm)

# Energy loss | -1920 < p < -1880

Count

120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.60 \pm 0.26 \%$$

$$\mu = 481.0 \pm 1.4$$

$$\sigma = 31.8 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.53 \pm 0.33 \%$$

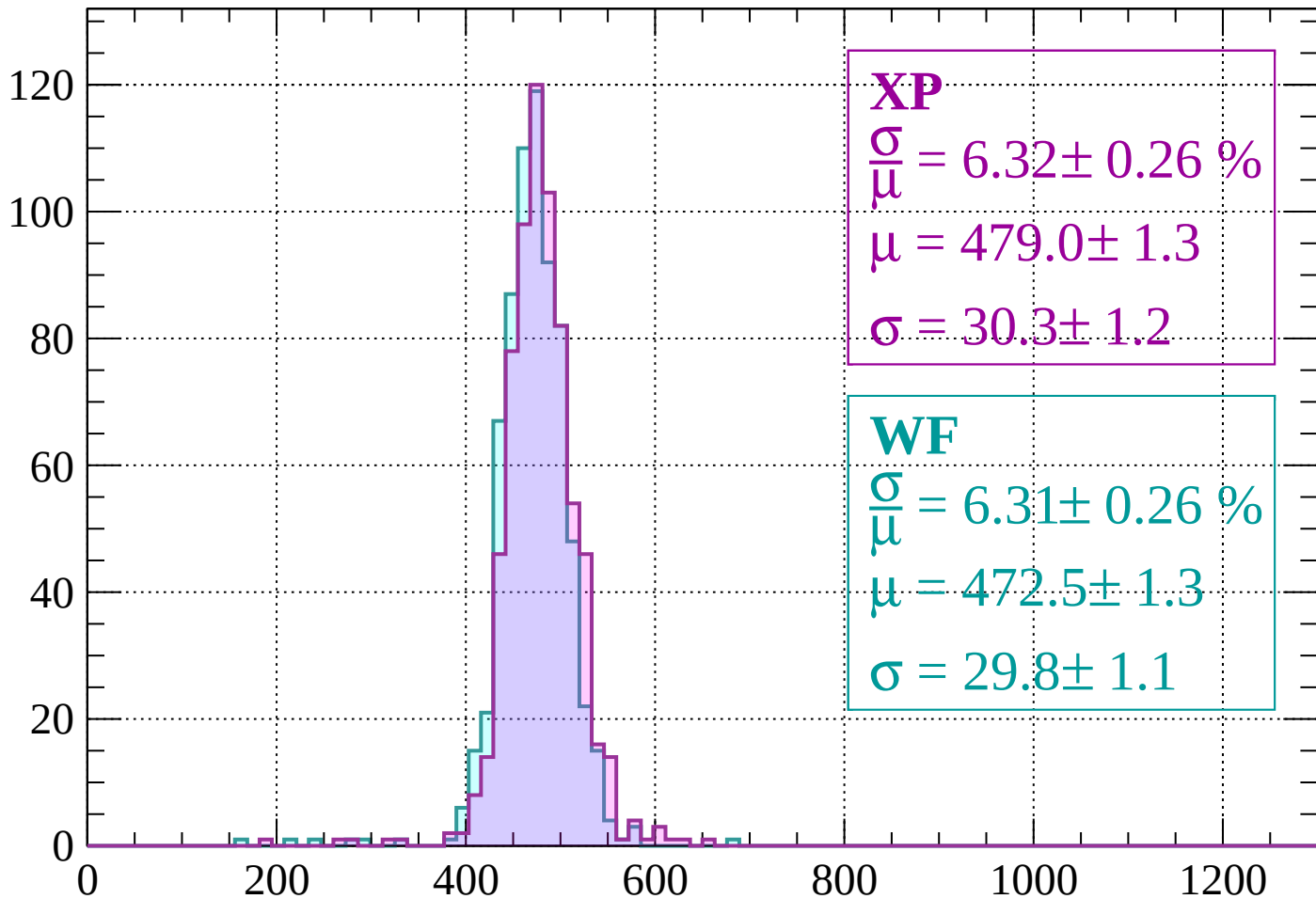
$$\mu = 474.5 \pm 1.4$$

$$\sigma = 31.0 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss |  $-1880 < p < -1840$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.32 \pm 0.26 \%$$

$$\mu = 479.0 \pm 1.3$$

$$\sigma = 30.3 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.31 \pm 0.26 \%$$

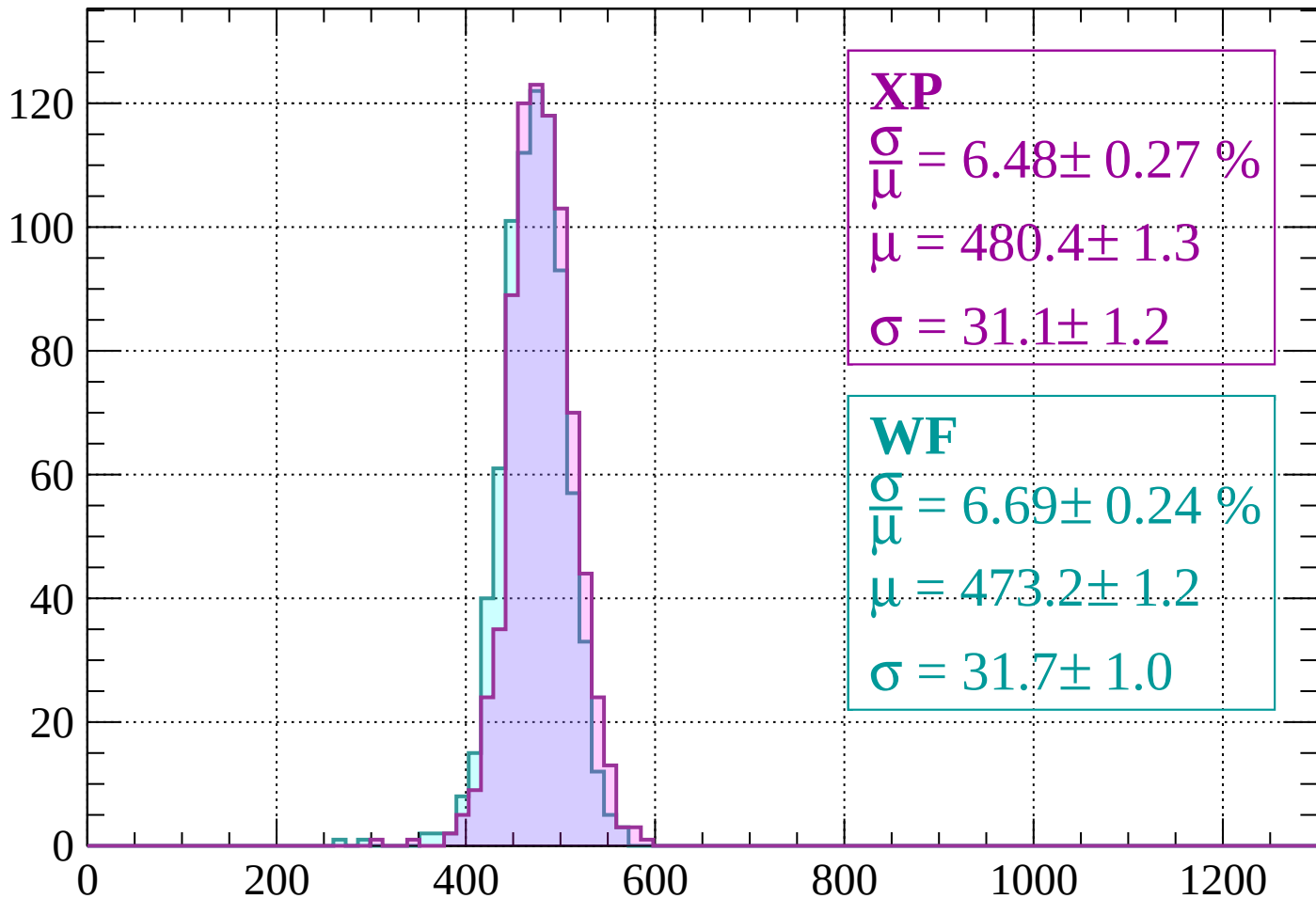
$$\mu = 472.5 \pm 1.3$$

$$\sigma = 29.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-1840 < p < -1800$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.48 \pm 0.27 \%$$

$$\mu = 480.4 \pm 1.3$$

$$\sigma = 31.1 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.69 \pm 0.24 \%$$

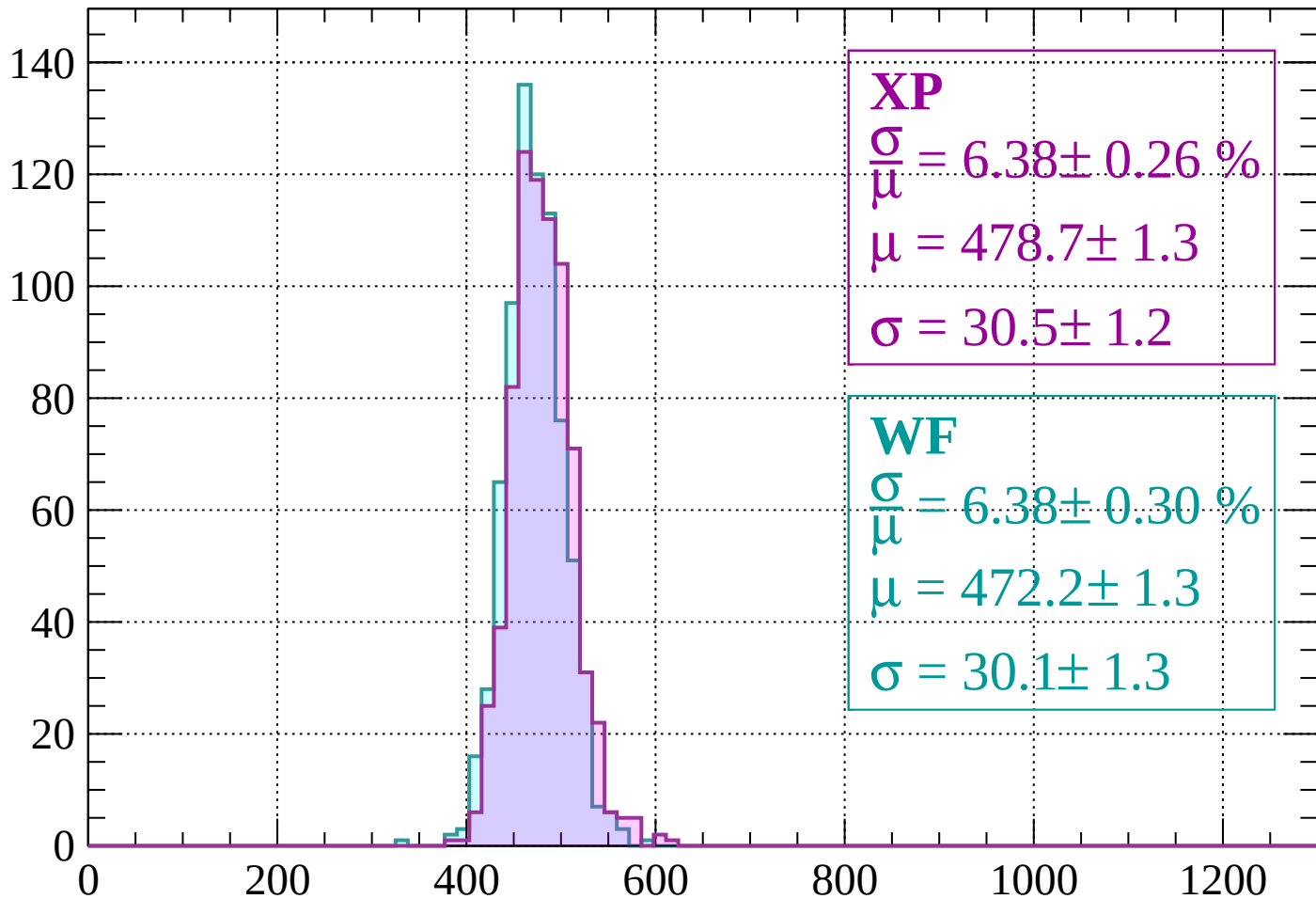
$$\mu = 473.2 \pm 1.2$$

$$\sigma = 31.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $-1800 < p < -1760$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.38 \pm 0.26 \%$$

$$\mu = 478.7 \pm 1.3$$

$$\sigma = 30.5 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.38 \pm 0.30 \%$$

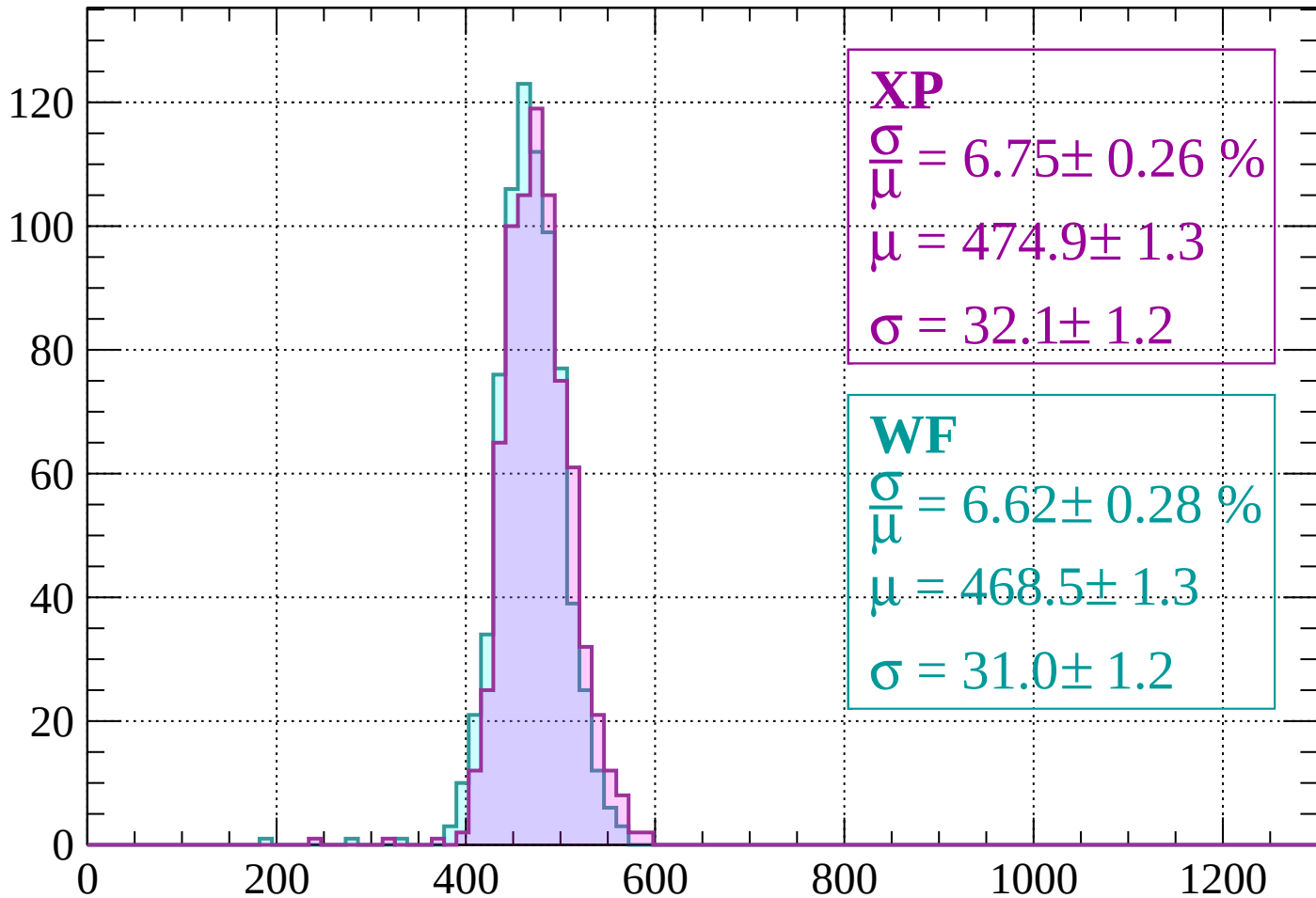
$$\mu = 472.2 \pm 1.3$$

$$\sigma = 30.1 \pm 1.3$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1760 < p < -1720$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.75 \pm 0.26 \%$$

$$\mu = 474.9 \pm 1.3$$

$$\sigma = 32.1 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.62 \pm 0.28 \%$$

$$\mu = 468.5 \pm 1.3$$

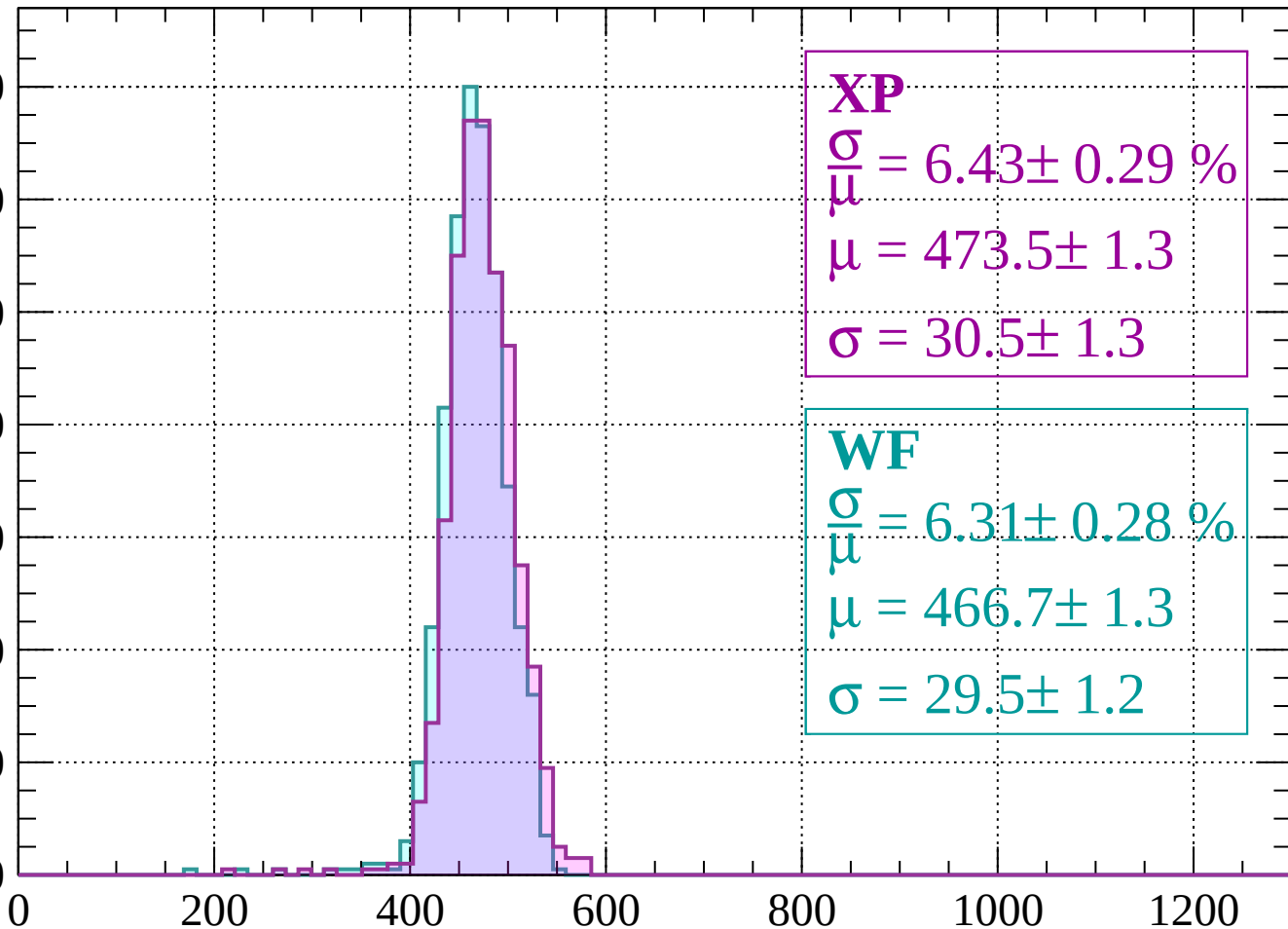
$$\sigma = 31.0 \pm 1.2$$

$dE/dx$  (ADC counts/cm)

# Energy loss | -1720 < p < -1680

Count

140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.43 \pm 0.29 \%$$

$$\mu = 473.5 \pm 1.3$$

$$\sigma = 30.5 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.31 \pm 0.28 \%$$

$$\mu = 466.7 \pm 1.3$$

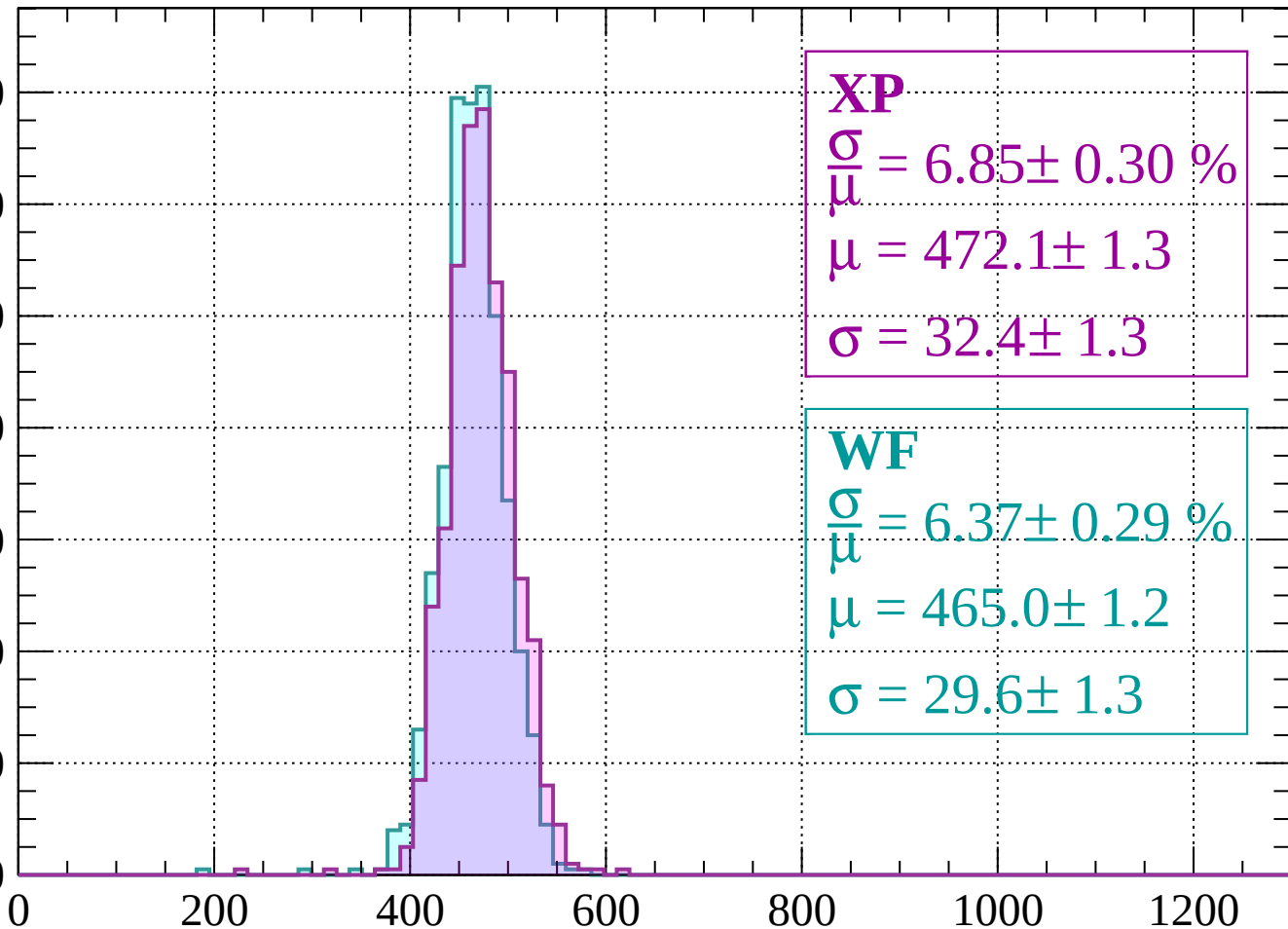
$$\sigma = 29.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss |  $-1680 < p < -1640$

Count

140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.85 \pm 0.30 \%$$

$$\mu = 472.1 \pm 1.3$$

$$\sigma = 32.4 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.37 \pm 0.29 \%$$

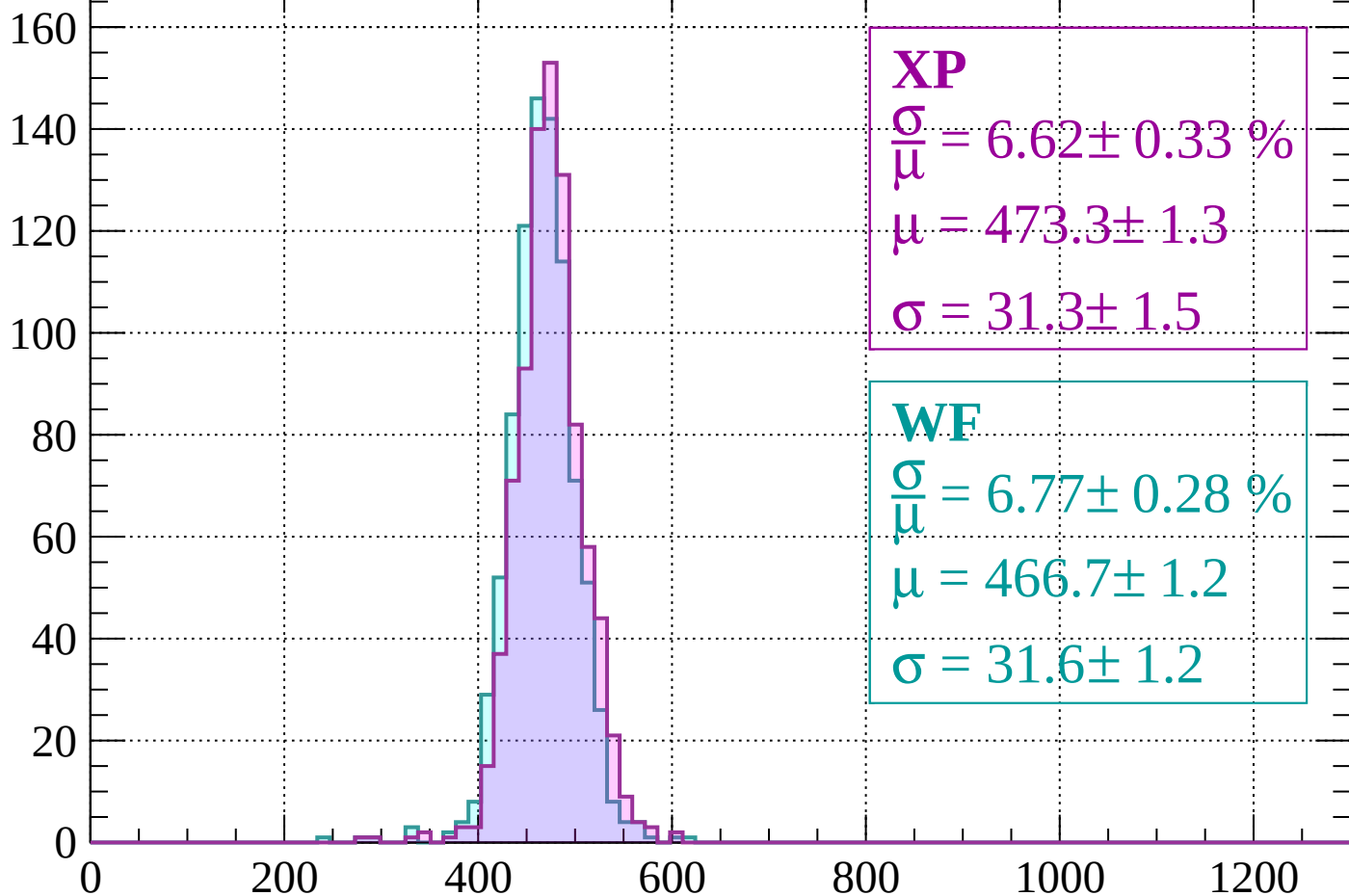
$$\mu = 465.0 \pm 1.2$$

$$\sigma = 29.6 \pm 1.3$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1640 < p < -1600$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.62 \pm 0.33 \%$$

$$\mu = 473.3 \pm 1.3$$

$$\sigma = 31.3 \pm 1.5$$

**WF**

$$\frac{\sigma}{\mu} = 6.77 \pm 0.28 \%$$

$$\mu = 466.7 \pm 1.2$$

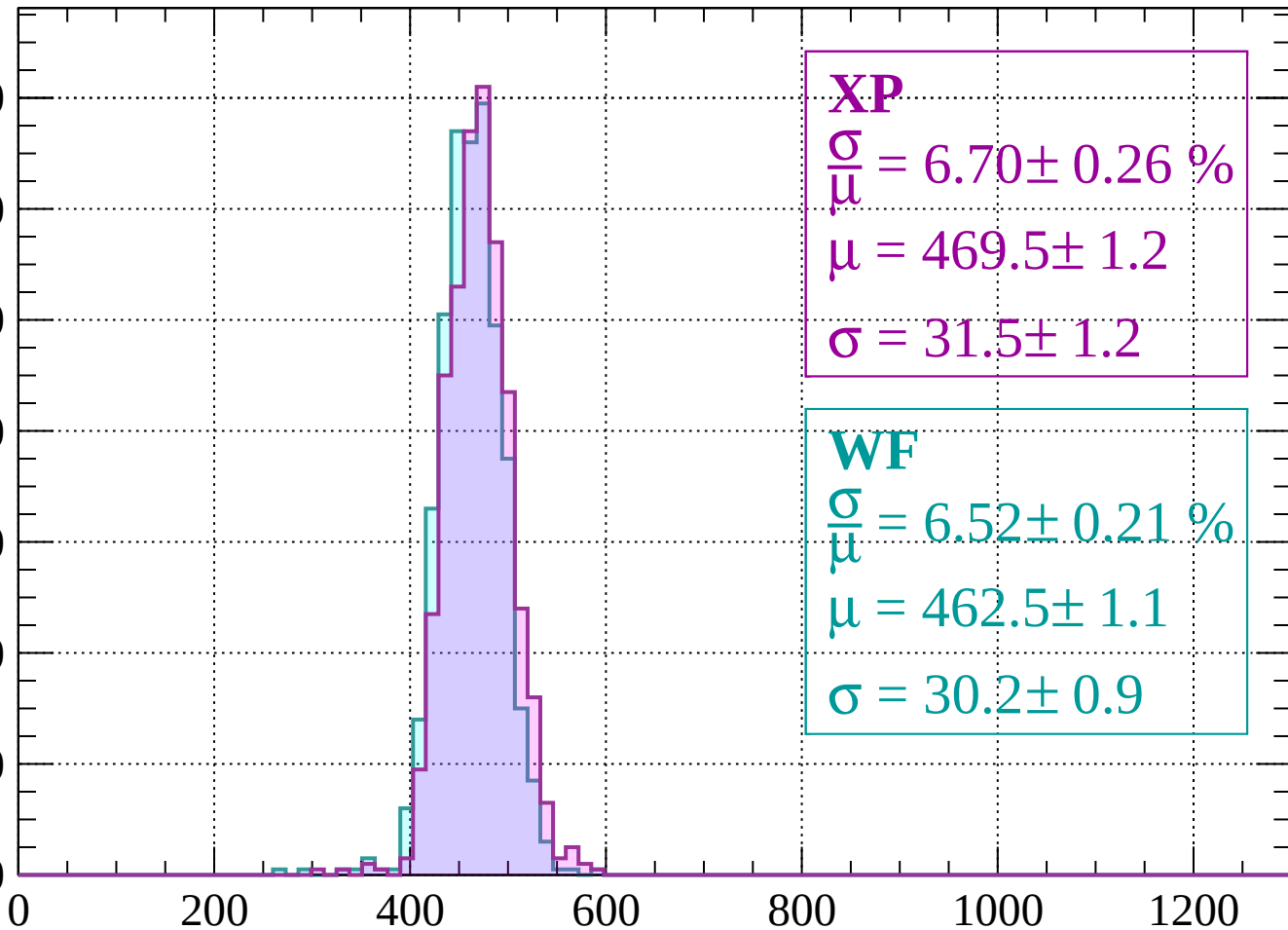
$$\sigma = 31.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss |  $-1600 < p < -1560$

Count

140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.70 \pm 0.26 \%$$

$$\mu = 469.5 \pm 1.2$$

$$\sigma = 31.5 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.52 \pm 0.21 \%$$

$$\mu = 462.5 \pm 1.1$$

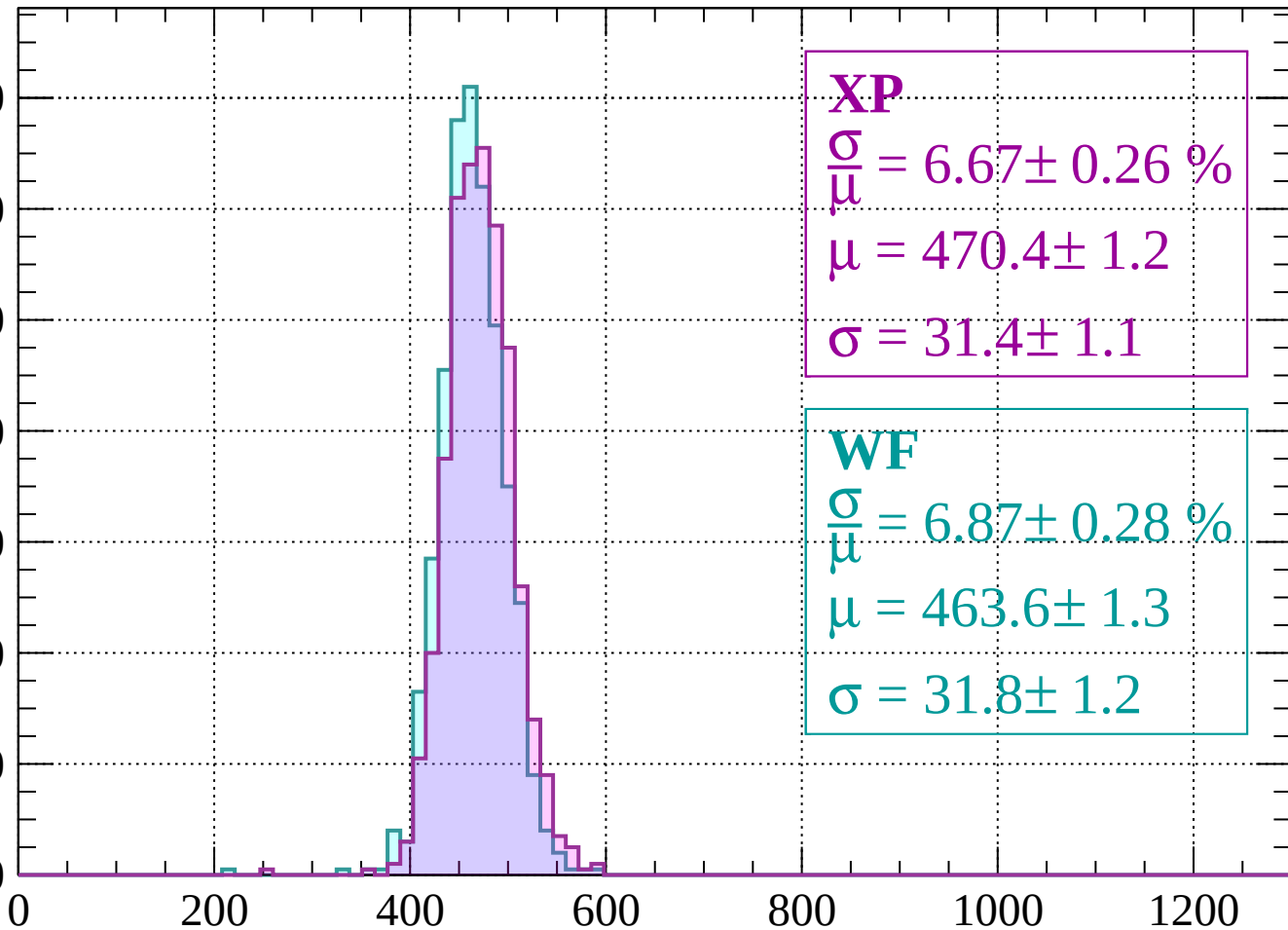
$$\sigma = 30.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $-1560 < p < -1520$

Count

140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.67 \pm 0.26 \%$$

$$\mu = 470.4 \pm 1.2$$

$$\sigma = 31.4 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 6.87 \pm 0.28 \%$$

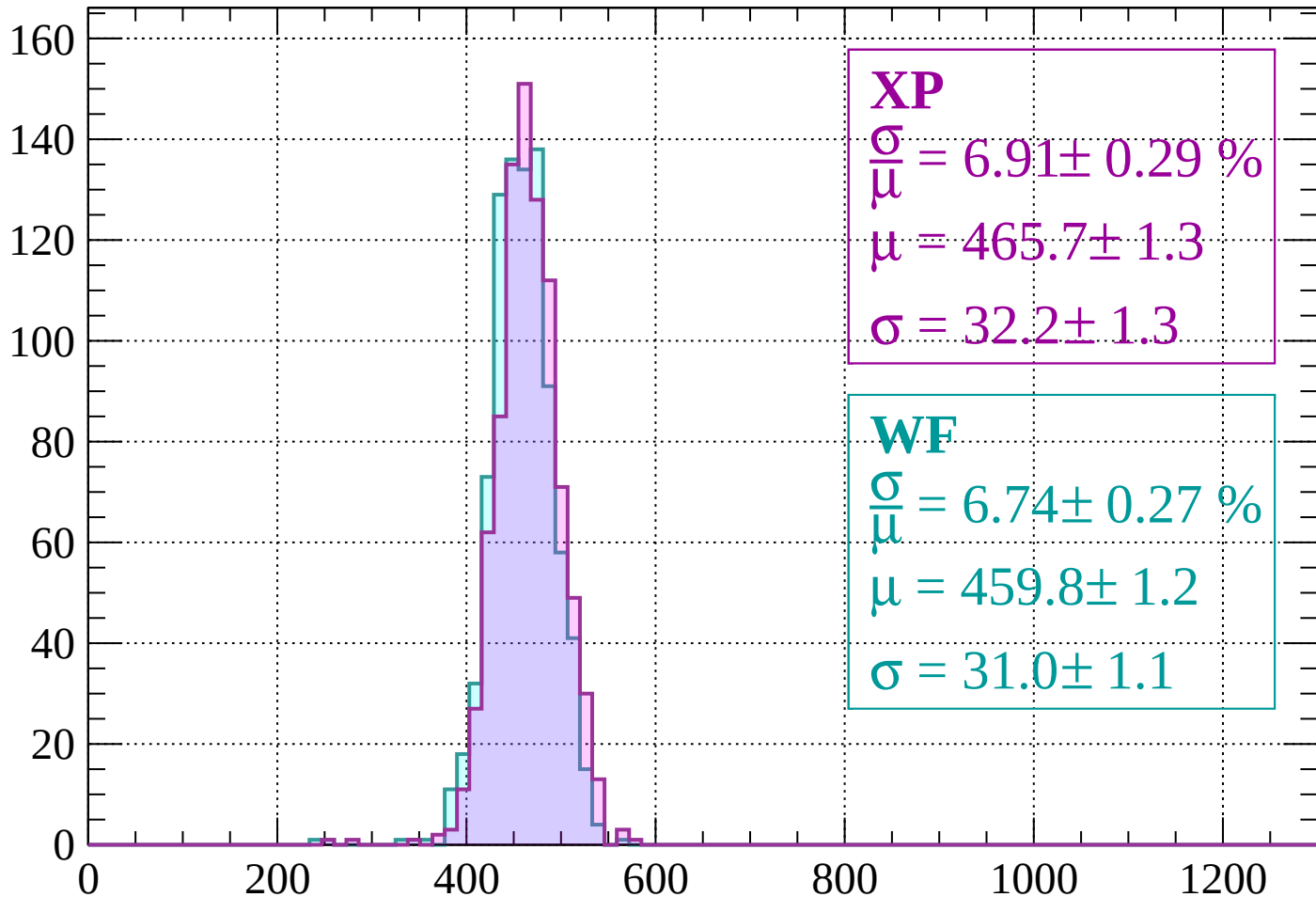
$$\mu = 463.6 \pm 1.3$$

$$\sigma = 31.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss |  $-1520 < p < -1480$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.91 \pm 0.29 \%$$

$$\mu = 465.7 \pm 1.3$$

$$\sigma = 32.2 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.74 \pm 0.27 \%$$

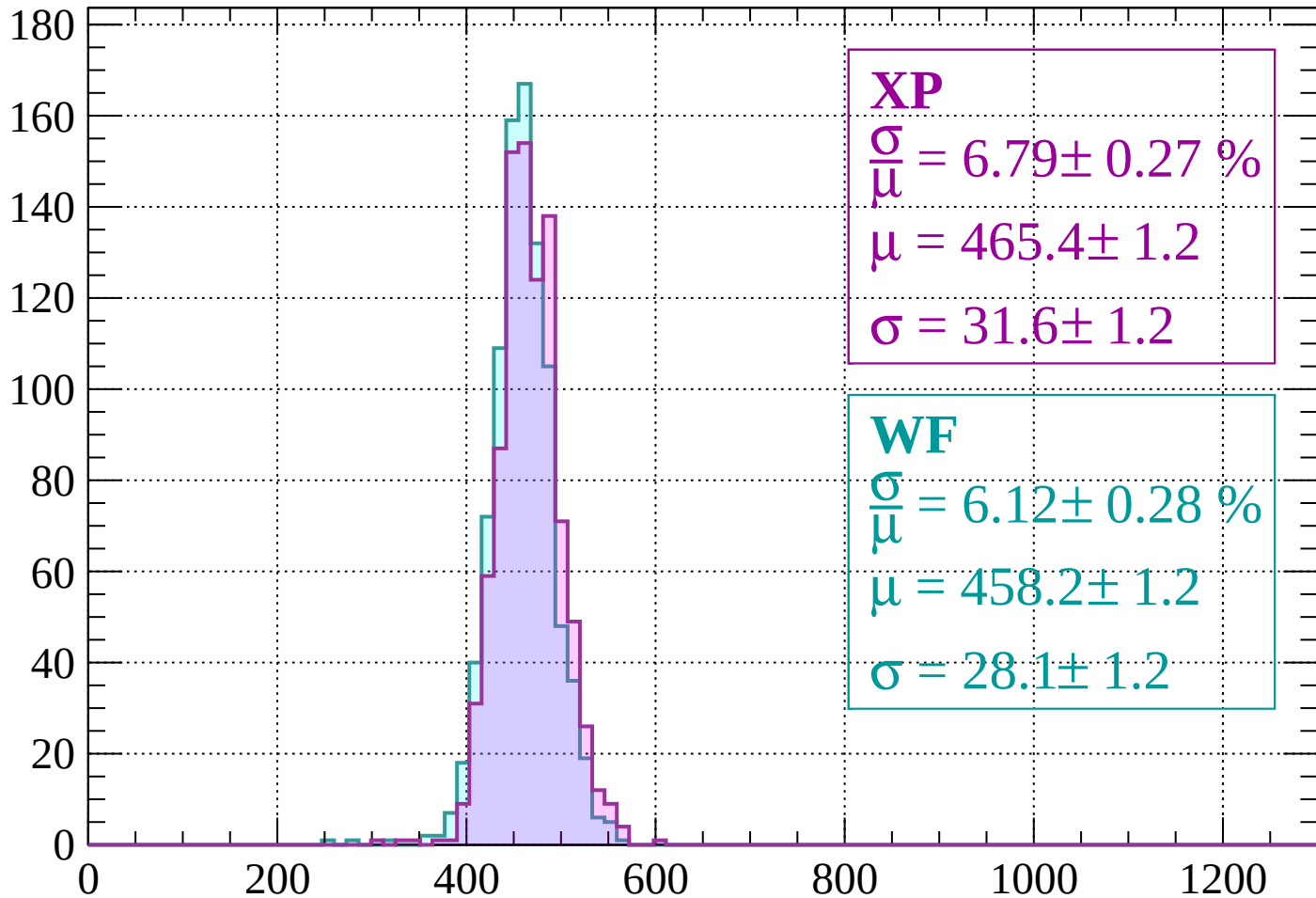
$$\mu = 459.8 \pm 1.2$$

$$\sigma = 31.0 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-1480 < p < -1440$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.79 \pm 0.27 \%$$

$$\mu = 465.4 \pm 1.2$$

$$\sigma = 31.6 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.12 \pm 0.28 \%$$

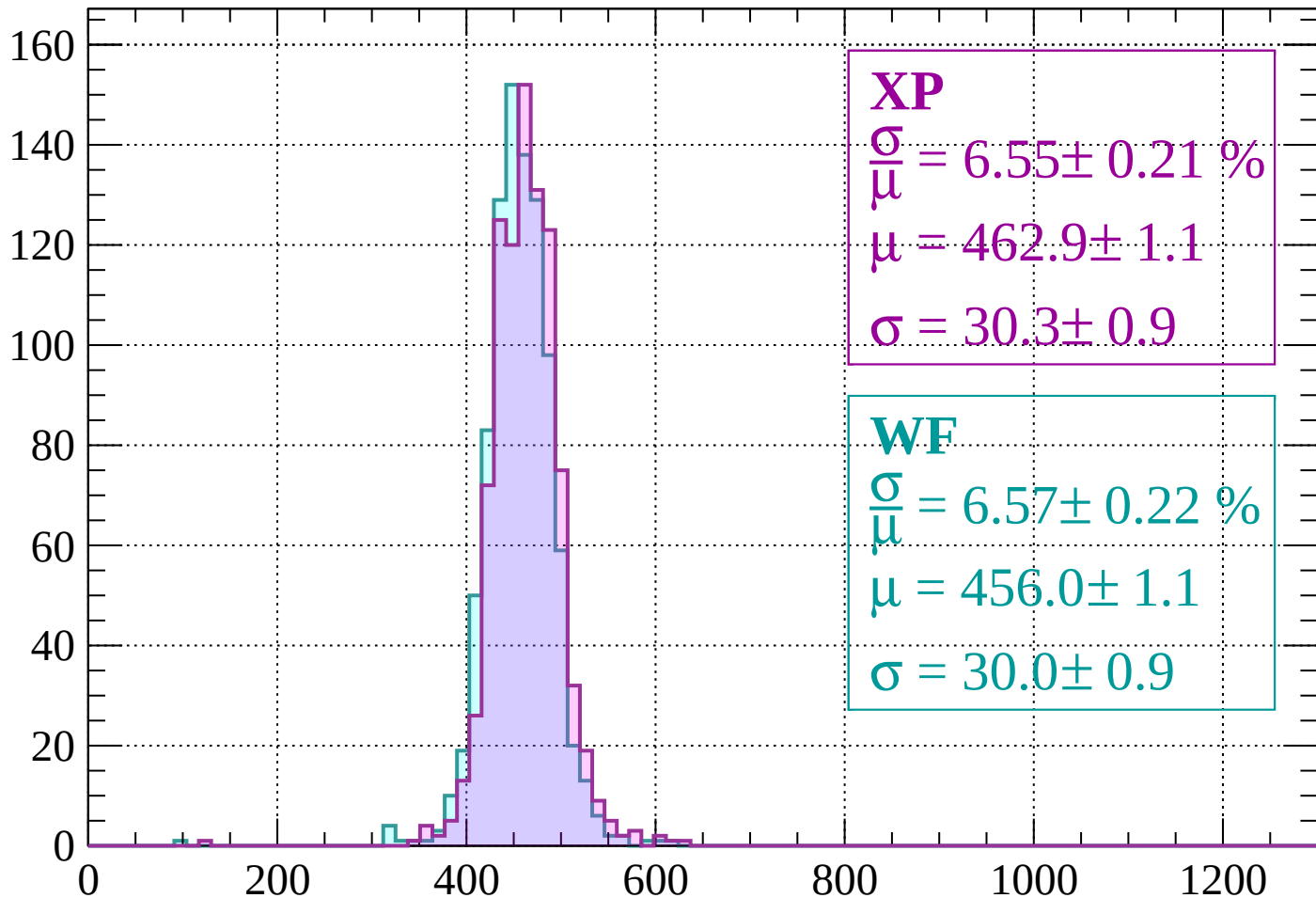
$$\mu = 458.2 \pm 1.2$$

$$\sigma = 28.1 \pm 1.2$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1440 < p < -1400$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.55 \pm 0.21 \%$$

$$\mu = 462.9 \pm 1.1$$

$$\sigma = 30.3 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.57 \pm 0.22 \%$$

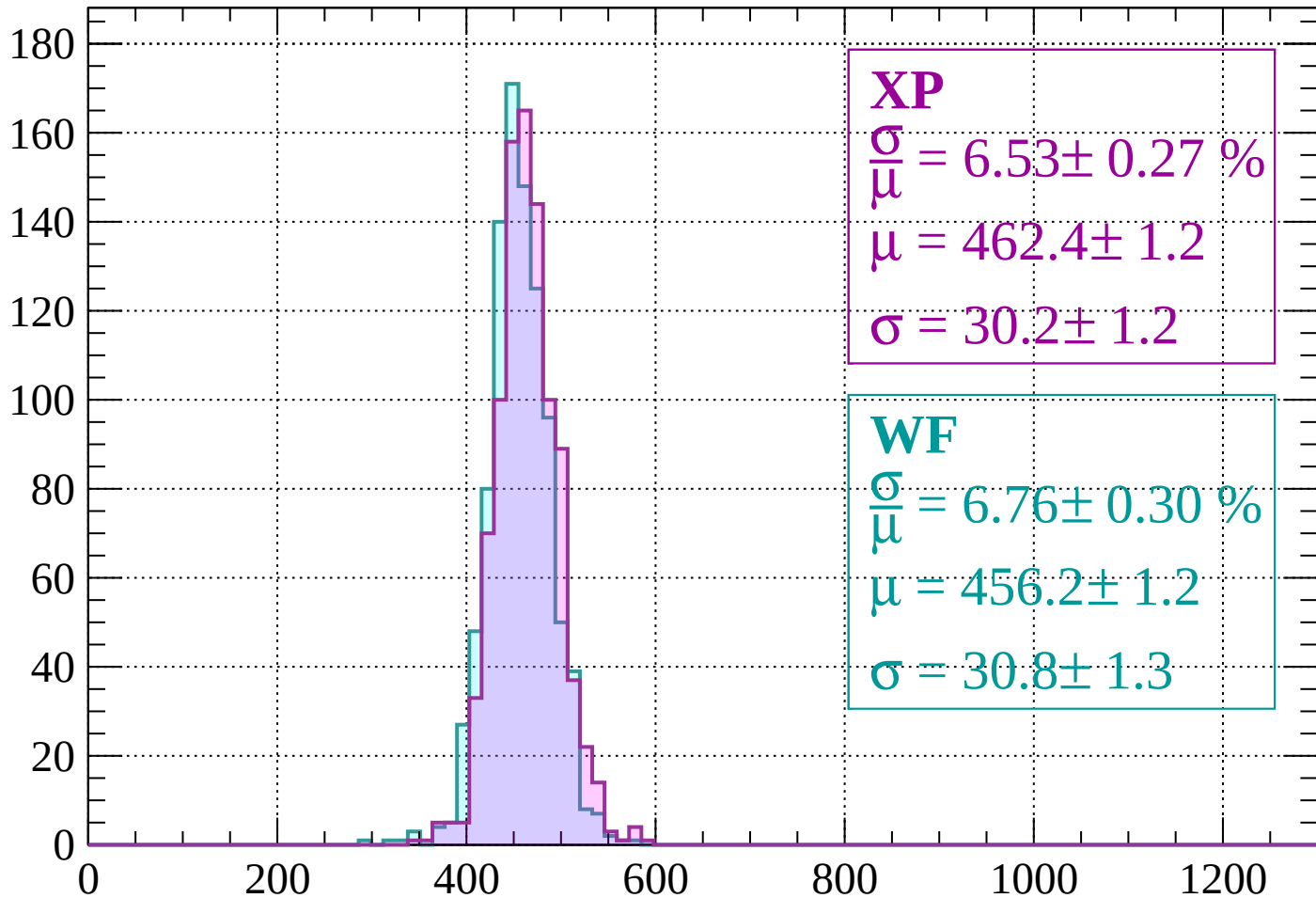
$$\mu = 456.0 \pm 1.1$$

$$\sigma = 30.0 \pm 0.9$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1400 < p < -1360$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.53 \pm 0.27 \%$$

$$\mu = 462.4 \pm 1.2$$

$$\sigma = 30.2 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.76 \pm 0.30 \%$$

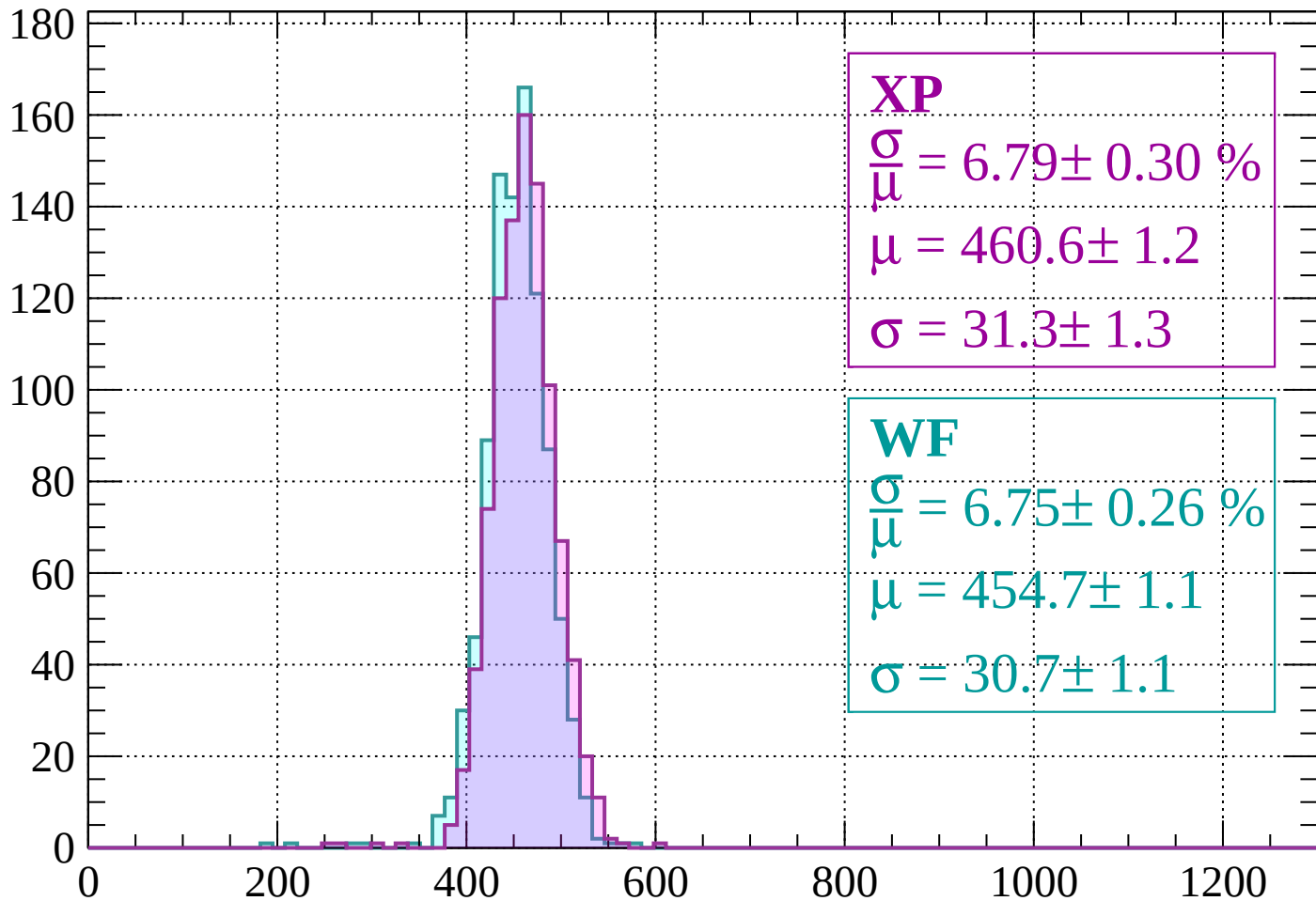
$$\mu = 456.2 \pm 1.2$$

$$\sigma = 30.8 \pm 1.3$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1360 < p < -1320$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.79 \pm 0.30 \%$$

$$\mu = 460.6 \pm 1.2$$

$$\sigma = 31.3 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.75 \pm 0.26 \%$$

$$\mu = 454.7 \pm 1.1$$

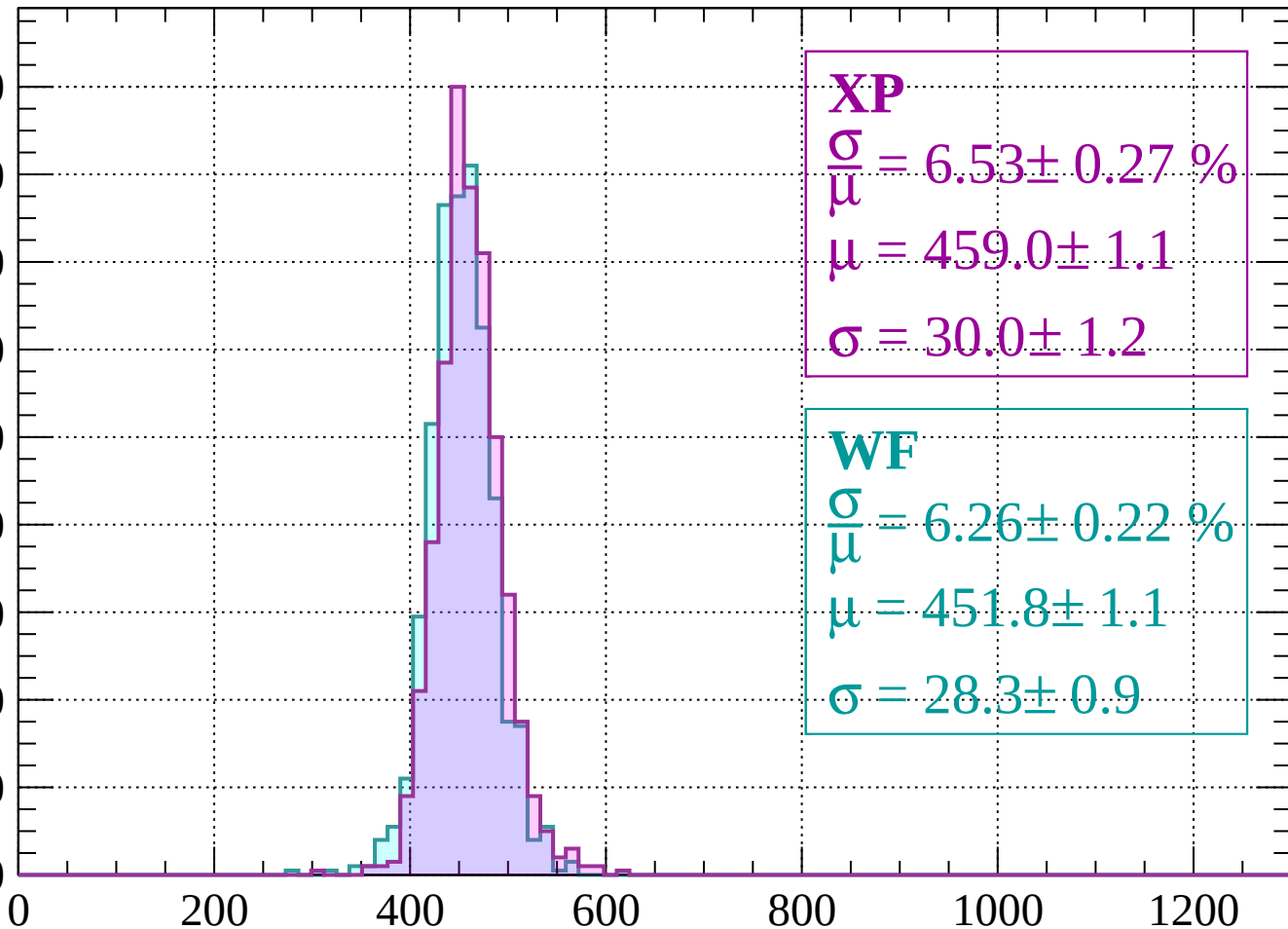
$$\sigma = 30.7 \pm 1.1$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1320 < p < -1280$

Count

180  
160  
140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.53 \pm 0.27 \%$$

$$\mu = 459.0 \pm 1.1$$

$$\sigma = 30.0 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.26 \pm 0.22 \%$$

$$\mu = 451.8 \pm 1.1$$

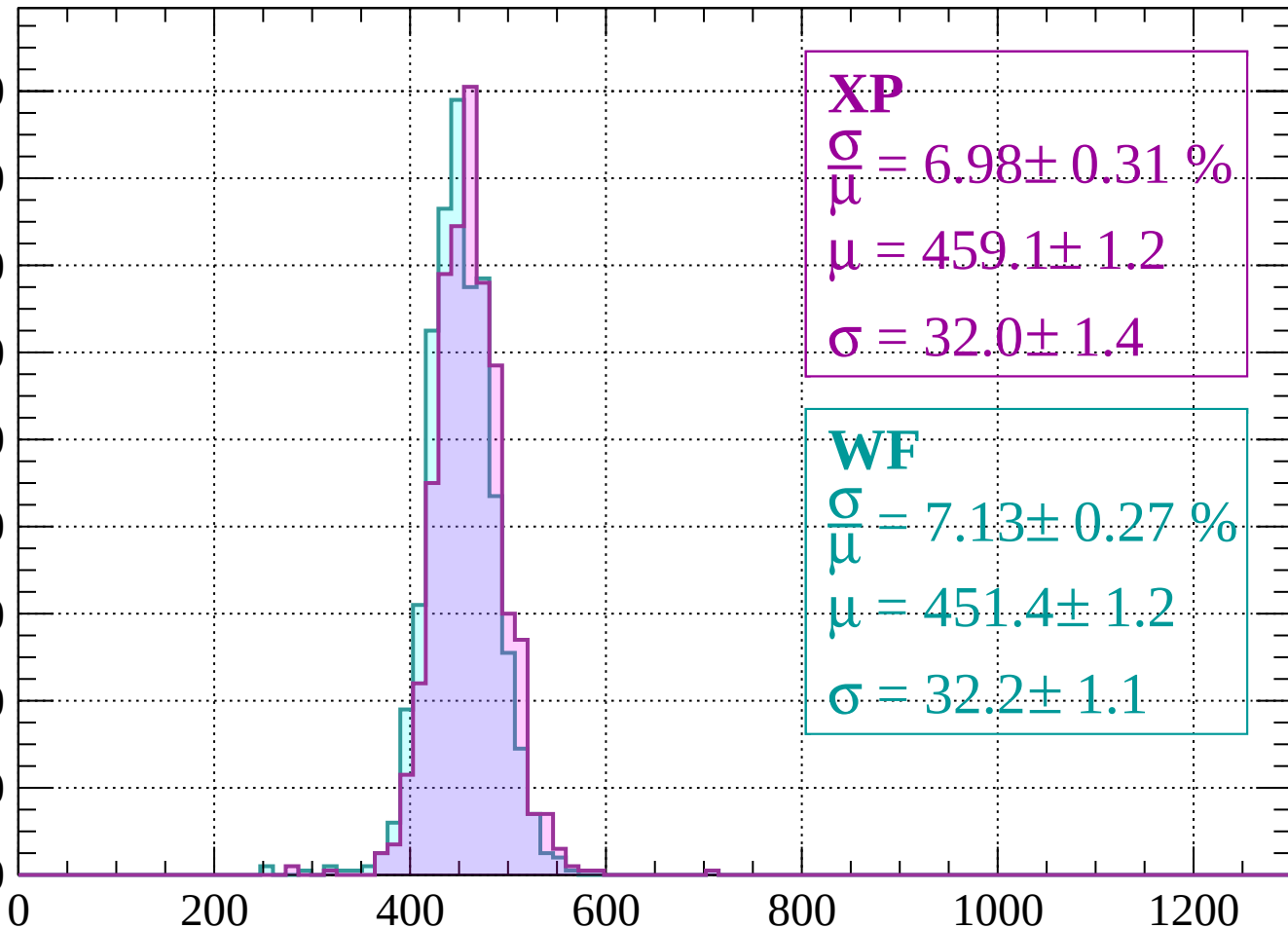
$$\sigma = 28.3 \pm 0.9$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1280 < p < -1240$

Count

180  
160  
140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.98 \pm 0.31 \%$$

$$\mu = 459.1 \pm 1.2$$

$$\sigma = 32.0 \pm 1.4$$

**WF**

$$\frac{\sigma}{\mu} = 7.13 \pm 0.27 \%$$

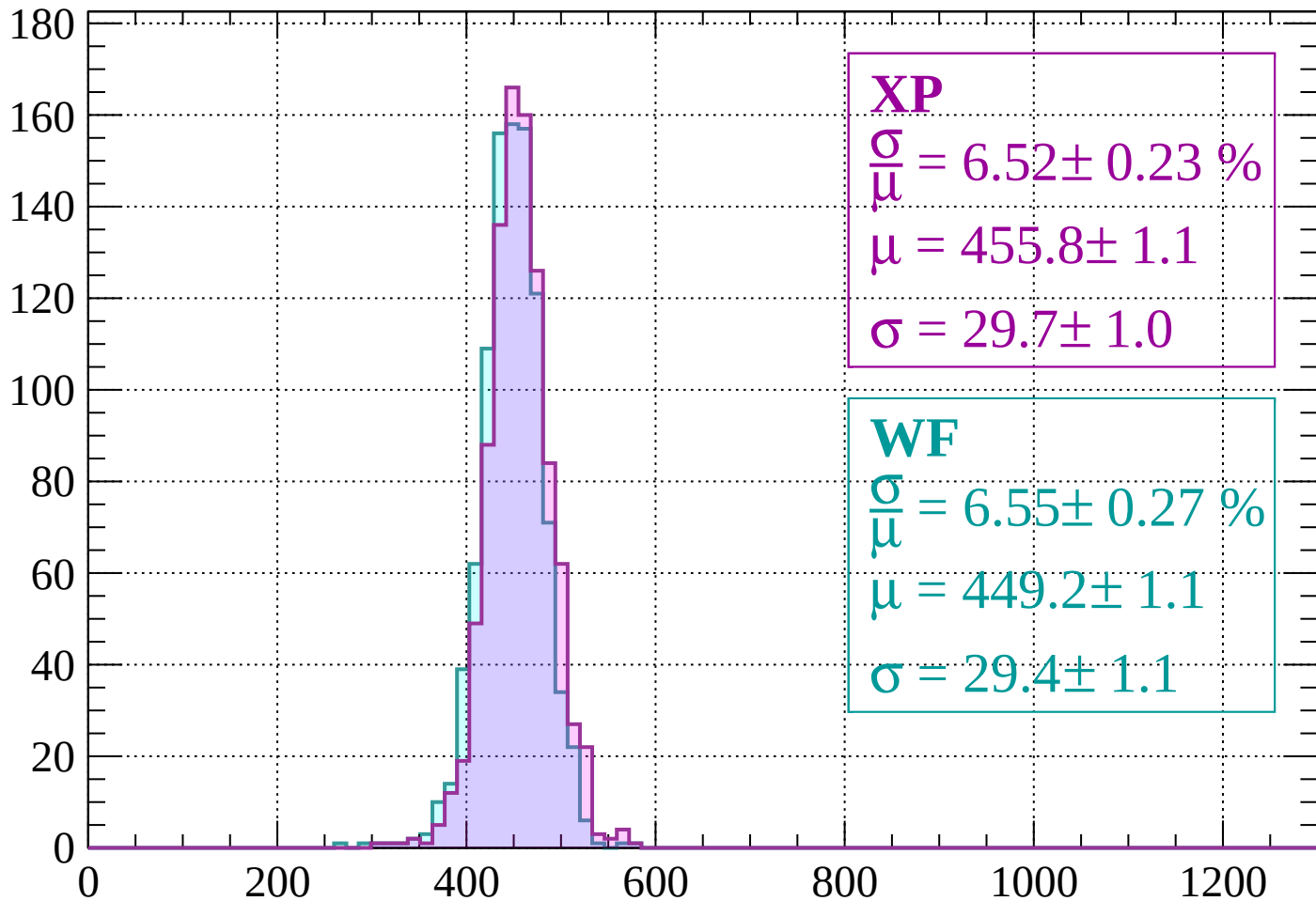
$$\mu = 451.4 \pm 1.2$$

$$\sigma = 32.2 \pm 1.1$$

dE/dx (ADC counts/cm)

# Energy loss | -1240 < p < -1200

Count



**XP**

$$\frac{\sigma}{\mu} = 6.52 \pm 0.23 \%$$

$$\mu = 455.8 \pm 1.1$$

$$\sigma = 29.7 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 6.55 \pm 0.27 \%$$

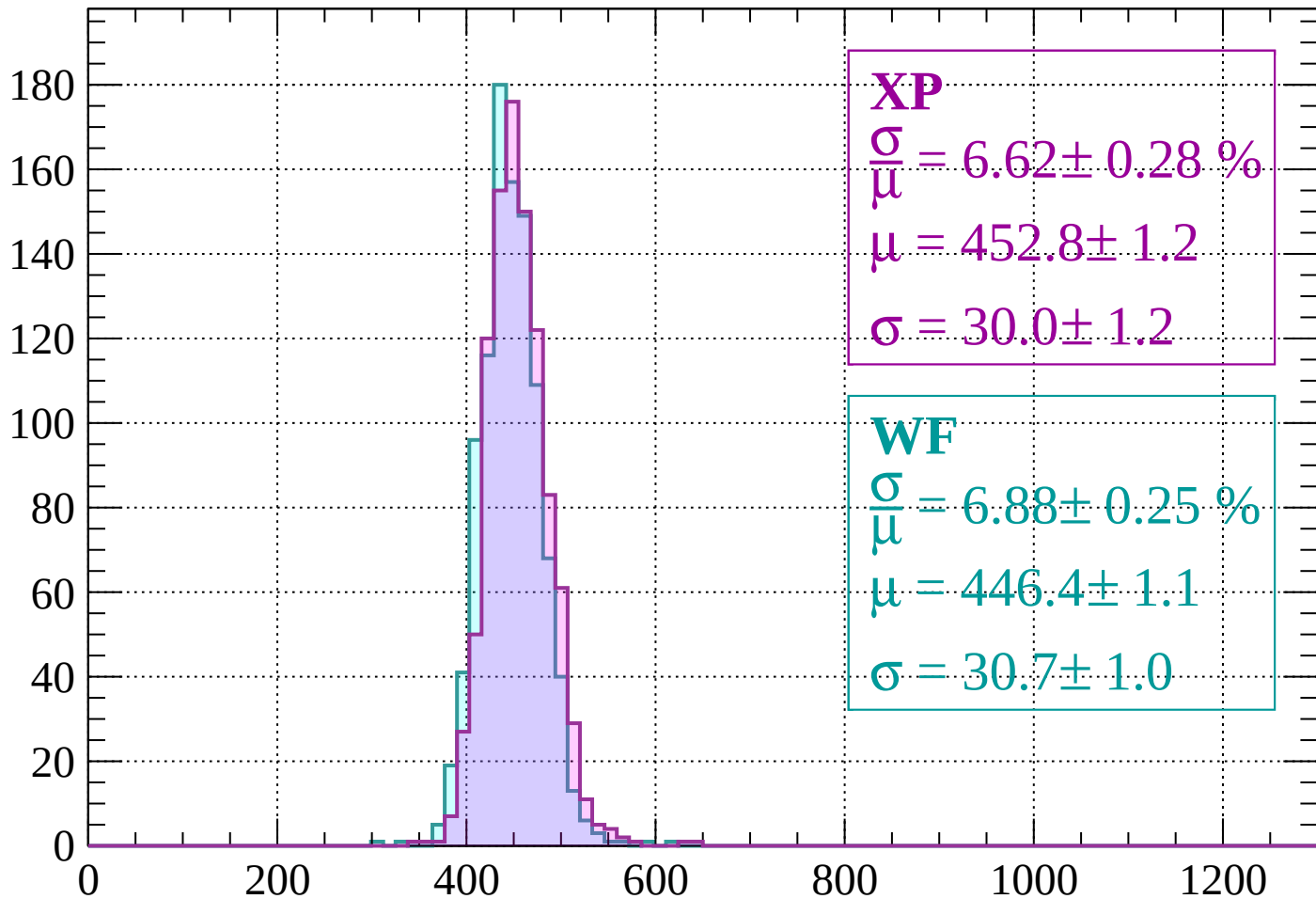
$$\mu = 449.2 \pm 1.1$$

$$\sigma = 29.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-1200 < p < -1160$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.62 \pm 0.28 \%$$

$$\mu = 452.8 \pm 1.2$$

$$\sigma = 30.0 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.88 \pm 0.25 \%$$

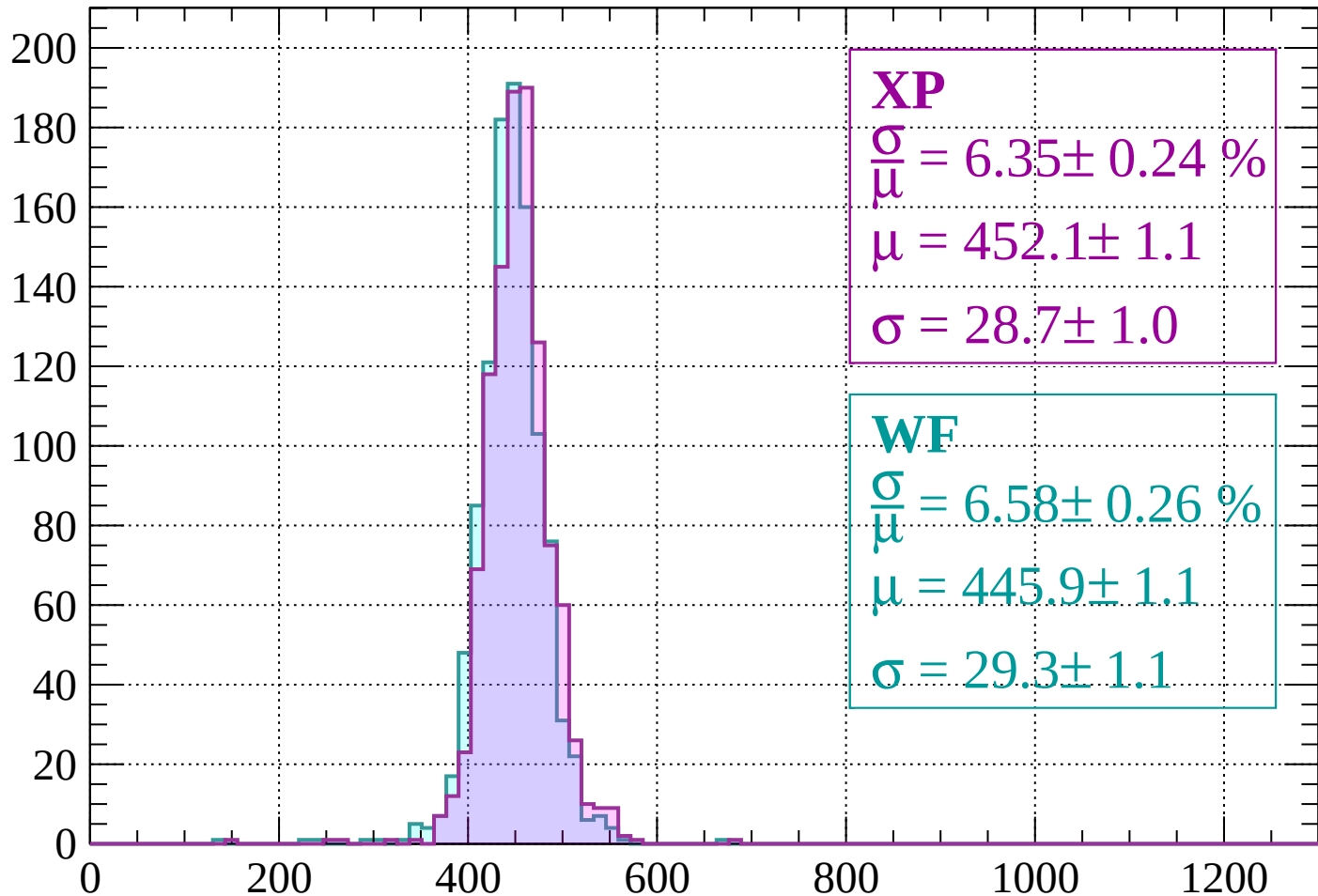
$$\mu = 446.4 \pm 1.1$$

$$\sigma = 30.7 \pm 1.0$$

$dE/dx$  (ADC counts/cm)

# Energy loss | -1160 < p < -1120

Count



**XP**

$$\frac{\sigma}{\mu} = 6.35 \pm 0.24 \%$$

$$\mu = 452.1 \pm 1.1$$

$$\sigma = 28.7 \pm 1.0$$

**WF**

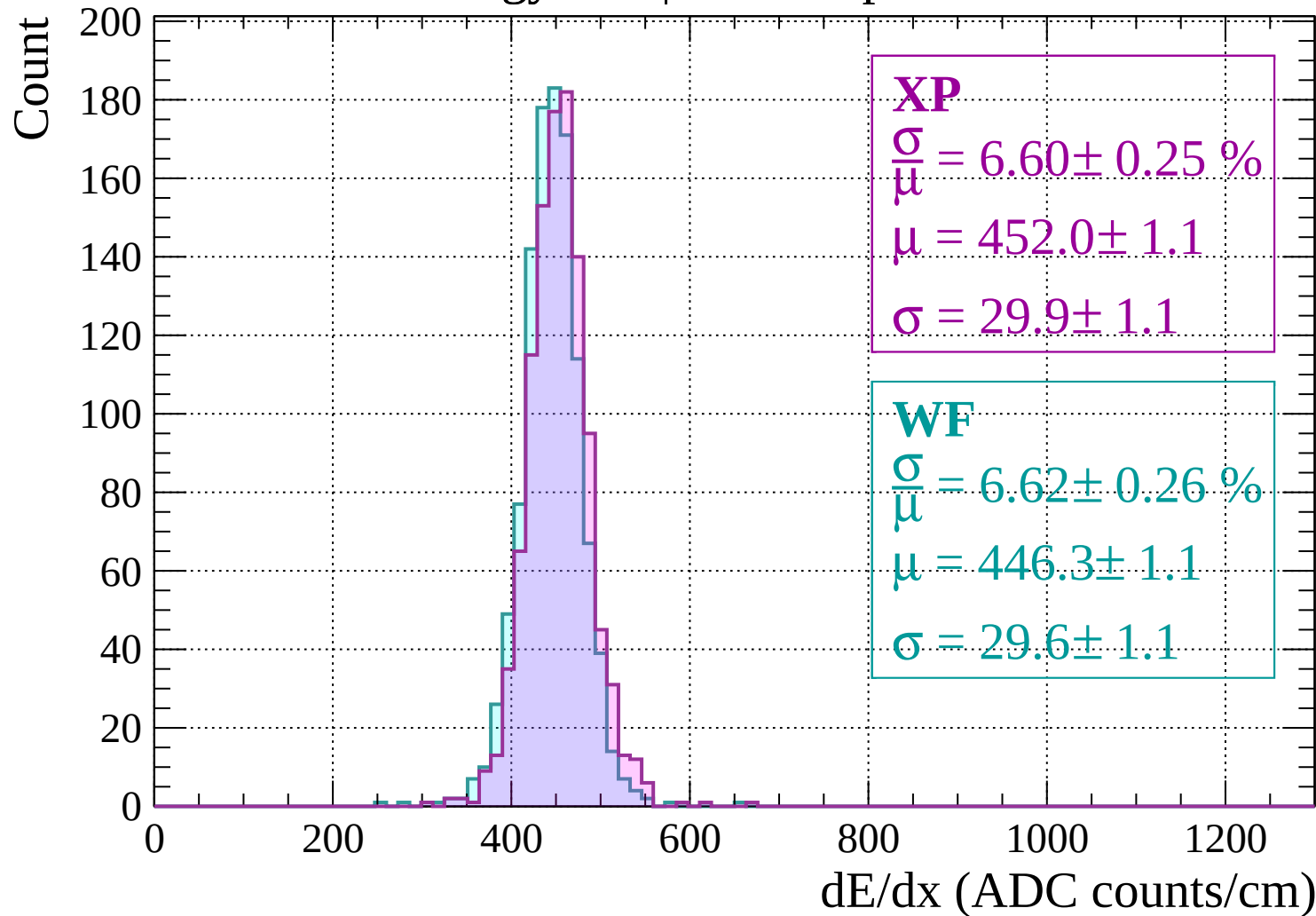
$$\frac{\sigma}{\mu} = 6.58 \pm 0.26 \%$$

$$\mu = 445.9 \pm 1.1$$

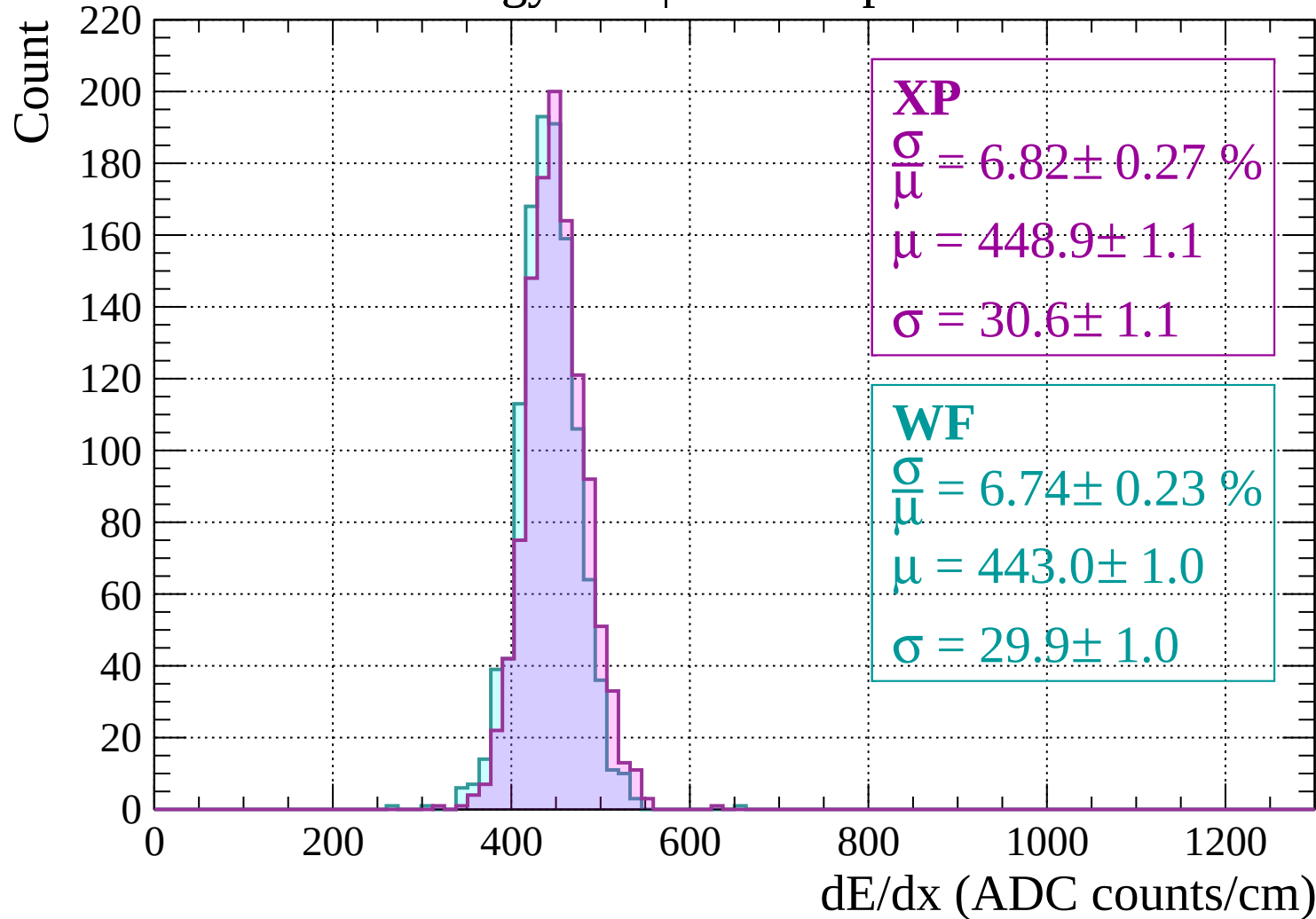
$$\sigma = 29.3 \pm 1.1$$

dE/dx (ADC counts/cm)

# Energy loss | -1120 < p < -1080

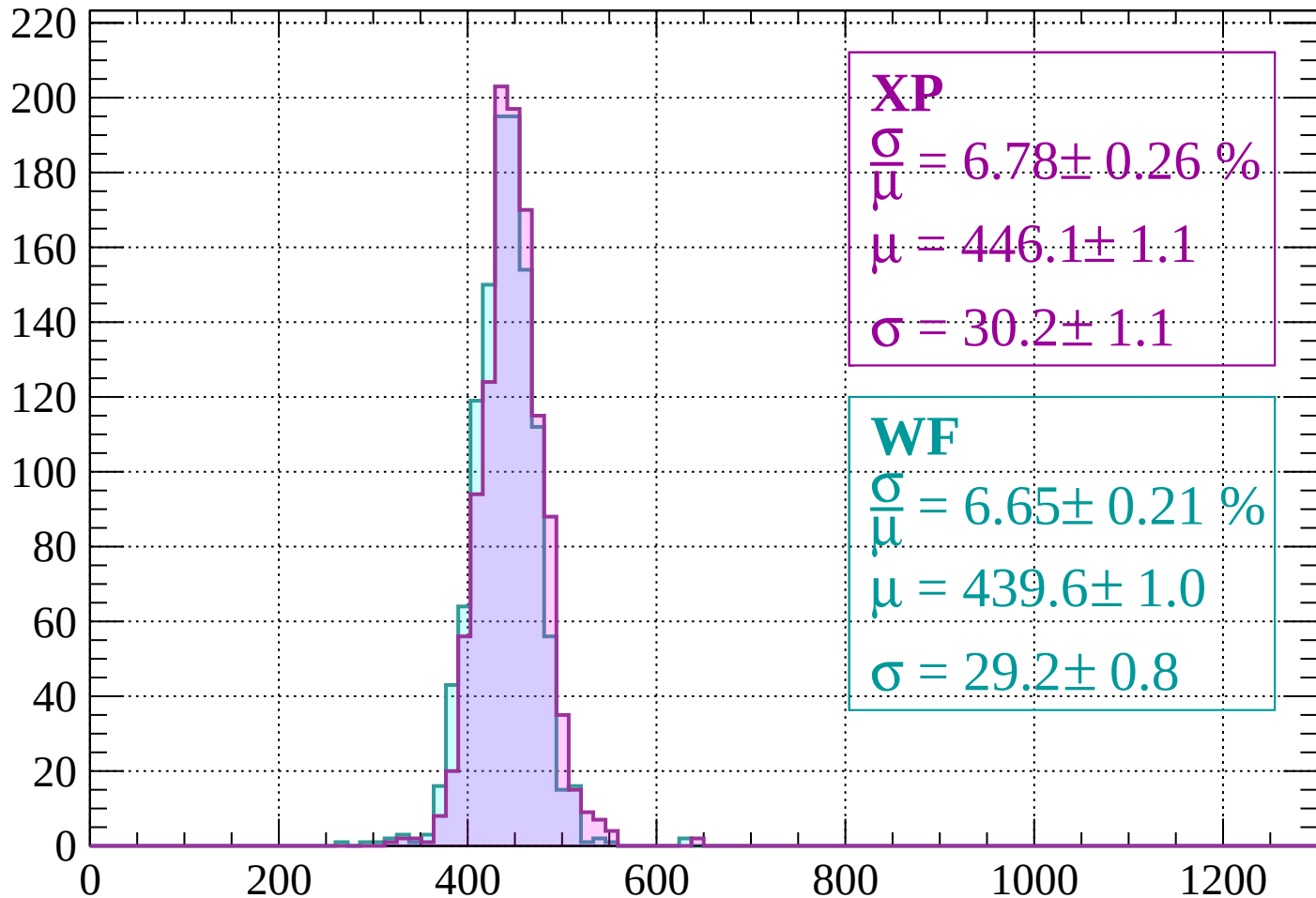


Energy loss |  $-1080 < p < -1040$



Energy loss |  $-1040 < p < -1000$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.78 \pm 0.26 \%$$

$$\mu = 446.1 \pm 1.1$$

$$\sigma = 30.2 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 6.65 \pm 0.21 \%$$

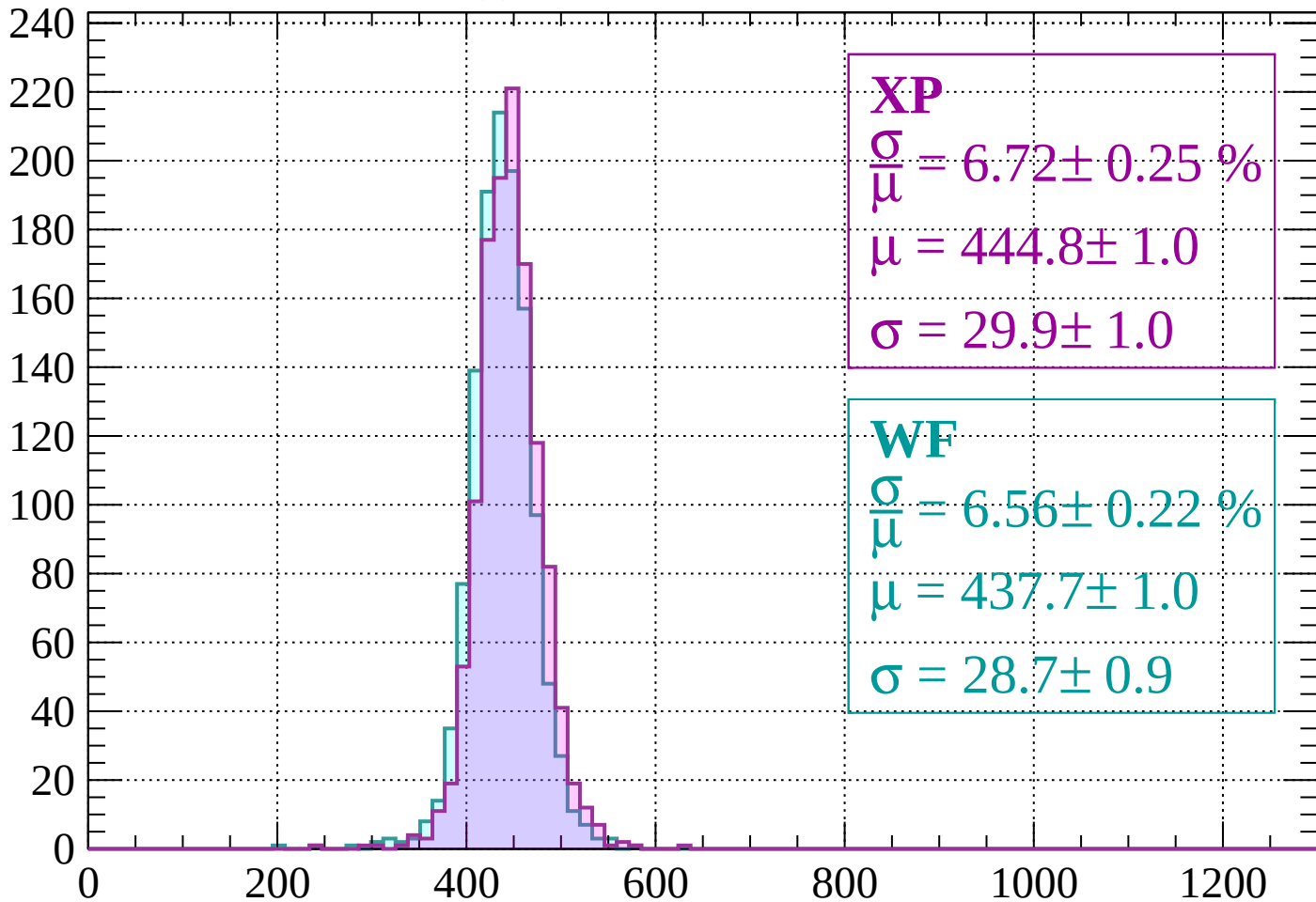
$$\mu = 439.6 \pm 1.0$$

$$\sigma = 29.2 \pm 0.8$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1000 < p < -960$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 6.72 \pm 0.25 \%$$

$$\mu = 444.8 \pm 1.0$$

$$\sigma = 29.9 \pm 1.0$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 6.56 \pm 0.22 \%$$

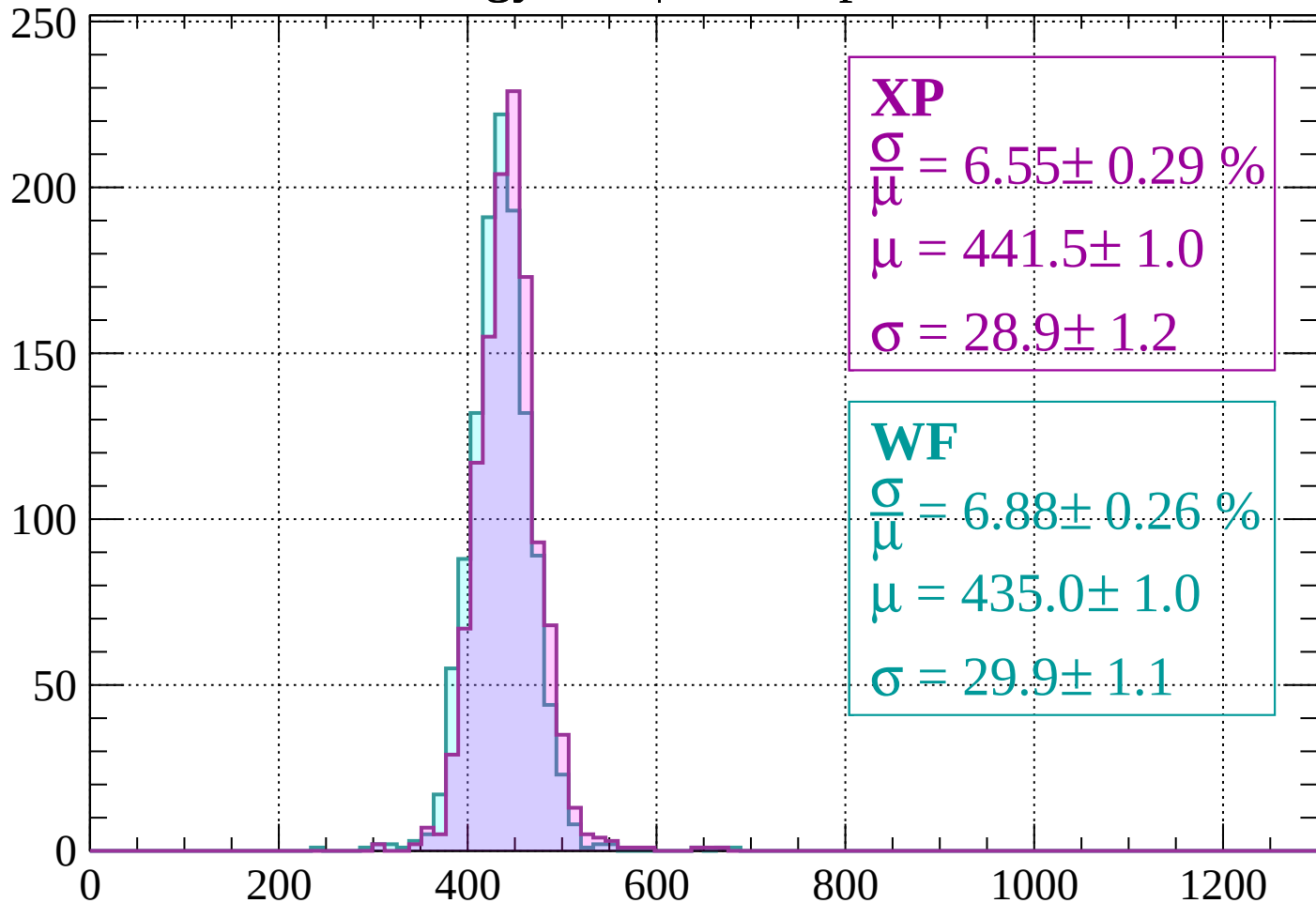
$$\mu = 437.7 \pm 1.0$$

$$\sigma = 28.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $-960 < p < -920$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.55 \pm 0.29 \%$$

$$\mu = 441.5 \pm 1.0$$

$$\sigma = 28.9 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.88 \pm 0.26 \%$$

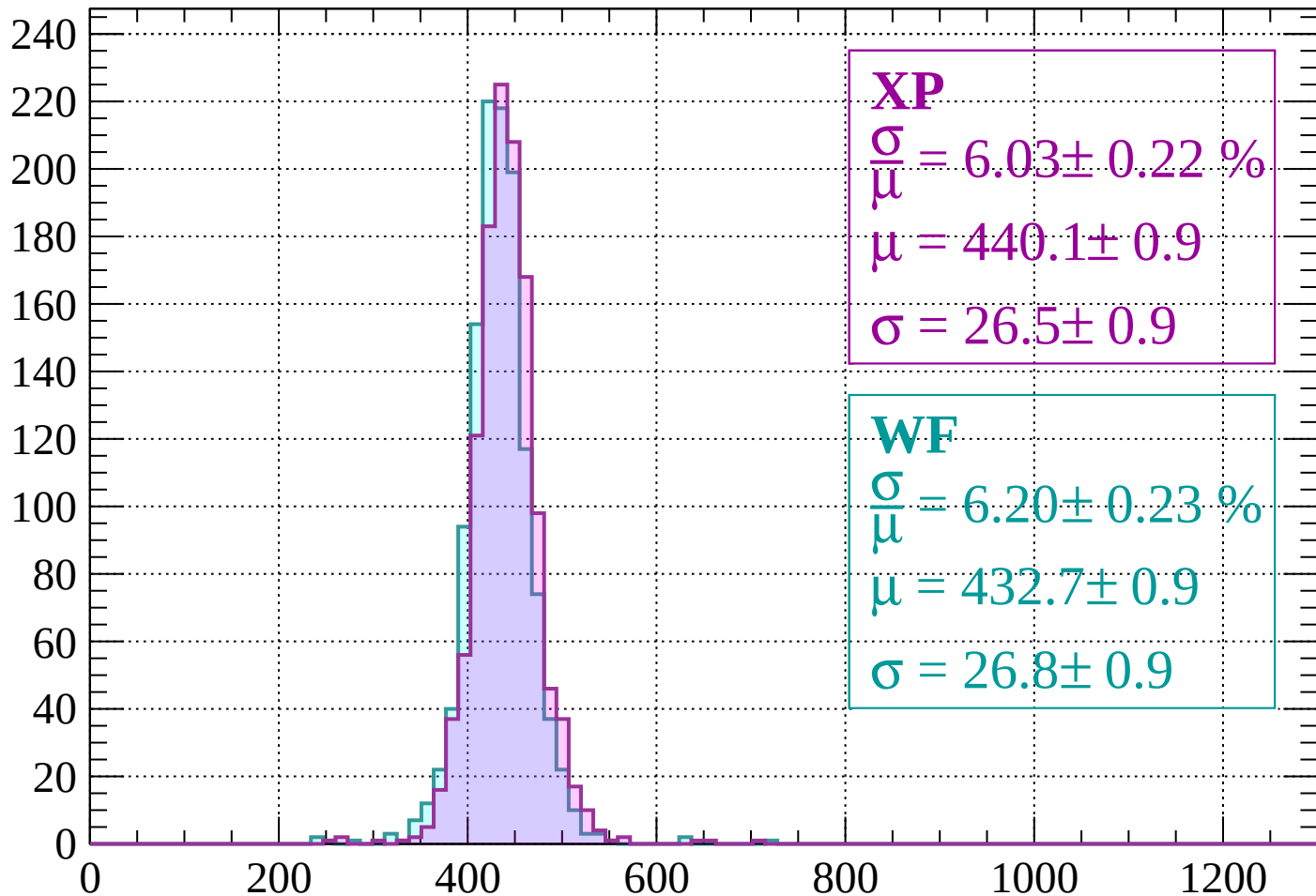
$$\mu = 435.0 \pm 1.0$$

$$\sigma = 29.9 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-920 < p < -880$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.03 \pm 0.22 \%$$

$$\mu = 440.1 \pm 0.9$$

$$\sigma = 26.5 \pm 0.9$$

**WF**

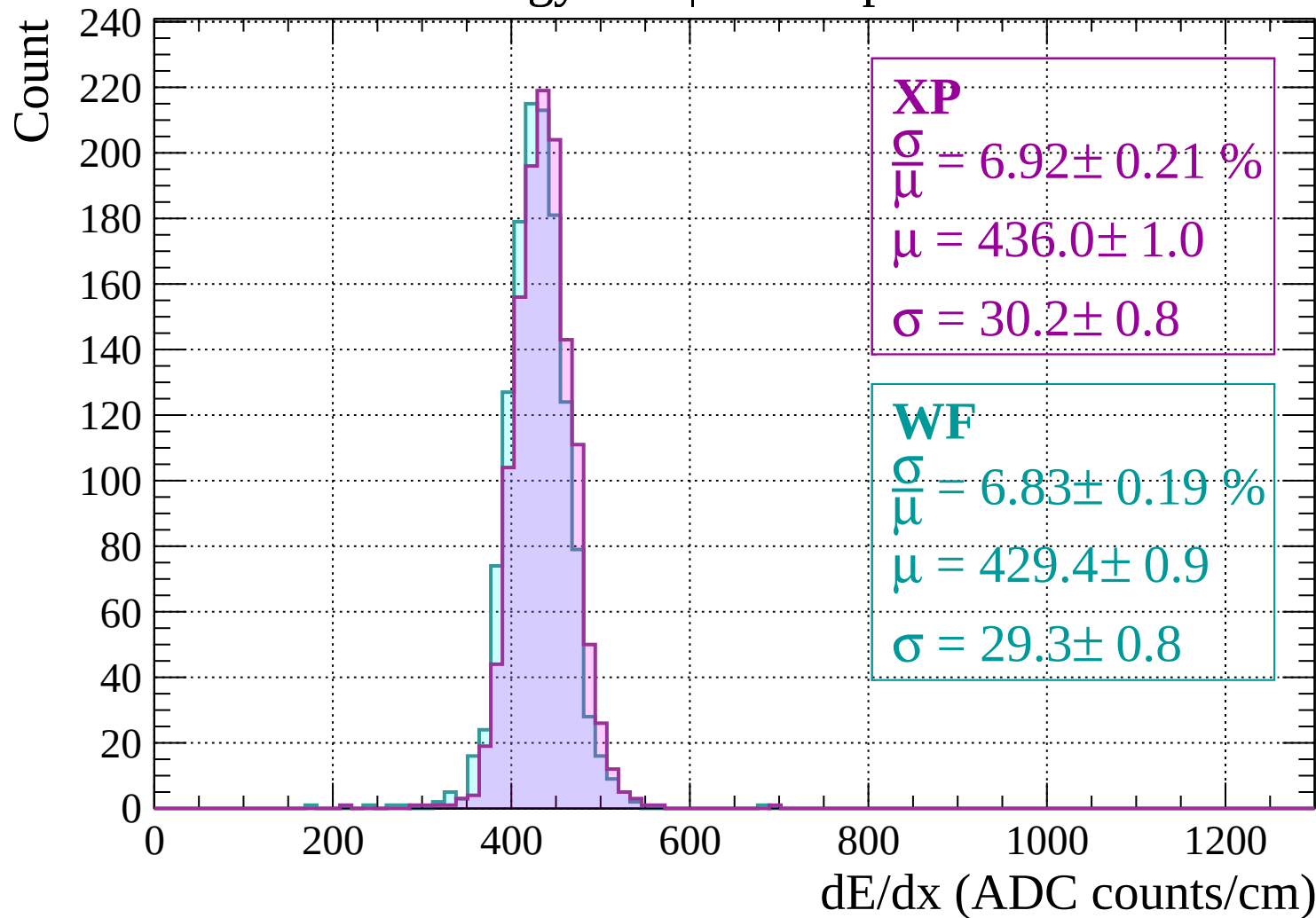
$$\frac{\sigma}{\mu} = 6.20 \pm 0.23 \%$$

$$\mu = 432.7 \pm 0.9$$

$$\sigma = 26.8 \pm 0.9$$

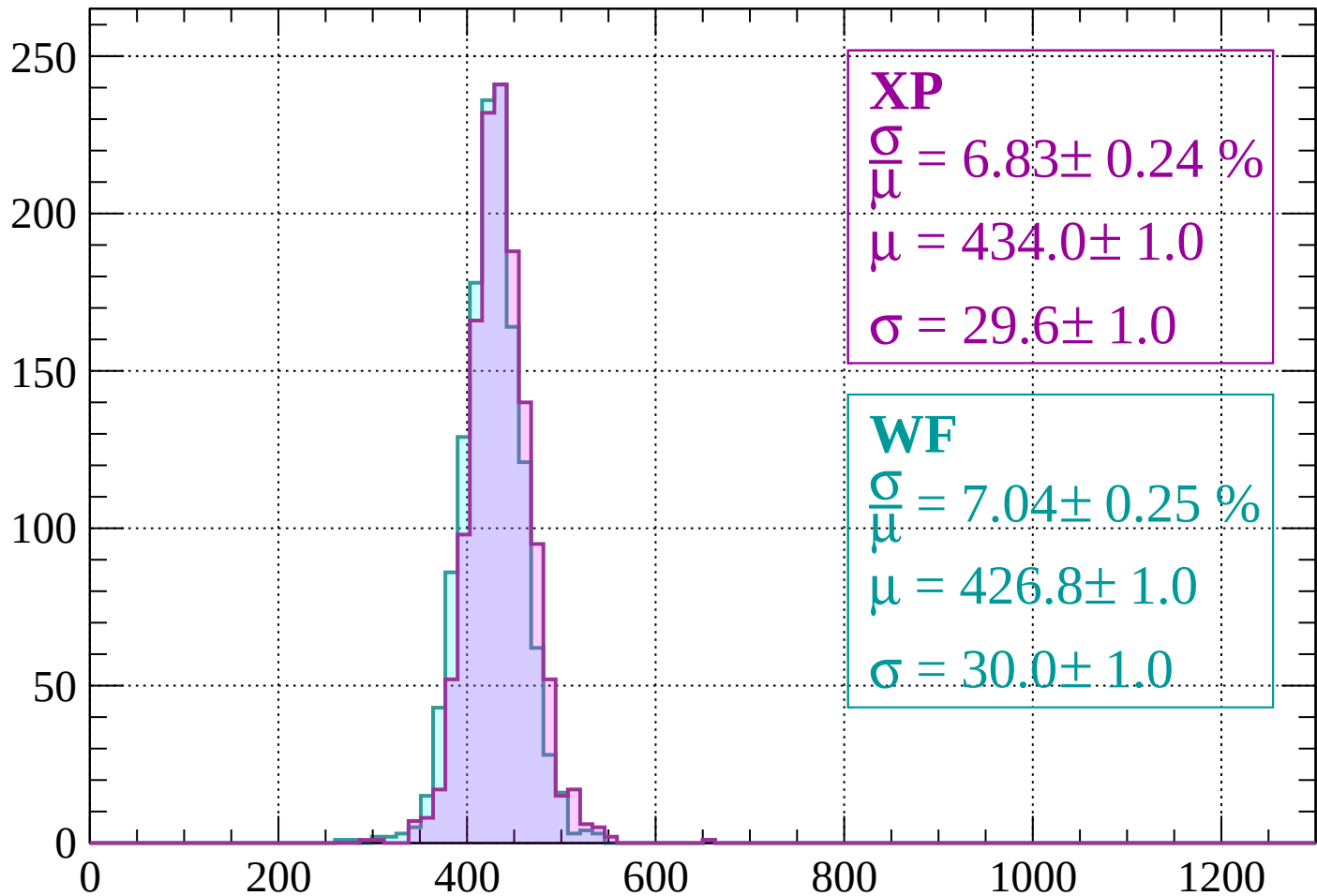
$dE/dx$  (ADC counts/cm)

Energy loss |  $-880 < p < -840$



Energy loss |  $-840 < p < -800$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.83 \pm 0.24 \%$$

$$\mu = 434.0 \pm 1.0$$

$$\sigma = 29.6 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 7.04 \pm 0.25 \%$$

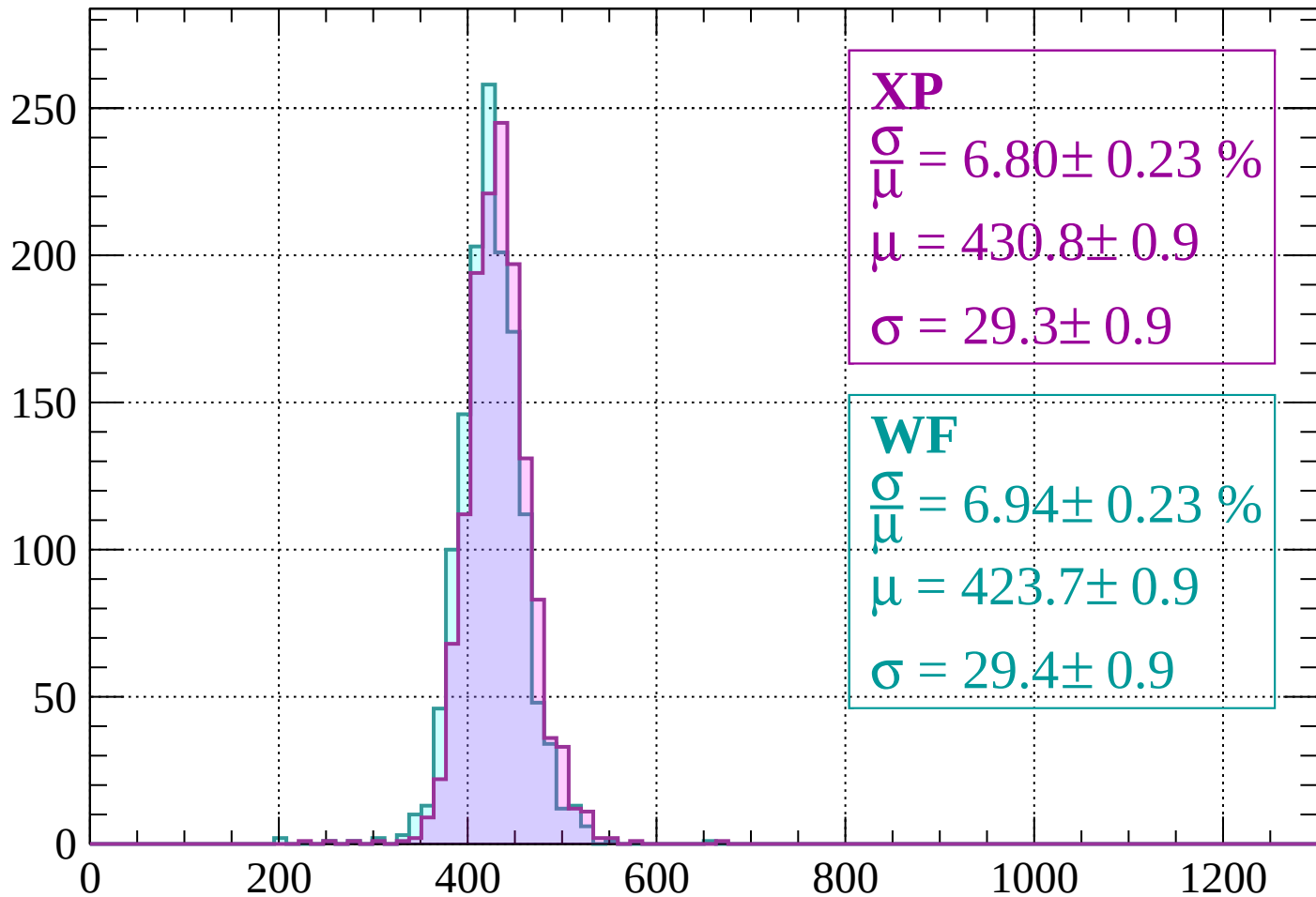
$$\mu = 426.8 \pm 1.0$$

$$\sigma = 30.0 \pm 1.0$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-800 < p < -760$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.80 \pm 0.23 \%$$

$$\mu = 430.8 \pm 0.9$$

$$\sigma = 29.3 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.94 \pm 0.23 \%$$

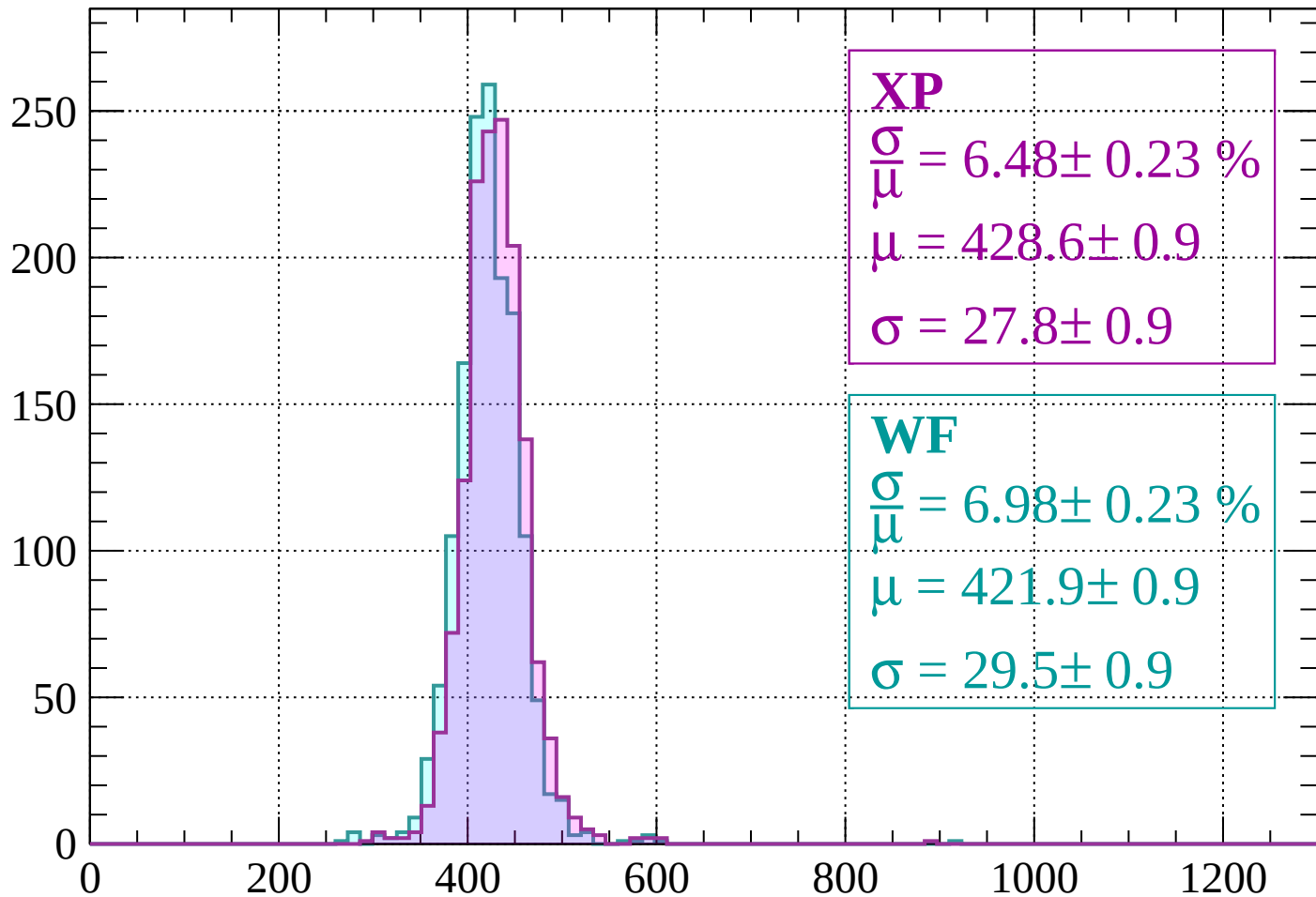
$$\mu = 423.7 \pm 0.9$$

$$\sigma = 29.4 \pm 0.9$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-760 < p < -720$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.48 \pm 0.23 \%$$

$$\mu = 428.6 \pm 0.9$$

$$\sigma = 27.8 \pm 0.9$$

**WF**

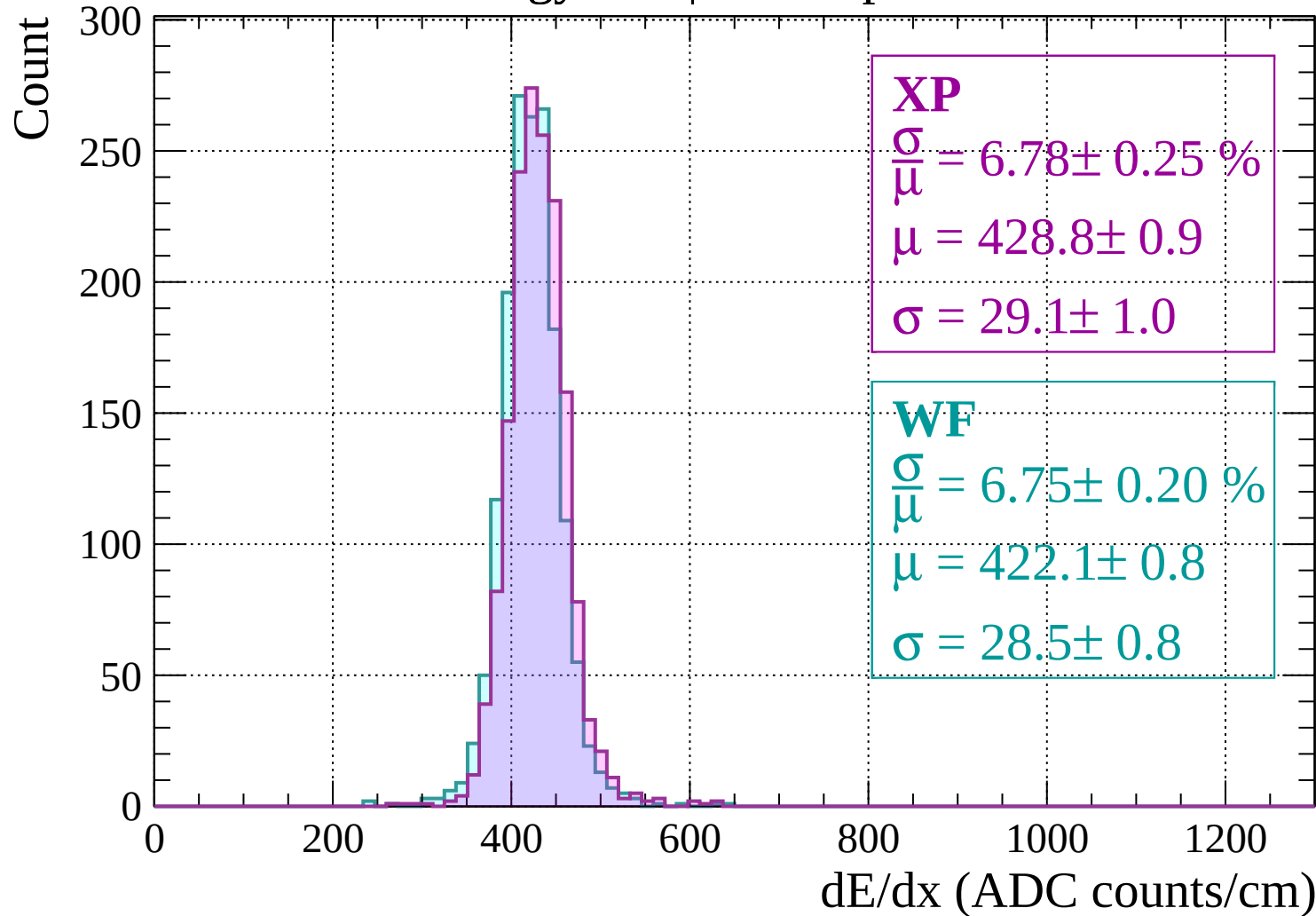
$$\frac{\sigma}{\mu} = 6.98 \pm 0.23 \%$$

$$\mu = 421.9 \pm 0.9$$

$$\sigma = 29.5 \pm 0.9$$

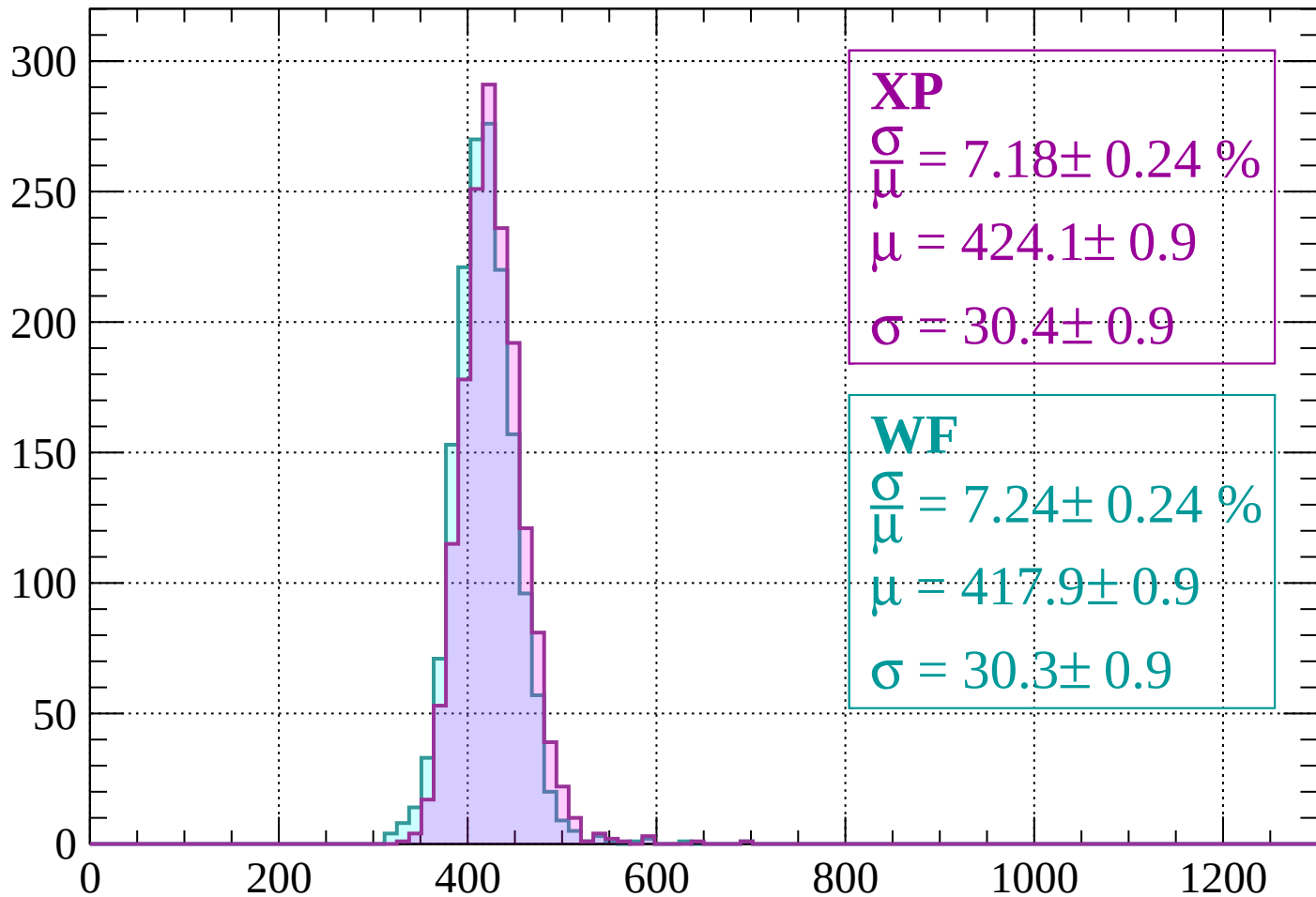
$dE/dx$  (ADC counts/cm)

Energy loss |  $-720 < p < -680$



Energy loss |  $-680 < p < -640$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.18 \pm 0.24 \%$$

$$\mu = 424.1 \pm 0.9$$

$$\sigma = 30.4 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.24 \pm 0.24 \%$$

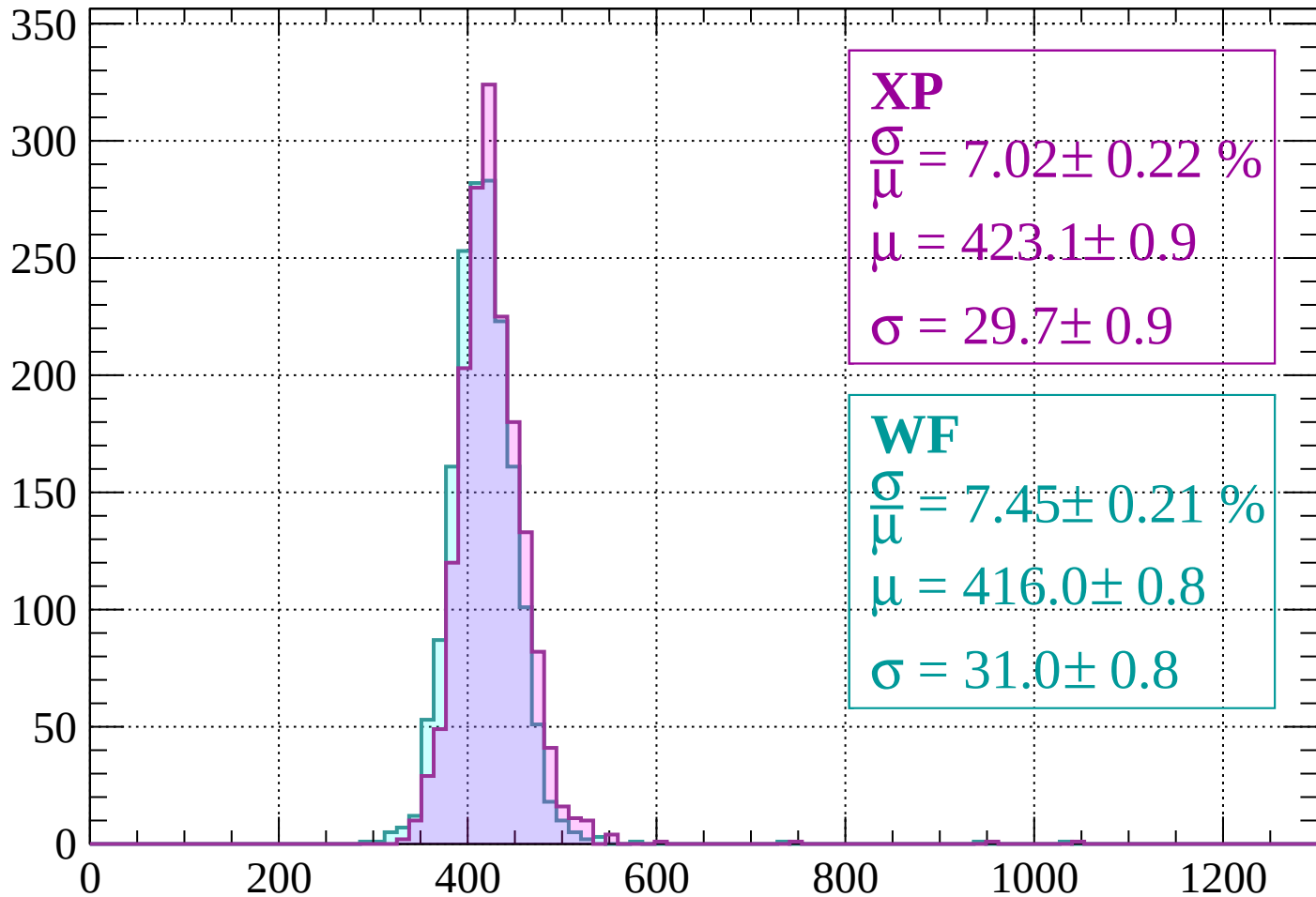
$$\mu = 417.9 \pm 0.9$$

$$\sigma = 30.3 \pm 0.9$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-640 < p < -600$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.02 \pm 0.22 \%$$

$$\mu = 423.1 \pm 0.9$$

$$\sigma = 29.7 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.45 \pm 0.21 \%$$

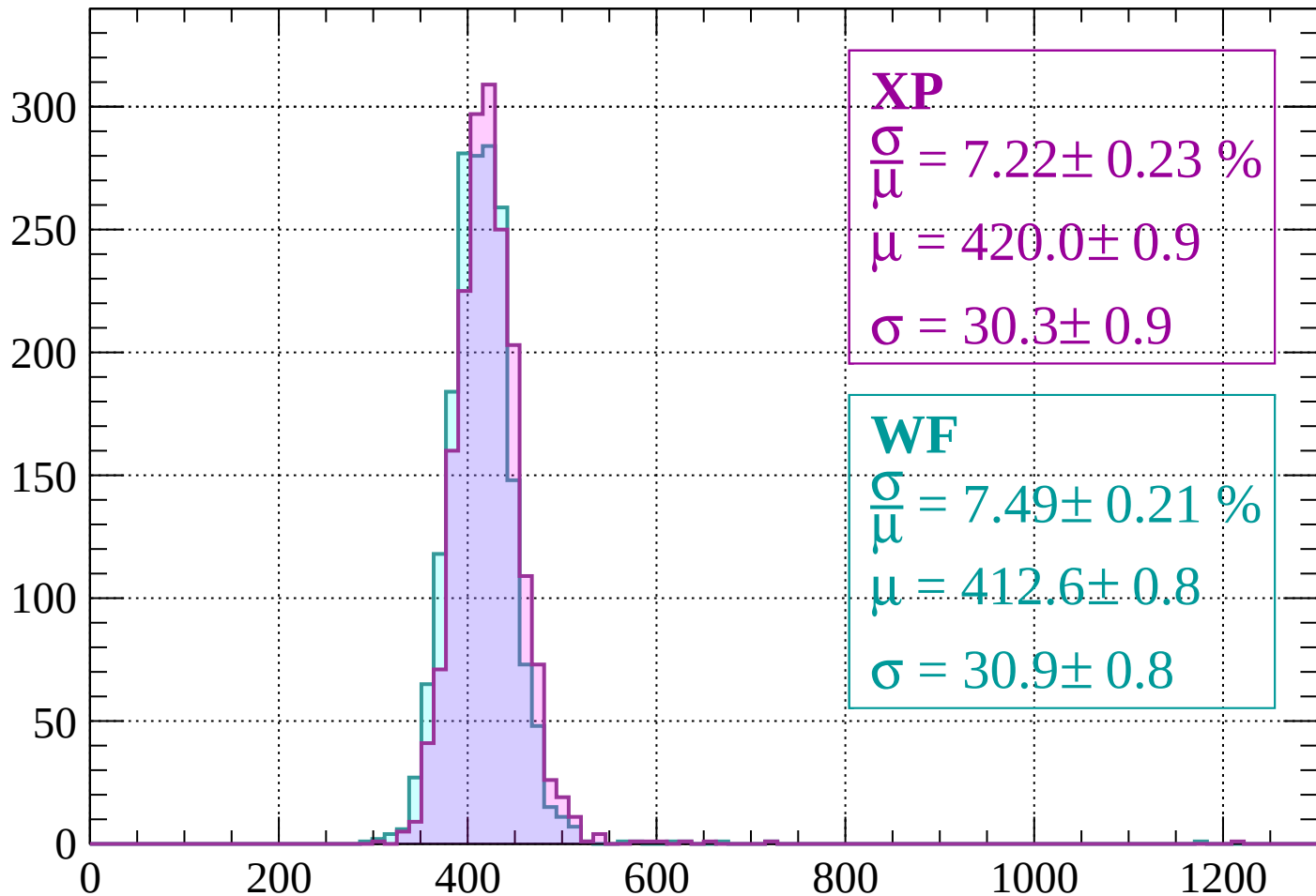
$$\mu = 416.0 \pm 0.8$$

$$\sigma = 31.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-600 < p < -560$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.22 \pm 0.23 \%$$

$$\mu = 420.0 \pm 0.9$$

$$\sigma = 30.3 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.49 \pm 0.21 \%$$

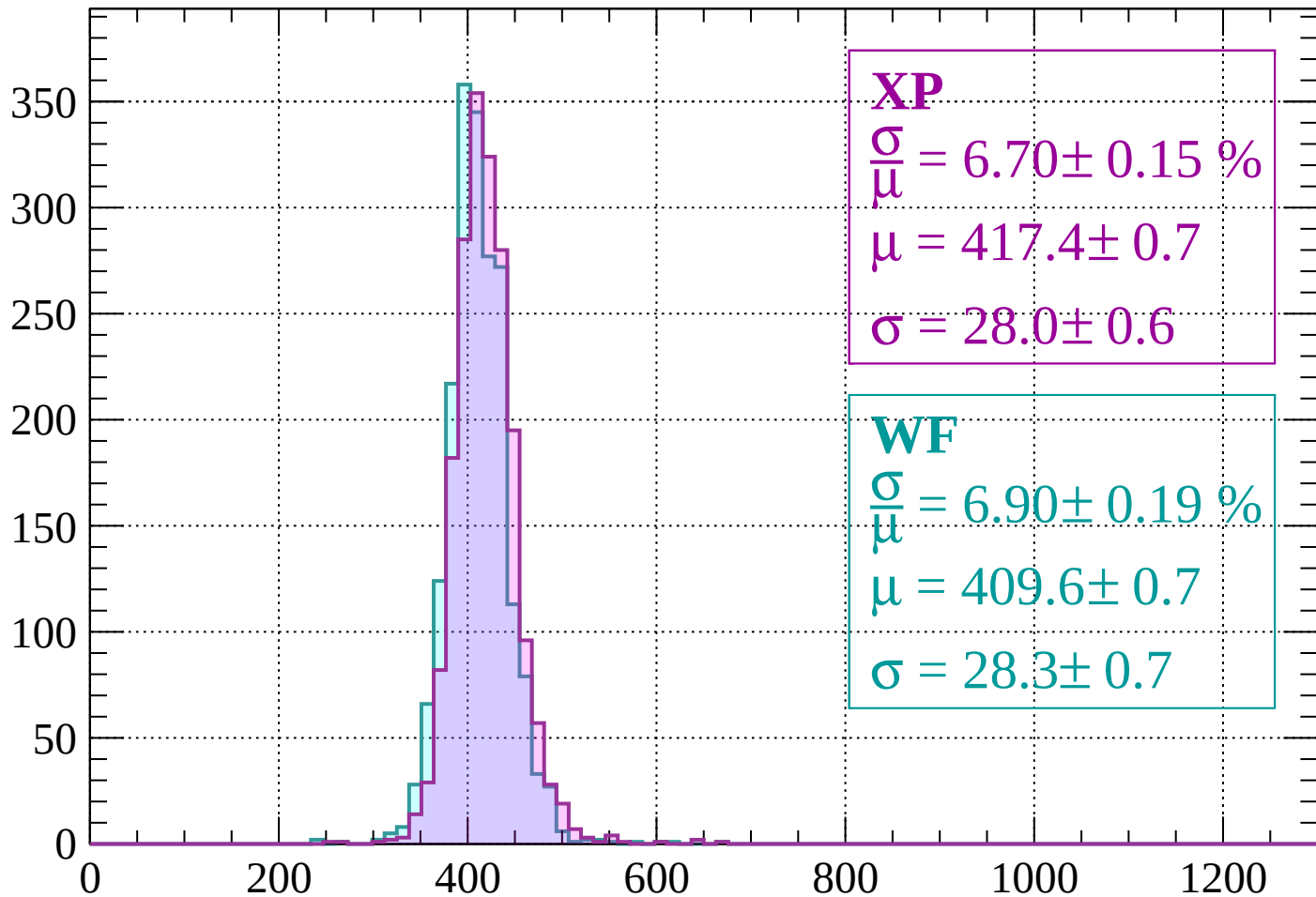
$$\mu = 412.6 \pm 0.8$$

$$\sigma = 30.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-560 < p < -520$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.70 \pm 0.15 \%$$

$$\mu = 417.4 \pm 0.7$$

$$\sigma = 28.0 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 6.90 \pm 0.19 \%$$

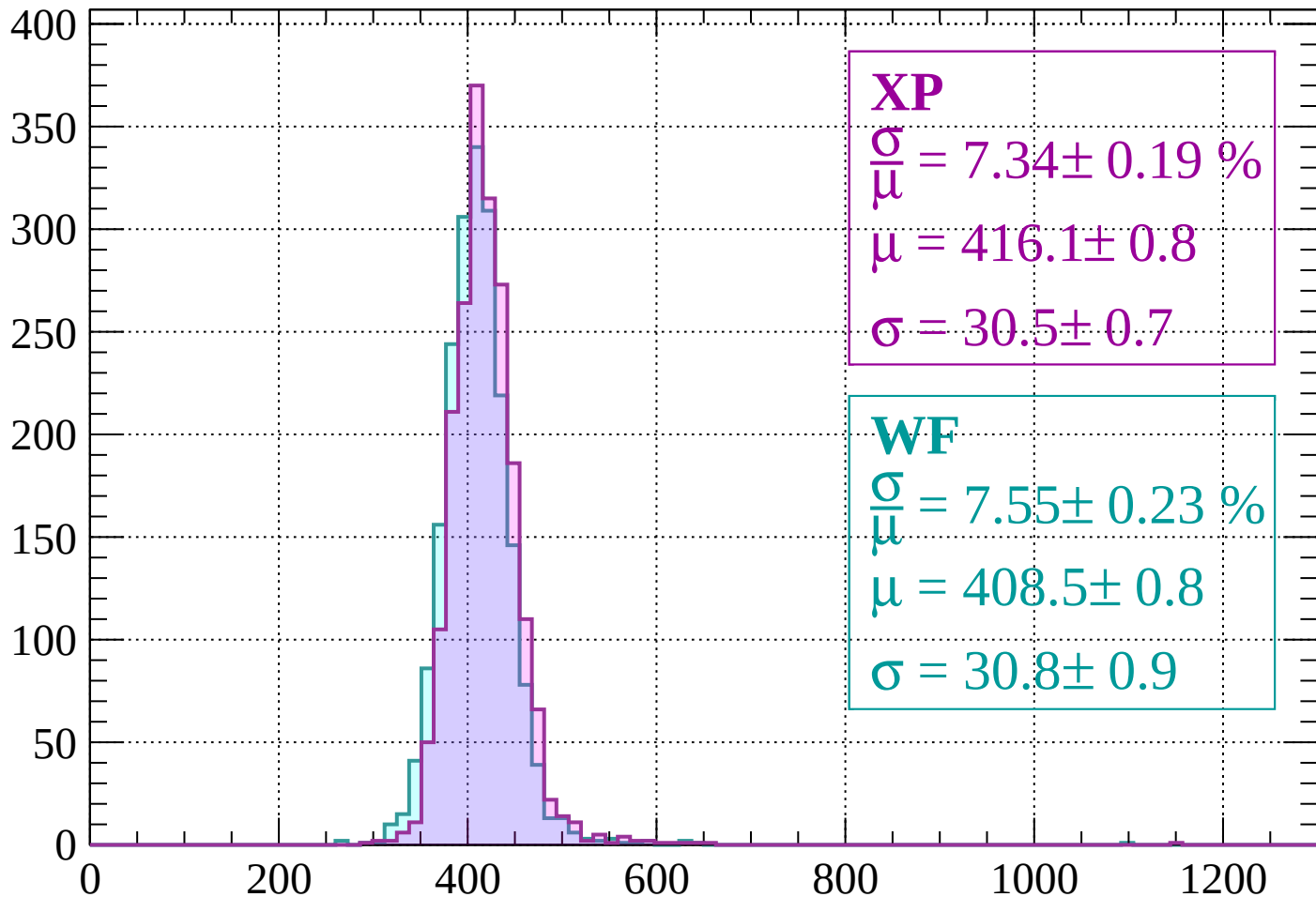
$$\mu = 409.6 \pm 0.7$$

$$\sigma = 28.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-520 < p < -480$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.34 \pm 0.19 \%$$

$$\mu = 416.1 \pm 0.8$$

$$\sigma = 30.5 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 7.55 \pm 0.23 \%$$

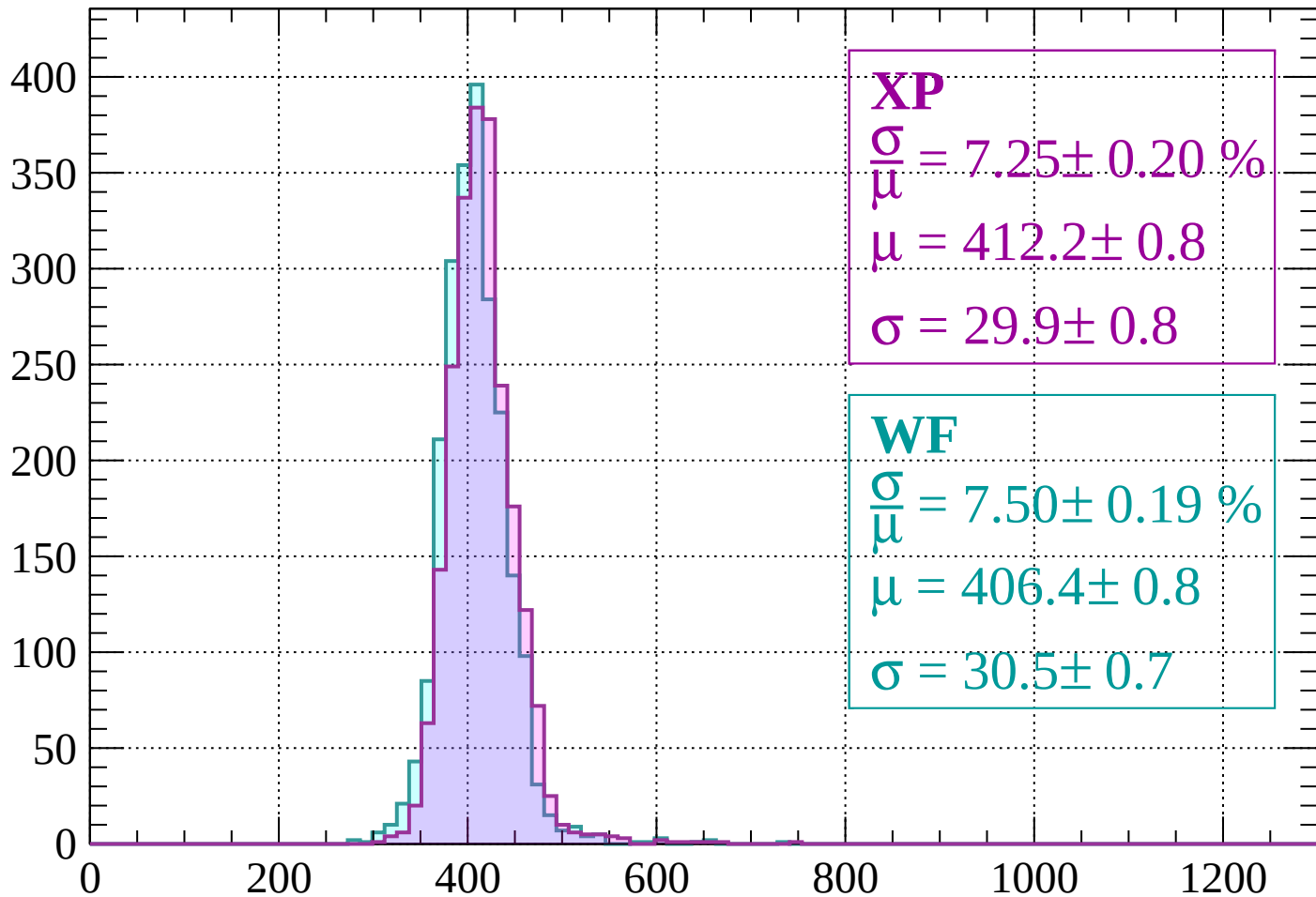
$$\mu = 408.5 \pm 0.8$$

$$\sigma = 30.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $-480 < p < -440$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 7.25 \pm 0.20 \%$$

$$\mu = 412.2 \pm 0.8$$

$$\sigma = 29.9 \pm 0.8$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 7.50 \pm 0.19 \%$$

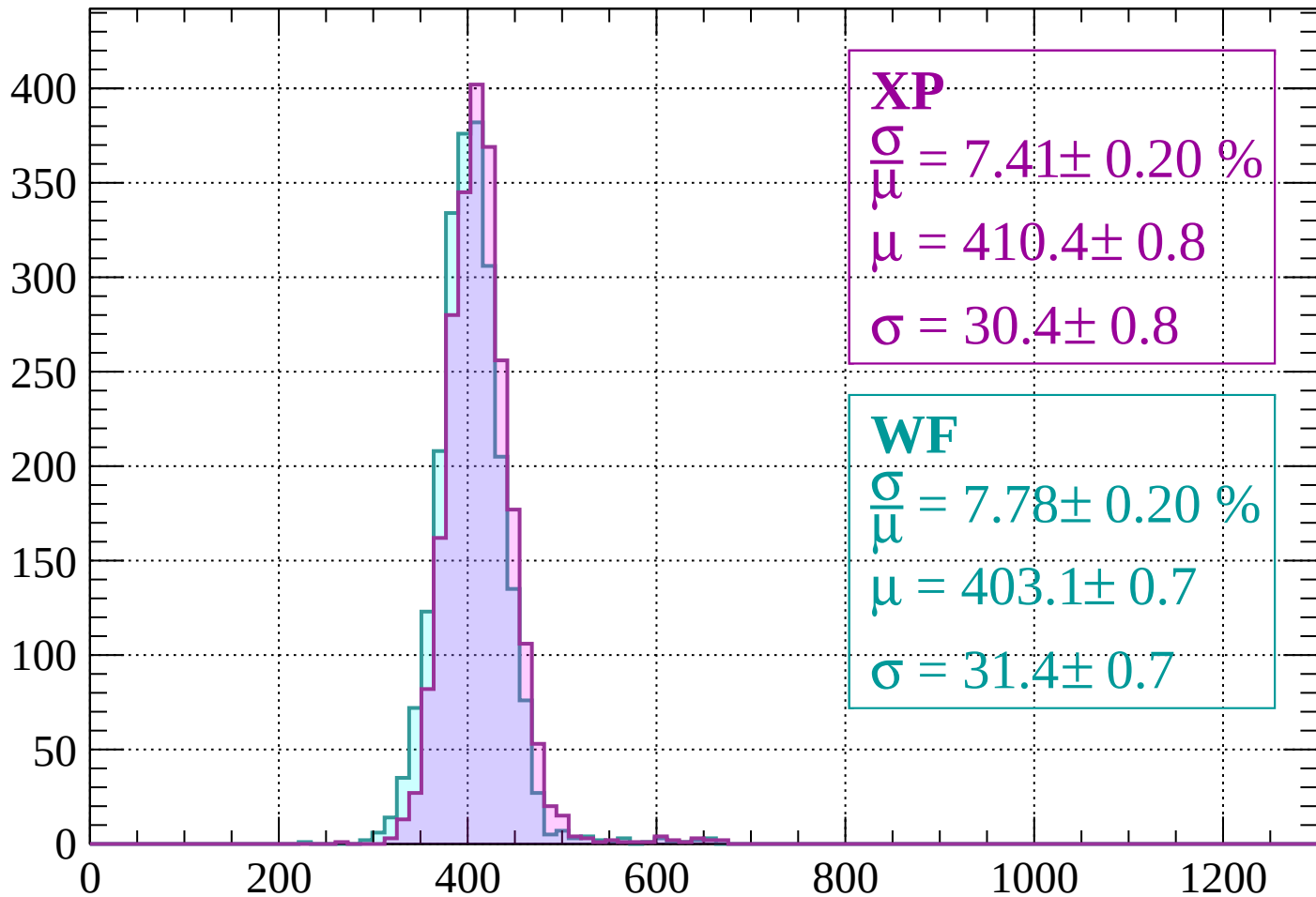
$$\mu = 406.4 \pm 0.8$$

$$\sigma = 30.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-440 < p < -400$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.41 \pm 0.20 \%$$

$$\mu = 410.4 \pm 0.8$$

$$\sigma = 30.4 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.78 \pm 0.20 \%$$

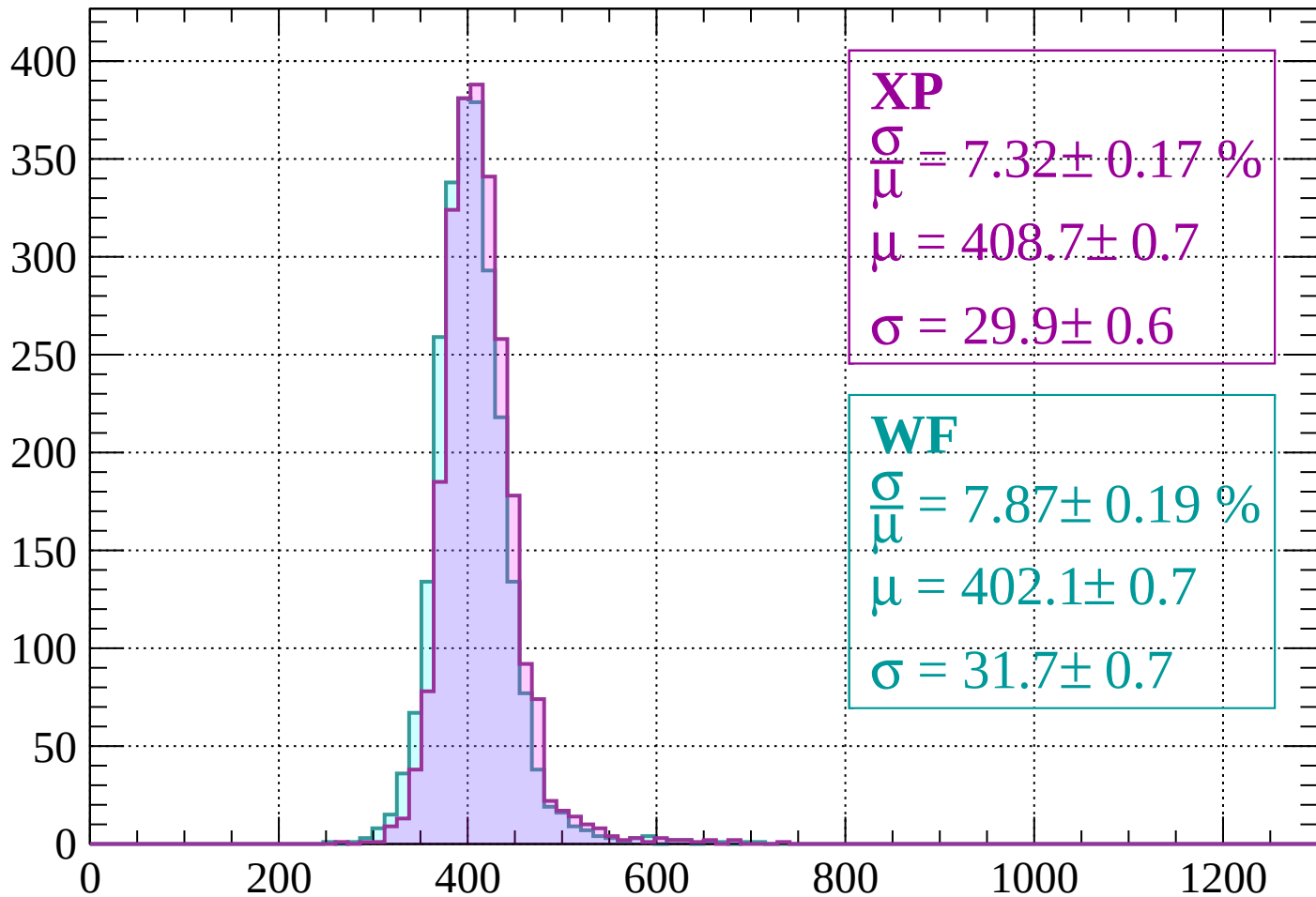
$$\mu = 403.1 \pm 0.7$$

$$\sigma = 31.4 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-400 < p < -360$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.32 \pm 0.17 \%$$

$$\mu = 408.7 \pm 0.7$$

$$\sigma = 29.9 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 7.87 \pm 0.19 \%$$

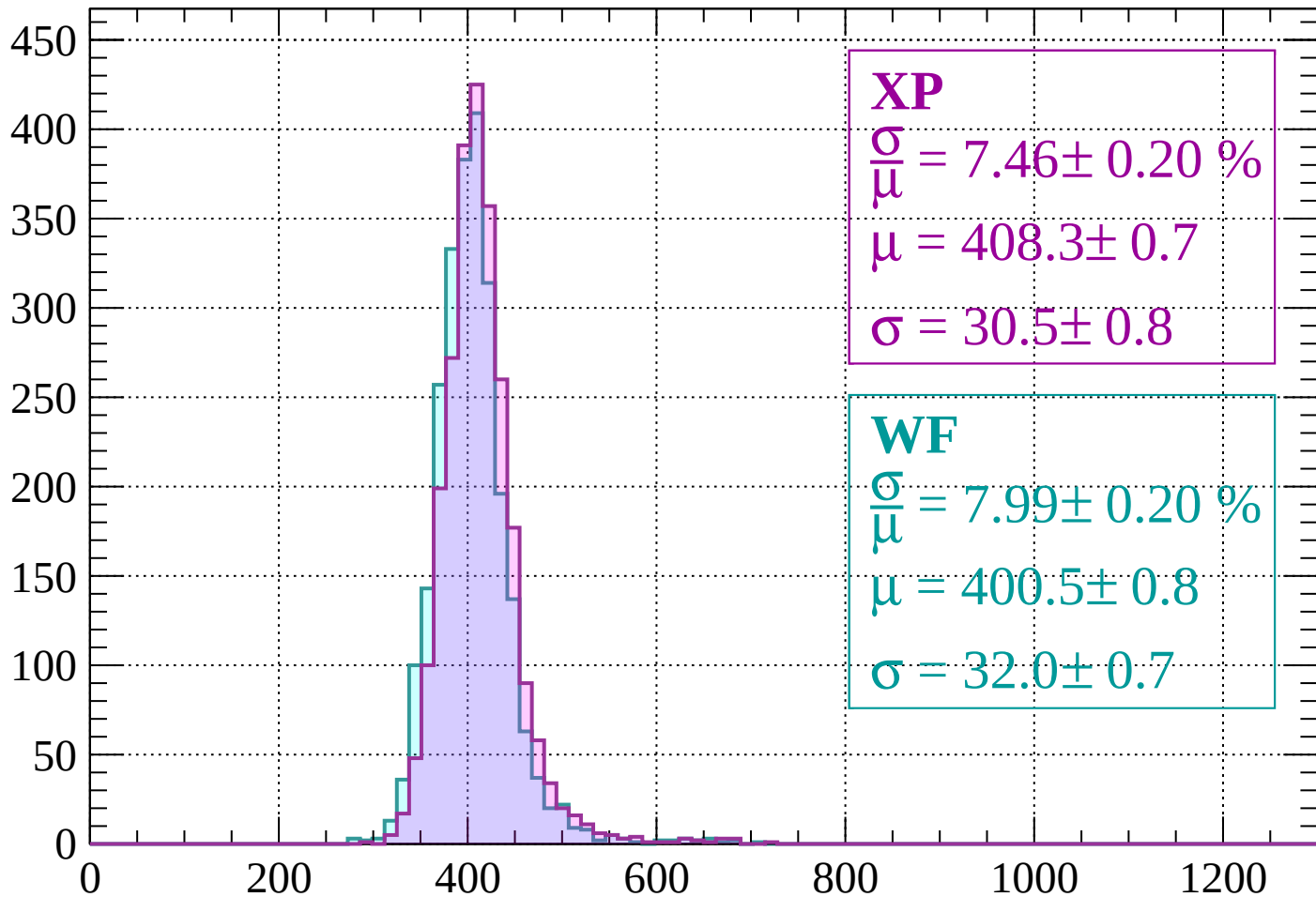
$$\mu = 402.1 \pm 0.7$$

$$\sigma = 31.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-360 < p < -320$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.46 \pm 0.20 \%$$

$$\mu = 408.3 \pm 0.7$$

$$\sigma = 30.5 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.99 \pm 0.20 \%$$

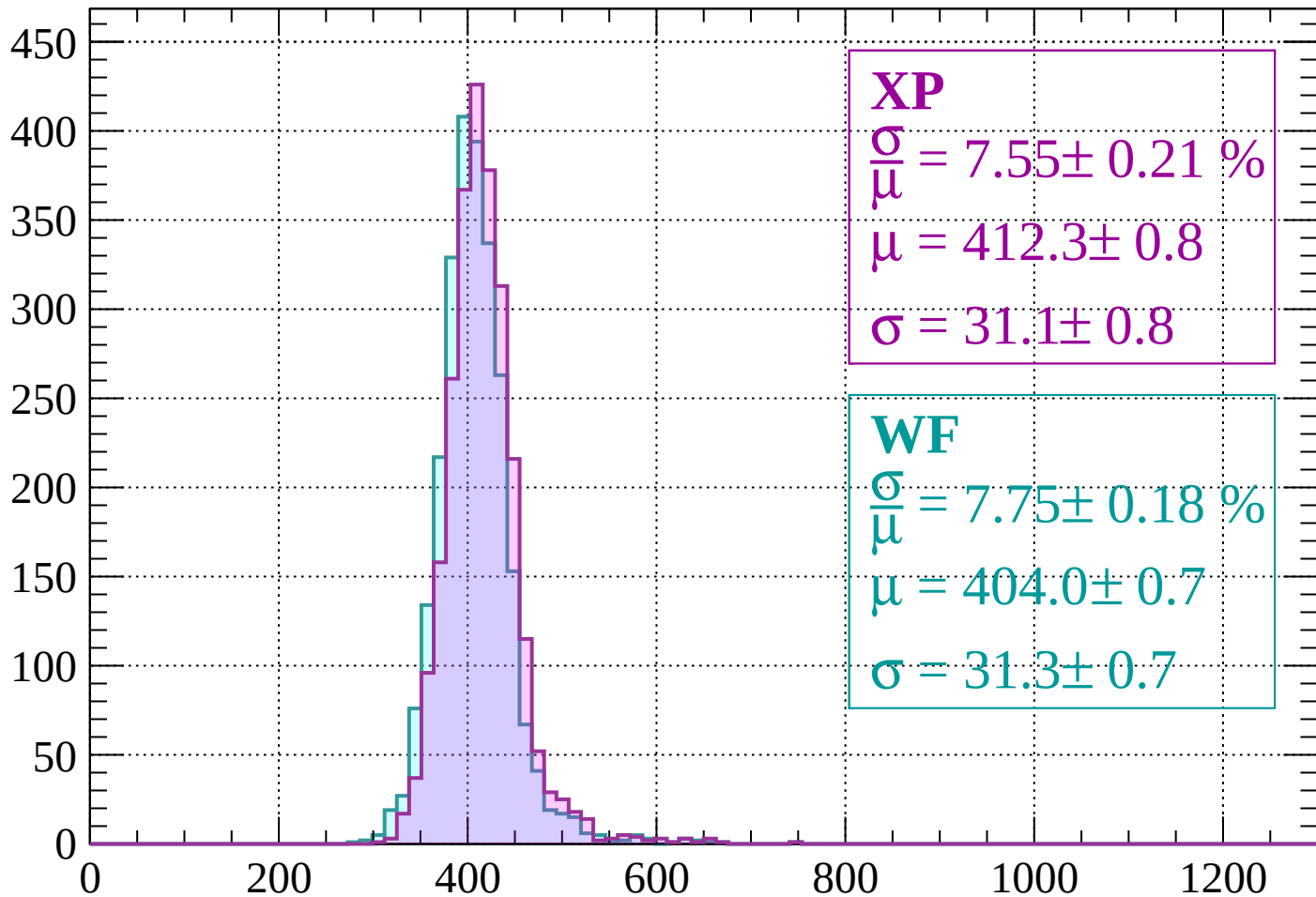
$$\mu = 400.5 \pm 0.8$$

$$\sigma = 32.0 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-320 < p < -280$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.55 \pm 0.21 \%$$

$$\mu = 412.3 \pm 0.8$$

$$\sigma = 31.1 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.75 \pm 0.18 \%$$

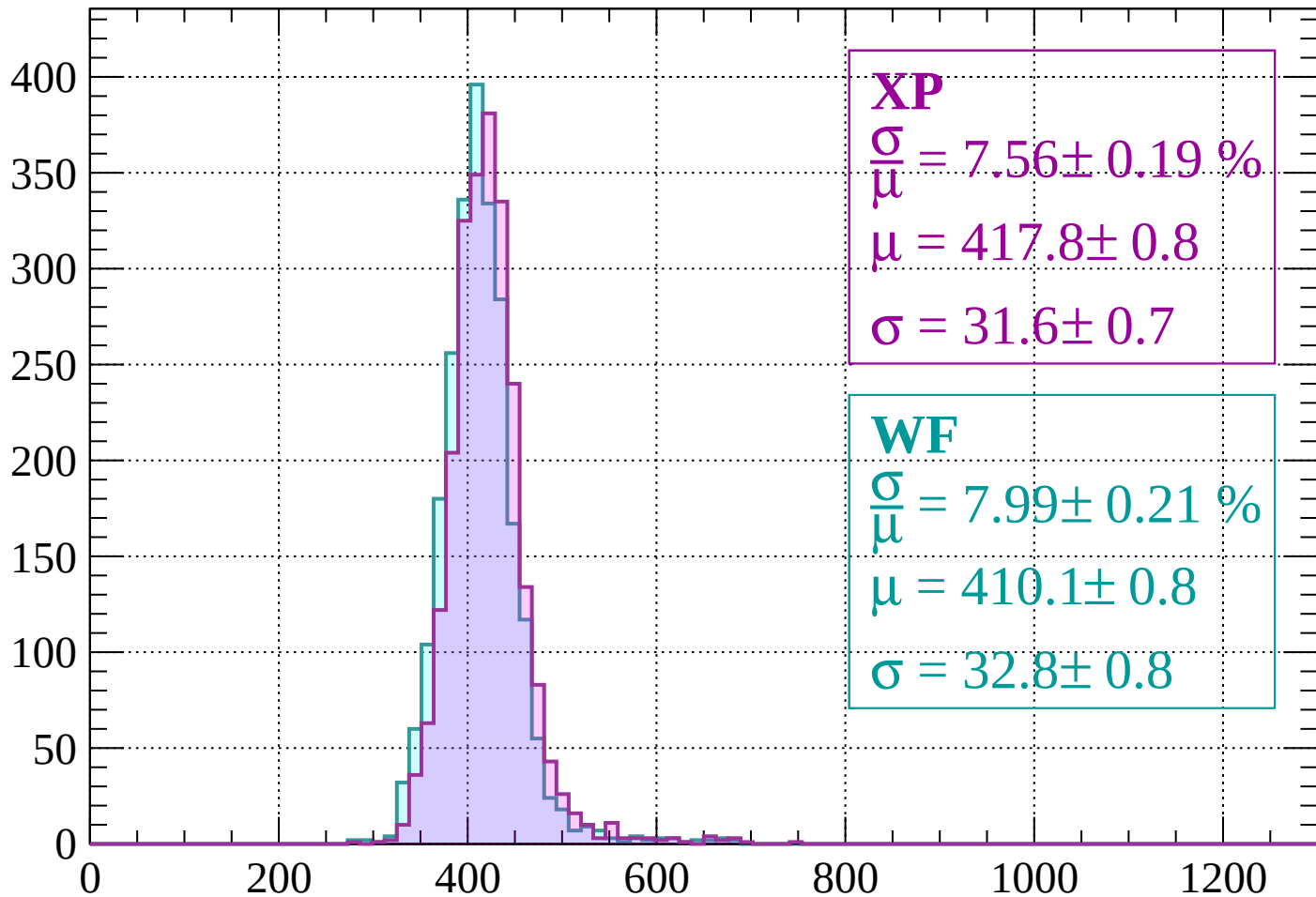
$$\mu = 404.0 \pm 0.7$$

$$\sigma = 31.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-280 < p < -240$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.56 \pm 0.19 \%$$

$$\mu = 417.8 \pm 0.8$$

$$\sigma = 31.6 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 7.99 \pm 0.21 \%$$

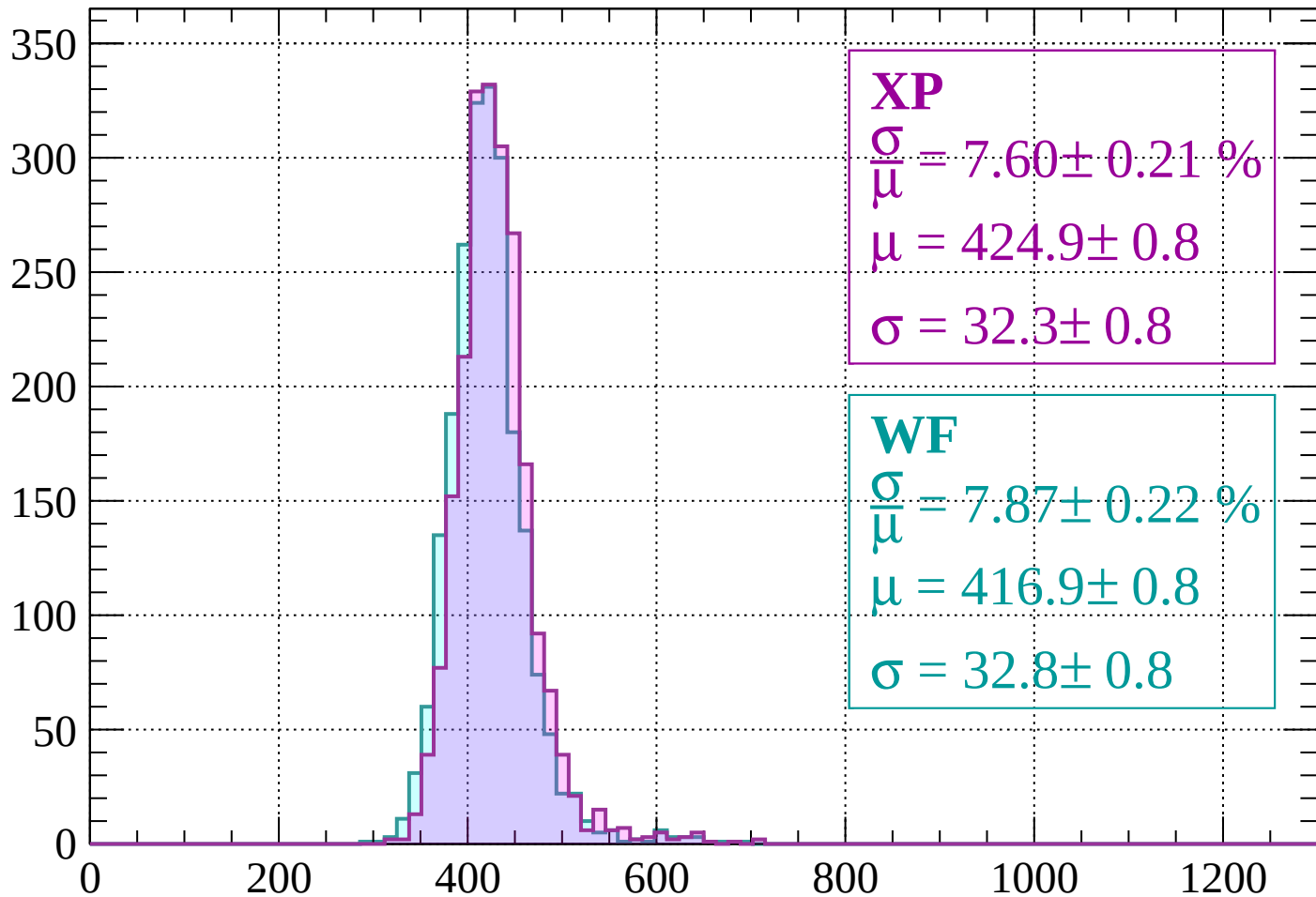
$$\mu = 410.1 \pm 0.8$$

$$\sigma = 32.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-240 < p < -200$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.60 \pm 0.21 \%$$

$$\mu = 424.9 \pm 0.8$$

$$\sigma = 32.3 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.87 \pm 0.22 \%$$

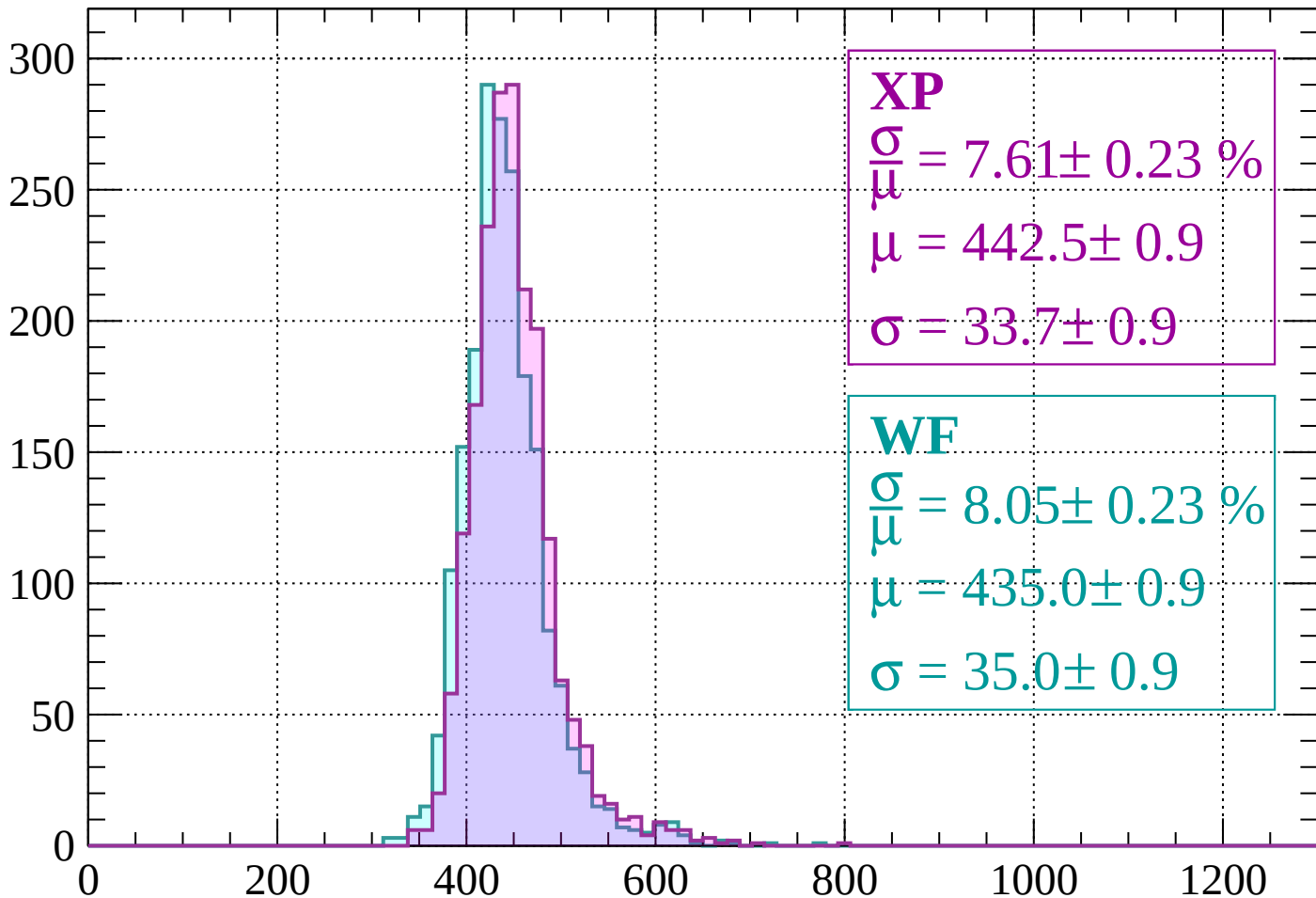
$$\mu = 416.9 \pm 0.8$$

$$\sigma = 32.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-200 < p < -160$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.61 \pm 0.23 \%$$

$$\mu = 442.5 \pm 0.9$$

$$\sigma = 33.7 \pm 0.9$$

**WF**

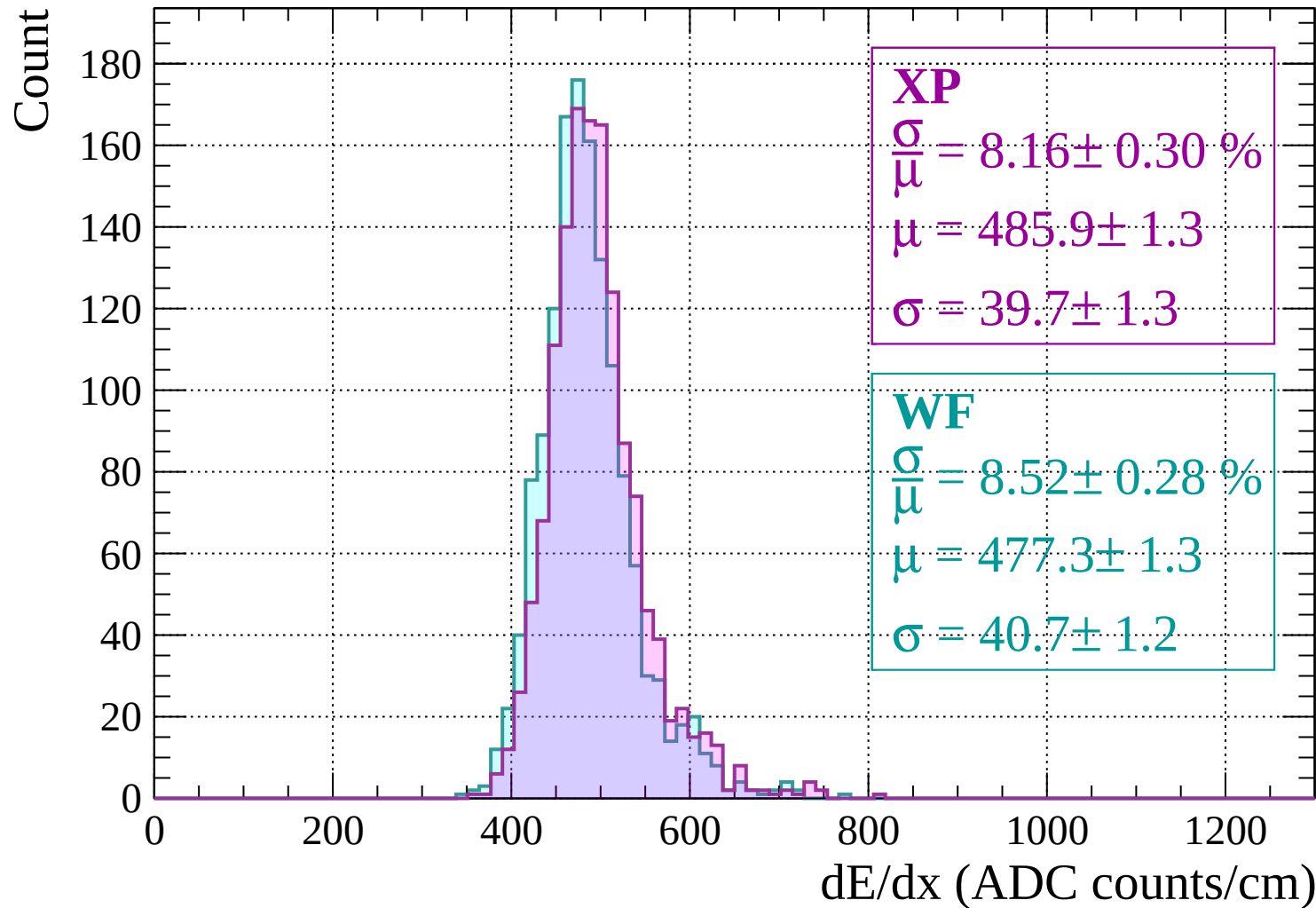
$$\frac{\sigma}{\mu} = 8.05 \pm 0.23 \%$$

$$\mu = 435.0 \pm 0.9$$

$$\sigma = 35.0 \pm 0.9$$

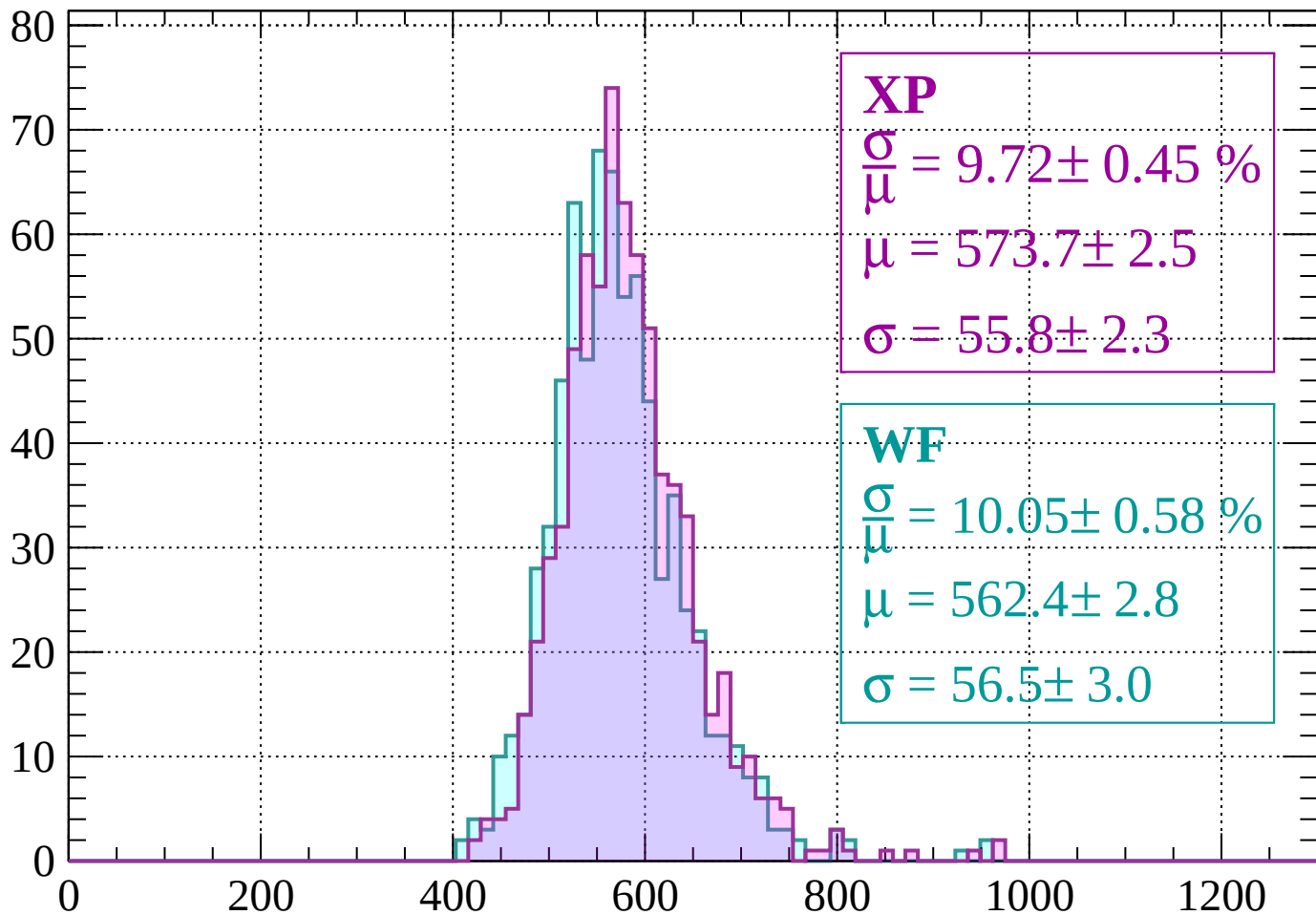
dE/dx (ADC counts/cm)

# Energy loss | $-160 < p < -120$



Energy loss |  $-120 < p < -80$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.72 \pm 0.45 \%$$

$$\mu = 573.7 \pm 2.5$$

$$\sigma = 55.8 \pm 2.3$$

**WF**

$$\frac{\sigma}{\mu} = 10.05 \pm 0.58 \%$$

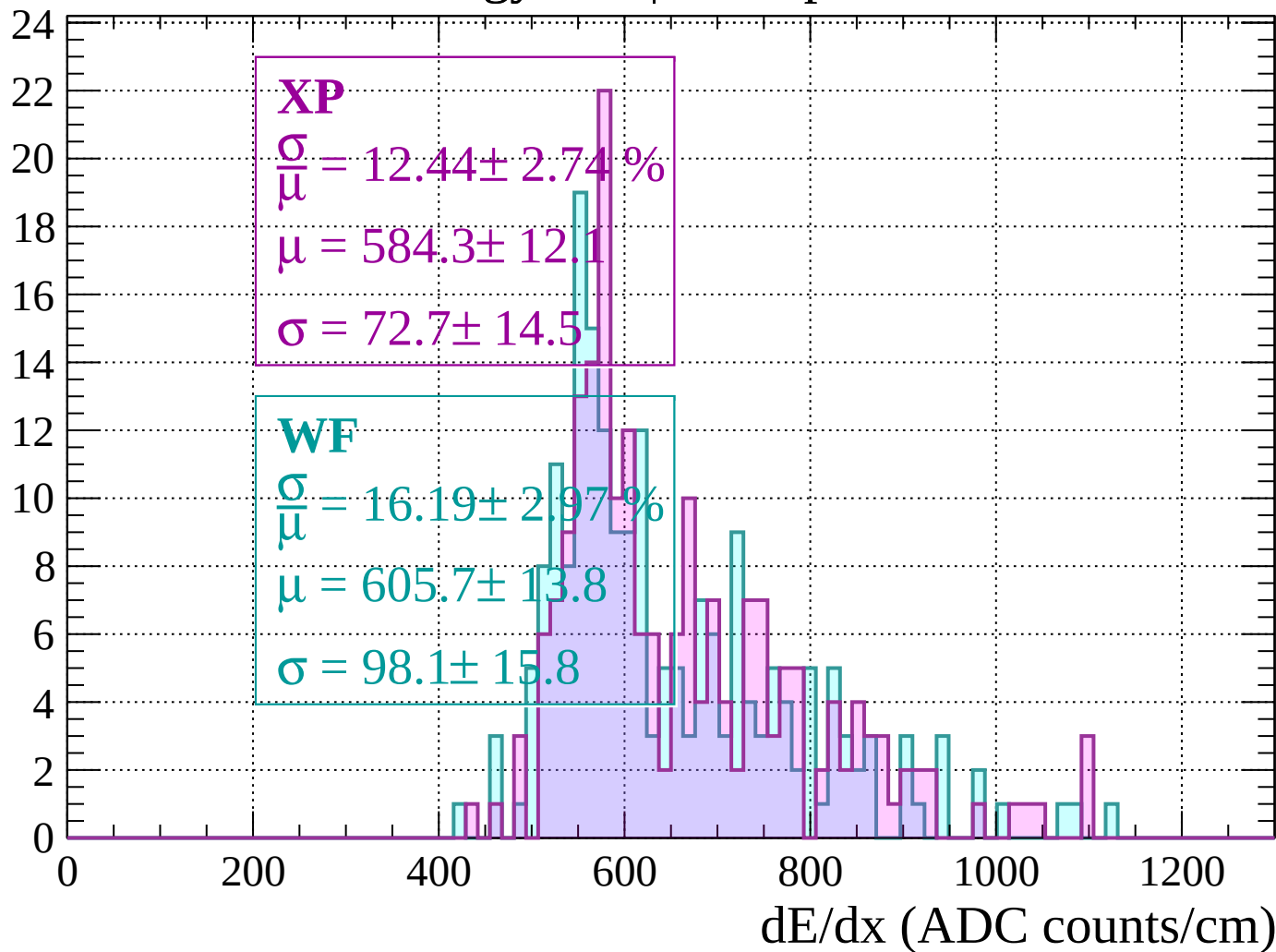
$$\mu = 562.4 \pm 2.8$$

$$\sigma = 56.5 \pm 3.0$$

dE/dx (ADC counts/cm)

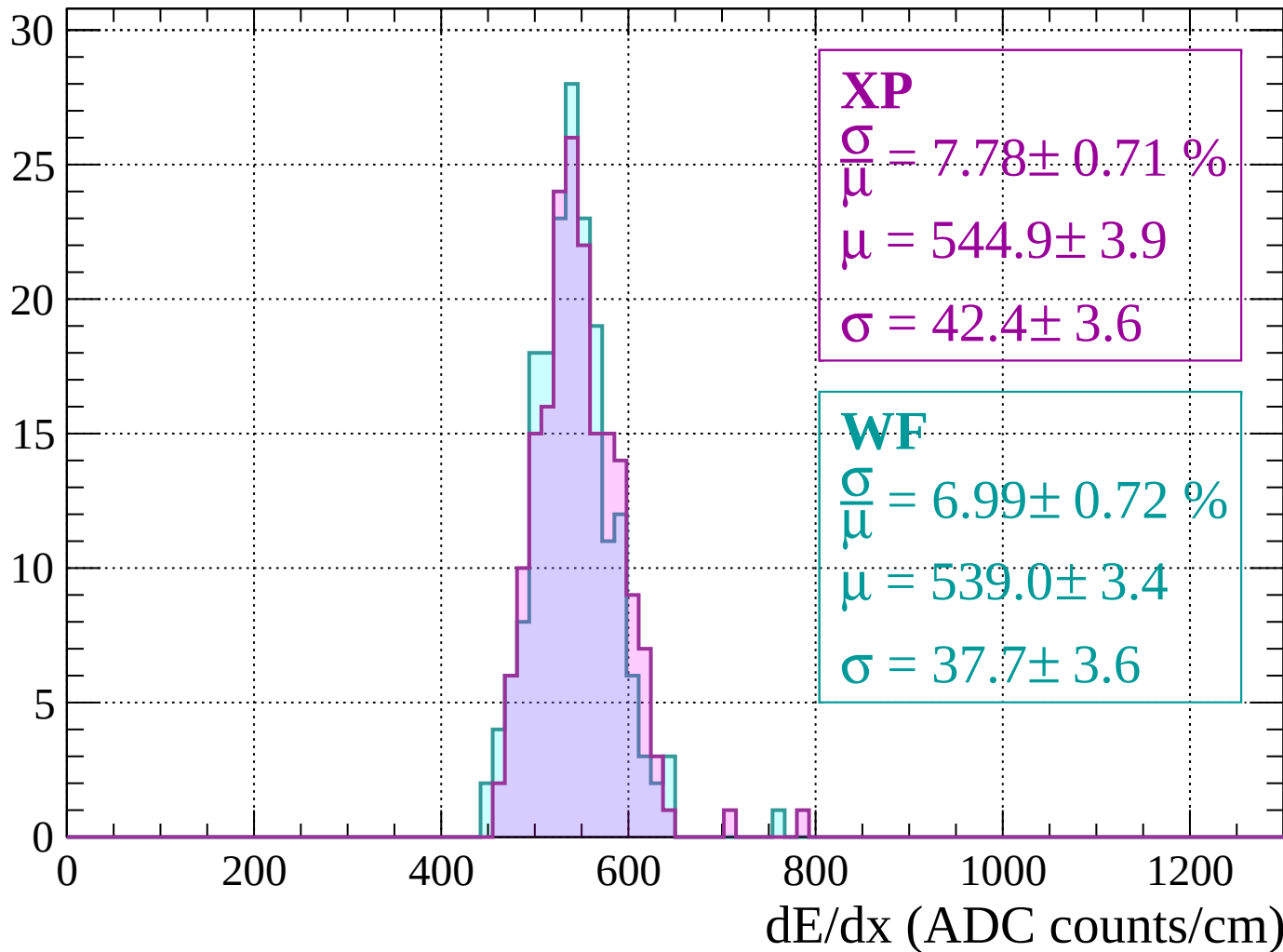
# Energy loss | -80 < p < -40

Count



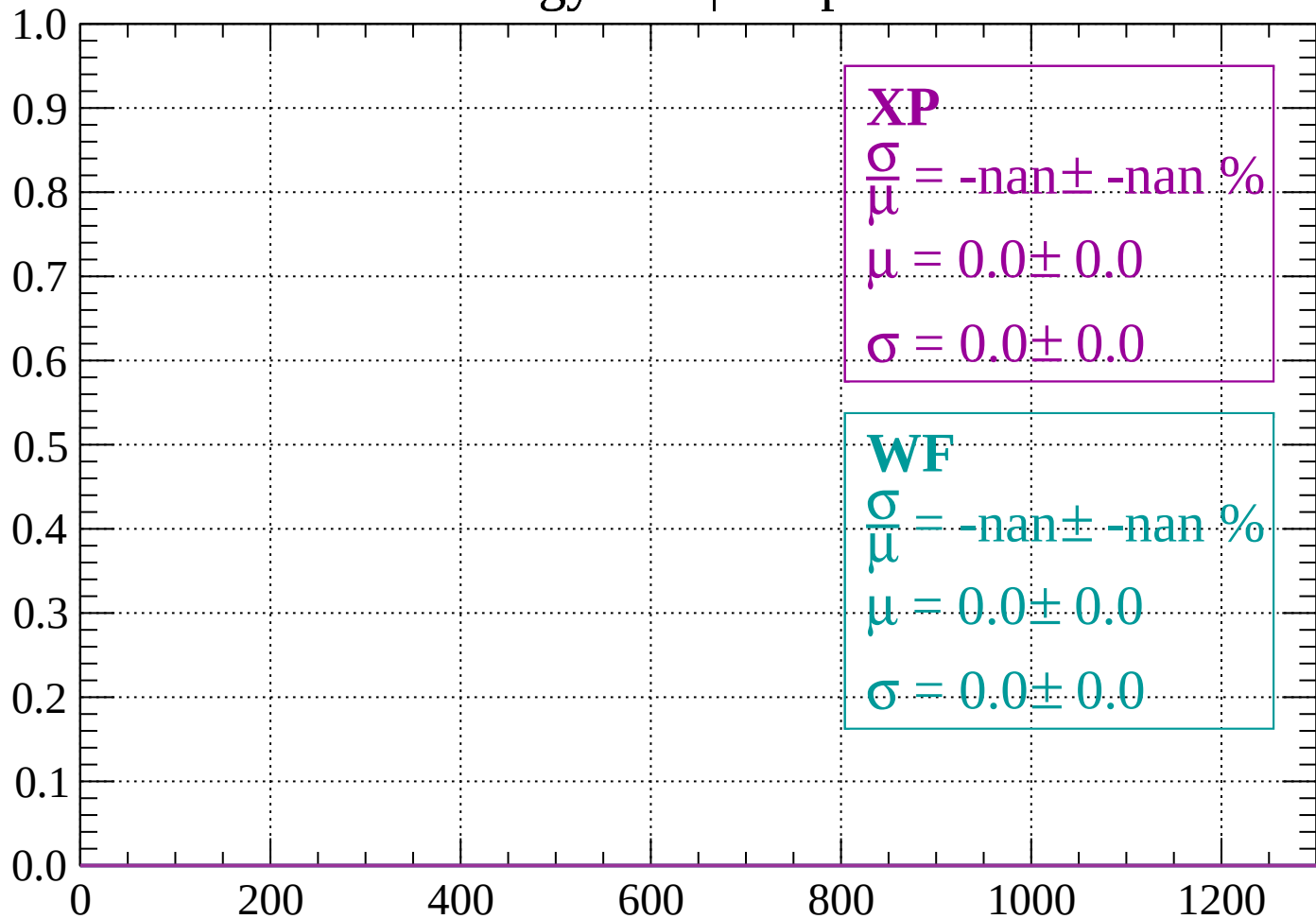
Energy loss |  $-40 < p < 0$

Count



Energy loss |  $0 < p < 40$

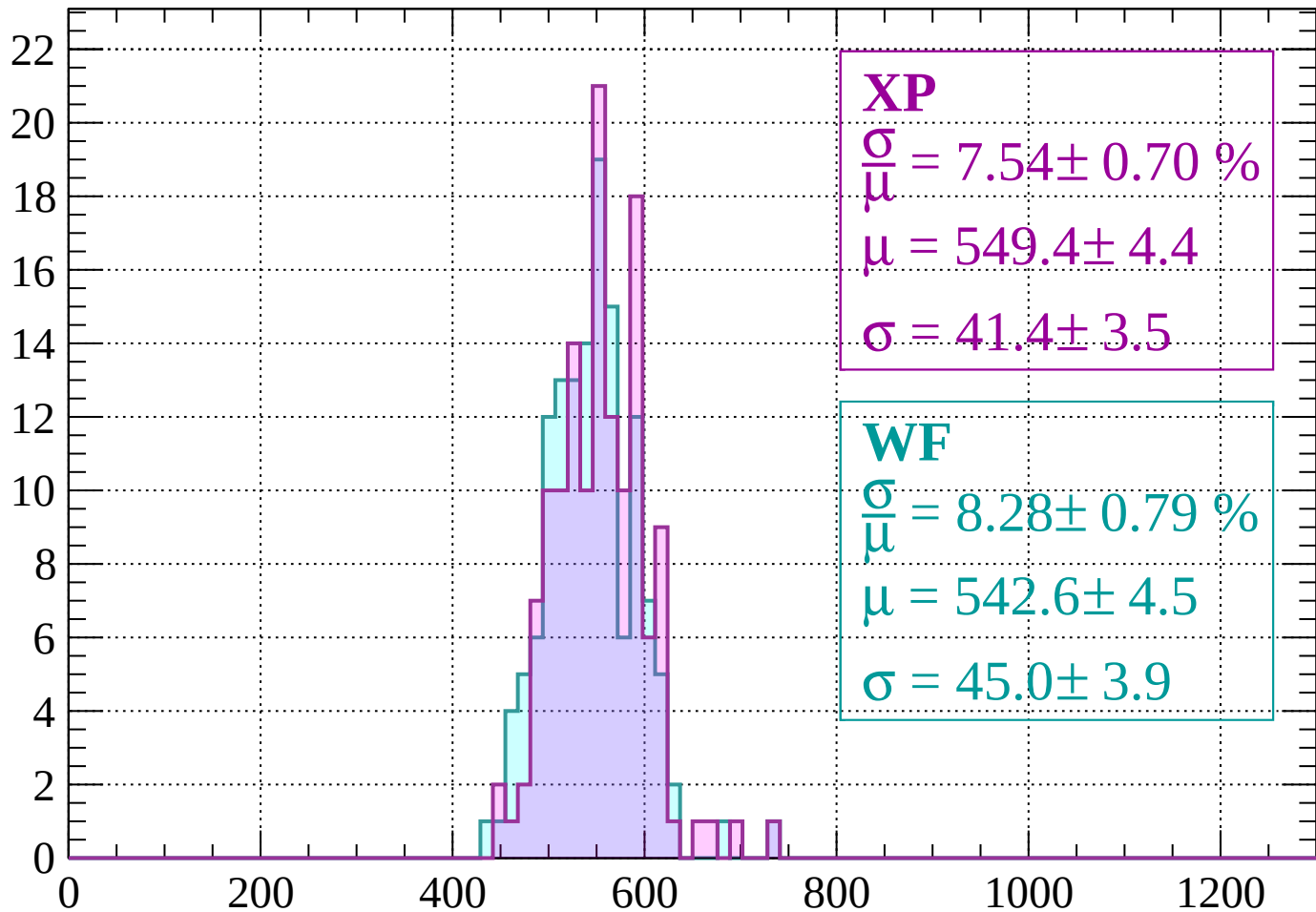
Count



dE/dx (ADC counts/cm)

Energy loss |  $40 < p < 80$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.54 \pm 0.70 \%$$

$$\mu = 549.4 \pm 4.4$$

$$\sigma = 41.4 \pm 3.5$$

**WF**

$$\frac{\sigma}{\mu} = 8.28 \pm 0.79 \%$$

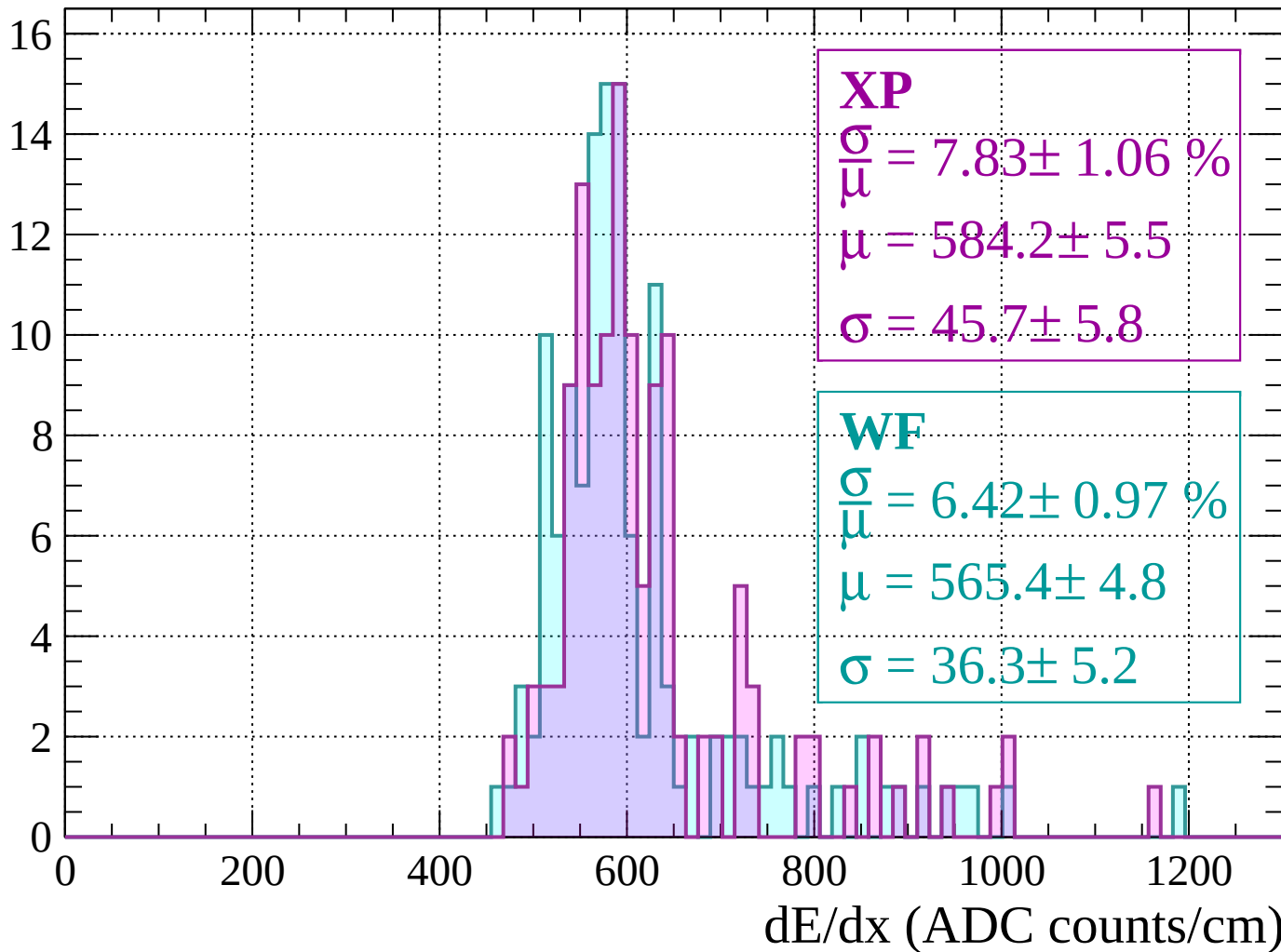
$$\mu = 542.6 \pm 4.5$$

$$\sigma = 45.0 \pm 3.9$$

$dE/dx$  (ADC counts/cm)

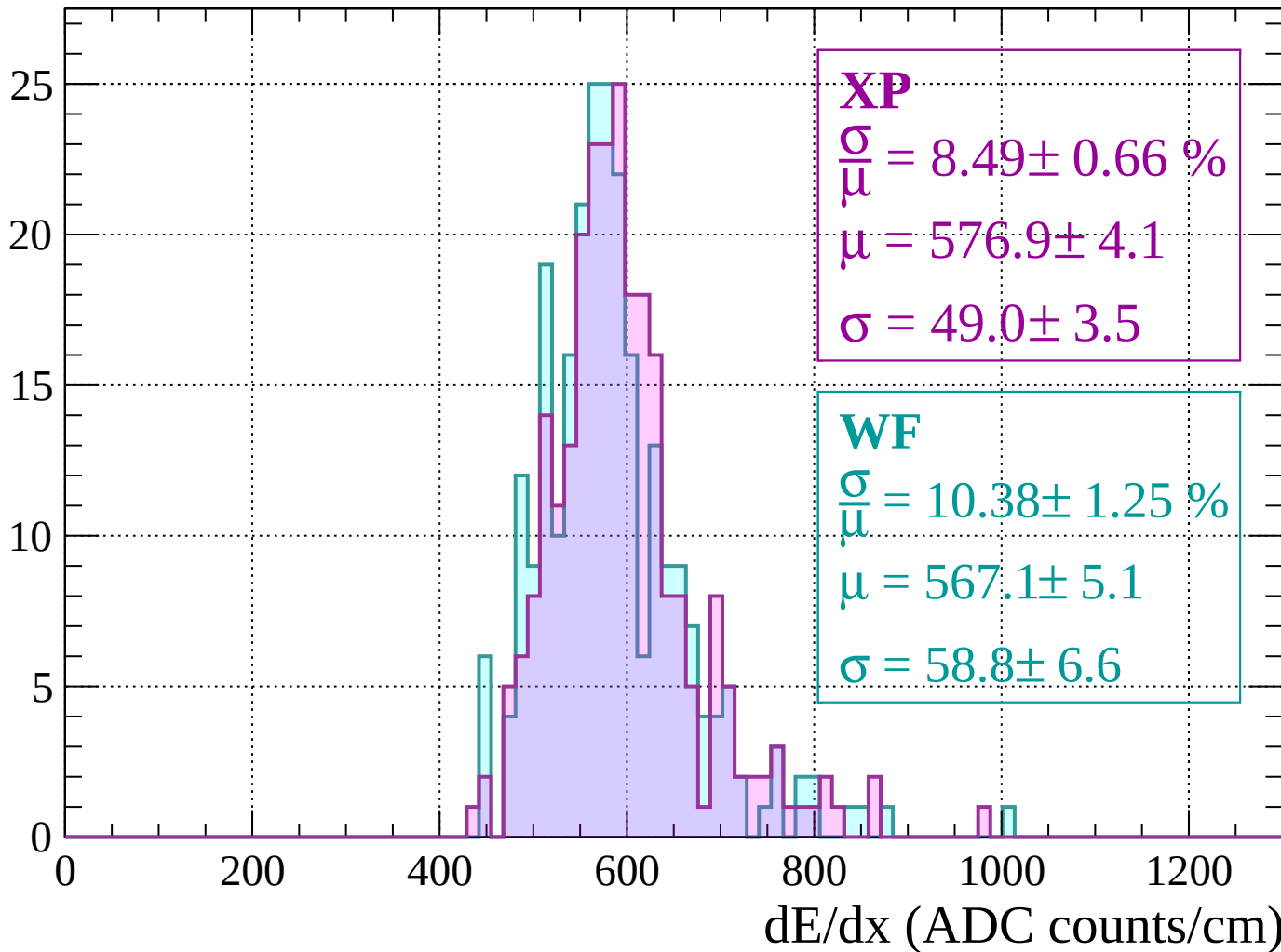
Energy loss |  $80 < p < 120$

Count



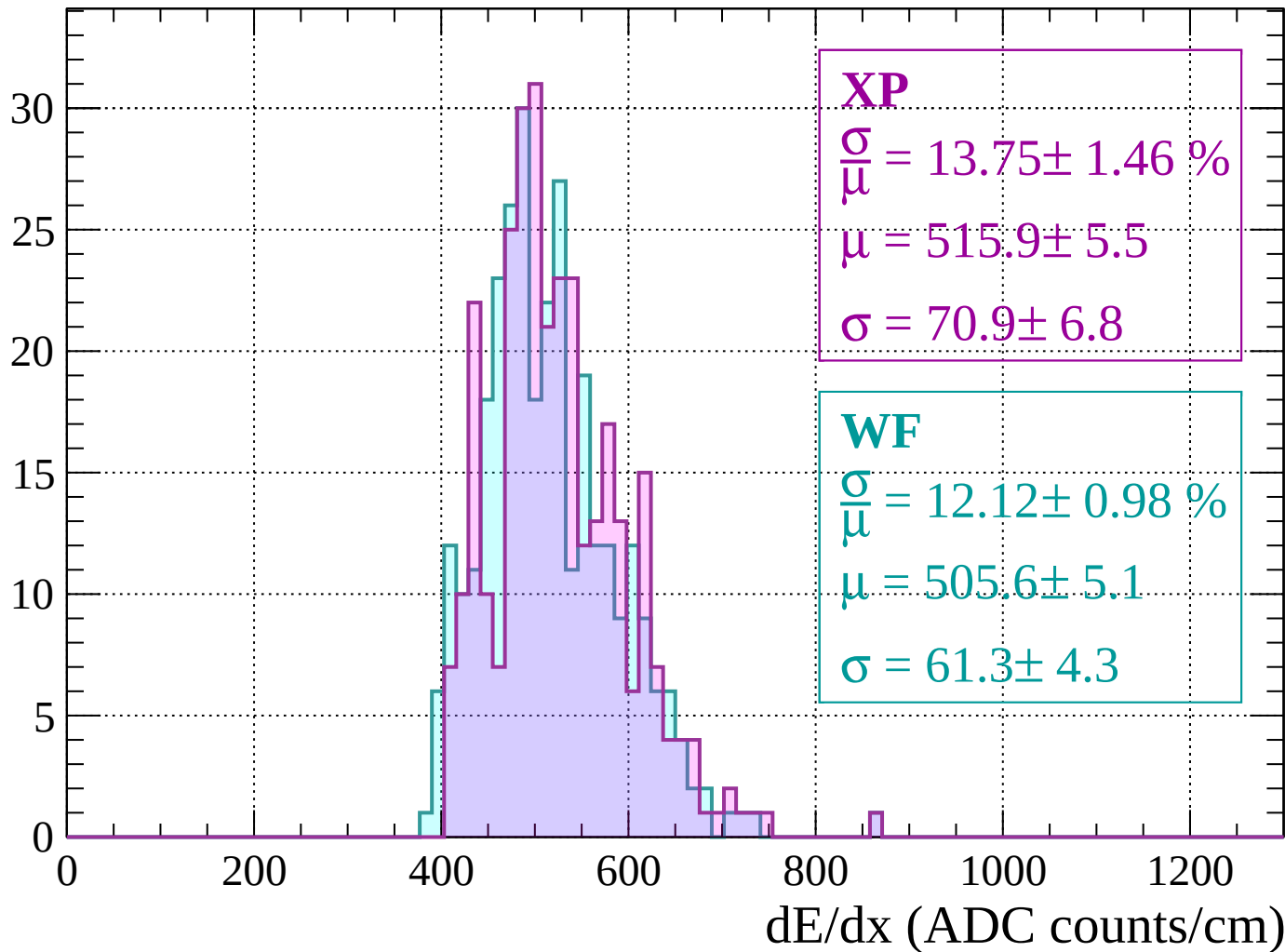
Energy loss |  $120 < p < 160$

Count



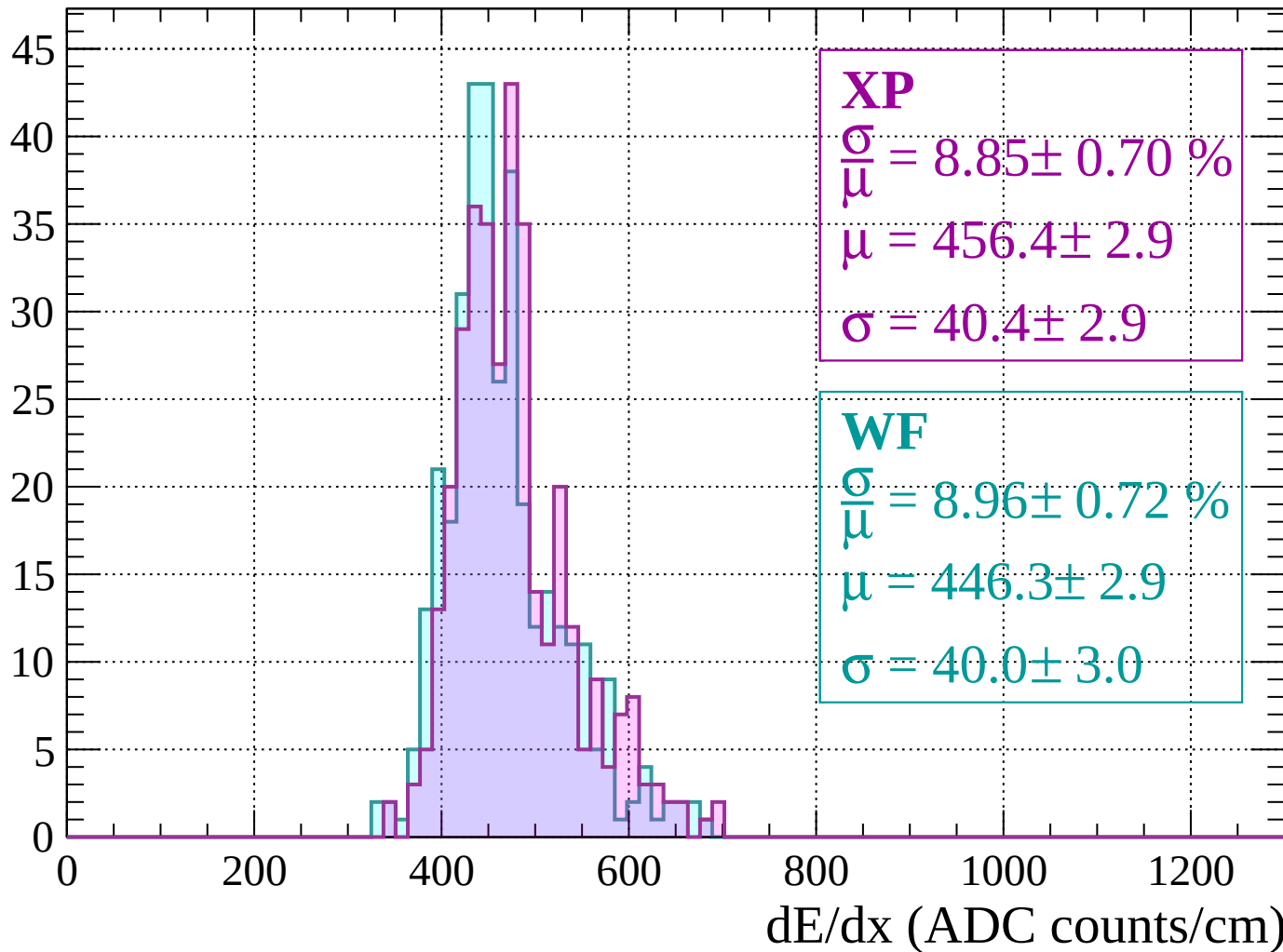
Energy loss |  $160 < p < 200$

Count



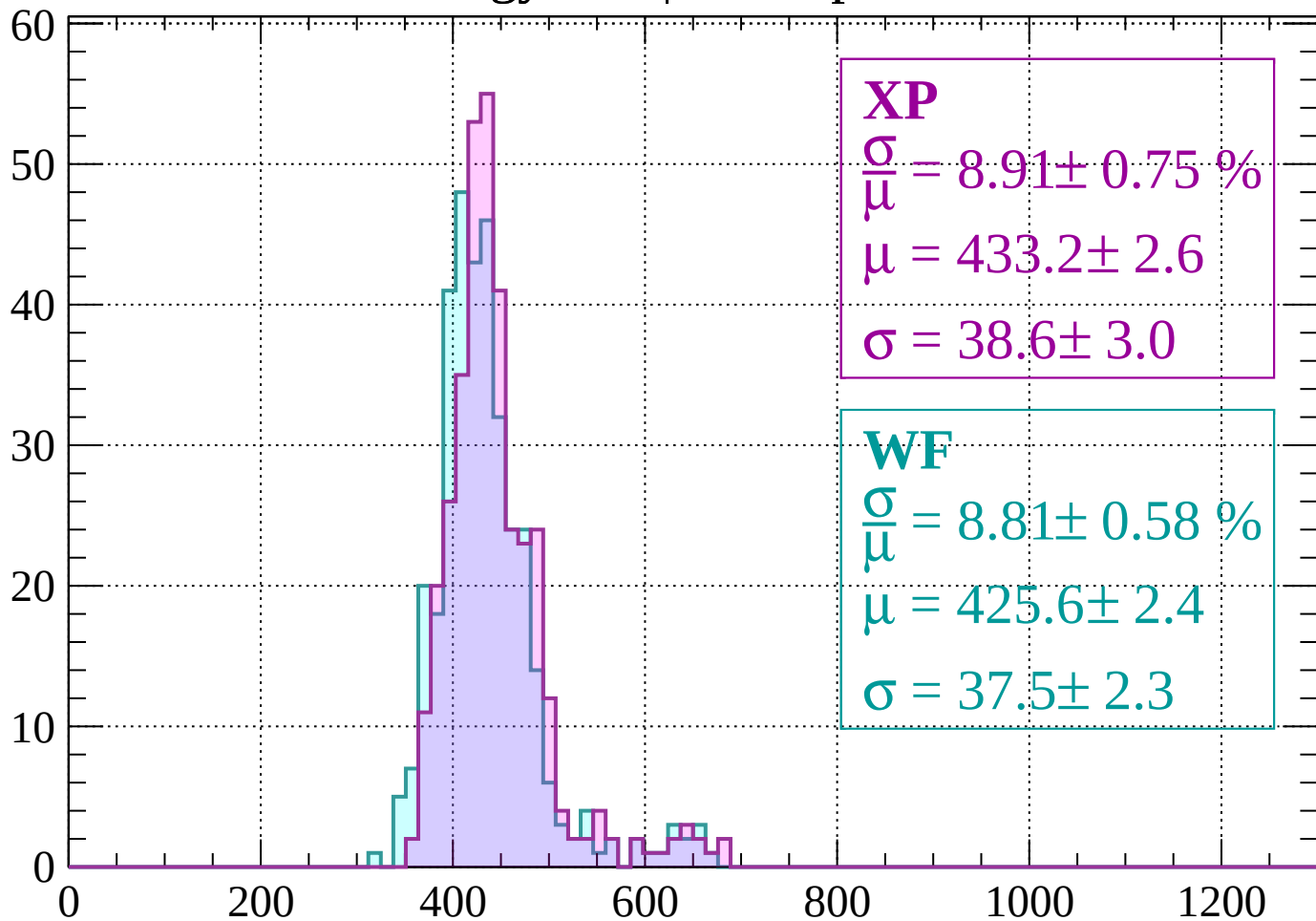
Energy loss |  $200 < p < 240$

Count



Energy loss |  $240 < p < 280$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 8.91 \pm 0.75 \%$$

$$\mu = 433.2 \pm 2.6$$

$$\sigma = 38.6 \pm 3.0$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 8.81 \pm 0.58 \%$$

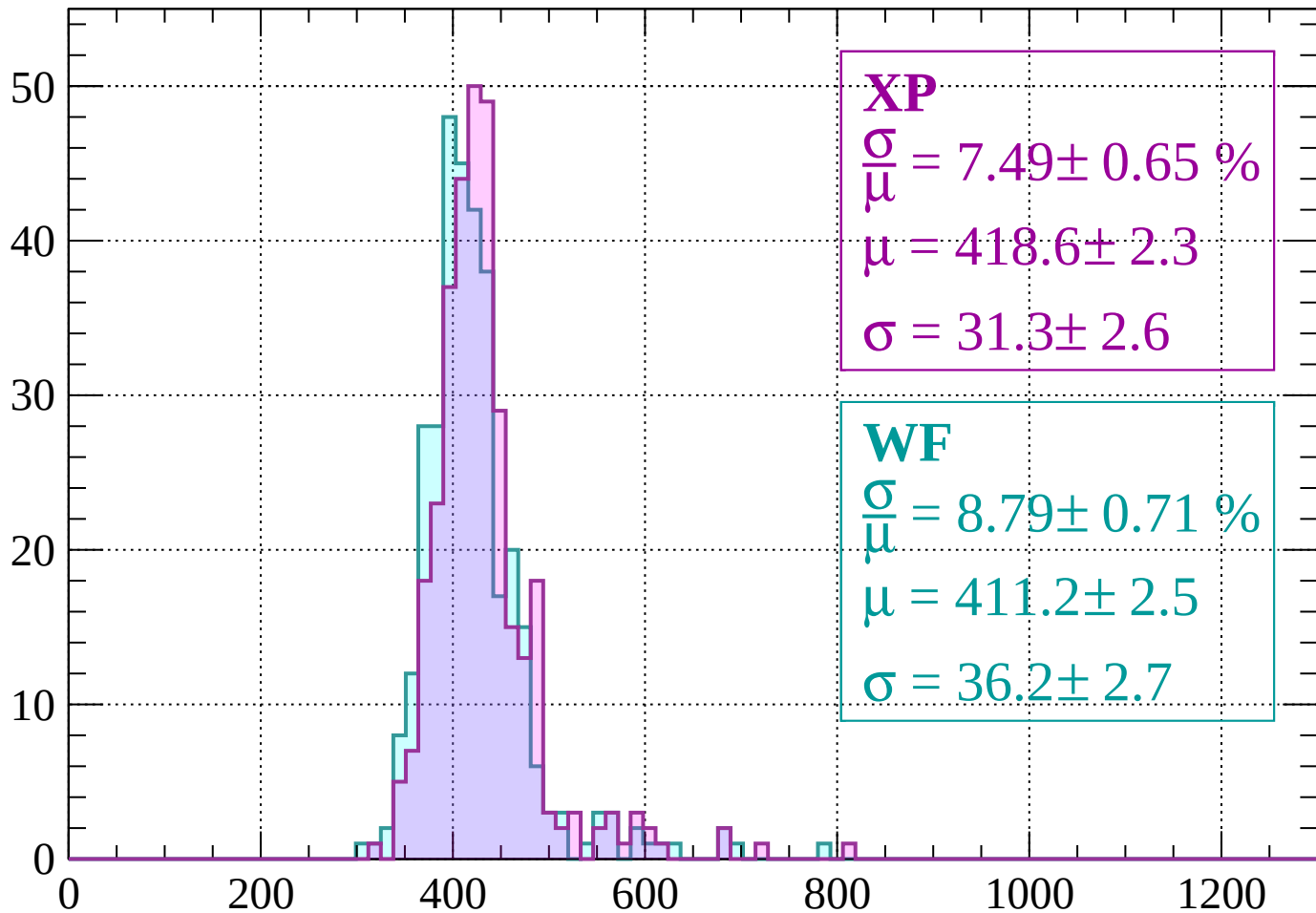
$$\mu = 425.6 \pm 2.4$$

$$\sigma = 37.5 \pm 2.3$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $280 < p < 320$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 7.49 \pm 0.65 \%$$

$$\mu = 418.6 \pm 2.3$$

$$\sigma = 31.3 \pm 2.6$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 8.79 \pm 0.71 \%$$

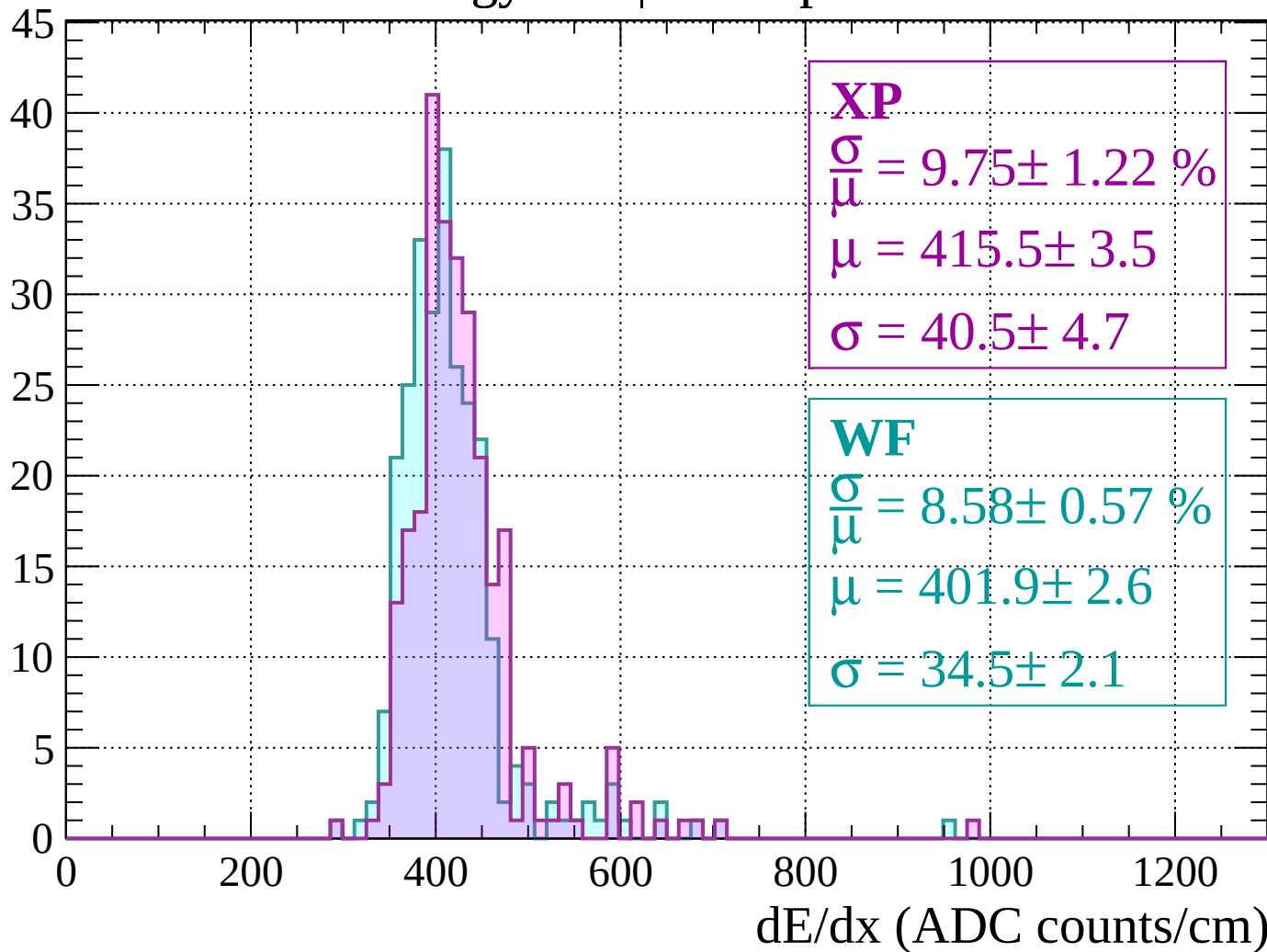
$$\mu = 411.2 \pm 2.5$$

$$\sigma = 36.2 \pm 2.7$$

$dE/dx$  (ADC counts/cm)

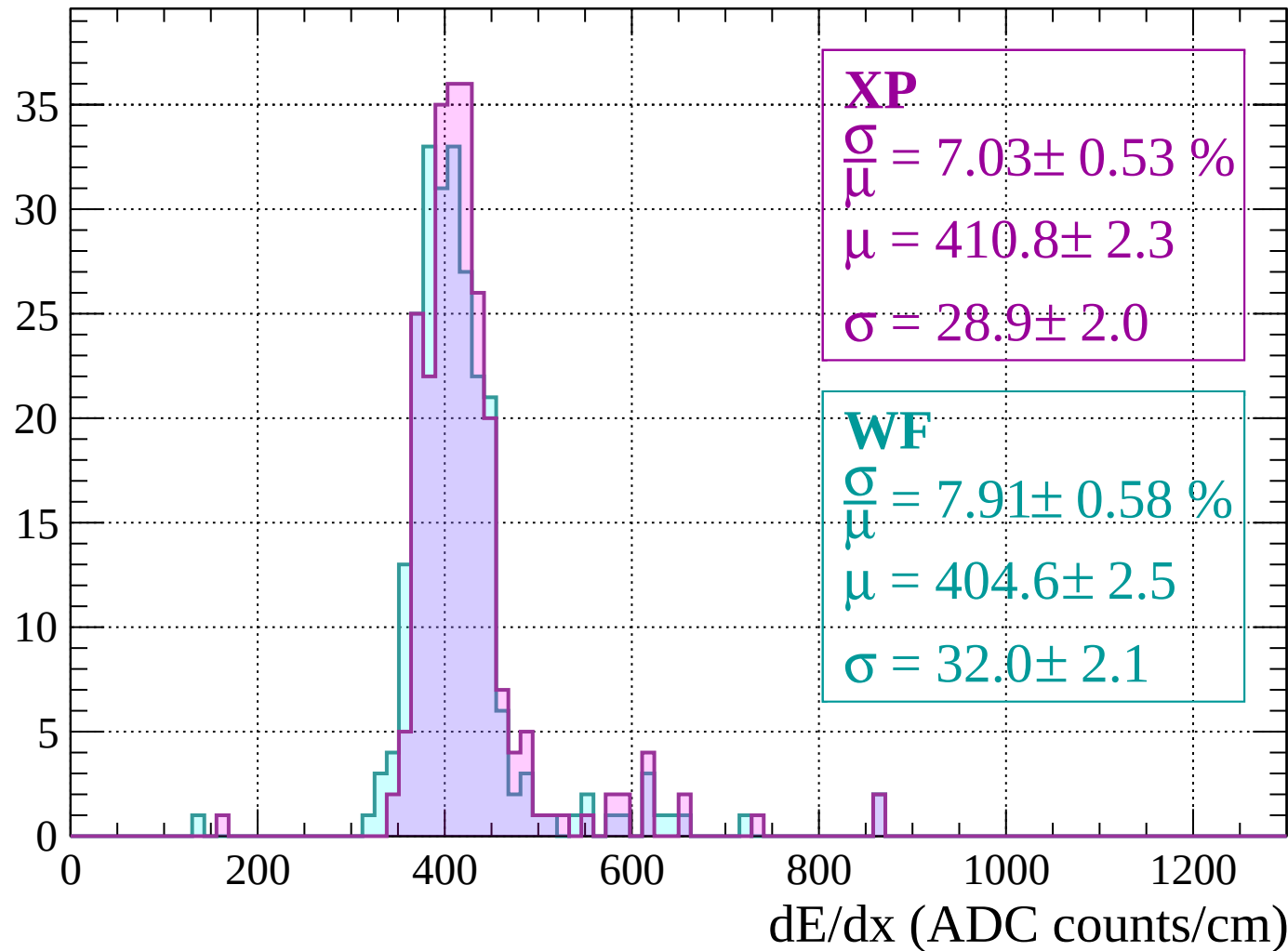
Energy loss |  $320 < p < 360$

Count



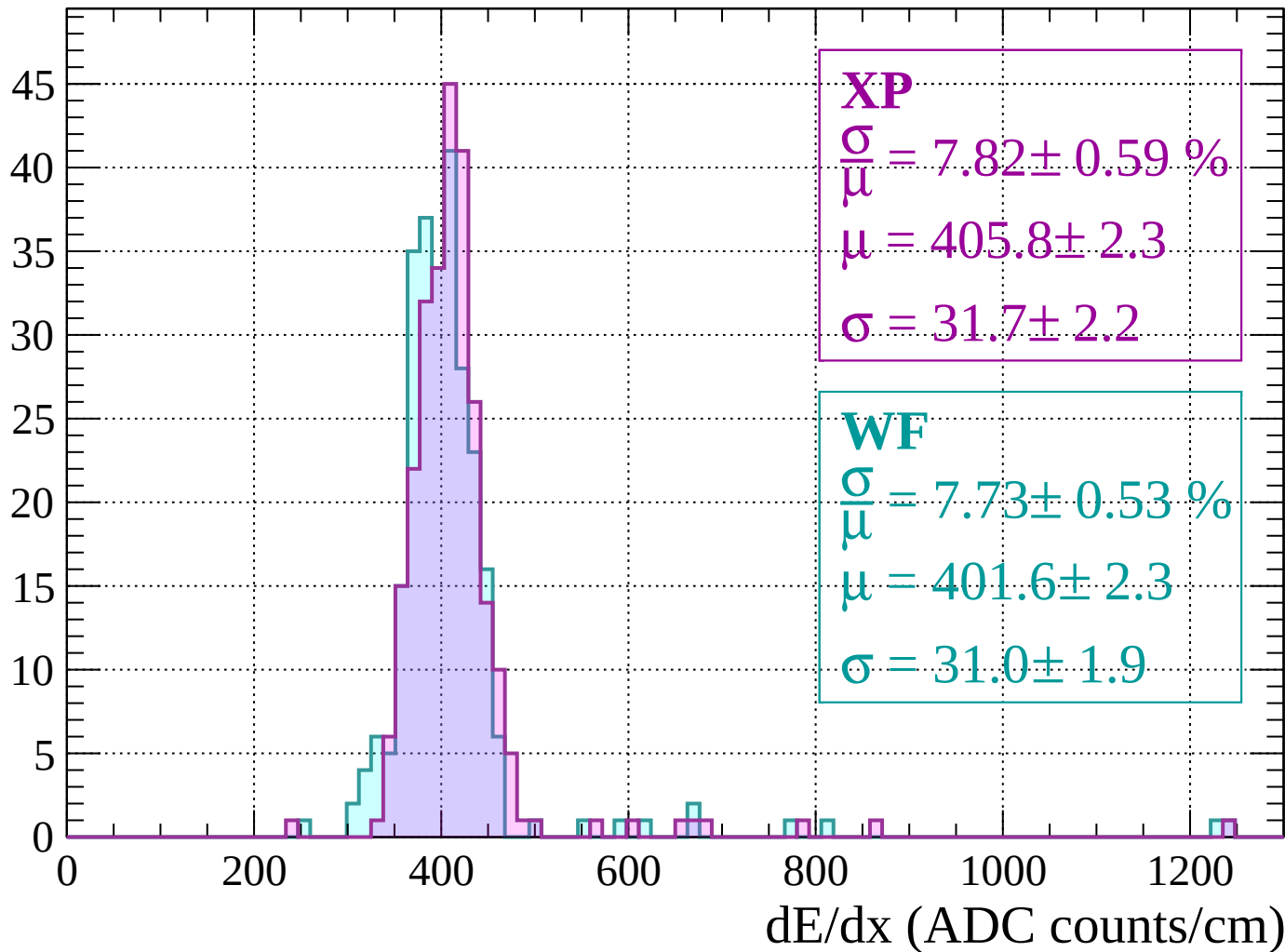
Energy loss |  $360 < p < 400$

Count



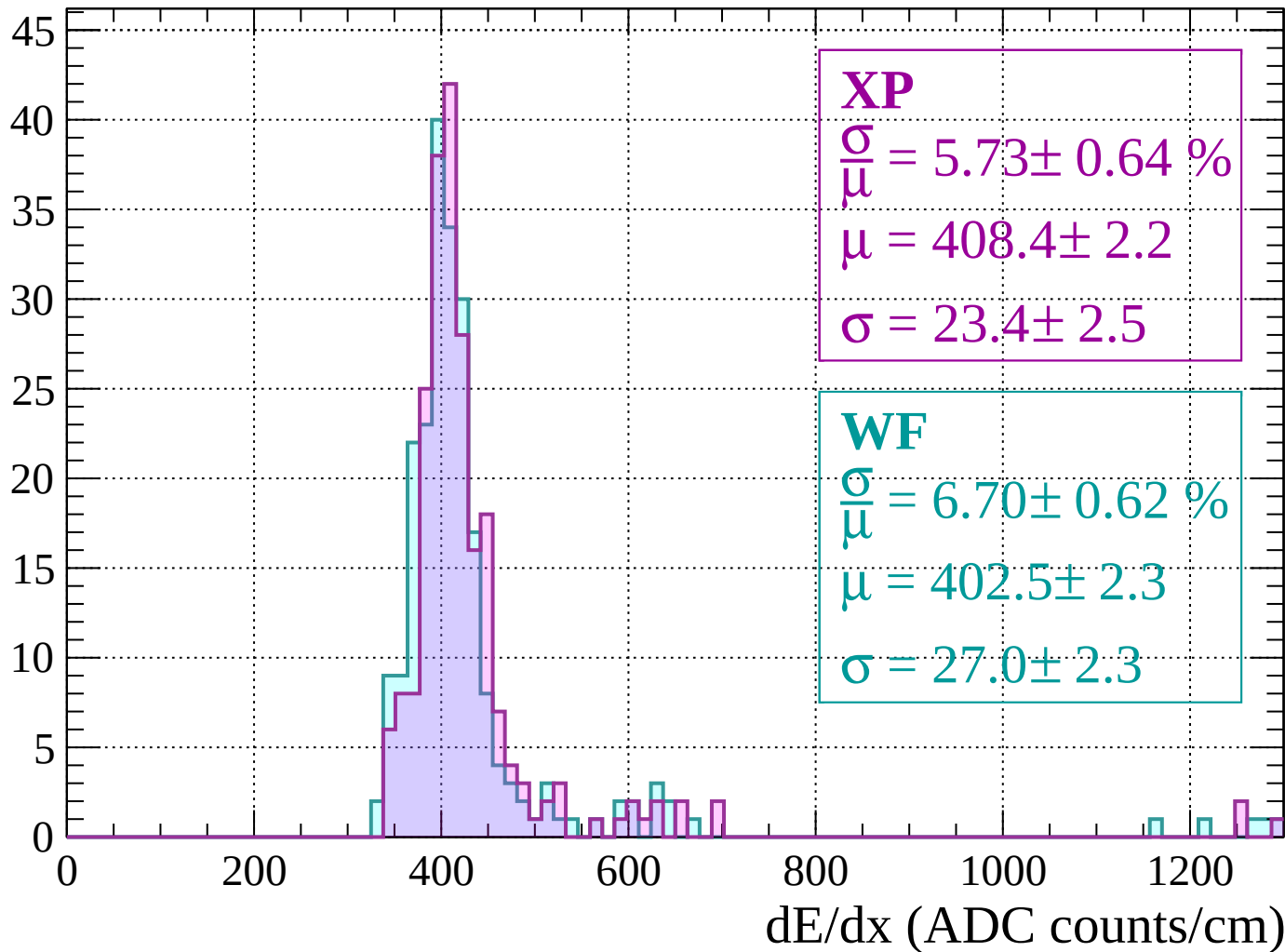
Energy loss |  $400 < p < 440$

Count



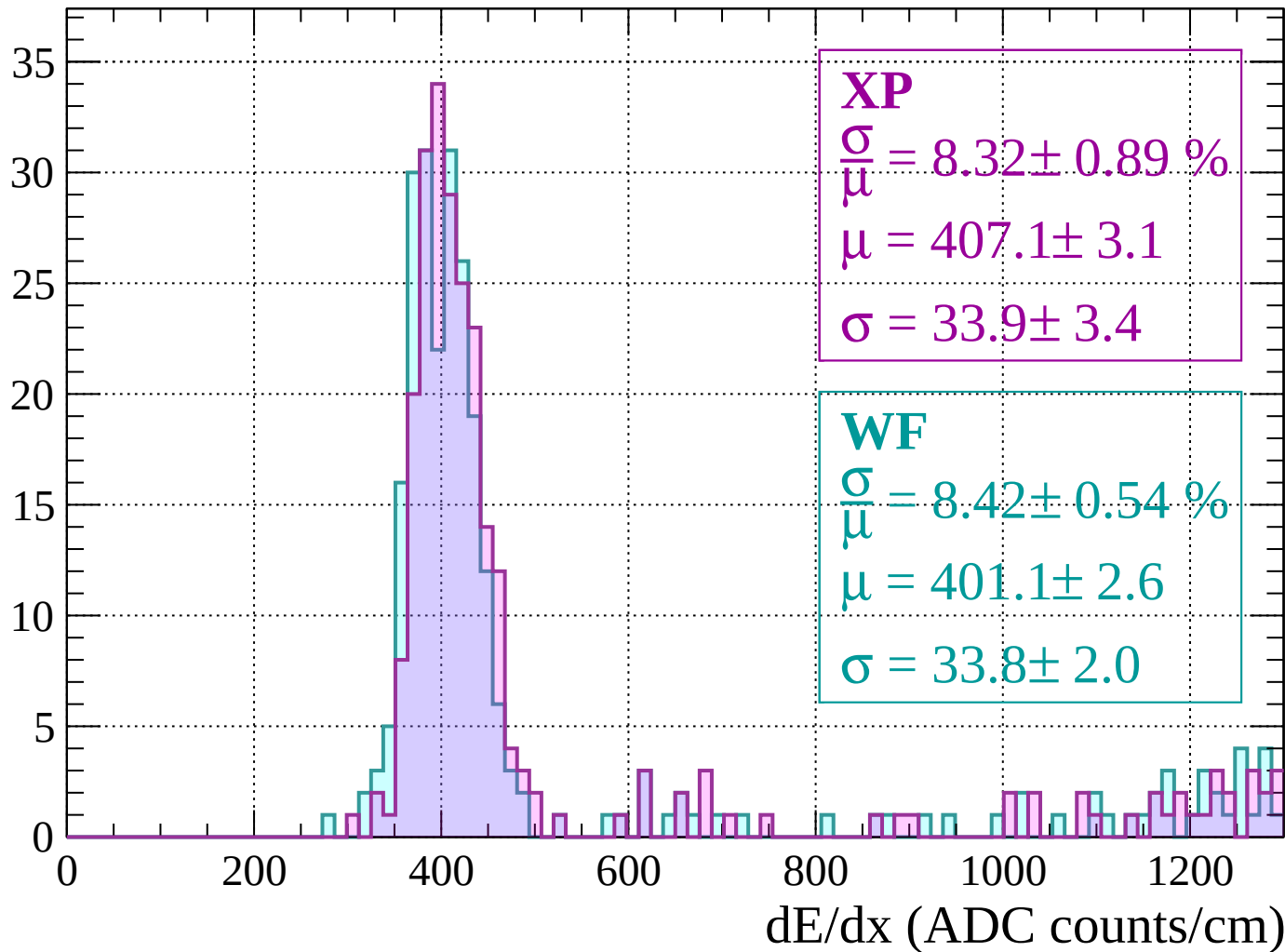
Energy loss |  $440 < p < 480$

Count



Energy loss |  $480 < p < 520$

Count



Energy loss |  $520 < p < 560$

Count

**XP**

$$\frac{\sigma}{\mu} = 7.75 \pm 0.53 \%$$

$$\mu = 413.5 \pm 2.5$$

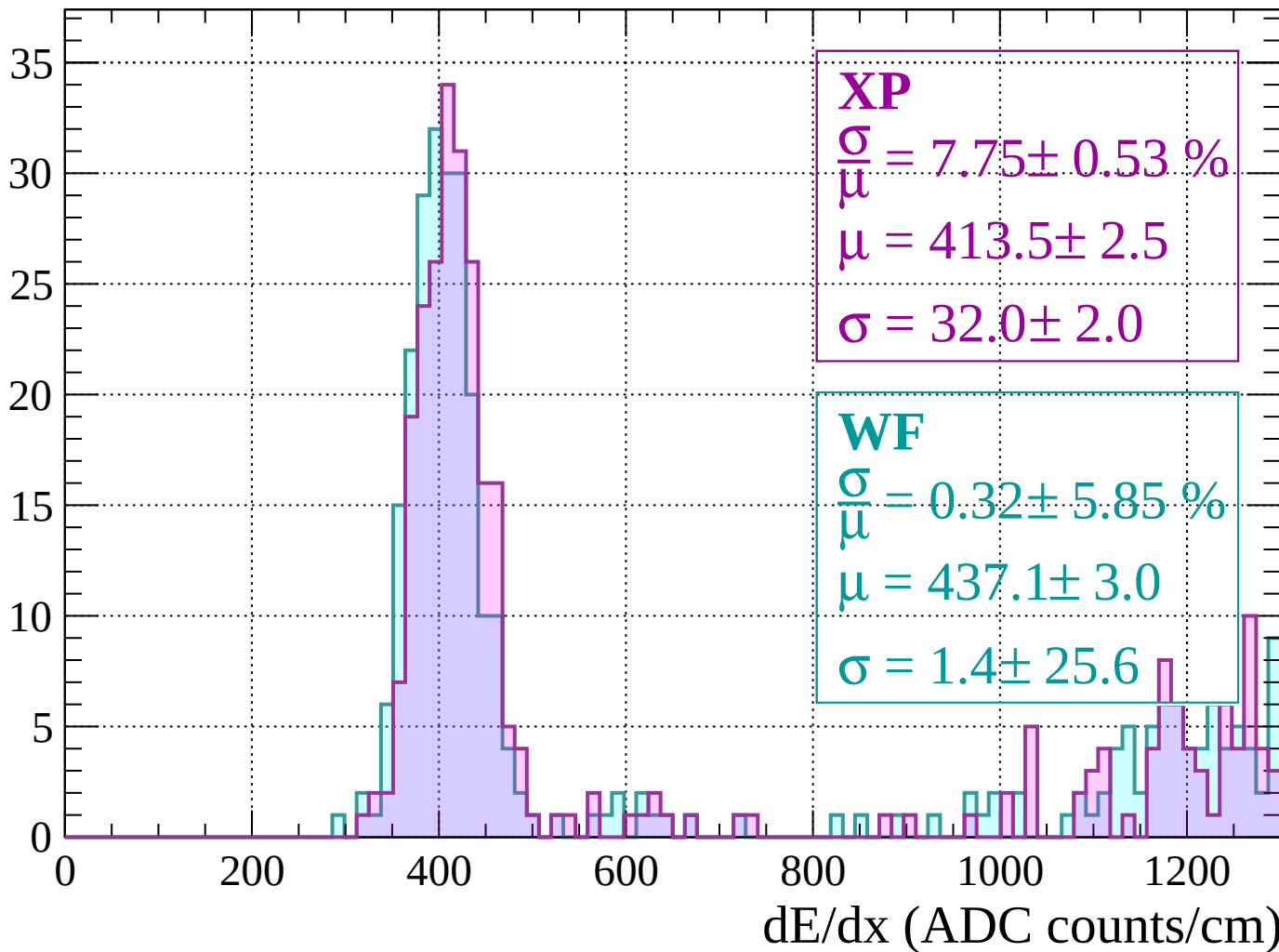
$$\sigma = 32.0 \pm 2.0$$

**WF**

$$\frac{\sigma}{\mu} = 0.32 \pm 5.85 \%$$

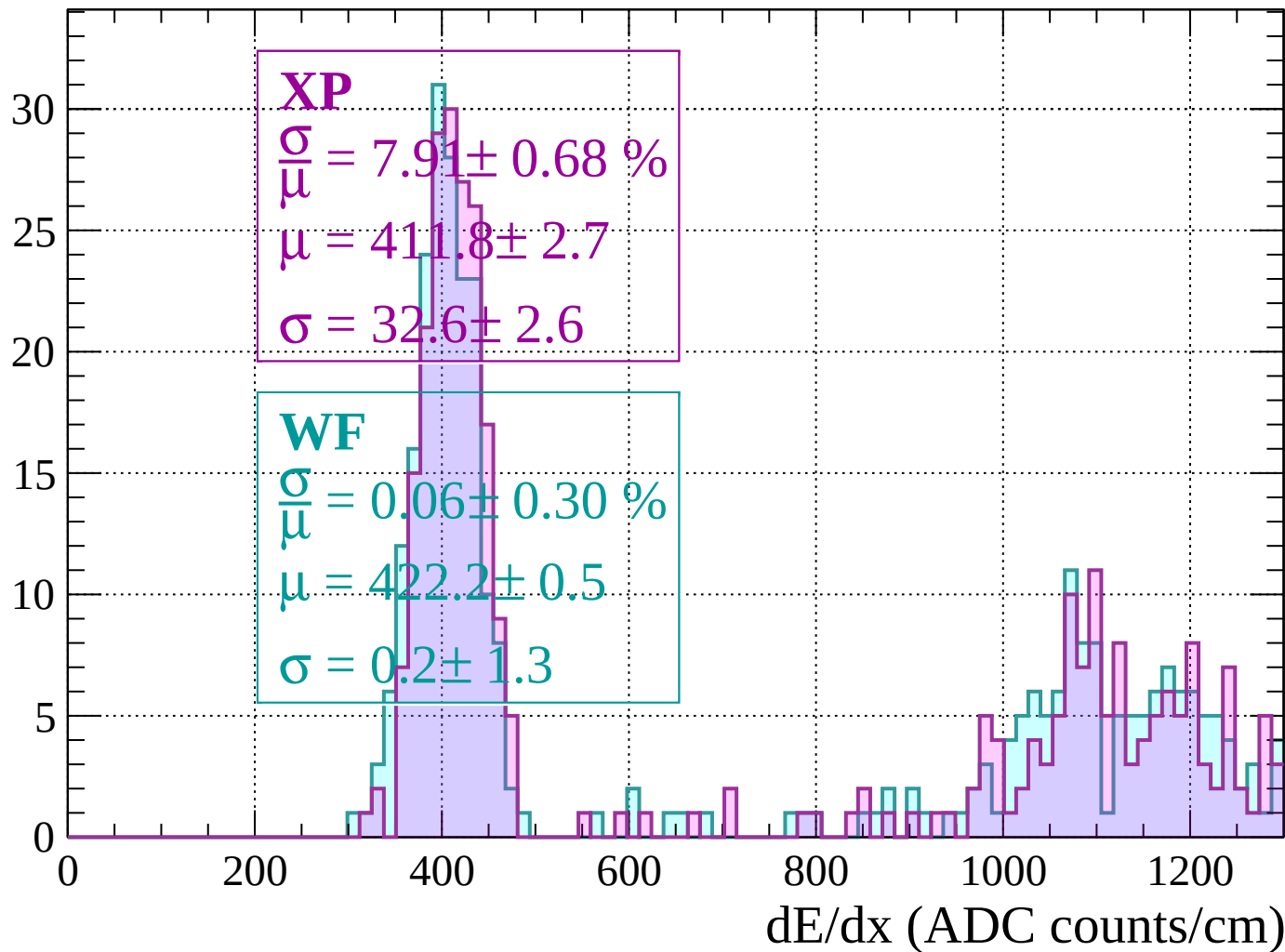
$$\mu = 437.1 \pm 3.0$$

$$\sigma = 1.4 \pm 25.6$$



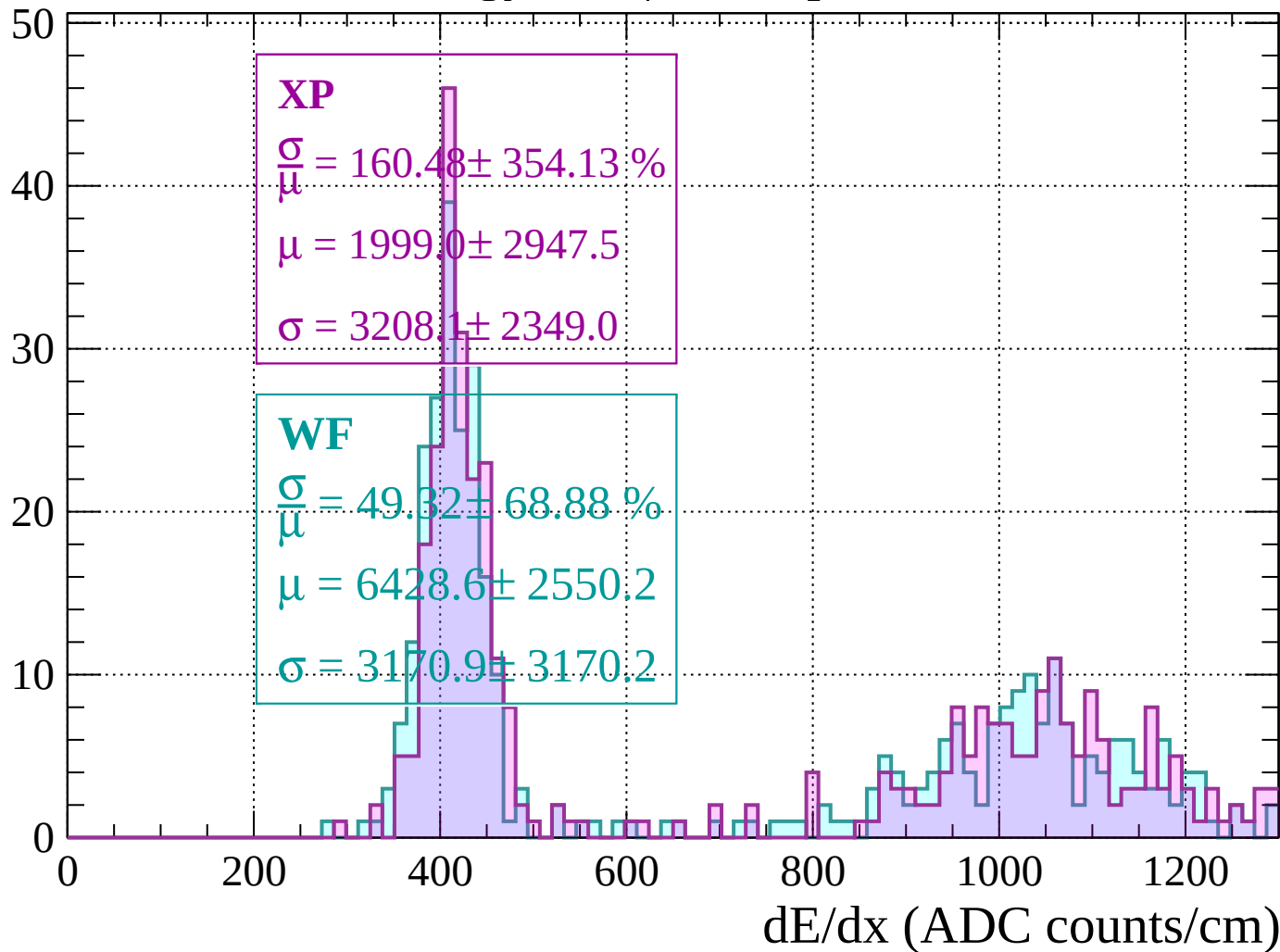
Energy loss |  $560 < p < 600$

Count



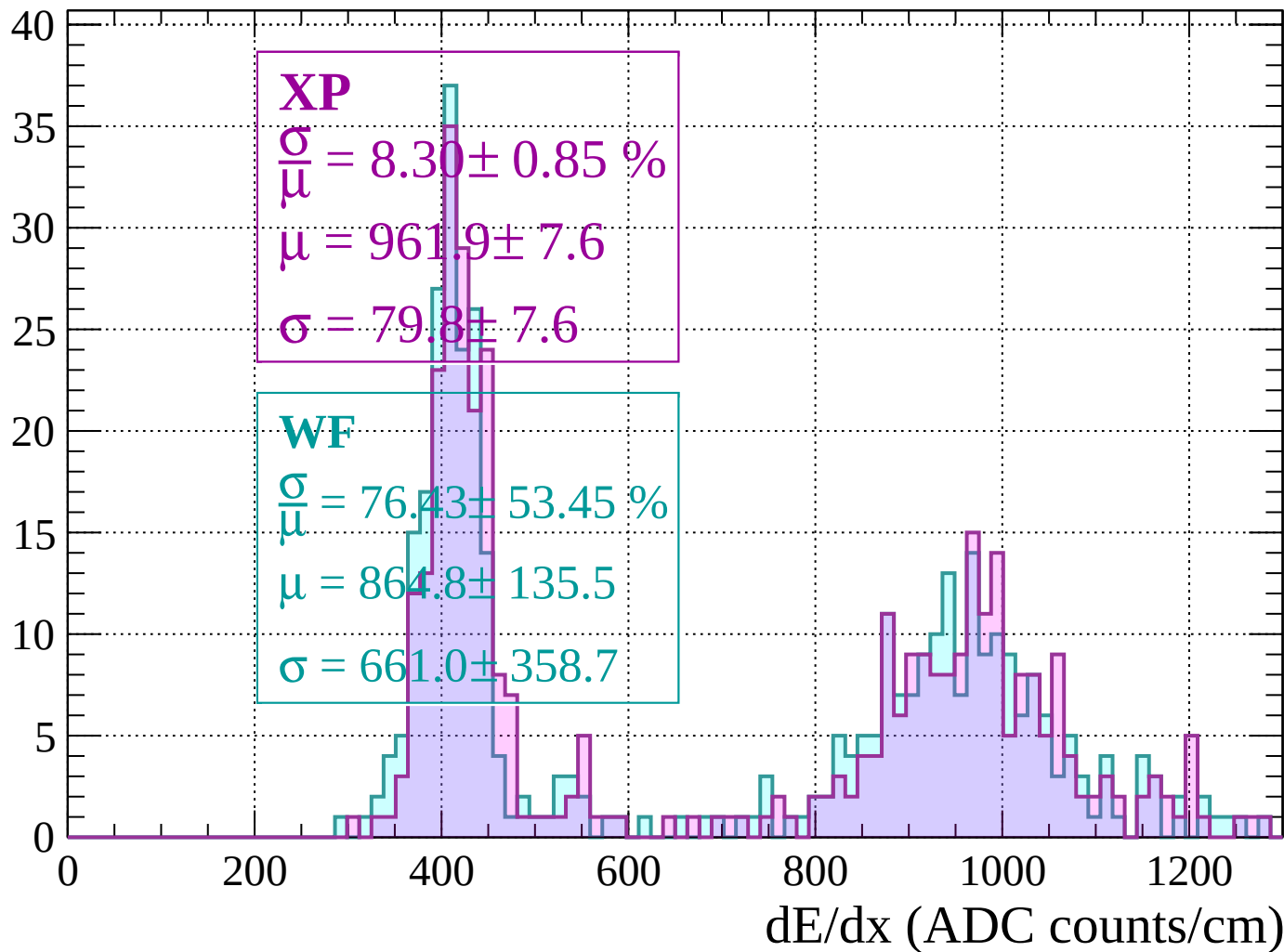
# Energy loss | $600 < p < 640$

Count



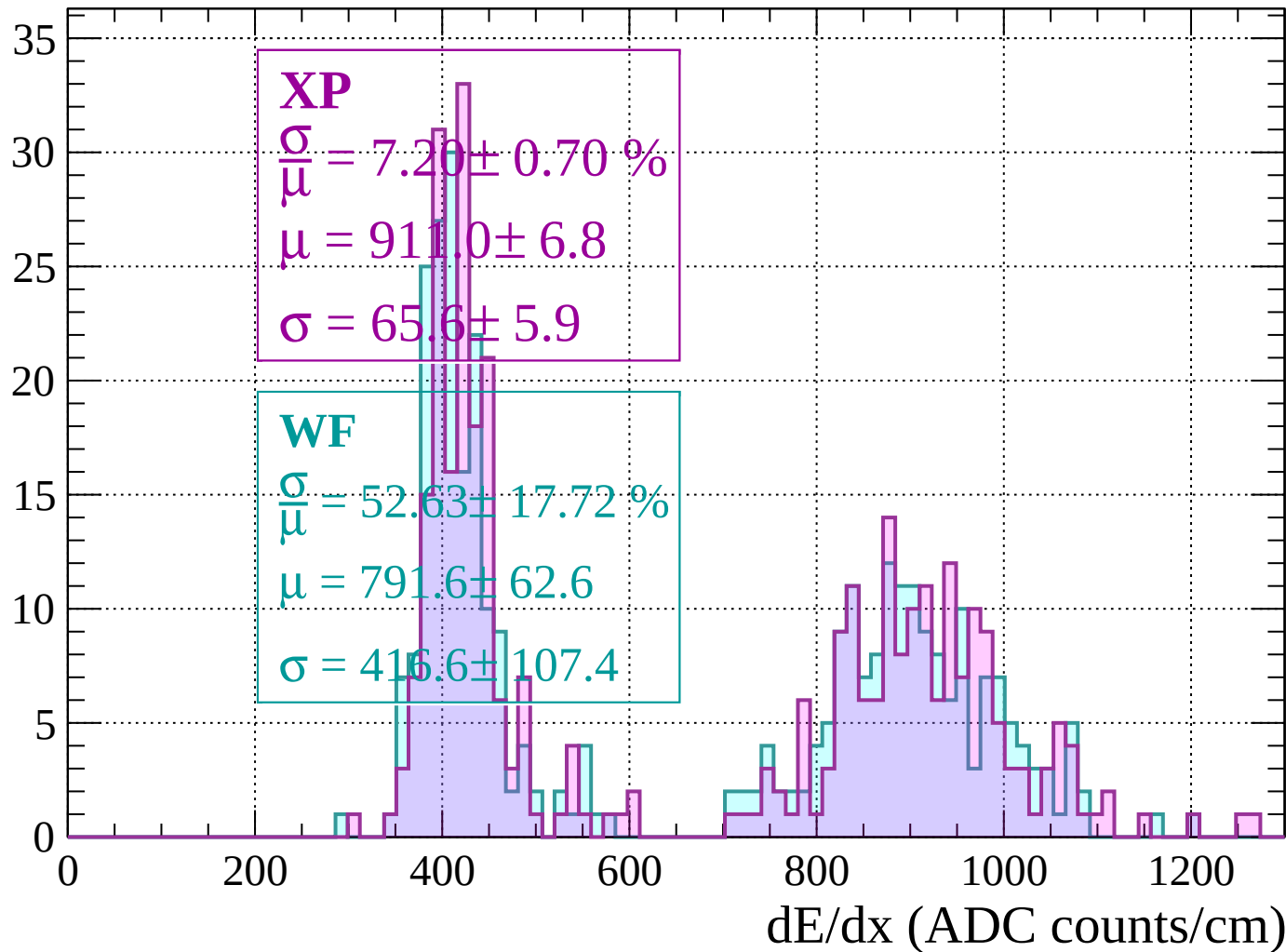
# Energy loss | $640 < p < 680$

Count



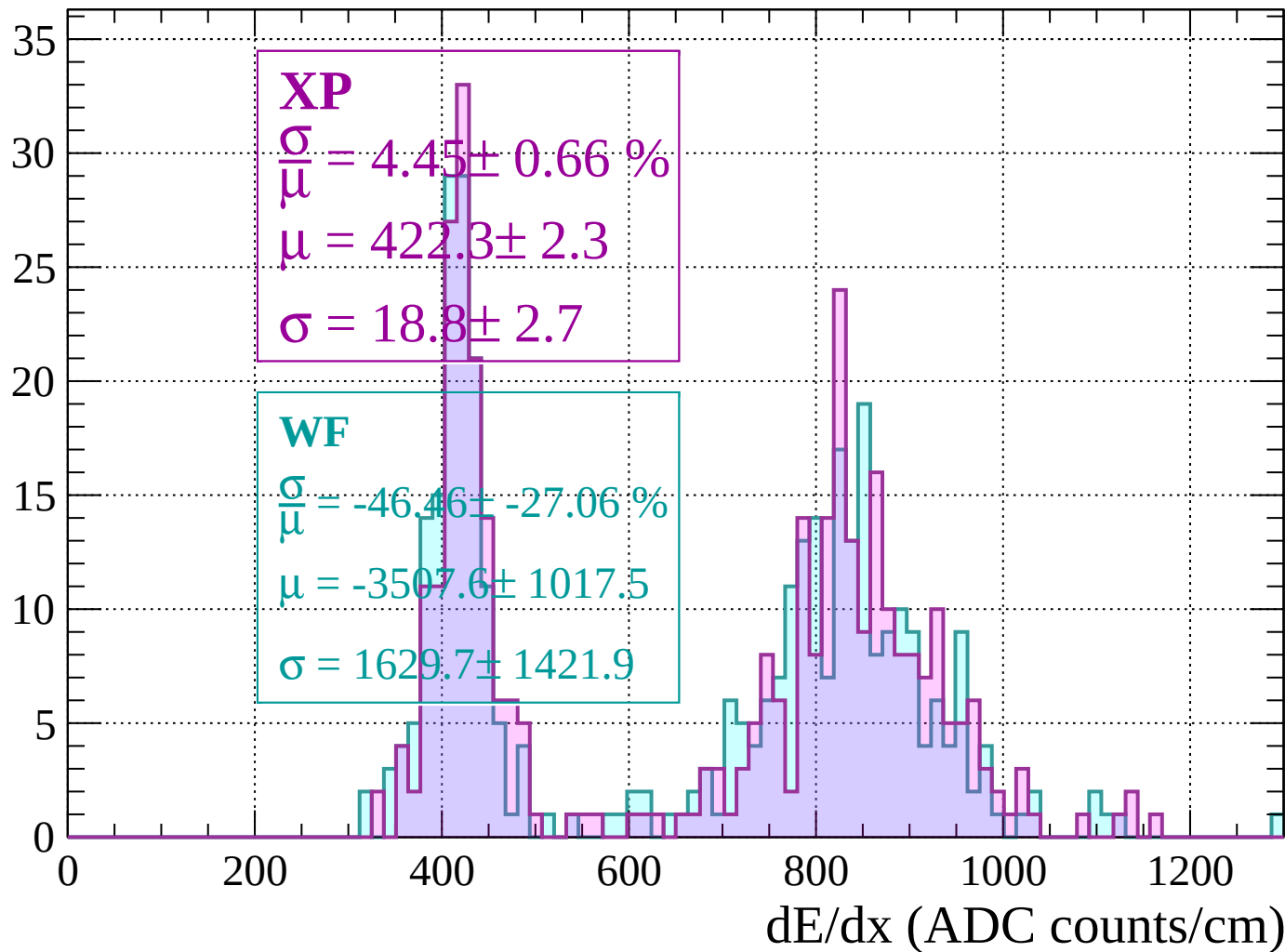
Energy loss |  $680 < p < 720$

Count



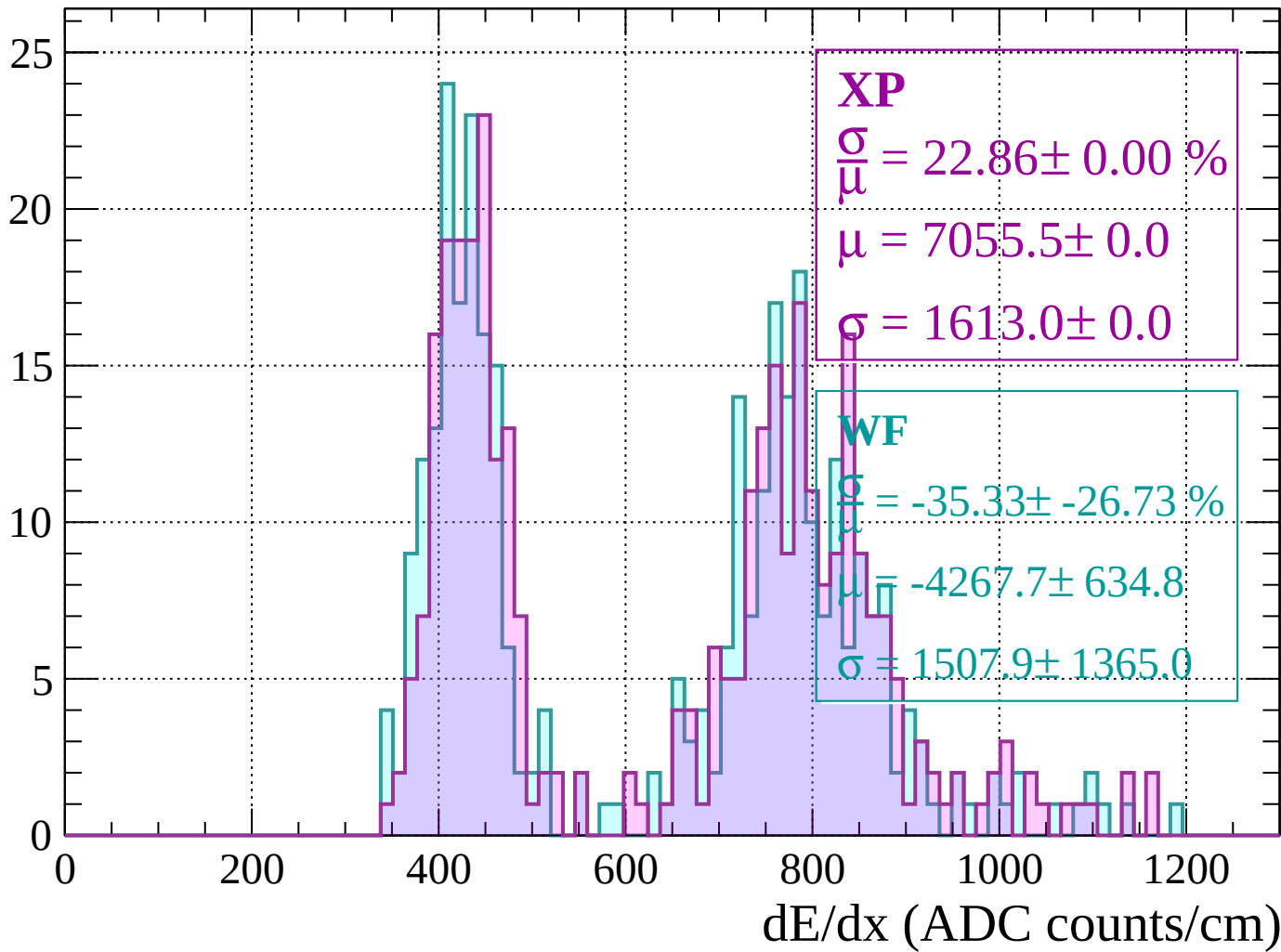
Energy loss |  $720 < p < 760$

Count



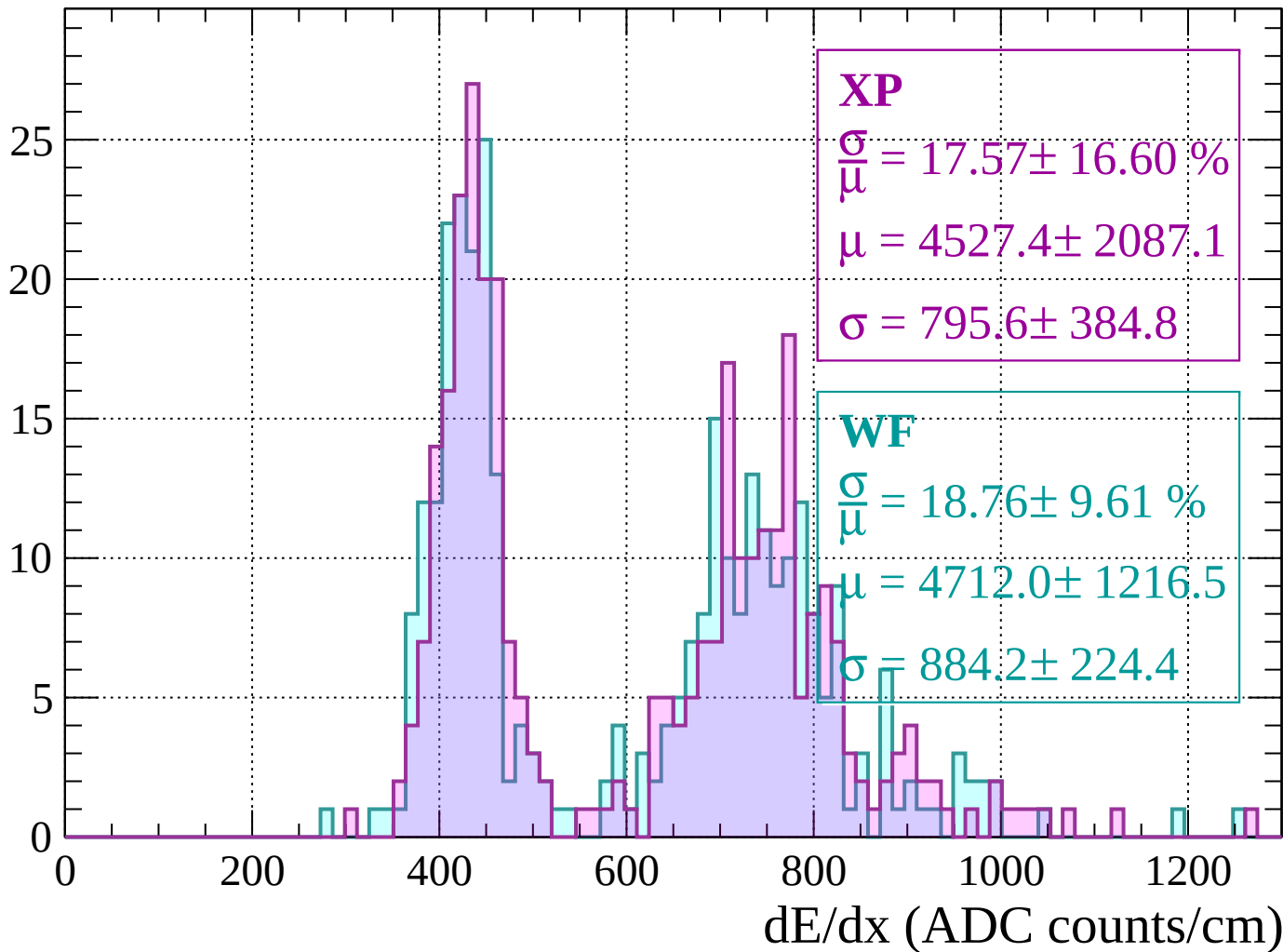
Energy loss |  $760 < p < 800$

Count



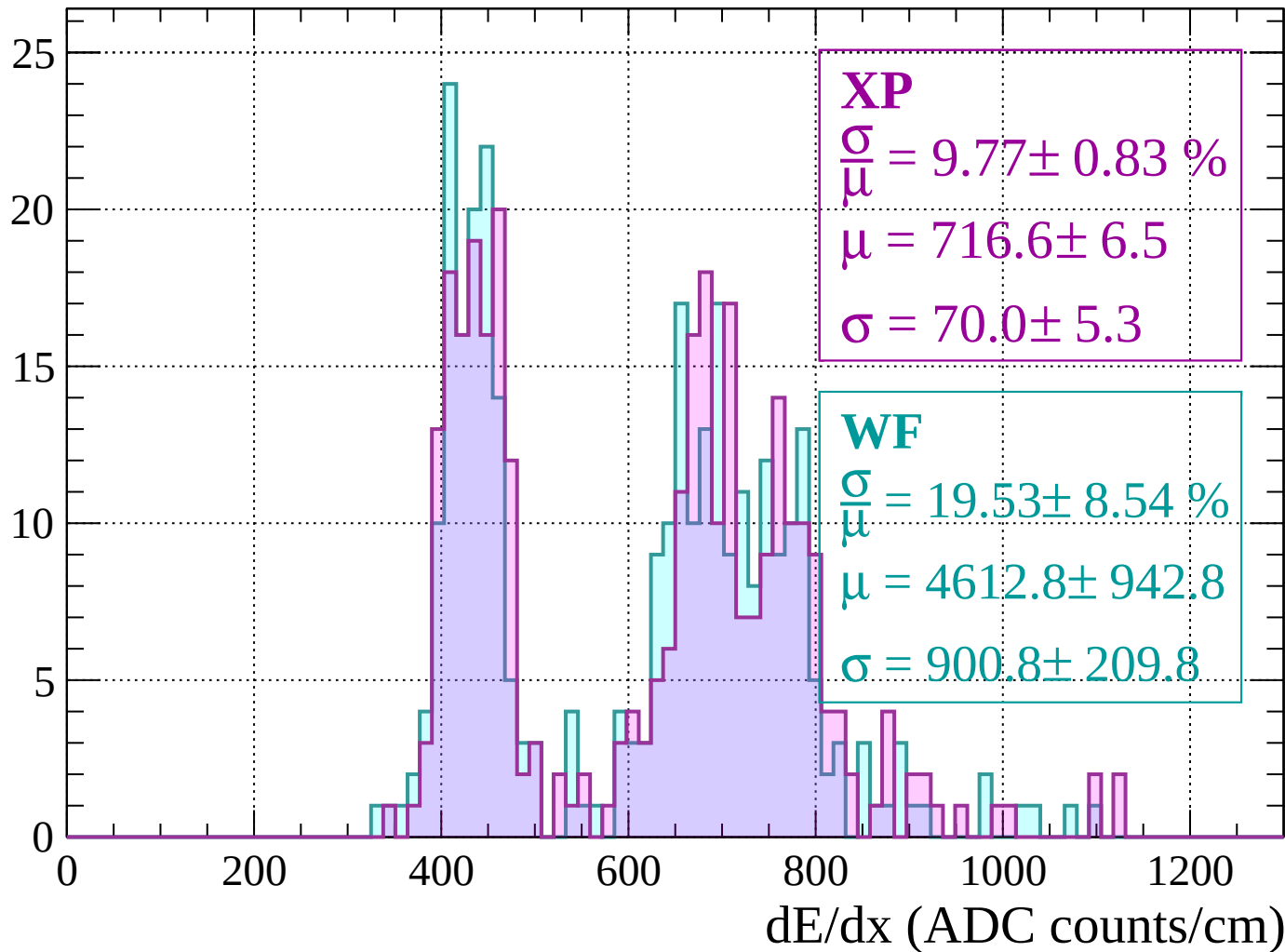
Energy loss |  $800 < p < 840$

Count



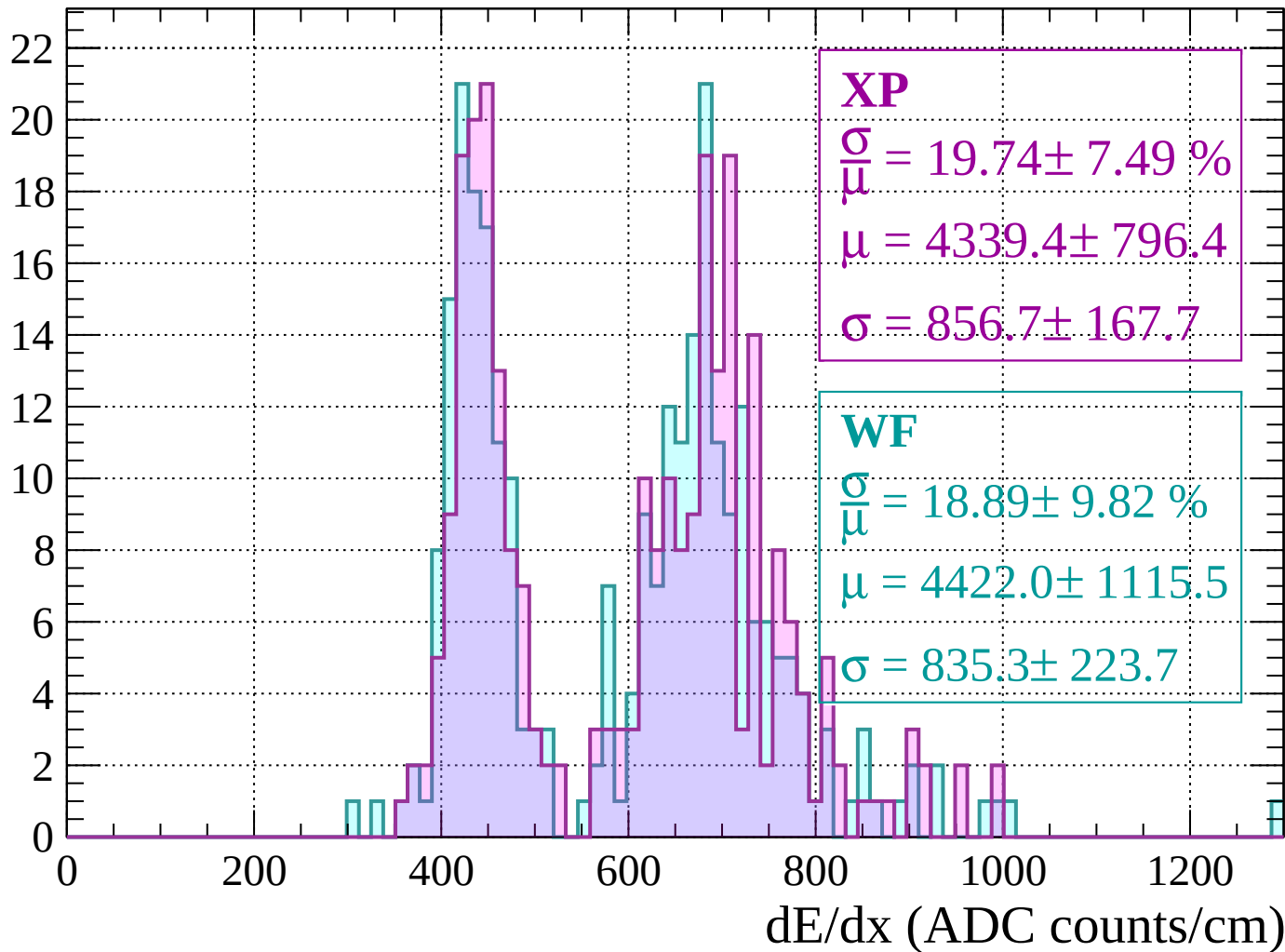
Energy loss |  $840 < p < 880$

Count



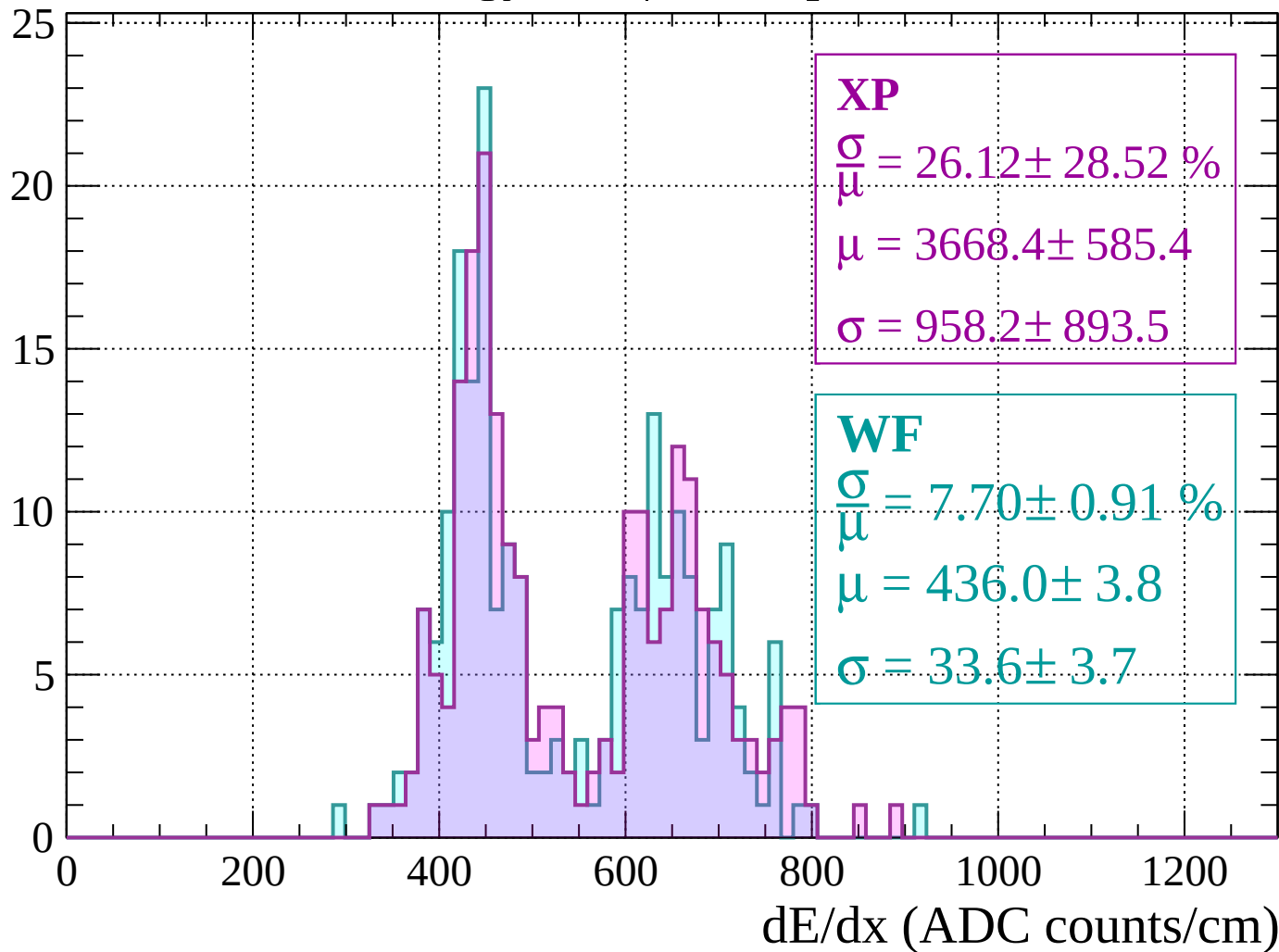
Energy loss |  $880 < p < 920$

Count



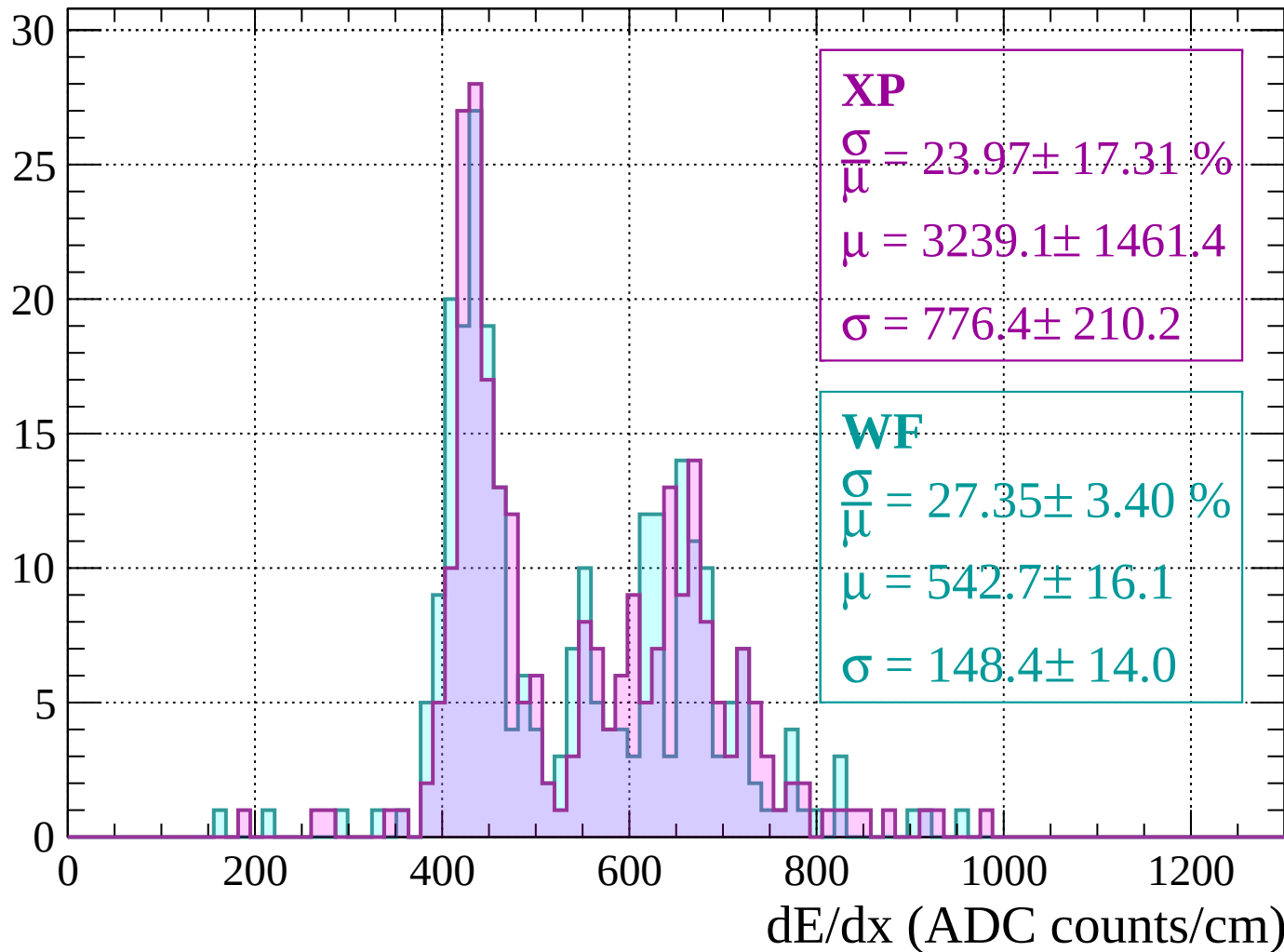
# Energy loss | $920 < p < 960$

Count



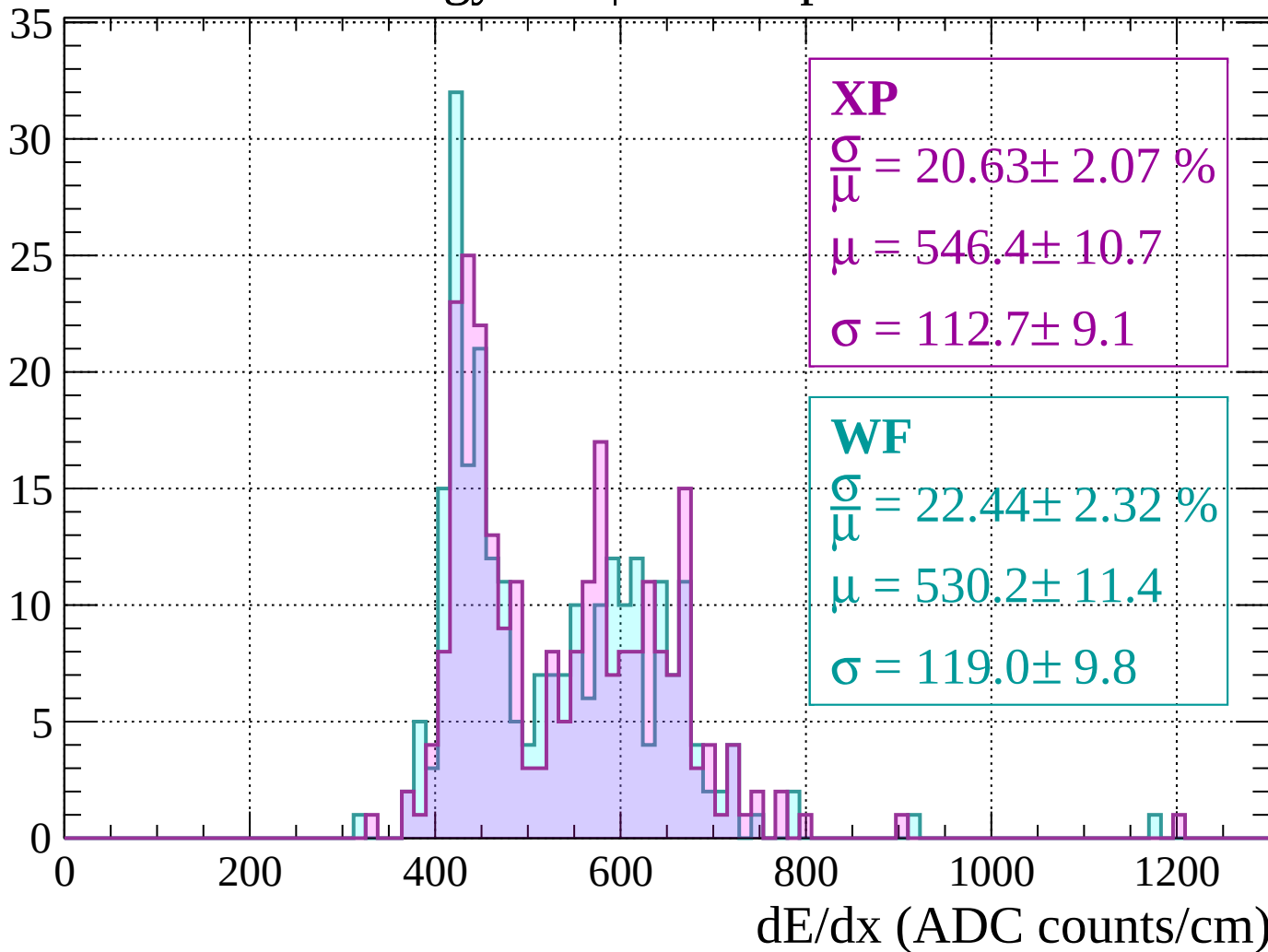
# Energy loss | $960 < p < 1000$

Count



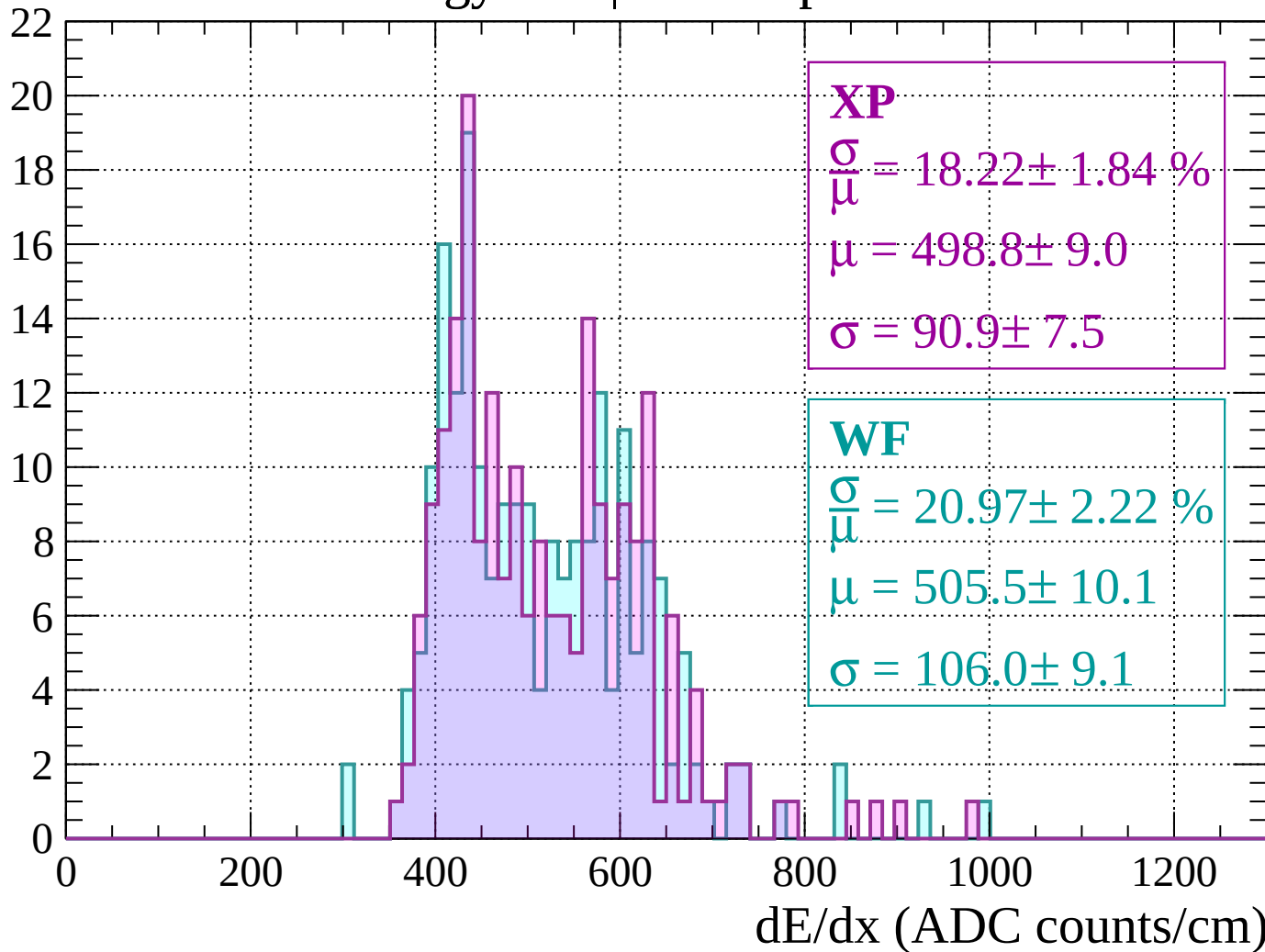
Energy loss |  $1000 < p < 1040$

Count



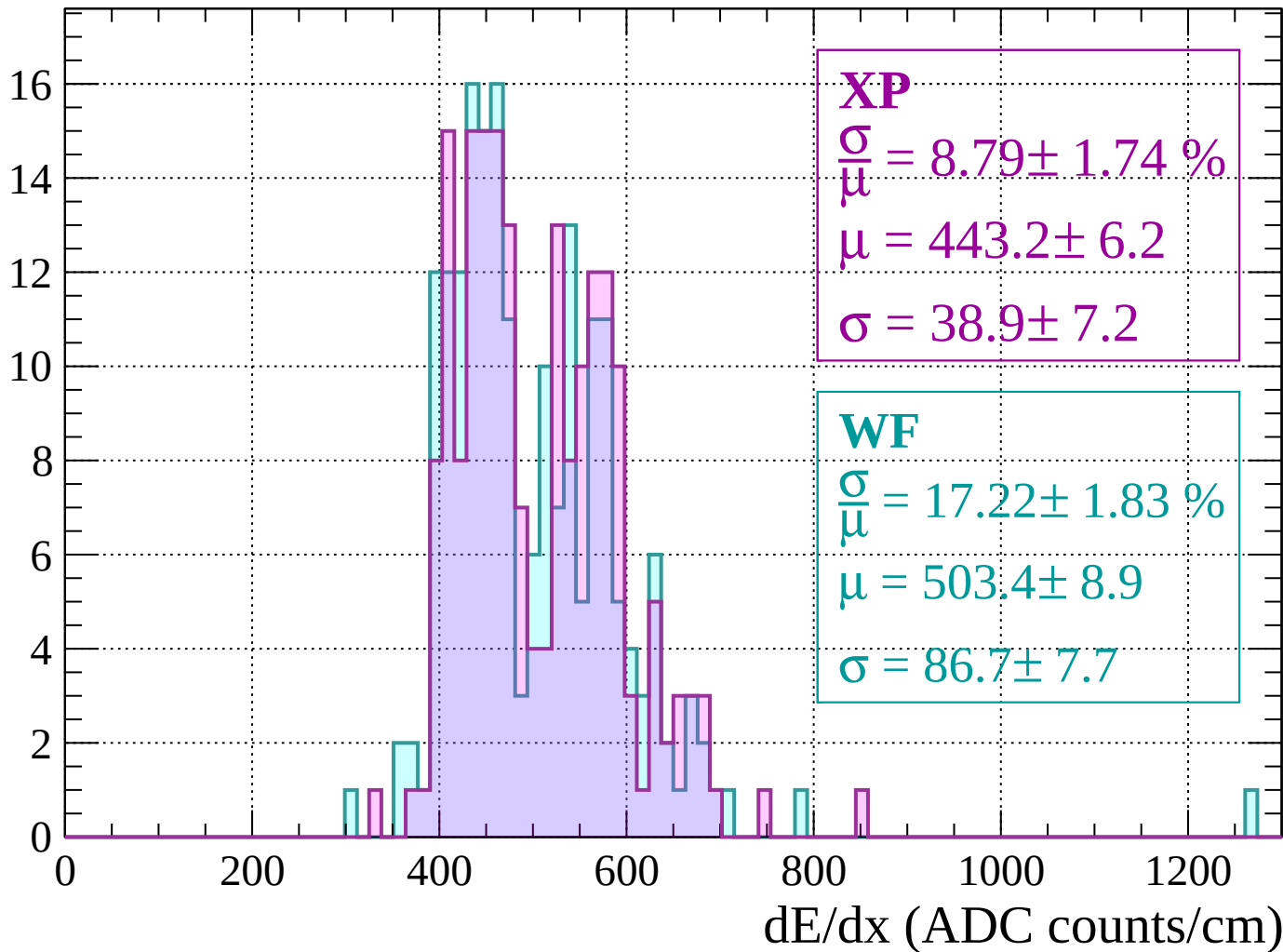
Energy loss |  $1040 < p < 1080$

Count



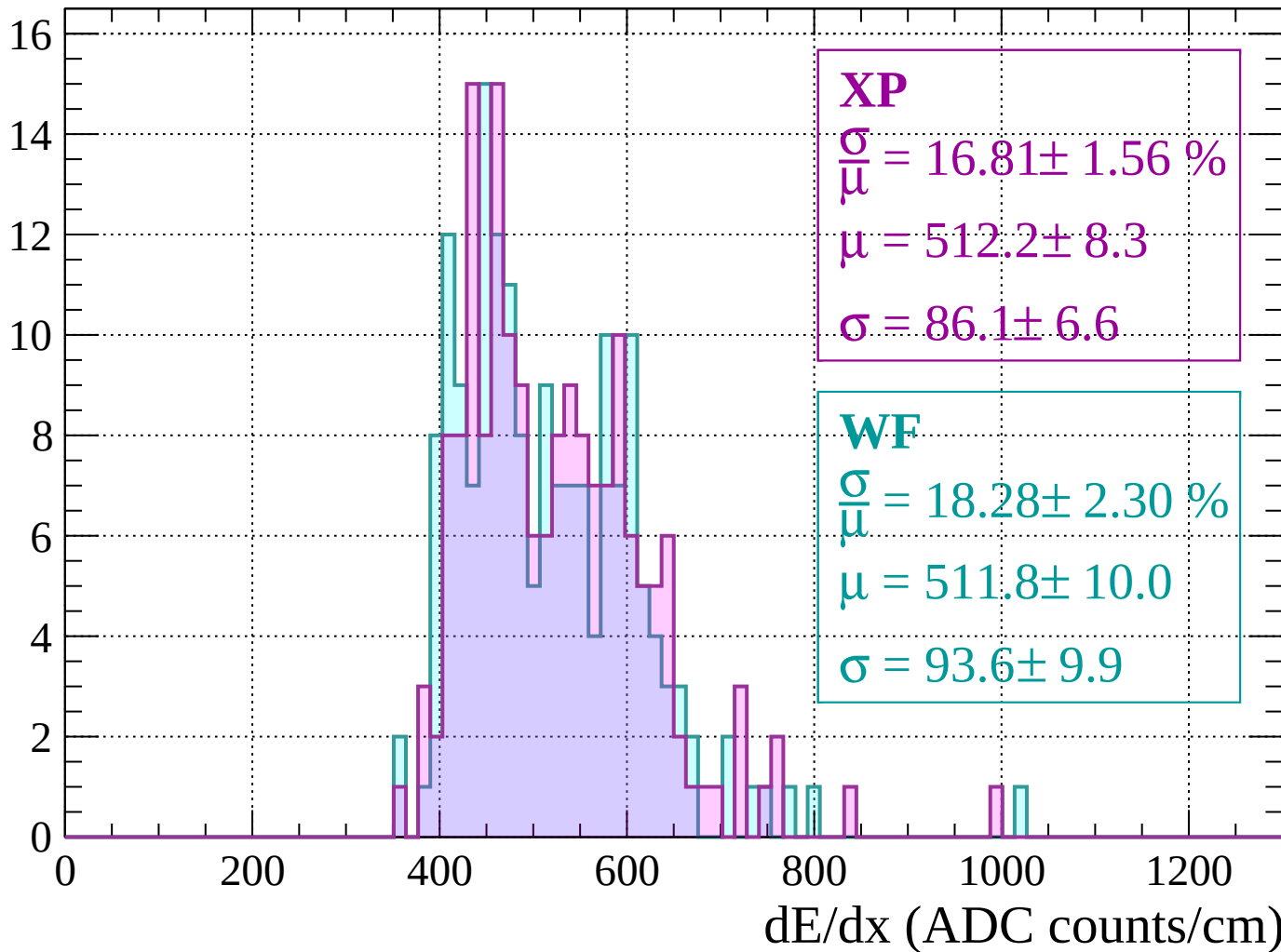
Energy loss |  $1080 < p < 1120$

Count



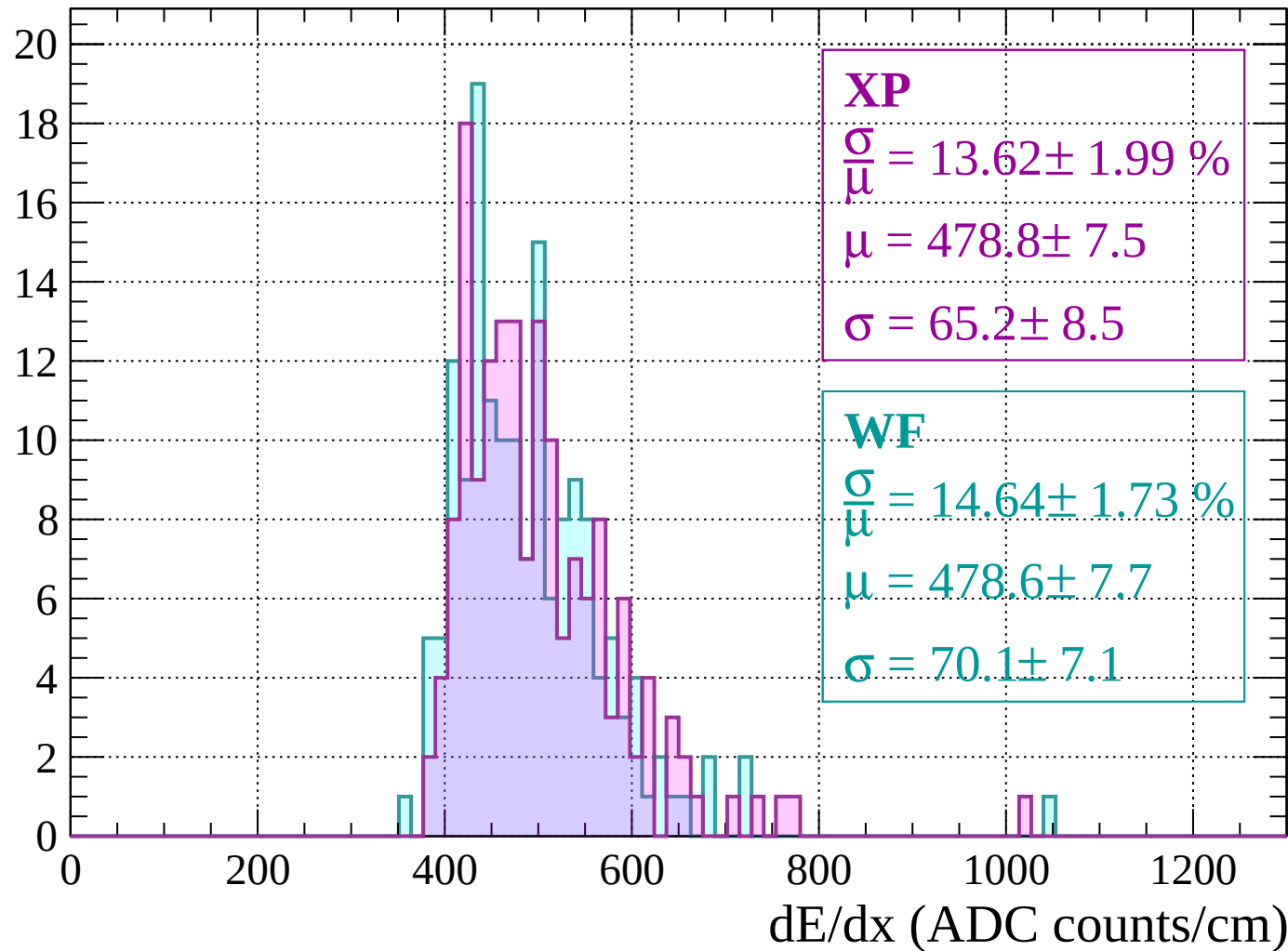
Energy loss |  $1120 < p < 1160$

Count



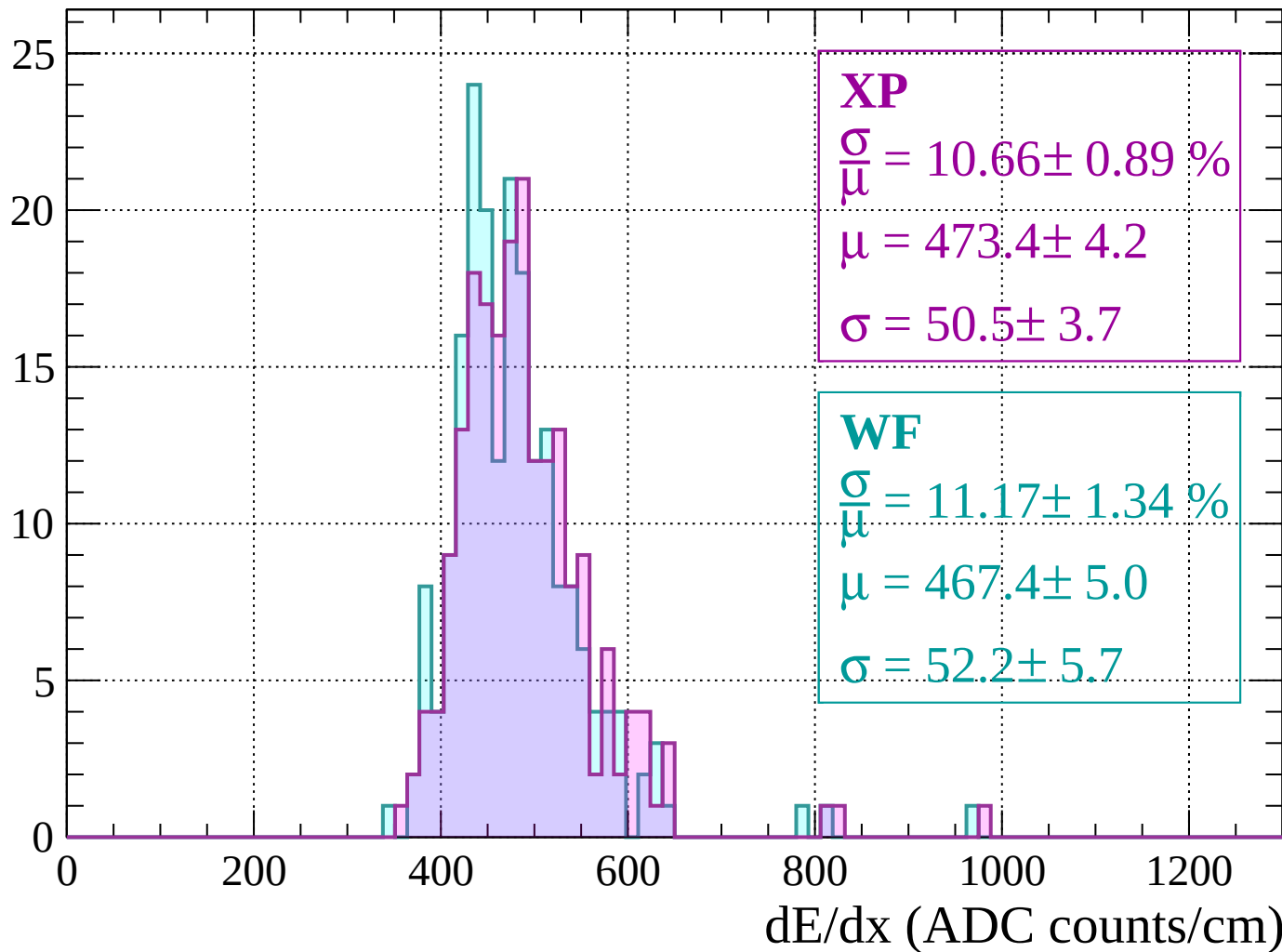
Energy loss |  $1160 < p < 1200$

Count



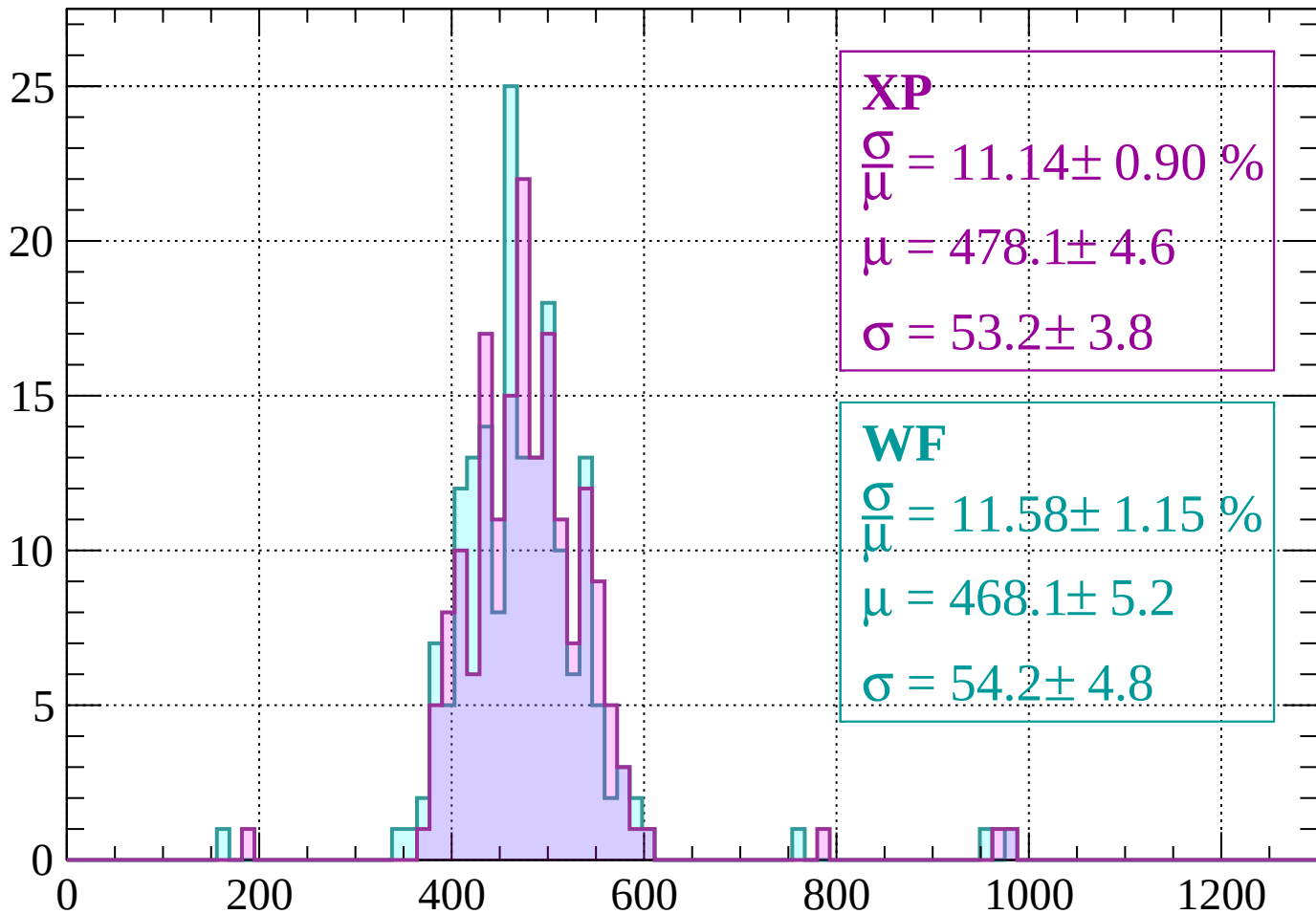
Energy loss |  $1200 < p < 1240$

Count



Energy loss |  $1240 < p < 1280$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.14 \pm 0.90 \%$$

$$\mu = 478.1 \pm 4.6$$

$$\sigma = 53.2 \pm 3.8$$

**WF**

$$\frac{\sigma}{\mu} = 11.58 \pm 1.15 \%$$

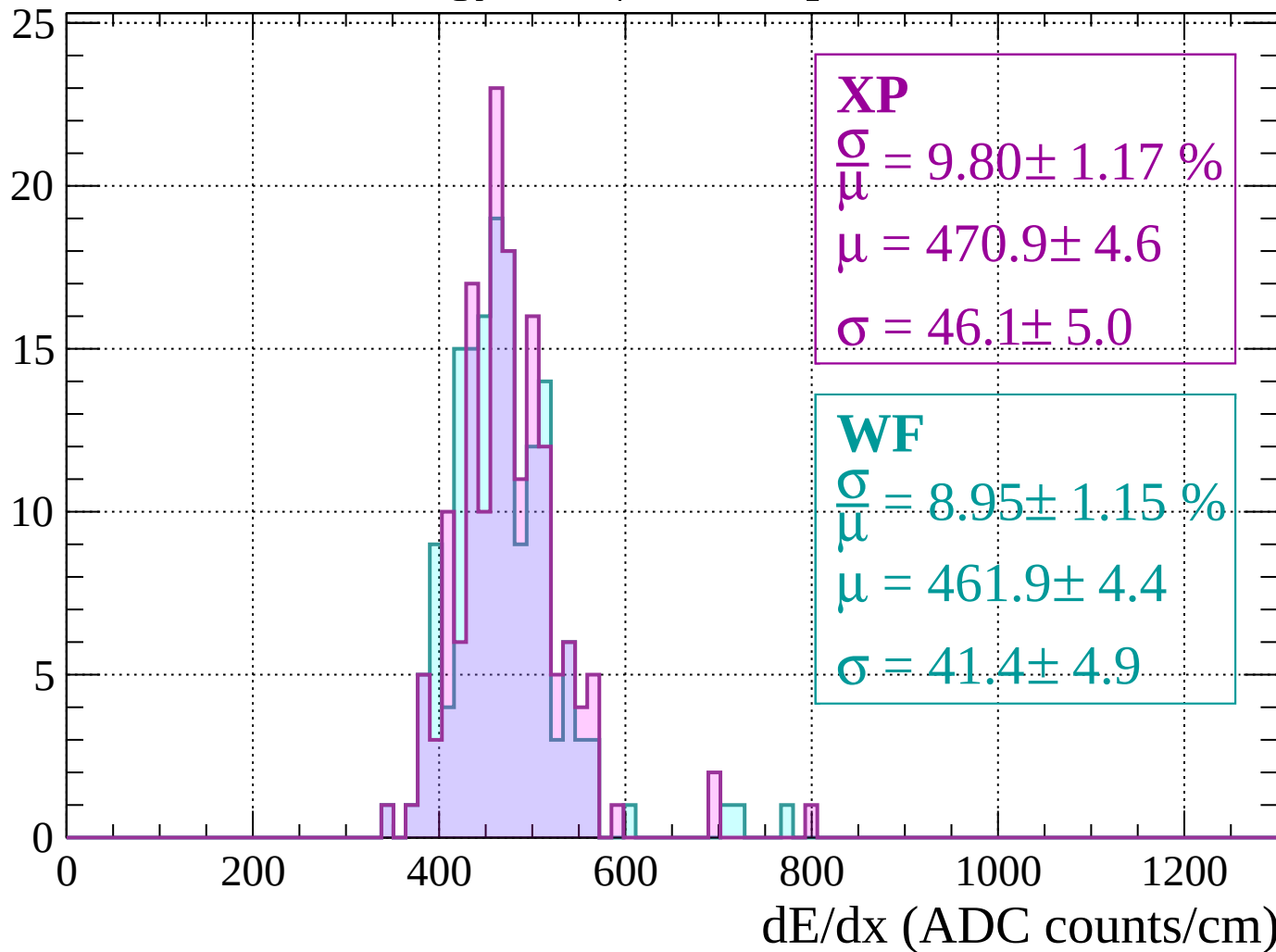
$$\mu = 468.1 \pm 5.2$$

$$\sigma = 54.2 \pm 4.8$$

$dE/dx$  (ADC counts/cm)

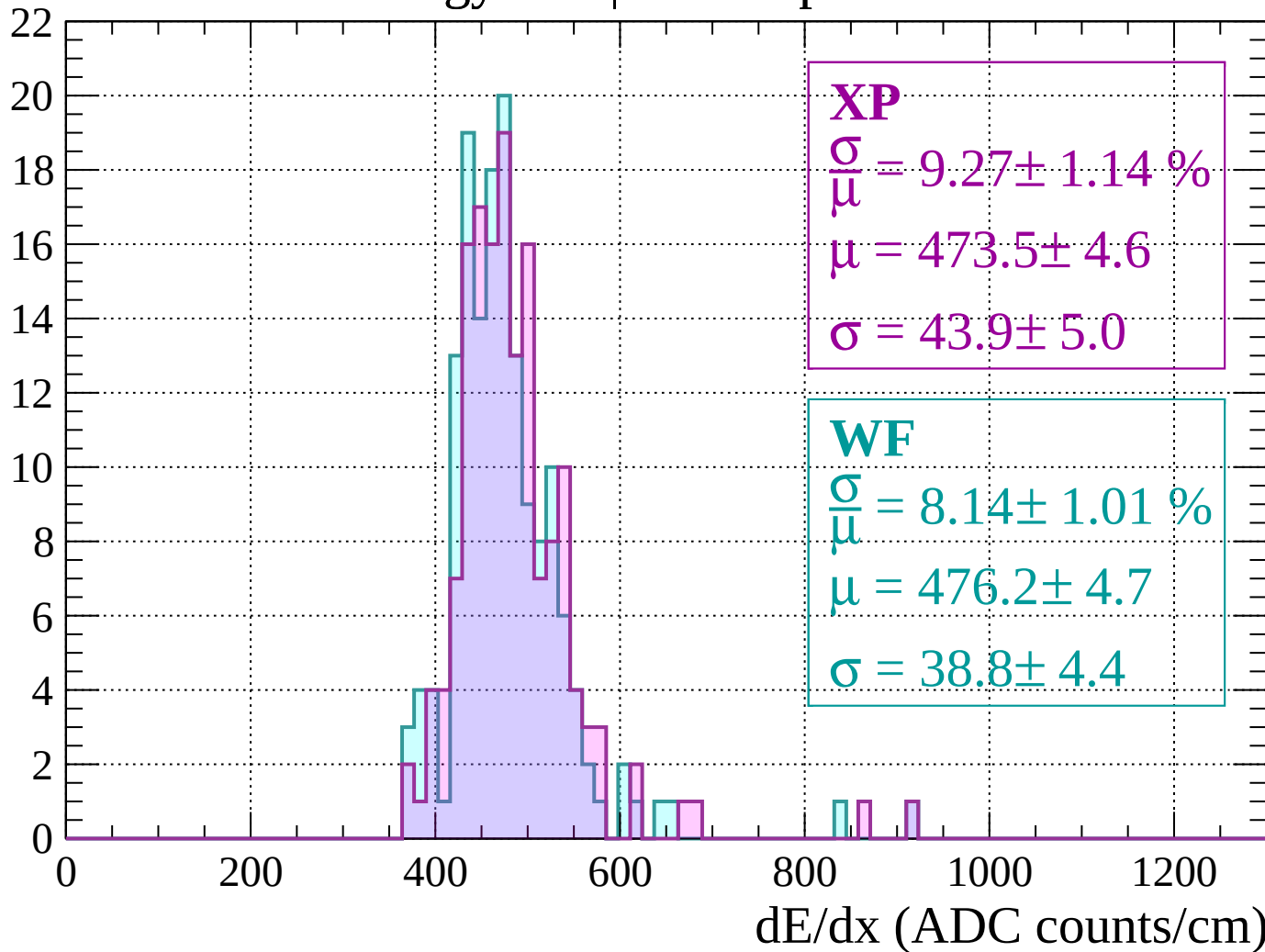
Energy loss |  $1280 < p < 1320$

Count



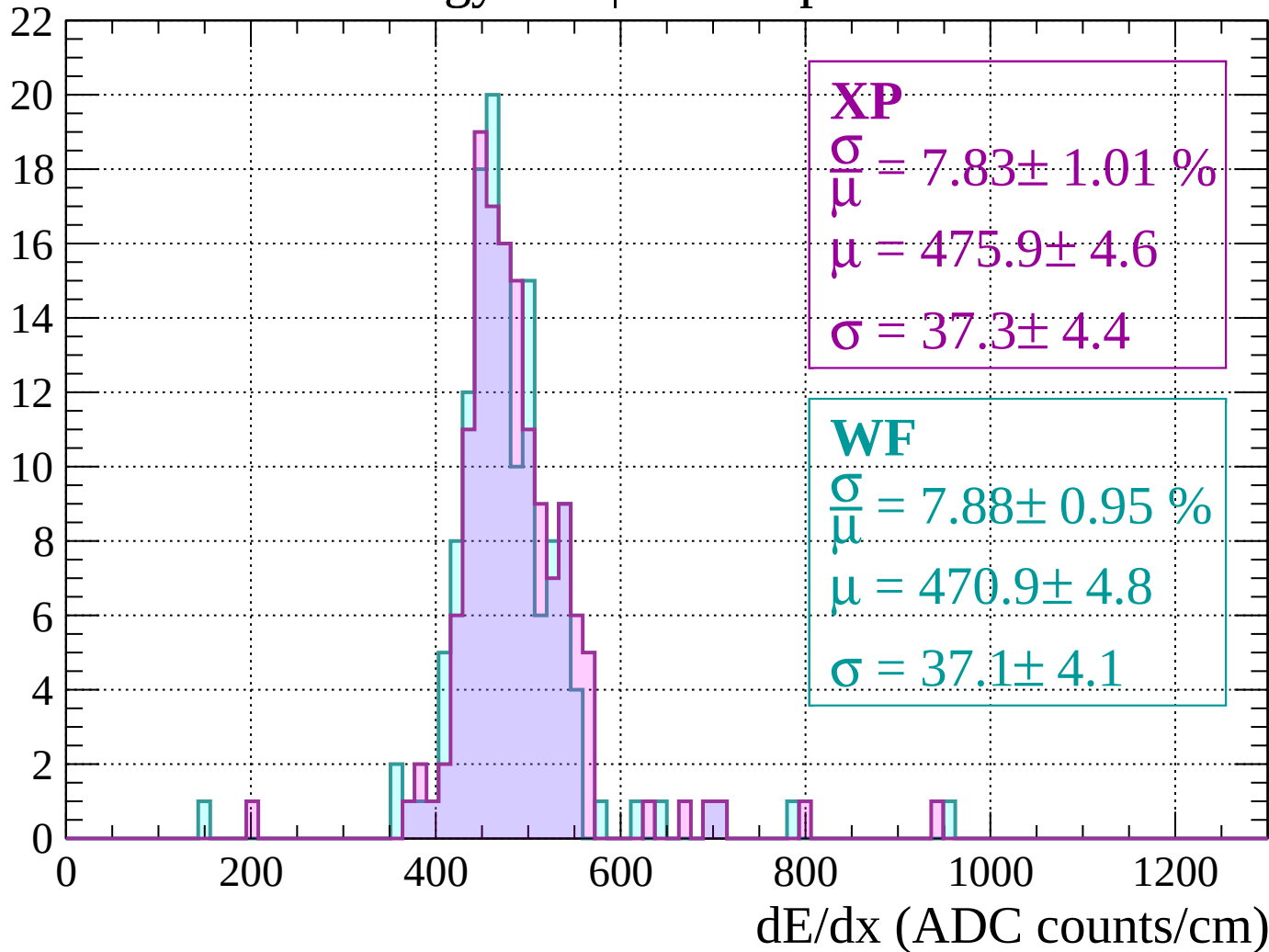
Energy loss |  $1320 < p < 1360$

Count



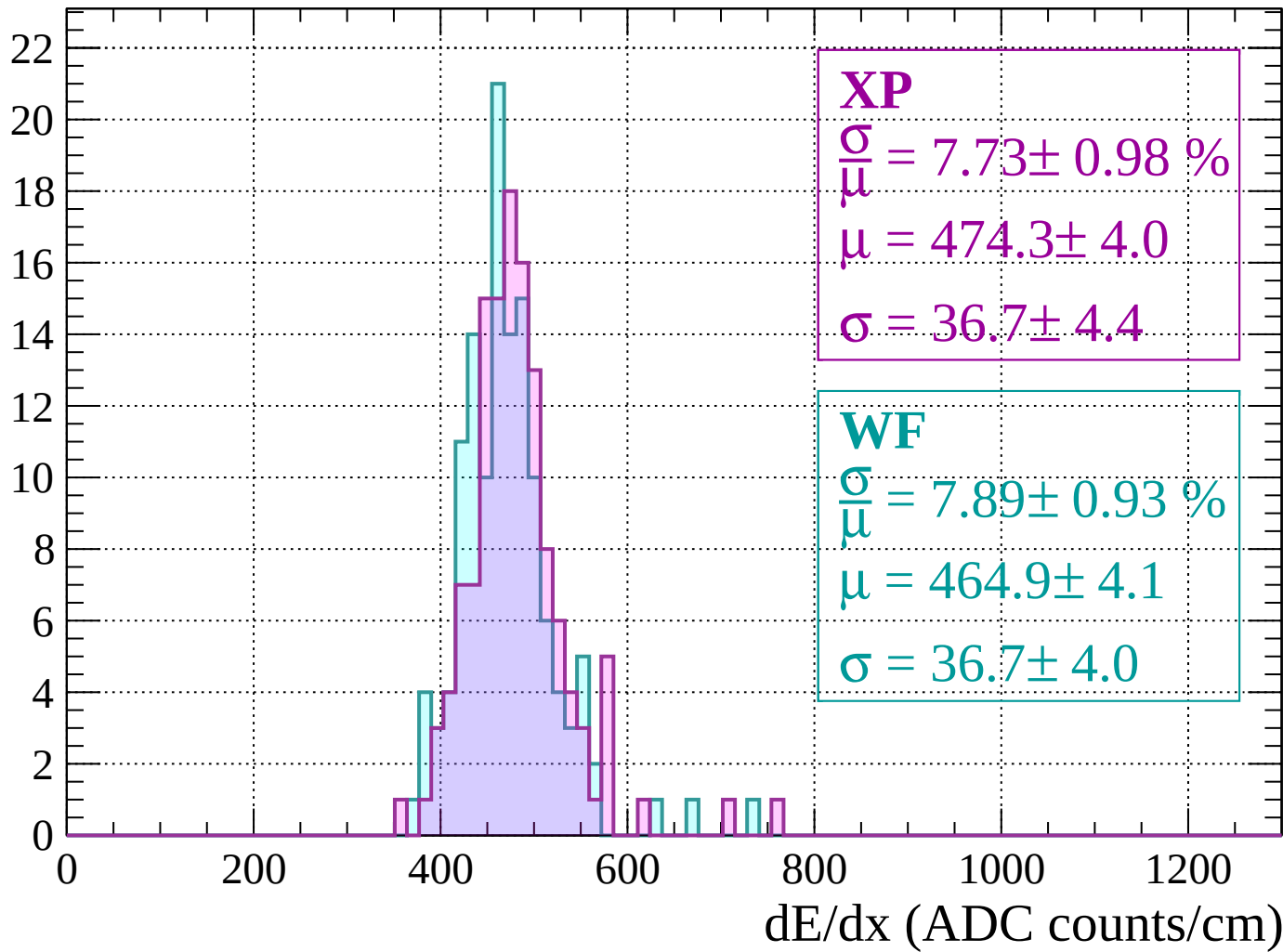
Energy loss |  $1360 < p < 1400$

Count



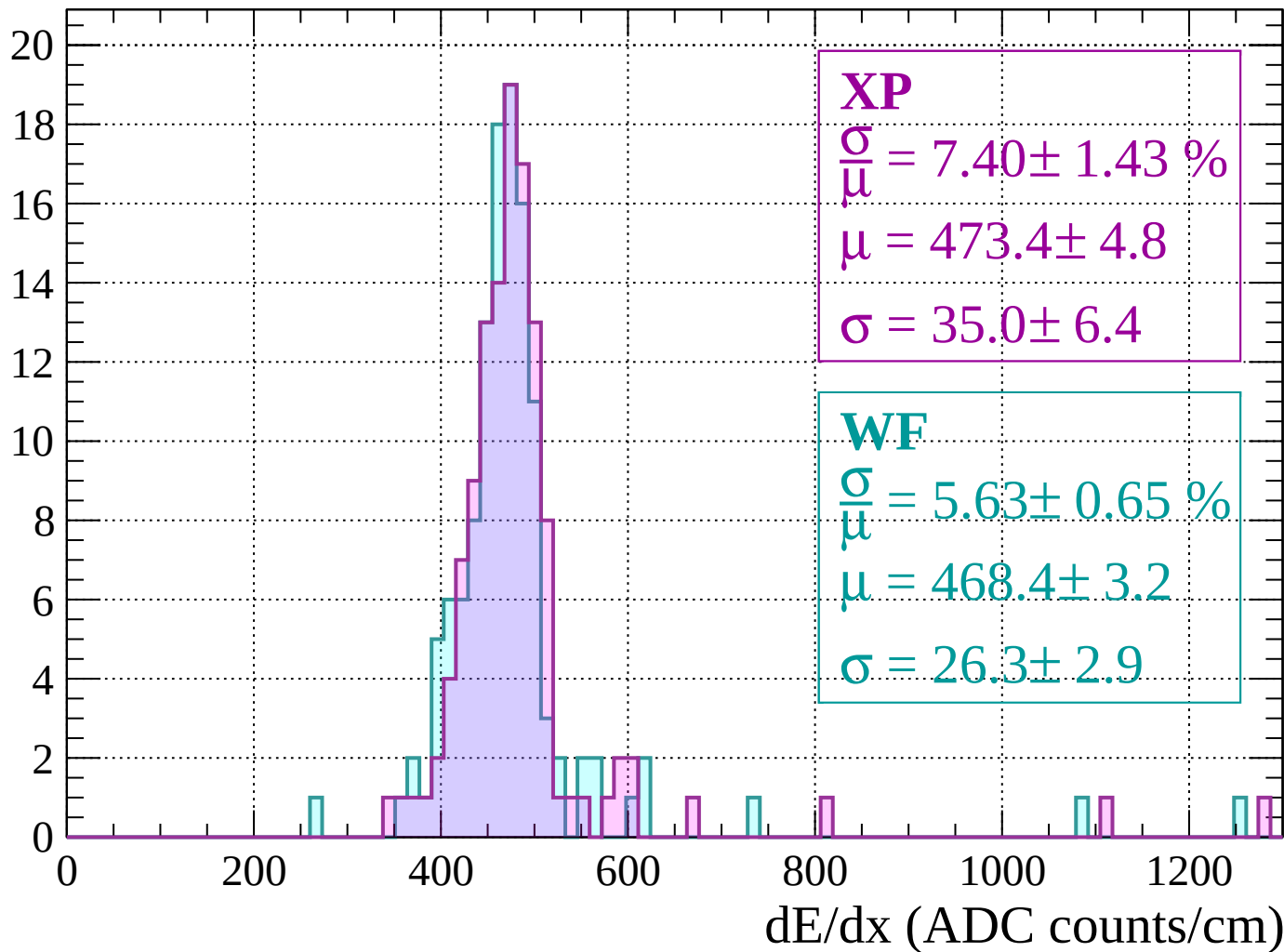
Energy loss |  $1400 < p < 1440$

Count



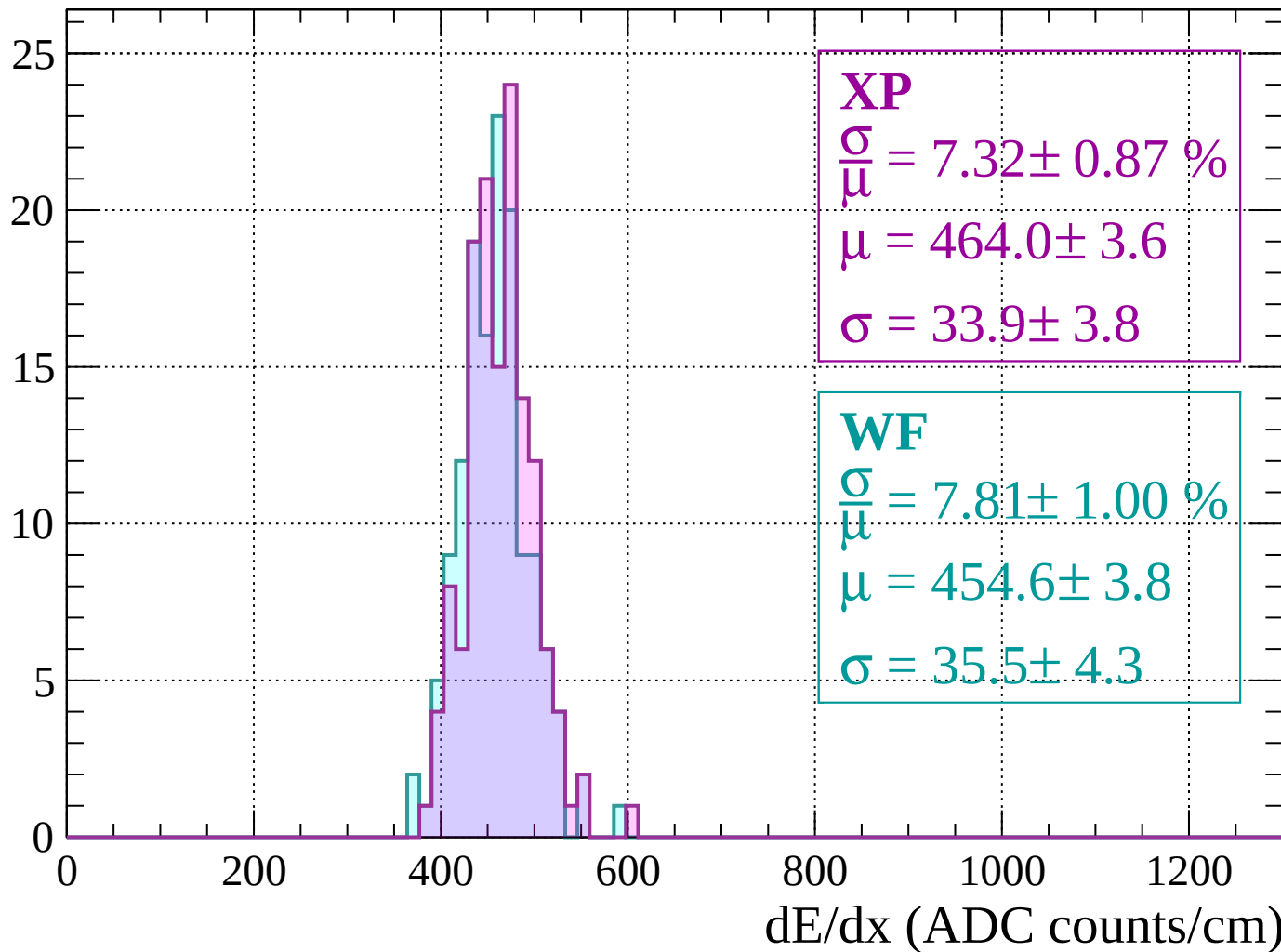
Energy loss |  $1440 < p < 1480$

Count



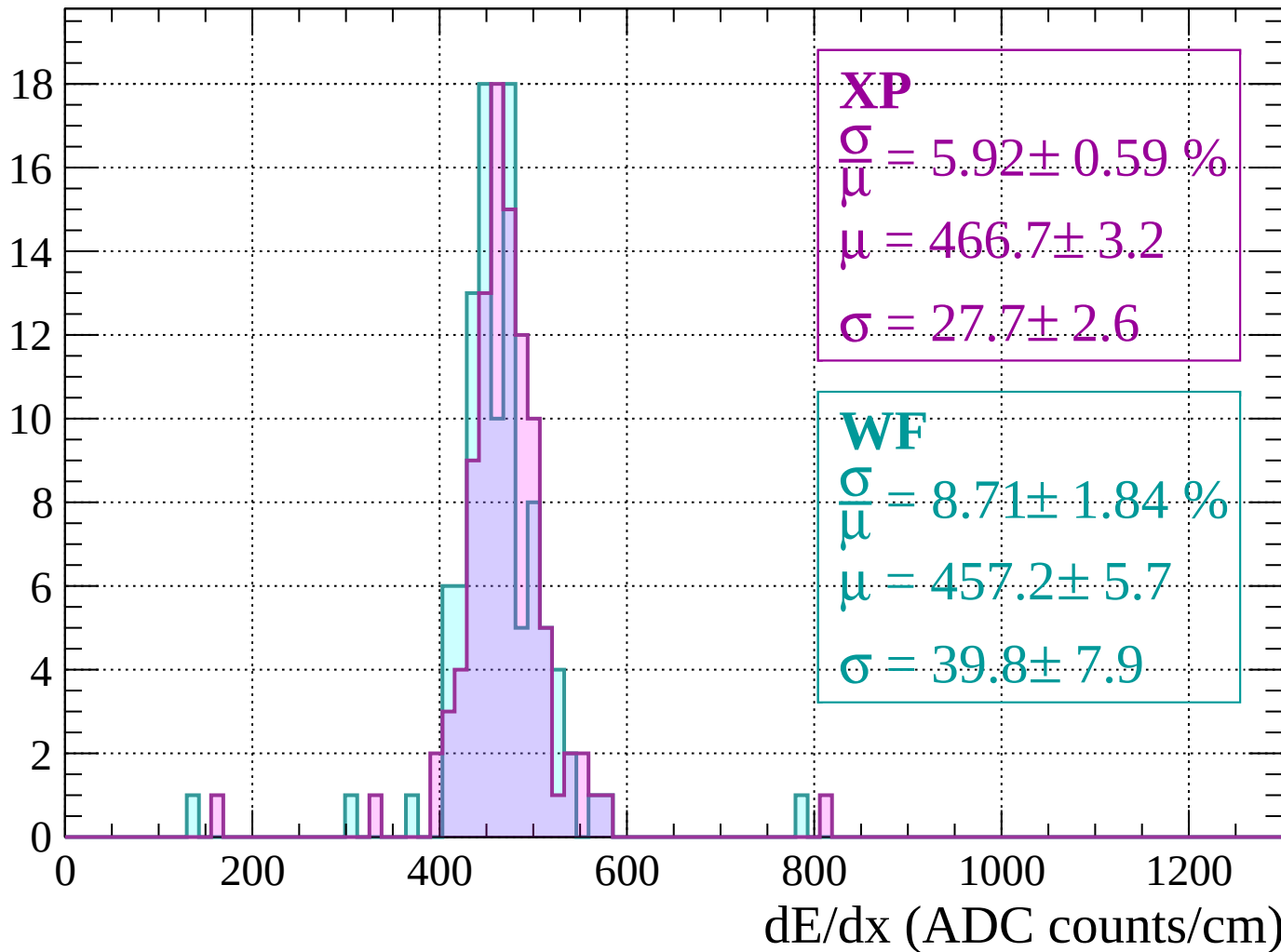
Energy loss |  $1480 < p < 1520$

Count



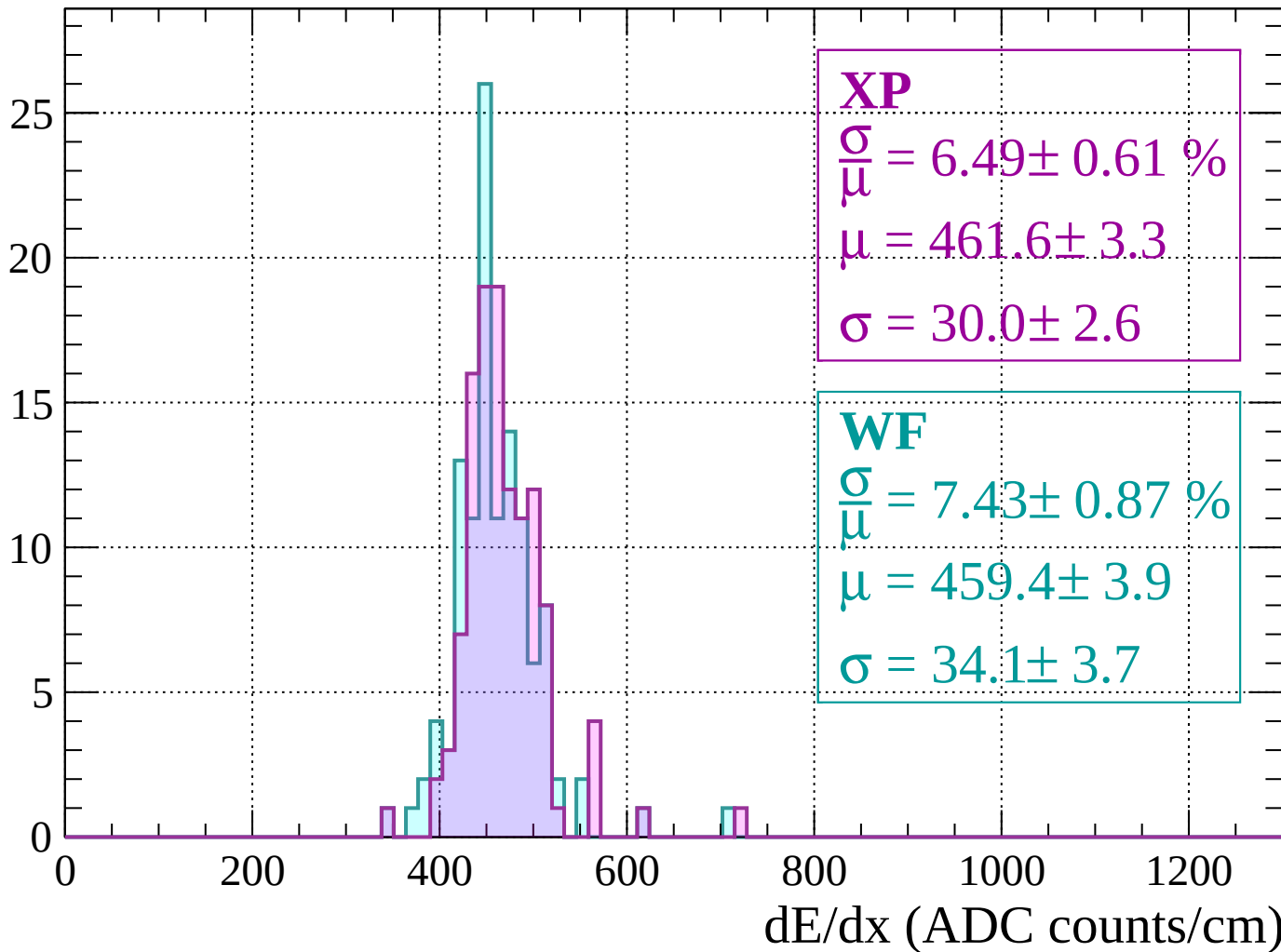
Energy loss |  $1520 < p < 1560$

Count



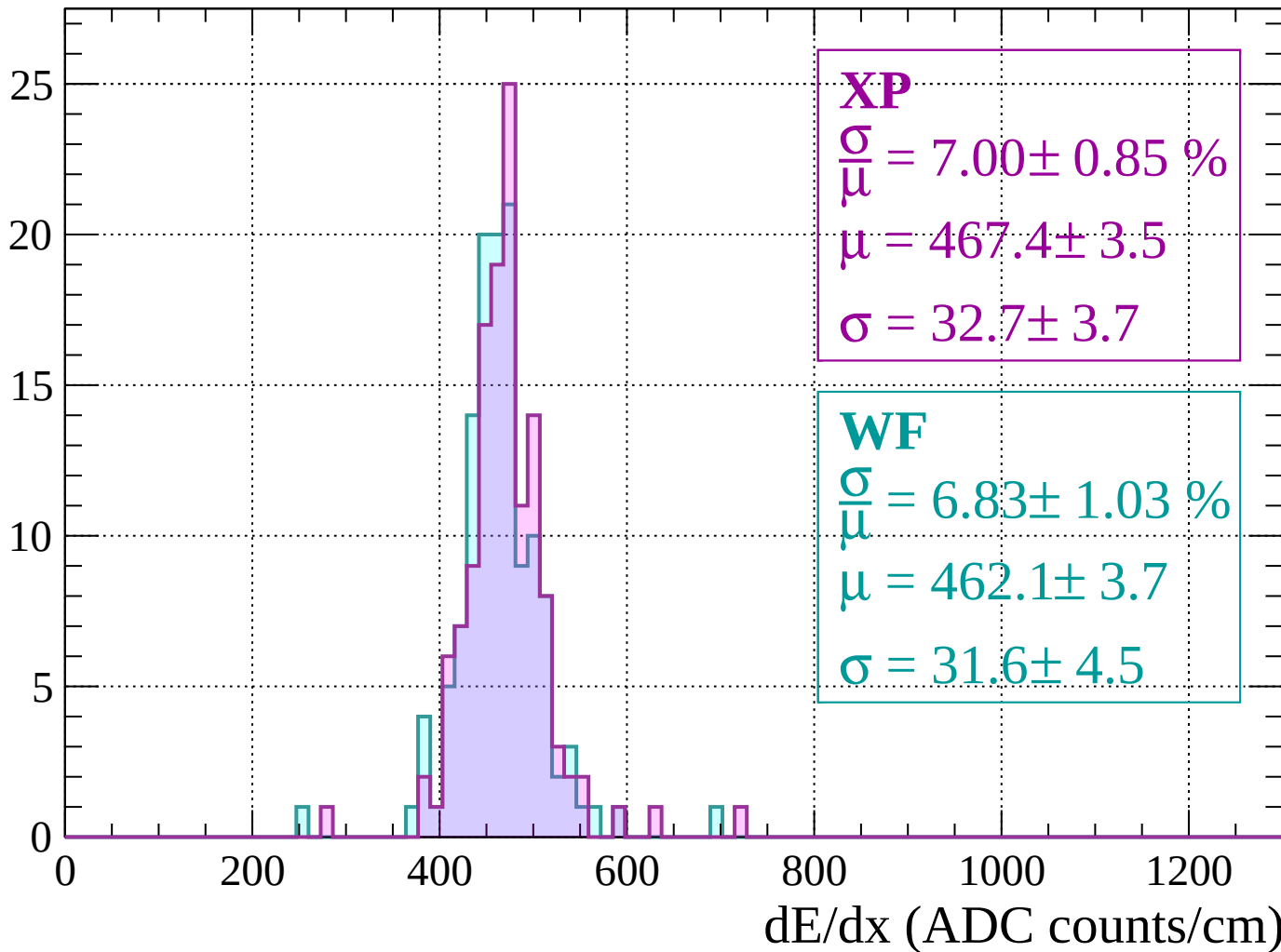
Energy loss |  $1560 < p < 1600$

Count



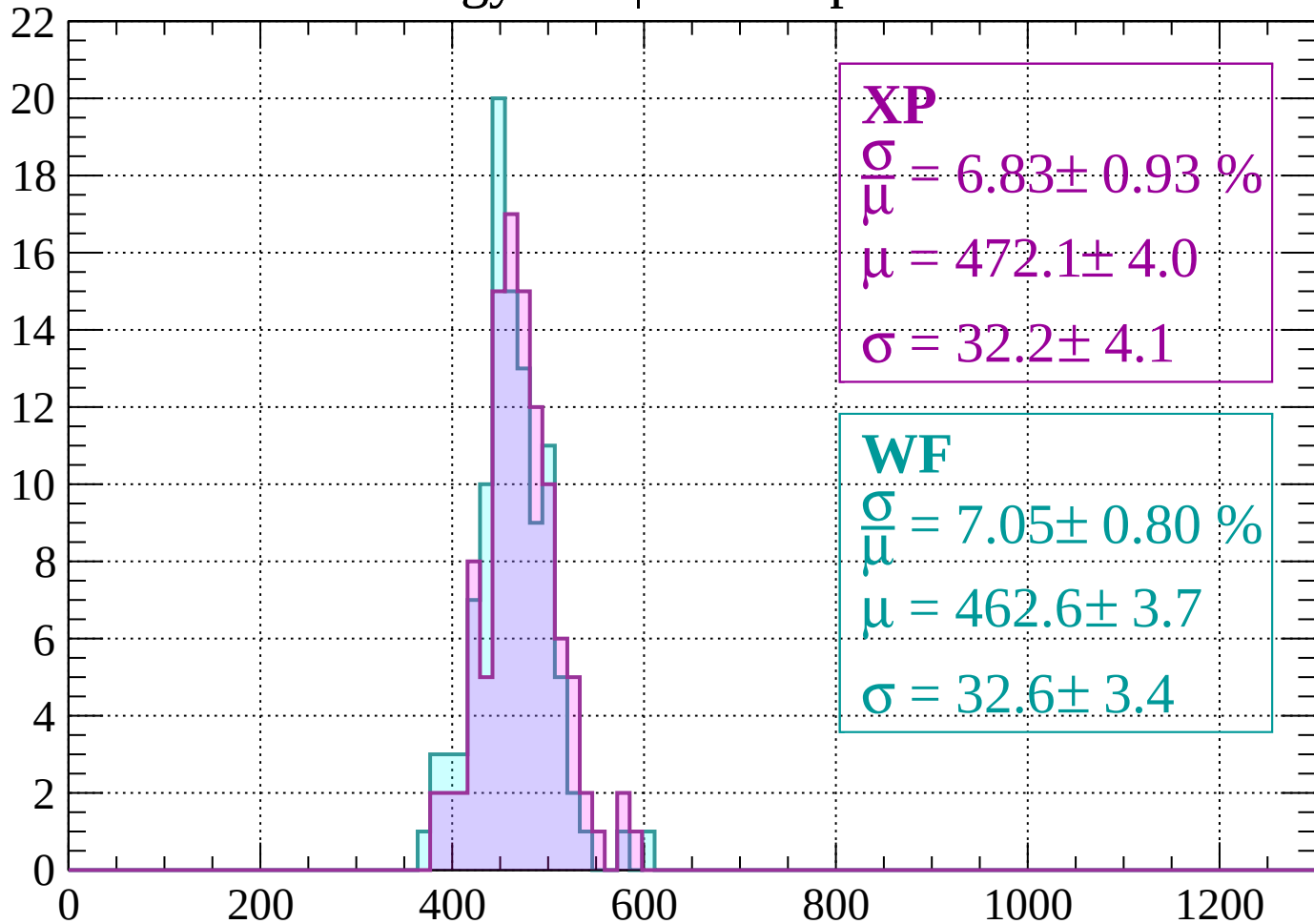
Energy loss |  $1600 < p < 1640$

Count



Energy loss |  $1640 < p < 1680$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.83 \pm 0.93 \%$$

$$\mu = 472.1 \pm 4.0$$

$$\sigma = 32.2 \pm 4.1$$

**WF**

$$\frac{\sigma}{\mu} = 7.05 \pm 0.80 \%$$

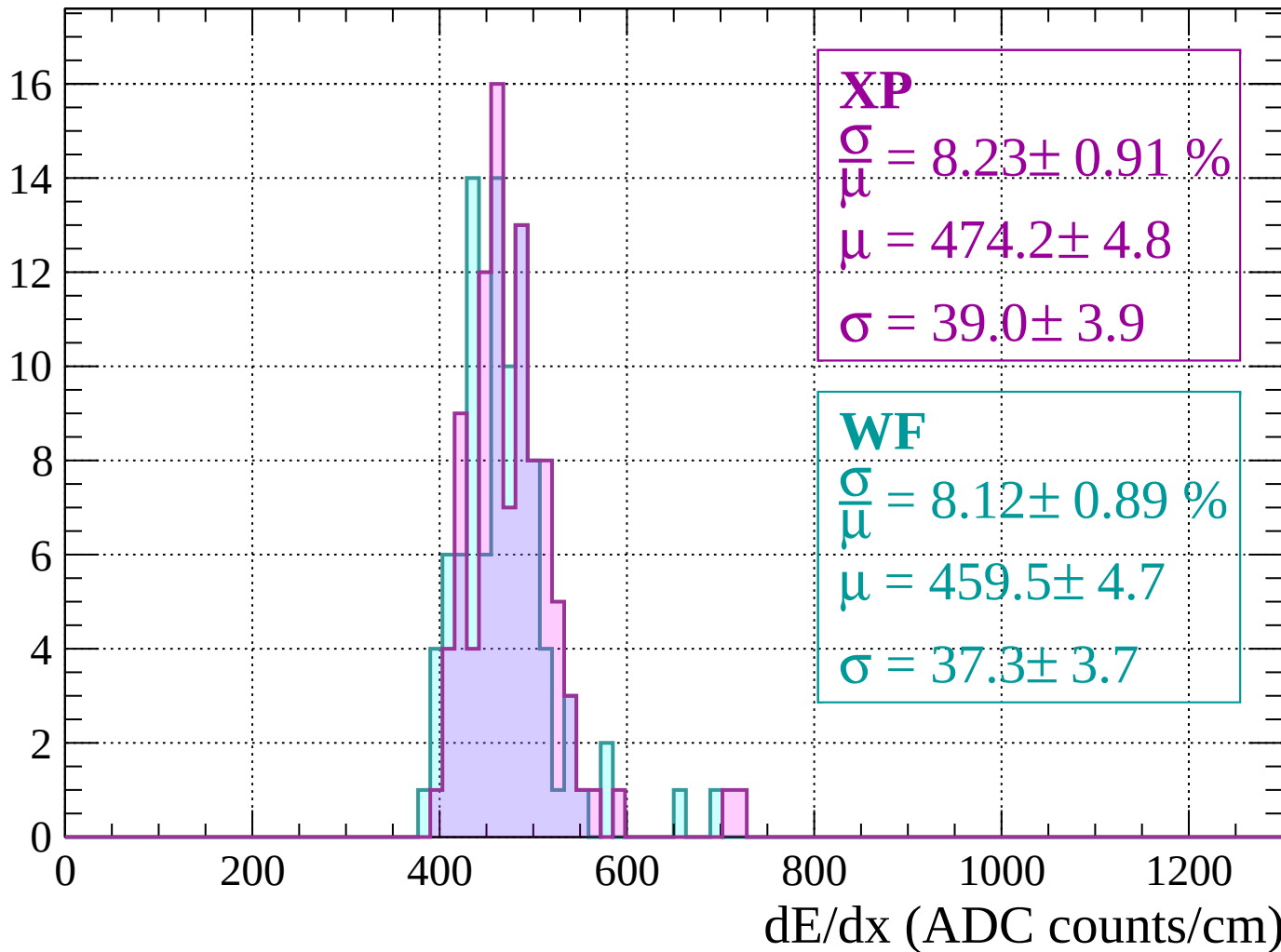
$$\mu = 462.6 \pm 3.7$$

$$\sigma = 32.6 \pm 3.4$$

$dE/dx$  (ADC counts/cm)

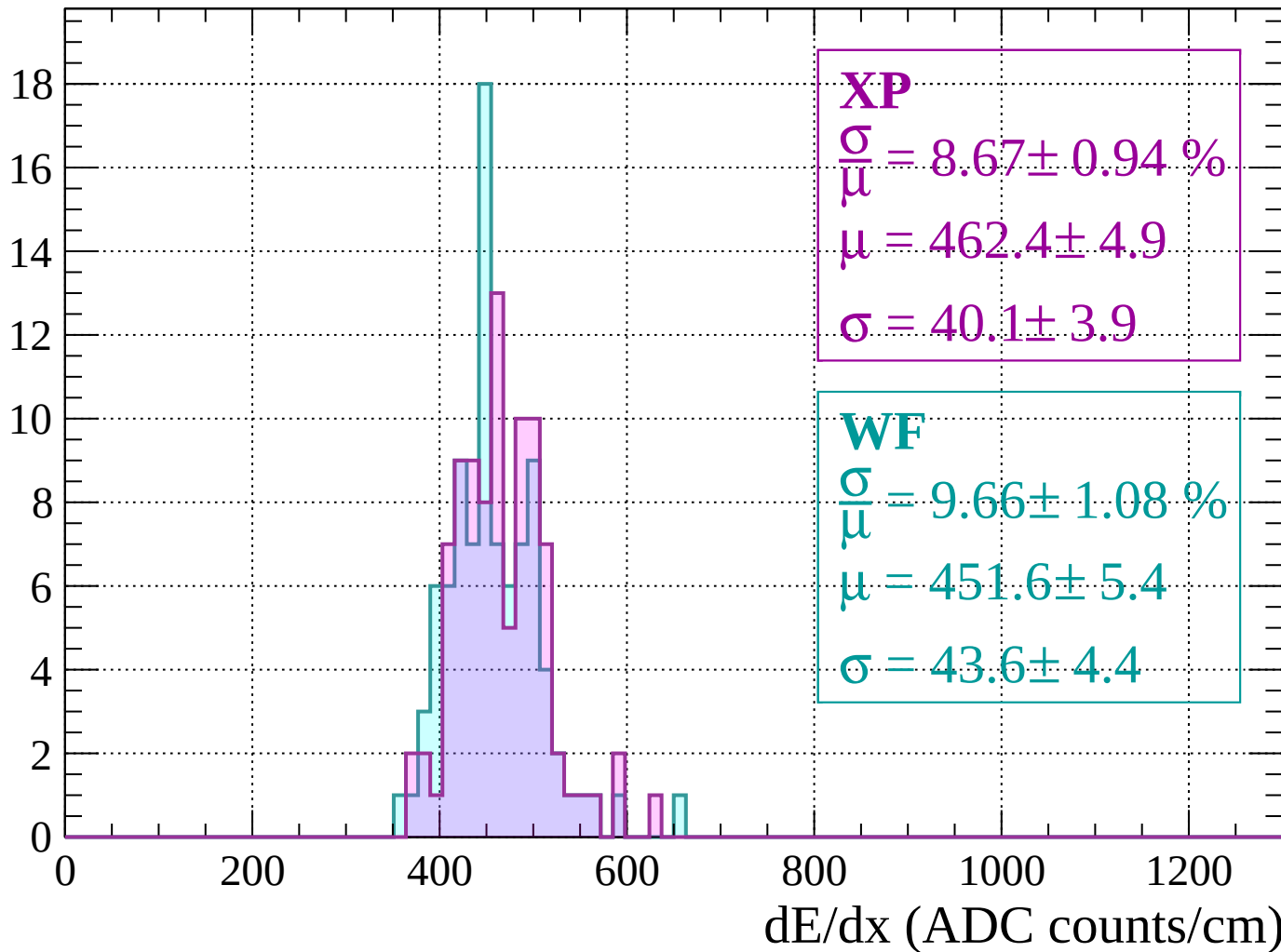
Energy loss |  $1680 < p < 1720$

Count



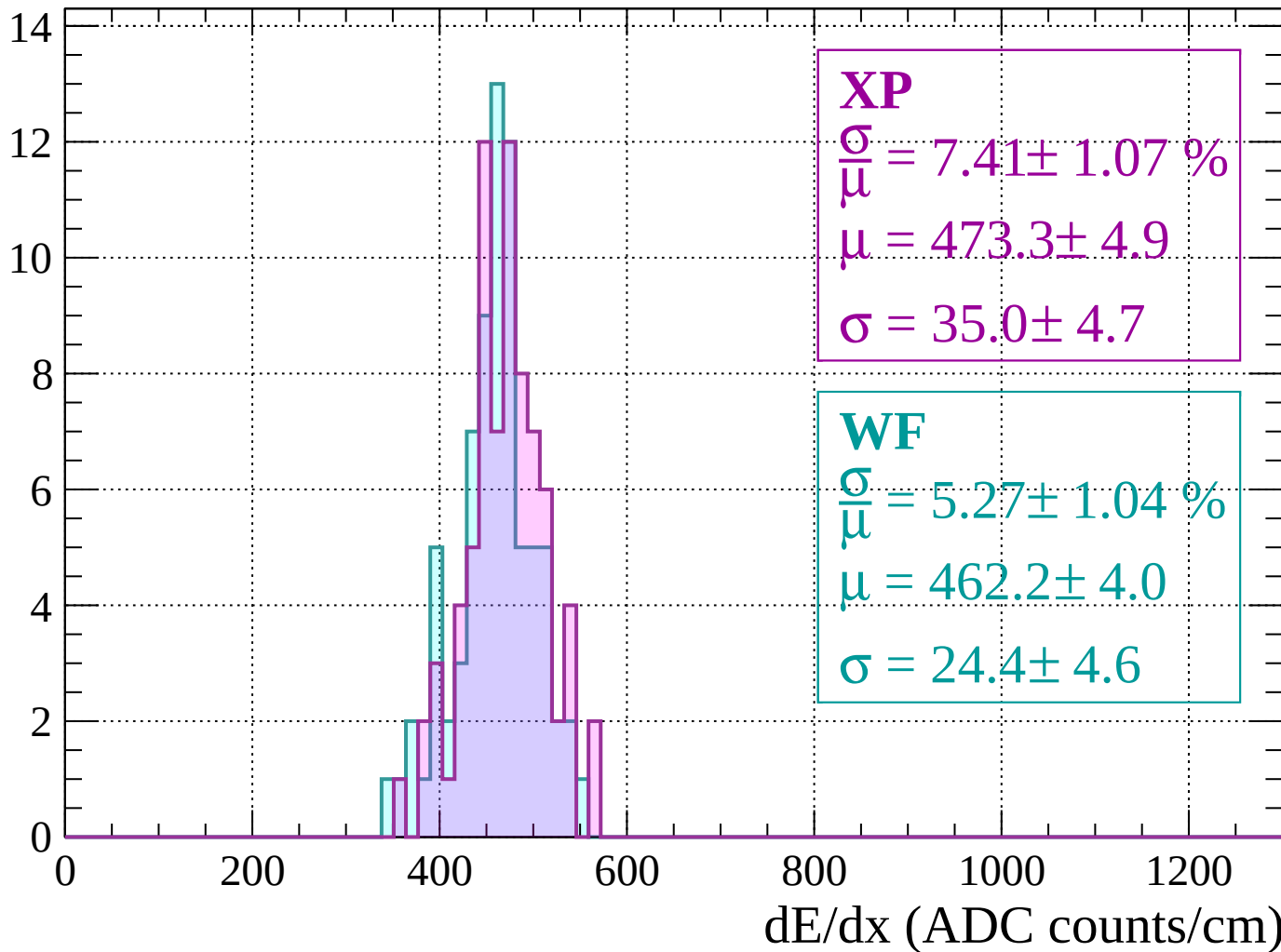
Energy loss |  $1720 < p < 1760$

Count



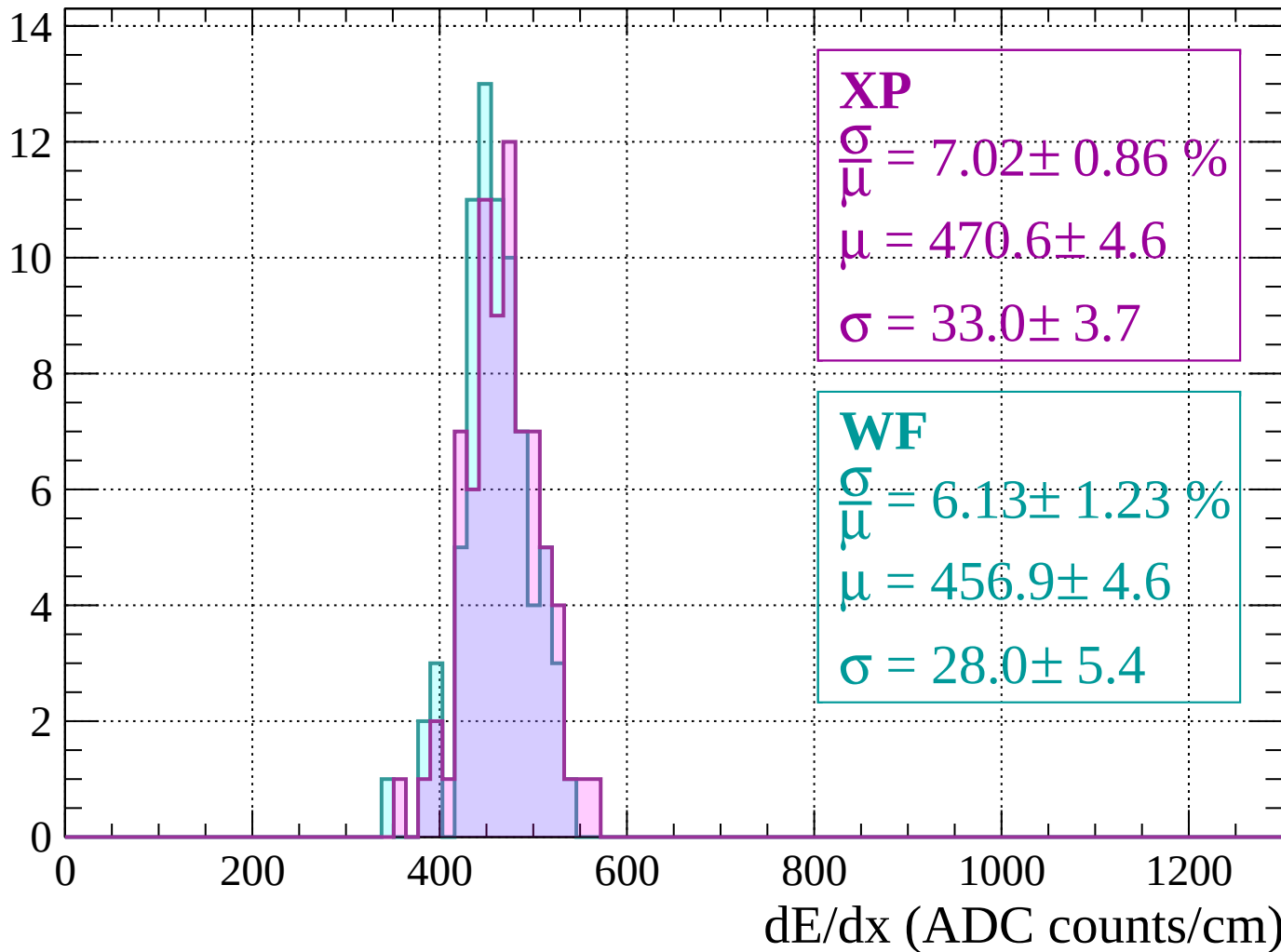
Energy loss |  $1760 < p < 1800$

Count



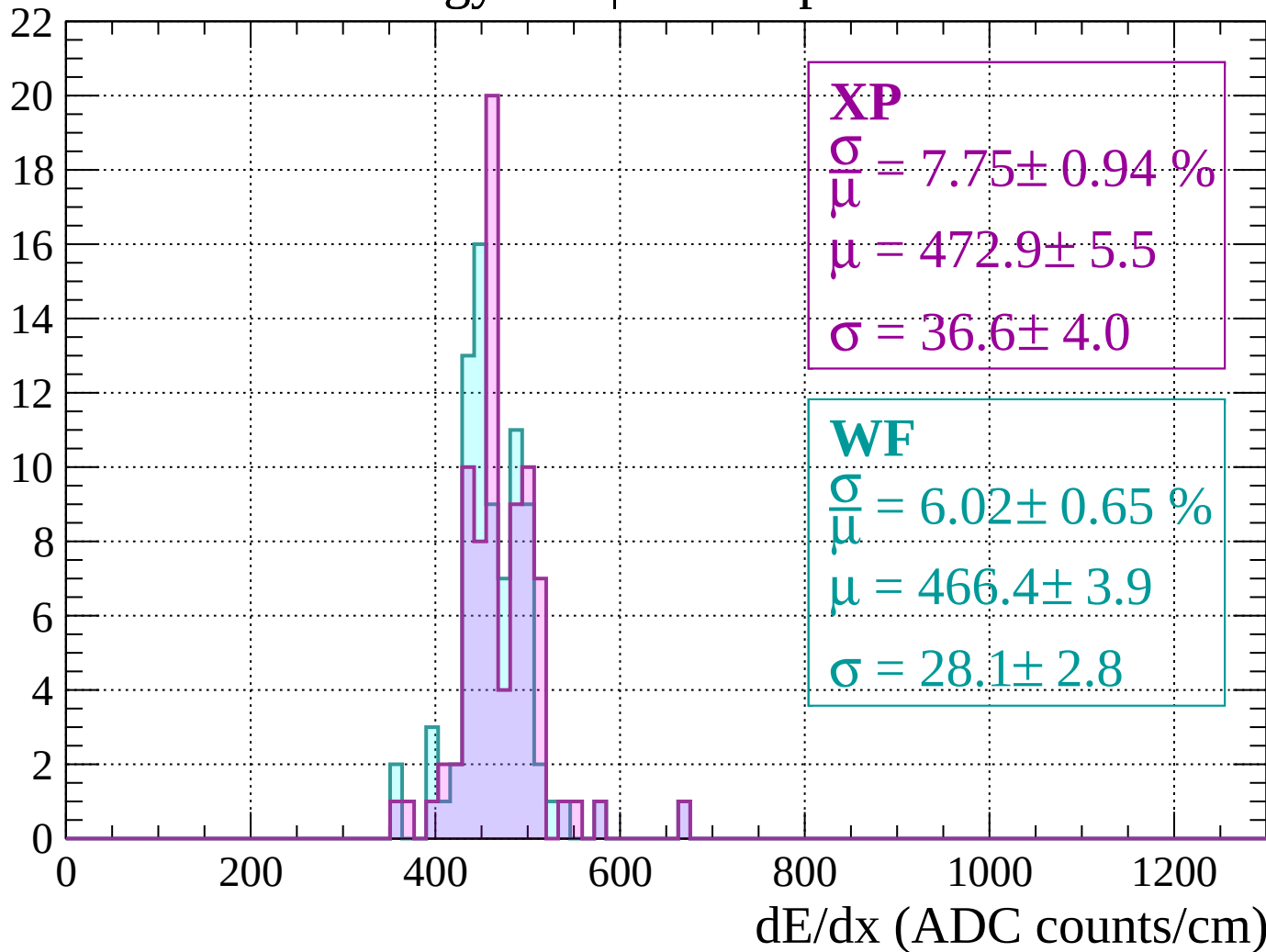
Energy loss |  $1800 < p < 1840$

Count



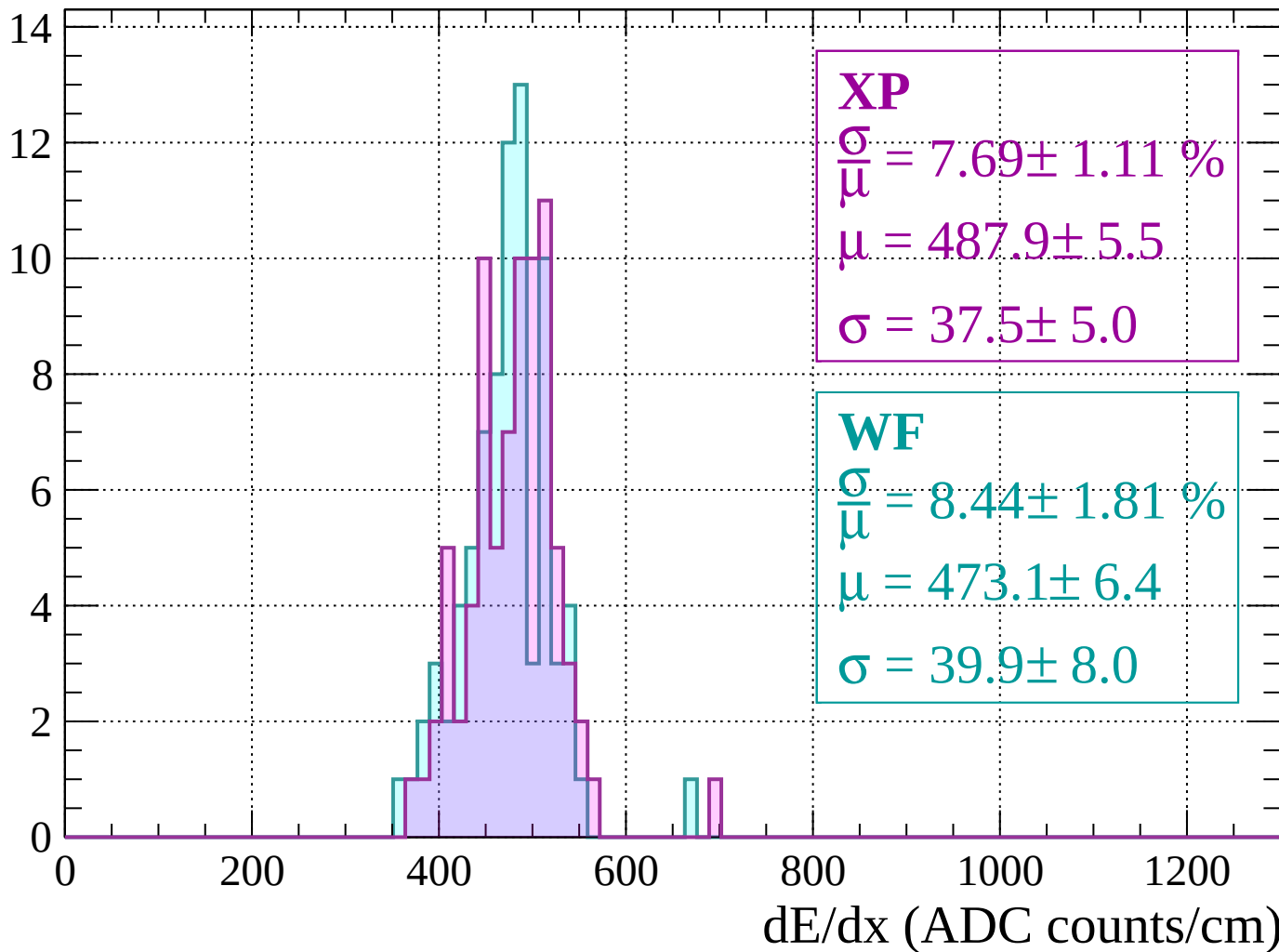
Energy loss |  $1840 < p < 1880$

Count



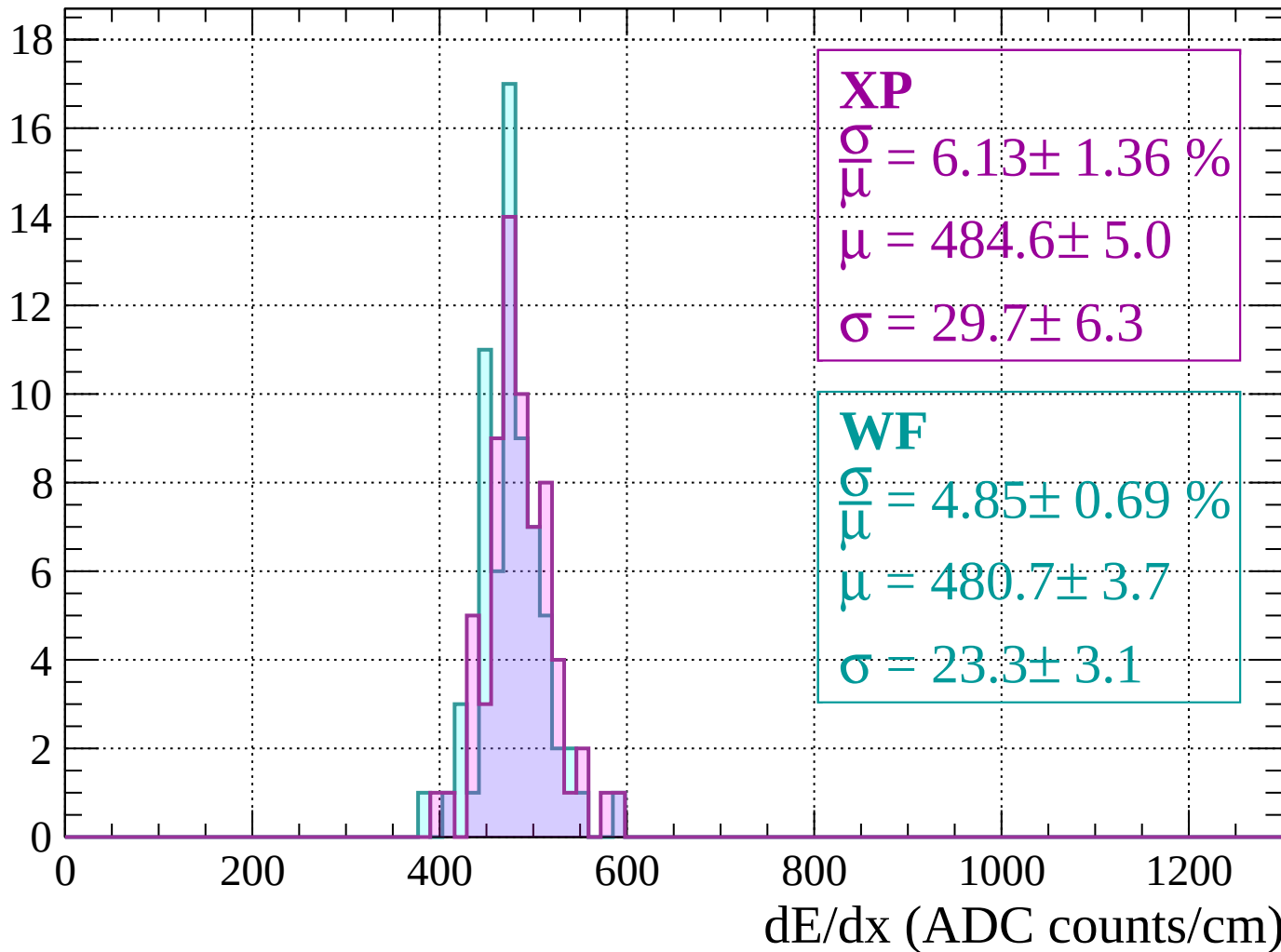
Energy loss |  $1880 < p < 1920$

Count



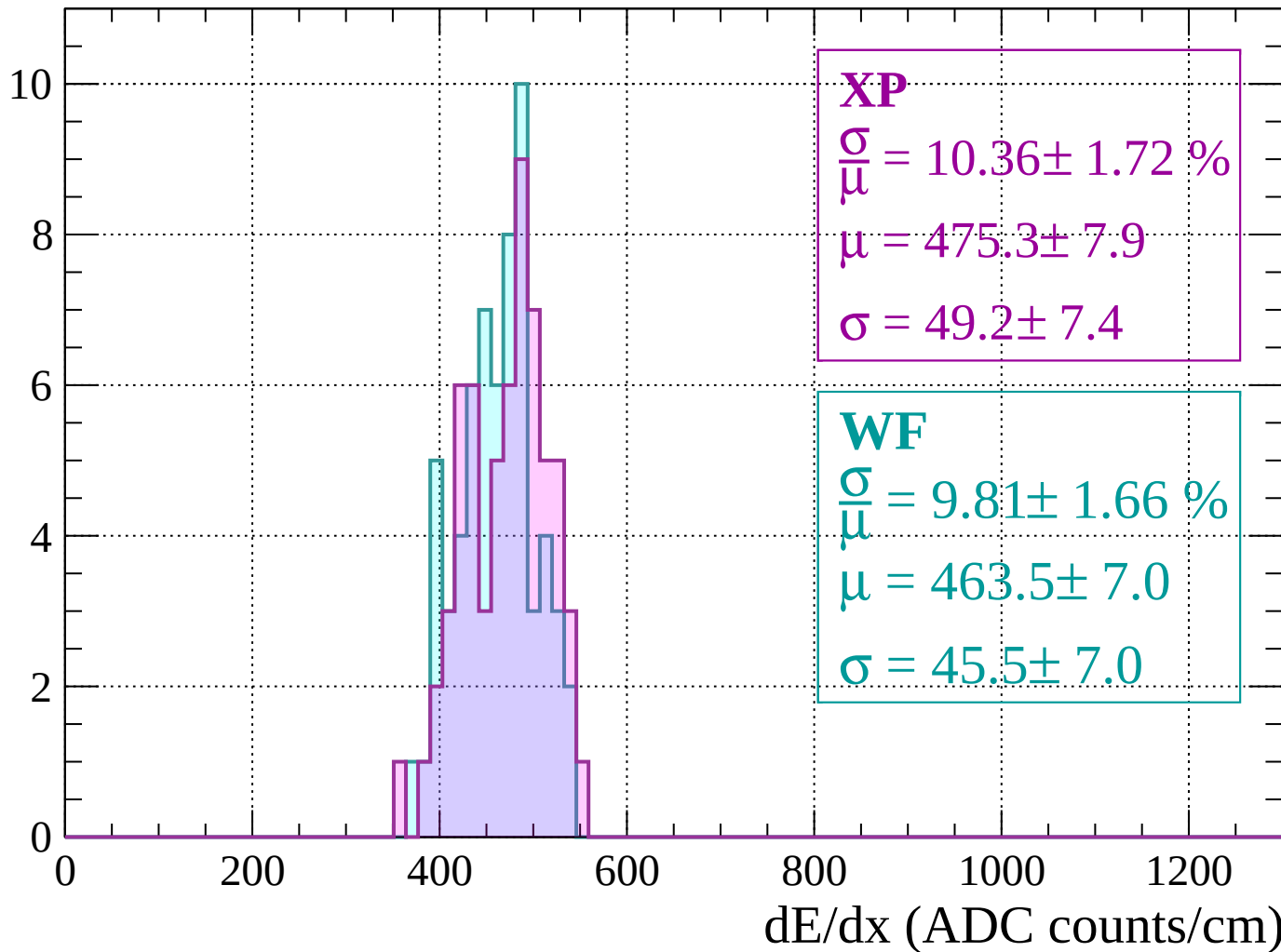
Energy loss |  $1920 < p < 1960$

Count



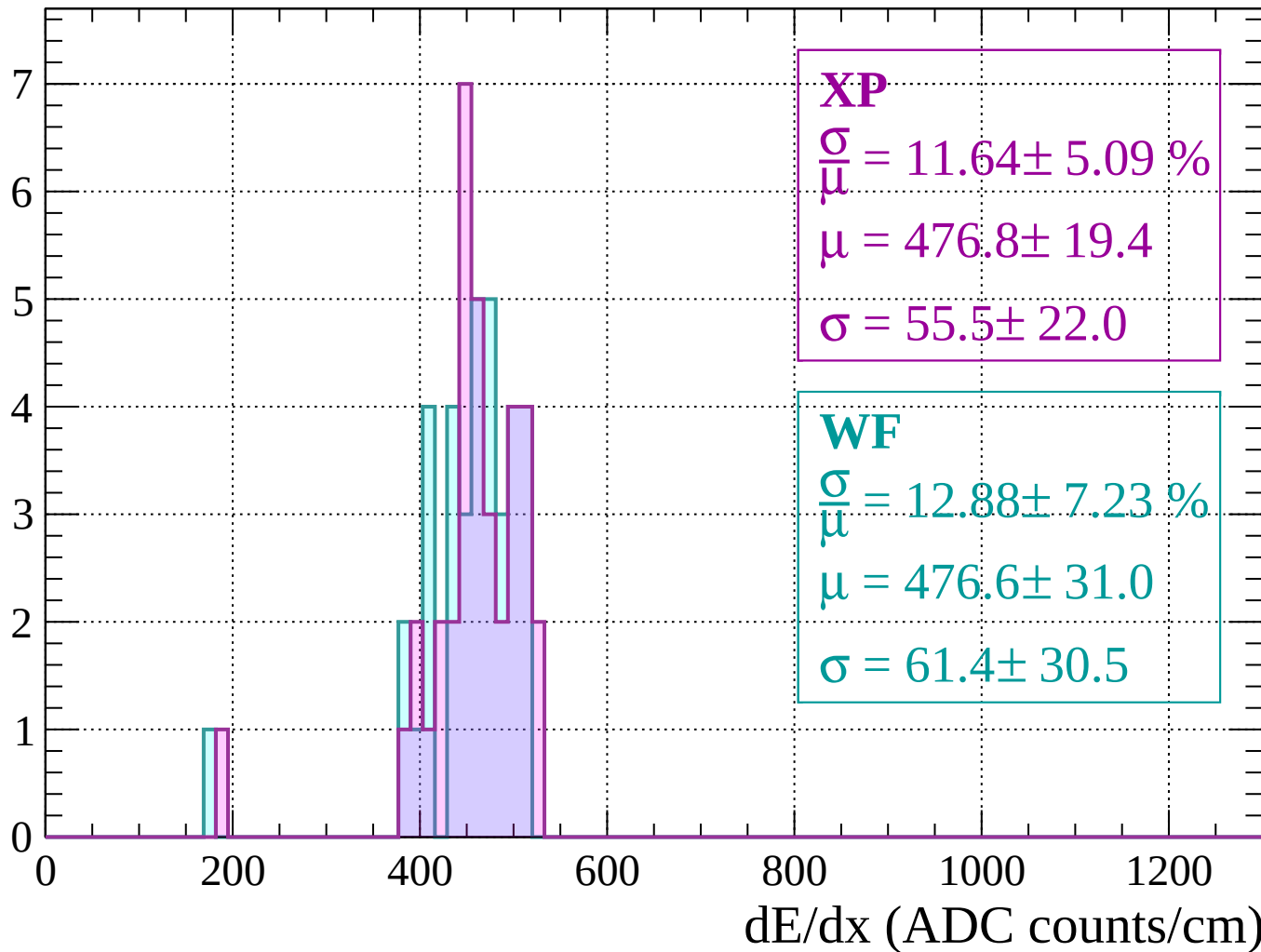
Energy loss |  $1960 < p < 2000$

Count



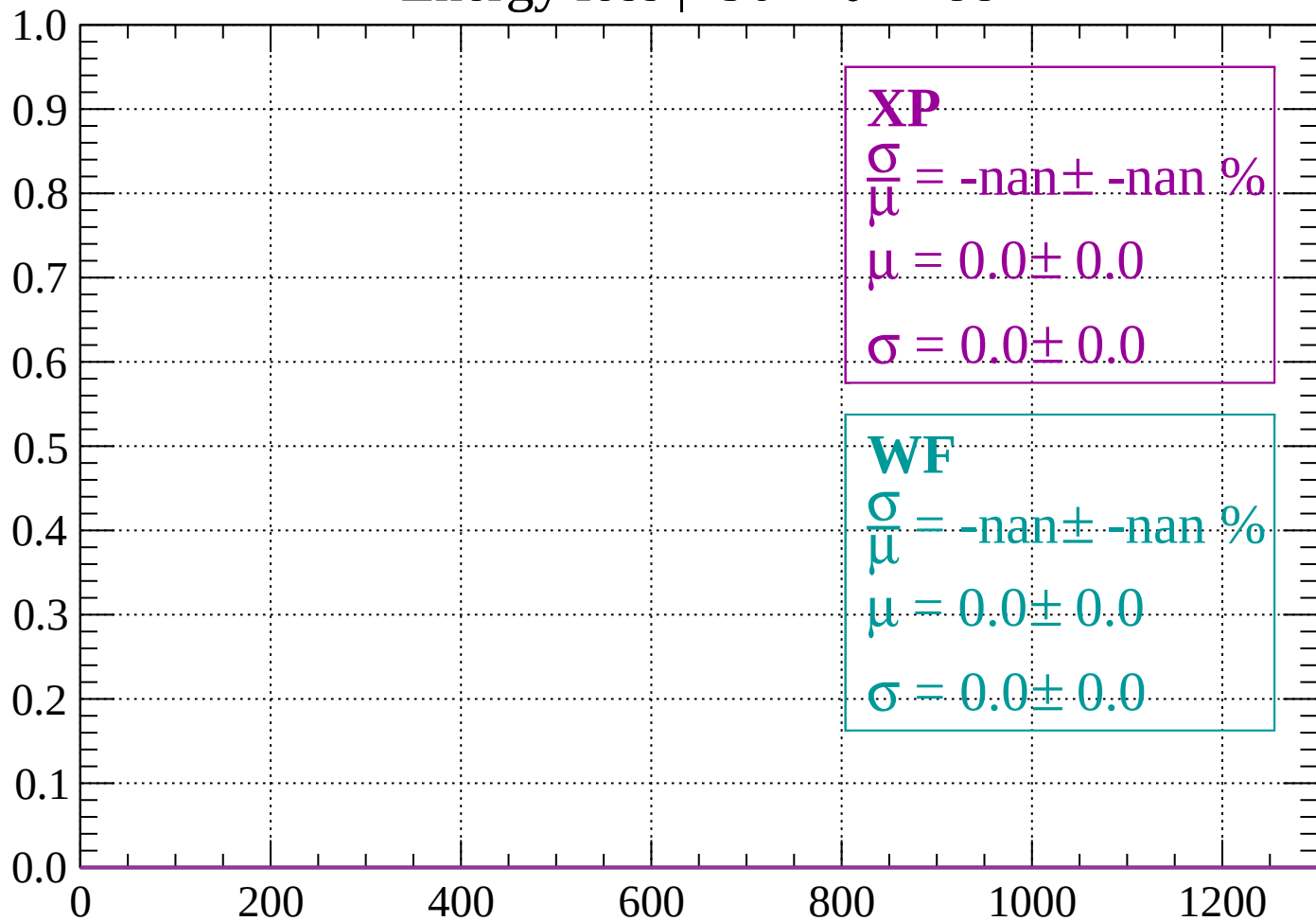
Energy loss |  $2000 < p < 2040$

Count



Energy loss |  $-90 < \theta < -88$

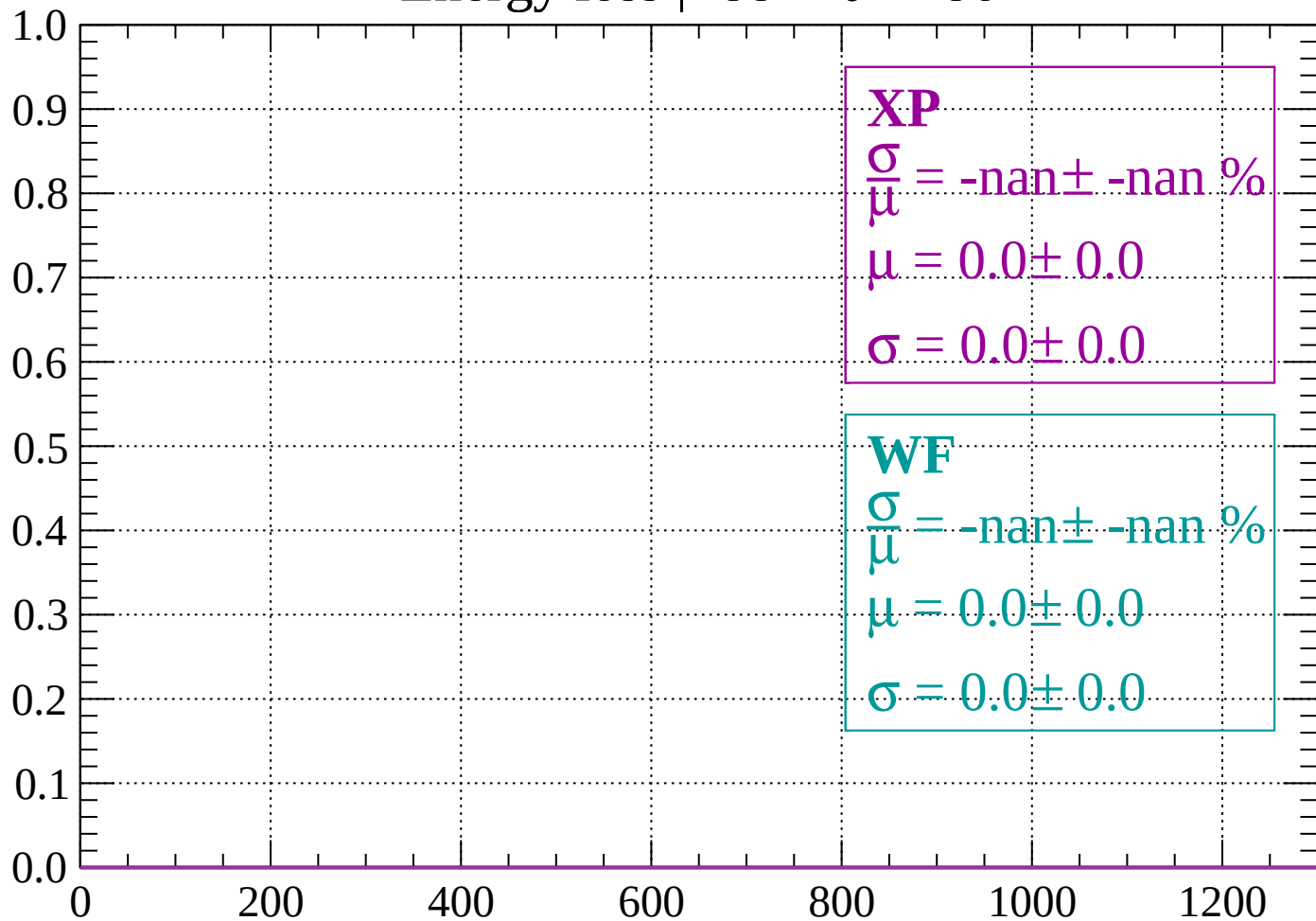
Count



dE/dx (ADC counts/cm)

Energy loss |  $-88 < \theta < -86$

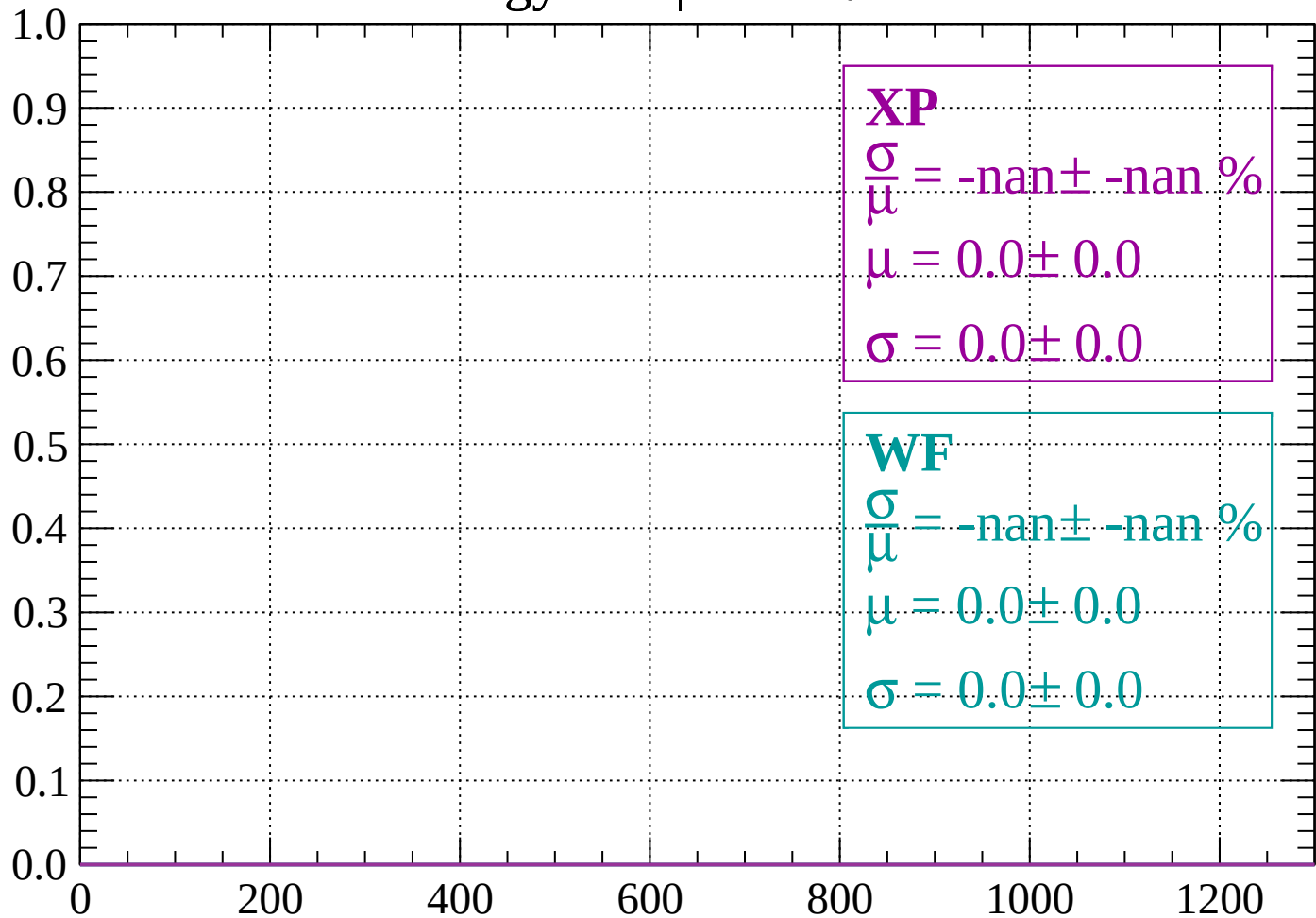
Count



dE/dx (ADC counts/cm)

Energy loss |  $-86 < \theta < -84$

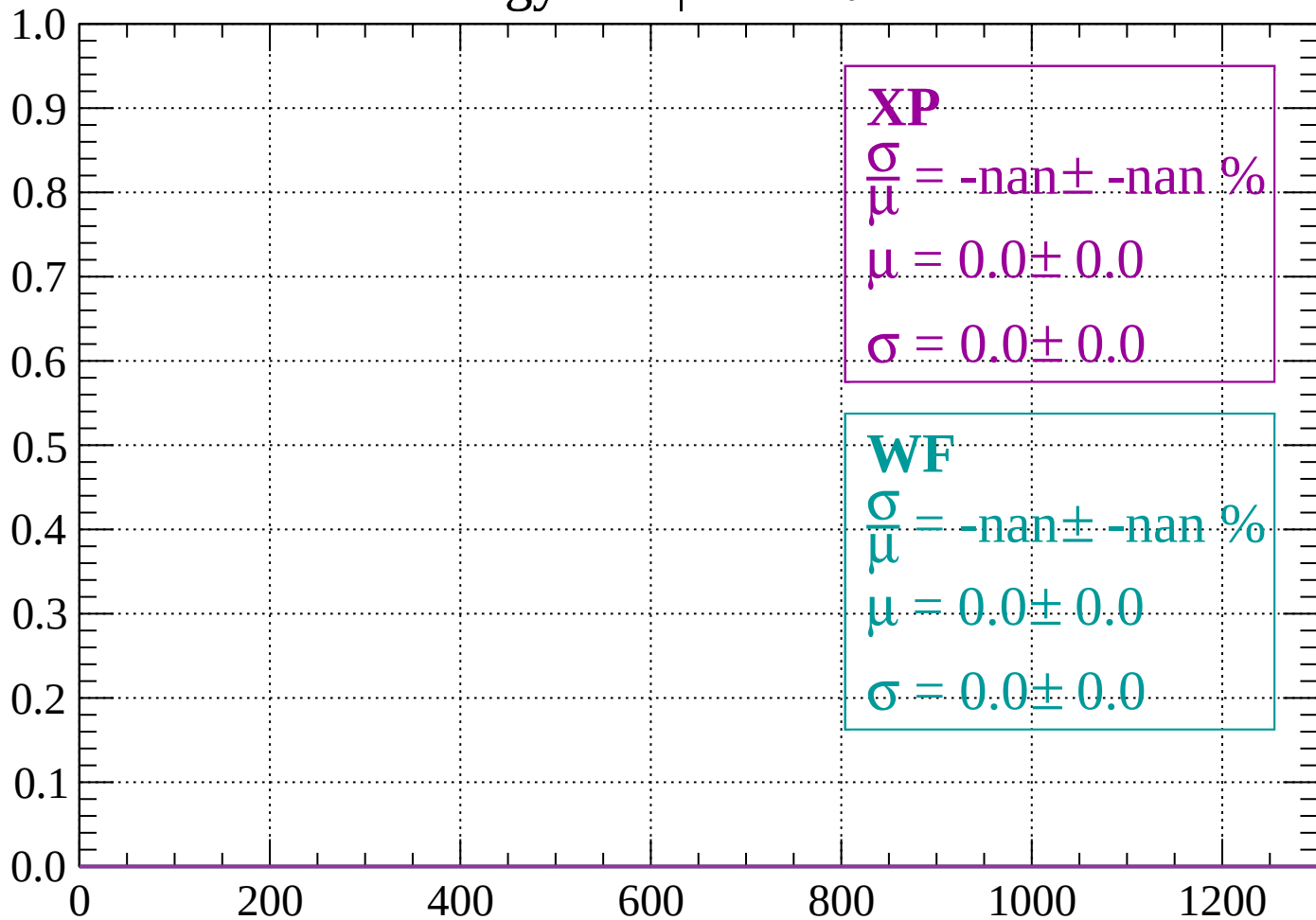
Count



dE/dx (ADC counts/cm)

Energy loss |  $-84 < \theta < -82$

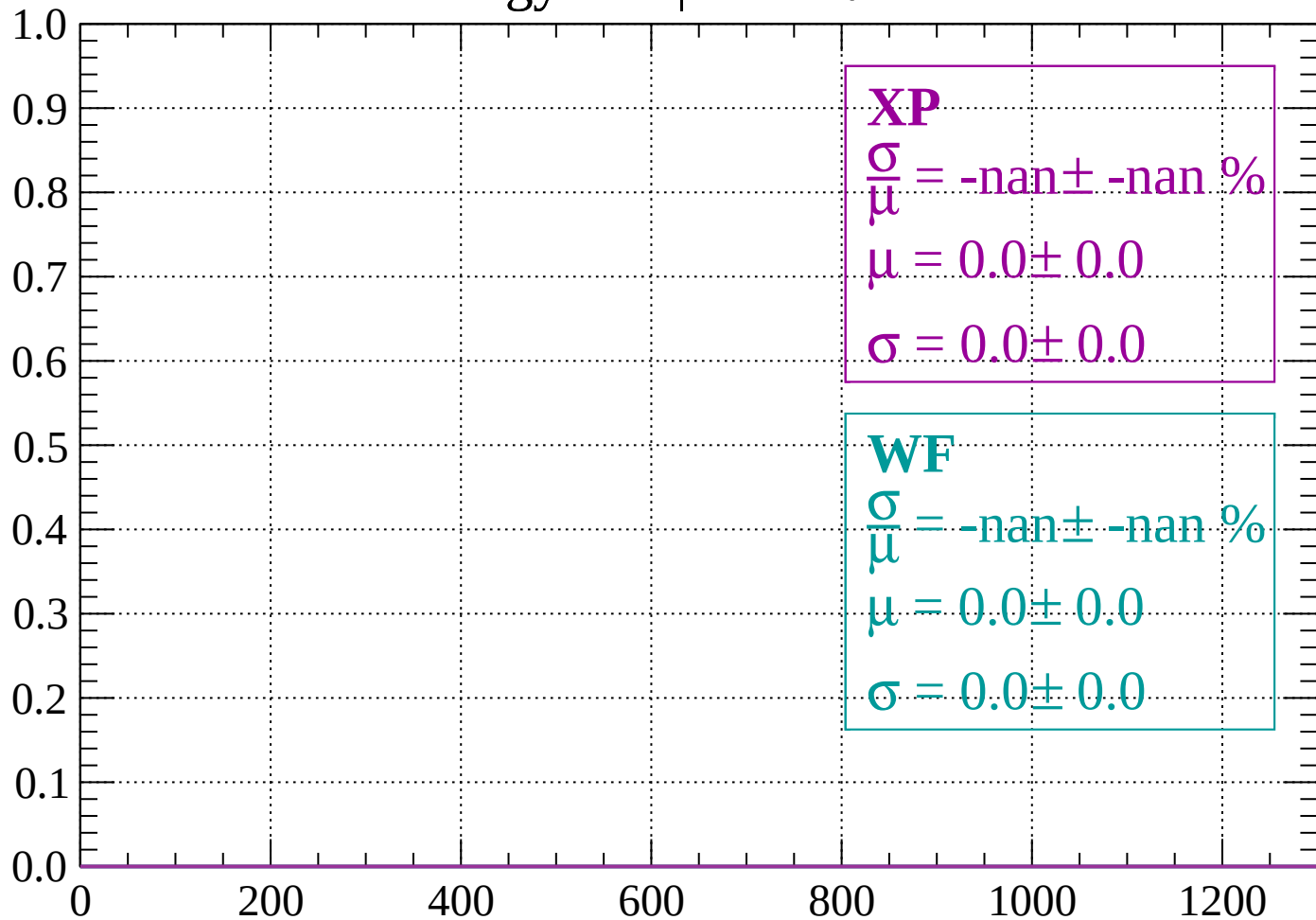
Count



dE/dx (ADC counts/cm)

Energy loss |  $-82 < \theta < -80$

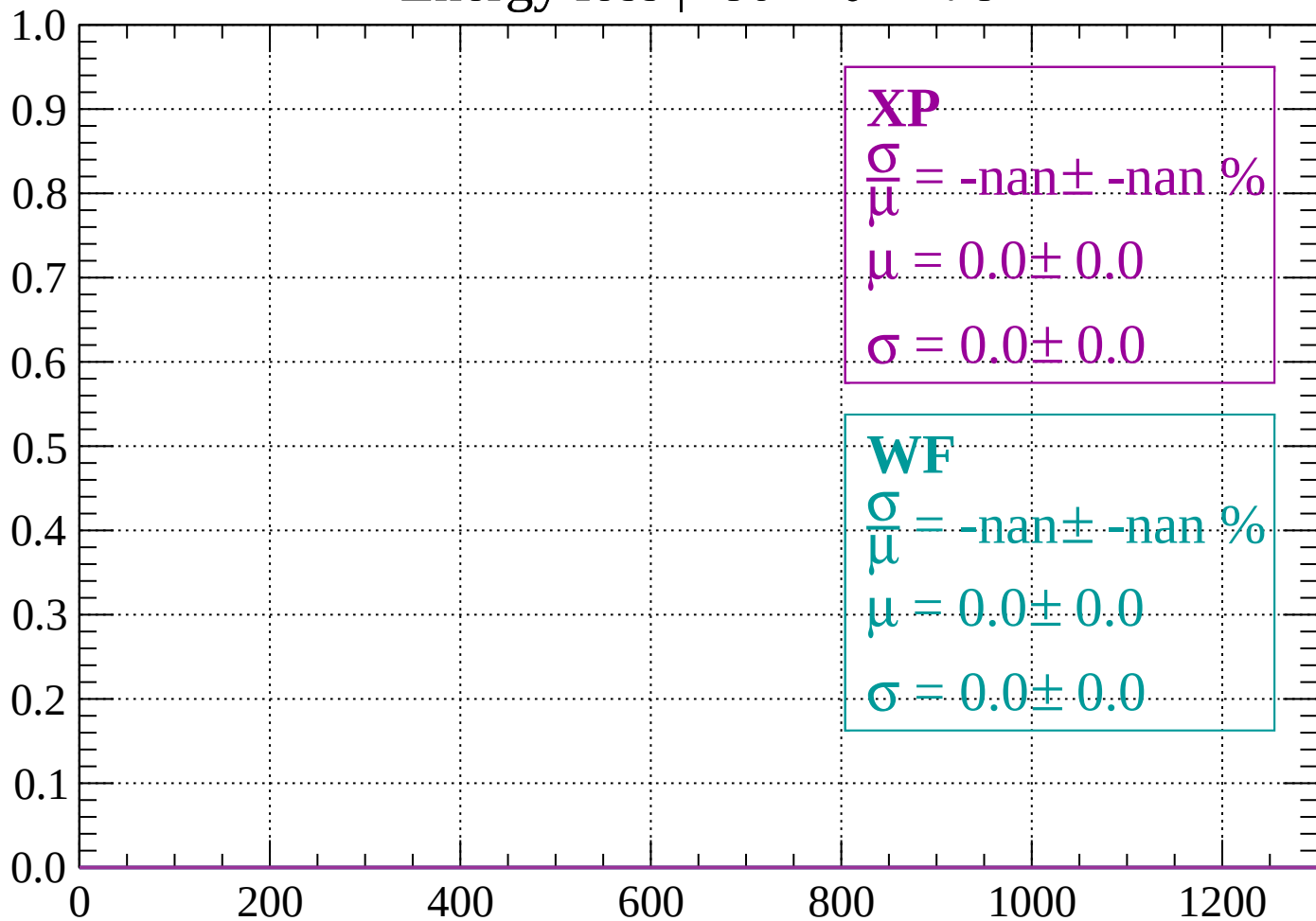
Count



dE/dx (ADC counts/cm)

Energy loss |  $-80 < \theta < -78$

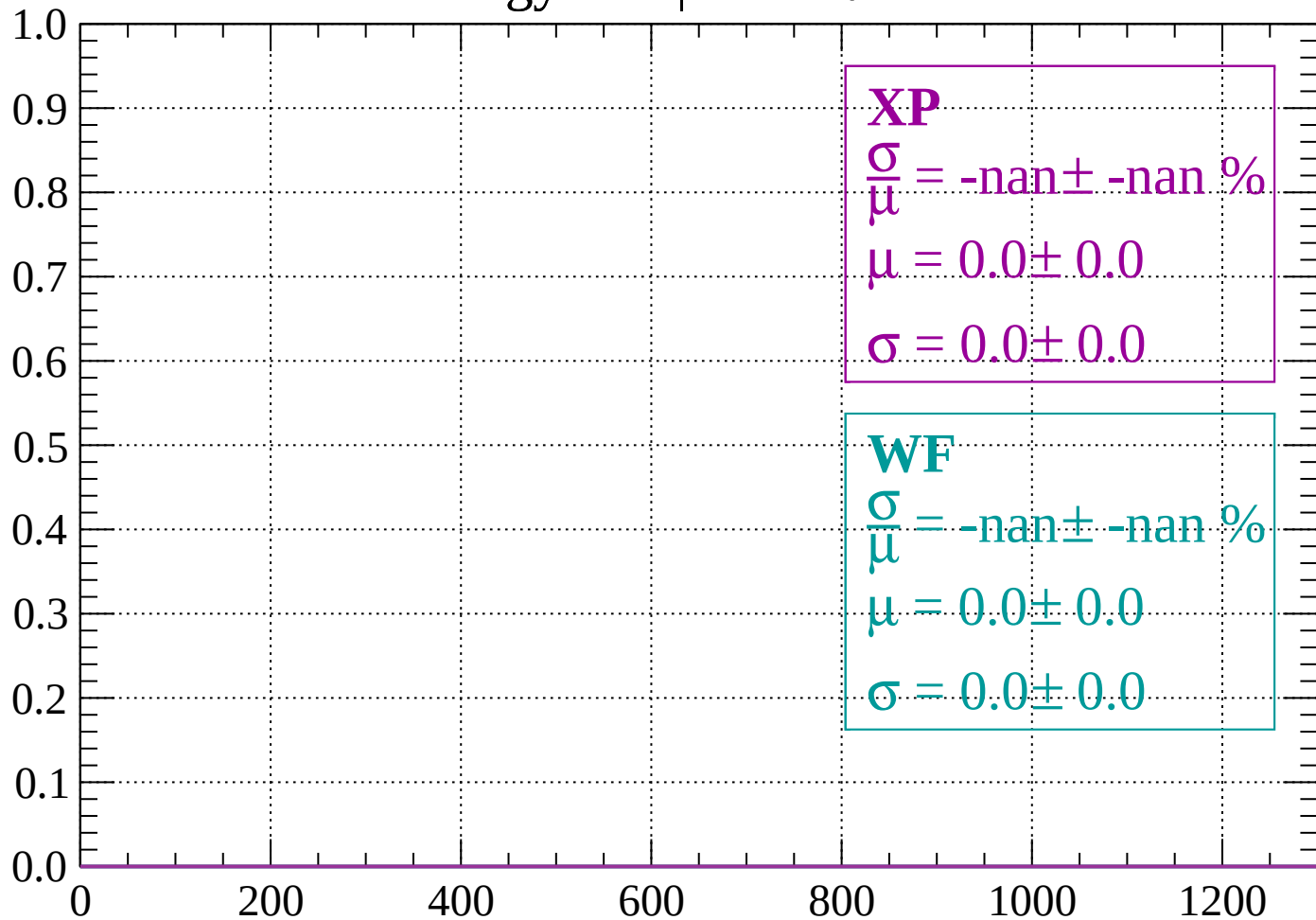
Count



dE/dx (ADC counts/cm)

Energy loss |  $-78 < \theta < -76$

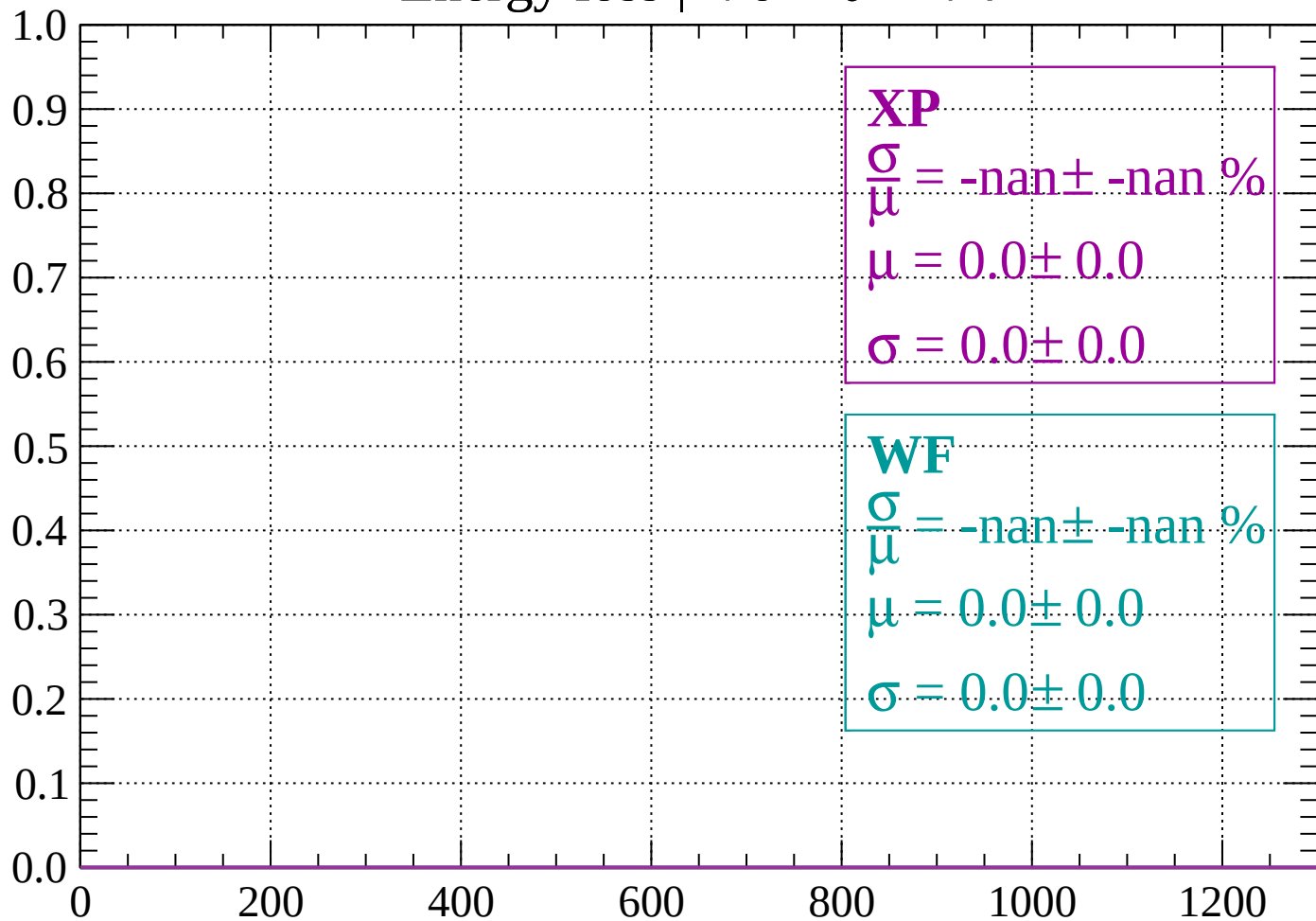
Count



dE/dx (ADC counts/cm)

Energy loss |  $-76 < \theta < -74$

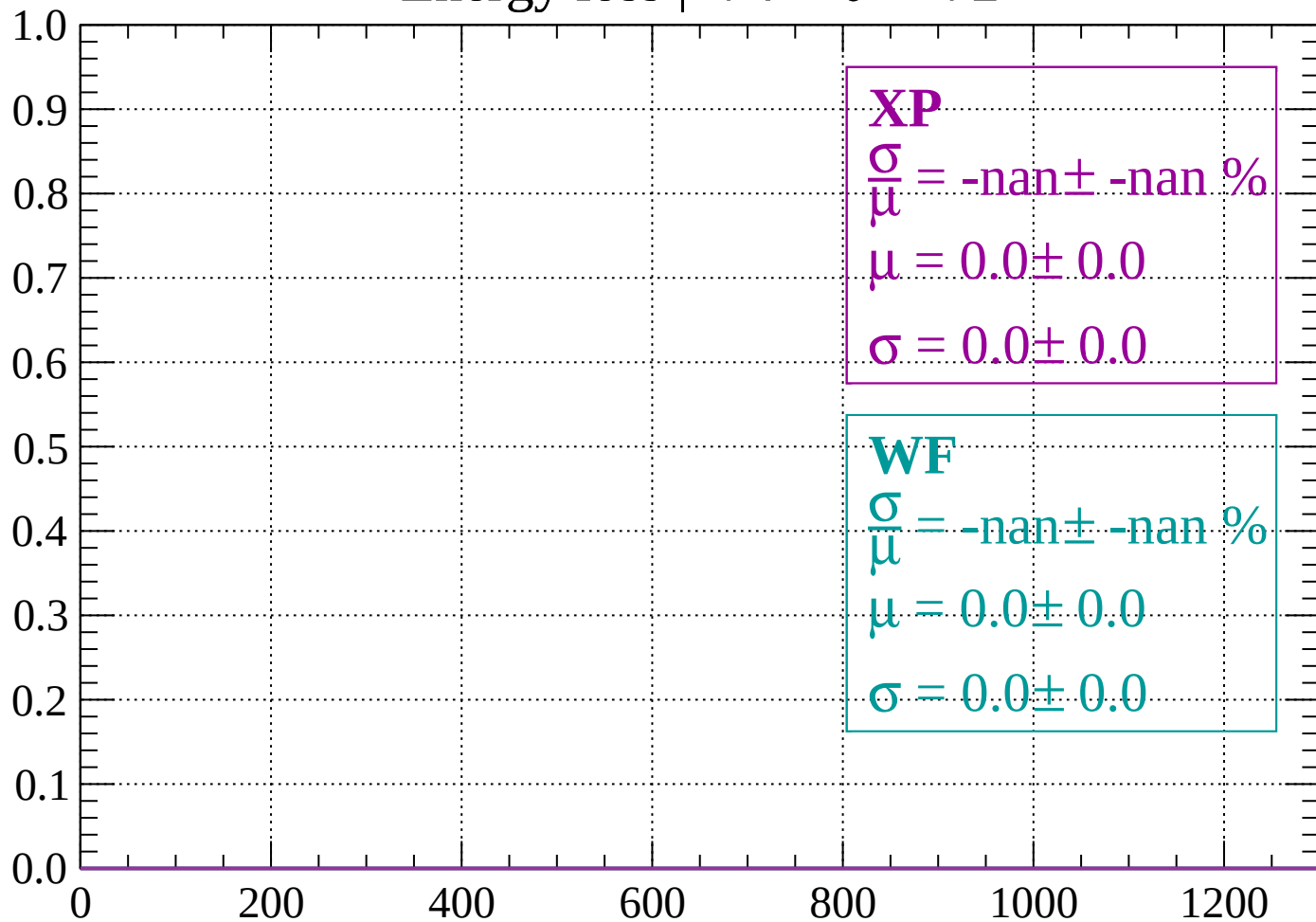
Count



dE/dx (ADC counts/cm)

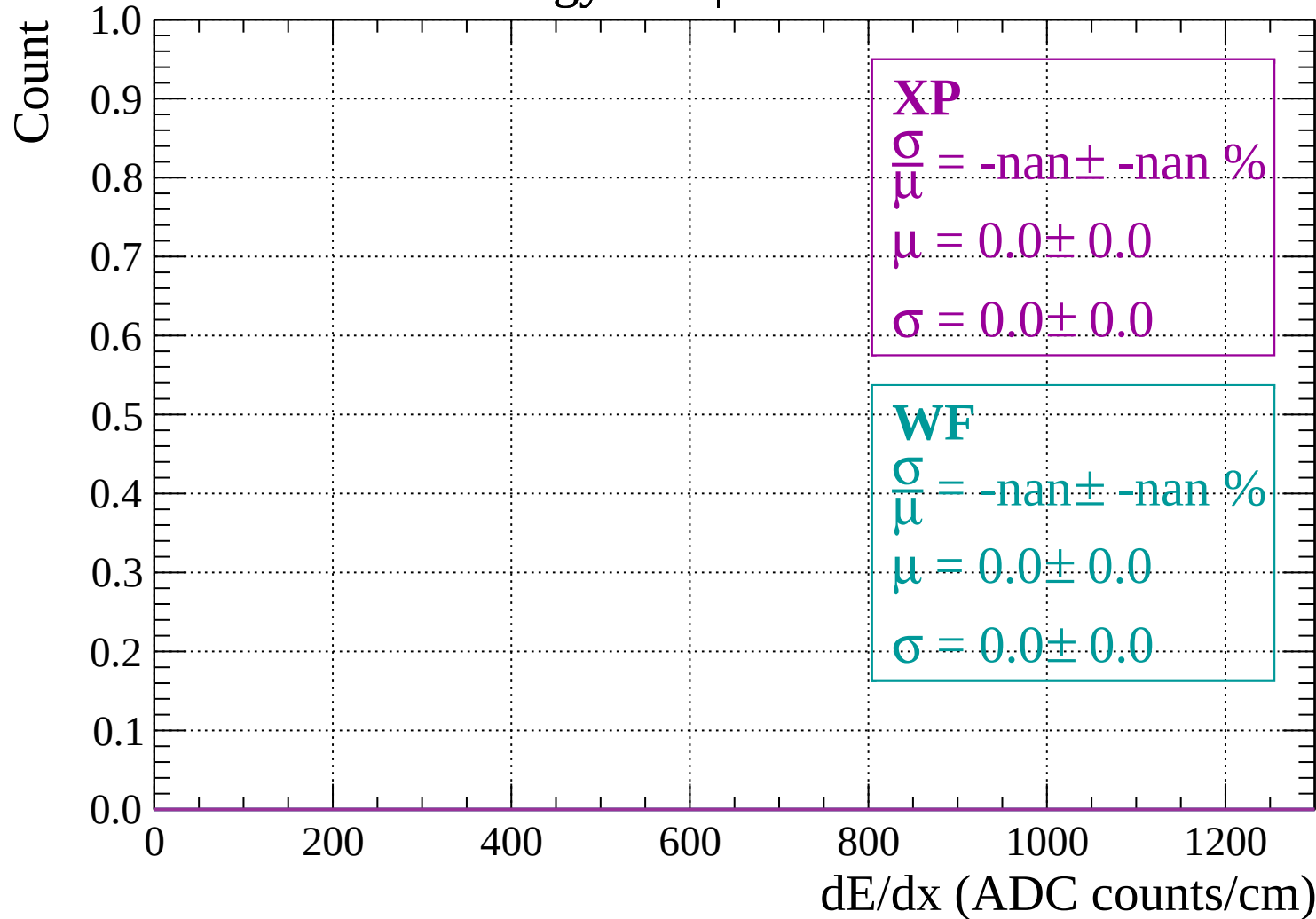
Energy loss |  $-74 < \theta < -72$

Count



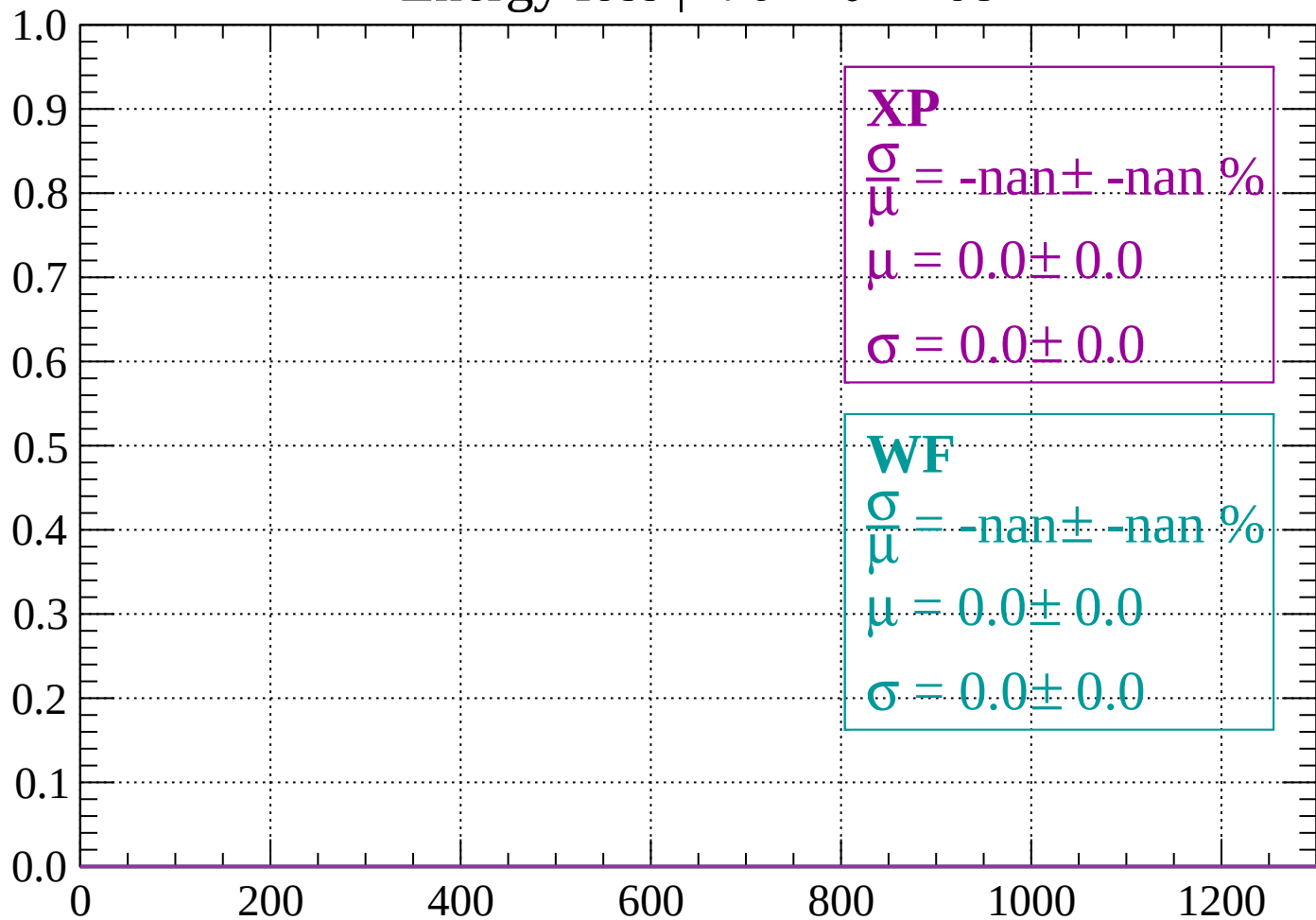
dE/dx (ADC counts/cm)

# Energy loss | $-72 < \theta < -70$



# Energy loss | $-70 < \theta < -68$

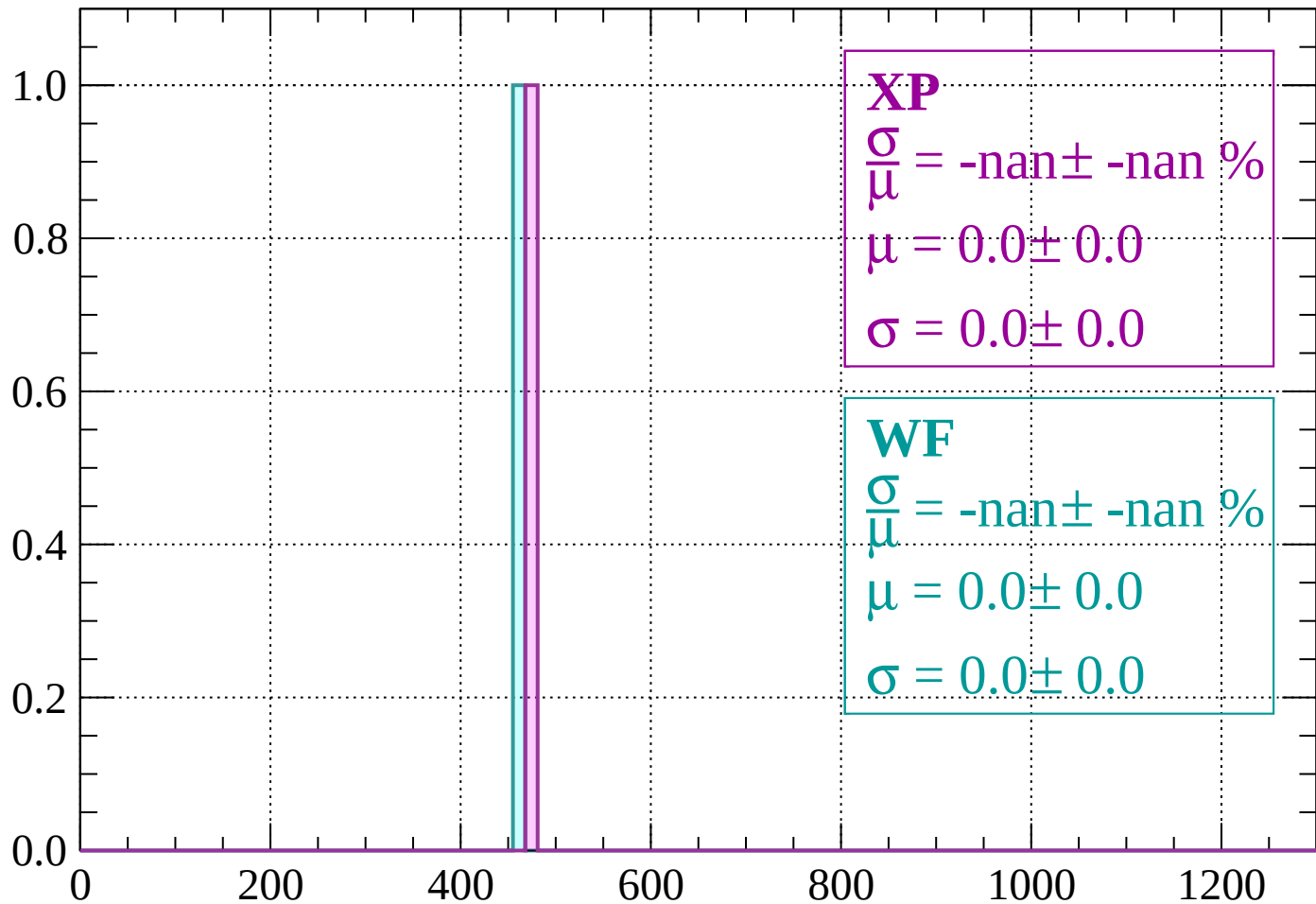
Count



dE/dx (ADC counts/cm)

Energy loss |  $-68 < \theta < -66$

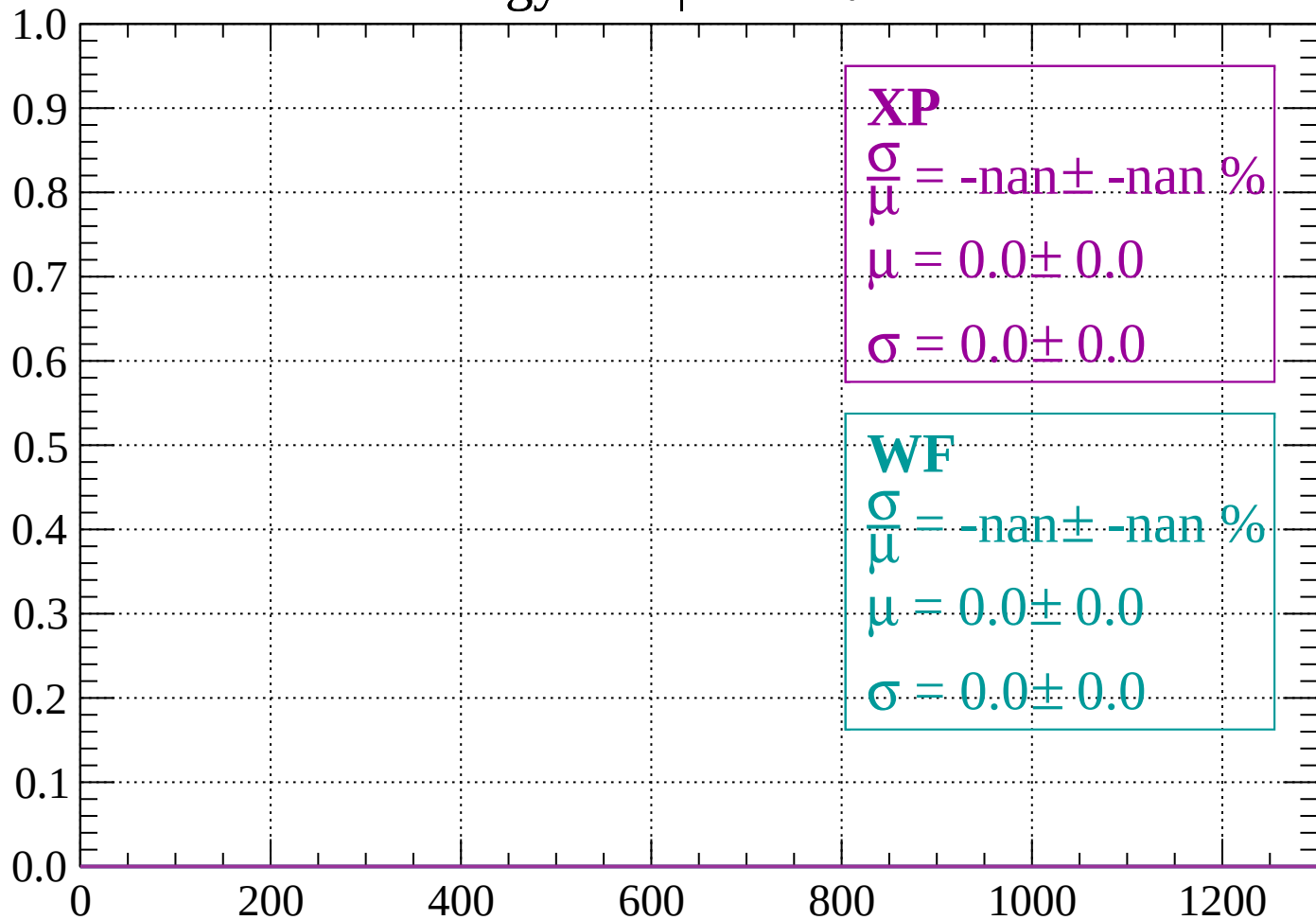
Count



$dE/dx$  (ADC counts/cm)

Energy loss |  $-66 < \theta < -64$

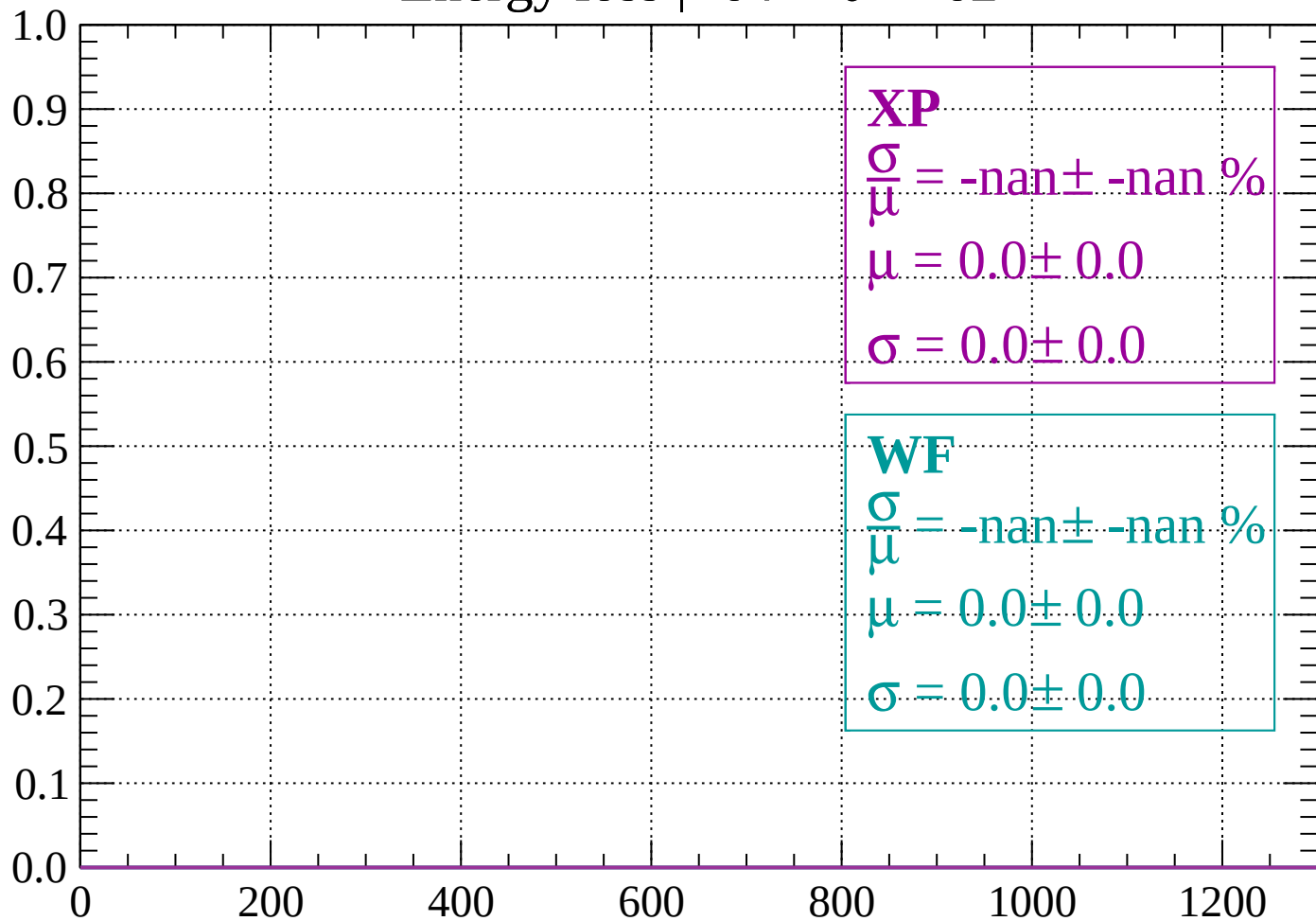
Count



dE/dx (ADC counts/cm)

Energy loss |  $-64 < \theta < -62$

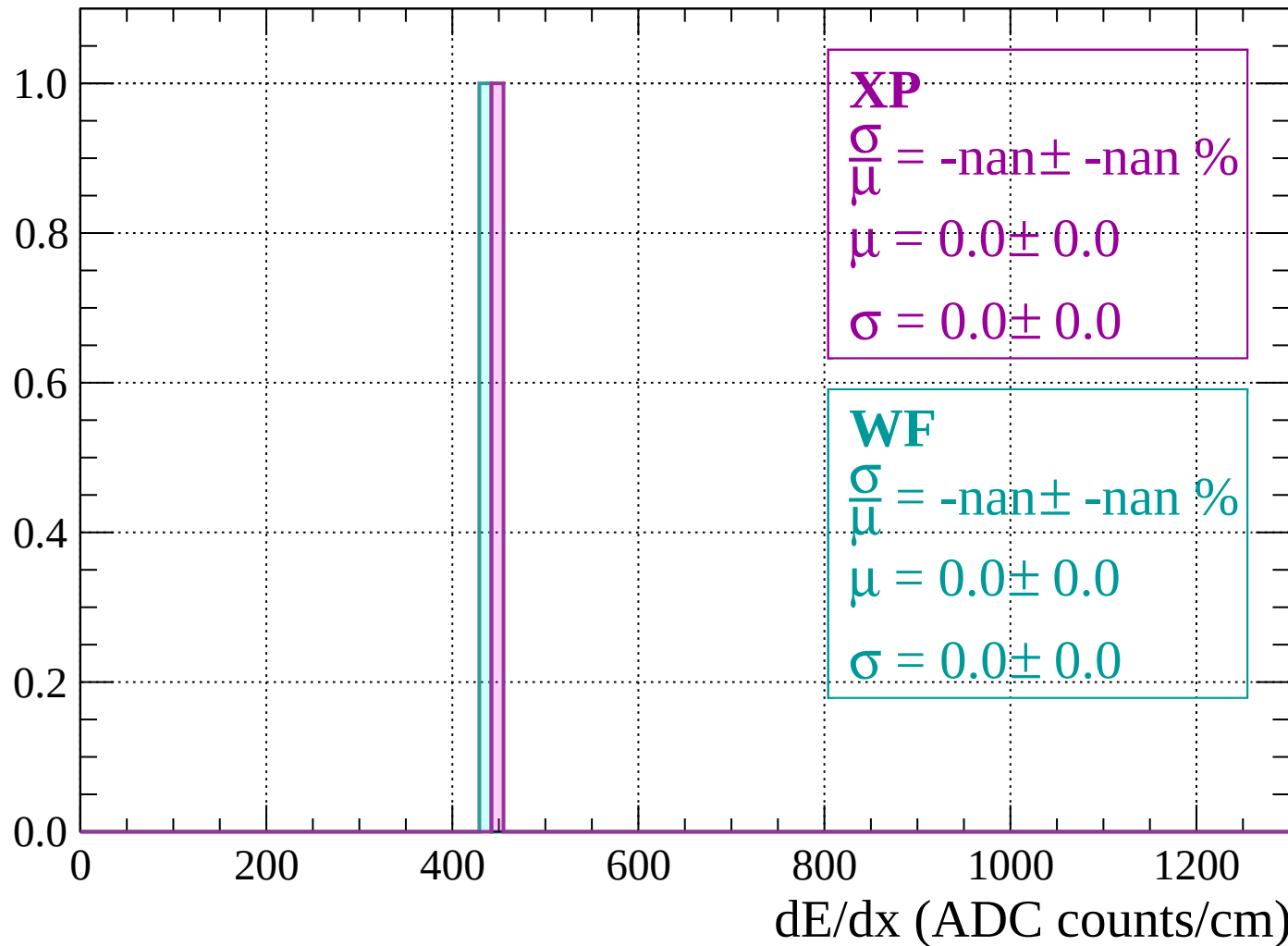
Count



dE/dx (ADC counts/cm)

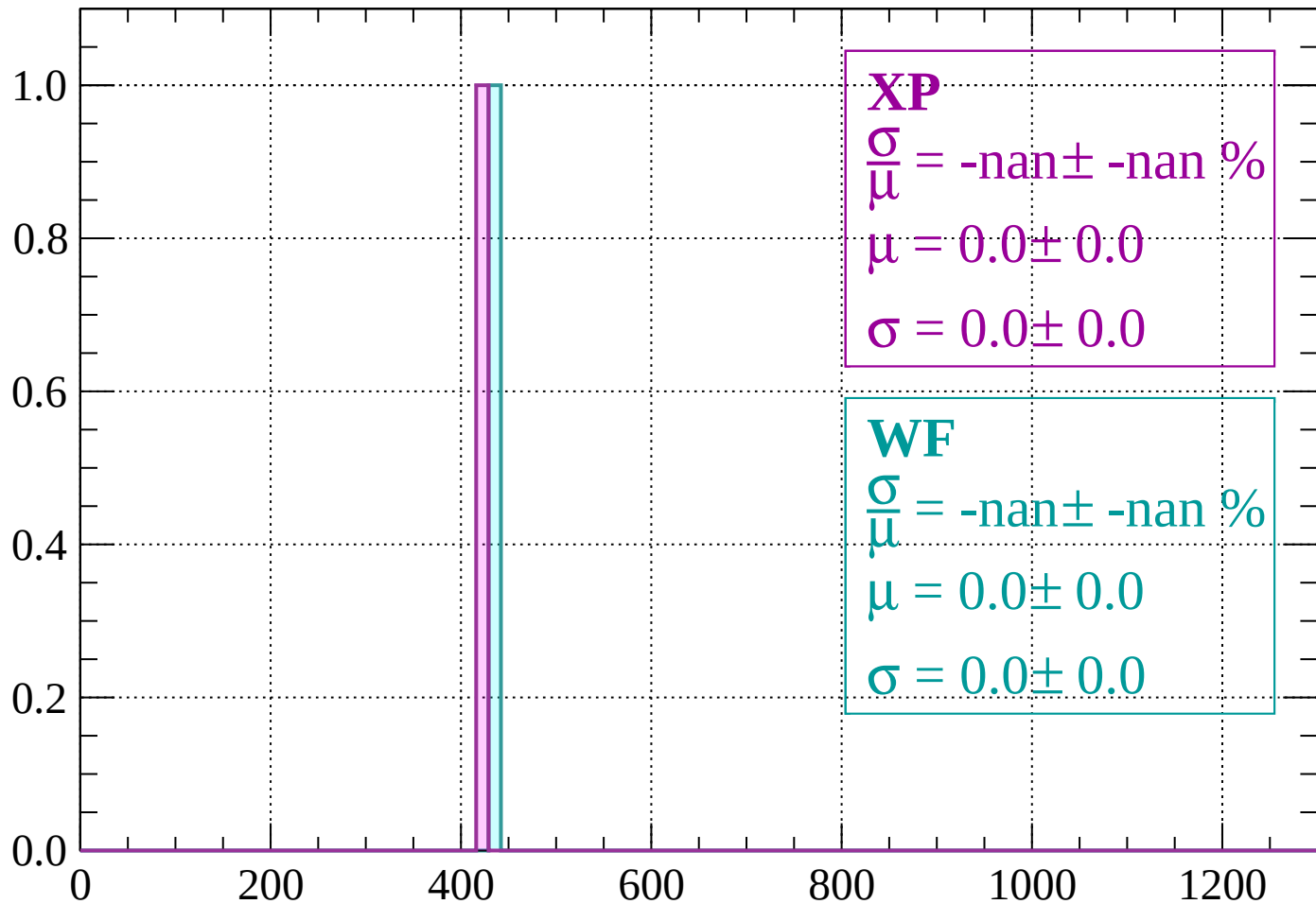
Energy loss |  $-62 < \theta < -60$

Count



Energy loss |  $-60 < \theta < -58$

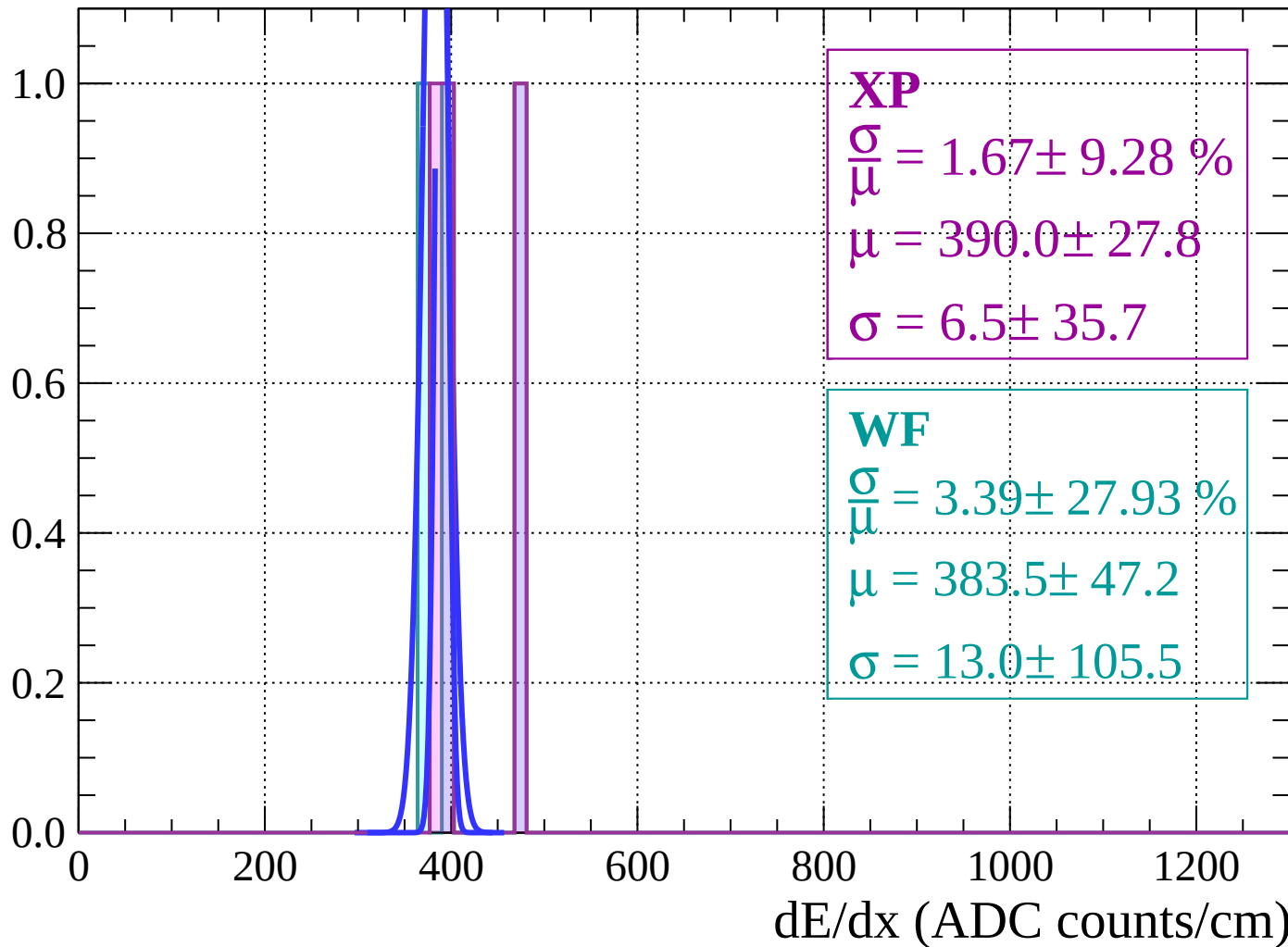
Count



$dE/dx$  (ADC counts/cm)

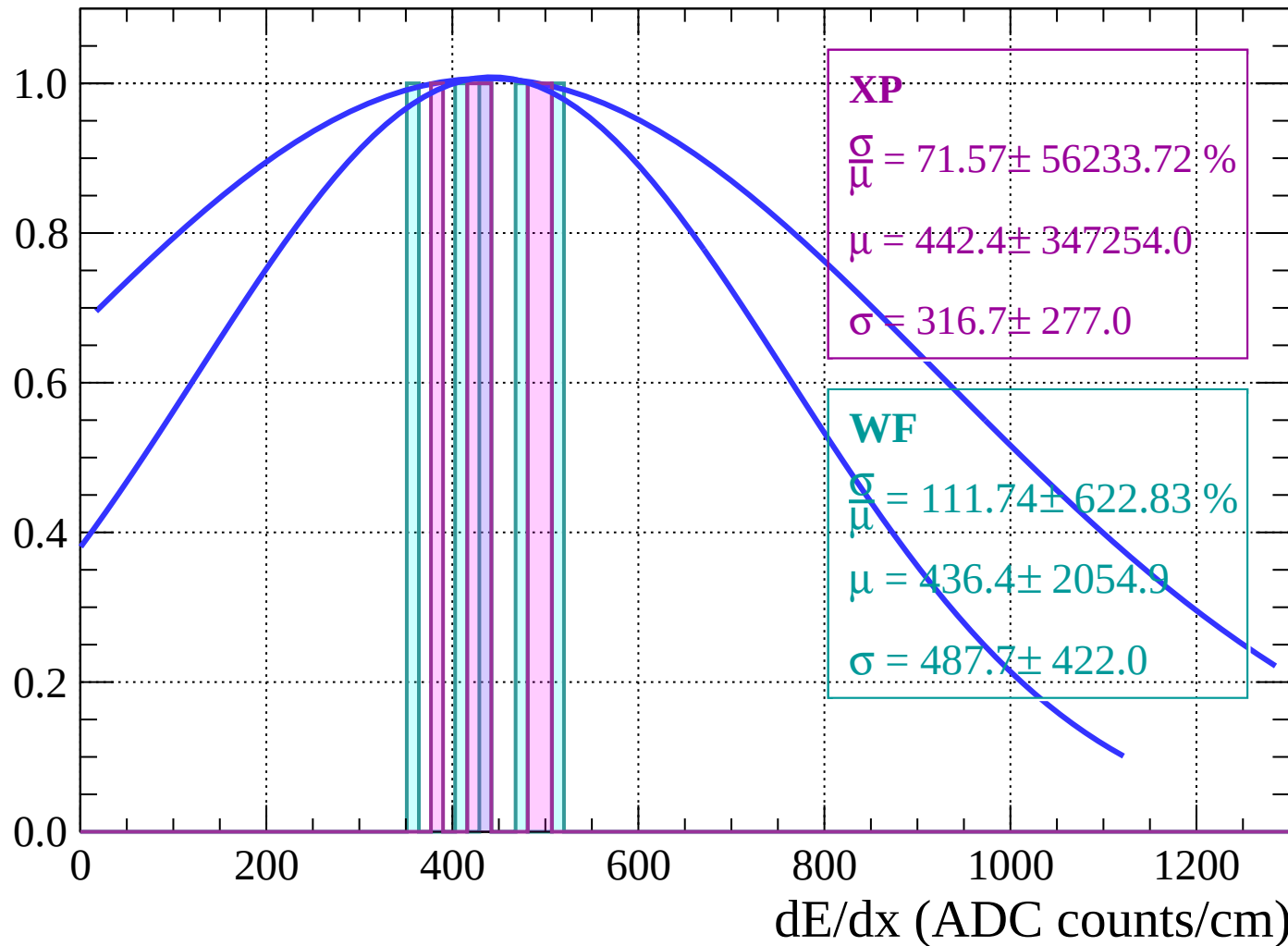
Energy loss |  $-58 < \theta < -56$

Count



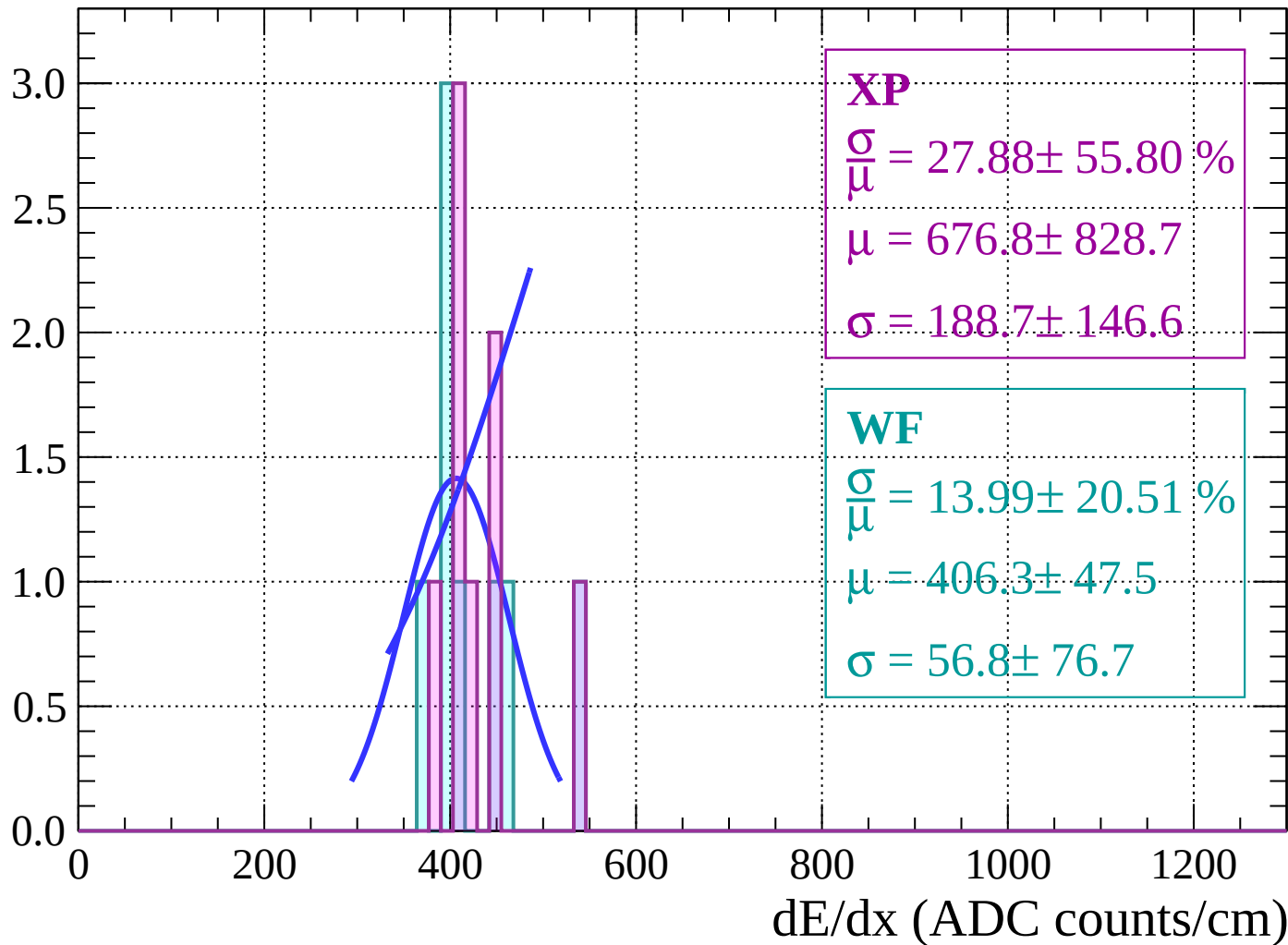
Energy loss  $|-56 < \theta < -54$

Count



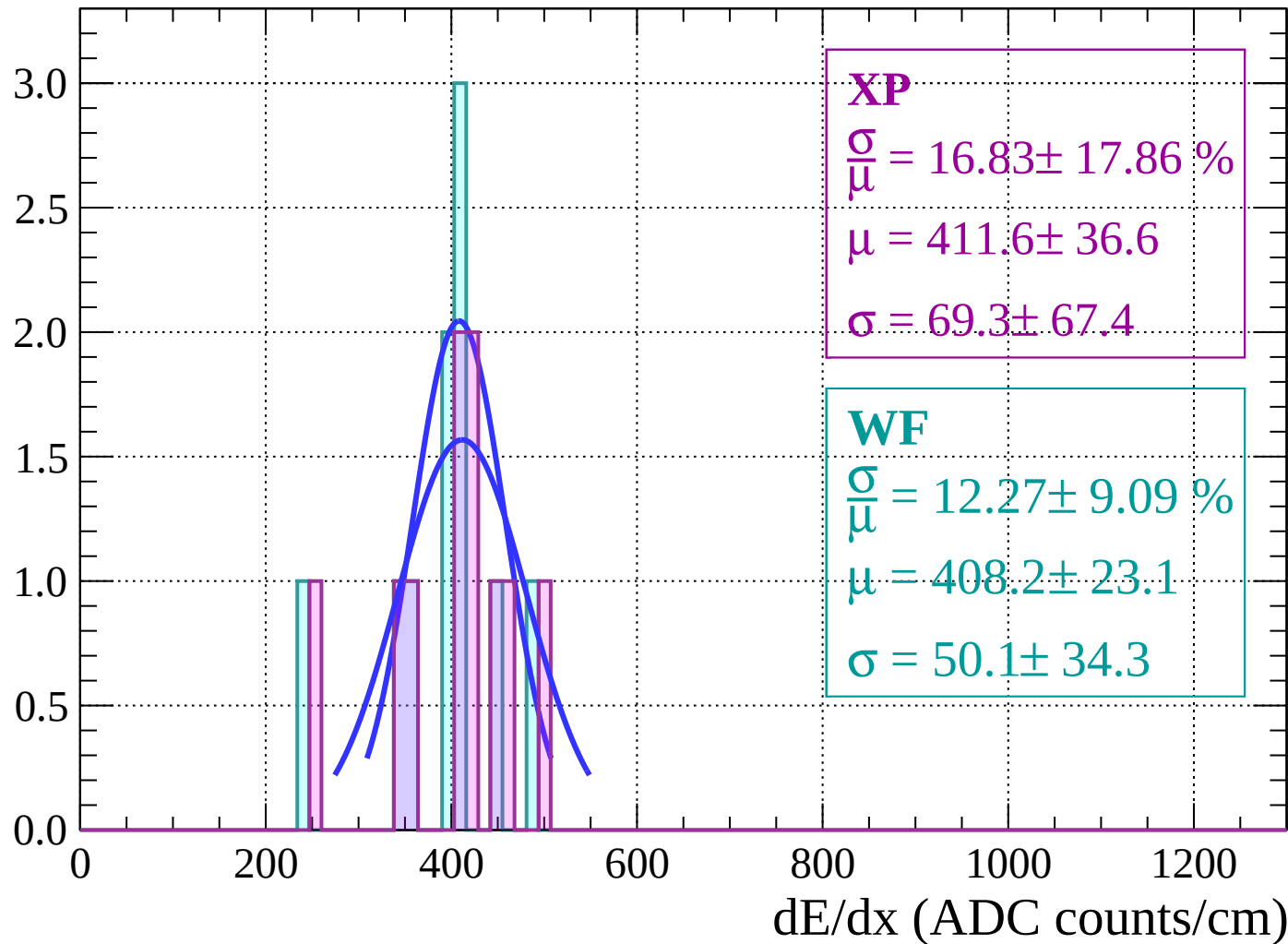
Energy loss |  $-54 < \theta < -52$

Count



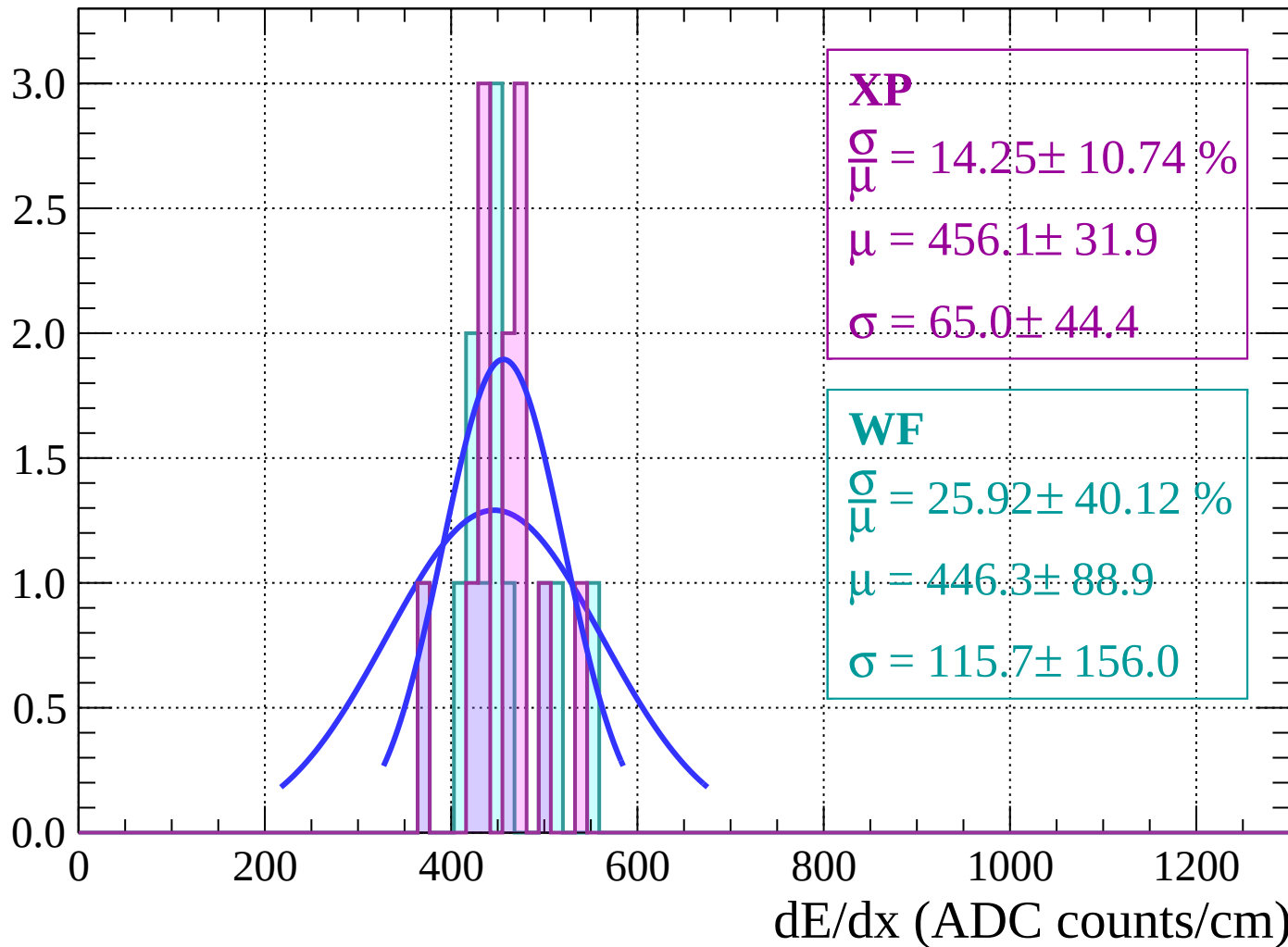
Energy loss |  $-52 < \theta < -50$

Count



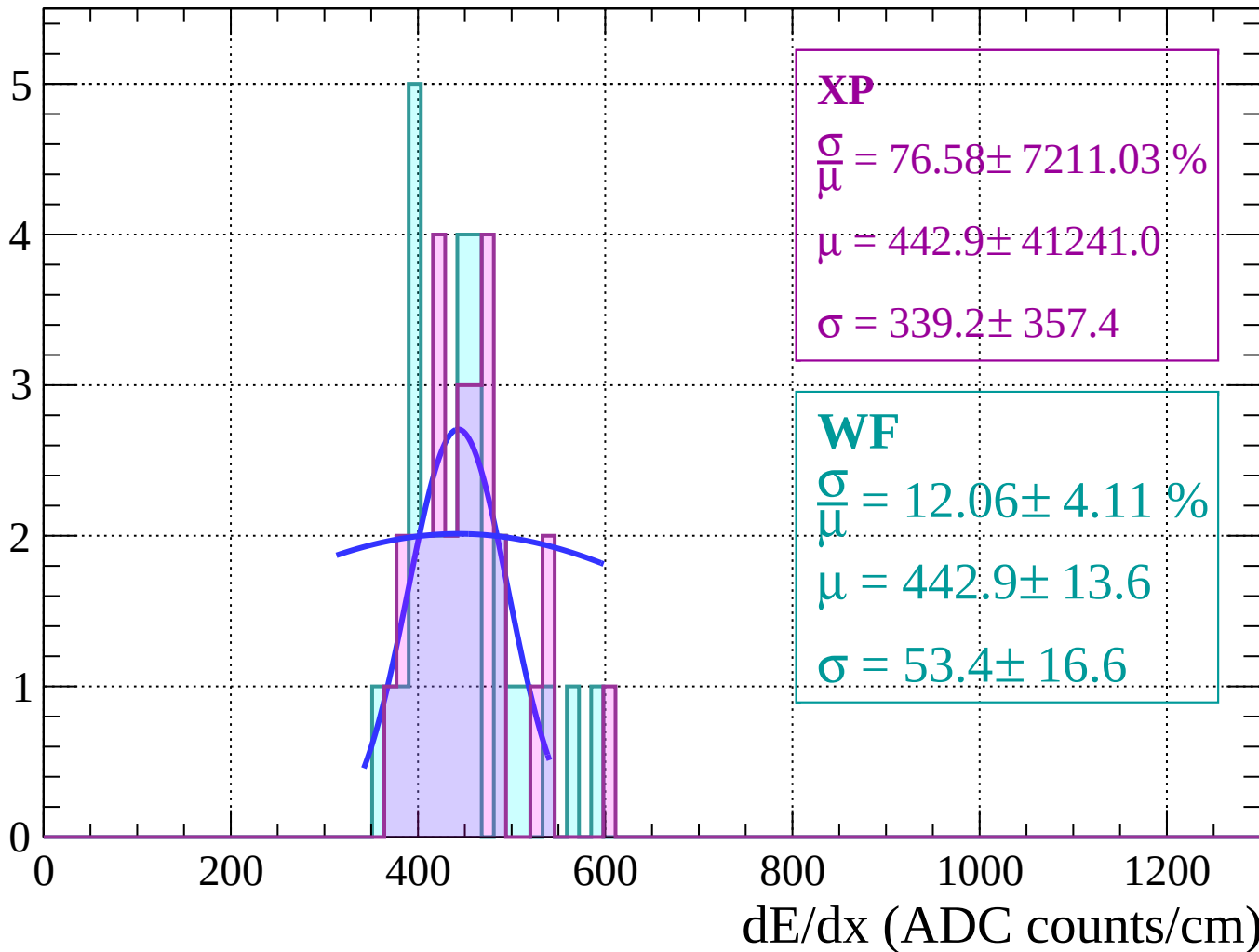
Energy loss |  $-50 < \theta < -48$

Count



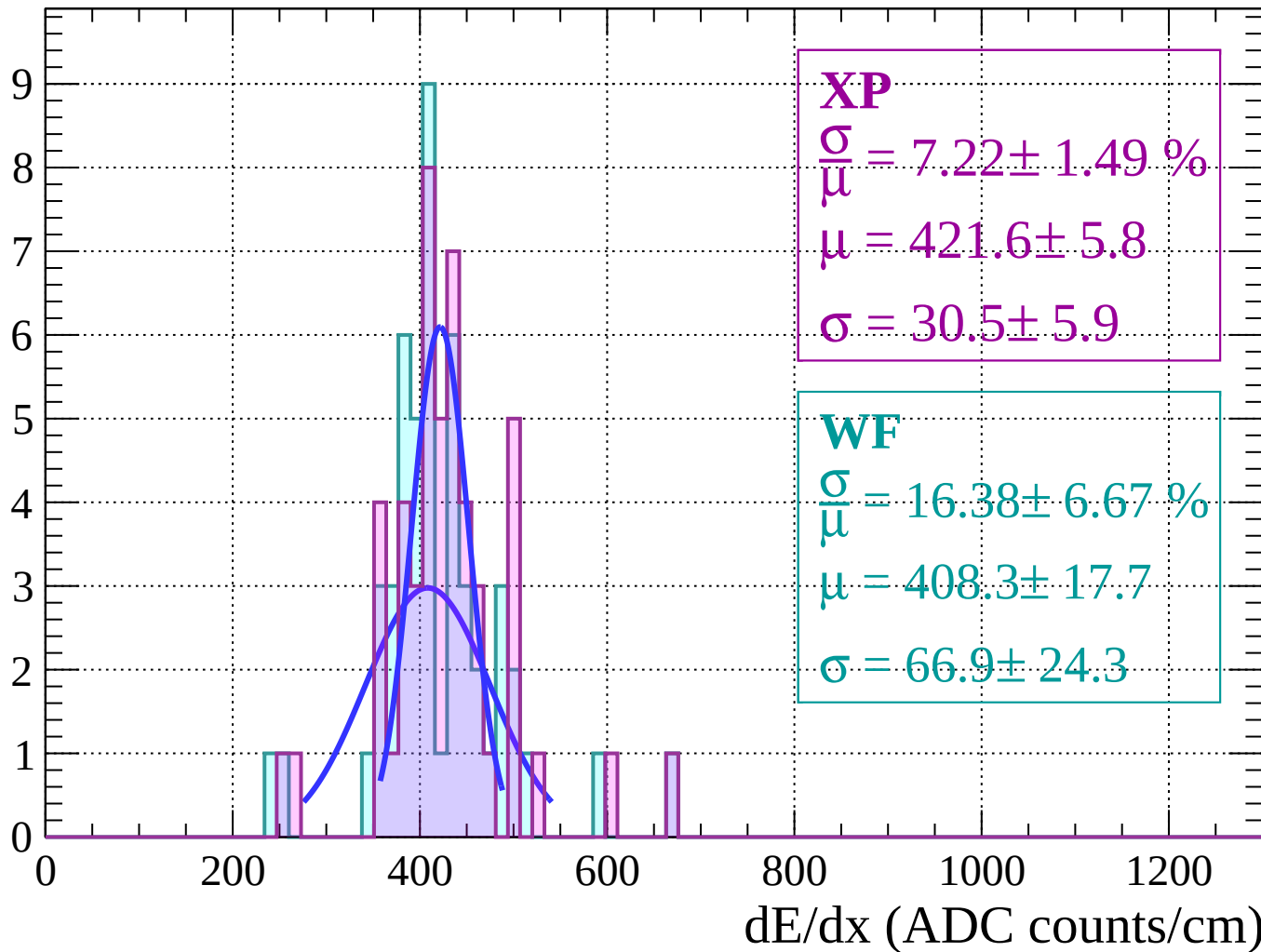
Energy loss |  $-48 < \theta < -46$

Count



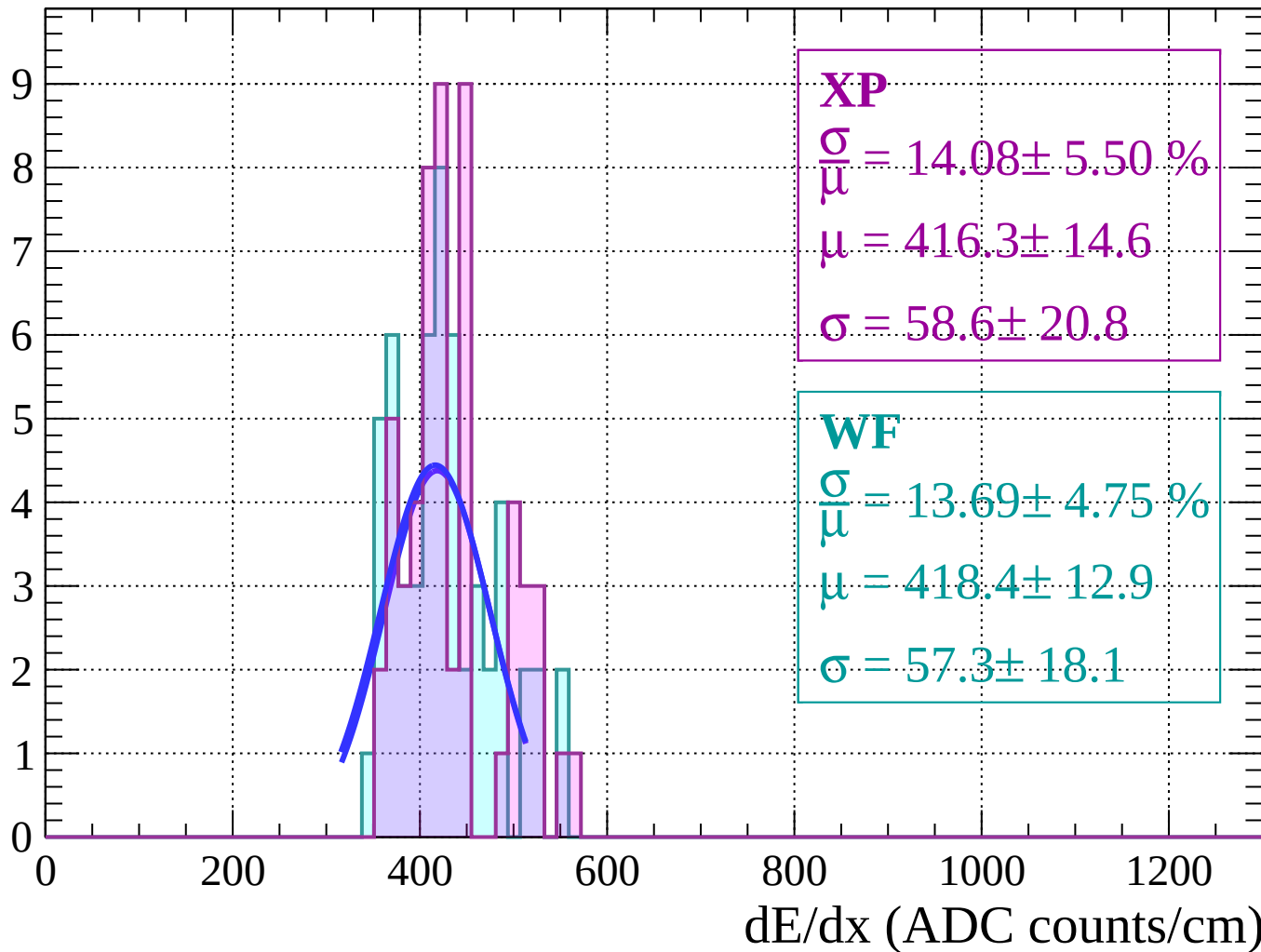
Energy loss |  $-46 < \theta < -44$

Count



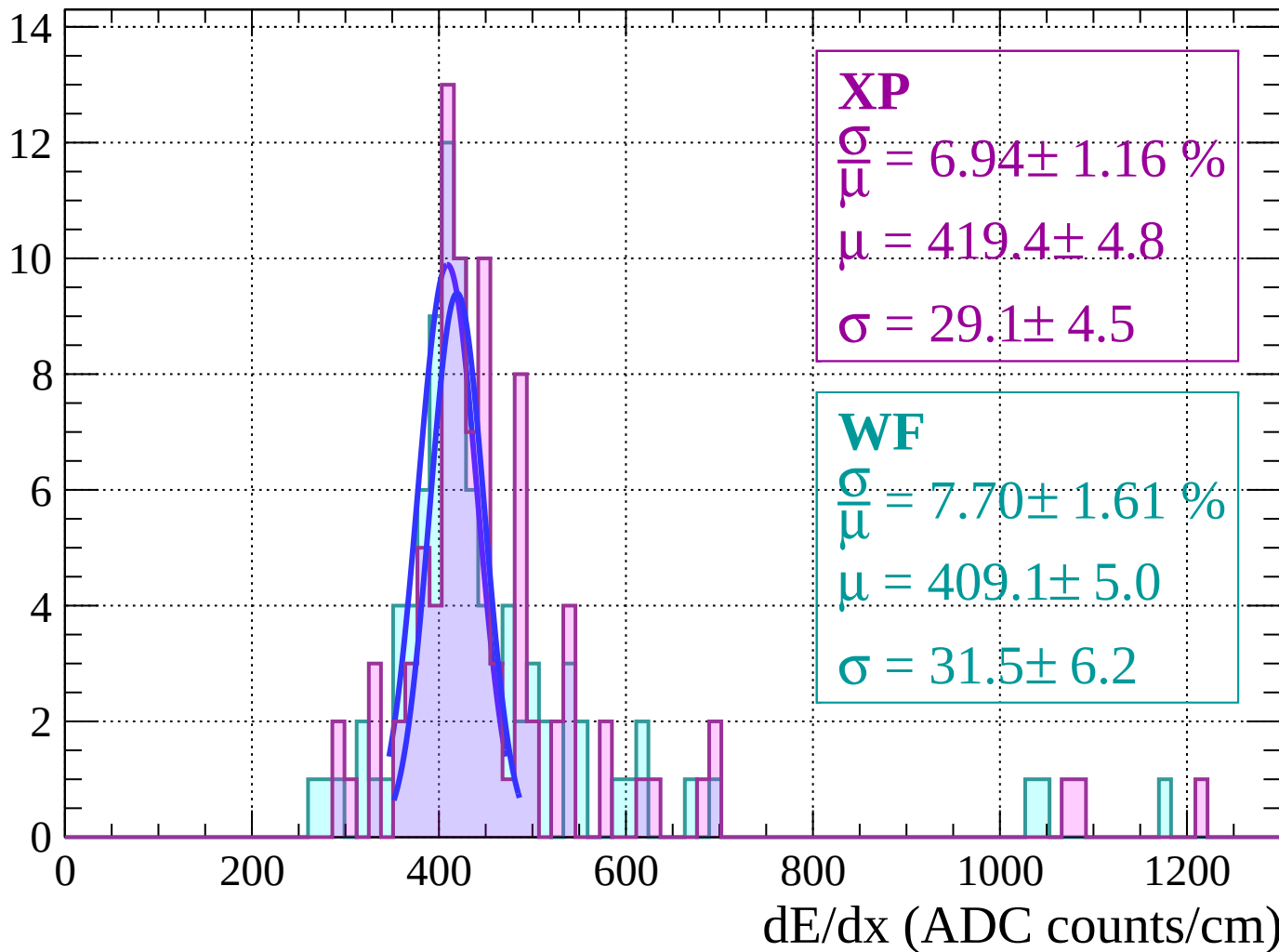
Energy loss |  $-44 < \theta < -42$

Count



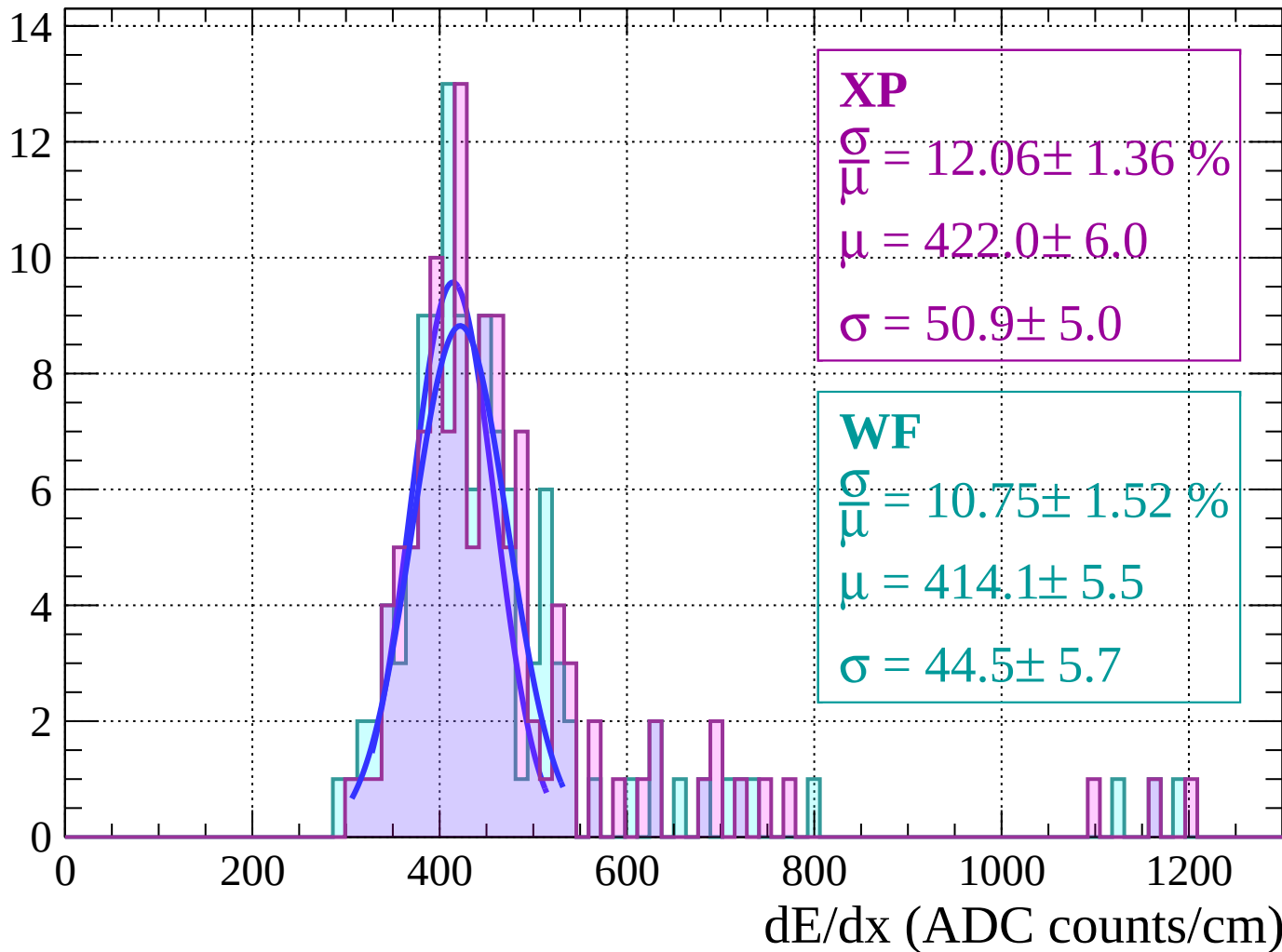
Energy loss |  $-42 < \theta < -40$

Count



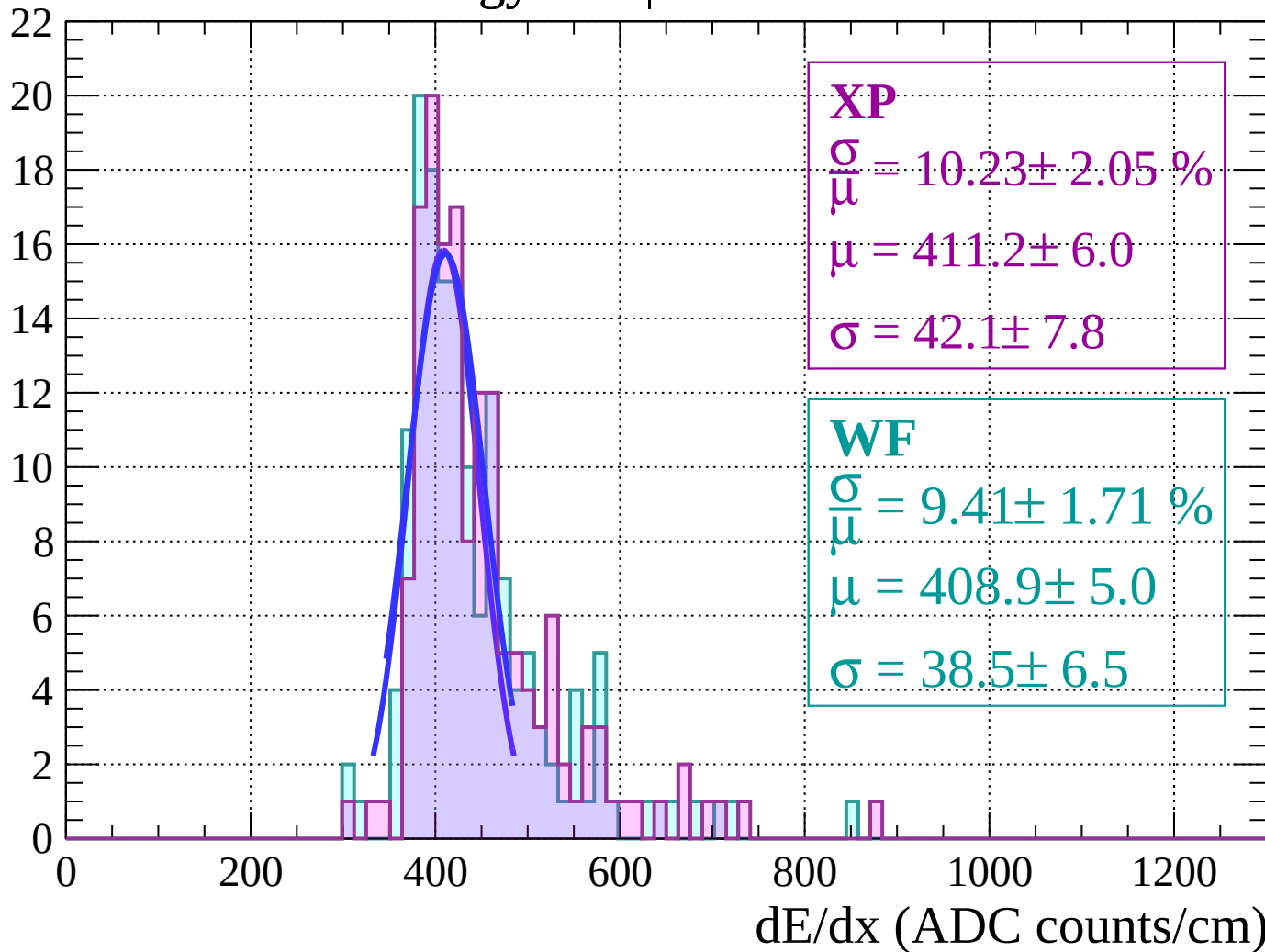
Energy loss |  $-40 < \theta < -38$

Count



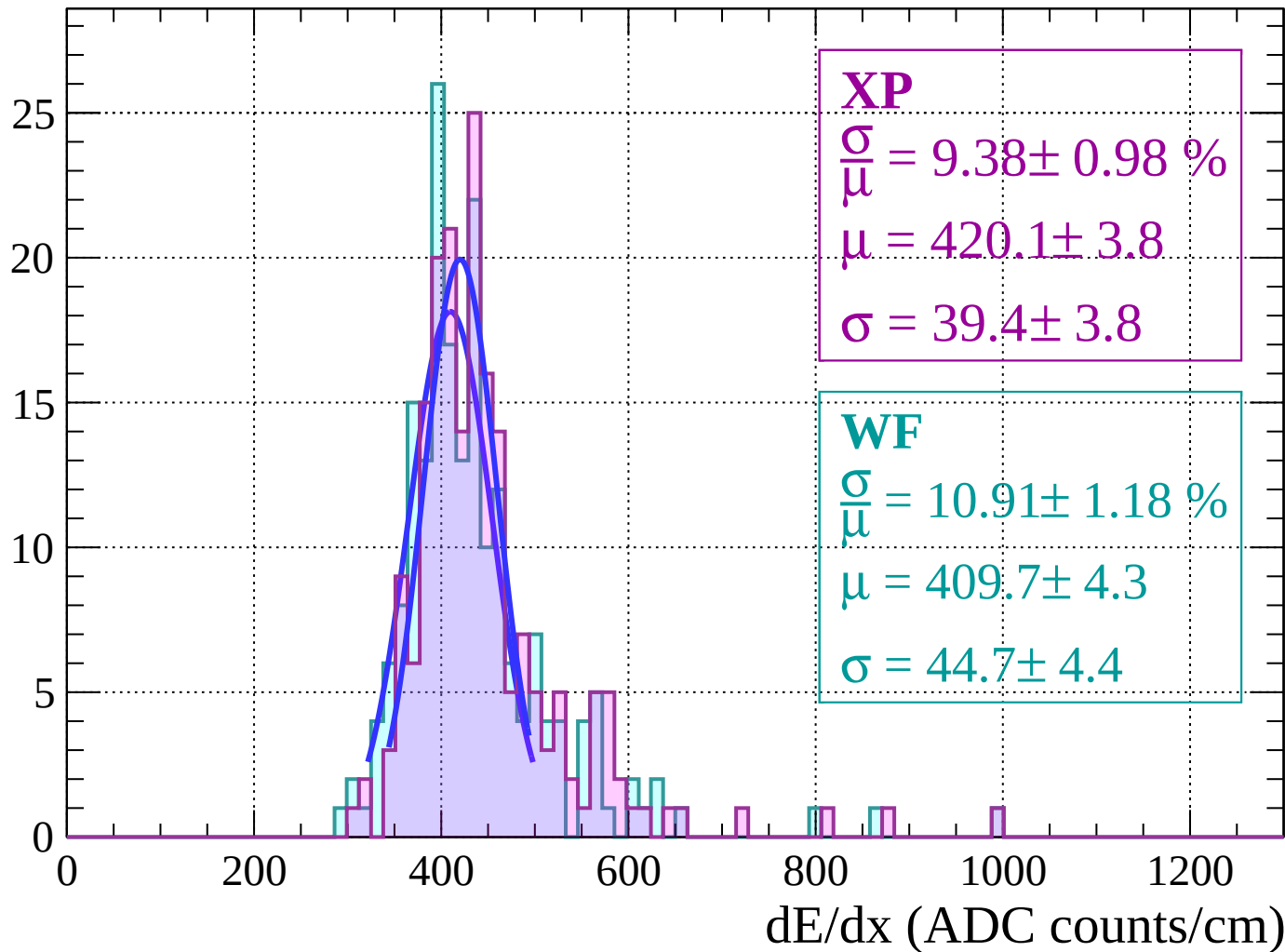
Energy loss |  $-38 < \theta < -36$

Count



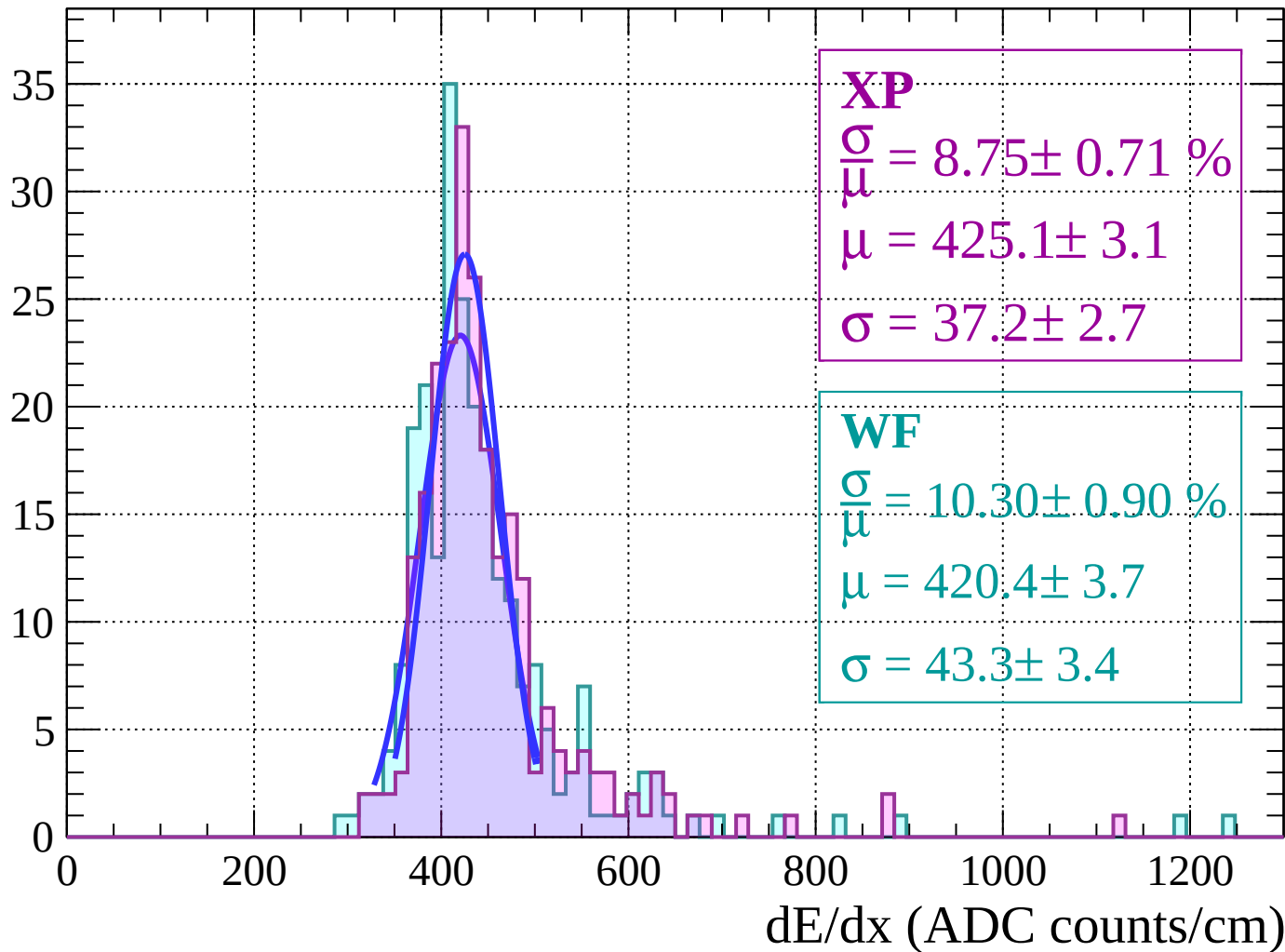
Energy loss  $|-36 < \theta < -34$

Count



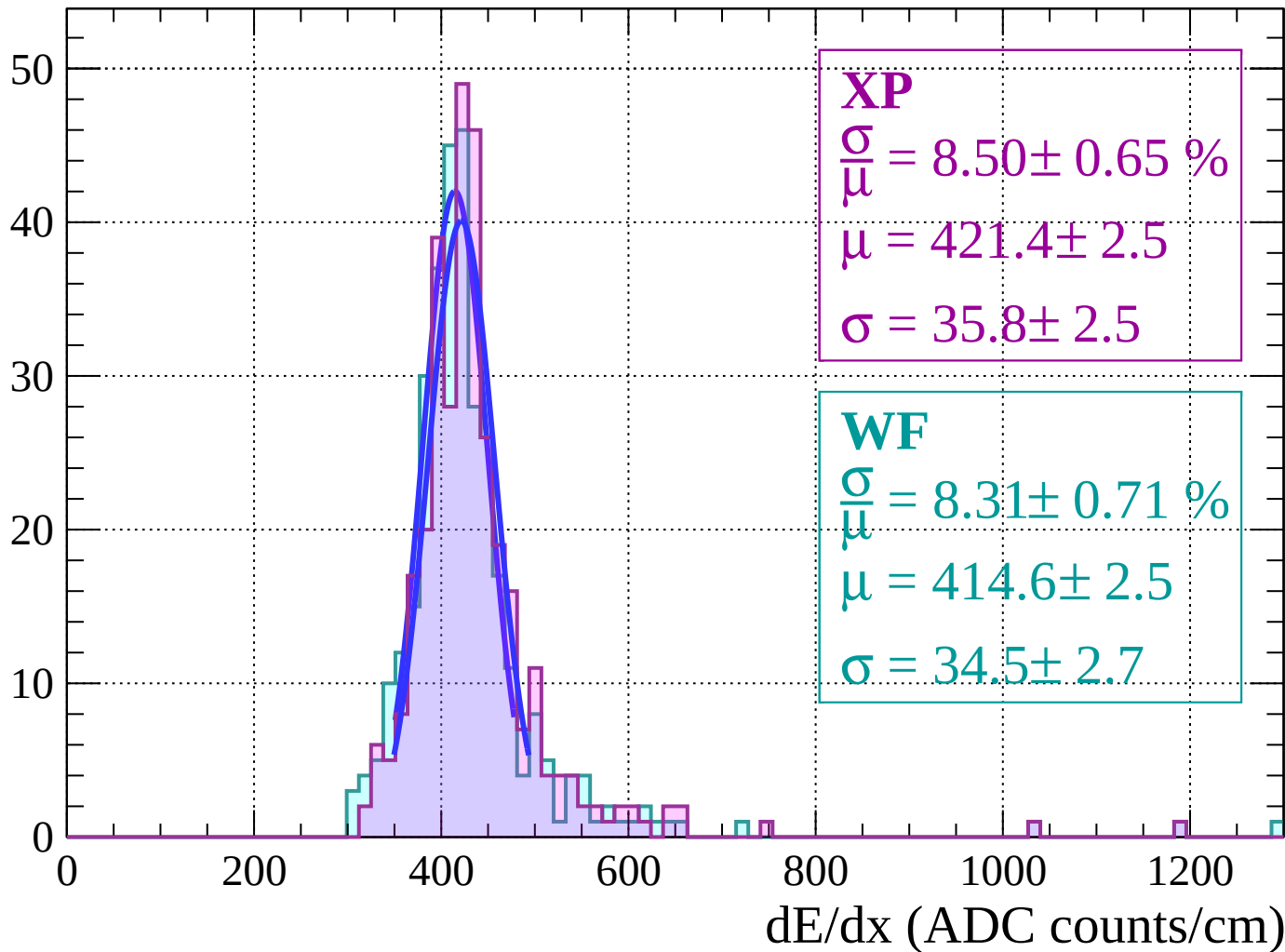
Energy loss  $|-34 < \theta < -32$

Count



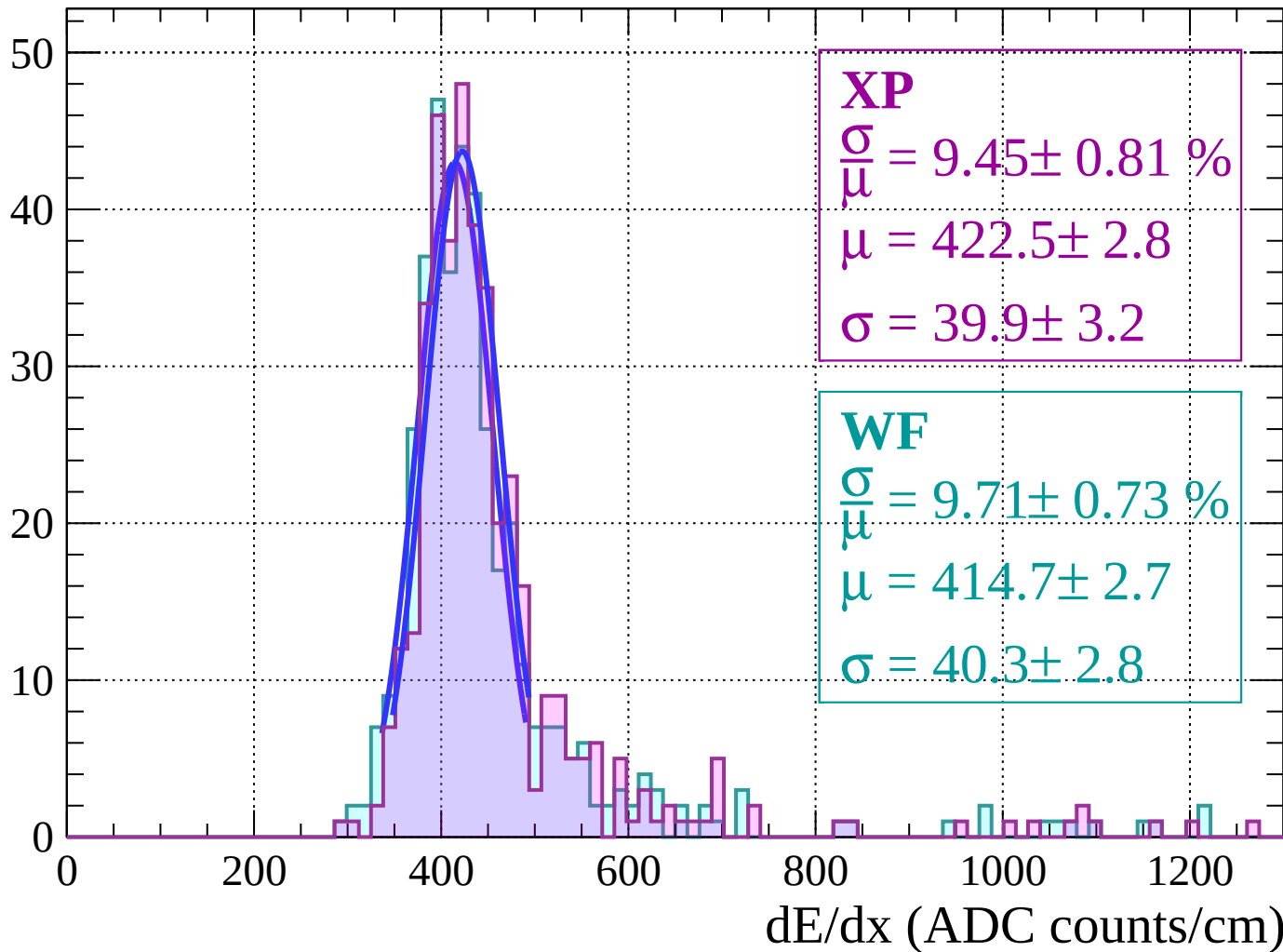
Energy loss |  $-32 < \theta < -30$

Count



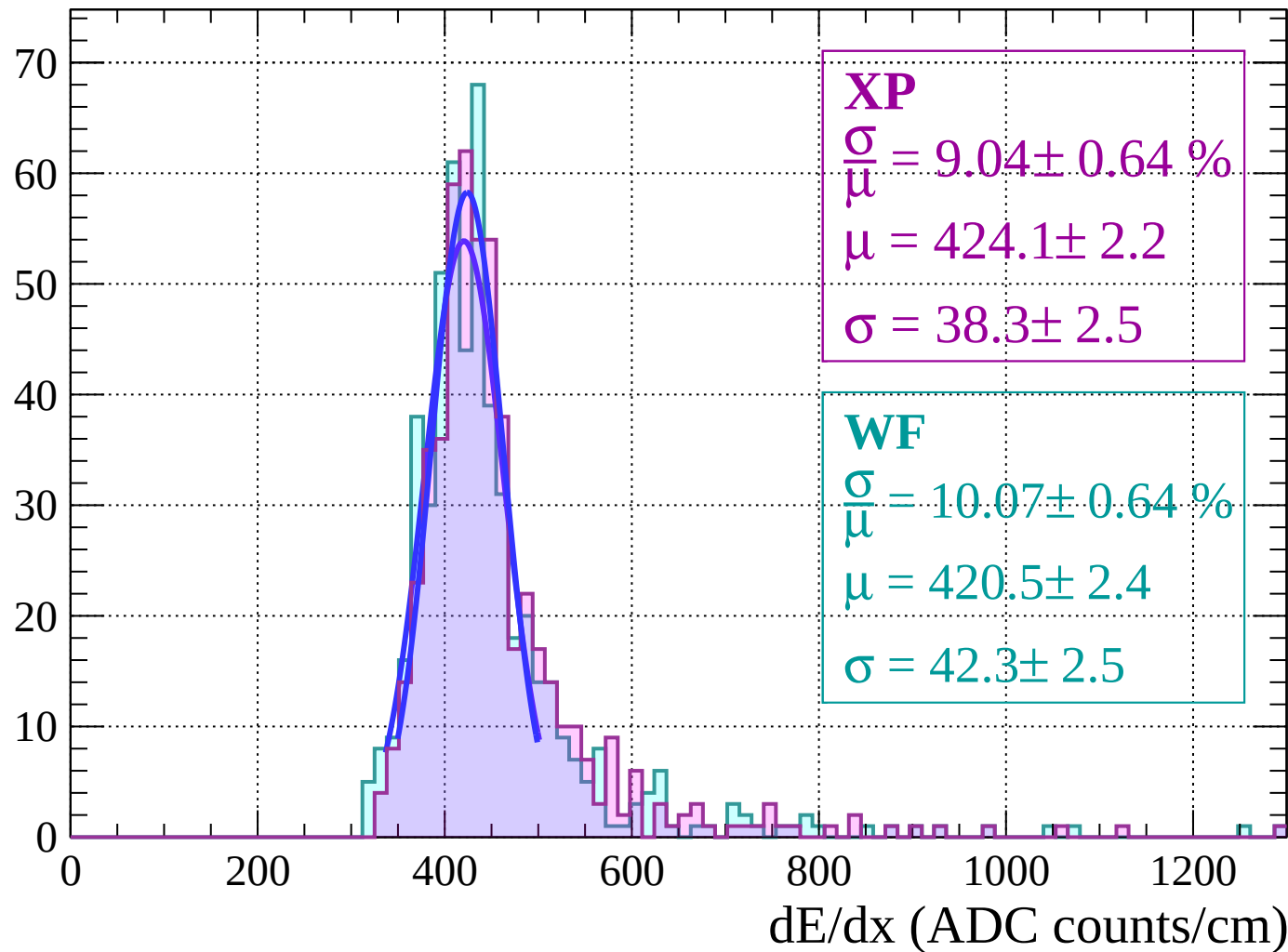
Energy loss |  $-30 < \theta < -28$

Count



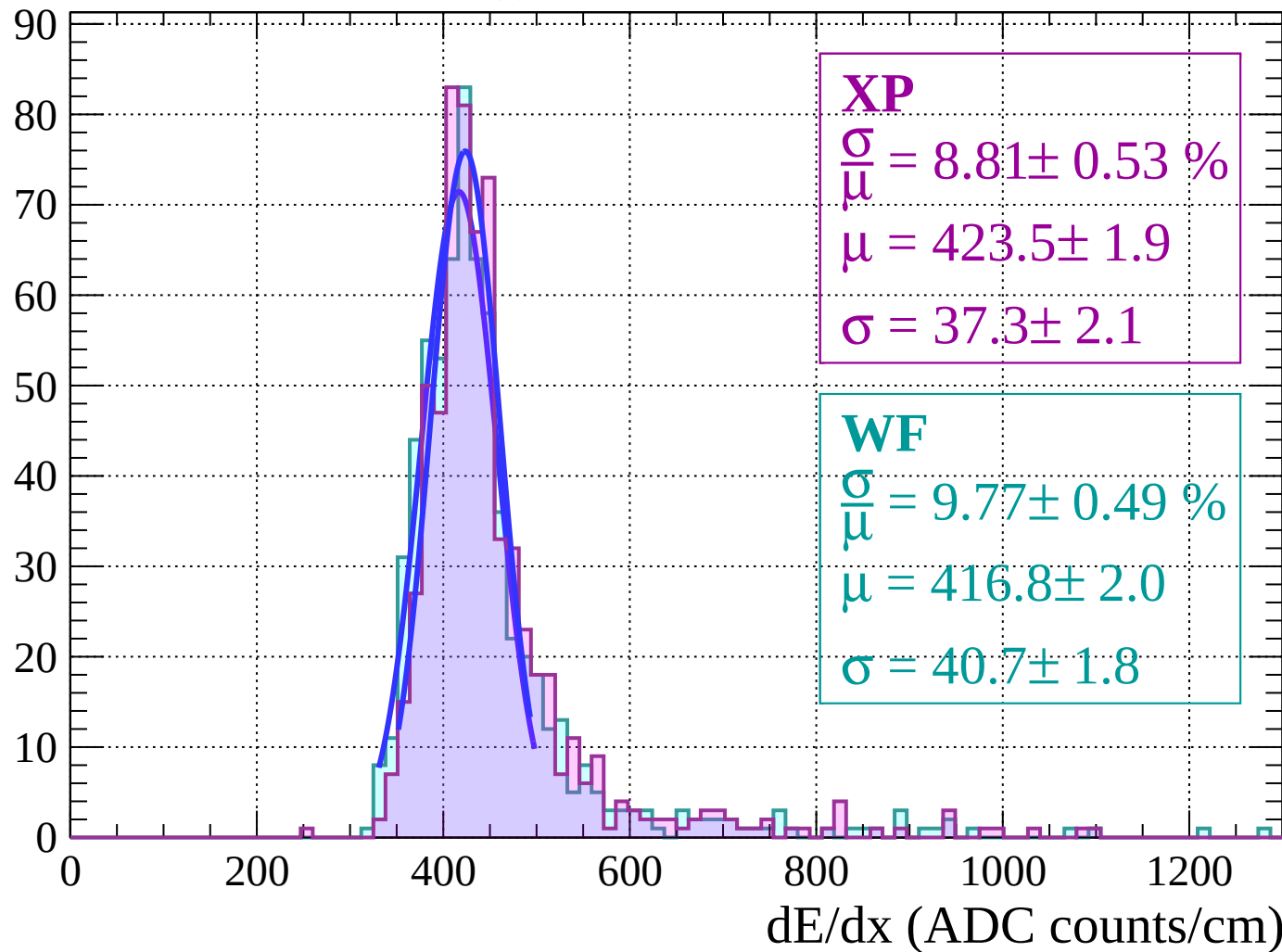
Energy loss  $|-28 < \theta < -26$

Count



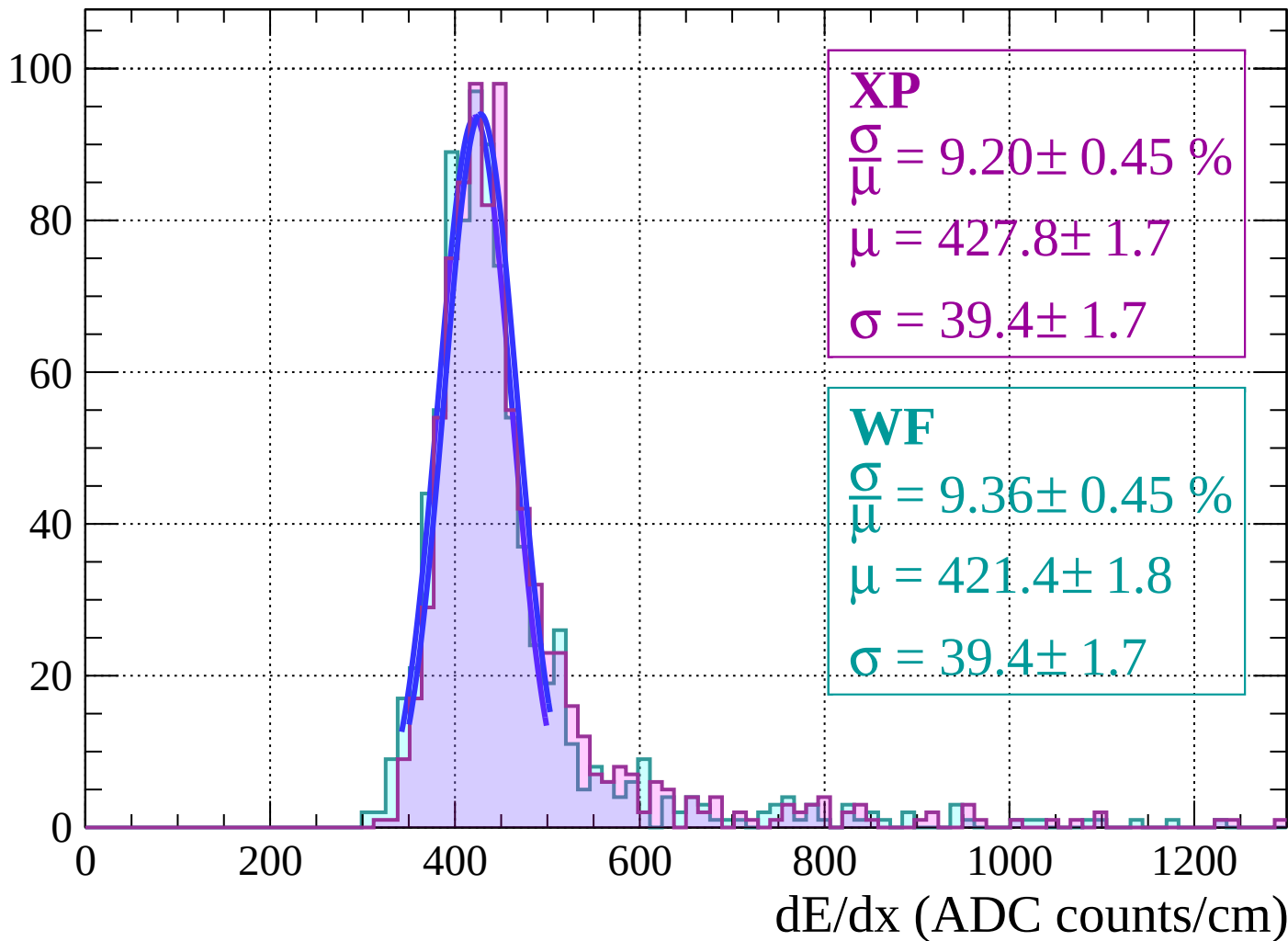
Energy loss |  $-26 < \theta < -24$

Count



Energy loss  $|-24 < \theta < -22$

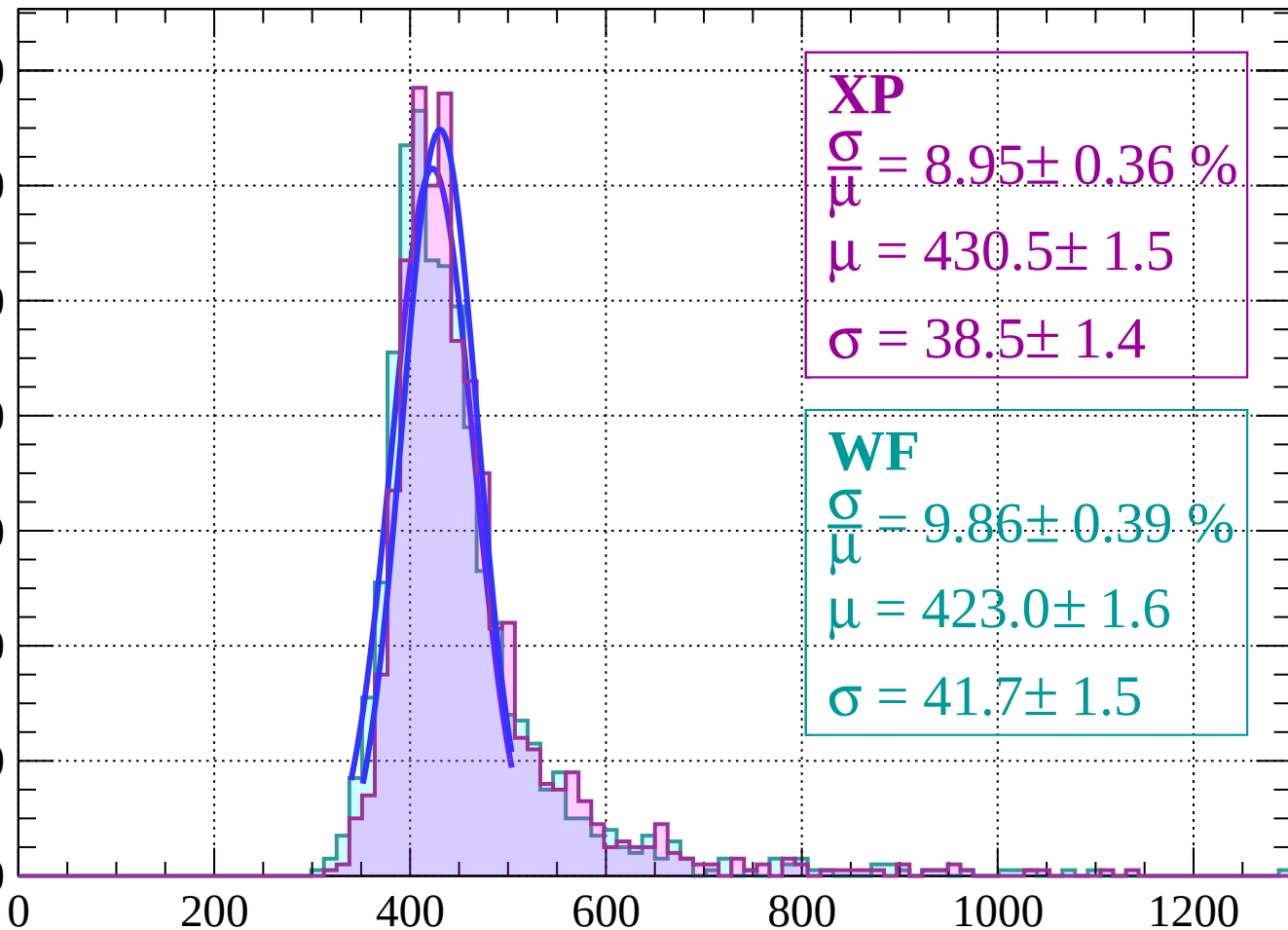
Count



Energy loss  $|-22 < \theta < -20$

Count

140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 8.95 \pm 0.36 \%$$

$$\mu = 430.5 \pm 1.5$$

$$\sigma = 38.5 \pm 1.4$$

**WF**

$$\frac{\sigma}{\mu} = 9.86 \pm 0.39 \%$$

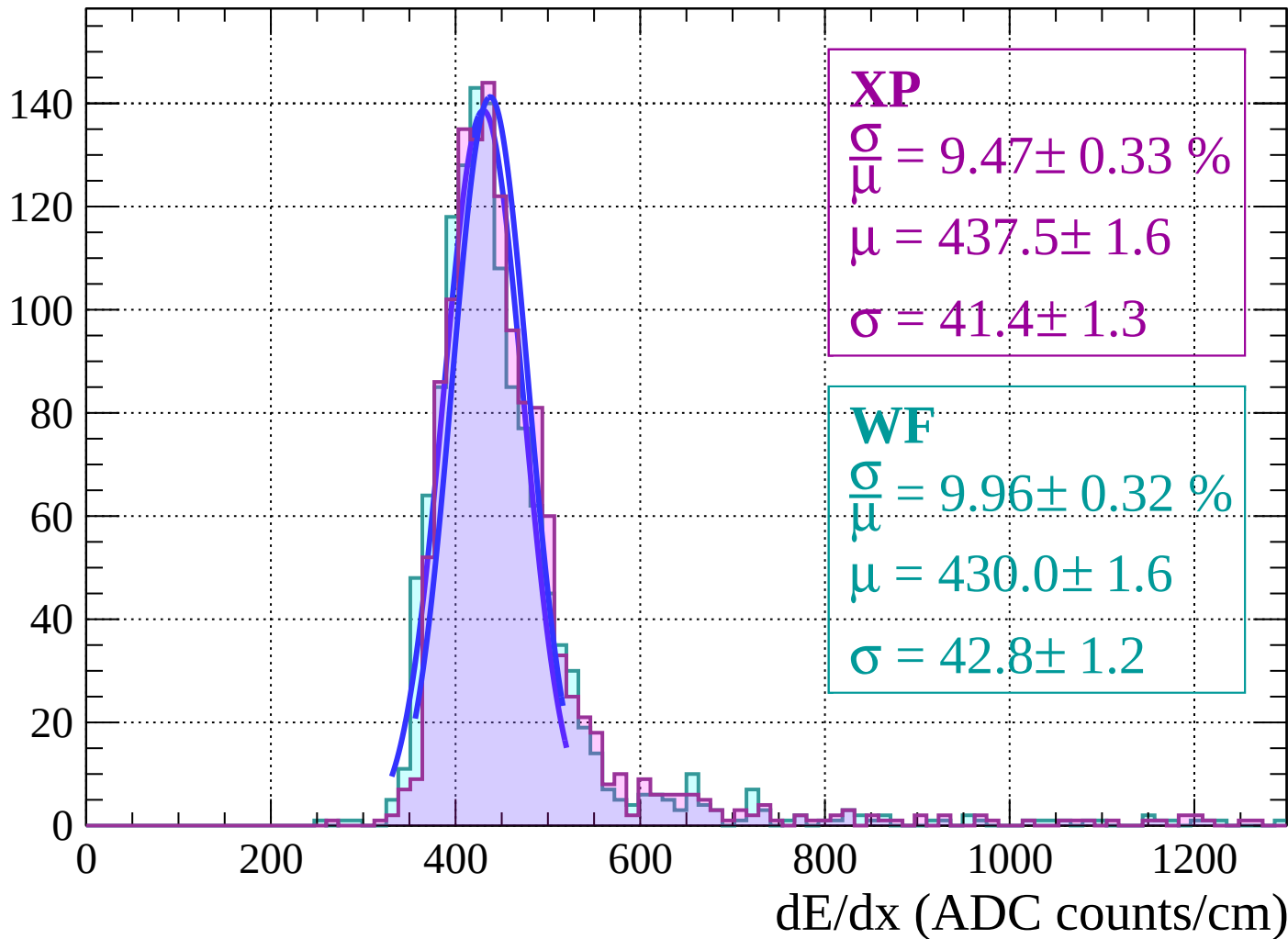
$$\mu = 423.0 \pm 1.6$$

$$\sigma = 41.7 \pm 1.5$$

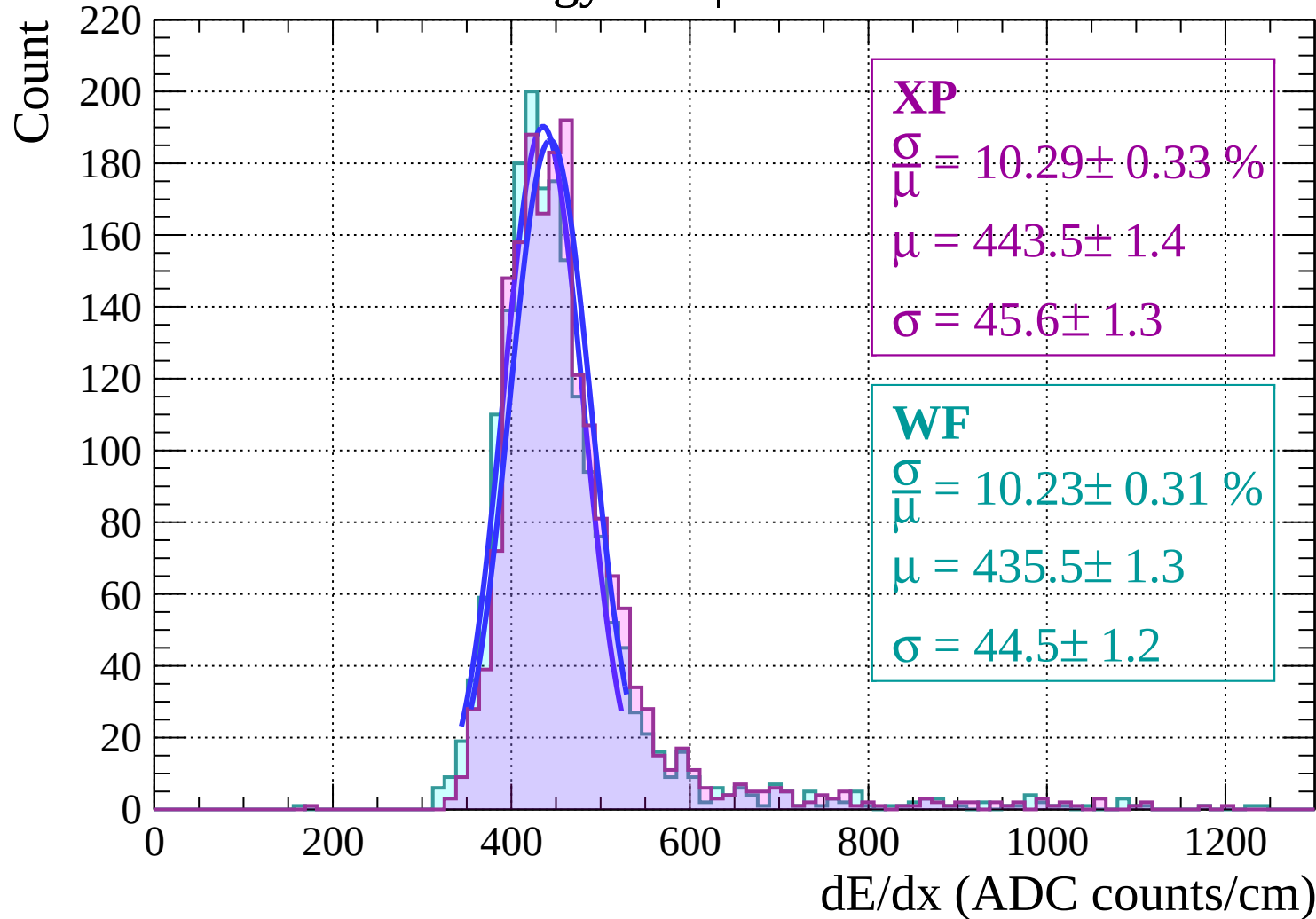
dE/dx (ADC counts/cm)

Energy loss |  $-20 < \theta < -18$

Count

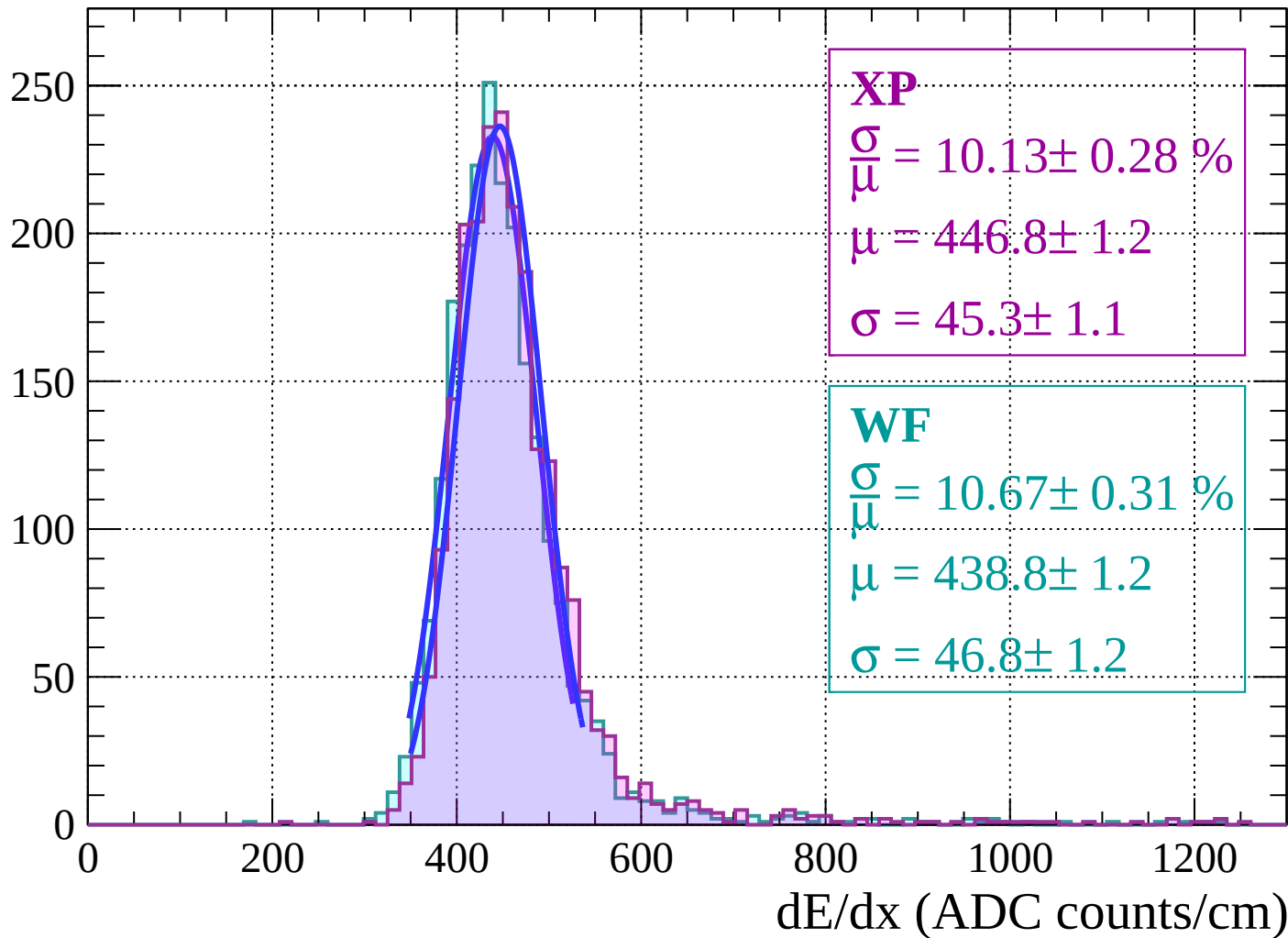


# Energy loss | $-18 < \theta < -16$

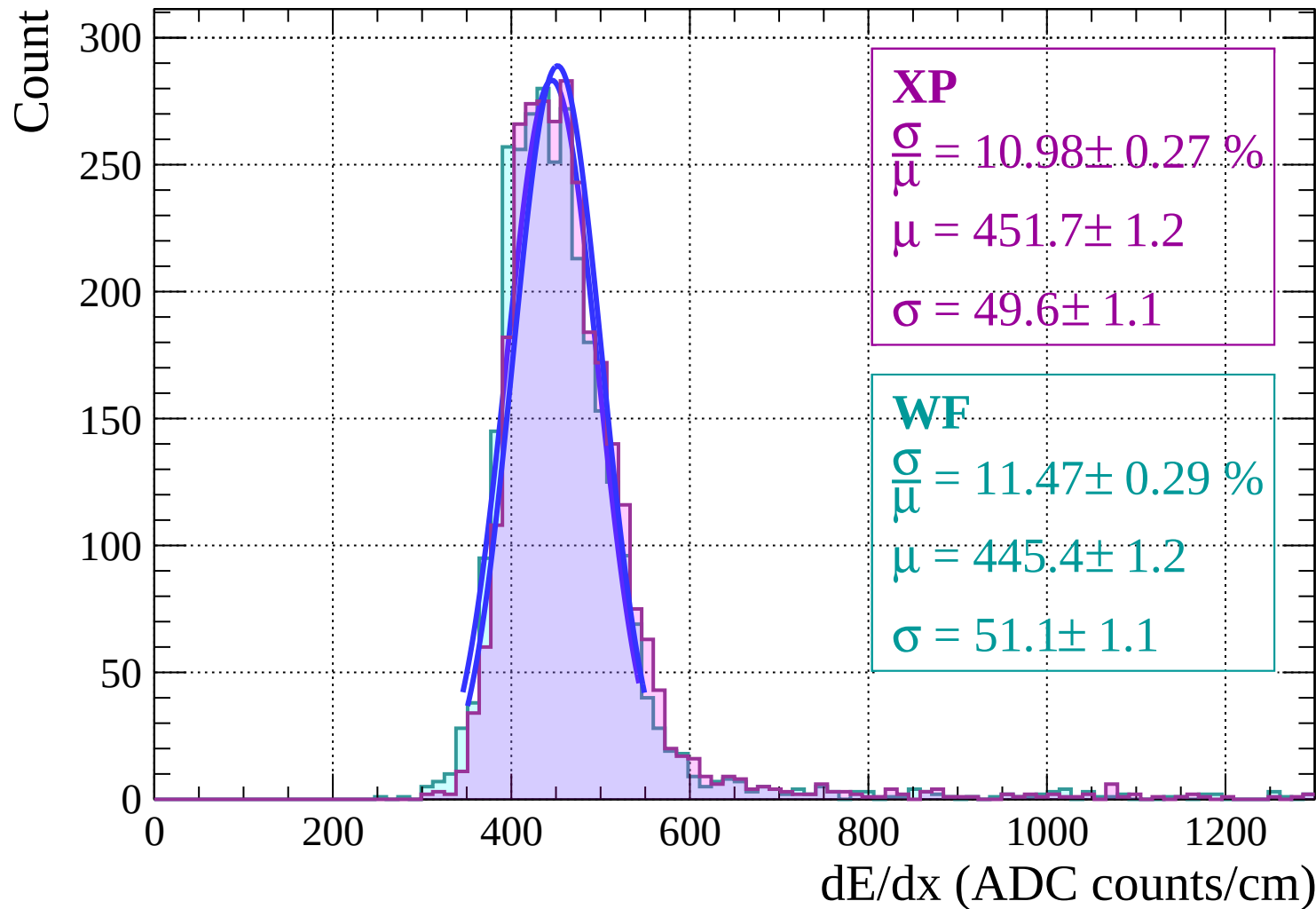


Energy loss |  $-16 < \theta < -14$

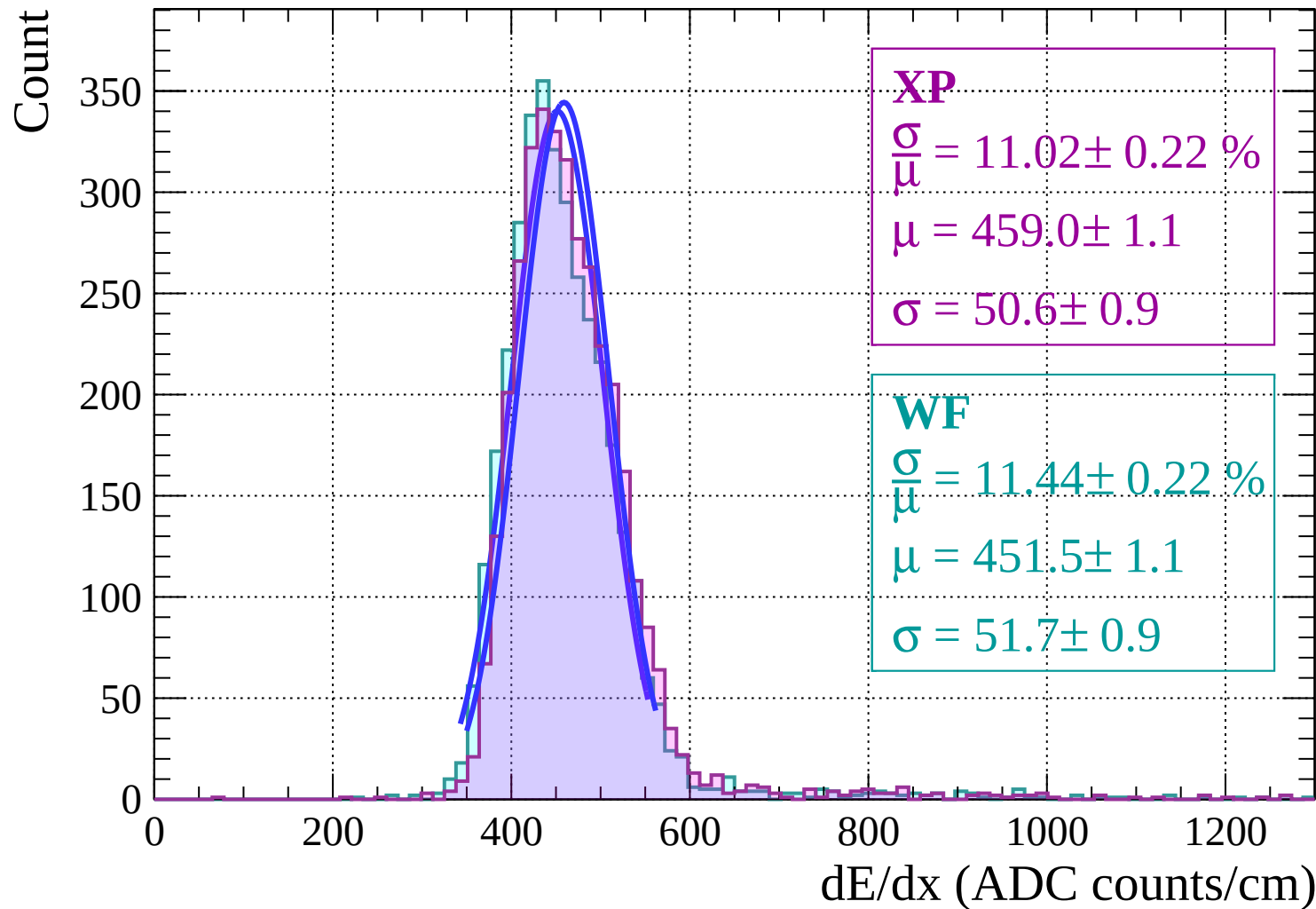
Count



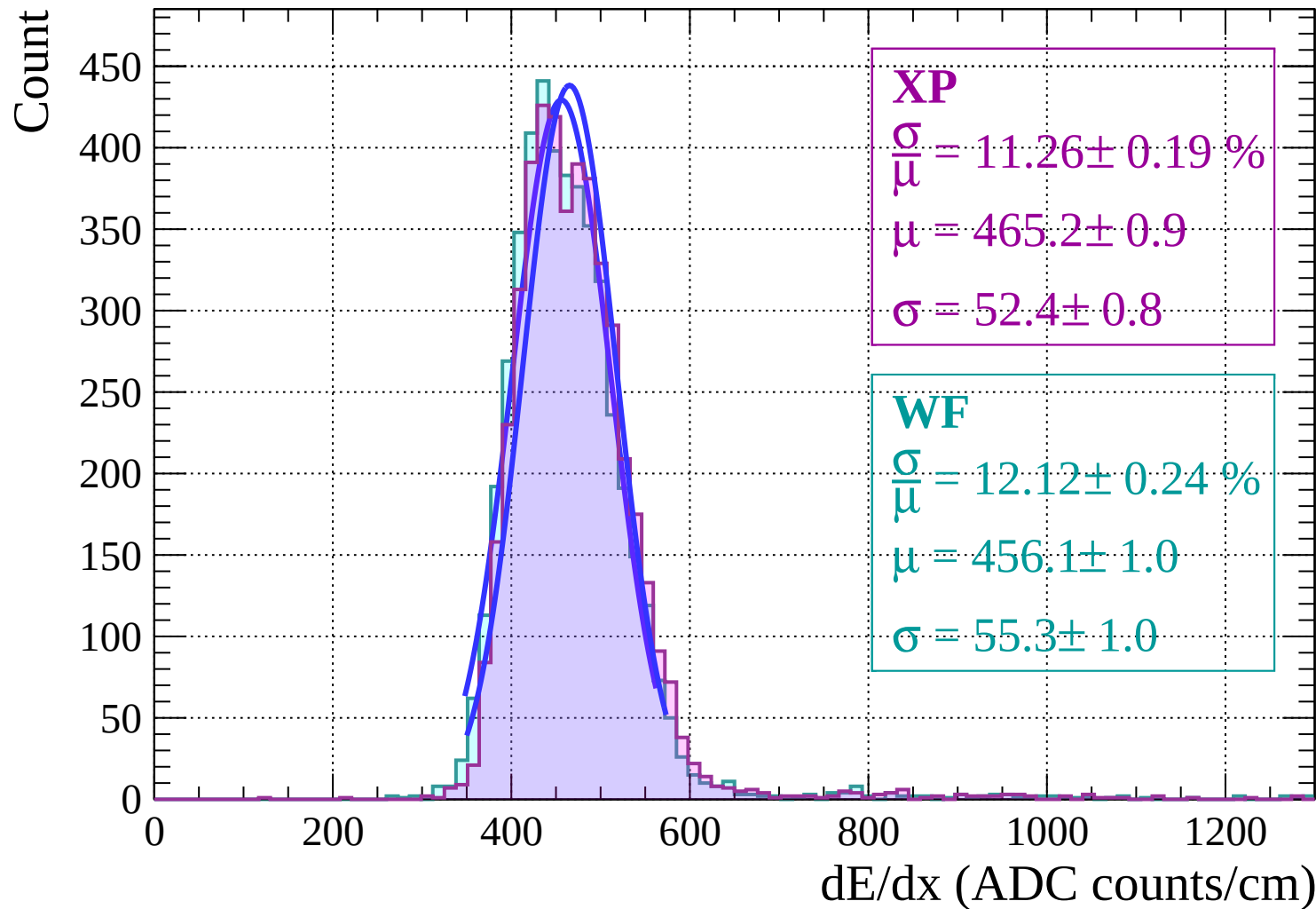
# Energy loss | $-14 < \theta < -12$



# Energy loss | $-12 < \theta < -10$

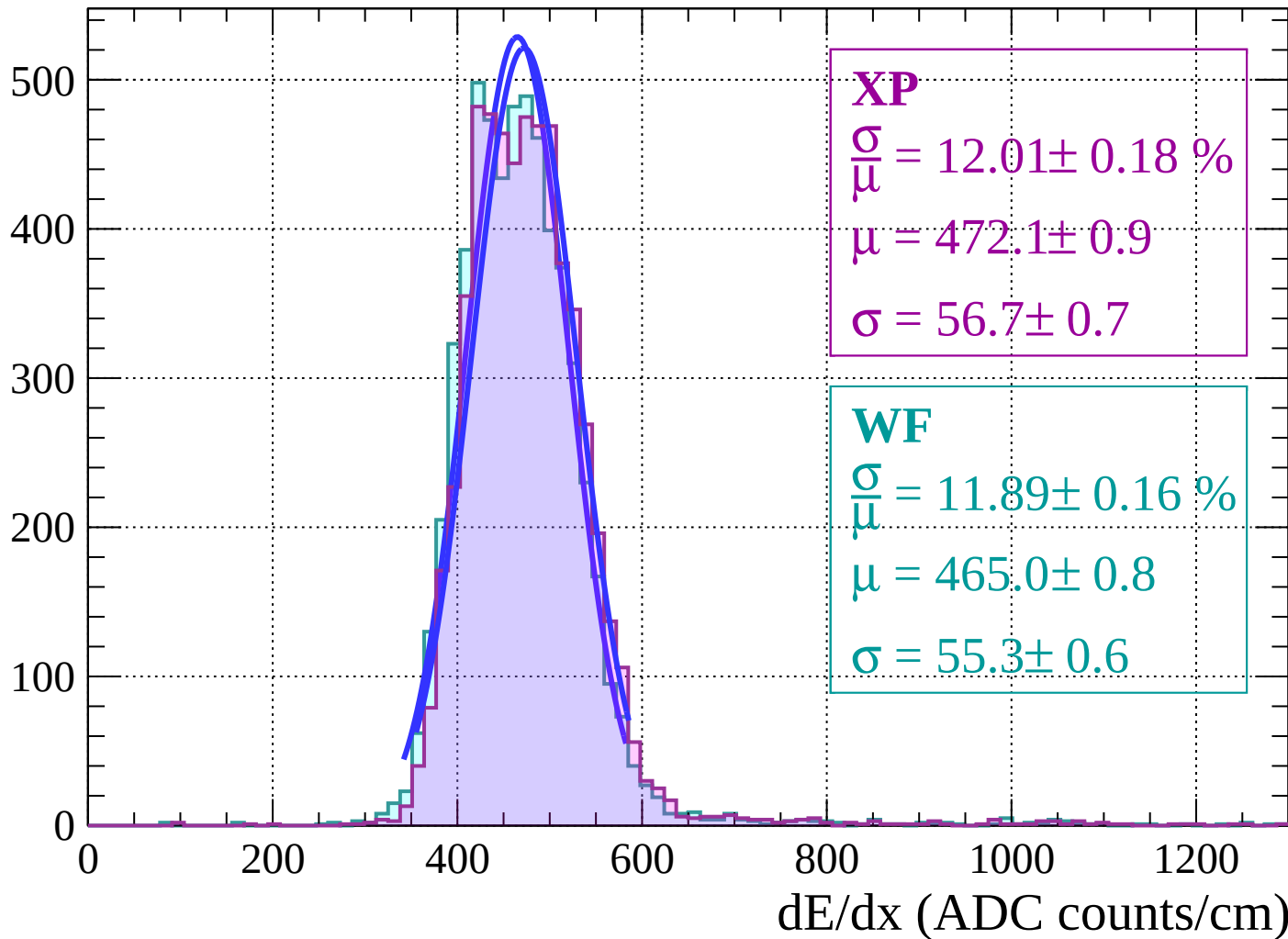


Energy loss |  $-10 < \theta < -8$



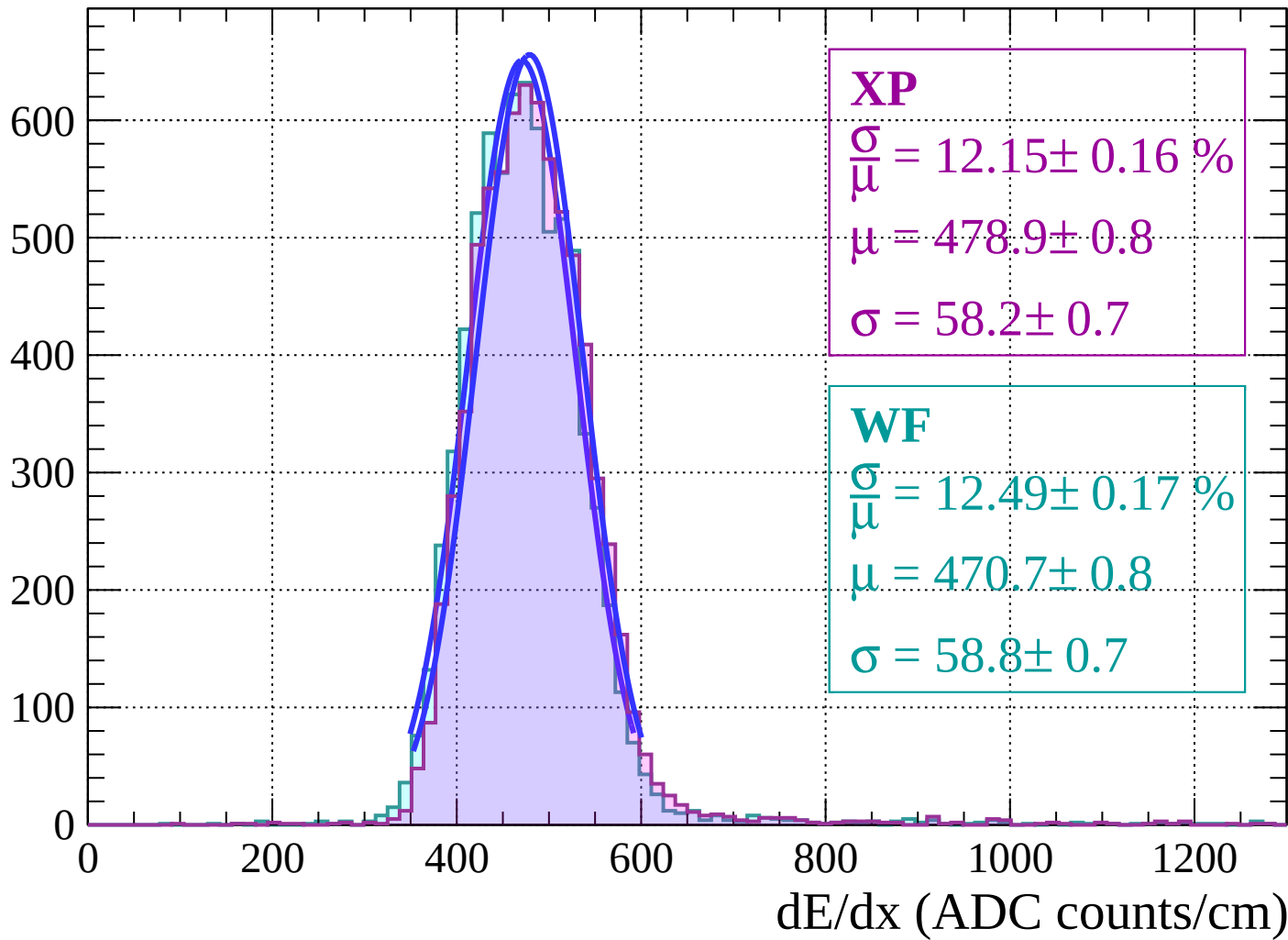
Energy loss  $-8 < \theta < -6$

Count

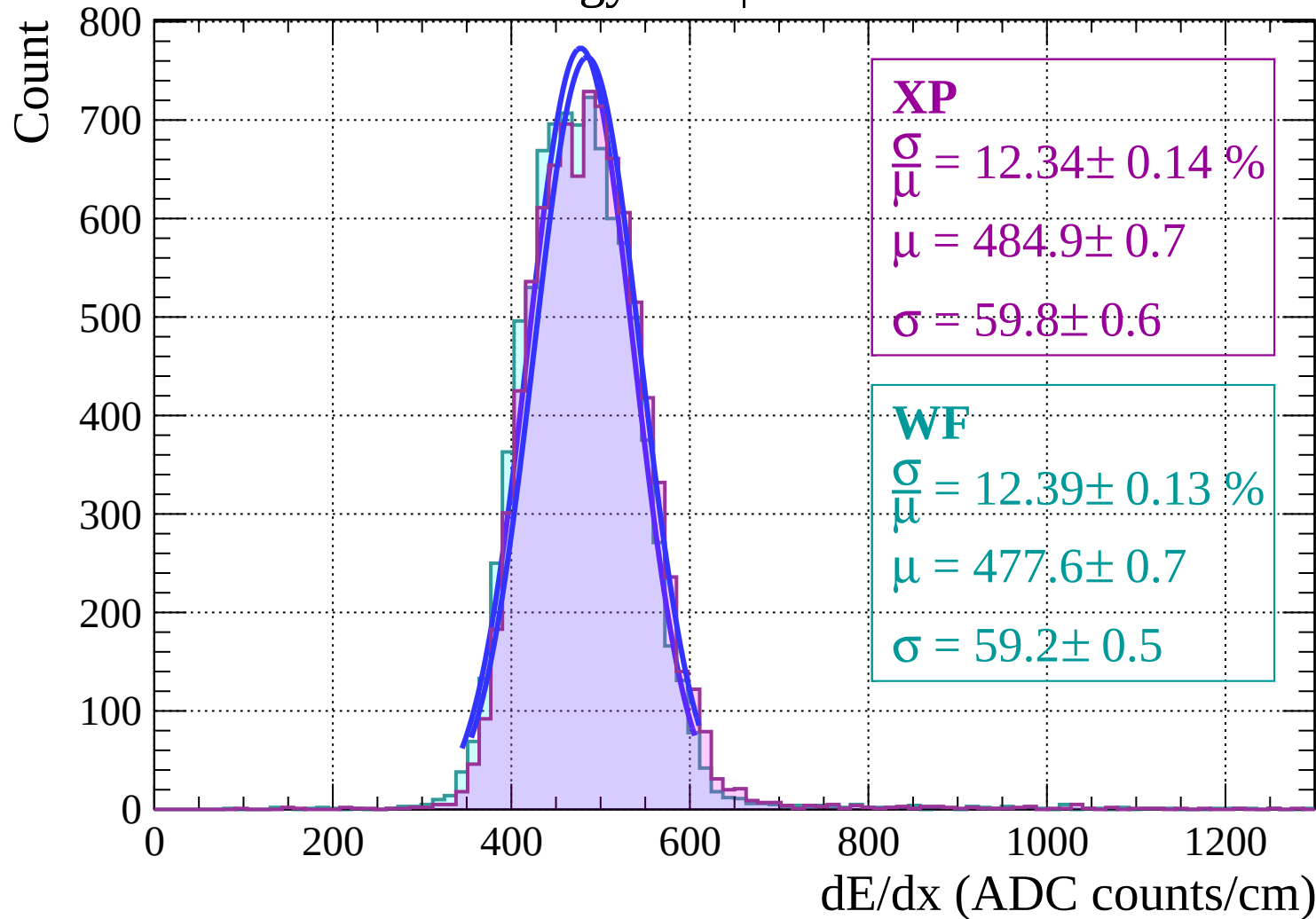


Energy loss  $-6 < \theta < -4$

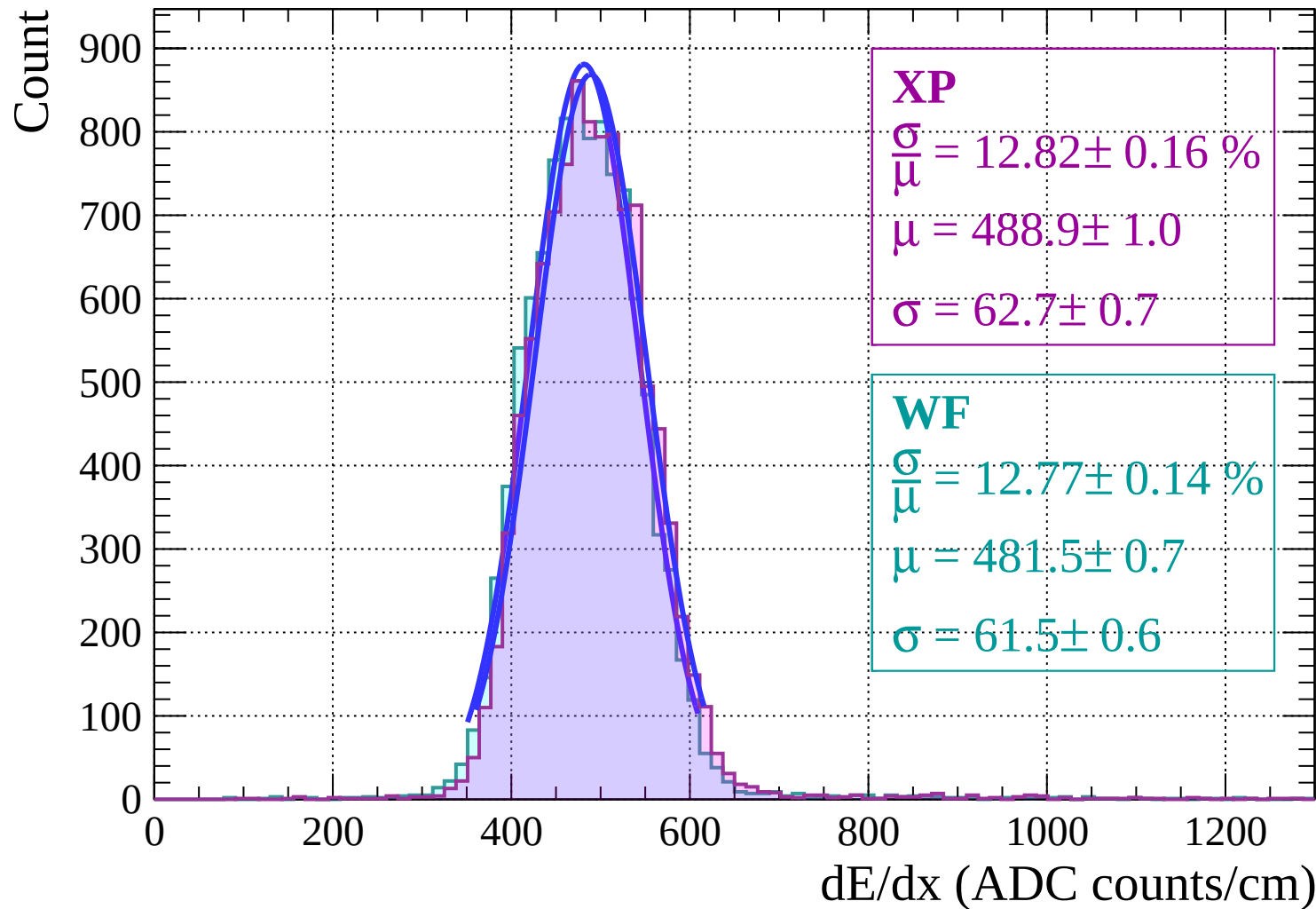
Count



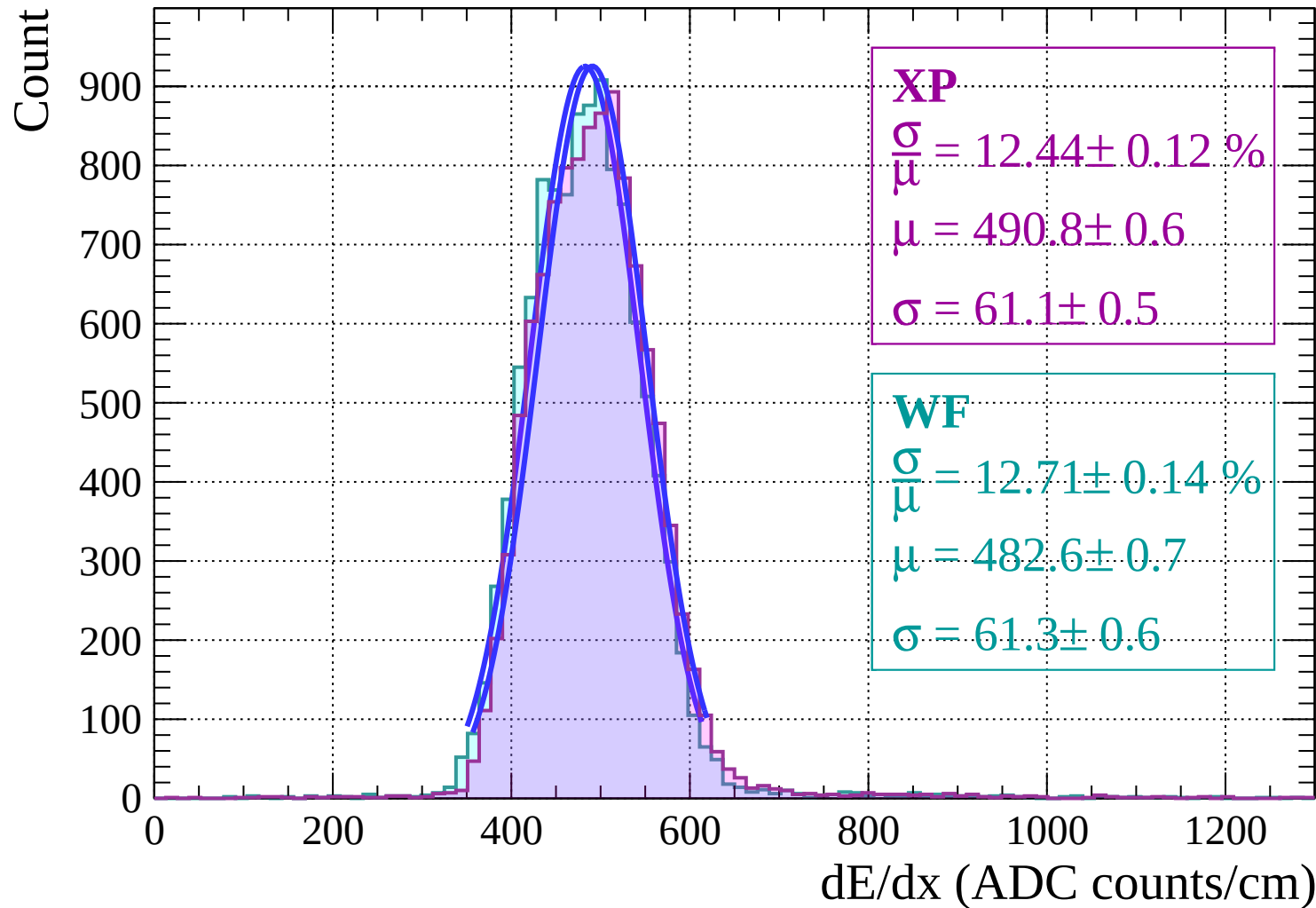
Energy loss |  $-4 < \theta < -2$



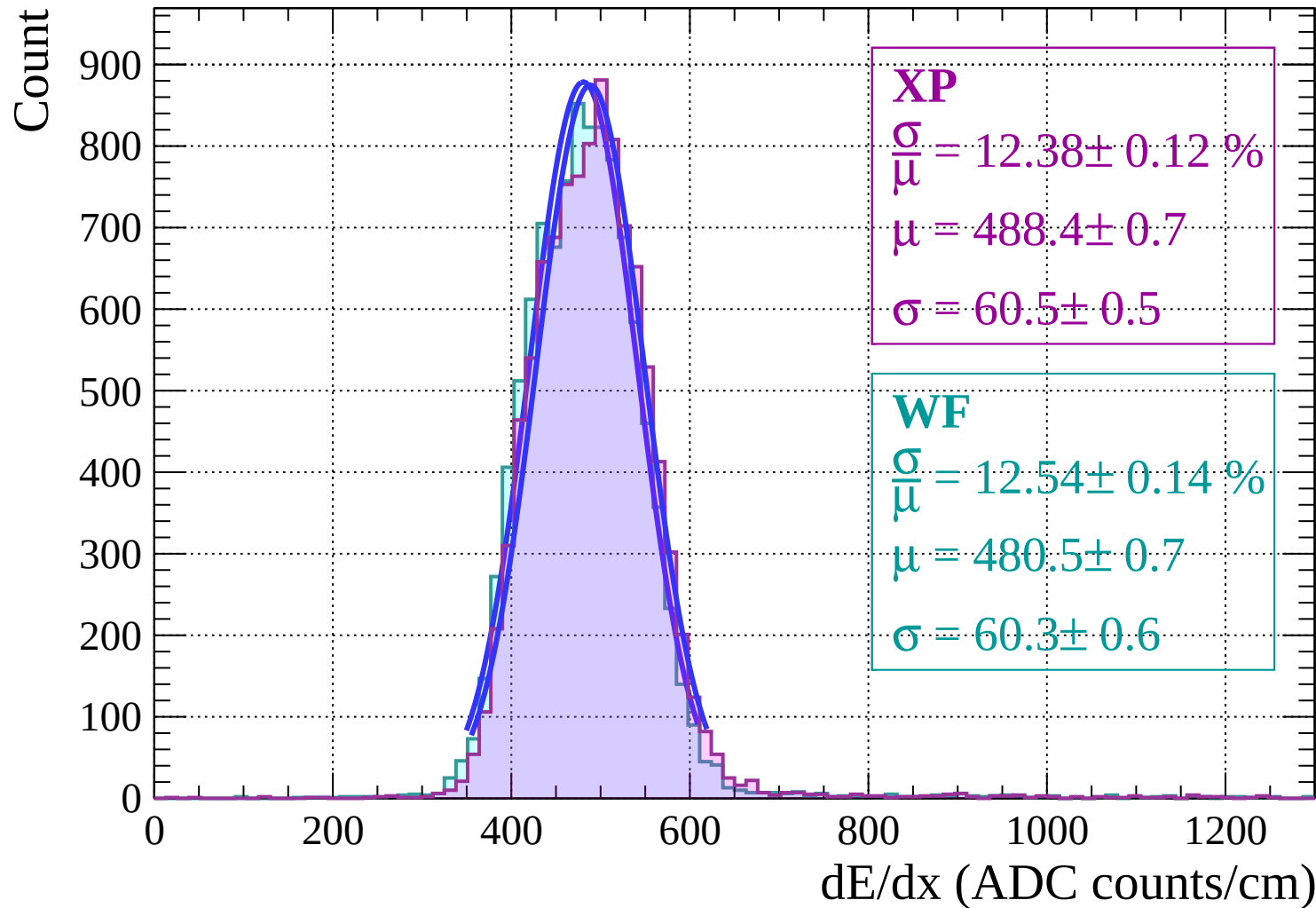
# Energy loss | $-2 < \theta < 0$



# Energy loss | $0 < \theta < 2$

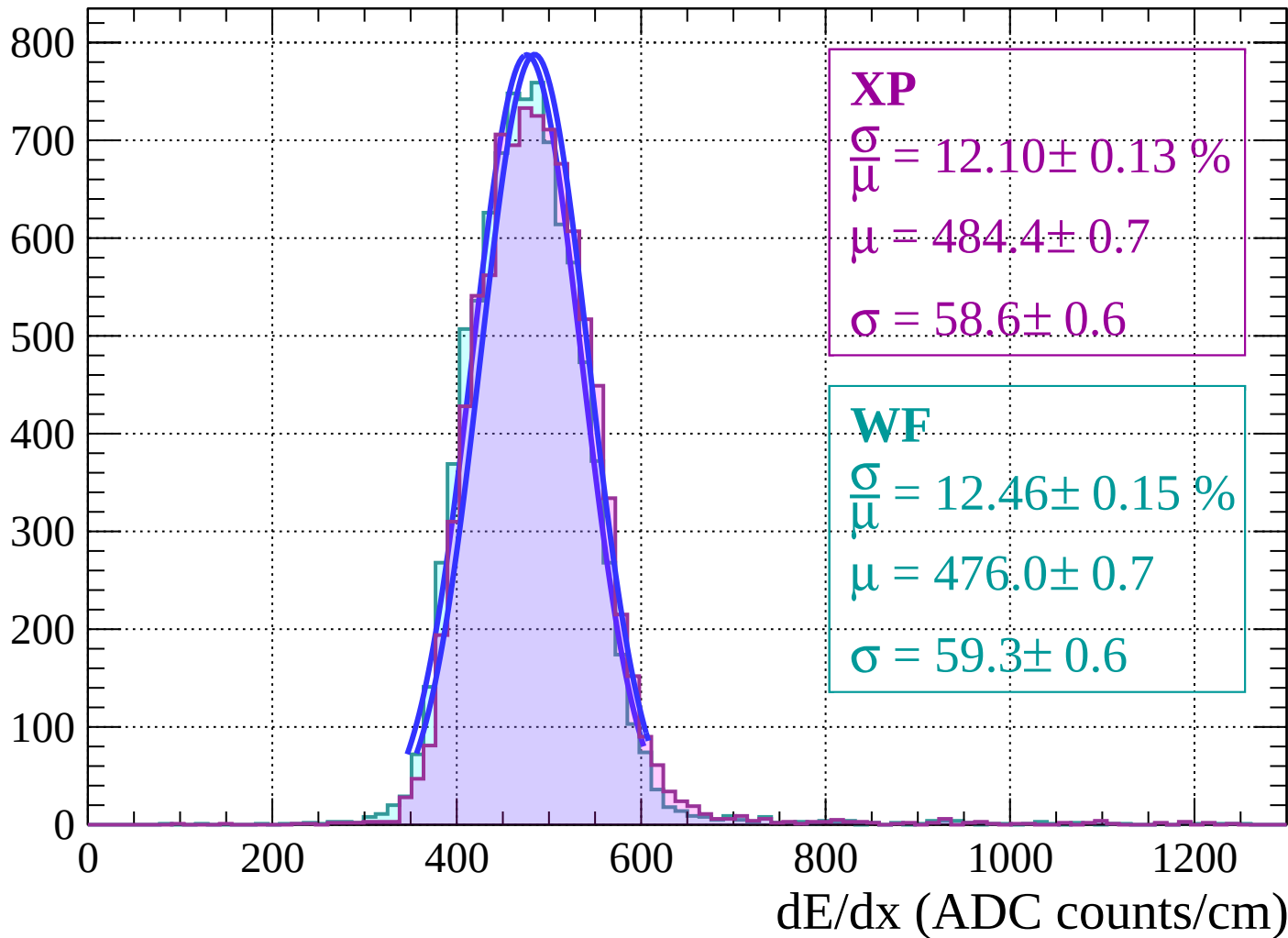


Energy loss |  $2 < \theta < 4$



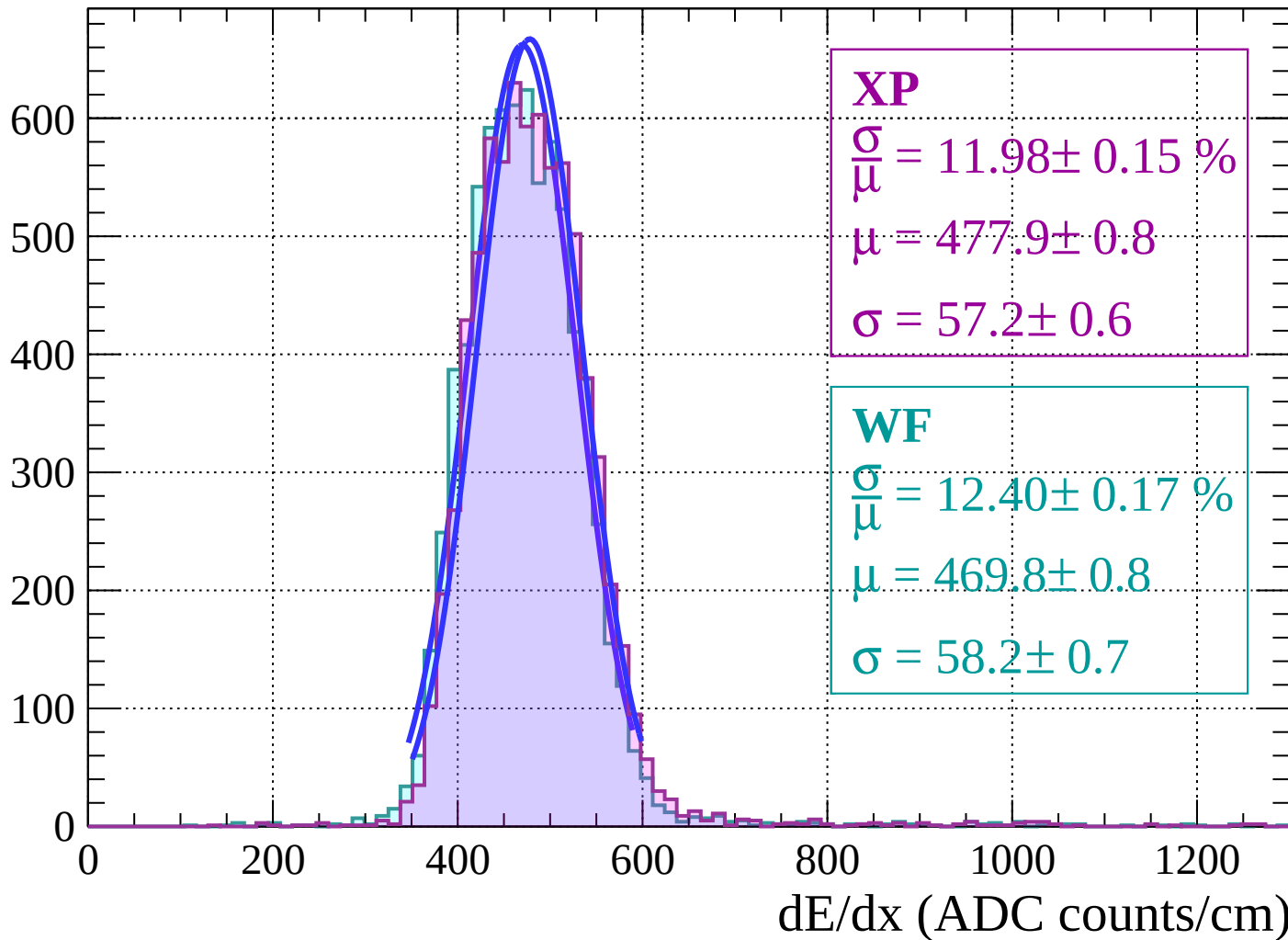
# Energy loss | $4 < \theta < 6$

Count



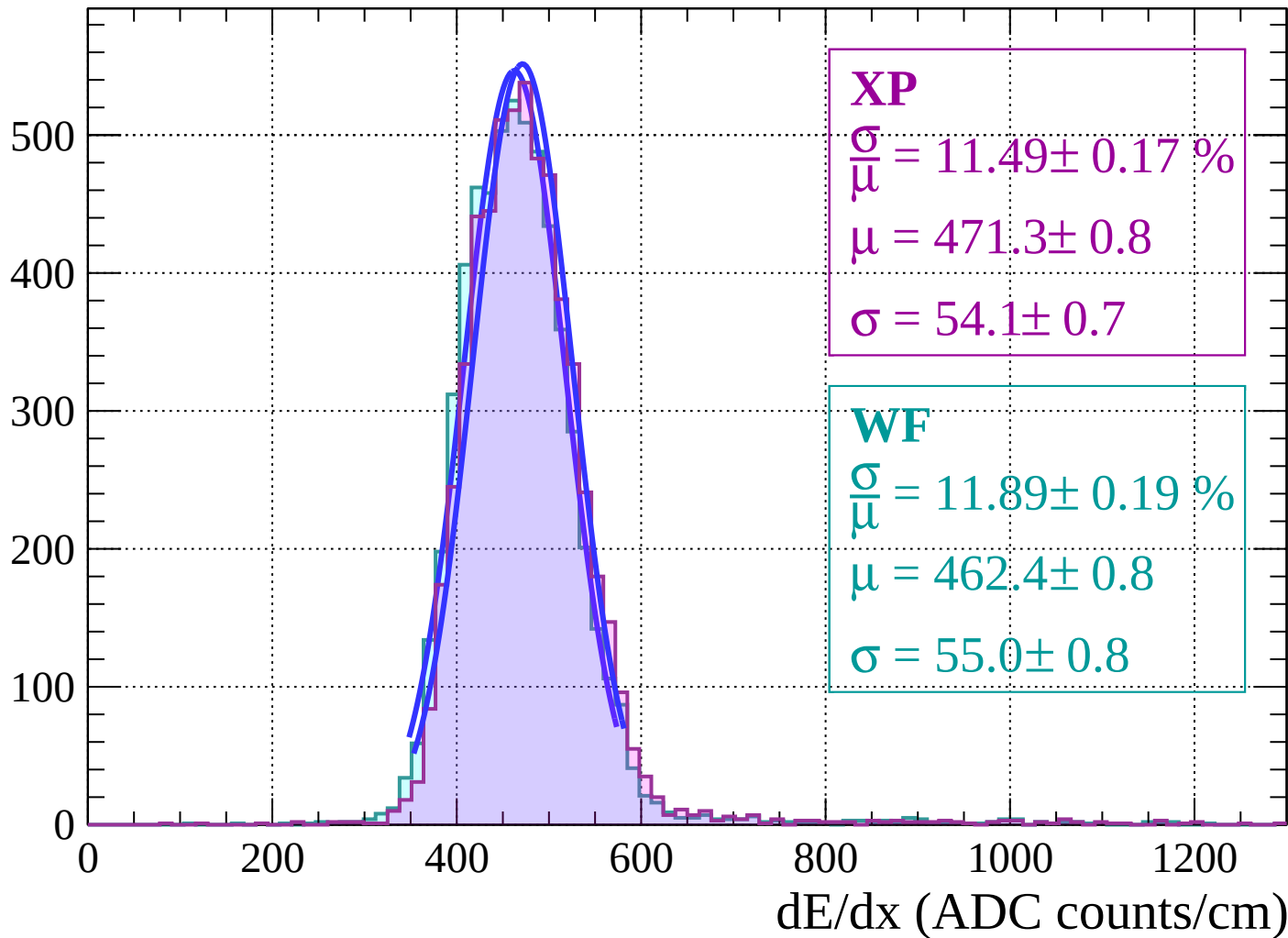
Energy loss |  $6 < \theta < 8$

Count



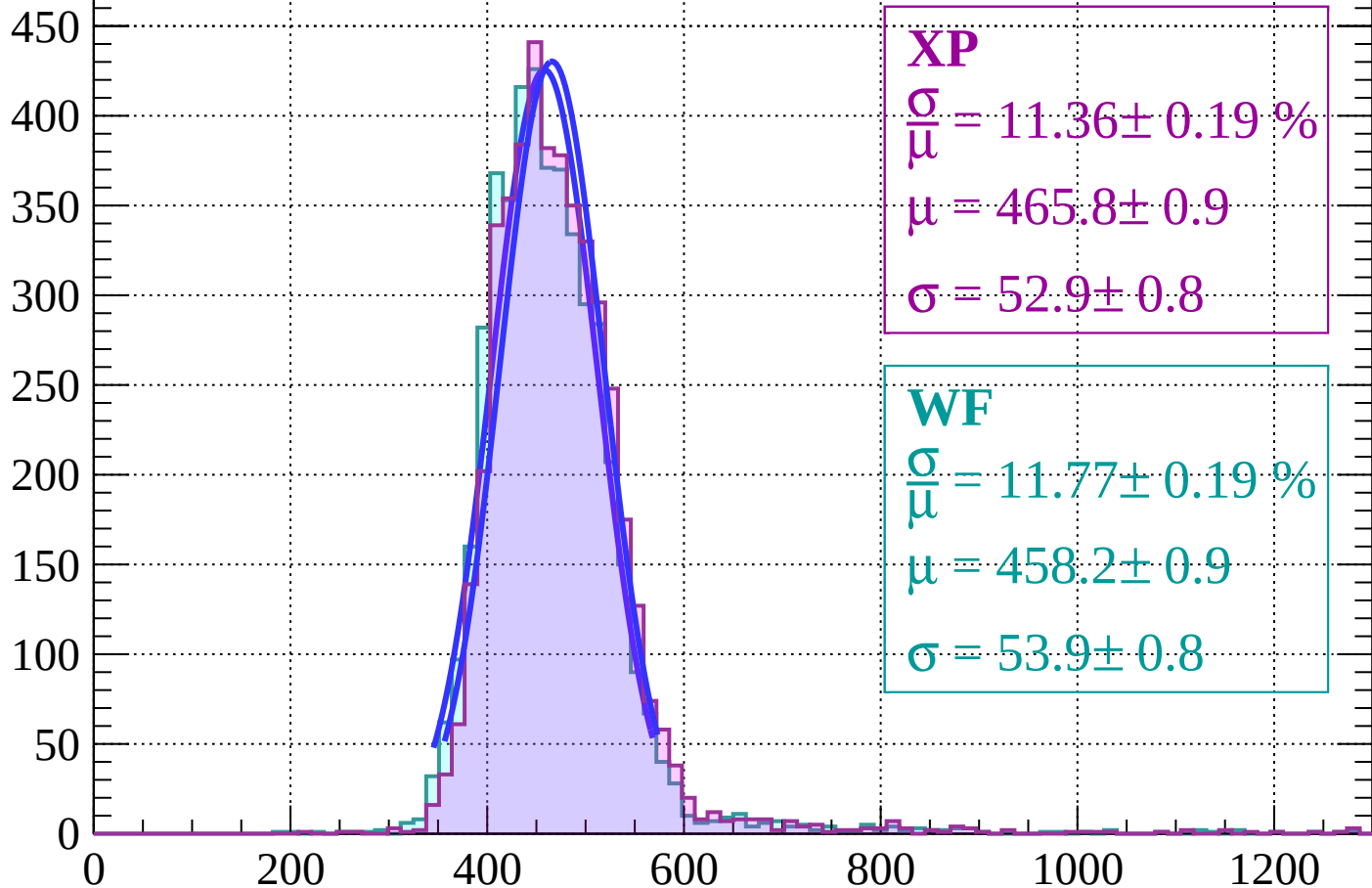
Energy loss |  $8 < \theta < 10$

Count



Energy loss |  $10 < \theta < 12$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.36 \pm 0.19 \%$$

$$\mu = 465.8 \pm 0.9$$

$$\sigma = 52.9 \pm 0.8$$

**WF**

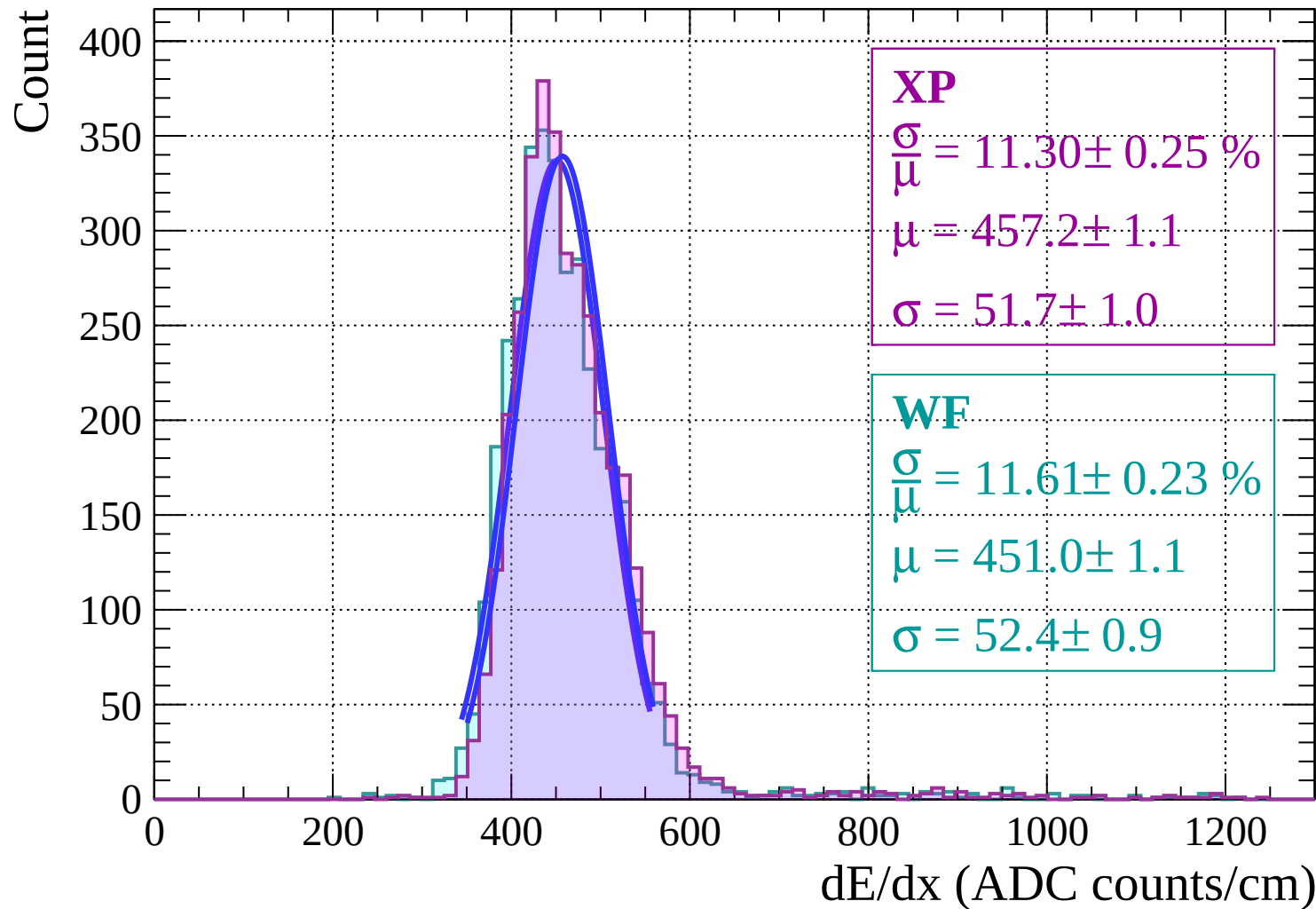
$$\frac{\sigma}{\mu} = 11.77 \pm 0.19 \%$$

$$\mu = 458.2 \pm 0.9$$

$$\sigma = 53.9 \pm 0.8$$

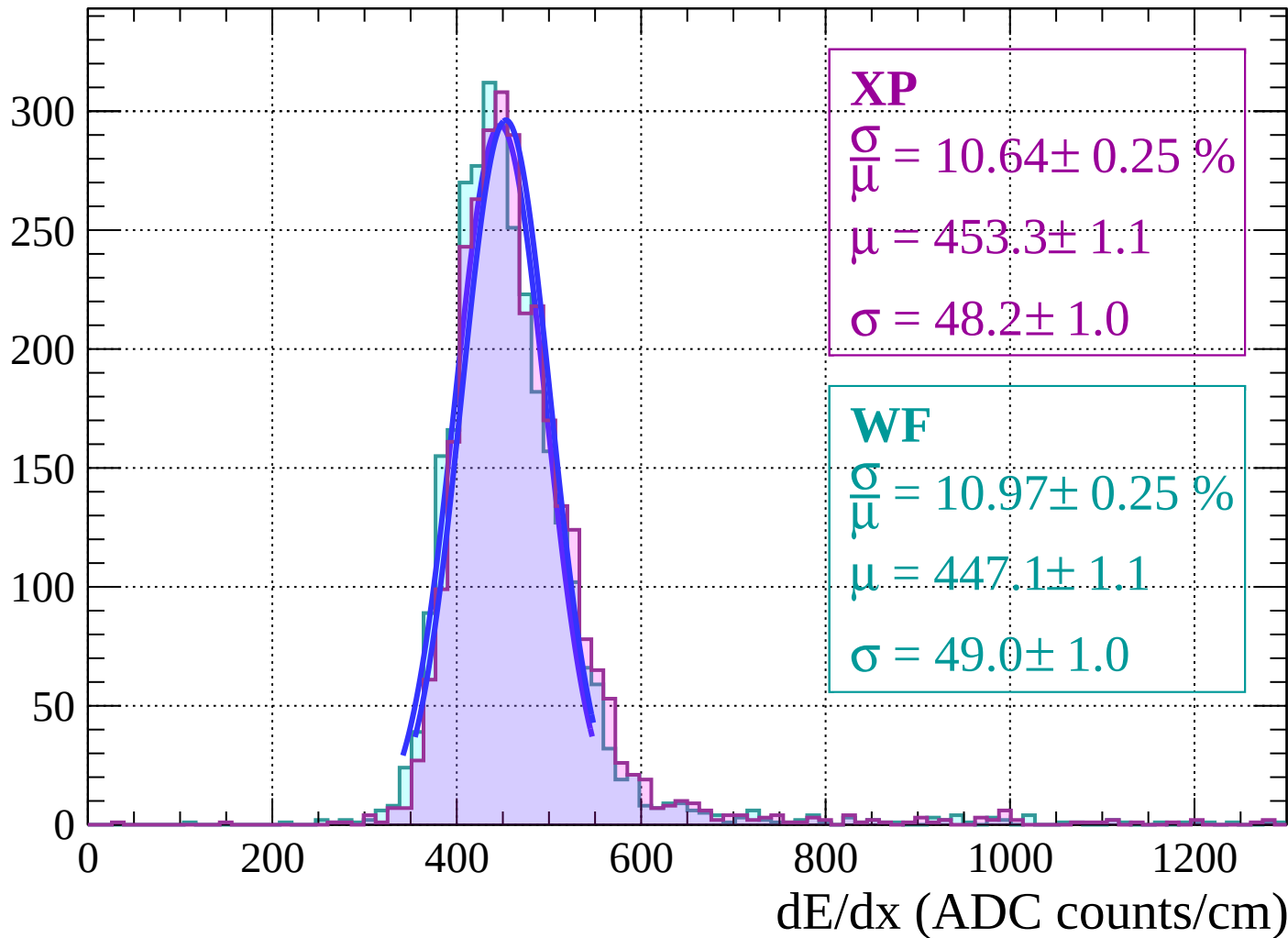
dE/dx (ADC counts/cm)

# Energy loss | $12 < \theta < 14$



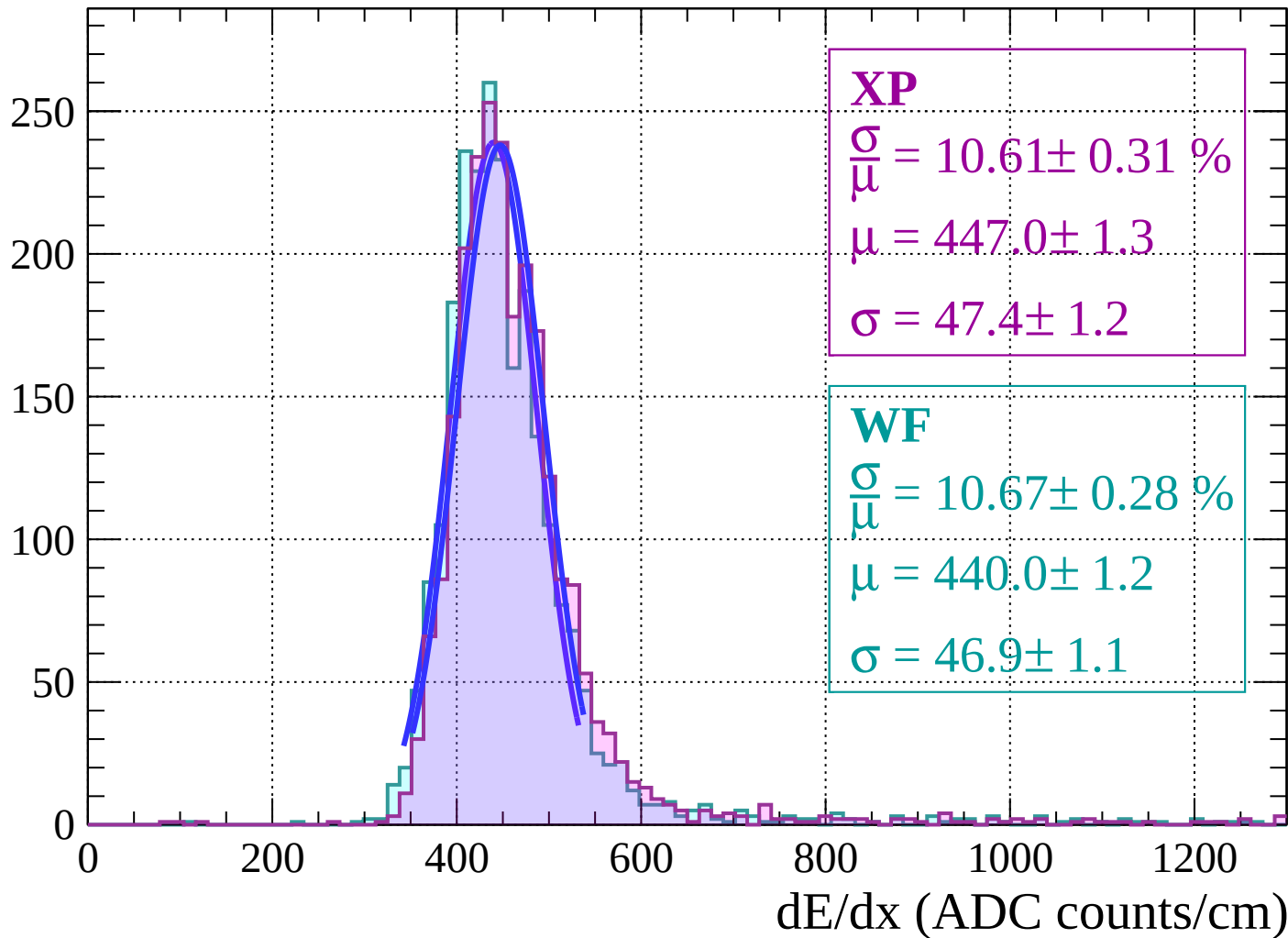
Energy loss |  $14 < \theta < 16$

Count



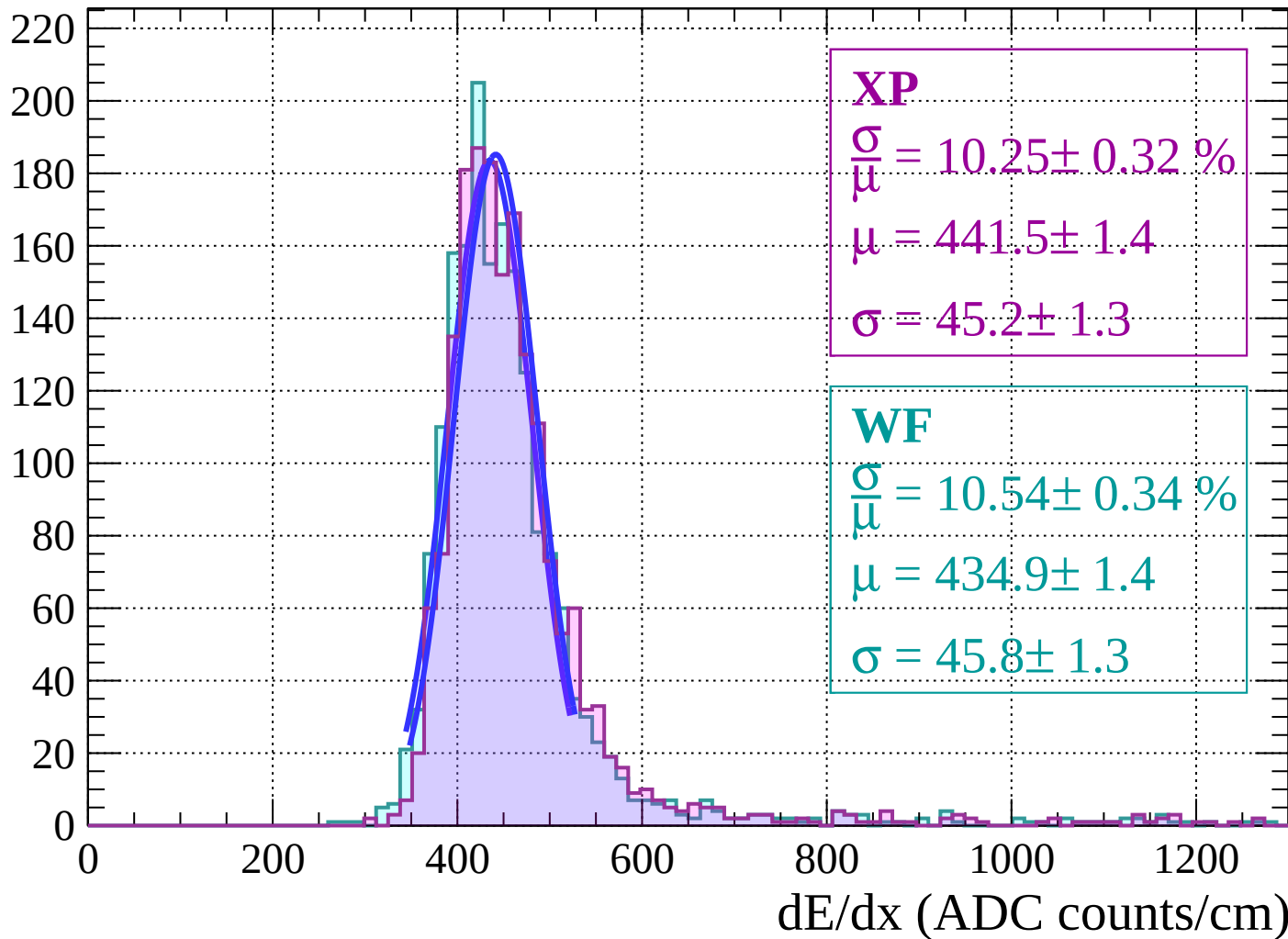
Energy loss |  $16 < \theta < 18$

Count



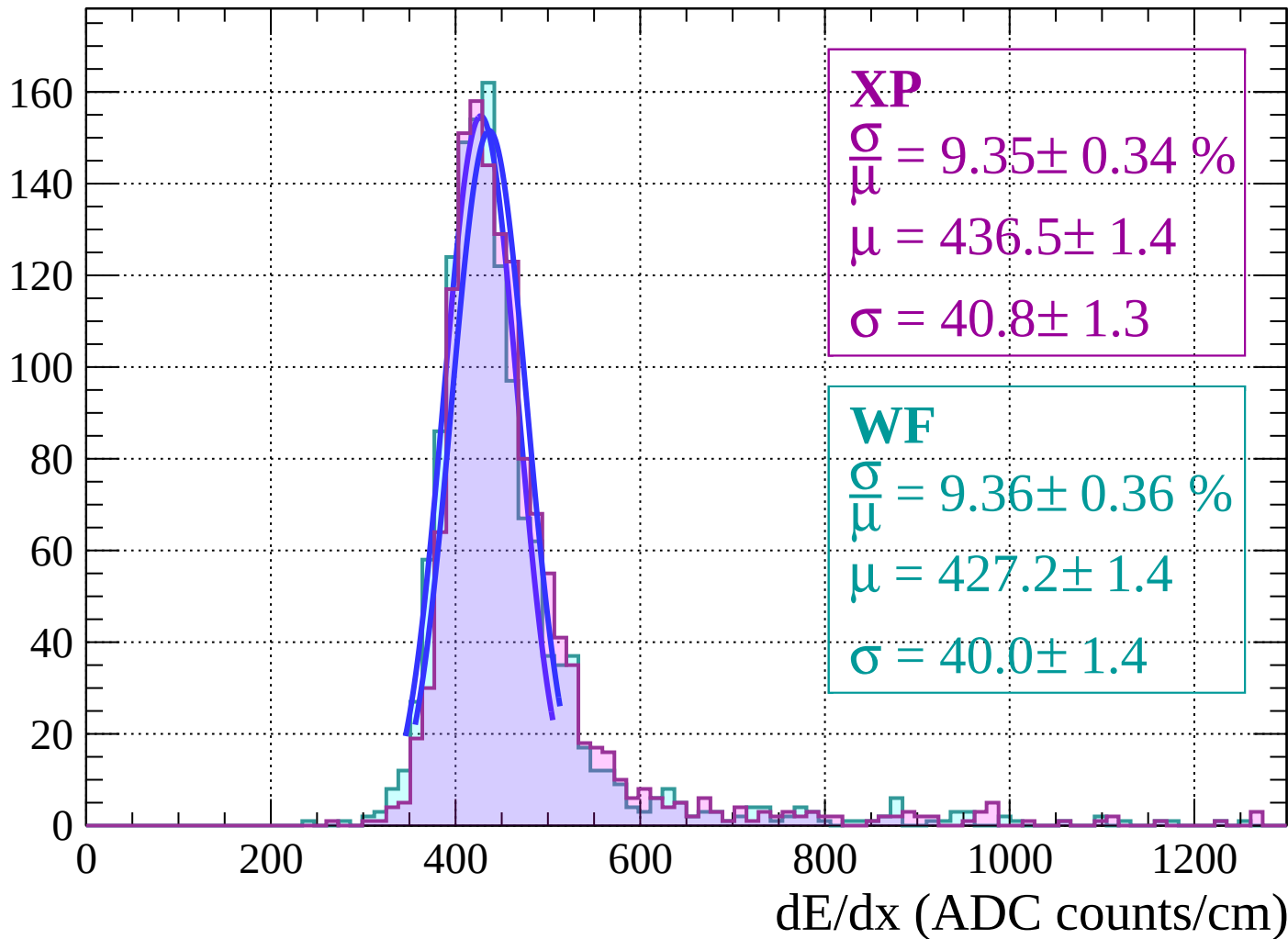
# Energy loss | $18 < \theta < 20$

Count



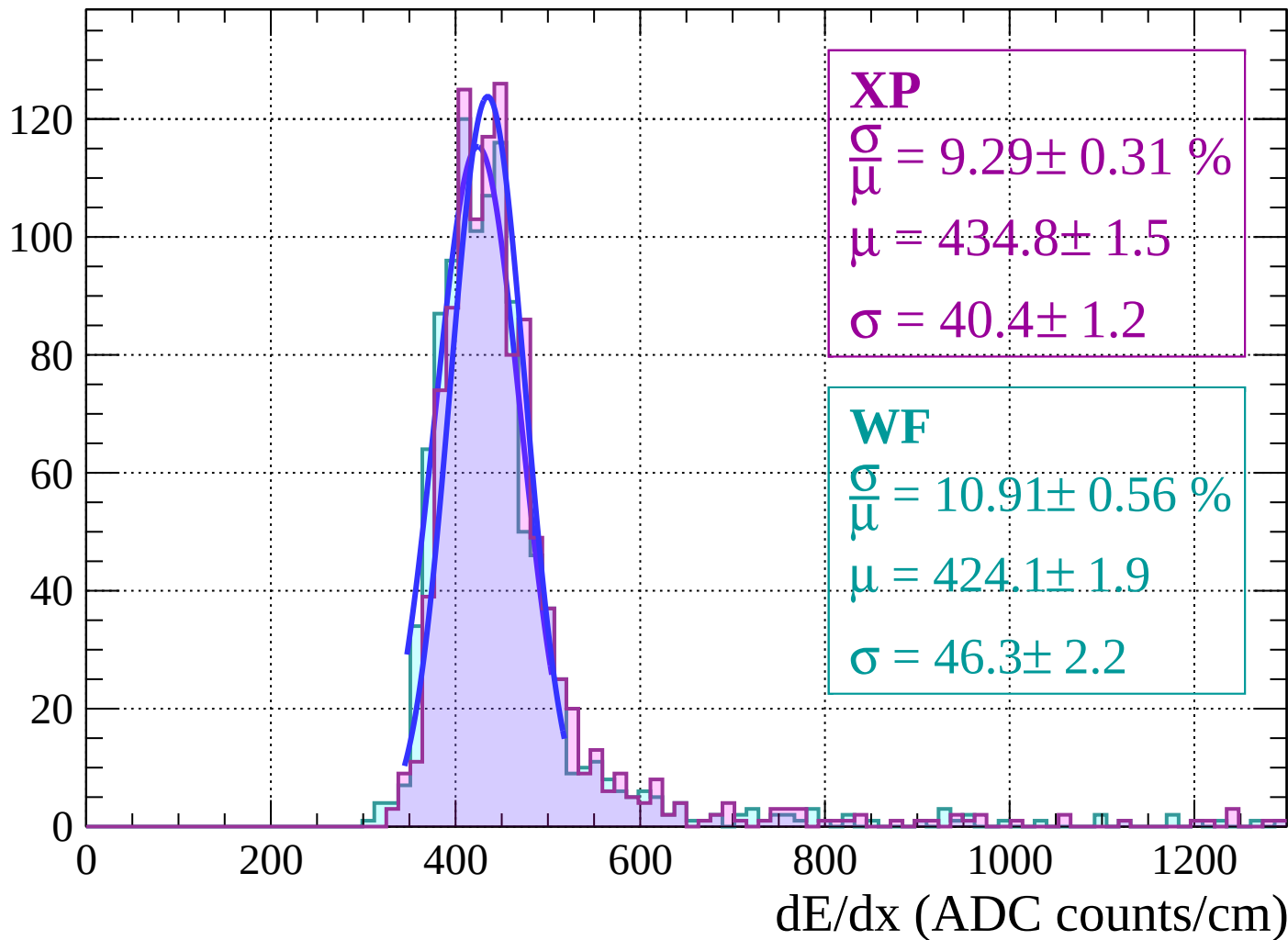
Energy loss |  $20 < \theta < 22$

Count



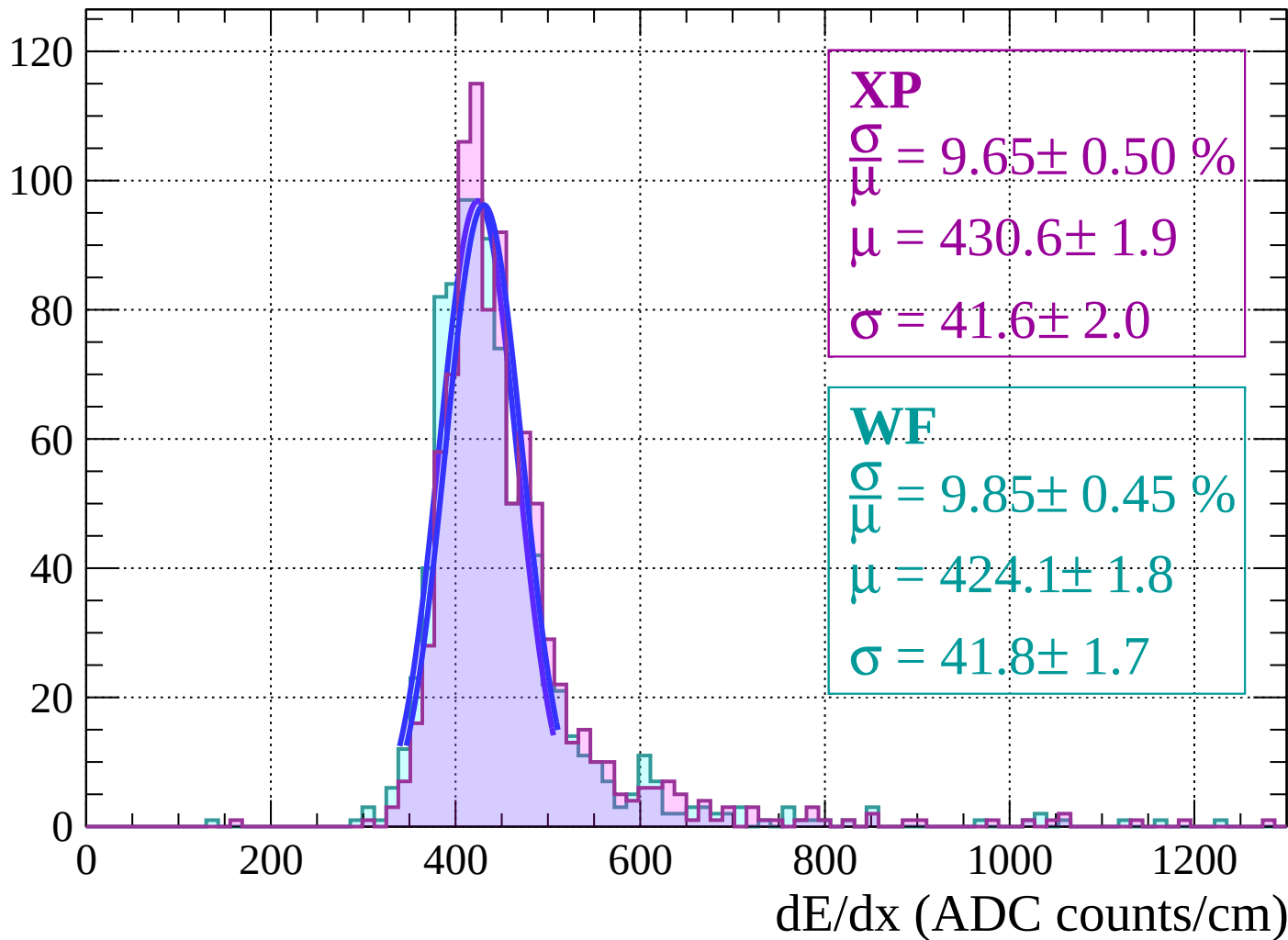
Energy loss |  $22 < \theta < 24$

Count



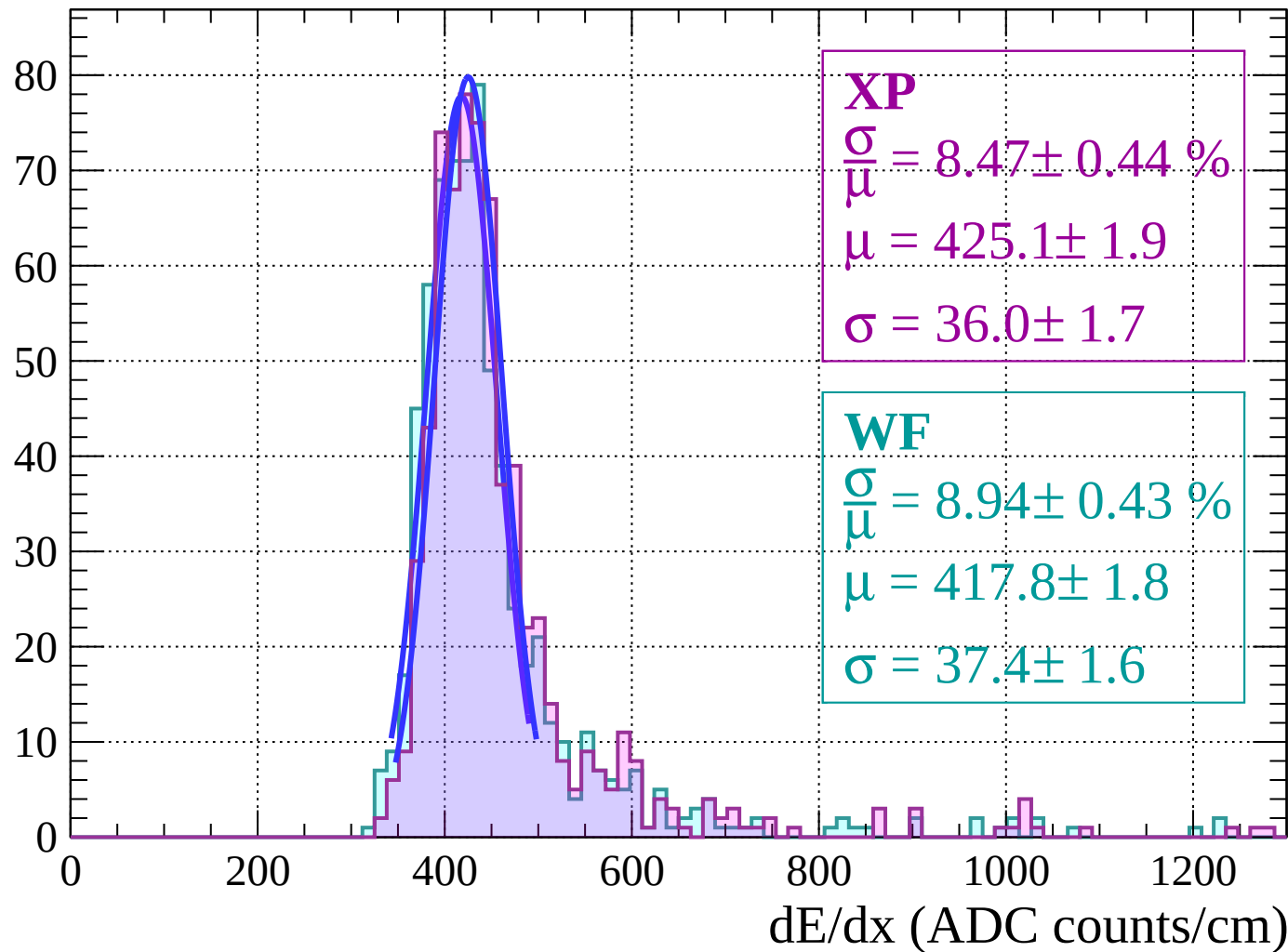
Energy loss |  $24 < \theta < 26$

Count



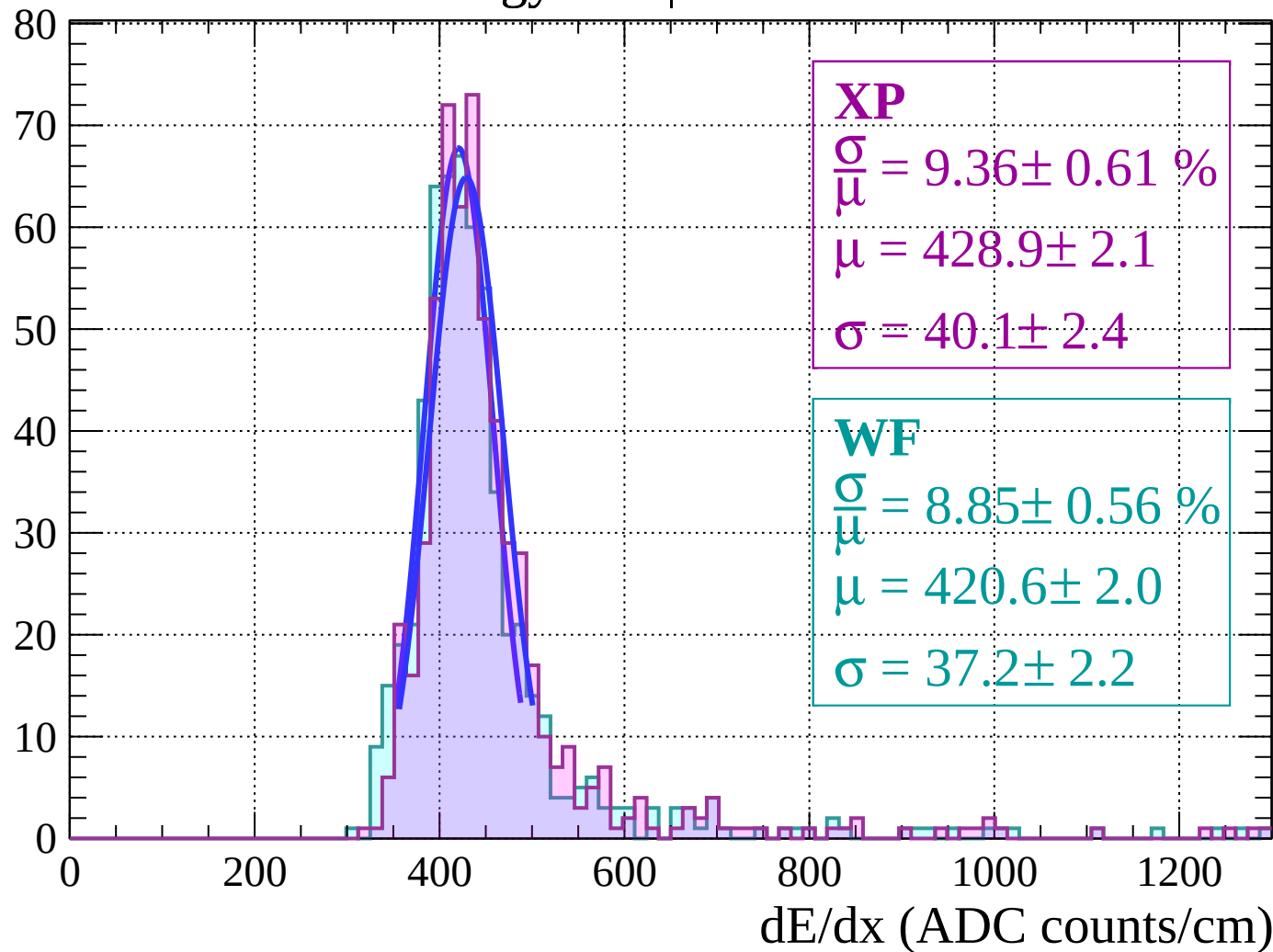
Energy loss |  $26 < \theta < 28$

Count



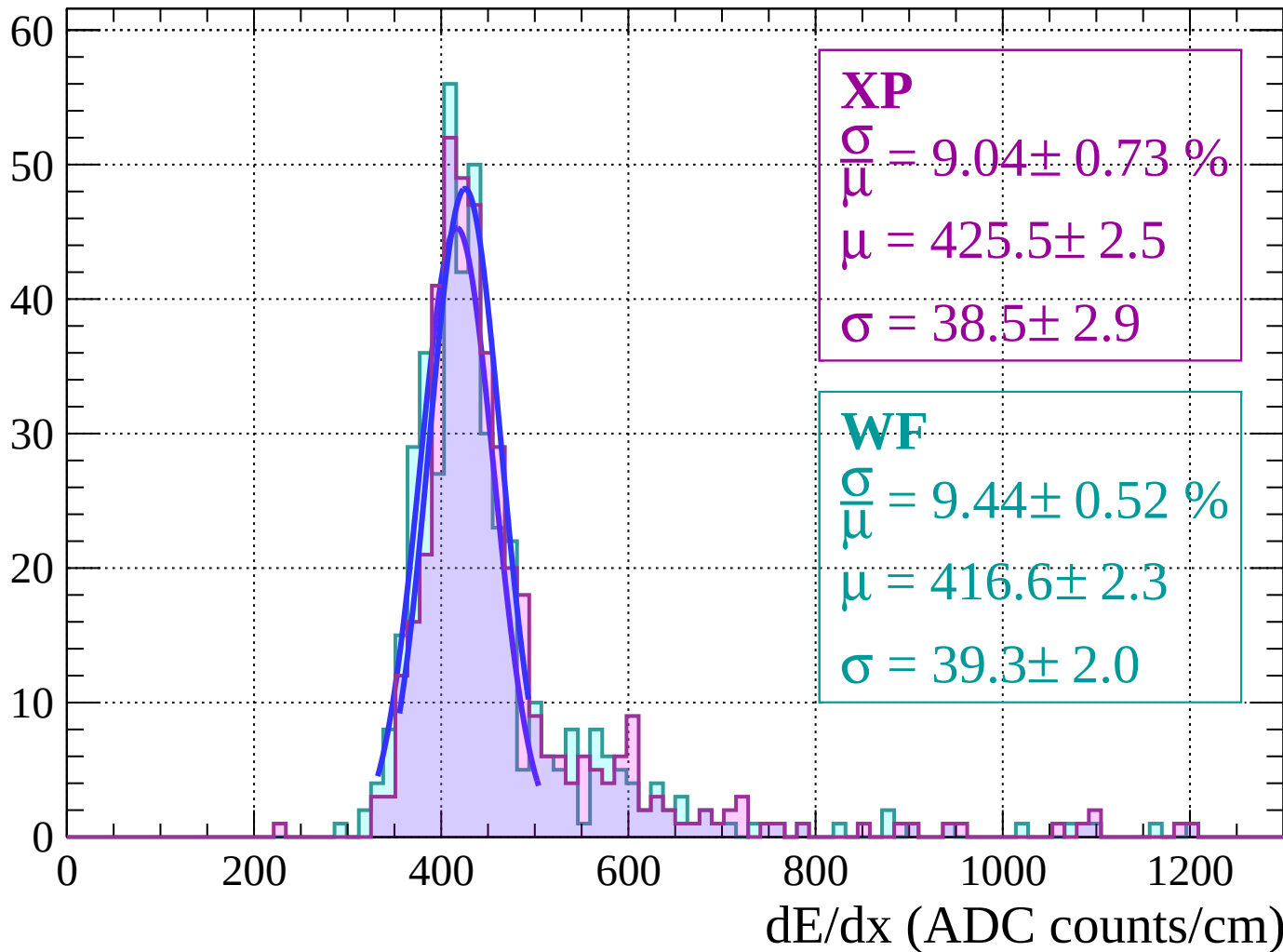
Energy loss |  $28 < \theta < 30$

Count



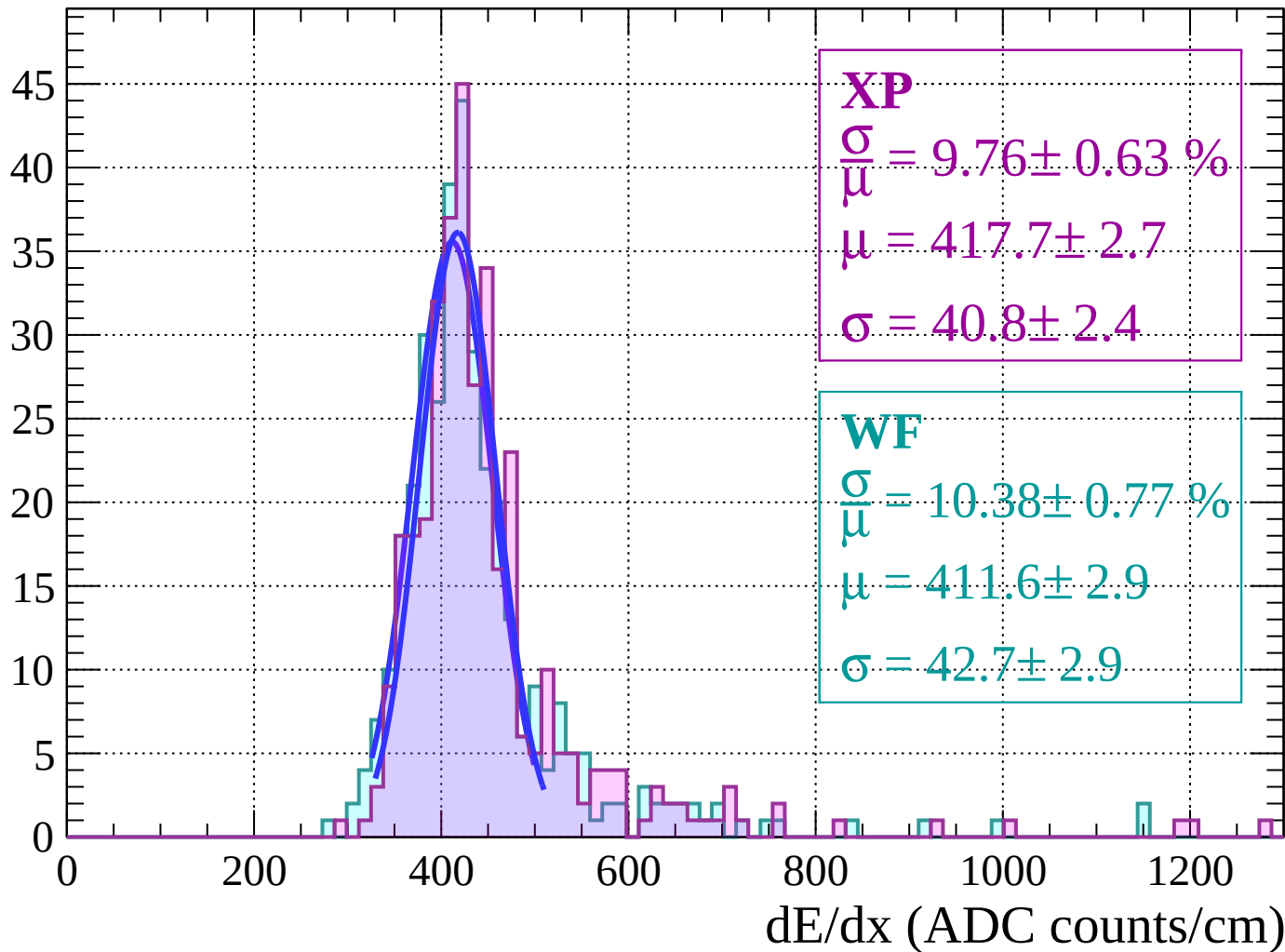
Energy loss |  $30 < \theta < 32$

Count



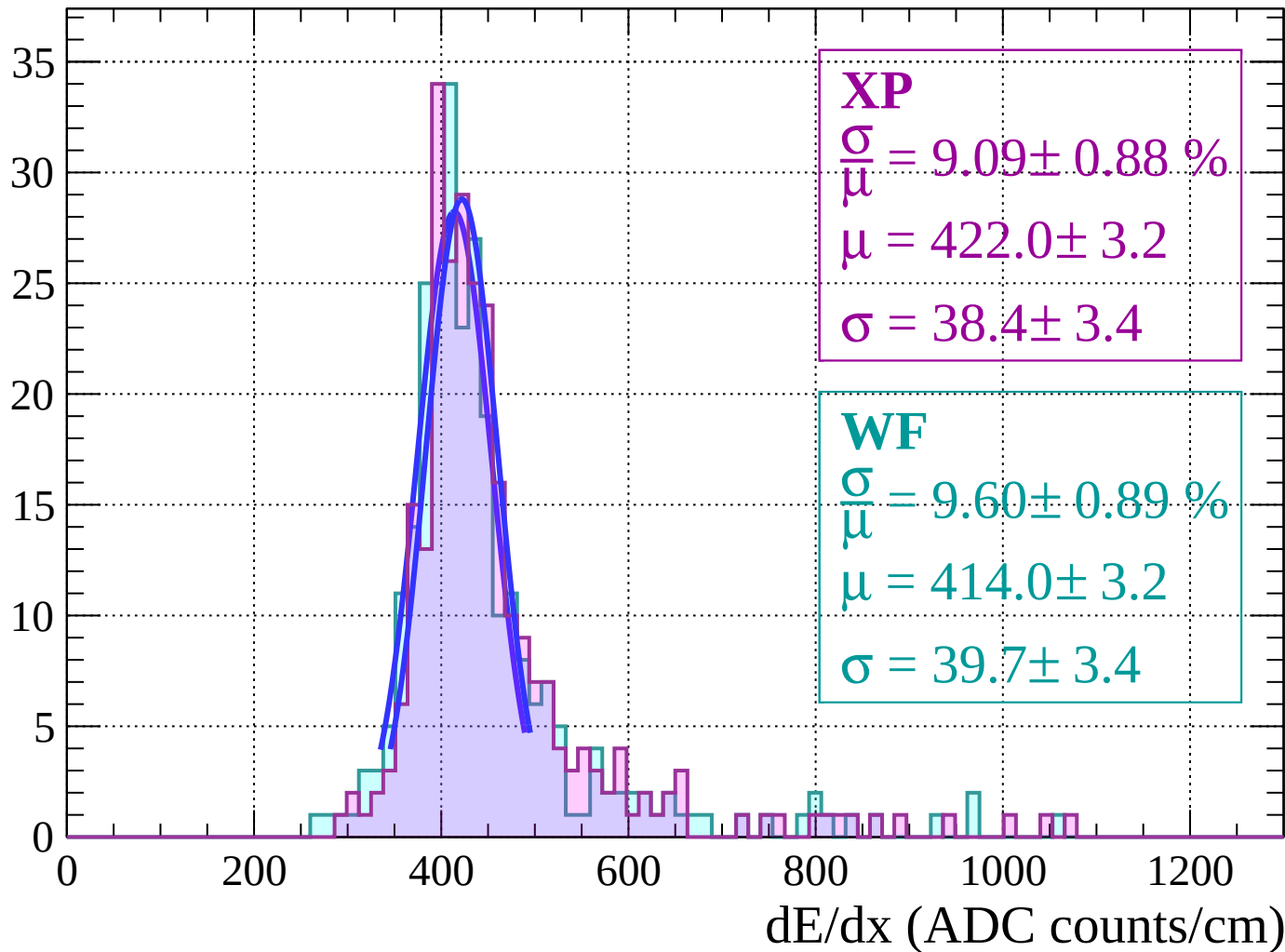
Energy loss |  $32 < \theta < 34$

Count



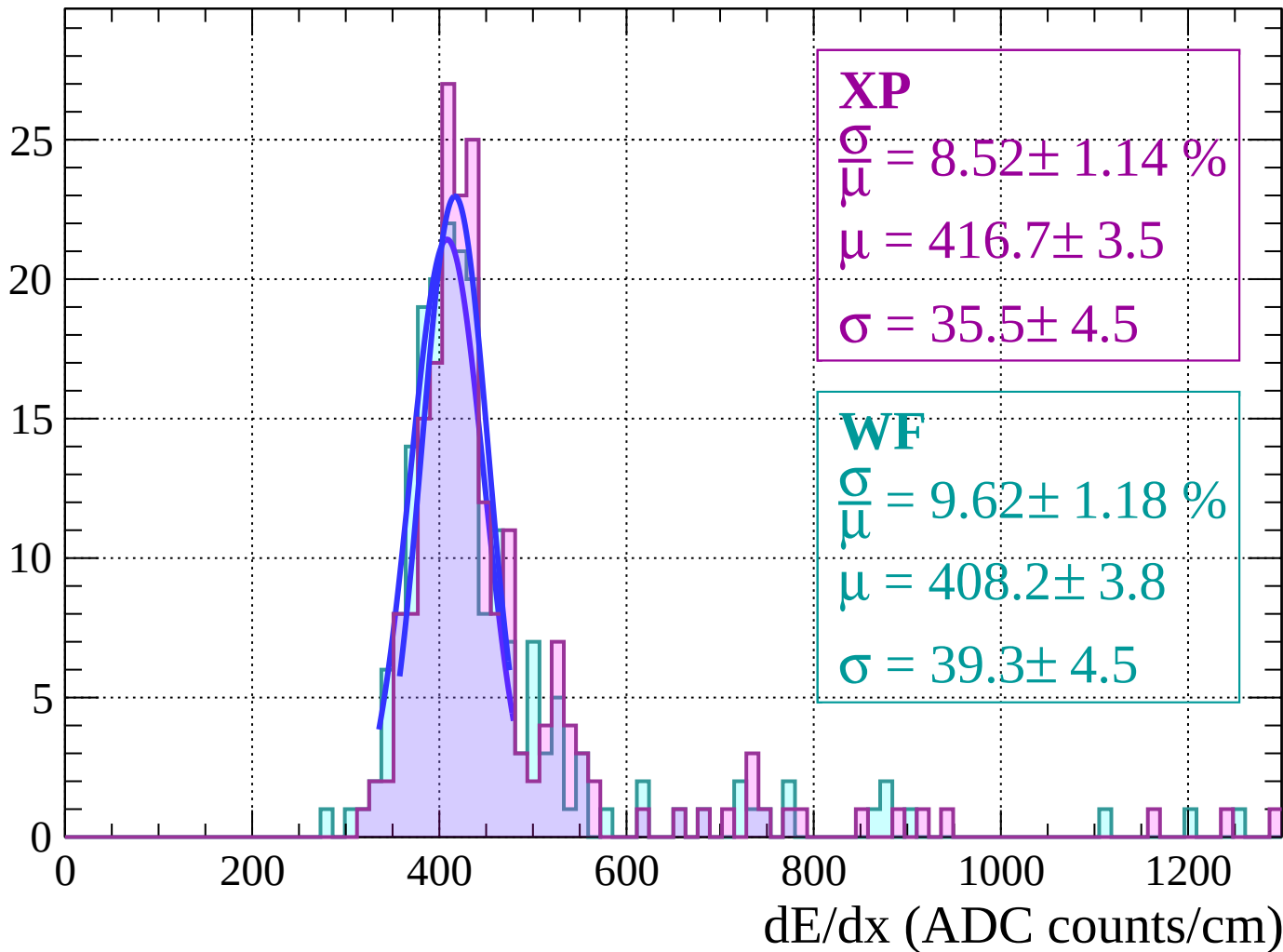
Energy loss |  $34 < \theta < 36$

Count



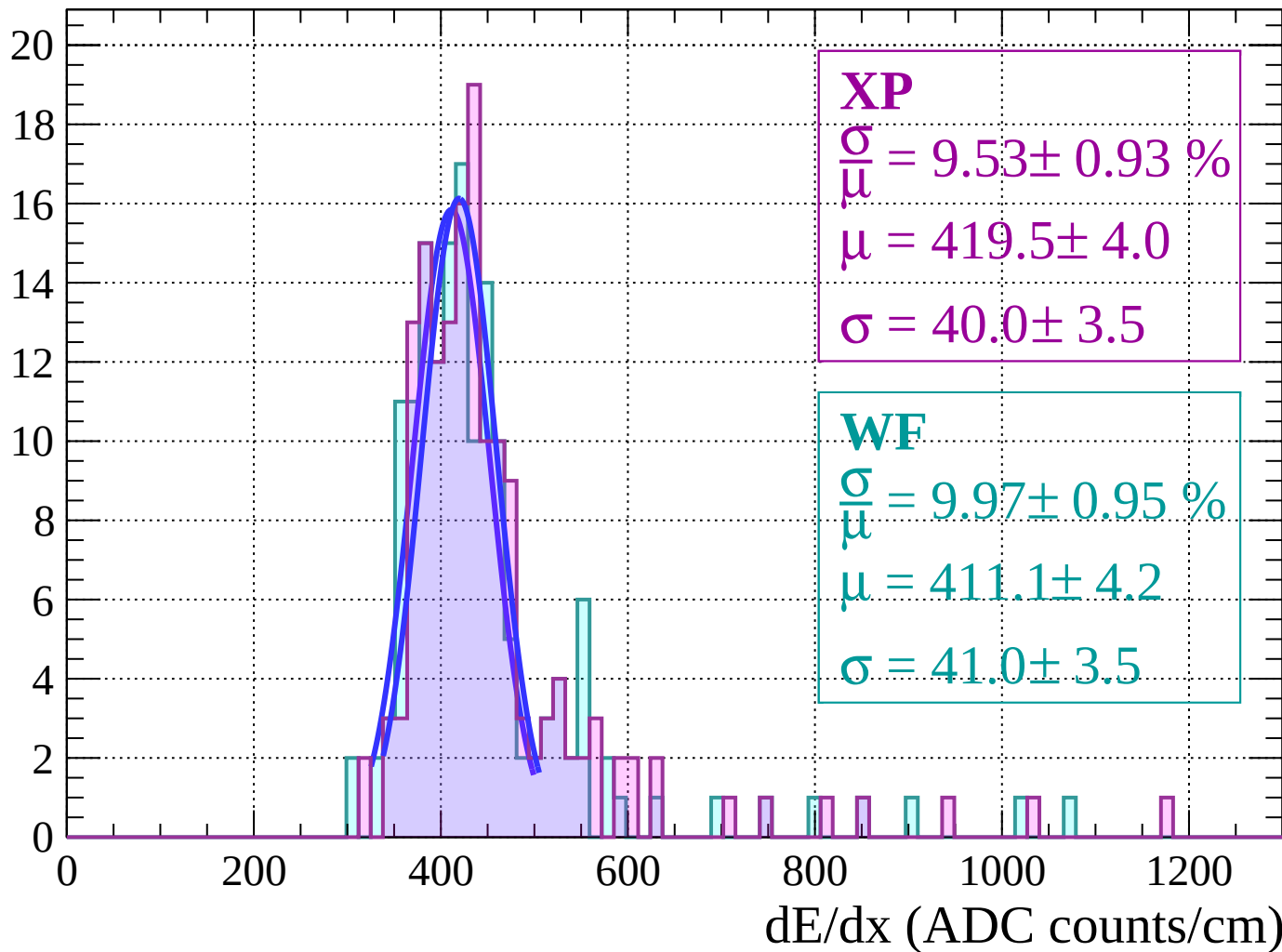
Energy loss |  $36 < \theta < 38$

Count



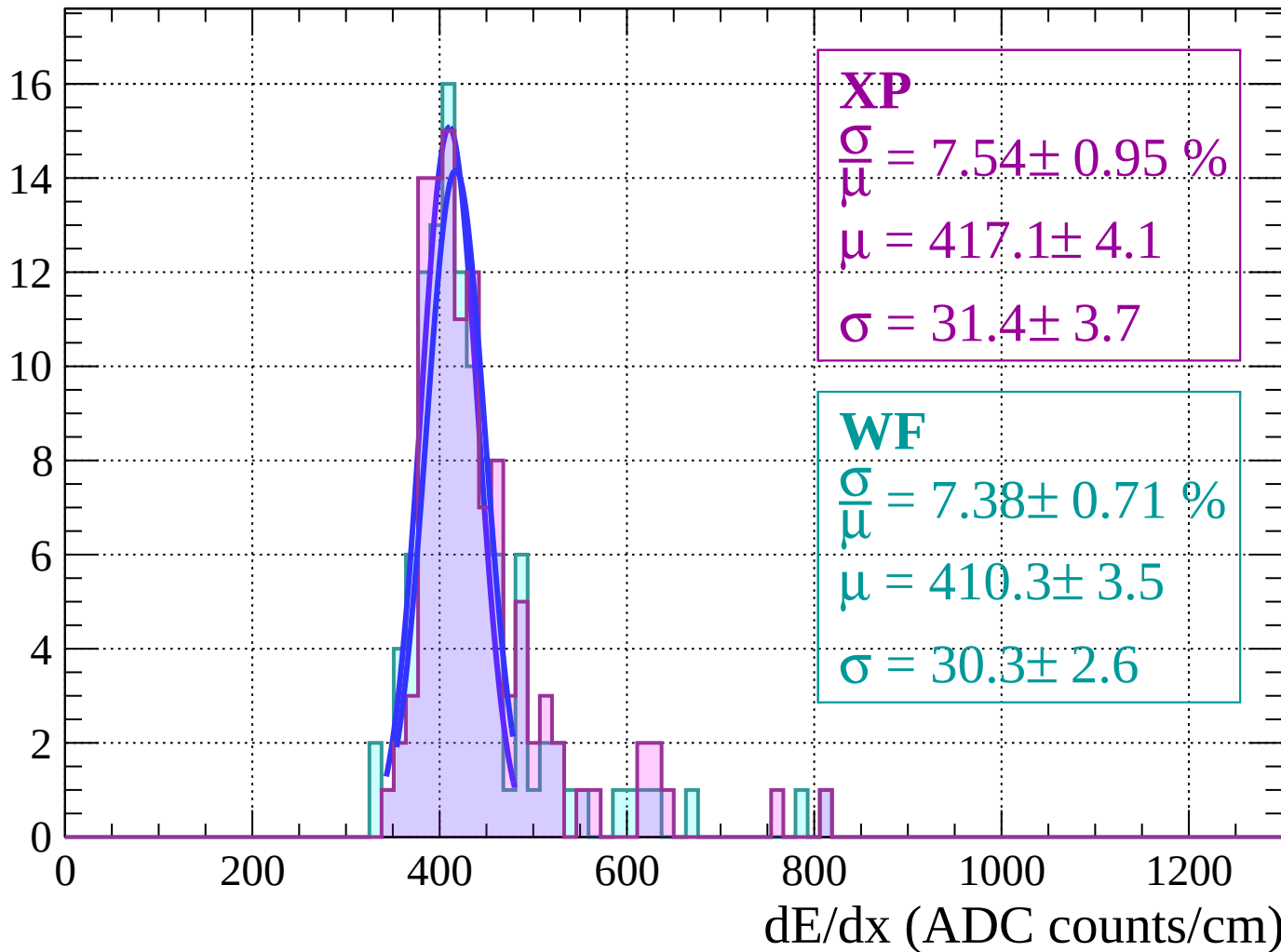
Energy loss |  $38 < \theta < 40$

Count



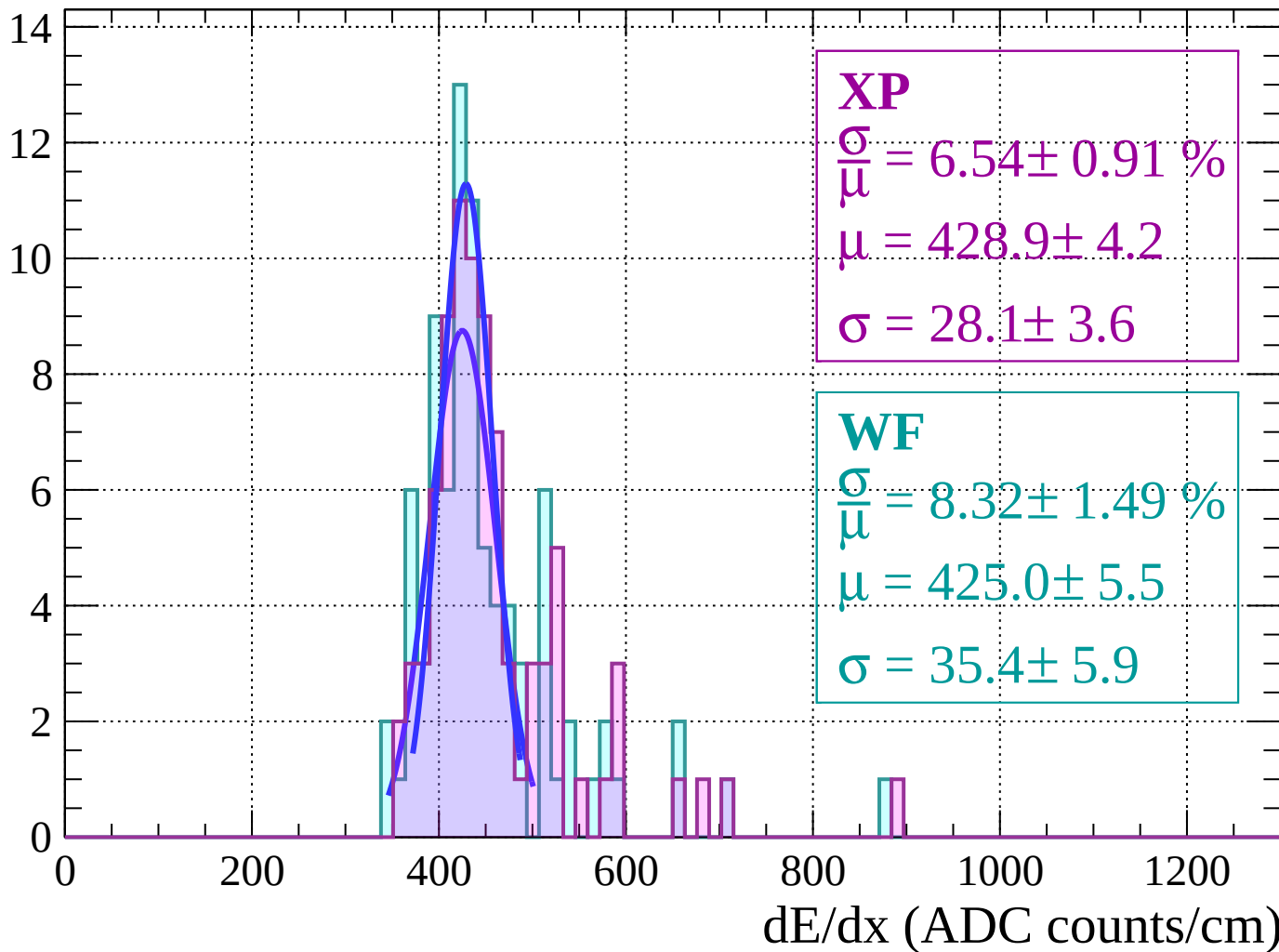
Energy loss |  $40 < \theta < 42$

Count



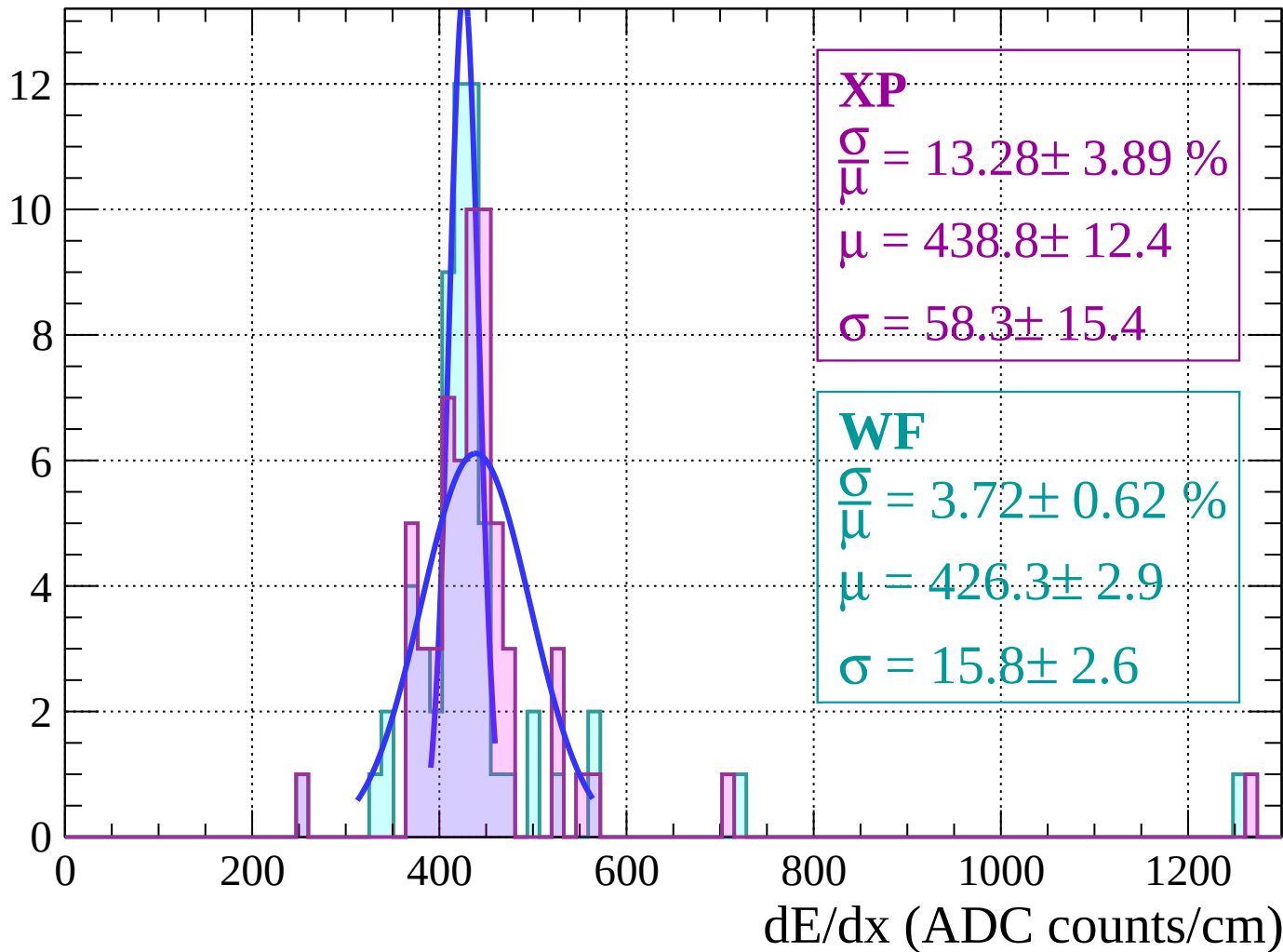
Energy loss |  $42 < \theta < 44$

Count



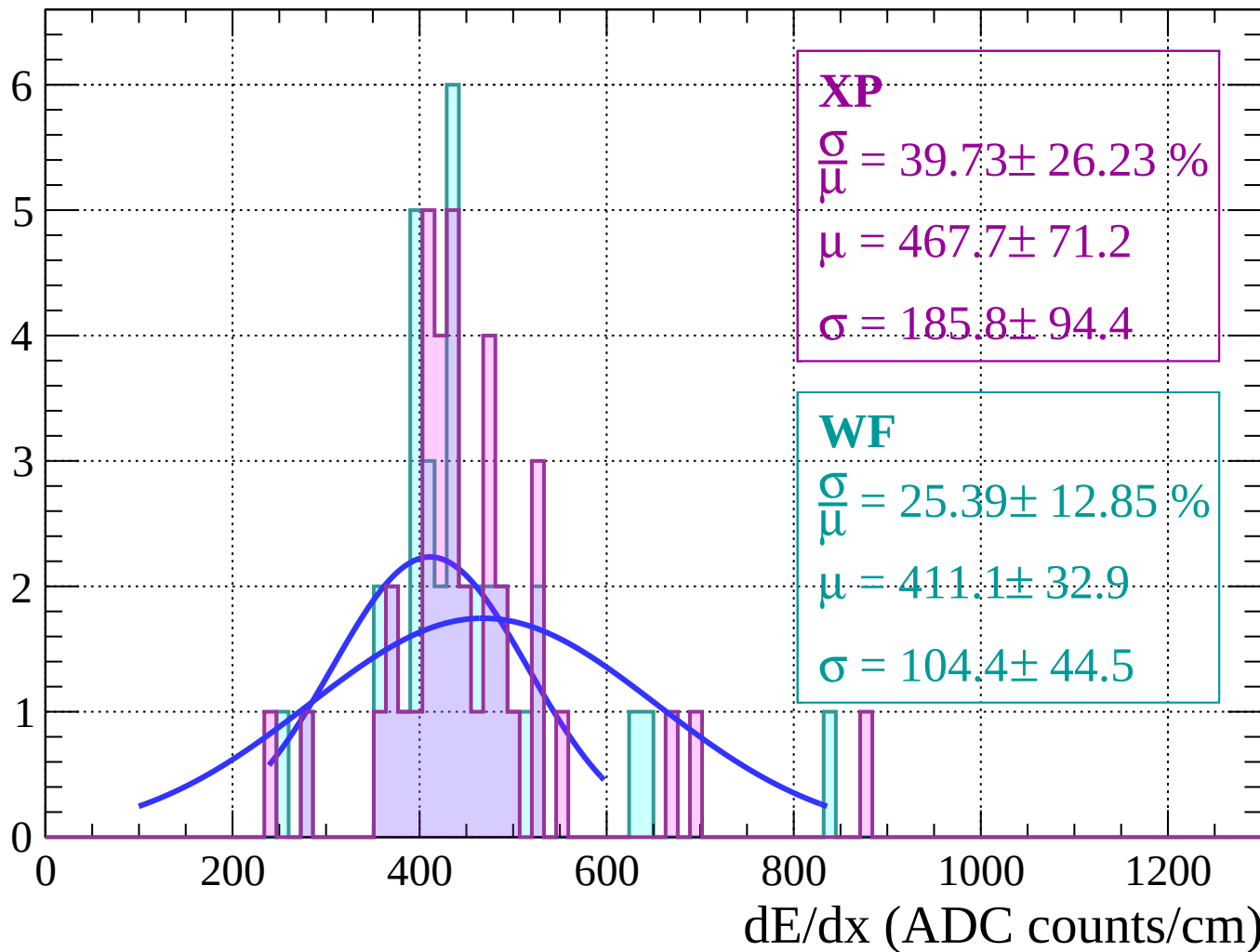
Energy loss |  $44 < \theta < 46$

Count



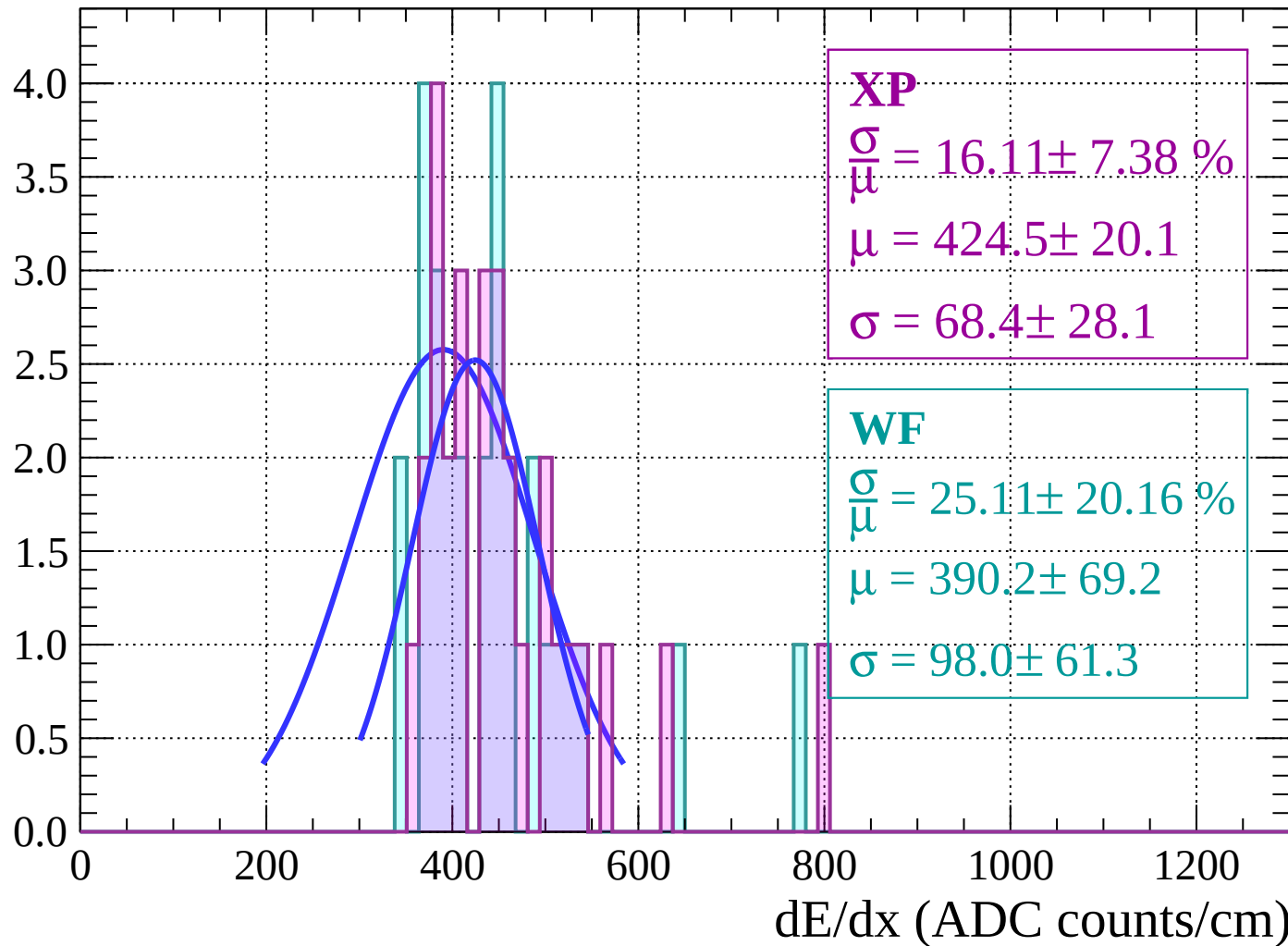
Energy loss |  $46 < \theta < 48$

Count



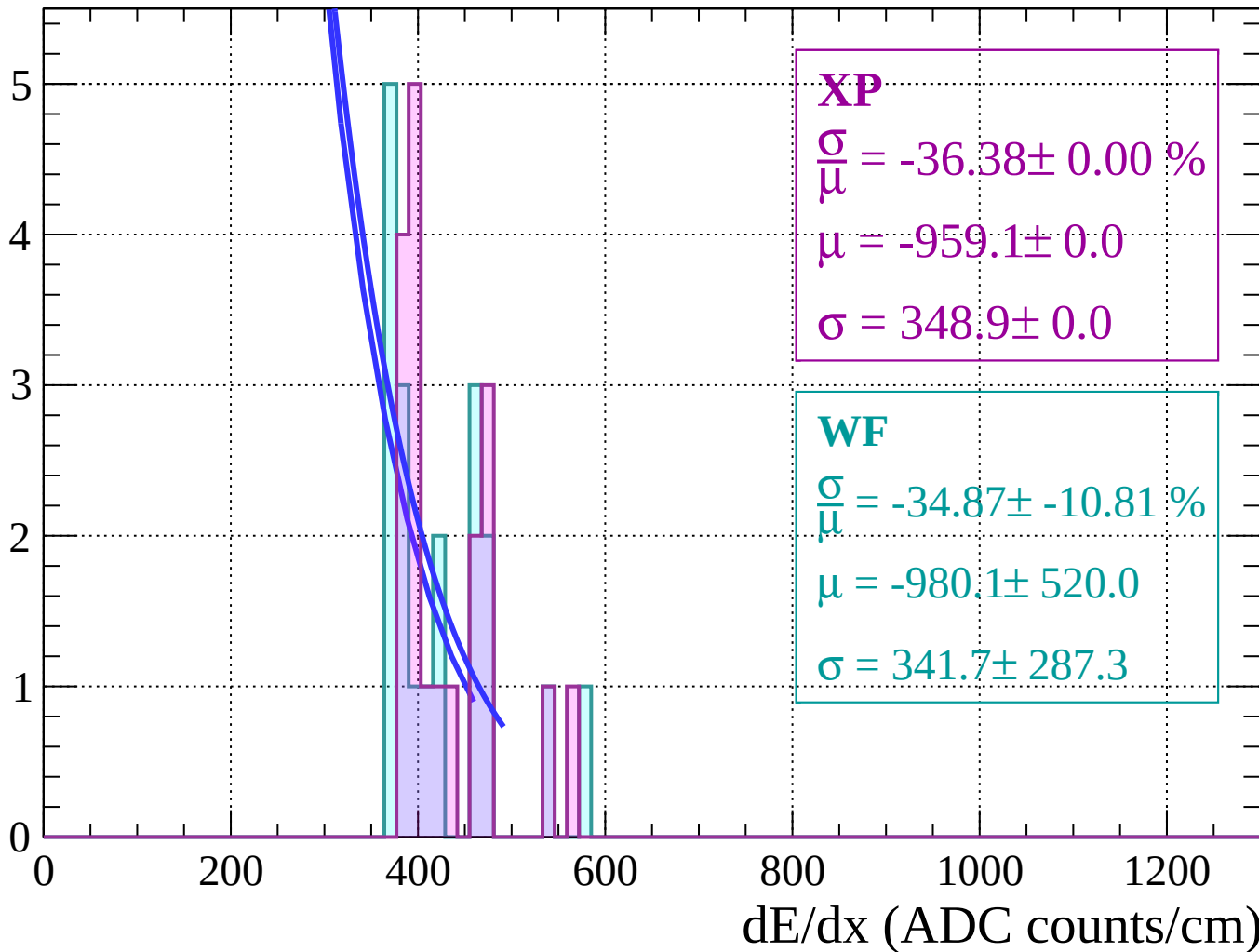
Energy loss |  $48 < \theta < 50$

Count



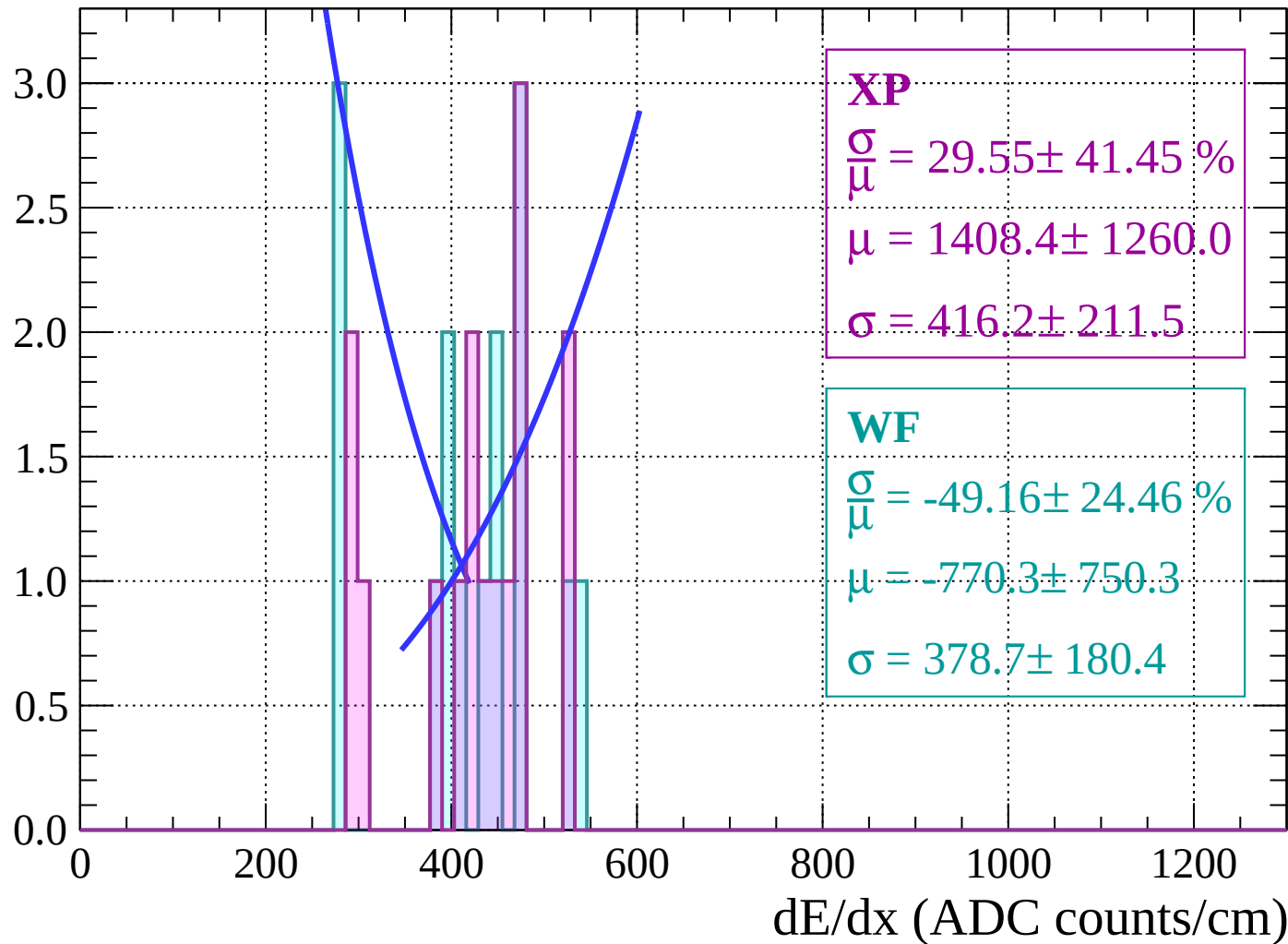
Energy loss |  $50 < \theta < 52$

Count



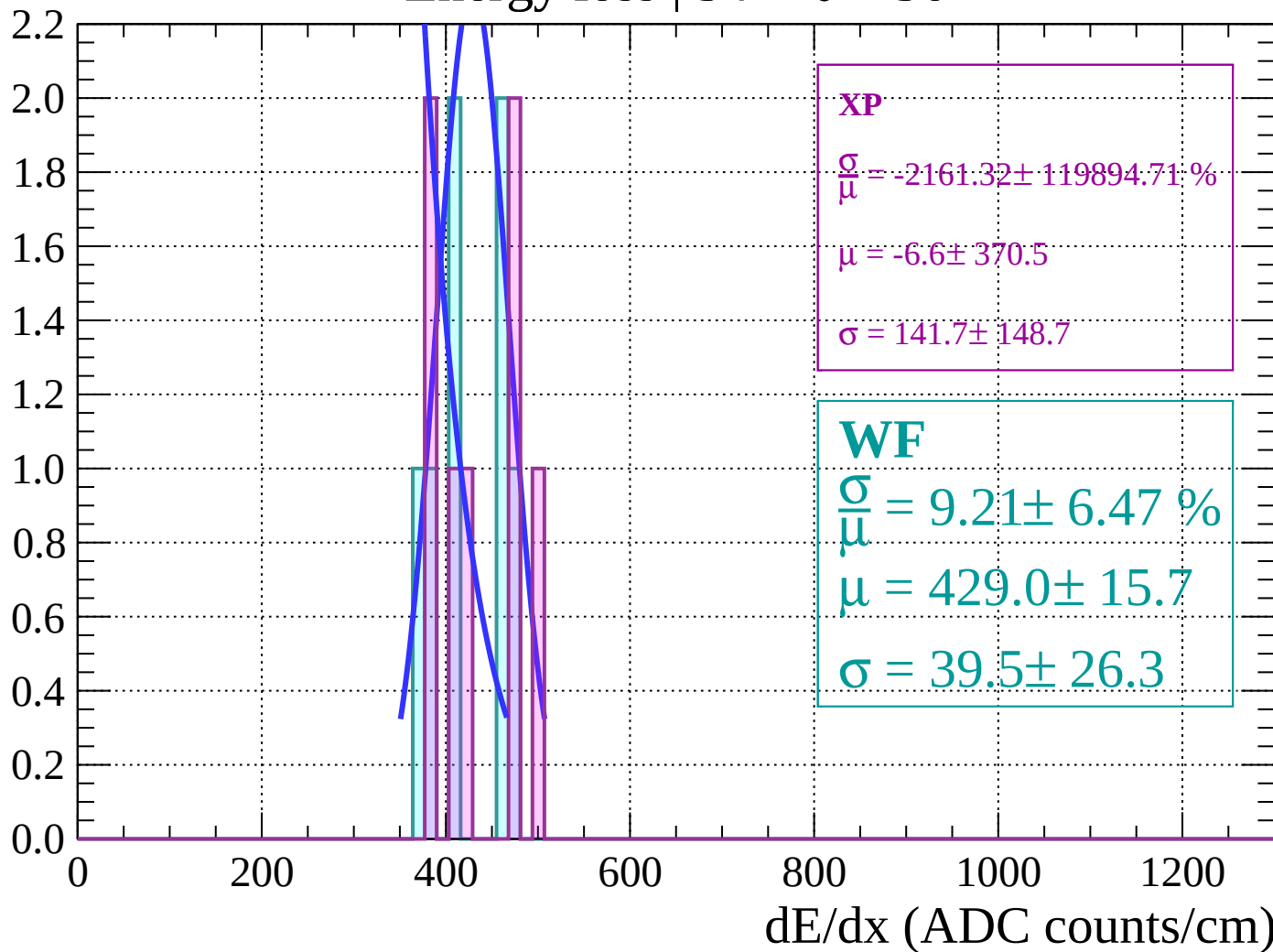
Energy loss |  $52 < \theta < 54$

Count



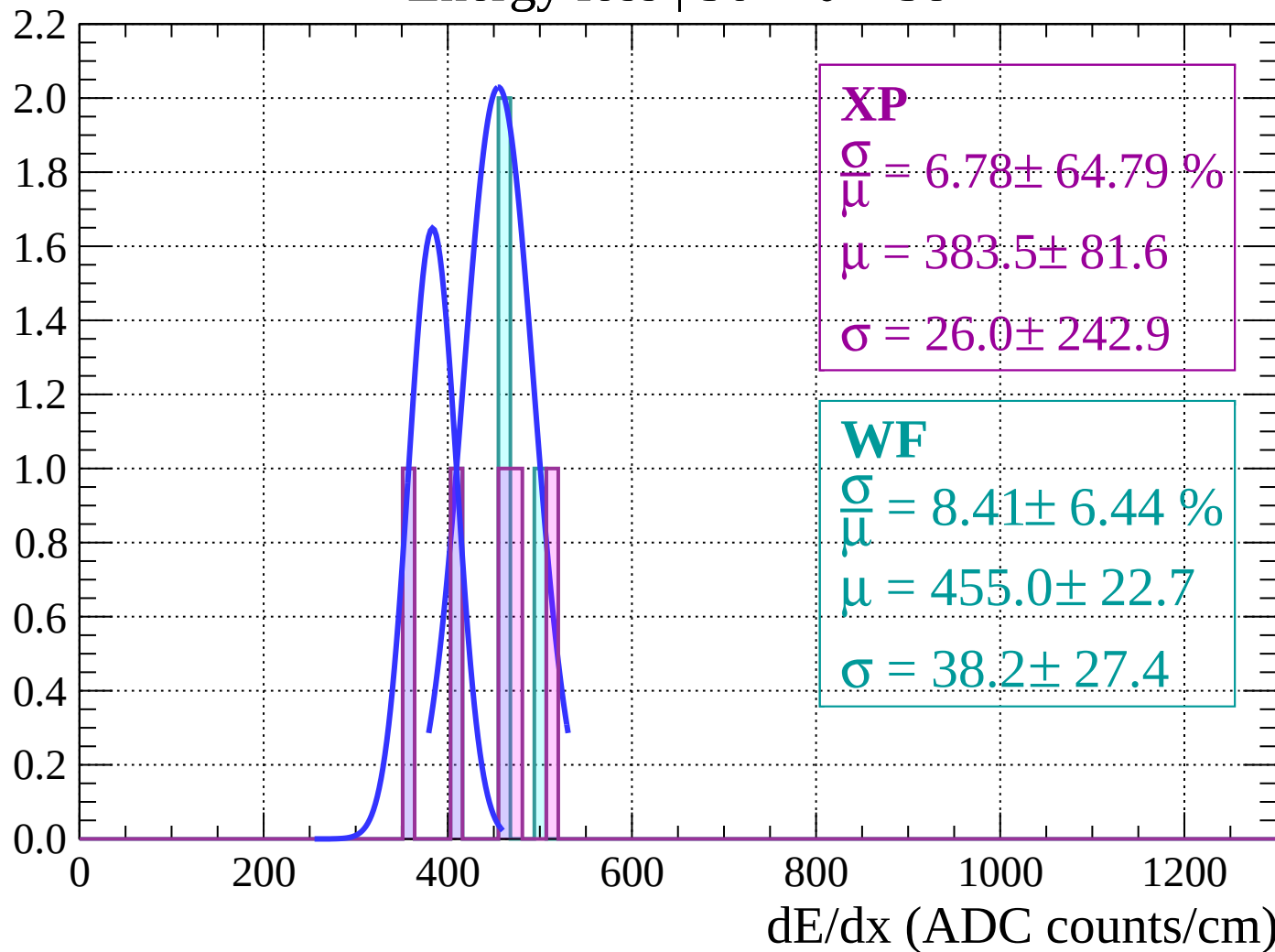
Energy loss |  $54 < \theta < 56$

Count



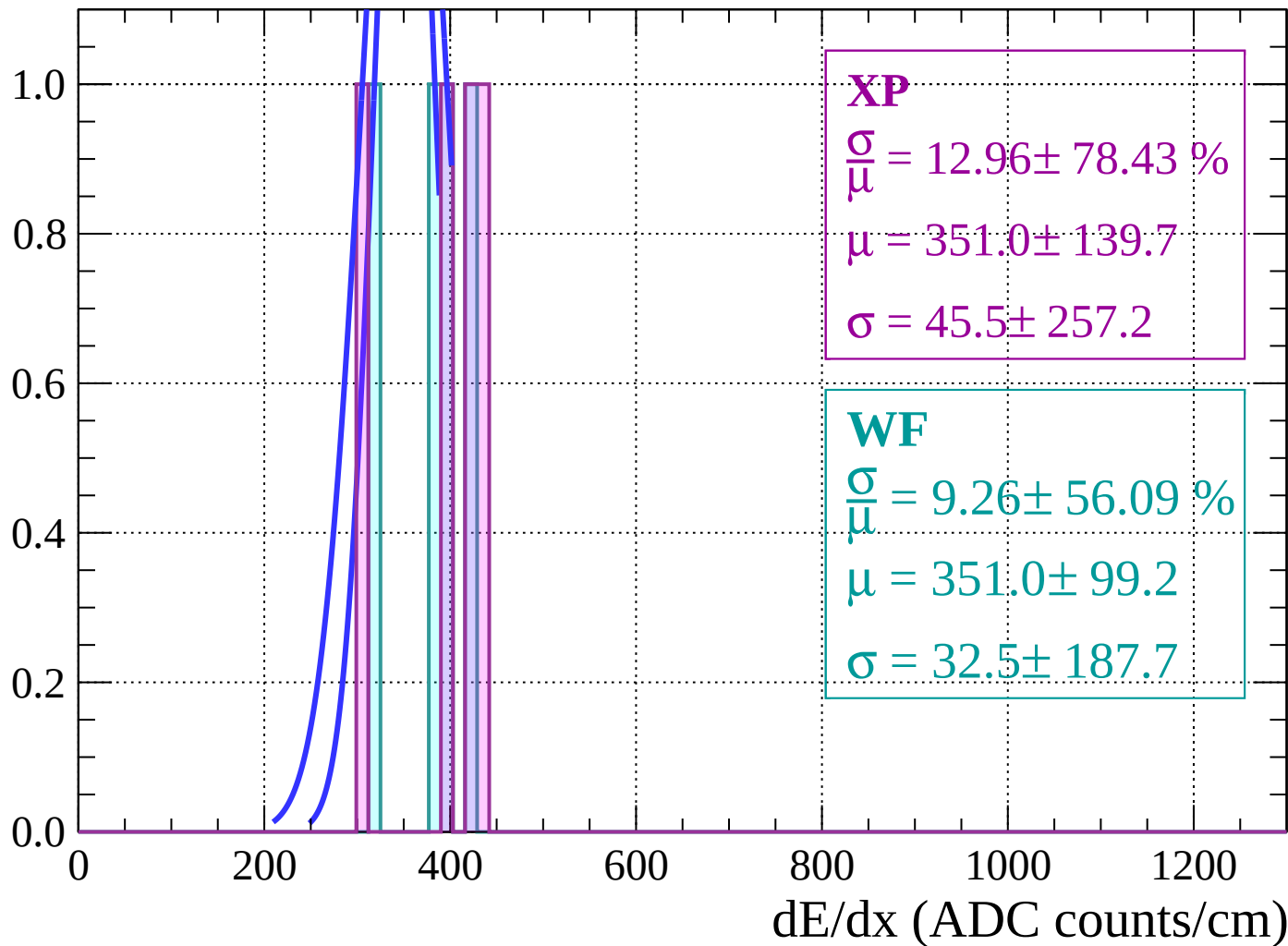
Energy loss |  $56 < \theta < 58$

Count



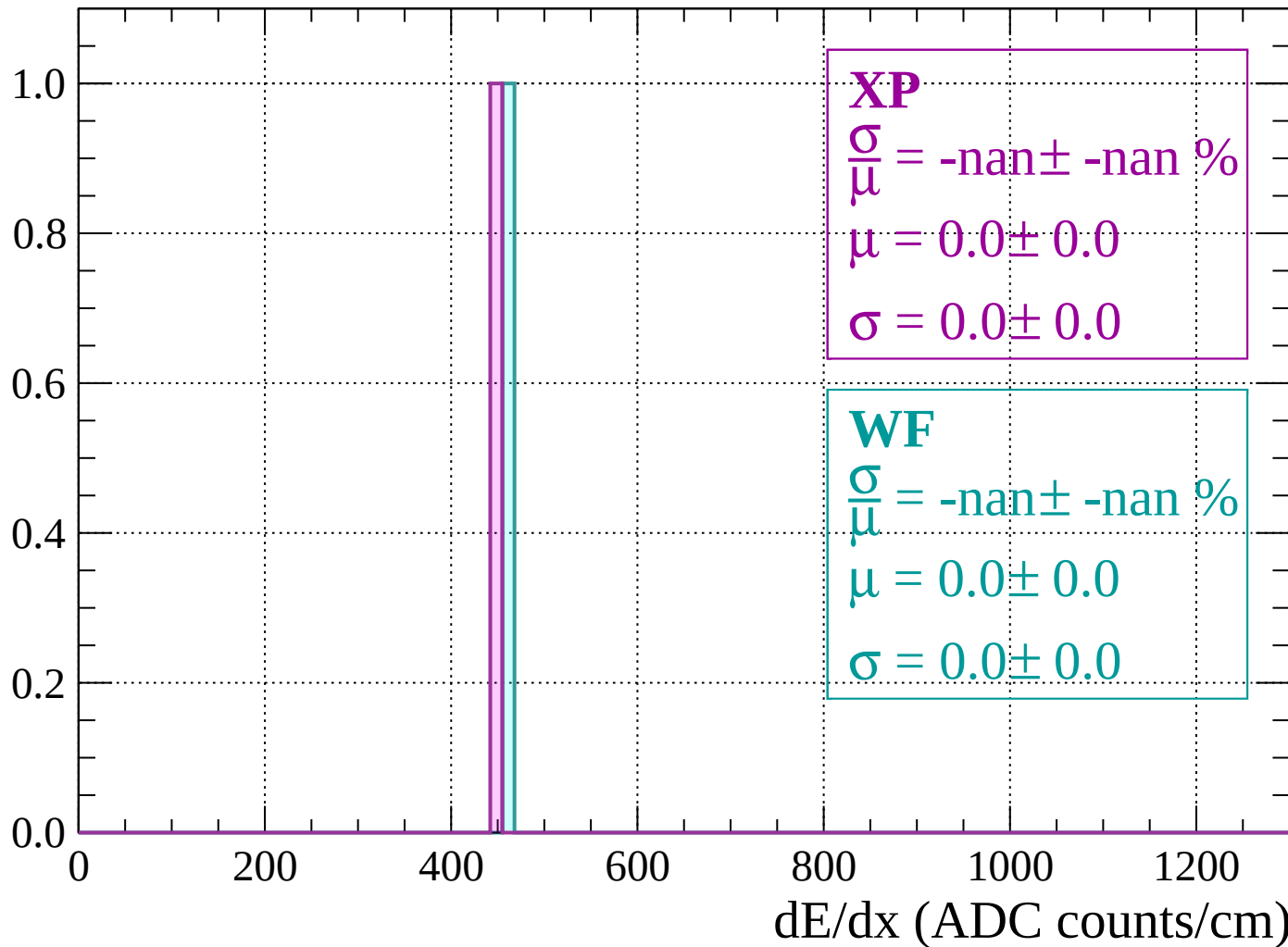
Energy loss |  $58 < \theta < 60$

Count



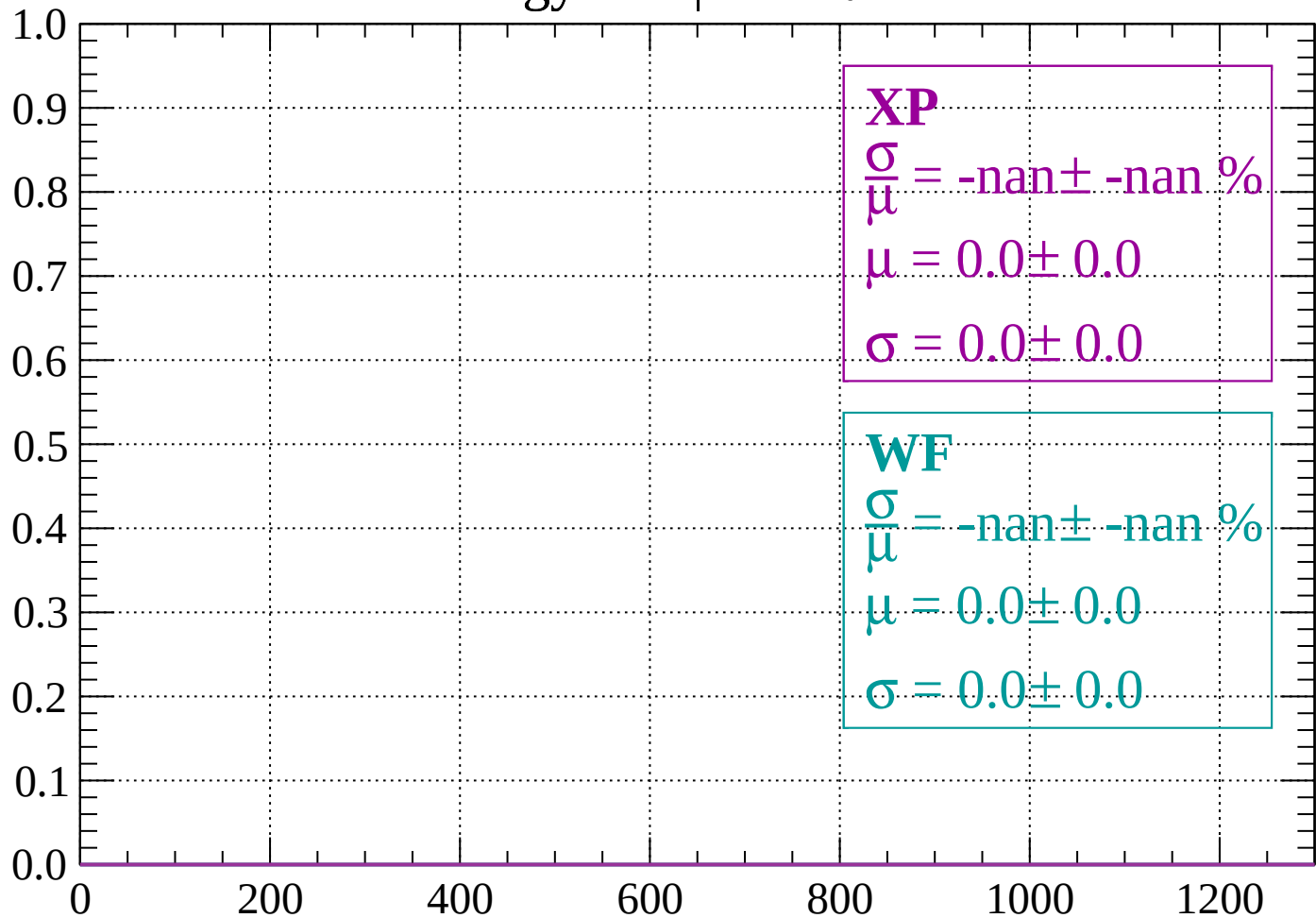
Energy loss |  $60 < \theta < 62$

Count



Energy loss |  $62 < \theta < 64$

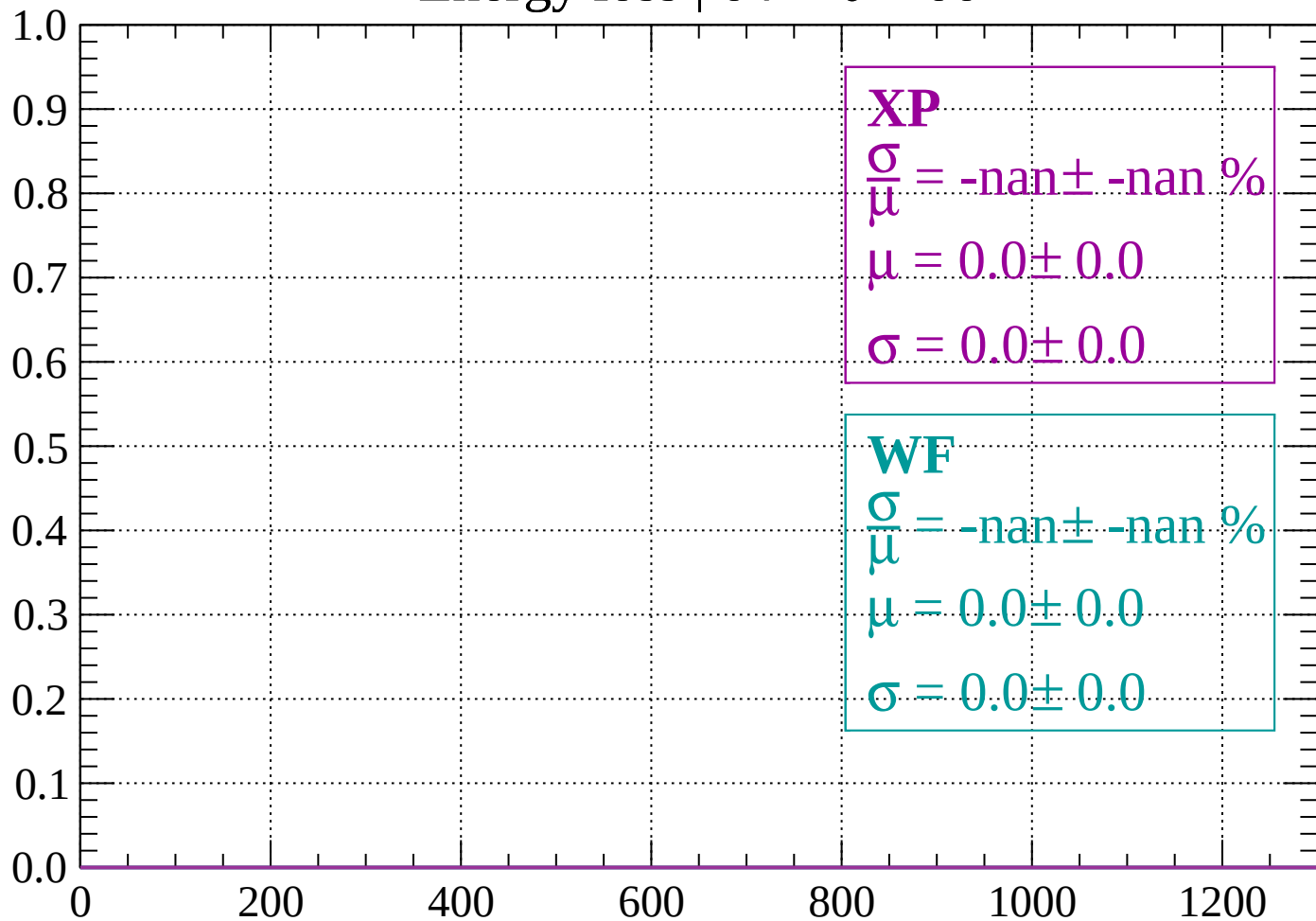
Count



dE/dx (ADC counts/cm)

Energy loss |  $64 < \theta < 66$

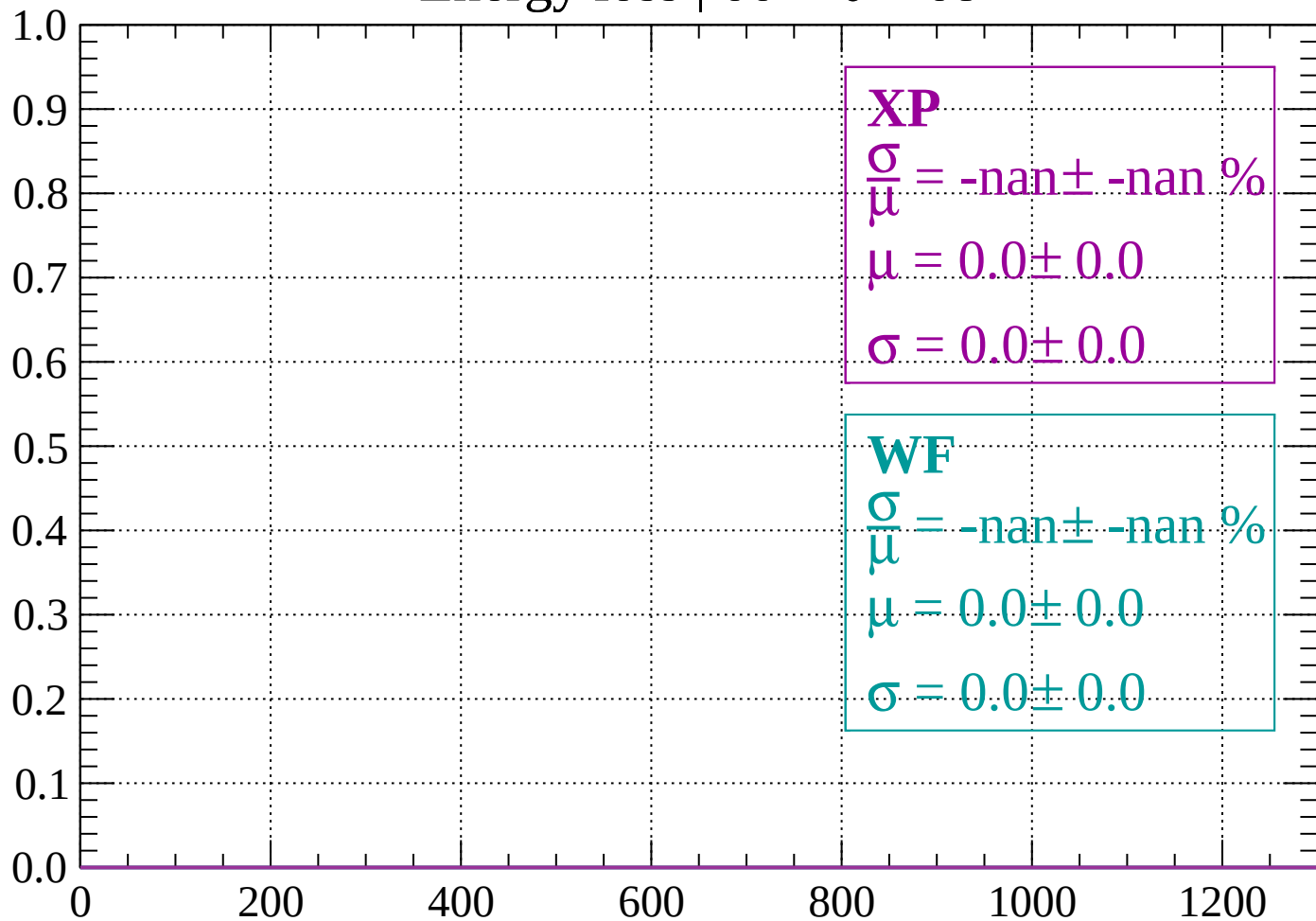
Count



dE/dx (ADC counts/cm)

Energy loss |  $66 < \theta < 68$

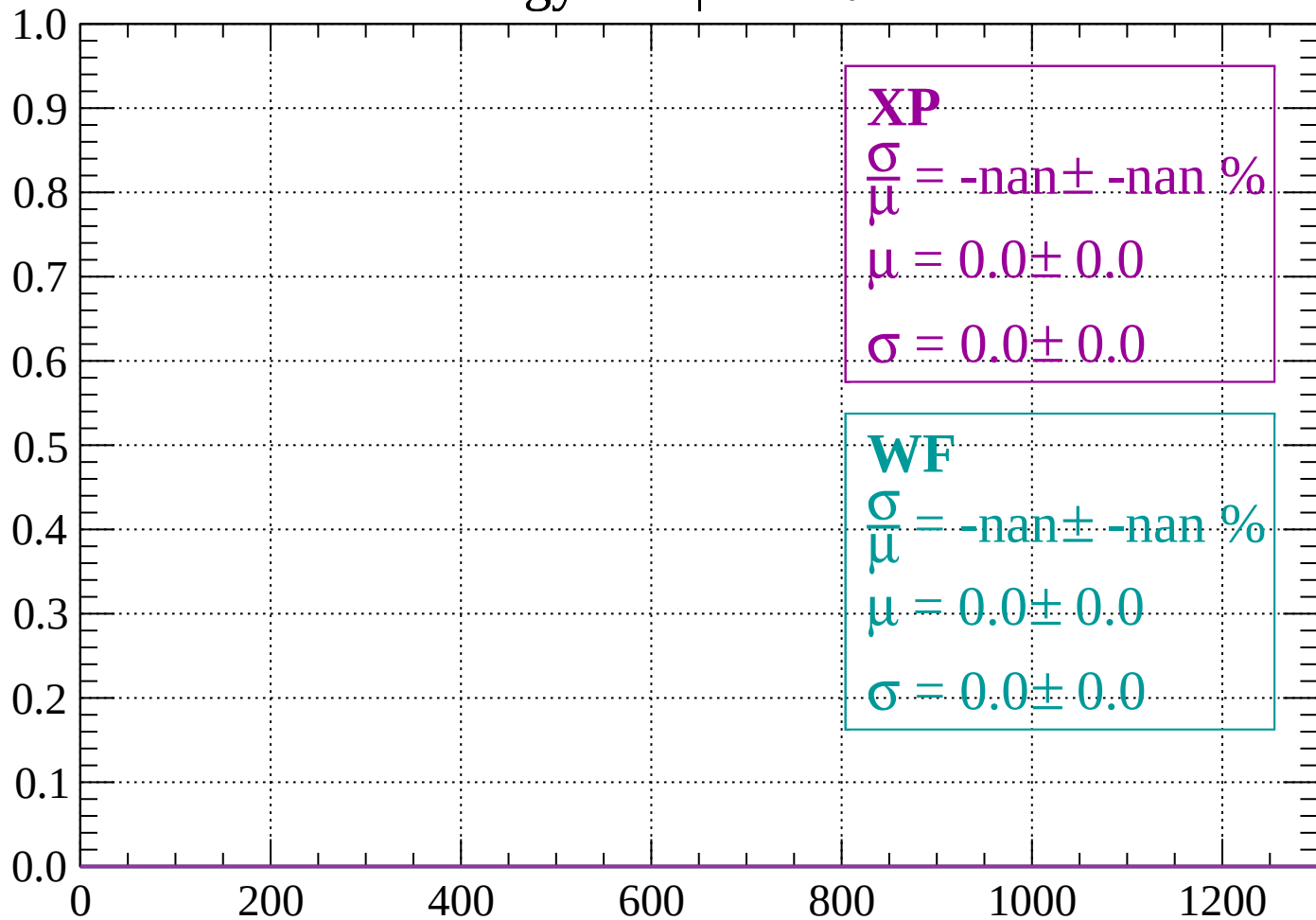
Count



dE/dx (ADC counts/cm)

Energy loss |  $68 < \theta < 70$

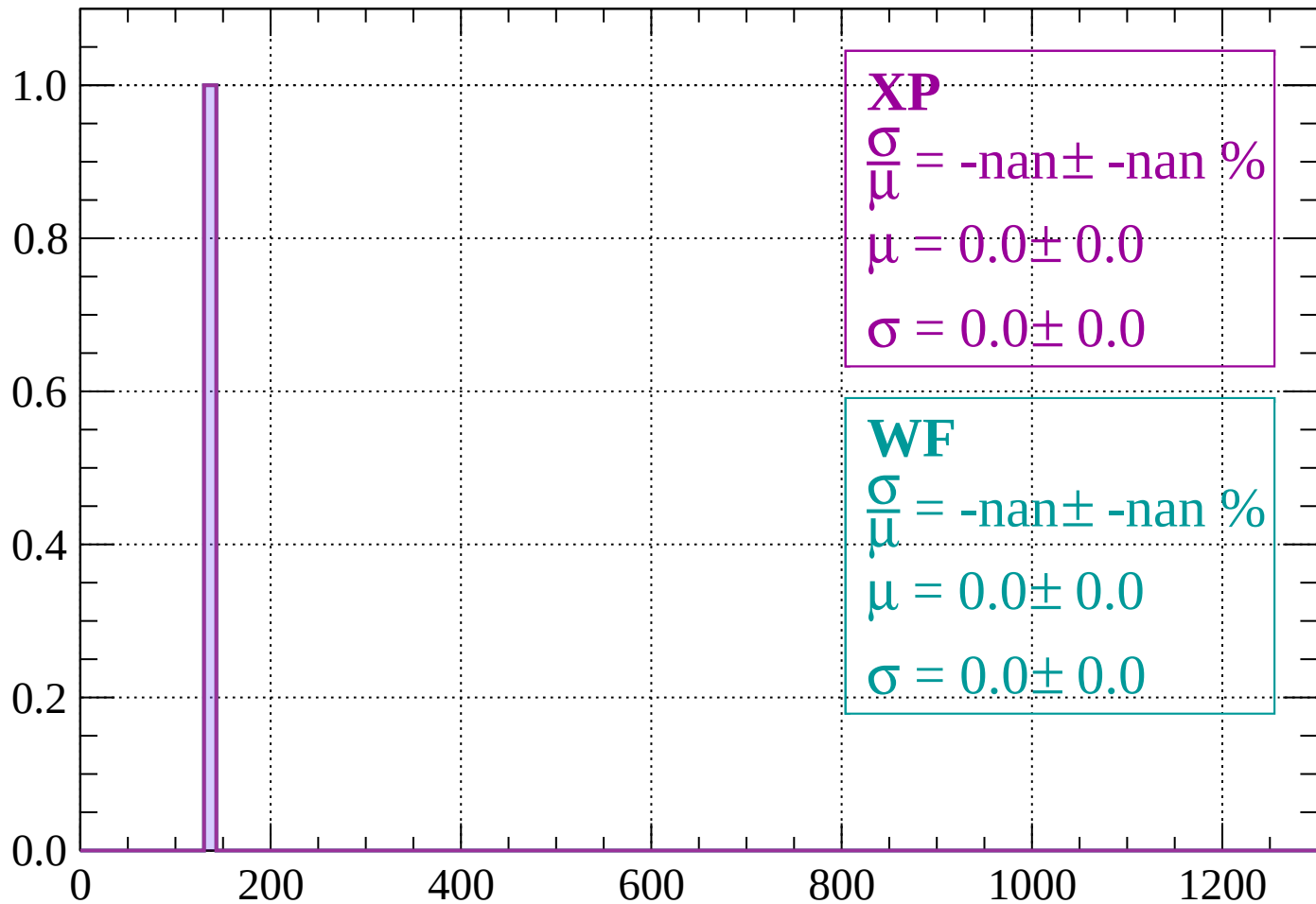
Count



dE/dx (ADC counts/cm)

Energy loss |  $70 < \theta < 72$

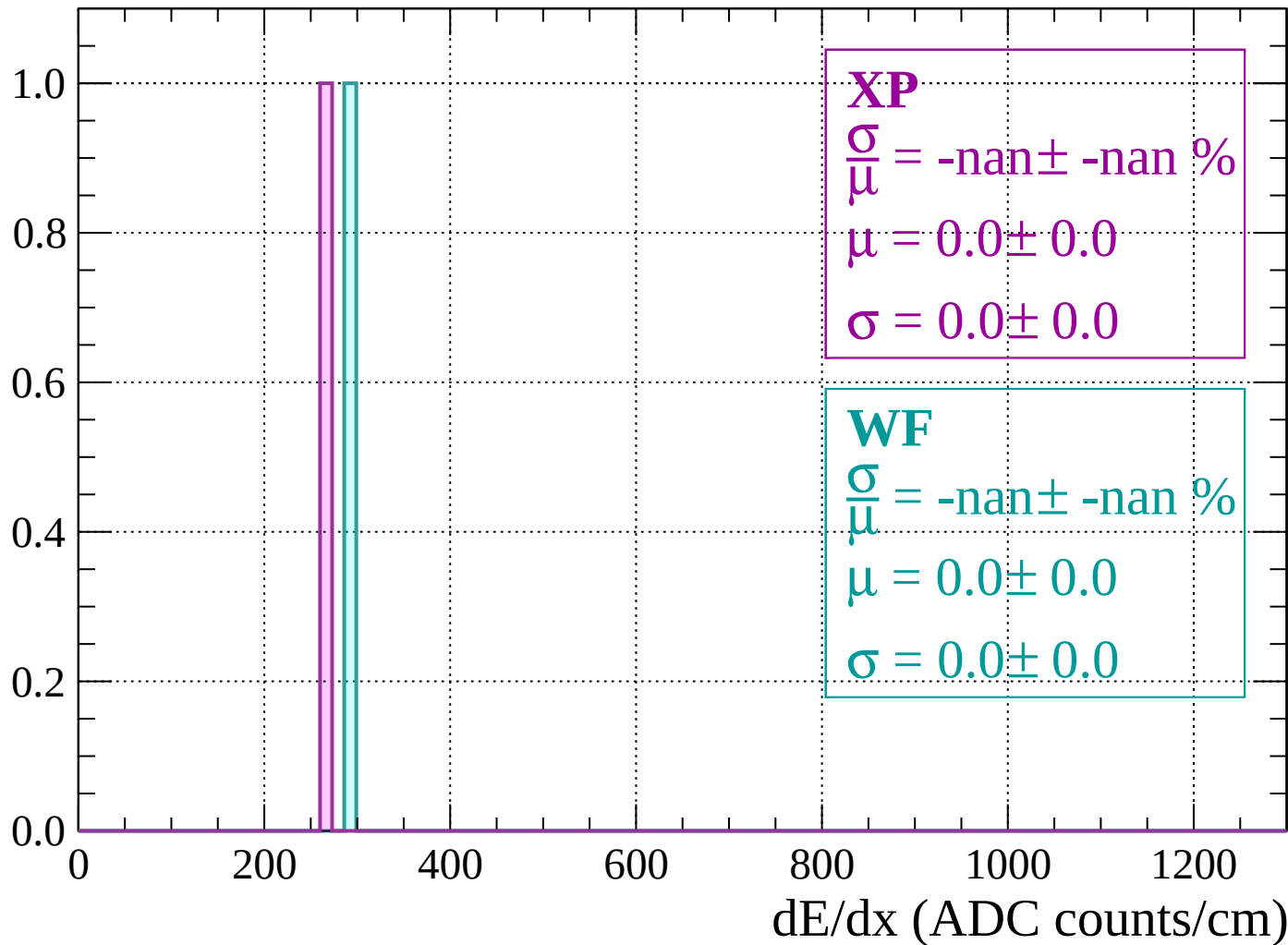
Count



$dE/dx$  (ADC counts/cm)

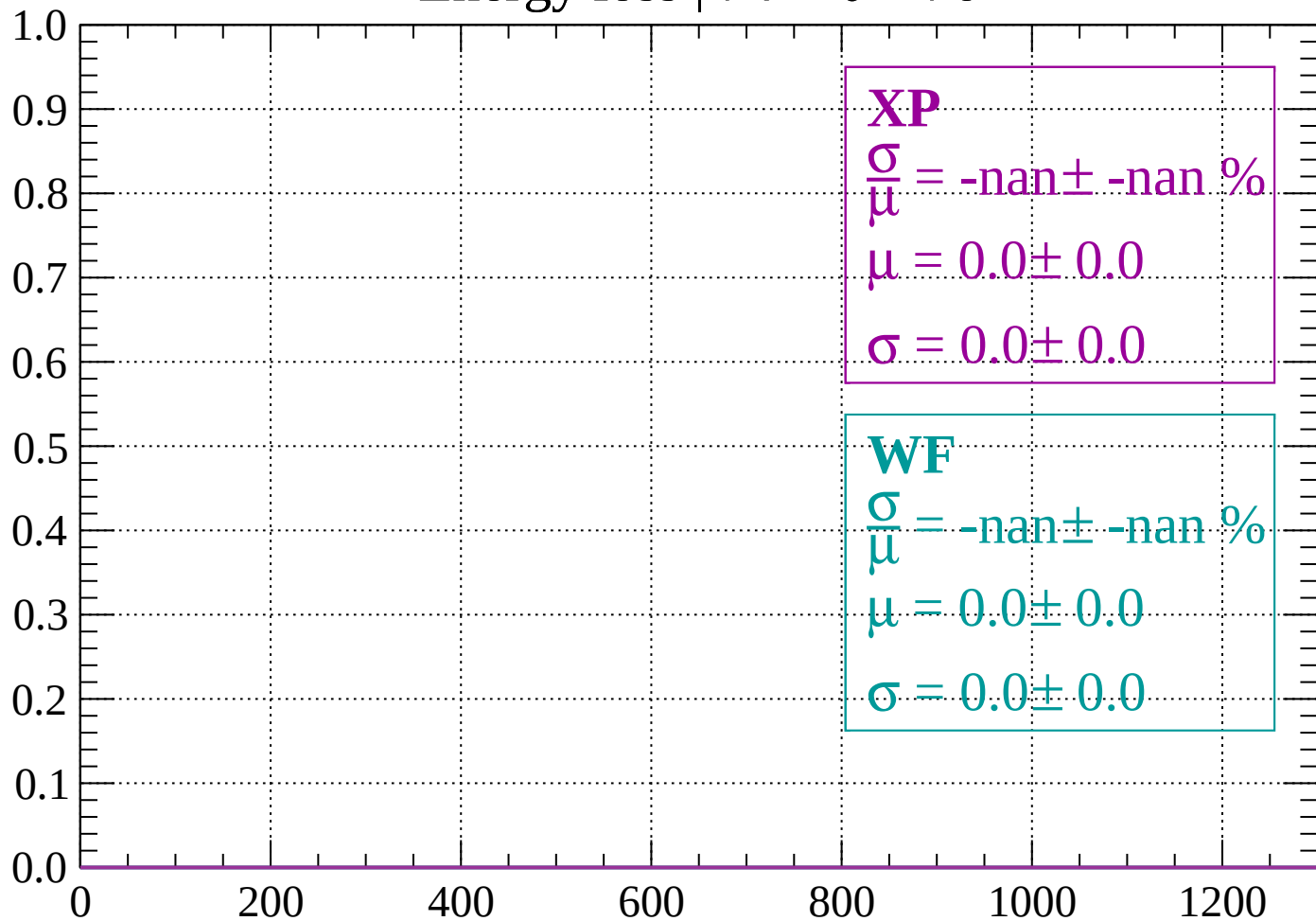
Energy loss |  $72 < \theta < 74$

Count



Energy loss |  $74 < \theta < 76$

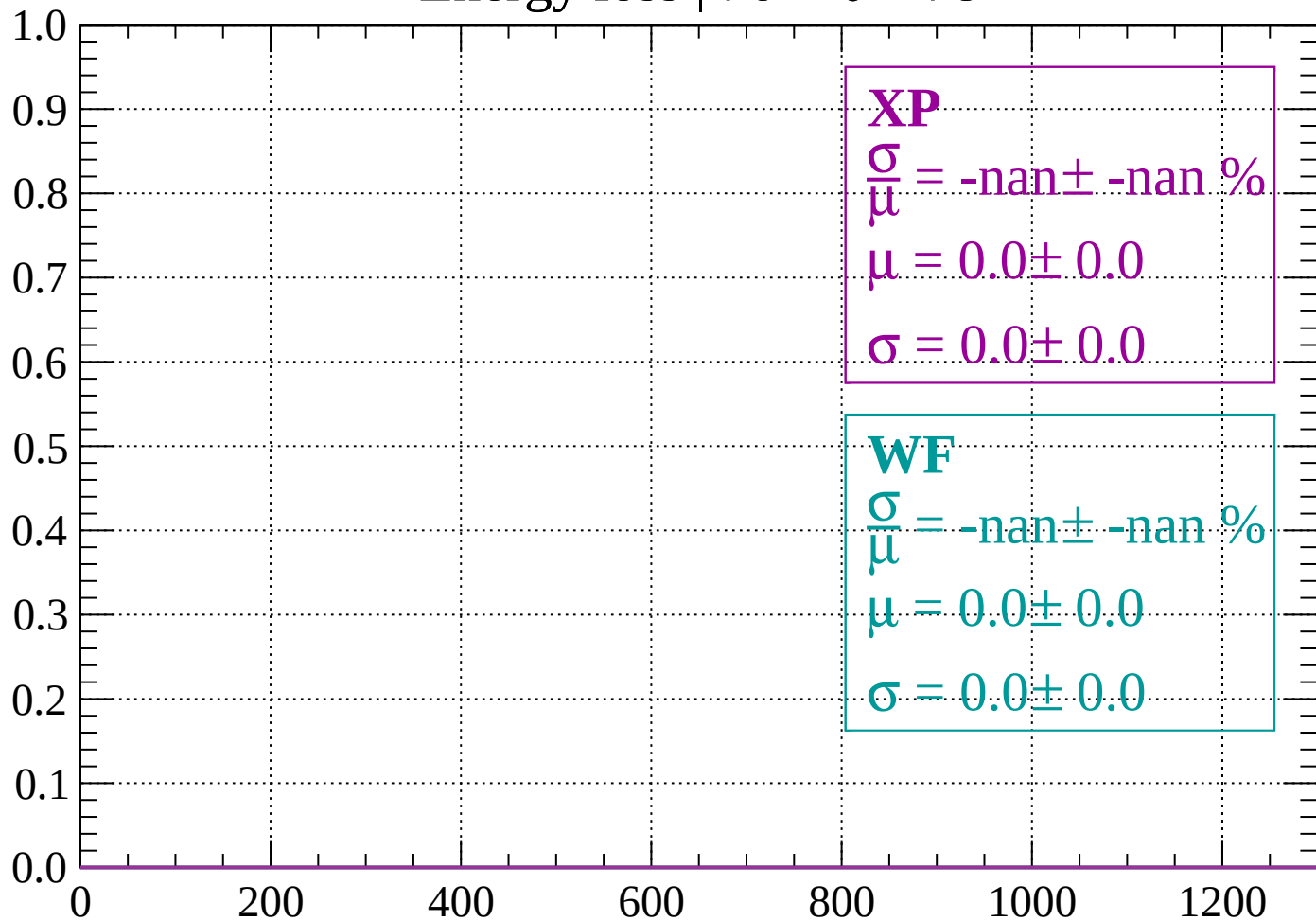
Count



dE/dx (ADC counts/cm)

Energy loss |  $76 < \theta < 78$

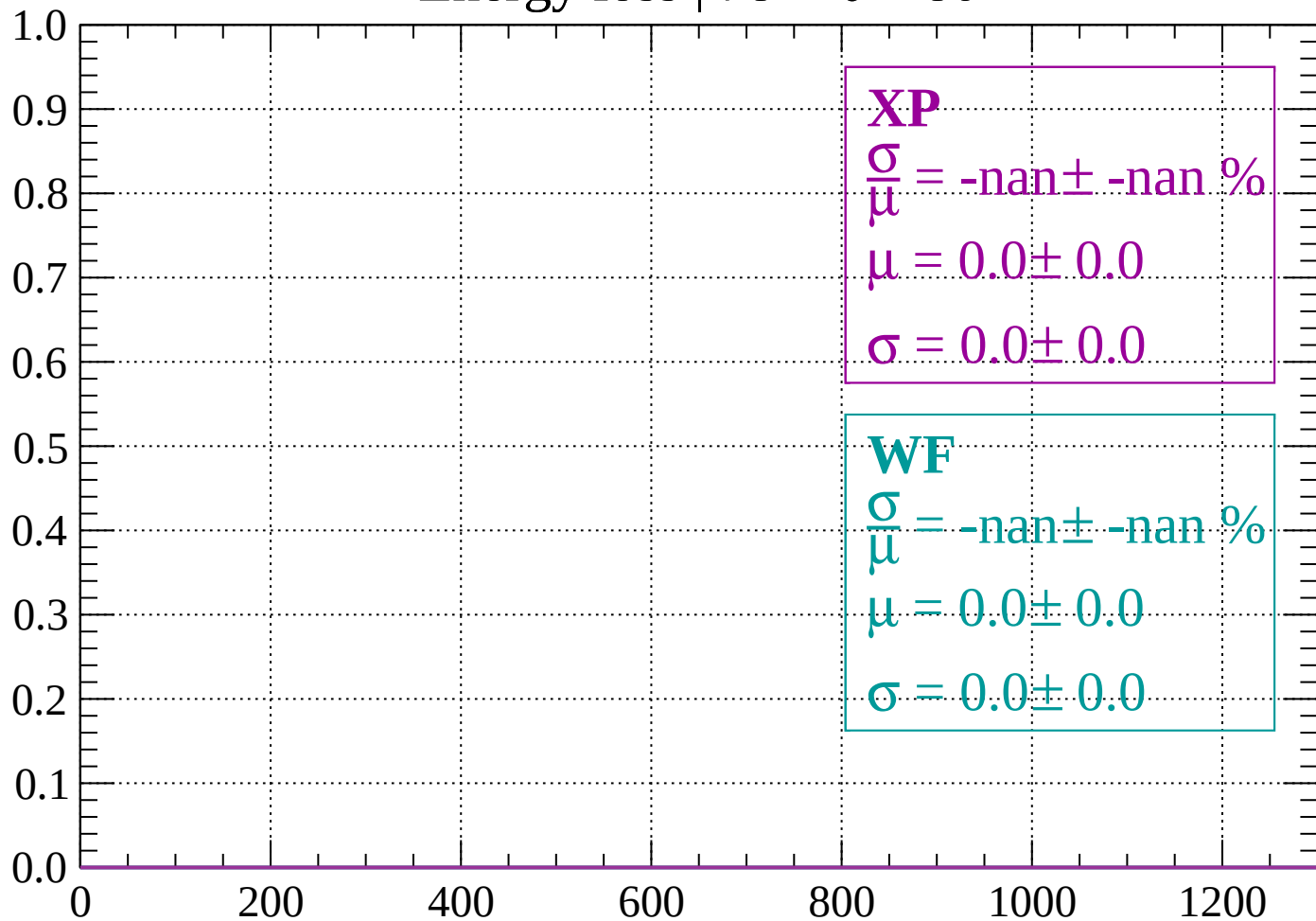
Count



dE/dx (ADC counts/cm)

Energy loss |  $78 < \theta < 80$

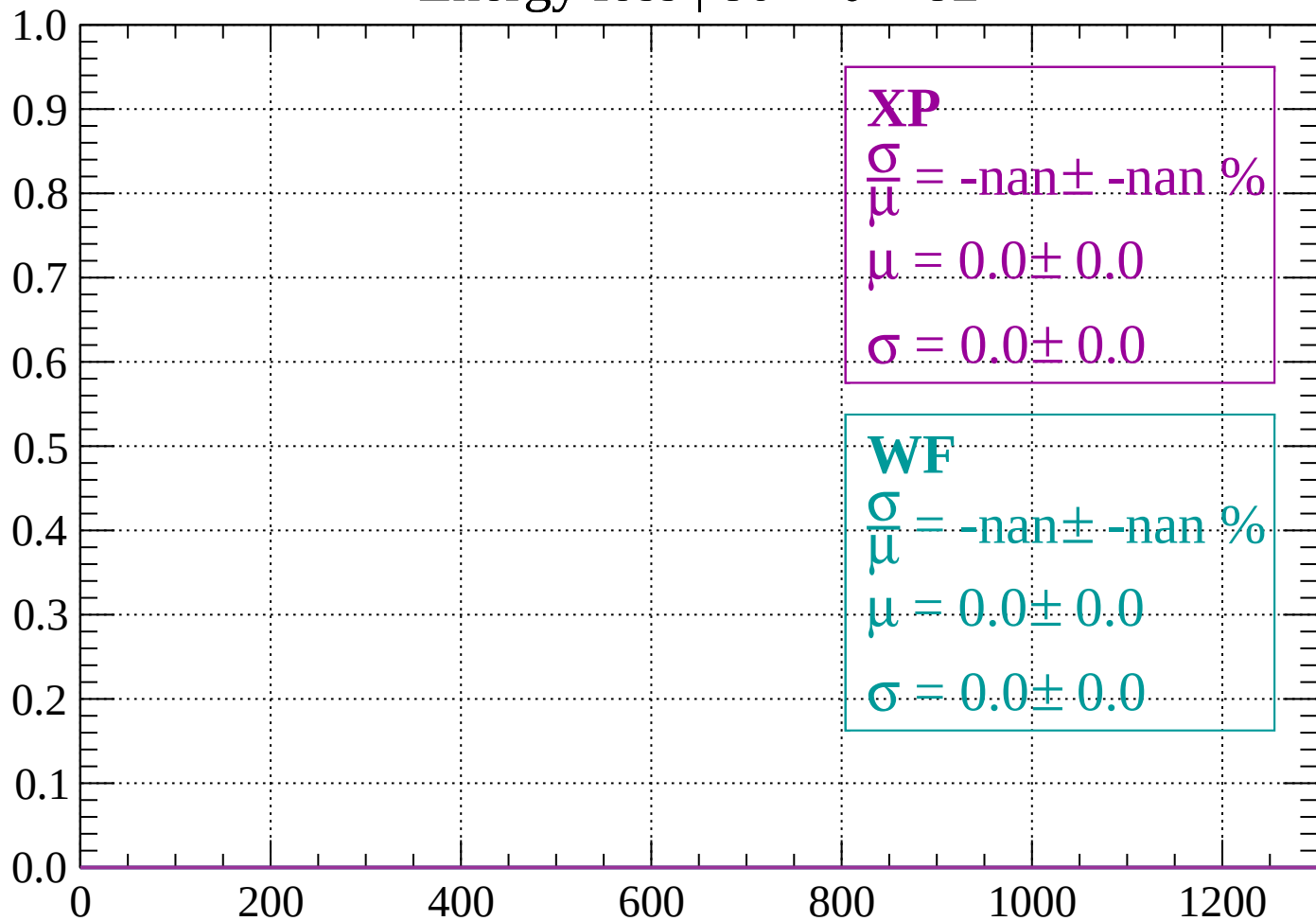
Count



dE/dx (ADC counts/cm)

Energy loss |  $80 < \theta < 82$

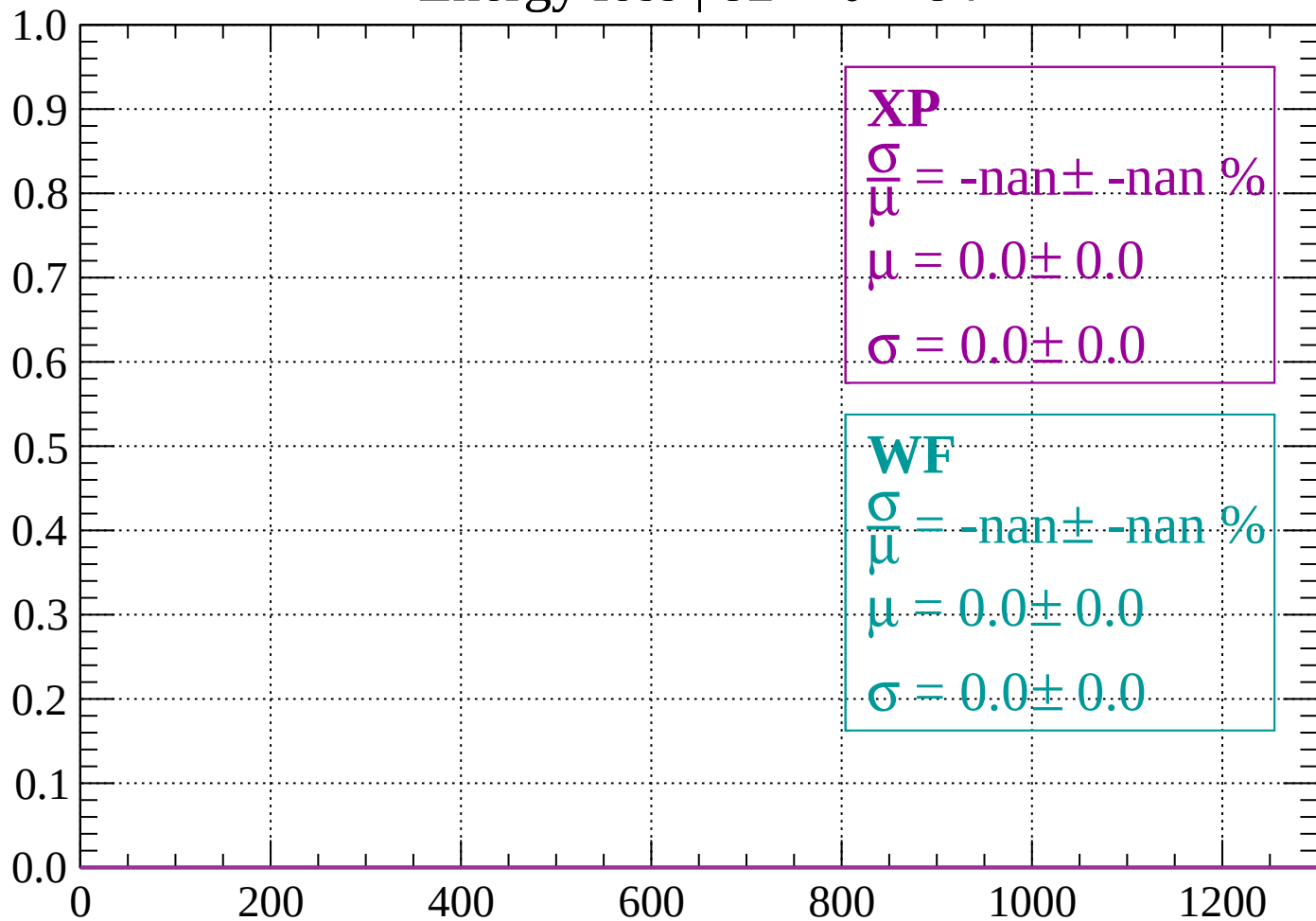
Count



dE/dx (ADC counts/cm)

Energy loss |  $82 < \theta < 84$

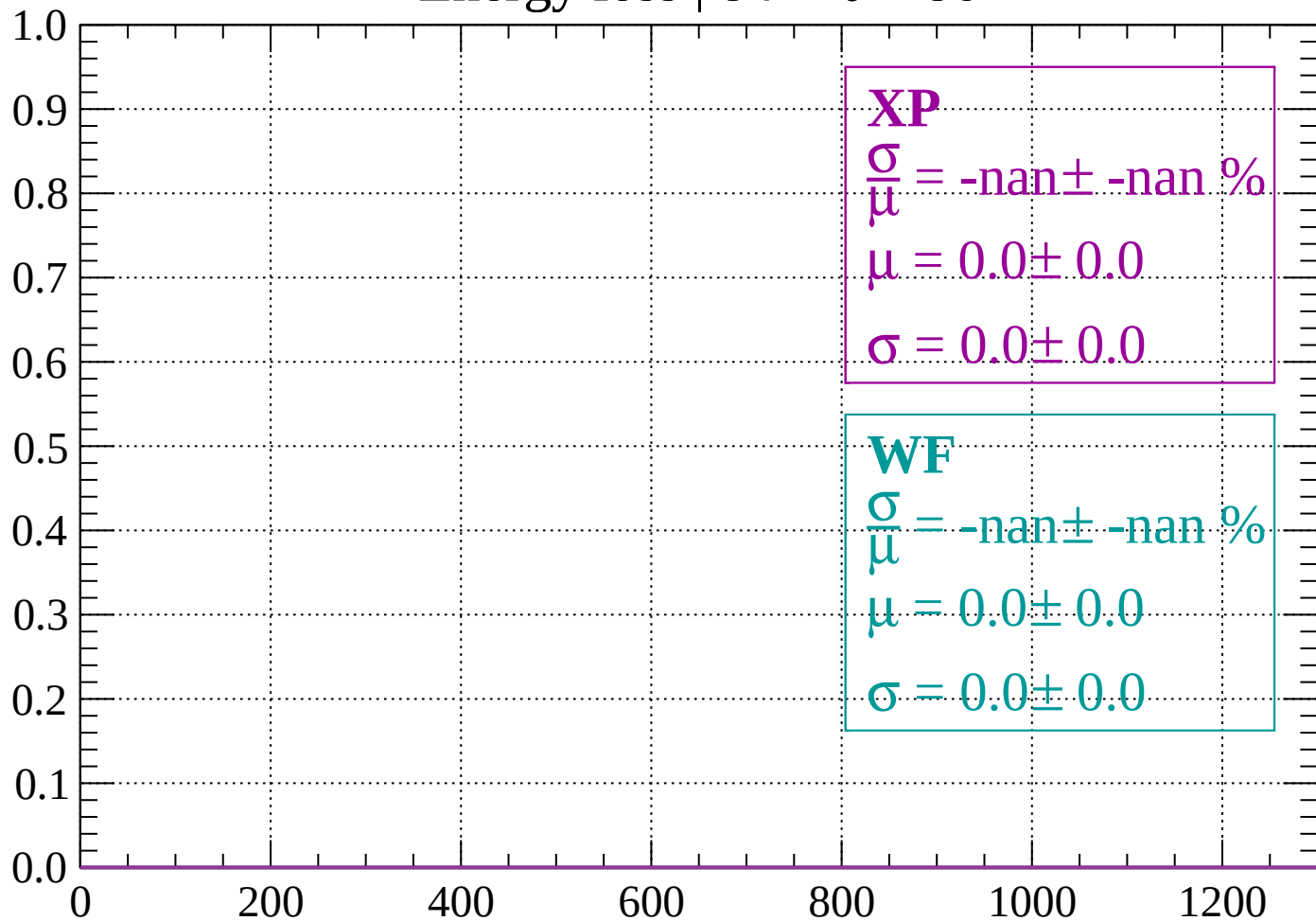
Count



dE/dx (ADC counts/cm)

Energy loss |  $84 < \theta < 86$

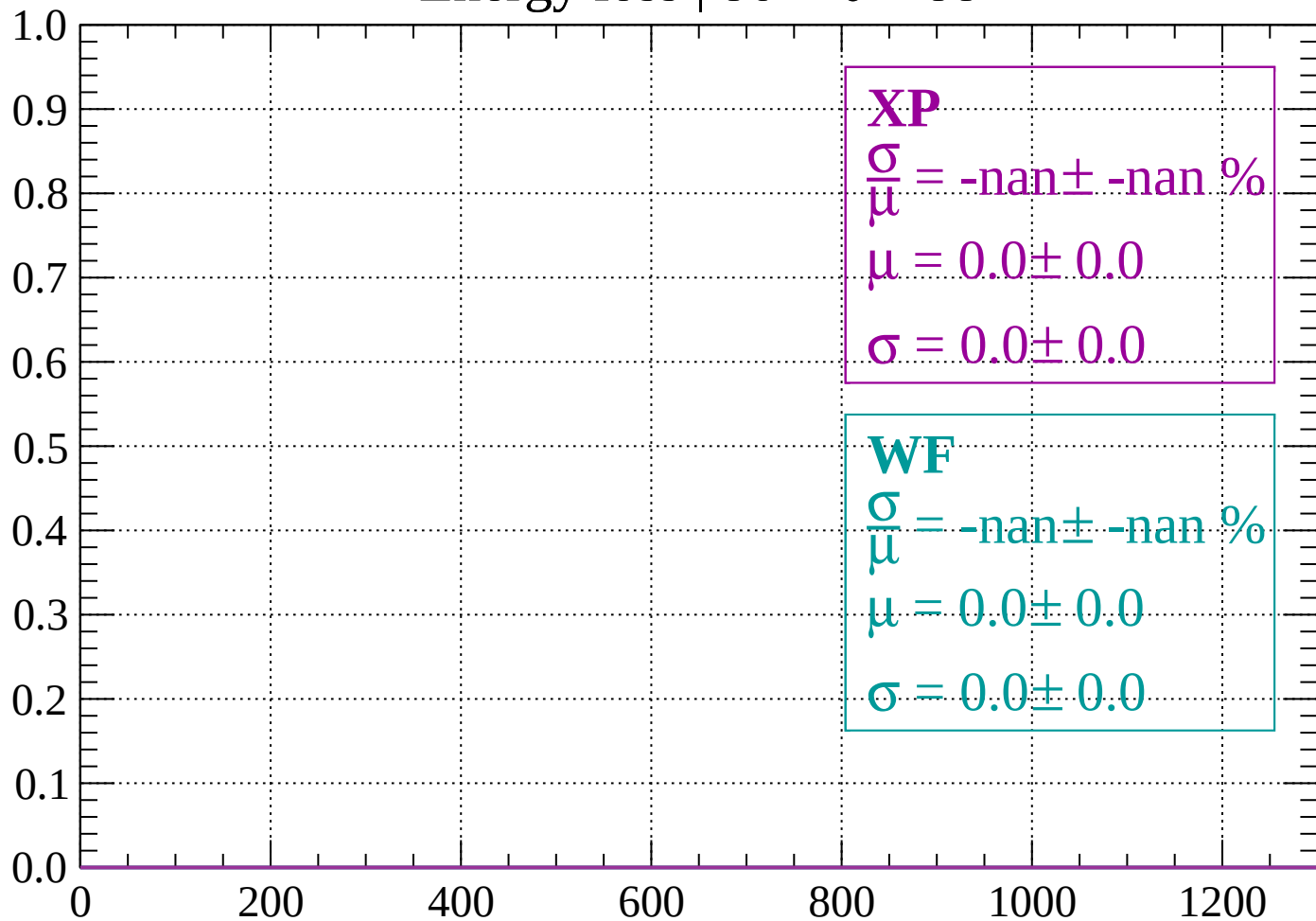
Count



dE/dx (ADC counts/cm)

Energy loss |  $86 < \theta < 88$

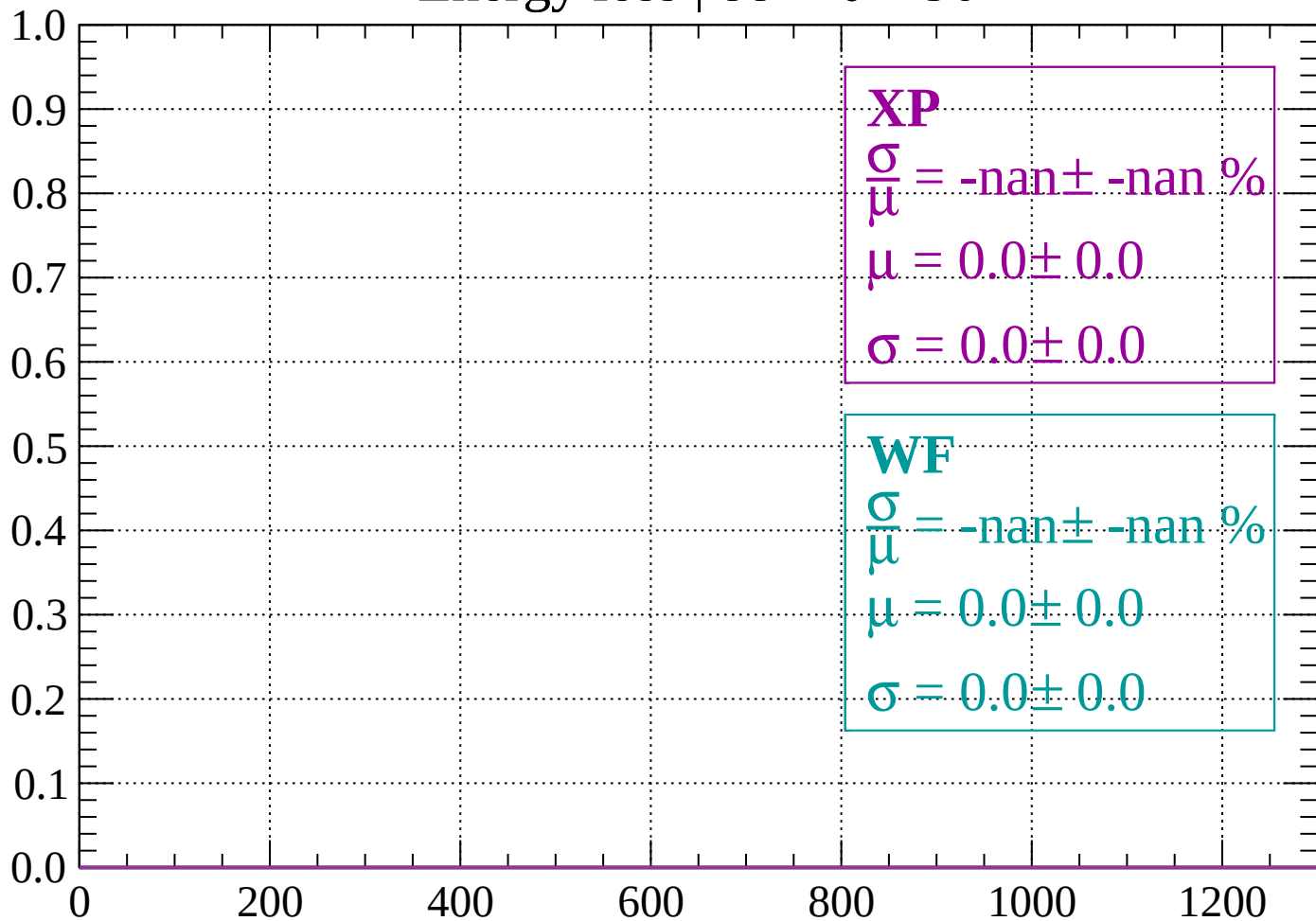
Count



dE/dx (ADC counts/cm)

Energy loss |  $88 < \theta < 90$

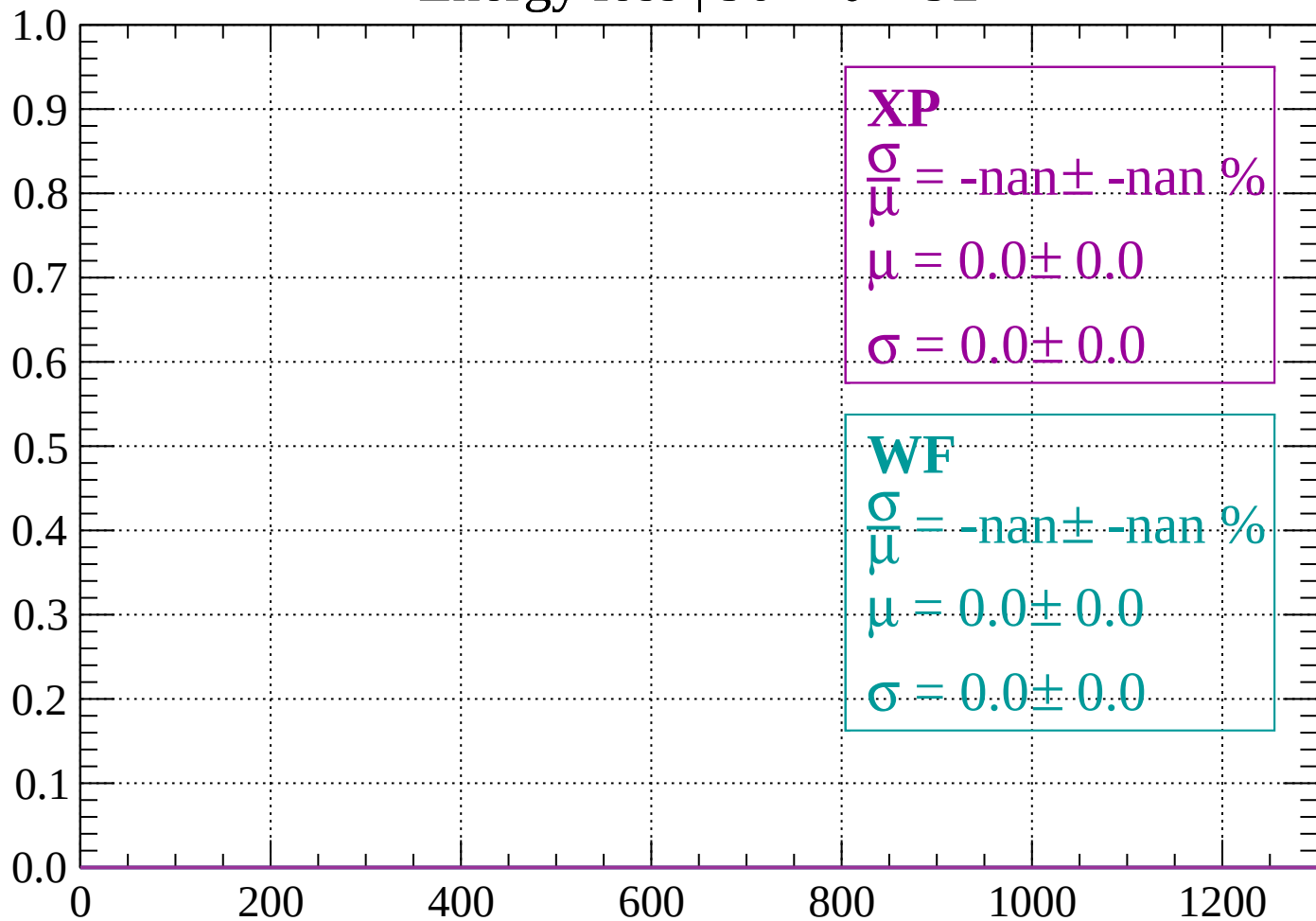
Count



dE/dx (ADC counts/cm)

Energy loss |  $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

